NAVFAC

Naval Facilities Engineering Systems Command

NAVFAC MIDLANT, Public Works Department, Maine

eProjects Work Order No.: 1740579

SPECIFICATIONS

B60: PRODUCTION FACILITY REPAIRS - SIOP

At the

Portsmouth Naval Shipyard, Kittery, Maine

LICENSED ARCHITECT
JOSHUA
M.
RILEY
No. 04278
THE STATE OF NEW HAMPSHIRE
10-01-25

Volume 1 of 3

PREPARED BY:

NAVFAC Midlant PWD ME
Building 59
Portsmouth Naval Shipyard
Kittery, Maine 03801-2032

REQUEST FOR PROPOSAL PREPARED BY:

Architectural: Josh Riley, R.A.
Civil: Dan Philipps, P.E.
Structural: David N. Martin, P.E., S.E.
Mechanical: Rick Creswell, P.E.
Electrical: Dale Lincoln, P.E.
Fire Protection: Jeff DeMaine, P.E.
Environmental: N/A
Estimating: Toscano Clements Taylor Cost Consultants, LLC

Scott Hughes, R.A. Digitally signed by Scott Hughes, R.A.
Date: 2025.12.10 09:52:31 -05'00'

Submitted By: Scott Hughes, R.A.	Signature	Date: October 1, 2025

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Date: 2025.12.11 13:01:12-05'00'

Approved By PME	Signature	Date:

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Date: 2025.12.11 13:05:21 -05'00'

Design Manager:	Signature	Date:

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HAVING PARTICIPATED IN THE DESIGN OF EPROJECT NO. 1740579, PORTSMOUTH NAVAL SHIPYARD, KITTERY MAINE, AND HAVING THOROUGHLY REVIEWED THE COMPLETED PROJECT DOCUMENTS, I DECLARE THAT THE FACILITY DESIGN INCLUDED HEREIN COMPLIES WITH ALL APPLICABLE UNITED STATES CODES AND LAWS.

10/01/2025

LICENSED ARCHITECT
JOSHUA
M.
RILEY
No. 04278
THE STATE OF NEW HAMPSHIRE
10-01-25

SIGNATURE | DATE | PROFESSIONAL SEAL

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02/11, CHG 1: 08/14

PART 1 GENERAL

1.1 SUMMARY

This Section lists the drawings for the project pursuant to Contract Clause "DFARS 252.236-7001, Contract Drawings, Maps and Specifications."

1.2 CONTRACT DRAWINGS

Contract drawings are as follows:

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DWG. NO	NAVFAC DWG NO.	PWD ME DWG NO.	TITLE
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HM102			FIRST FLOOR HAZARDOUS MATERIALS PART PLAN - WEST
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CG101			GRADING, DRAINAGE AND EROSION CONTROL PLAN
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DWG. NO	NAVFAC DWG NO.	PWD ME DWG NO.	TITLE
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SD106			SECOND FLOOR FRAMING REMOVALS PLAN
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AE223			BUILDING SECTIONS 3
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AE302			SECTION DETAILS
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AE411			ENLARGED ELEVATOR PLANS AND SECTIONS
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AE622			WINDOW REPAIR AND RESTORATION NOTES AND SCHEDULE
AE623			WINDOW REPAIR AND RESTORATION WINDOW TYPES
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AE640			ROOM FINISH SCHEDULE
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AE703			FIRST FLOOR MEZZANINE REFLECTED CEILING PLAN
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PD103			FIRST FLOOR PLUMBING REMOVALS PLAN - WEST
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MP101			FIRST FLOOR MECHANICAL PIPING PLAN - EAST
MP102			FIRST FLOOR MECHANICAL PIPING PLAN - WEST
MP103			FIRST FLOOR MEZZANINE MECHANICAL PIPING PLAN - EAST
MP104			FIRST FLOOR MEZZANINE MECHANICAL PIPING PLAN - WEST
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M-401			ENLARGED MECHANICAL PART PLANS
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DWG. NO	NAVFAC DWG NO.	PWD ME DWG NO.	TITLE
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FA105			ATTIC - FIRE ALARM
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DWG. NO	NAVFAC DWG NO.	PWD ME DWG NO.	TITLE
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DWG. NO	NAVFAC DWG NO.	PWD ME DWG NO.	TITLE
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EL103			FIRST FLOOR MEZZANINE LIGHTING PART PLAN - EAST
EL104			FIRST FLOOR MEZZANINE LIGHTING PART PLAN - WEST
EL105			SECOND FLOOR LIGHTING PLAN
EL106			SECOND FLOOR MEZZANINE LIGHTING PLAN
EL107			ATTIC LIGHTING PART PLAN - WEST
EL701			LIGHTING FIXTURE SCHEDULE AND DETAILS
EL702			LIGHTING FIXTURE DETAILS 1
ET101			FIRST FLOOR TECHNOLOGY AND SECURITY PART PLAN - EAST
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ET103			FIRST FLOOR MEZZANINE TECHNOLOGY AND SECURITY PART PLAN - EAST
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ET105			SECOND FLOOR TECHNOLOGY AND SECURITY PLAN
ET106			B59 FIRST FLOOR TECHNOLOGY PLAN
ET501			TELECOMMUNICATIONS RISER DIAGRAMS
ET502			TELECOMMUNICATIONS DETAILS
EY501			SECURITY SYSTEM RISER AND DETAILS

1.3 BORING LOGS

The Government does not guarantee that borings indicate actual conditions, except for the exact locations and the time that they were made. Subsurface data, not specified or indicated, have been obtained by the Government at this station. The data are available for examination by prospective bidders in the office of the Contracting Officer.

1.4 SUBSURFACE DATA

Subsurface data, not specified or indicated, have been obtained by the Government at the station. The data are available for examination by prospective bidders in the office of the Contracting Officer.

-- End of Document --

SECTION 01 11 00.00 22
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SUMMARY OF WORK (PWD ME)
02/24

PART 1 GENERAL

This Section applies to Design-Bid-Build projects at the Portsmouth Naval Shipyard and AOR facilities.

1.1 DIVISION 01 SPECIFICATION REVIEW MEETING

The Contractor must coordinate a Division 01 Specification Review Meeting, to be held within 5 working days after contract award. The meeting must include a discussion to develop a mutual understanding of the project's Division 01 specification requirements including, but not limited to, administrative requirements, QC Plan, and requirements and expectations for the contractor's work sequencing and preparation plan. Refer to 01 30 00.00 22 Administrative Requirements (PWD ME) Division 01 Specification Review meeting paragraph for additional information.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

Work Sequencing and Preparation Plan; G

Salvage Plan; G

SD-11 Closeout Submittals

RZ Checklist; G

1.3 PERIOD OF PERFORMANCE

Base Bid: The Government has estimated that the onsite contract work performance period 608 calendar days. The mobilization date is contingent upon the completion of current ongoing projects which have an estimated completion time of late summer 2026.

a. Option 1: Furniture, Fixtures, and Equipment (FF&E) - The onsite contract work performance period is 90 calendar days and is Concurrent with Base Bid period performance.

b. Option 2: Industrial Equipment - The onsite contract work performance period is 365 calendar days and is Concurrent with Base Bid period of performance.

c. Option 3 - Radon Fans: Government to award/execute Option upon final testing of passive radon system if results indicate the requirement for active system. The onsite contract work performance period is 40 calendar days.

d. Option 4: Electronic Security Systems (ESS) - Onsite contract work performance period is 90 calendar days and is Concurrent with Base Bid period performance.

e. Option 5: Additional Window Restoration - The onsite contract work performance period is 186 calendar days and is concurrent with base bid.

f. Option 6: Not applicable.

g. Option 7: Additional Annex Repairs - The onsite contract work performance period is 186 calendar days and is Concurrent with Base Bid period performance.

h. Option 8: Additional Finishes - The onsite contract work performance period is 186 calendar days and is Concurrent with Base Bid period performance.

i. Option 9: Electrical Service Infrastructure - Government to award/execute Option within 365 days of Contract Award date. The onsite contract work performance period is 186 calendar days and is Concurrent with Base Bid period performance.

j. Option 10: Resinous Flooring - The onsite contract work performance period is 186 calendar days and is Concurrent with Base Bid period performance.

k. Option 11: Communication Service Routing - The onsite contract work performance period is 186 calendar days and is Concurrent with Base Bid period performance.

1.3.1 Contractor Mobilization After Contract Award

The Government will provide the Contractor with a 75 calendar day notice when the ongoing project contract(s) will be completing and the Contractor will be able to commence work. A Preconstruction meeting will be held 30 calendar days prior to mobilization.

Within 60 calendar days of award the Government and the Contractor must hold a Post Award Meeting to review timelines and review proposed path forward.

1.4 WORK COVERED BY CONTRACT DOCUMENTS

1.4.1 Project Description

The work includes structural upgrades and repairs and upgrading life safety, egress, and accessibility components in the building to meet current code requirements. The project includes providing an elevator, adding building wide sprinkler system, and fire alarm system. The project scope includes hazardous materials abatement, structural replacement/repairs, power distribution and system upgrades, building envelope repairs, interior space renovations, utility upgrades, and incidental related work. Phasing and coordination with Shipyard operations is paramount and mission critical elements need to remain fully operational throughout the duration of the project. Building 60 is an historic facility and requires special coordination to have limited impact on the historic fabric and character defining features of the structure. Work must comply with the SECRETARY OF THE INTERIOR'S STANDARDS FOR THE

TREATMENT OF HISTORIC PROPERTIES. Cross disciplinary coordination drawings are required for review and approval by the Contracting Officer and the PWD-ME Cultural Resources Manager prior to commencement of work. Refer to Section 01 30 10.00 22 COORDINATION PROCEDURES AND COORDINATION DRAWINGS (PWD ME).

Project Options Include:

Option 1: Furniture, Fixtures, and Equipment (FF&E)
Option 2: Industrial Equipment
Option 3: Radon Fans
Option 4: Electronic Security Systems (ESS)
Option 5: Additional Window Restorations
Option 6: Not Applicable
Option 7: Additional Annex Repairs
Option 8: Additional Finishes
Option 9: Electrical Service Infrastructure
Option 10: Resinous Flooring
Option 11: Communication Service Routing

1.4.2 Location

The work is located at the Portsmouth Naval Shipyard (PNSY), approximately as indicated. The exact location will be shown by the Contracting Officer. Note the "Project Site" is the specific area of construction within the limits of work as defined by the contract documents and any associated off site storage and/or manufacturing/fabrication facilities.

1.5 PROJECT SEQUENCING REQUIREMENTS

Provide Work Sequencing and Preparation Plan indicating how the work is to be accomplished and incorporating Government directives. The intent of this Plan is to show the Government that a complete understanding of the project requirements are understood including subcontractors. The Plan must be carefully prepared and must include every aspect of the project including an outage plan coordinated with the construction schedule that will be required to support the construction and how it is to be accomplished with an absolute minimum of disruption to Government operations. No work may begin on the project until this plan is reviewed and approved by the Contracting Officer. The following is the proposed sequencing for implementation of the project's construction:

a. Construction sequencing plans have been developed in coordination with the Government and are provided within the drawing set. Development of an independent construction phasing plan and construction schedule is the responsibility of the Contractor and must be prepared in coordination with the Contracting Officer and in accordance with Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS). The independent construction phasing plan and construction schedule must be prepared to avoid impacting production operations on the Portsmouth Naval Shipyard.

b. The Government requires and will provide escorts for the Contractor while working inside Building 59. Coordinate with the Contracting Officer to schedule escorts. All work within Building 59 require the presence of an escort. The project schedule developed in accordance with Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS) will be used to coordinate required escorts. Notify

Contracting Officer weekly of escorts required for the subsequent three weeks of work.

1.6 CONTRACTOR PERSONNEL

Provide the following personnel. Other specification sections may require engaging additional personnel not included in the listing below. Restrictions stated here supersede conflicting requirements in other sections. As used in the Restrictions column, "may serve" is conditional upon the person meeting the qualifications specified for each position.

Title	Specifications	Section Title	Restrictions
Project Manager	01 30 00.00 22	ADMINISTRATIVE REQUIREMENTS (PWD ME)	Cannot serve as any of the other personnel listed here.
Superintendent	01 30 00.00 22	ADMINISTRATIVE REQUIREMENTS (PWD ME)	Cannot serve as any of the other personnel listed here.
Designated Project Scheduler	01 32 17.00 20	NETWORK ANALYSIS SCHEDULES (NAS)	
Site Safety and Health Officer (SSHO)	01 35 26.00 22	GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME)	Cannot serve as any of the other personnel listed here, except competent persons.
Competent Persons	01 35 26.00 22	GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME)	Designate in writing.
Archaeologist	01 44 00.00 22	ARCHAEOLOGICAL MONITORING (PWD ME)	
Contractor's Quality Control Manager (QCM or CQCM)	01 45 00.00 22	QUALITY CONTROL (PWD ME)	Cannot serve as Project Manager, Superintendent, SSHO, QC Specialists, or Special Inspector.
QC Specialist - Mechanical	01 45 00.00 22	QUALITY CONTROL (PWD ME)	Cannot serve as Special Inspector.
QC Specialist - Fire Protection (FPQC)	01 45 00.00 22	QUALITY CONTROL (PWD ME)	Cannot serve as Special Inspector for Spray-applied fireproofing nor firestopping.
Certified Industrial Hygienist (CIH)	01 45 00.00 22	QUALITY CONTROL (PWD ME)	
Special (Structural) Inspector (SI)	01 45 35.00 22	SPECIAL INSPECTIONS (PWD ME)	Must be 3rd party. Cannot serve as QCM.

Title	Specifications	Section Title	Restrictions
Associate Special Inspector - Firestopping	01 45 35.00 22	SPECIAL INSPECTIONS (PWD ME)	Cannot serve as FPQC.
Environmental Manager	01 57 19.00 22	TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME)	Cannot serve as Superintendent or SSHO.
Environmental Inspector(s)	01 57 19.00 22, 3.1.2	TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME)	Required to be PE.
Lead Commissioning Specialist (CxC)	01 91 00.15 22	BUILDING COMMISSIONING (PWD ME)	Must be 3rd party.
Commissioning Specialist-Mechanical (CxS-M)	01 91 00.15 22	BUILDING COMMISSIONING (PWD ME)	Must be 3rd party.
Commissioning Specialist-Electrical (CxS-E)	01 91 00.15 22	BUILDING COMMISSIONING (PWD ME)	Must be 3rd party.

1.7 PROJECT ENVIRONMENTAL GOALS

Distribute copies of the Environmental Goals to each subcontractor and the Contracting Officer. The overall goal for design, construction, and operation is to produce a project that meets the functional program needs and incorporates sustainability principles. Specifically:

a. Avoid site degradation and erosion. Minimize offsite environmental impact.

b. Use the minimum amount of energy, water, and materials feasible to meet the design intent. Select energy and water efficient equipment and strategies.

c. Use environmentally preferable products and decrease toxicity level of materials used.

d. Optimize operational performance (through commissioning efforts) in order to ensure energy efficient equipment operates as intended. Consider the durability, maintainability, and flexibility of building systems.

e. Manage construction site and storage of materials to ensure no negative impact on the indoor environmental quality of the building.

f. Reduce construction waste through reuse, recycling, and supplier take-back.

1.8 EXISTING WORK

Remove or alter existing work in such a manner as to prevent injury or damage to any portions of the existing work which remain.

Repair or replace portions of existing conditions which have been altered/damaged during construction operations to match existing or adjoining work, as approved by the Contracting Officer. At the completion of operations, existing conditions must be in a condition equal to or better than that which existed before new work started.

Refer to Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME).

1.9 LOCATION OF UNDERGROUND UTILITIES

Verify location of underground utilities in accordance with Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME).

1.9.1 Notification Prior to Excavation

Notify the Contracting Officer at least fourteen (14) calendar days prior to starting excavation work. Refer to Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME) Attachment B for Dig Safe requirements.

1.10 GOVERNMENT-FURNISHED MATERIAL AND EQUIPMENT

The Government will furnish the following materials and equipment for installation by the Contractor: See Equipment (Q) sheets for Government provided industrial equipment.

1.10.1 Delivery Schedule

Notify the Contracting Officer in writing at least 45 calendar days in advance of the date on which the materials and equipment are required.

1.10.2 Delivery Location

The materials and equipment will be delivered to Building 60.

1.11 SECURE NETWORK PROVIDERS (NETWORK N, NETWORK C, NETWORK P, AND NETWORK S)

1.11.1 Secure Network Providers Contractor Access

Allow the Secure Network Providers Contractor(s) access to the facility towards the end of construction (finishes 90% complete, rough-in 100% complete, Inside Plant (ISP)/Outside Plant (OSP) infrastructure in place) to provide equipment in the telecommunications room and spaces and to make final connections. Coordinate and facilitate joint use of building spaces with the Secure Network Providers Contractor(s) during the final stages of each phase of construction. Coordinate with the Contracting Officer to facilitate a coordination meeting between the secure network providers contractor(s). After initial meeting, within one week, incorporate the effort of additional coordination into the construction schedule as required to support maintaining the Contract duration.

1.12 SALVAGE MATERIAL AND EQUIPMENT

Items designated to be salvaged remain the property of the Government. Salvage and store items on site until the completion of project unless directed otherwise by the Contracting Officer. Stored items must be affixed with label indicating contents, original location, and project name.

Items to be salvaged and stored for reinstallation must be stored within the Contractor's laydown areas or within another Government approved location.

Provide a salvage plan, listing material and equipment to be salvaged, and their storage location. Refer to Section 02 42 91 REMOVAL AND SALVAGE OF HISTORIC CONSTRUCTION MATERIAL. Maintain property control records for material or equipment designated as salvage. Provide a system for property control in the salvage plan. Store and protect salvaged materials and equipment until disposition by the Contracting Officer. Update the plan with storage locations once finalized. Access to the salvaged material and equipment must be provided to the Government upon request.

Prior to the completion of the project, coordinate with the Contracting Officer for final storage location(s) of the salvaged materials to be turned over to the Government. Dispose of materials if directed to by the Contracting Officer.

1.13 BENEFICIAL OCCUPANCY REQUIREMENTS

All work elements identified as "Required for Facility Delivery (BOD)" included in Section 01 30 00.00 22 Appendix A RZ Checklist must be completed and accepted by the Contractor's QC Manager prior to submitting the form to the NAVFAC PWD ME Construction Manager for NAVFAC PWD ME signatures.

The Contractor must submit the completed RZ Checklist with the required documentation no later than ten (10) working days prior to requesting the Government take Beneficial Occupancy.

Failure of the Contractor to verify all work and provide the required documentation will delay the processing of acceptance of Beneficial Occupancy.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

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SECTION 01 14 00.00 22
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WORK RESTRICTIONS (PWD-ME)
11/22, CHG 4: 02/25

PART 1 GENERAL

This Section applies to Design-Bid-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

1.2 DEFINITIONS

a. "Tier 1Infrastructure" include shipyard facilities (utilities and structures) that require additional work restrictions and procedures when working on or in the vicinity of. Coordinate with the Contracting Officer for a list of Tier 1 Infrastructure associated with the scope of the project. This list will be communicated through written means.

b. Existing Infrastructure" includes all infrastructure not specified above. The contractor is required to follow all contract requirements when working in/around Existing Infrastructure and is not exempt from submittal requirements or from following three phases of control to manage construction activities.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

List of Contact Personnel; G

Outage Plan; G

Tier 1 Infrastructure Coordination Drawings; G

1.4 SPECIAL SCHEDULING REQUIREMENTS (PNSY)

a. Work requiring interruption (outage) of any Portsmouth Naval Shipyard roads, crane rail, railroads, and/or utility services must be coordinated/aligned with stakeholders prior to submission of the Outage package for final 15 calendar day approval. See Paragraph herein entitled TEMPORARY OUTAGE PROCESS PROCESS for Outage timing and instructions.

b. Permission to interrupt any Portsmouth Naval Shipyard roads, crane rail, railroads, and/or utilities or services must be submitted to the Contracting Officer in writing a minimum of 30 calendar days (20 working days) prior to the desired date of interruption. See Paragraph herein entitled TEMPORARY OUTAGE PROCESS.

c. When the project is located in and adjacent to restricted area. The Contractor is not allowed in these areas without continuous Government escorts. See the Scope of Work to determine if the project limits include any security Island in or adjacent to the work area.

d. Coordinate the work with the sequencing/phasing requirements outlined in the Scope of Work Memo and as outline in the Construction Drawings.

1.5 CONTRACTOR ACCESS AND USE OF PREMISES (PNSY)

1.5.1 Portsmouth Naval Shipyard Regulations

Notify the Contracting Officer 30 calendar days in advance for any wide loads. Contact the Contracting Officer to determine if there are other access limitations at the Portsmouth Naval Shipyard.

As part of the requests, provide the specific dates, hours, location(s), type(s) of work to be performed, Contract number, and project title. Based on the justification provided, the Contracting Officer may approve work outside regular hours.

Working on Weekends/Holidays: Any request to work on a weekend must be submitted to the Contracting Officer no later than two (2) working days prior to the requested work weekend/Holiday.

Ensure that Contractor personnel employed on the Portsmouth Naval Shipyard become familiar with and obey Portsmouth Naval Shipyard regulations including safety, fire, traffic and security regulations. Keep within the limits of the work and avenues of ingress and egress. Wear appropriate personal protective equipment (PPE) in designated areas. Do not enter any restricted areas unless required to do so and until cleared for such entry. Ensure all Contractor equipment, include delivery vehicles, are clearly identified with their company name.

1.5.1.1 Subcontractors and Personnel Contacts

Provide a list of contact personnel of the Contractor and subcontractors including addresses and telephone numbers for use in the event of an emergency. As changes occur and additional information becomes available, correct and change the information contained in previous lists.

1.5.1.2 Identification Badges and Installation Access

a. Application for and use of badges will be as directed. Obtain access to the Portsmouth Naval Shipyard by submitting the SECNAV Form 5512-NSNPT Jul 2019 (Attachment C) a minimum of 15 working days in advance for vetting and approval, or by obtaining passes each day from the Portsmouth Naval Shipyard's Pass and Identification/Security Office. One-day passes, issued through the Portsmouth Naval Shipyard's Pass and Identification Office, will be furnished without charge. Report any instances of lost or stolen badges to the Contracting Officer immediately.

Prime contractor is responsible for recovering any badges that are no longer required, including from workers who are no longer employed by prime or subs, and immediately turning badges into the Pass/ID Office. Failure to maintain accountability of issued badges could cause delays in future badge requests. Contracts are warned that badges are checked for currency and any badge not used within 30 days will be disabled.

b. All Contractors who possess a Commercial Access Control (CAC) System Card are also required to submit an access request (5512) to support a valid purpose to be on the Portsmouth Naval Shipyard. A second form of valid ID may also be required, if requested, upon arrival at the Portsmouth Naval Shipyard's Entrance Gate. All personnel without CAC cards will need two (2) forms of approved identification for access to the Portsmouth Naval Shipyard.

c. One-Day Passes: Participation in the vetting process is mandatory, unless a valid emergency exists. Then it is possible to utilize an escort from the department effected. If the Contractor chooses to not participate in the vetting process, they will not be allowed on the Portsmouth Naval Shipyard. The Government will not be responsible for any cost or lost time associated with obtaining daily passes or added vehicle inspections.

d. PNS has an online appointment system. Appointments must be made using https://kiosk.na4.qless.com/kiosk/app/home/191.

1.5.1.3 Additional Personnel Requirements

1.5.1.3.1 General Construction and Finish Work

General construction and finish work must be performed by U.S. firms using U.S. citizens. General construction includes construction activities such as building sitework, utilities, foundations, structure, and enclosure or shell, including doors, windows and façade work. Finish Work includes construction activities such as insulation, floor, partition, and ceiling systems; cabinet work; conveyor systems; specialties; building furnishings, fixtures, and equipment; and mechanical and electrical services and equipment including those specialized for fire protection, security, communication, control, energy conservation, safety, comfort, convenience, and similar purposes.

1.5.1.3.2 Electronic Security Systems Equipment

Electronic Security Systems equipment such as processing control units, workstations, field panels, sensors, high security locks, card readers, cable installation, and system programming, testing and training must be performed by U.S. companies using U.S. citizens who have been subjected to a trustworthiness determination].

1.5.2 Radiological

1.5.2.1 Radiological Indoctrination (PNSY)

All personnel working at the Portsmouth Naval Shipyard are required to view a 15 minute video briefing on radiological postings and controls in use at the Portsmouth Naval Shipyard. The briefing will be given at the Pass Office prior to issue of security badges and vehicle passes.

Any employee who disregards, alters, moves, or otherwise tampers with a radiological posting, or who disobeys a radiological instruction, may be removed from the Portsmouth Naval Shipyard and denied future access.

1.5.2.2 Yellow Materials (PNSY)

Do not use organge gloves or yellow colored materials at the Portsmouth Naval Shipyard for the following purposes: Protective clothing, hoods, sheeting, tarpaulins, polyethylene bottles or other containers, tapes, bags, banding, identification marks on tools, boundary markers, ribbons, vent ducts, temporary erosion control devices, survey ribbon, etc. (Yellow material should not be used when alternate colors are available and suitable for use unless authorized by Shipyard procedures. Contact the Contracting Officer for a list of yellow items that have been approved for use on the Portsmouth Naval Shipyard.) Dispose of generated yellow colored waste off of the Portsmouth Naval Shipyard. Portsmouth Naval Shipyard refuse containers must not be used for disposal of yellow colored waste materials. Yellow colored items such as described above are of special significance within the Portsmouth Naval Shipyard and are subject to strict controls. Yellow colored Contract generated debris must be bagged in non-translucent containers, and promptly removed from the Portsmouth Naval Shipyard.

1.5.2.3 Smoke Detectors (PNSY)

Ionization type smoke detectors and duct smoke detectors contain radioactive material and are prohibited from use on the Portsmouth Naval Shipyard. Photoelectric smoke detectors are the only type authorized for use on the Portsmouth Naval Shipyard.

1.5.2.4 Radioactive Sources (PNSY) (Troxler for Earth Work)

All Contracts involving radiation generating devices must conform to the requirements listed in Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME) and U.S. Army Corps of Engineers Safety Manual EM 385-1-1. All requirements are to be submitted to the Contracting Officer at least 14 days prior to commencement of operations involving radiation generating devices. A requirements checklist will be provided by NAVFAC (COTs) Contractor Oversight Technician.

1.5.2.5 Laser Control

Comply with laser safety requirements under 21 CFR 1040 and ANSI 2136.1-1986 for any work under this Contract utilizing lasers.

1.5.2.6 Energy Conservation

In cooperation with Government representatives, participate in an active program directed toward the efficient use of energy.

1.5.2.7 Fire Prevention (PNSY)

Require employees to become familiar with fire prevention regulations within the Portsmouth Naval Shipyard to include the proper method of turning in a fire alarm, storage of flammable and combustible materials, and control of combustible waste and trash. Any HOT WORK (welding, burning, grinding, cutting, etc.) requires a HOT WORK PERMIT prior to commencing such work. This permit is obtained from the Portsmouth Naval Shipyard's Fire Department via the Contracting Officer and must be submitted at least three(3) working days prior to commencing any Hot Work.

1.5.2.8 Identification and Control of Seamed (Welded) Pipe and Tubing (PNSY)

Submarine Safety regulations prohibit the use of seamed (welded) pipe or tubing within the Portsmouth Naval Shipyard, unless such pipe or tubing is identified and controlled so as to prevent its inadvertent substitution for seamless pipe or tubing. The following requirements apply and will be strictly enforced:

Any seamed (welded) copper-nickel, carbon steel, carbon-moly steel, stainless steel, nickel-chromium-iron alloy, or nickel-copper pipe or tubing intended for use on the Portsmouth Naval Shipyard must be identified in the following manner PRIOR TO DELIVERY TO THE PORTSMOUTH NAVAL SHIPYARD:

Use a lead-free white paint, to mark a 24-inch long stripe and the word "welded" alternately along the entire length of the pipe or tubing. Apply a one-half inch wide stripe unless the size of the pipe or tubing requires use of a narrower stripe.

Maintain positive control over seamed pipe or tubing until worked into place or removed from the Portsmouth Naval Shipyard.

Seamless pipe or tubing may be substituted for any seamed (welded) pipe or tubing specified in the technical specifications.

The above requirements do not apply to square or rectangular tubing, copper or brass pipe or tubing, or to piping or tubing which has been incorporated into equipment or fixtures prior to delivery to the Portsmouth Naval Shipyard.

1.5.2.9 Pesticide and Herbicide Control

Do not apply pesticides nor herbicides unless specifically required by this Contract. Where application of pesticides or herbicides is required, provide the submittals required by the specification and obtain written approval prior to any application. The Contracting Officer will be required to review and approve pesticides or herbicides submittals.

1.5.2.10 Smoking Policy

In accordance with NAVFAC policy, smoking is prohibited inside all buildings and other facilities except those areas specifically identified as smoking areas (e.g., smoking shelters). Smoking is not permitted within 20 feet of air intakes, doorways, or windows.

1.5.3 Occupied and Existing Buildings

Work will be in an existing unoccupied building and around existing buildings which are occupied. Do not enter buildings without prior approval of the Contracting Officer.

1.5.4 Utility Cutovers and Interruptions

a. Make utility cutovers and interruptions after normal working hours or on Saturdays, Sundays, and Government holidays. Conform to procedures required in paragraph WORK OUTSIDE REGULAR HOURS.

b. Ensure that new utility lines are complete, except for the connection, before interrupting existing service.

c. Interruption to water, sanitary sewer, storm sewer, telephone service, electric service, air conditioning, heating, fire alarm, compressed air, and steam are considered utility cutovers pursuant to the paragraph WORK OUTSIDE REGULAR HOURS. Such interruptions are limited to various timeframes, as noted on the Outage Drawing, Sheets G-008 and G-009.

d. Operation of Portsmouth Naval Shipyard Utilities: Do not operate nor disturb the setting of control devices in the station utilities system, including water, sewer, electrical, and steam services. The Government will operate the control devices as required for normal conduct of the work. Notify the Contracting Officer giving reasonable advance notice when such operation is required.

e. Connection to Existing Sanitary Sewer Line: Provide positive verification that the existing line conveys sanitary sewer; verify line is not incorrectly connected to a storm drain.

1.5.5 Temporary Outage Process

a. Outage Requirements

The Outage process includes development of a plan and outage package, vetting and aligning the Outage Plan and associated impacts with stakeholders, and final outage submittal and Government approval.

Temporary outages are required when normal system operation is or could be impacted for utility, facility, fire lane, road, and ground level track. This situation can occur when inspecting, repairing, or upgrading, utility systems and equipment, buildings and structures (facilities), and areas on and within 10' adjacent to ground level trackage (that extends below pavement depth). These temporary outages may affect one or several buildings, facilities, production processes, and geographical locations. Although most outage requirements are addressed in this section, the entire specification must be reviewed for additional outage requirements.

Outages involving Tier 1 Shipyard Infrastructure require a specific Risk Analysis and Risk Mitigation Plan as outlined in Paragraph herein entitled TIER 1 SHIPYARD INFRASTRUCTURE REQUIREMENTS. The risk mitigation analysis and risk mitigation plan must be submitted to the Contractor Officer's Representative for review and approval prior to the submission of outage request.

Significant outages required to execute the work may be outlined on project Drawings. The Drawings do not include all the outages that will be required to support the execution of all work. If not already provided in the project drawing package, the contractor will coordinate with the PWD ME Design Manager (DM)/Appropriate Technical code, via the ET as liaison, to develop an outage submittal package and a target outage date. The Contractor is responsible for the final outage submittal package.

All Outages packages must be submitted to the Back Shift Rep via the Contracting Officer's Representative for final stakeholder concurrence only after coordination/alignment with stakeholders on scope, impact and target outage date, and at least 15 calendar days in advance of the target outage date. The Final Government Outage concurrence period is 15 calendar days. Any Outage requests denied due to incomplete stakeholder alignment or information required to support the requested outages, will not be justification for a delay claim. The Contractor must review any planned outage with the NAVFAC Construction Management Team/stakeholders prior to submitting the outage request.

The Contracting Officer's Representative will provide the Outage Request Form and the local Outage Instruction.

The Outage Coordination Meeting described in Item C below is intended to provide a forum to review the outage plan and information required to support any outage reviews.

At a minimum, the outage submittal package must address:

1) Specific location/system requiring the outage.

2) Duration of outage.

3) Scope of work required for the outage.

4) Outage work plan, including organization responsible for each step to accomplish the outage.

5) Scope of outage impact, identifying specific affected systems and stakeholders.

6) Hazardous energy control.

7) For Tier 1 Shipyard Infrastructure (see paragraph herein entitled TIER 1 SHIPYARD INFRASTRUCTURE REQUIREMENTS) a Tier 1 Infrastructure Risk Analysis and Risk Mitigation plan is required.

8) Adequate sketches that clearly depict location, situation, and relevant information/interfaces.

9) Include LOTO boundaries and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1, Section 11.A.02 (Isolation). Some examples of energy isolation devices and procedures are highlighted in EM 385-1-1, Section 12.D. In accordance with EM 385-1-1, Section 12.A.01, where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy. These activities include, but are not limited to, a review of HECP and HEC procedures, as well as applicable

Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1, Section 11.A.02 and NFPA 70E, work on energized electrical circuits m for Outage must not be performed without prior Government authorization. Government permission is considered through the outage process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is not feasible.

b. Additional Outage Requirements:

The Three Week Look Ahead schedule must include upcoming outages planned for that three week period. Outages planning must start early in the construction process to allow adequate coordination and alignment with affected stakeholders and to integrate timing with critical shipyard operations.

The Work Sequencing and Preparation Plan must address how outages will be accomplished with an absolute minimum of disruption to Government operations.

Provide an Outage Schedule that indicates all the required Outages that will be required to support the Construction. See Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS).

c. Outage Coordination Meeting:

Outage Coordination Meetings are to be held as necessary, prior to submittal of an outage request package. The meetings are intended to develop a mutual understanding of the administrative and technical scope and requirements needed to support development of the outage request package, and execution of the outage.

This meeting should include applicable personnel from the Prime Contractor, the Prime and Subcontractors to perform the work, the Contracting Officer, PWD-ME DM, Designer of Record, the NAVFAC PWD ME UEM representatives, and applicable stakeholders.

All parties must fully coordinate and be aligned with the activities and dates required to support the outages.

See also Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS paragraph OUTAGE COORDINATION MEETING and 01 35 26.00 22 Governmental Safety Requirements (PWD ME) paragraph UTILITY OUTAGE SAFETY AND COORDINATION REVIEW MEETING.

d. Outage Safety Review Meeting

During the Outage Safety Review meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HECP training in accordance with EM 385-1-1. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the Subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

See also Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS paragraph OUTAGE SAFETY REVIEW MEETING and 01 35 26.00 22 Governmental Safety Requirements (PWD ME) paragraph UTILITY OUTAGE SAFETY AND COORDINATION REVIEW MEETING.

1.6 WORKING HOURS (PNSY)

Regular working hours must consist of a period established by the Contracting Officer between 7 AM and 3:30 PM, Monday through Friday, excluding Government holidays. The regular working hours must be confirmed with the Contracting Officer.

1.6.1 Work Outside Regular Hours (PNSY)

Work outside regular working hours requires Contracting Officer approval. Provide written requests fifteen (15) Calendar days prior to such work to allow arrangements to be made by the Government for inspecting the work in progress and to allow scheduling of full time escorts in the building(s) if required. During periods of darkness, the different parts of the work must be lighted in a manner approved by the Contracting Officer.

As part of the requests, provide the specific dates, hours, location(s), type(s) of work to be performed, Contract number, and project title. Based on the justification provided, the Contracting Officer may approve work outside regular hours.

Working on Weekends/Holidays: Any request to work on a weekend must be submitted to the Contracting Officer no later than two (2) working days prior to the requested work weekend/Holiday.

1.7 FIRE PROTECTION

1.7.1 Compliance (PNSY)

Comply with COE EM 385-1-1, NFPA 241, and Portsmouth Naval Shipyard fire regulations. Obtain approval no later than two (2) working days from the Portsmouth Naval Shipyard Fire Chief via the Contracting Officer prior to commencement of hot work operations.

1.7.2 Fire Lanes

1.7.2.1 Fire Lane Width

Fire lanes shall be a minimum of Twenty (20) feet in width.

1.7.2.2 Vertical Clearance for Fire Lane

Fire lanes shall have a minimum of Thirteen (13) feet Six (6) inches nominal vertical clearance, which shall be provided and maintained over the full width of the fire lane.

1.7.2.3 Angle of Approach/Departure

Minimum inside turn radius shall be Twenty-Five (25) feet and minimum outside turn radius shall be Fifty (50) feet.

1.7.2.4 Waiver Request for Fire Lane size and dimensions

Due to existing infrastructure, some requirements of this section may not be achievable. The Contractor must submit a Waiver Request to the Fire Department for review. Additional provisions may be required for the approval of the waiver.

1.7.2.5 Parking in or Blocking Fire Lanes

Parking in or blocking fire lanes is expressly prohibited for any reason without prior Fire Department approval. Vehicles left unattended in fire lanes are subject to towing. Contractors will be issued a Non-Compliance Notice for vehicles left unattended in fire lanes.

a. First Offense: The Contractor will remove the vehicle from the Project site immediately and the vehicle will not be allowed to return to the Project site for Seven (7) Days. The contractor will perform a safety stand down within three (3) days of the incident and the Fire Department shall be invited to attend the safety stand down.

b. Second Offense: Contractor and Sub-Contractors vehicles will be permanently barred from the Project site and a Non-Compliance Notice will be issued. Individuals performing deliveries in support of the Project will be allowed but are expected to be on site for the shortest duration required to perform the delivery, and the driver/operator shall remain in the immediate vicinity of the vehicle at all times.

1.7.2.6 Approval to Temporarily Block a Roadway/Fire Lane

The Contractor may request approval to temporarily block a roadway/fire lane as part of a Road Outage to complete work associated with the Project. Approval may only be given by the Fire Chief or his/her assigned representatives. Arrange through the Contracting Officer with the Fire Chief for scheduling the temporary blockage prior to submitting the outage package. Final approval will be through the outage process. Refer to paragraph herein entitled FACILITY/UTILITY/SERVICE OUTAGE & INTERRUPTIONS PROCESS for the outage process. Add the following to the outage package:

a. Alternative analysis that discusses whether a practical alternative to the outage exists. The analysis must address the activity purpose, need, and why the activity cannot be completed by:

   (1) Utilizing, managing, or expanding the work area to avoid the fire lane.
   (2) Reducing the size, scope, configuration or density of the activity as proposed (phasing).

b. Illustration showing the location of contractor equipment during the proposed outage.

c. Provisions for emergency access (i.e. a steel plate shall be kept with 10 feet of the excavation.

d. Additional provisions may be required by the Fire Department on a site by site basis.

1.8 SECURITY REQUIREMENTS

Employees and representatives performing work under this Contract are required to be United States citizens. If naturalized, the individual must present his/her naturalization papers to the Security Officer for inspection. Foreign born personnel must present evidence of citizenship regardless of citizenship of parents, as required by immigration laws.

1.8.1 Access to the Portsmouth Naval Shipyard (PNSY)

Contract Clause "FAR 52.204-2, Security Requirements and Alternate II" and the following apply:

Access to areas designated as restricted NAVSEA spaces will require an escort by a Government Representative. Notify the Contracting Officer at least 14 Calendar Days in advance of the date access is required.

Obtain security badges and vehicle passes to enter the Portsmouth Naval Shipyard at the Portsmouth Naval Shipyard's Pass/Security Office. Furnish proof that employees are U.S. citizens to obtain badges to enter the Portsmouth Naval Shipyard.

Complete Department of Homeland Security Form I-9; Employment Eligibility Verification for each employee and furnish proof that employees are U.S. citizens to obtain badges to enter Portsmouth Naval Shipyard.

Vehicles must not be equipped with cameras, including but not limited to dash mounted or any manufacturer safety features (lane assist, backup, etc.). Cameras can be temporarily covered and disabled, however a government escort will be required to accompany the vehicle while on base and escorts will be limited and in some cases not available. Project delays claims due to coordination efforts regarding vehicle cameras will not be entertained by the Government.

1.8.2 Application and Issue of Security Badges

"Temporary" Security Badges will be issued to Contractor personnel requiring access to the shipyard for less than 90 days. Badge(s) will be issued upon satisfactory proof of U.S. citizenship, in the form of an original or certified birth certificate, passport, or naturalization papers. A picture ID is required in addition to satisfactory proof of citizenship. See Attachment B.

"Permanent" (photo) Standard Access Control Badges will be issued to Contractor personnel requiring access to the shipyard for 90 or more days. Contractor personnel will be required to complete an authorization application form for local record check, and a personal information sheet. The forms will be furnished to the Contractor following award of any contract resulting from this solicitation, at time of pre-performance or pre-construction conference."

In the event access is required to Contract work areas not permitted by the level of security badge issued, such need must be demonstrated and an escort obtained.

STANDARD ACCESS CONTROL BADGES MUST BE ATTACHED TO THE OUTER GARMENT AND

DISPLAYED ABOVE THE WAISTLINE AT ALL TIMES WHILE ON THE PORTSMOUTH NAVAL SHIPYARD.

PERSONNEL MUST NOT ENTER AREAS FOR WHICH THEY HAVE NOT BEEN CLEARED. WHERE A NEED HAS BEEN DEMONSTRATED TO ENTER SUCH AREAS, PERSONNEL WILL BE UNDER CONSTANT ESCORT BY PERSONNEL WHO HAVE BEEN CLEARED. FAILURE TO ADHERE TO POSTED SECURITY REQUIREMENTS MAY RESULT IN REMOVAL OF THE EMPLOYEE FROM THE PORTSMOUTH NAVAL SHIPYARD WITH FUTURE ACCESS DENIED.

1.8.3 Application and Issue of Vehicle Passes (PNSY)

Vehicle passes will be issued upon satisfactory proof of a valid Operator's License, Vehicle Insurance, and State Vehicle Registration. Temporary passes will be issued for short term or single trip requirements on a case by case basis. All vehicles permitted to enter or park on the Portsmouth Naval Shipyard must comply with the Portsmouth Naval Shipyard's traffic and parking regulations and must only park in assigned areas, which may or may not be in the vicinity of the site of the Contract work. Do not park a vehicle in such a manner that crane tracks, railroad tracks, and vehicle access routes are blocked (including associate clearance zones). Vehicles left unattended which are blocking such access routes are subject to towing and loss of vehicle passes. Parking on the Portsmouth Naval Shipyard may be in excess of one-half mile from the worksite.

1.8.4 Application and Issue of Vehicle Passes for Entry into Portsmouth Naval Shipyard's Controlled Industrial Areas (CIA)

Vehicular access to the CIA must be minimized and all vehicles must comply with the following requirements:

Vehicles must visibly display a CIA vehicle entry pass and inspection pass from the Commercial Vehicle Inspection Station (CVIS), Building 386. CIA passes will only be issued to company owned or leased vehicles, rental vehicles rented in the company name, or privately owned vehicles the company has certified in writing, to be necessary in the performance of Contracted work. CIA passes will be issued on weekends and holidays at Building 29, from the Watch Supervisor. Personnel not possessing the level security badge required for CIA access must be accompanied by a properly badged escort to obtain the CIA vehicle pass.

Vehicles will only be allowed in the CIA for the transportation of tools, parts, and materials to and from the worksite. An exception to this policy, employees may be transported to and from the worksite after full vehicle inspection at CVIS if a specific security plan has been developed and approved by the Portsmouth Naval Shipyard Security Officer.

Parking of privately owned or company owned vehicles not utilized in the execution of construction activities (e.g. not a working truck or specialty vehicle) within the CIA or in any project laydown space is prohibited.

For Vehicles remaining in the CIA for more than 24 hours: Obtain a CIA Overnight Parking Permit from the PWD ME Construction ET. Vehicles without this pass are subject to being towed. Passes will only be issued to those demonstrating a valid reason as determined by Portsmouth Naval Shipyard Security.

Vehicles must not be equipped with cameras, including but not limited to dash mounted or any manufacturer safety features (lane assist, backup, etc.). Cameras can be temporarily covered and disabled, however a government escort will be required to accompany the vehicle while on base and escorts will be limited and in some cases not available. Project delays claims due to coordination efforts regarding vehicle cameras will not be entertained by the Government.

1.8.5 Application and Issue of Crane Passes (PNSY)

Comply with EM 385-1-1. For Crane Passes at the Portsmouth Naval Shipyard to be valid, the Certificate of Compliance (FORM 16-1) must be completed and accepted.

1.8.6 Return of Badges and Vehicle Passes

Ensure all vehicle access permits and personnel badges are returned to the Portsmouth Naval Shipyard Security Officer when the need has ended. Account in writing for each missing pass or badge prior to final payment being made on the Contract. Any vehicle passes issued must be returned to Portsmouth Naval Shipyard Security daily.

1.8.7 Security Responsibilities (PNSY)

Employees must not transport, drink, or have in their possession any alcoholic beverages. Possession of any controlled substances without a physician's prescription is also prohibited. Any employee appearing to be under the influence of intoxicating liquor or narcotics will be apprehended by Portsmouth Naval Shipyard Police, escorted off of the Portsmouth Naval Shipyard, and turned over to the local Police Department.

Any vehicle found to contain controlled substances, including usable residue, may be seized and impounded. Within 24 hours of the work day following any vehicle seizure, the Portsmouth Naval Shipyard Police will have determined whether forfeiture of the vehicle is required. If not, the vehicle will be returned to the owner or authorized agent. If the vehicle is determined to be appropriate for forfeiture, the Portsmouth Naval Shipyard's Legal Officer will notify the Drug Enforcement Administration of such seizure and impoundment, for initiation of forfeiture proceedings pursuant to Title 2l, U.S. Code, Section 88l. Such actions may be taken regardless of whether the owner/operator of the vehicle had knowledge of the presence of drugs in the vehicle. The Government may pursue criminal or other disciplinary actions pursuant to Title l8, U.S. Code, Section l382.

Possession of firearms, ammunition, and/or explosives is prohibited. In the event explosives are required for construction work, specific handling requirements and approvals must be obtained from the Portsmouth Naval Shipyard Security Officer via the Contracting Officer.

Cameras, video equipment, or similar photographic equipment must not be introduced into nor removed from the Portsmouth Naval Shipyard. In the event such equipment is required for performance of Contract work, approvals must be obtained from the Portsmouth Naval Shipyard Security Officer via the Contracting Officer.

Weapons (firearms, personal knives with blades 2-1/2 inch long or greater, Mace, Pepper Spray etc.) are not permitted aboard the Portsmouth Naval Shipyard.

At a minimum, all Superintendents, QC Manager and SSHO (and anyone acting in this capacity) must have a CIA compliant mobile phone on their persons any time while work is occurring at the Contract jobsite(s). The phone number(s) associated with these phones must be provided during the pre-con meeting and work will not be allowed to proceed without one.

Laptops and any other computer, regardless of form factor, must not be introduced nor removed from the Portsmouth Naval Shipyard without authorization. Only business use computers are allowed. Personally owned computers are not authorized on the Portsmouth Naval Shipyard. If a computer is required to perform work on the Portsmouth Naval Shipyard, a Visitors Portable Computer Equipment Registration form must be completed and submitted for approval. These forms can be obtained and completed at the pass office upon entrance to the Portsmouth Naval Shipyard or through the Portsmouth Naval Shipyard POC.

In the event that a camera is required for the performance of the Contracted work, approval must be obtained from the C1120 Security Office. In the event of any deviation from the requirement to remove and epoxy the camera, authorization must be sought from C1120. All requests for camera use and deviations must come to C1120 through the Contracting Officer's Representative or Portsmouth Naval Shipyard Point of Contact (POC). Authorizations and approved computer registrations must remain with the devices and be presentable upon request. Devices without a documented approval or observed being used in a manner that violates the terms of the Visitor Portable Computer Equipment Registration agreement are subject to search and seizure.

Basic guidance for use of Portable Electronic Devices (PED) at Portsmouth Naval Shipyard: See Attachment 01 14 00.00 22_ PNSY Portable Electronic Devices Policy. Obtain Contracting Officer approval of their intent to use, of any portable electronic device which has the capability to record, copy, store, export and transmit data, photography, digital images, video or audible information, inside the CIA or any NAVSEA controlled spaces. Provide the manufacturer’s data describing the electronic capabilities of the device(s).

Use of wireless networking and communication on the Portsmouth Naval Shipyard requires prior authorization. Communication devices such as cellular hotspots and wireless routers must also be registered using the same Visitors Portable Computer Equipment Registration. This does not include air-cards and straight cellular data. It does include any device that transmits or broadcasts Wi-Fi, including cellular phones with the personal hot-spot feature enabled.

Broadband internet connections must be authorized prior to use on the Portsmouth Naval Shipyard. All requests for physical cable service from an internet service provider (ISP) must be authorized and coordinated through C109. Work through the COR to request broadband access.

Driver use of a hand-held cellular phone in a moving vehicle on the Portsmouth Naval Shipyard is prohibited. This prohibition does not include hands-free cellular phone devices. Hands-free devices include console/dash-mounted or otherwise secured cellular phones with integrated features such as voice-activation, speed dial, speakerphone or other similar technology for sending and receiving calls.

Driver use of any portable, personal listening device worn inside the aural canal, around or covering the driver's ear while operating a motor vehicle, is prohibited. Listening devices include wired or wireless earphones and headphones (including blue tooth or similar technology), and do not include hearing aids or devices designed and required for hearing protection.

The use of radar or laser detection devices to indicate the presence of speed recording instruments or to transmit simulated erroneous speeds is prohibited in accordance with OPNAVINST 5100.2H.

Indoctrinate personnel on access limitations to ensure security control is maintained as an integral part of their work pattern and habit.

Unescorted personnel found in security areas requiring a higher level clearance than the level represented by the badge displayed will be removed from the area with possible confiscation of security badges and vehicle passes.

1.8.8 Portable Electronic Devices (PED) Policy - PNSY

A PED is a portable electronic device that is easily transportable and has the capability to record, copy, store, export, or transmit data, photography, digital images, video or audible information. Examples of PEDs include, but are not limited to: cell phones, fitness trackers, smart watches, watches with input capability, tablet, laptops, media devices, Bluetooth consumer electronics, cellular transceivers, and portable diagnostic equipment.

a. PEDs are PROHIBITED within certain spaces on PNSY. Contractor must check with the Contracting Officer for project specific guidance. Examples of areas where PEDs are prohibited include:

   - Secure Rooms, Nuclear Work Area (NWA)
   - Controlled Nuclear Information Area (CNIA)
   - Occupied Dry Dock

b. PEDs are permitted in other areas on the Portsmouth Naval Shipyard, subject to the usage restrictions, as outlined below:

   - Security Islands
   - Controlled Industrial Area
   - All other PNS spaces not identified above

1.8.8.1 Bluetooth Wi-Fi Hotspots

Personal or contractor Bluetooth devices are authorized but are PROHIBITED from being connected to any Government Furnished Equipment. Wi-Fi hotspots must be approved by PNSY NAVSEA Code 109. Use of Personal Wi-Fi hotspots is PROHIBITED.

1.8.8.2 Photos/Videos

NO PHOTOGRAPHY OR VIDEOGRAPHY IS PERMITTED UNLESS YOU HAVE A PHOTOGRAPHY PERMIT APPROVED BY CODE 1120 AND A GOVERNMENT CAMERA DEVICE.

1.8.8.3 Violations

Discovery of PEDs in prohibited areas, or discovery of prohibited materials (including Government information or digital photographs that have not been authorized for public release) on a personally owned PED, may result in the immediate confiscation of the device by Code 1120 with referral to the appropriate authorities for investigation and, if warranted, personnel action or prosecution. The authority to confiscate a personal PED based on reasonable suspicion for review of images, recordings, or videos is the responsibility of Code 1120 or law enforcement personnel. Contractors and subcontractors found to be in violation of the Portsmouth Naval Shipyard PED Policy will have the violation reported to the Contracting Officer and may be subject to criminal charges. The penalties for any contractor or subcontractor personnel who are found to be in possession of a PED with prohibited capabilities in violation of this policy will be evaluated on a case by case basis, but at a minimum, the Standard Access Control Badge (SACB) and Defense Biometric Identification System card will be onfiscated. Access to the installation will be denied and a Security Deficiency Log (SDL) will be issued. If the violation is more severe (e.g., PED with prohibited capabilities found with pictures inside of a NAVSEA controlled space, etc.) or repeat in nature violations, contractor or subcontractor personnel may be permanently barred from access to NAVSEA controlled spaces.

1.8.9 Access to Unclassified Information

Access to U.S. Navy technical information manuals, documents, drawings, plans, specifications, and other information (e.g., photos, presentations, renderings, papers, etc.) is Government property and restricted to an official need-to-know basis. Handle, control, and safeguard to prevent oral, visual, and documentary disclosure to the public, to foreign sources, and to personnel not having an official need-to-know.

1.8.10 Disclosure of Information

a. Do not release to anyone outside the Contractor's organization any unclassified information, regardless of medium (e.g., hard copies, computer files, film, tape, document), pertaining to any part of this Contract or any program related to this Contract, unless the Contracting Officer has given prior written approval.

b. Requests for approval under paragraph (a) must identify the specific information to be released, the medium to be used, and the purpose for the release. Submit the request to the Contracting Officer at least 10 business days before the proposed date for release.

c. The Contractor agrees to include a similar requirement, including this paragraph (c), in each subcontract under this Contract. Subcontractors must submit requests for authorization to release through the Prime Contractor to the Contracting Officer.

1.8.11 Portsmouth Naval Shipyard Operations Security Statement

During the course of this Contract, in addition to those restrictions, instructions, and guidelines delineated in the Contract Statement of Work, Contract Data Requirements List (CDRL), and/or other references provided, adhere to the following minimum requirements in support of the Portsmouth Naval Shipyard (PNS) OPSEC Program:

Introduction of personal electronic devices, laptops, tablet PCs, cellular phones, cameras, recording devices, and data recording/storage devices into Government spaces is STRICTLY controlled and forbidden in some cases. Company issued equipment required for the performance of work must be approved by the PNS Code 1120 Security Office. Photography and recording is not allowed except for official use and by permit only. Photographs will be reviewed by Code 1123.2 to ensure sensitive information is not revealed.

Do not discuss Government operations in public or over unprotected or unencrypted communications. Do not post to company websites, publications, newsletters or other media, any images, data, or information that reveal sensitive Government operations, personnel, equipment, and/or classified or controlled unclassified information. When in doubt, company press releases related to this Contract must be coordinated through the Contracting Officer's Representative (COR) or Technical Point of Contact, as applicable, and in conjunction with PNS Public Affairs Office (PAO).

Due to observation of events, operations, physical changes, etc. which may reveal National Security Information or Naval Nuclear Propulsion Information (NNPI), specific restrictions are needed to preclude unintentional release of this information to unauthorized parties. (Unauthorized disclosure and transfer of National Security Information is punishable under 18 USC § 793.) Therefore, do not disclose to unauthorized third parties or post to unofficial sites (including Social Networking sites) any images, data or information, or observed events that reveal sensitive Government operations, personnel, equipment, including, but not limited to:

Tactics, techniques and procedures, production or work schedules, any visible or concealed modifications, upgrades, additions to vessels, weapons or equipment; increases, changes, or decreases in work/deployment frequency or Government personnel, vehicle, vessel movements; specialized equipment orders, deliveries, shipments, etc. Unauthorized disclosures and attempts to solicit this type of information by unauthorized third parties or others not affiliated with this Contract must be reported to PNS Code 1120 Security and/or PNS Police Department. Non-Disclosure requirements remain in effect during the duration of this Contract and indefinitely thereafter.

Government issued badges and identification must be removed and/or concealed from plain sight when off PNS and must not be left in vehicles or unprotected. Badges and passes may not be duplicated, copied, or loaned to others. Lost or stolen identification badges, vehicle passes, etc., must be immediately reported to PNS Code 1120 Security Office and/or PNS Police Department.

It is strongly recommended the Contractor mark and protect related internal production schedules, deliverables, inventories and shortages, and identified vulnerabilities related to production of Government material. Internal company markings (e.g., Business Sensitive, etc.) are appropriate for identifying the aforementioned as sensitive information. All Government information must be destroyed at Contract termination of returned to the Government at the Government's discretion.

1.8.12 MARINE ACTIVITIES (NOT APPLICABLE)

1.8.13 Construction Vehicles

Do not utilize any vehicle that will exceed an HS20 wheel load. The use of "off road", "Utility", "ATV", or any vehicles which cannot be legally operated on State roadways or highways is prohibited.

1.9 EMPLOYEE PARKING

Refer to EMPLOYEE PARKING PLAN paragraph in Section 01 50 00.00 22 TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS (PWD ME)

1.10 TIER 1 INFRASTRUCTURE REQUIREMENTS

Construction activities directly impacting or within three feet of Tier 1 shipyard infrastructure are required to meet the following: Develop, submit, and make available a Tier 1 Infrastructure Coordination Drawings by area identifying clearly for all contract personnel the locations of the listed infrastructure and the three-foot buffer zone. Revision to the Tier 1 Infrastructure Coordination Drawings will be required throughout the life of the project as utilities are installed and removed and as new structures are constructed. Tier 1 Infrastructure Coordination Drawings should be developed specific to scopes of work and site locations.

All construction activities adjacent to Tier 1 infrastructure is required to have an approved Work Authorization Form completed and signed off by OICC management. OICC Management includes Commanding Officer, Deputy, Operations Officer, and Construction Division Director. The following is required to be completed as part of that submittal:

a. Executive Summary and Work Plan

b. Calculations, if required to show no impact to structures

c. Required submittals completed and approved

d. Activity Hazard Analysis

e. Utility Location and Outage Process completed

f. Preparatory and Initial Meetings scheduled

Provide any supplemental information necessary (crane and equipment plans, environmental plans, etc). Provide Work Authorization Form two weeks prior to commencement of work.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

ARE YOU A CONTRACTOR, VENDOR, OR SUPPLIER WHO NEEDS BASE ACCESS?

IF YES, please read below:

Effective 14 August 2017, NCACS credentials will no longer be accepted for base access. Please read below:

- If you applied for an NCACS credential **PRIOR to 17 April 2017**, your NCACS credential will remain in effect for base access through 14 August 2017, when NCACS credentials will no longer be accepted.

- If you applied for an NCACS credential **BETWEEN 17 April and 31 May 2017** you will also be required to obtain a DBIDS credential in order to obtain base access.

- **AFTER 31** May no new NCACS applications will be accepted. All new contractors, vendors and/or suppliers requesting base access will be required to obtain a DBIDS credential.

- **AFTER 14 August 2017 only DBIDS credentials will be accepted for base access.**

- There is no cost to obtaining a DBIDS credential.

FOR MORE INFORMATION VISIT:

https://www.cnic.navy.mil/om/dbids.html

DBIDS ACCESS REQUEST FORM (NAVSHIPYD PTSMH 5500)

1. FIRST NAME:
2. LAST NAME:
3. COMPANY NAME:
4. TITLE:
5. PHONE NUMBER:
DBIDS INFORMATION:
ARE YOU THE PRIMARY DBIDS COMPANY:
DO YOU SUB TO A GC: ☐ No ☐ Yes If yes who:
COMPANY INFORMATION:
COMPANY NAME:
COMPANY ADDRESS:
COMPANY PHONE NUMBER:
PRIMARY POINT OF CONTACT:
FACILITY: *PORTSMOUTH NAVAL SHIPYARD*
ACCESS TIMEFRAME REQUIRED: : ☐ 0600-1800, ☐ 1400-2300, ☐ 24/7
SPONSOR INFORMATION (MUST BE SHIPYARD PERSONNEL):
SPONSOR ORGANIZATION:
SPONSOR OR NAME:
SPONSOR PHONE NUMBER:
SPONSOR TITLE:
SPONSOR EMAIL ADDRESS:
CONFIRM SPONSOR EMAIL ADDRESS:
CONTRACT NUMBER:
CONTRACT EXPIRATION DATE:
NUMBER OF EMPLOYEE REGISTERING AT THIS FACILITY:

All applications will be submitted to CNI_DBIDS@navy.mil

If you have any questions please contact CAPT Capozzi at 207-438-2147.

installation.

(2) Persons requesting access who are not in possession of an approved, government issued credential listed in paragraph 1204.b. shall provide an unexpired document for identity proofing purposes listed in Table 12-1. Any fraudulent information passed during the process may lead to prosecution under appropriate legal authorities.

(3) Authorized government representatives shall, prior to acceptance, visually and tactilely (by touch or feel) inspect documents for evidence of tampering, alteration, or other indications of falsified/fraudulent documents. Authorized government representatives will not accept documents that appear to be fraudulent, forged, or counterfeit and follow the CO's directed standards for response actions to include detention of persons attempting to provide fraudulent documents. Indications of tampered documents include:

(a) Strange text, fonts, slightly altered text, incomplete letters, misaligned words, strange spacing and errors in punctuation or spelling.

(b) Texture or physical indication the photograph has been glued over the original.

Table 12-1. List of Identity Proofing Documents

Documents that Establish Identity
1. U.S. Passport or Passport Card
2. Permanent Resident Card or Alien Registration Receipt Card (Form I-551)
3. Foreign passport that contains a temporary I-551 stamp or temporary I-551 printed notation on a machine-readable immigrant visa (MRIV)
4. Employment Authorization Document (Card) that contains a photograph (Form I-766)
5. In the case of a nonimmigrant alien authorized to work for a specific employer incident to status: (a) Foreign passport; and (b) Form I-94 or Form I-94A has the following: (1) Bearing the same name as the passport; and (2) An endorsement of the alien's nonimmigrant status, as long as the period of endorsement has not yet expired and the proposed employment is not in conflict with any restrictions or limitations identified on the form.
6. Driver's license or ID card issued by a RealID Act compliant state or outlaying possession of the U.S., provided it contains

a photograph and biographic information such as name, date of birth, gender, height, eye color, and address. Licenses or IDs possessing "NOT APPLICABLE FOR FEDERAL PURPOSES" will not be accepted.
7. State-issued Enhanced Driver's licenses
8. Driver's license issued by the U.S. Department of State
9. Border Crossing Card (From DSP-150)
10. Identification card issued by Federal, State, or local government agencies, provided it contains a photograph and biographic information such as name, date of birth, gender, height, eye color, and address.
11. Veteran Health Identification Card (VHIC) issued by the Department of Veterans Affairs
12. Department of Homeland Security "Trusted Traveler" Cards (Global Entry, NEXUS, SENTRI, FAST)
13. U.S. Certificate of Naturalization or Certificate of Citizenship (Form N-550)
14. School identification card with a photograph
15. Persons under the age of 18 who are unable to present a document listed above may present one of the below documents. (a) School record or report card (b) Day care or nursery school record (c) Birth certificate (original or certified copy)
16. Native American Tribal Photo ID cards
17. U.S. Coast Guard Merchant Mariner Credential (MMC) or Merchant Mariner's Documents (MMD)
18. Other documents that may be provide for identity proofing purposes, but must be accompanied by a second form of ID with photograph and biographical information.
a. Social Security Number card
b. Original or certified copy of a birth certificate issued by a state, county, municipal authority, or outlying possession of the U.S. bearing an official seal.
c. Certification of birth Abroad issued by the U.S. Department of State (Form FS-545)
d. Certification of Report of Birth issued by the U.S. Department of State (Form DS-1350
e. Voter's Registration Card

d. Outside the Continental United States (OCONUS)

(1) NSF will utilize appropriate identity proofing credentials such as a passport or nationally issued identity card, or other COCOM approved ID.

(2) ISO will codify approved non-U.S. identity proofing

DEPARTMENT OF THE NAVY LOCAL POPULATION ID CARD/BASE ACCESS PASS REGISTRATION

PRIVACY ACT STATEMENT:

AUTHORITY: 10 U.S.C. 113, Secretary of Defense; DoD Directive 1000.25, DoD Personnel Identity Protection (PIP) Program; DoD Instruction 5200.08, Security of DoD Installations and Resources and the DoD Physical Security Review Board (PSRB); DoD 5200.08-R, Physical Security Program; DoD Directive 5200.27, Acquisition of Information Concerning Persons and Organizations not Affiliated with the Department of Defense (Exception to policy memos); Directive-Type Memorandum (DTM) 09-012, Interim Policy Guidance for DoD Physical Access Control; DTM 14-005, DoD Identity Management Capability Enterprise Services Application (IMESA) Access to FBI National Crime Information Center (NCIC) Files; and E.O. 9397 (SSN), as amended; OPNAVINST 5530.14E, Navy Physical Security and Law Enforcement Program; Marine Corps Order P5530.14, Marine Corps Physical Security Program Manual; SORNNM05512-2 Badge and Access Control System Records and DMDC 16, Identity Management Engine for Security and Analysis (IMESA): http://dpcld.defense.gov/Privacy/SORNsIndex

PURPOSE(S): To control physical access to Department of Defense (DoD), Department of the Navy (DON) or U.S. Marine Corps Installations/Units controlled information, installations, facilities, or areas over which DoD, DON, or U.S. Marine Corps has security responsibilities by identifying or verifying an individual through the use of biometric databases and associated data processing/information services for designated populations for purposes of protecting U.S./Coalition/allied government/national security areas of responsibility and information; to issue badges, replace lost badges, and retrieve passes upon separation; to maintain visitor statistics; collect information to adjudicate access to facility; and track the entry/exit times of personnel.

ROUTINE USE(S): To designated contractors, Federal agencies, and foreign governments for the purpose of granting Navy officials access to their facility.

DISCLOSURE: Providing registration information is voluntary. Failure to provide requested information may result in denial of access to benefits, privileges, and DoD installations, facilities and buildings.

IDENTITY PROOFING AND APPLICANT INFORMATION

1. LAST NAME:	2. FIRST NAME:	3. MIDDLE NAME:	4. NAME SUFFIX: ☐ Jr. ☐ Sr. ☐ I ☐ II ☐ III ☐ IV

5. RACE (*Check one or more*): ☐ AMERICAN INDIAN or ALASKA NATIVE ☐ ASIAN ☐ BLACK or AFRICAN AMERICAN ☐ HISPANIC OR LATINO ☐ NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER ☐ WHITE

6. GENDER (*Check one*): ☐ MALE ☐ FEMALE	7. DATE OF BIRTH:	8. CITY OF BIRTH:	9. STATE OF BIRTH:	10. BIRTH COUNTRY:

11. US CITIZEN (*Check*): ☐ YES ☐ NO	12. DUAL CITIZENSHIP: ☐ YES ☐ NO CITIZENSHIP IF OTHER THAN US *(Country)* :

U.S. Citizen Minimum Documentation Required:
By Birth - Social Security No and/or State ID/Drivers License.
Naturalized - Certification Number, Petition Number, Date, Place and Court, United States passport number, Social Security No and/or State ID/Drivers License.
Derived - Parent's certification number, Social Security No and/or State ID/Drivers License.
Alien Minimum Documentation Required:
Registration Number, Expiration date, Date of entry, Port of entry.

13. IDENTITY SOURCE DOCUMENTS PRESENTED:	14. DOCUMENT NUMBER:	15. ISSUED BY STATE/COURT:	16. ISSUED BY COUNTRY:	17. ISSUED:	18. EXPIRES:
☐ Social Security No.			United States		
☐ State ID/Drivers License			United States		
☐ Passport No.					
☐ Certification Number and Petition Number					
☐ Derived - Parent's Certification Number:			United States		
☐ Alien Registration No.			United States		
		Date of Entry:		Port of Entry:	

OTHER APPROVED IDENTITY SOURCE DOCUMENTS:

☐					
☐					

19. WEIGHT (*Pounds*):	20. HEIGHT (*Inches*):	21. HAIR COLOR (*Check one*): ☐ Blond ☐ Brown ☐ Black ☐ Gray ☐ Red ☐ White ☐ Silver ☐ Auburn ☐ Bald	22. EYE COLOR *(Check one):* ☐ Brown ☐ Green ☐ Blue ☐ Hazel ☐ Black ☐ Gray ☐ Violet ☐ Unknown

23. HOME ADDRESS (*Include city, state, zip code*):	HOME PHONE (*Include Area Code*):
24. BASE SPONSOR'S NAME:	SPONSOR PHONE (*Include Area Code*):

EMPLOYMENT ACTIVITY INFORMATON

25. EMPLOYER NAME AND ADDRESS *(Include city/state/zip code):*	EMPLOYER PHONE *(Include Area Code)*:
26. SUPERVISOR NAME AND ADDRESS *(Include city/state/zip code)*:	SUPERVISOR PHONE*(Include Area Code)*:

27. Check the applicable box for WORK HOURS box or check the OTHER box and enter the work hours, then check the applicable box for WORK DAYS:

WORK HOURS: ☐ 0600-1800 ☐ 0800-1700 ☐ OTHER 24/7 WORK DAYS: ☐ SN ☐ M ☐ T ☐ W ☐ TH ☐ F ☐ ST

PRIOR FELONY CONVICTIONS

28. Have you ever been convicted of a Felony? ☐ YES ☐ NO _______ *Initial*

REQUIREMENT TO RETURN LOCAL POPULATION ID CARD

29. I understand that I am required to return my Local Population Identification Card to the Base Pass Office when it expires or if my employment is terminated for any reason. ________ *(initial)*

AUTHORIZATION AND RELEASE AND CERTIFICATION

30. I hereby authorize the DOD/DON and other authorized Federal agencies to obtain any information required from the Federal government and/or state agencies, including but not limited to, the Federal Bureau of Investigation (FBI), the Defense Security Service (DSS), the U.S. Department of Homeland Security (DHS).

I have been notified of DON right to perform minimal vetting and fitness determination as a condition of access to DON installation/facilities. I understand that I may request a record identifier; the source of the record and that I may obtain records from the State Law Enforcement Office as may be available to me under the law. I also understand that this information will be treated as privileged and confidential information.

I release any individual, including records custodians, any component of the U.S. Government or the individual State Criminal History Repository supplying information, from all liability for damages that may result on account of compliance, or any attempts to comply with this authorization. This release is binding, now and in the future, on my heirs, assigns, associates, and personal representative(s) of any nature. Copies of this authorization that show my signature are as valid as the original release signed by me.

FALSE STATEMENTS ARE PUNISHABLE BY LAW AND COULD RESULT IN FINES AND/OR IMPRISONMENT UP TO FIVE YEARS.

BEFORE SIGNING THIS FORM, REVIEW IT CAREFULLY TO MAKE SURE YOU HAVE ANSWERED ALL QUESTIONS FULLY AND CORRECTLY.

I DECLARE UNDER PENALTY OF PERJURY THAT THE STATEMENTS MADE BY ME ON THIS FORM ARE TRUE, COMPLETE AND CORRECT.

DATE _______________ SIGNATURE ____

FINAL DETERMINATION ON YOUR ACCESS: The Base Commanding Officer has final authority for determination on granting physical access to DON controlled installations/facilities under his/her jurisdiction.

BELOW COMPLETED BY BASE REGISTRAR PERSON CONDUCTING IDENTY PROOFING and NCIC CHECK

31. INFORMATION VERIFIED BY:	32. ENTERED IN C/S SYSTEM BY:	33. PASS ISSUE DATE:	34. PASS EXPIRATION DATE:

35. NCIC CHECK PERFORMED BY:	36. RESULTS OF NCIC CHECK:	37. RESULTS OF LOCAL RECORDS CHECK:
	☐ NO RECORDS ☐ RECORD IDENTIFIER	☐ NO RECORDS ☐ RECORD IDENTIFIER
	RECORD NUMBER:	RECORD NUMBER:

Office of Under Secretary of Defense Directive-Type Memorandum (DTM) 09-012, "Interim Policy Guidance for DoD Physical Access Control," December 8, 2009. DTM 09-012 requires that DoD installation government representatives query the National Crime Information Center (NCIC) and Terrorist Screening Database to vet the claimed identity and to determine the fitness of non-federal government and non-DoD-issued card holders (i.e. visitors) who are requesting unescorted access to a DoD installation. The minimum criteria to determine the fitness of a visitor is: 1) not on a terrorist watch list; 2) not on an DoD installation debarment list; and 3) not on a FBI National Criminal Information Center (NCIC) felony wants and warrants list. Additionally, SECNAV Memo, Policy for Sex Offender Tracking and Assignment and Access Restrictions within the Department of the Navy, of 7 Oct 08 and OPNAVINST 1752.3 established the Navy's policy on sex offenders, requiring Region Commanders (REGCOMs) and Installation Commanding Officers (COs) to prohibit sex offender access to DoN facilities and Navy owned, leased or PPV housing. This form describes the authority and purpose to collect and share the required information; and identifies the applicant/visitor and sponsor; and authorizes the DoD to perform the minimum vetting and fitness determination criteria. A favorable response on the vetting and fitness determination is required to receive access to DOD-controlled installation/facilities.

Instruction for completing the Local Population Access Registration Form

INSTRUCTIONS: Please complete all information in black ink (printed) or by typing. By voluntarily providing your Personal Information, you agree to the following terms and restrictions:

RESTRICTIONS: Local Population Identification Card/Base Access Pass may only be used by person to whom they are issued and for the specific business/purpose issued. Applicants are reminded that soliciting (i.e., door-to-door sales) is prohibited on the base, and that such activity is grounds for cancellation of the Pass. Additionally, such action may result in debarment from the base and legal action. The Base Commanding Officer has discretion over specifying the period of validity for any Local Population ID Cards/Base Access Passes that are issued under his/her jurisdiction.

Review the Privacy At Statement that is printed at the top of the form

Block 1: Enter the Last Name.
Block 2: Enter the First Name.
Block 3: Enter the Middle Name.
Block 4: If applicable, check the box for Name Suffix.
Block 5: Check the applicable box for Race.
Block 6: Check the applicable box for Gender.
Block 7: Enter Date of Birth.
Block 8: Enter City of Birth.
Block 9: Enter State of Birth.
Block 10: Enter Country of Birth.
Block 11: Check the applicable box for US Citizenship.
Block 12: If not a US Citizen, enter the name of the Country of Citizenship.
Block 13: Two forms of identity source documents from the list of acceptable documents listed below must be presented to the base registrar with this completed form. Check the box for the type of Documents that will be presented for identity proofing. If the document type is not listed, use the two rows under Other Approved Identity Source Documents to enter the type of document(s) that you will present.
Block 14: Enter the Document Number located on the Identity Proofing Source document that was checked in Block 13.
Block 15: Enter the State that issued the Identity Source Document.
Block 16: Enter the Country that issued the Identity Source Document.
Block 17: Enter the Date that the Identity Source Document was issued.
Block 18: Enter the Date that the Identity Source Document will expire.
Block 19: Enter Weight in pounds.
Block 20: Enter Height in inches.
Block 21: Check the applicable box for Hair Color.
Block 22: Check the applicable box for Eye Color.
Block 23: Enter Home Address Including City, State, Zip Code, and Home Telephone Number.
Block 24: Enter Name of Registrant's Base Sponsor and Base Sponsor's Telephone Number.
Block 25: Enter Employer Name and address including City, State, Zip Code, and Employer's Telephone Number.
Block 26: Enter Supervisor's Name including City, State, Zip Code, and Supervisor's Telephone Number.
Block 27: Check the applicable box for Work Hours box or check the OTHER box and enter the work hours, then check applicable boxes for Work Days. Block 28: Check the applicable answer if you have been convicted of Felony and enter initials.
Block 28: Check the applicable box for felony conviction.
Block 29: Enter initials to accept terms for returning Local Population Identification Card.
Block 30: Sign and date the form to attest that the foregoing information is true and complete to best of your knowledge.

LIST OF ACCEPTABLE DOCUMENTS - All documents must not be expired.
Must present one selection from List A or a combination of one selection from List B and one selection from List C.

List A - Documents that Establish Identity and Employment Authorization

OR

1. U.S. Passport or U.S. Passport Card
2. Permanent Resident Card or Alien Registration Receipt Card (Form I-551).
3. Foreign passport that contains a temporary I-551 stamp or temporary I-551 printed notation on a machine-readable immigrant visa.
4. Employment Authorization Document that contains a photograph (Form I-766).
5. For a nonimmigrant alien authorized to work for a specific employer because of his or her status:
   a. Foreign Passport; and
   b. Form I-94 or Form I-94A that has the following:
      (1) The same name as the passport; and
      (2) An endorsement of the alien's nonimmigrant status as long as that period of endorsement has not yet expired and the proposed employment is not in conflict with and restrictions or limitations identified on form.
6. Passport from the Federal States of Micronesia (FSM) or the Republic of the Marshal Islands (RM) with Form I-94 or Form I-94A indicating nonimmigrant admission under the Compact of Free Association Between the United Stated and FSM or RM.

List B - Documents that Establish Identity

AND

1. Driver's license or ID card issued by a State or outlying possession of the United States provided it contains a photograph or information such as name, date of birth, gender, height, eye color, and address.
2. ID card issued by federal, state or local government agencies or entities, provided it contains a photograph or information such as name, date of birth, gender, height, eye color, and address.
3. School ID card with a photograph
4. Voter's registration card.
5. U.S. Military card or draft record.
6. Military dependent's ID card.
7. U.S. Coast Guard Merchant Mariner Card.
8. Native American tribal document.
9. Driver's license issued by a Canadian government authority.

For persons under age 18 who are unable to present a document listed above:

10. School record or report card.
11. Clinic, doctor, or hospital record.
12 Day-care or nursery school record.

List C - Documents that Establish Employment Authorization

1. A Social Security Account Number card, unless the card includes one of the following restrictions:
   (1) NOT VALID FOR EMPLOYMEMT
   (2) VALID FOR WORK ONY WITH INS AUTHORIZATION.
   (3) VALID FOR WORK ONLY WITH DHS AUTHORIZATION.
2. Certification of Birth Abroad issued by the Department of State (Form FS-545).
3. Certification of Birth issued by the Department of State (Form DS-1360).
4. Original or certified copy of birth certificate issued by a State, county, municipal authority or territory of the United States bearing an official seal.
5. Native American tribal document.
6. U.S. Citizen ID Card (Form I-197).
7. Identification Card for Use of Resident Citizen in the United States (Form I-179).
8. Employment authorization document issued by the Department of Homeland Security.

The remainder of the form will be completed by the Base Registrar Person conducting Identify Proofing process and NCIC check.

AGENCY DISCLOSURE STATEMENT:

The public reporting burden for this collection of information, OMB 0703-0061, is estimated to average ten (10) minutes per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding the burden estimate or burden reduction suggestions to the Department of Defense, Washington Headquarters Services, Executive Services, at whs.mc-alex.esd.mbx.dd-dod-information-collections@mail.mil. Respondents should be aware that notwithstanding any other provision of law, no person shall be subject to any penalty for failing to comply with a collection of information if it does not display a currently valid OMB control number.

PLEASE DO NOT RETURN RESPONSE TO THE ABOVE ADDRESS.
Responses should be sent to the Base Registrar.

NAVFAC – PWD MAINE
OUTAGE REQUEST FORM

CIRCLE ONE:			
RAILROAD	STEAM	PARKING	ROAD/PARTIAL ROAD
CRANES	FRESH/HOT WATER	FIRE ALARM	MISC
HIGHBAY CRANE	ELECTRICAL	SPRINKLER SYSTEM	OTHER

Date: **Project Task Number:**

Project Manager Name

Contractor Name:

Contact Phone Number:

<u>DESCRIPTION OF OUTAGE REQUIRED:</u>

Location of Outage:

System Requiring Outage:
Date & Time of Outage:
Duration of Outage:

Contractor Signature: _____

Rep Signature:
Date Final Outage Approval Required:

NOTE: IF A ROAD OUTAGE, IS A RAM ROUTE AFFECTED: YES NO (Circle One)

NOTE: For rail/crane outages and railroad track outages, use NAVSHIPYD PTSMHINST 11230.3 form.

DEPARTMENT OF THE NAVY
PORTSMOUTH NAVAL SHIPYARD
PORTSMOUTH, N. H. 03804-5000

IN REPLY REFER TO:
NAVSHIPYD PTSMHINST 5500.6A
1120

31 MAY 2024

NAVSHIPYD PTSMHINST 5500.6A

From: Commander, Portsmouth Naval Shipyard

Subj: PORTSMOUTH NAVAL SHIPYARD PORTABLE ELECTRONIC DEVICES POLICY

Ref: See Enclosure (1)

Encl: (1) References
(2) PED Functionality Matrix
(3) Use of Apple Face ID on Flank Speed-Enrolled PNS Issued Equipment

1. Purpose.

a. To establish procedures, authorizations, and prohibitions for Portable Electronic Device (PED) use at Portsmouth Naval Shipyard (PNS). Revision A provides additional clarification, restructures the instruction to be more based on the capabilities of the device vice the device itself, incorporates lessons learned, and adds the waiver for camera capable PEDs inside the Controlled Industrial Area (CIA).

b. Exclusions. This instruction **does not**:

(1) Apply to properly credentialed law enforcement officials (including, but not limited to, Naval Security Forces, Federal Fire, and Naval Criminal Investigative Service) in performance of their official duties. This exclusion applies to official gear and not personally owned gear.

(2) Apply to equipment exempted by higher guidance; specifically per reference (a). These include receive-only pagers, Global Positioning System (GPS) receivers, hearing aids, pacemakers, other implanted medical devices, or personal life support systems. All of these pieces of equipment are allowed in all Naval Sea Systems Command (NAVSEA) spaces.

(3) Alter or supersede the existing authorities and policies regarding the protection of Sensitive Compartmented Information (SCI), as directed in reference (b) and other laws and regulations.

(4) Alter or supersede the existing authorities and policies regarding the protection of Special Access Programs (SAP).

(5) Alter the responsibilities of the Deputy Administrator for Naval Reactors set forth in reference (c) and the Under Secretary for Nuclear Security, Department of Energy, set forth in reference (d).

2. Cancellation. NAVSHIPYD PTSMHINST 5500.6 of 17 November 2020.

3. Effective Date. 10 JUN 2024

4. Scope and Applicability. This instruction applies to PNS and PNS Detachment San Diego (PNSDETSD) personnel, tenant commands, contractors, and visitors who will need to gain access to a NAVSEA controlled space.

5. Background. Introduction of PEDs poses a risk to the protection of classified information and Controlled Unclassified Information (CUI), including Naval Nuclear Propulsion Information (NNPI) as defined in reference (e). Various capabilities of PEDs (e.g., cameras, microphones) could permit direct ex-filtration of information, indirect ex-filtration of information via compromising electromagnetic emanations, and introduction of malicious capabilities to Navy information systems. Due to the fast pace of evolving technology, it is impractical to develop policy regarding PEDs based on device type or form factor; instead, it is imperative that policy governing these devices is focused on their capabilities.

6. Definitions

a. PNS. PNS is defined hereinafter as Portsmouth Naval Shipyard in Kittery, ME and PNSDETSD in San Diego, CA.

b. NAVSEA Controlled Space. Any Nuclear Work Areas (NWAs), Controlled Nuclear Information Areas (CNIAs), Security Islands (SIs), or the CIA.

c. PED. A PED is a portable electronic device that is easily transportable and has the capability to record, copy, store, export, or transmit data, photography, digital images, video or audible information. Examples of PEDs include, but are not limited to: cell phones, fitness trackers, smart watches, watches with input capability, tablets, laptops, media devices, Bluetooth consumer electronics, cellular transceivers, and portable diagnostic equipment.

d. Mobile Device. A type of PED which is further defined in reference (f) as a portable computing device that has a small form factor such that it can easily be carried by a single individual; is designed to operate without a physical connection (e.g., wirelessly transmit or receive information); possesses local, non-removable data storage; and is powered on for extended periods of time with a self-contained power source. Mobile devices also include voice communication capabilities, onboard sensors that allow the device to capture information (e.g., photograph, audio, video, record, or determine location), or built-in features for synchronizing local data with remote locations. Examples include, but are not limited to, smart phones, tablets, and E-readers.

e. For the purposes of this instruction, PED and mobile device are synonymous. This instruction applies to all devices broadly meeting the PED definition and includes government owned, contractor owned, and personally owned PEDs within NAVSEA spaces.

f. Near Field Communication (NFC). NFC is defined as a short-range wireless communication system employing radio waves to enable a mobile device to interact with another device or card reader when within 10 cm (4 in) of each other.

g. Computer cellular wireless service is to be treated the same as phone based cellular wireless service with regard to spaces where its use is allowed or prohibited.

h. Personal Wearable Fitness Devices (PWFD). PWFDs are a type of PED with limited functionality for fitness tracking. Permitted PWFDs, as defined in reference (g), typically include Bluetooth, receive only GPS, accelerometer, altimeter, gyroscope, heart activity, vibration feature, and NFC capabilities.

7. Policy. PEDs are prohibited within spaces authorized for processing classified information, or within spaces where any classified discussion is taking place, except as otherwise authorized. PEDs are permitted in other areas subject to the usage restrictions as outlined below:

a. Government owned and issued PEDs (e.g. iPhones and laptops) used by PNS employees (Unit Identifier Code 39040) will be managed and controlled by Information Technology and Cybersecurity (Code 109). Government owned and issued PEDs (e.g. iPhones and laptops) used by other commands will be managed and controlled by those commands. All government PEDs (e.g. iPhones and laptops) will have a serialized tamper-evident label affixed to the device that identifies the organization that applied the label to indicate that the photographic and video capabilities are disabled.

b. PEDs may not be connected to Government information systems unless explicitly allowed in the information system's Authorization To Operate (ATO) documentation. Government PEDs may not be physically connected to, backed up on, or transfer government sensitive information to a personally owned device.

c. Personally owned PEDs **must not** store government sensitive information to include documents, notes, or working papers.

d. Personally owned peripherals (e.g. keyboards, mice, headphones) may not be connected to Government PEDs, with the exception of analog audio (headphones/speakers) or monitors connected via one-way (PED to monitor) analog or digital connections. Personally owned wireless peripherals (i.e. Bluetooth headphones, speakers, etc.) are authorized for use with personally owned PEDs. Contractor owned wireless peripherals are authorized for use with contractor owned PEDs.

e. Government laptops for PNS employees must be approved by Code 109. All government laptops must adhere to the requirements of reference (j) for use in areas where collateral classified information is processed.

f. PWFDs are authorized in all areas per reference (g) if it meets the following criteria.

(1) If commercially available in the U.S. or through U.S. military exchange, marketed primarily as fitness or sleep devices, and designated as Federal Communication Commission (FCC) Class B digital devices (denoted as FCC Class B certified, or FCC Class B exempt) as described in reference (k).

(2) If they have Bluetooth, receive only Global Positioning System, accelerometer, altimeter, gyroscope, heart activity, vibration feature, or NFC capabilities.

(3) If they receive and contain vendor-supplied software updates that do not add any features or capabilities prohibited in paragraph 7.g.(3).

g. PEDs in Classified Spaces. These spaces include, but are not limited to, open storage areas, NWAs, Controlled Access Areas (CAA), secure rooms, vaults, occupied dry docks, shipboard, and applicable restricted areas.

(1) Consistent with reference (l), in spaces where classified SECRET and CONFIDENTIAL collateral information is processed, transmitted, stored, or discussed, PED with limited capabilities are permitted:

(a) If commercially obtained in the U.S. or through U.S. military exchange.

(b) If assigned a Federal Communications Commission (FCC) identifier denoting compliance with the limits for a Class B digital device (e.g., consumer grade electronics including laptops, tablets, and cellphones, etc.) designated by the FCC, pursuant to Part 15 of the FCC Rules, per reference (m). Class B devices are designated by the FCC as suitable for use by the general public in residential and commercial, business and industrial environments.

(c) If they contain only government approved software and/or vendor-supplied software and receive only updates that do not add any features or capabilities prohibited in paragraph 7.g.(2).

(d) If they have any or all of the following allowed capabilities: Bluetooth, receive only Global Positioning System, accelerometer, altimeter, gyroscope, heart monitor, vibration, or NFC capabilities, but none of the prohibited capabilities in paragraph 7.g.(2) of this instruction.

(e) If they have password or pin protections enabled and have up-to-date anti-virus software protection installed, where those capabilities exist.

(2) The following capabilities are prohibited on PEDs in classified spaces:

(a) Cellular and Wi-Fi transceivers. NMCI unclassified laptops with embedded Wi-Fi are authorized for use in classified spaces per reference (j) which requires laptops be physically connected to the unclassified network before powering on or have Wi-Fi disabled prior to entry into the space.

(b) Photographic, video capture, recording, and transmission, microphone, and/or audio capture, recording, and transmission capabilities.

(c) Installed removable media unless otherwise authorized.

(d) Capability to perform radio frequency transmission at greater than 100 milliwatts (mW) Equivalent Isotropically Radiated Power as, where feasible, determined using FCC data.

(3) The following capabilities are prohibited on PWFDs in classified spaces:

(a) If they contain any external or conflicting hardware or software modifications, including the installation of third party apps.

(b) If they contain cellular or Wi-Fi, photographic, video capture/recording, microphone, or audio recording capabilities.

(c) In conjunction with USB accessories, including but not limited to Bluetooth dongles and charging cables.

(d) From being connected to any government information systems.

(4) If a space (e.g., individual office or conference room) is intermittently used for processing or discussing classified information, positive action must be taken to remove PEDs with prohibited capabilities from the space before classified information is processed or discussed. If classified information must be transported through areas where PEDs with prohibited capabilities are present, the classified information must be protected to ensure that the presence of PEDs do not inappropriately disclose the information.

(5) Per reference (n), the use of government owned PEDs is permitted in Naval Nuclear Propulsion Program Emergency Control Centers during drills and emergencies, regardless of their radio frequency capabilities.

(6) Use of PEDs in spaces processing information at a level higher than SECRET collateral is not permitted unless explicitly allowed in the information system's ATO documentation per reference (o).

h. <u>PEDs in U-NNPI Spaces</u>. Consistent with reference (p) in spaces where U-NNPI is processed, transmitted, stored, or discussed (e.g., SIs, CNIAs), PEDs are permitted as follows:

(1) For CNIA spaces (where U-NNPI information is processed or stored in its material form or physical manifestation (e.g., industrial processes, material)), personally owned PED with photographic, video capture, recording, and transmission, microphone, and/or audio capture, recording, and transmission capabilities are prohibited. Government and contractor owned PEDs are permitted in these areas provided they have their photographic and video capabilities (logically or physically) disabled, and they have been marked to identify that the capability is disabled as specified in paragraph 7.h.(1).(a) of this instruction.

(a) Organizations that logically disable photographic and video capabilities using mobile device management solutions must develop and maintain programs to mark PEDs that have been logically disabled. These programs must provide a serialized, tamper evident marking to annotate the photographic and video capabilities are disabled; the label must identify the organization that applied the label, and it must be visible on the PED even when contained within a case. All NAVSEA organizations should honor the markings of other NAVSEA organizations and contractors.

(b) Contractor owned PEDs, other than cell phones, that do not have markings to identify that the capability is disabled are required to have the PEDs reviewed by Security Services (Code 1123) and Code 109 (if applicable) to ensure the capabilities have been disabled. This will be documented via a laptop registration or camera permit.

(c) Personal and contractor owned cell phones are authorized inside of CNIA spaces if the camera(s) is (are) removed and the void filled with an epoxy or if it meets the requirements of paragraph 7.h.(1). Additionally, the camera application on the device must indicate that it is disabled.

(2) For SI spaces and inside the CIA, camera capable PEDs (unmodified) are permitted. Although these devices are permitted, use of these devices to take photographs, record video, or record audio is prohibited. This allowance for inside the CIA is provided by reference (q).

i. <u>Wi-Fi and Wi-Fi Hotspots</u>. Use of wireless hotspots must be coordinated with the Information Systems Security Manager (ISSM) to ensure usage does not interfere with official-use wireless capabilities or the effectiveness of wireless intrusion detection systems. The use of Wi-Fi hotspots is authorized as follows:

(1) Wi-Fi hotspots are not authorized in spaces in which classified material is being handled, processed, or discussed unless explicitly allowed by the information system's ATO documentation per reference (o). A Tempest evaluation conducted by a Certified Tempest Technical Authority must be done prior to installing Wi-Fi hotspots in a building that contains classified spaces.

(2) Wi-Fi hotspots are not authorized in areas directly adjacent to sensitive compartmented information spaces or Special Access Program (SAP) spaces without the prior authorization of the cognizant special security officer or SAP control officer.

(3) Navy enterprise provided wireless solutions, to include Wi-Fi hotspots and Wi-Fi hotspot capability on Navy enterprise provided PEDs (e.g., Navy issued Government cellphones and laptops), are authorized for use within NAVSEA per its Navy approved configuration, but use must be coordinated with the ISSM to de-conflict with wireless intrusion detection systems.

(4) Contractor owned and Government owned (but not provided as an approved Navy enterprise solution) Wi-Fi hotspots may be used within NAVSEA spaces, but must be coordinated with the ISSM.

(5) Personally owned Wi-Fi hotspots are not authorized. The Wi-Fi hotspot function on personally owned PEDs must be turned off while in NAVSEA spaces.

(6) Dual homing (connecting to both wireless and wired networks simultaneously) is prohibited unless explicitly allowed by the information system's ATO documentation per reference (o).

j. Classified PEDs. NAVSEA personnel may use approved classified PEDs per the requirements defined in reference (r). Requests for Department of Defense Mobility Classified Capability PEDs must be approved by the Command Information Office (SEA 00I) Enterprise Authorized Approving Official for the Defense Information Systems Agency (DISA) service requests.

k. In areas where PEDs are authorized, the use of recording or imaging capabilities built into the PED are subject to the restrictions in reference (i) and (p).

l. The capabilities of PEDs continue to evolve in response to a changing commercial marketplace. Supervisors should refer to enclosure (2), which explains PEDs by functionality and serves as an aid to reference existing PED policy.

m. Government PEDs (e.g., iOS devices) are authorized to use biometrics for authentication. Government PEDs are Flank Speed enrolled and have photography capabilities disabled, through managed security profiles. Enclosure (3) provides current amplifying guidance for biometric authentication on Flank Speed-enrolled devices.

n. Reasonable Accommodation. Government issued PEDs, including recording devices, may be approved for use through the reasonable accommodation (RA) process on a case by case basis for individuals with disabilities. The ISSM and Activity Security Manager (ASM) must review the PED as part of the RA interactive process to ensure the device does not introduce an unacceptable risk. Recording devices approved through the RA process must be issued a permit per reference (i) and (p), tracked for accountability, and disposed of per reference (s).

o. Exceptions:

(1) If a PED with prohibited capabilities is required to support operational requirements, approval from the ASM and ISSM is required. This request will be supported if the prohibited capability can be disabled in a verifiable and technically sound manner (e.g., logically or physically disabled, tamper proof tape over camera, escorted at all times, etc.).

(2) Approval from the appropriate Authorizing Official and Certified Tempest Technical Authority is required prior to authorizing exceptions for wireless capabilities prohibited by this instruction per reference (a).

(3) Medically prescribed android or iPhone devices paired with heart and glucose monitors have also been approved for use in classified spaces, provided the recording capabilities are disabled, does not receive or make calls, and are programmed with proprietary software as a controller for the medical devices only. These devices need to be reviewed by and authorization given by the ASM prior to being used inside of NAVSEA controlled spaces.

(4) All exceptions to this PED policy must be documented in writing, approved by the Director of Security, and include a discussion of the business need, pertinent risks, and how the risks are mitigated.

p. Violations:

(1) Discovery of PEDs in prohibited areas, or discovery of prohibited materials (including Government information or digital photographs that have not been authorized for public release on PEDs) on a personally owned PED, may result in the immediate confiscation of the device by Code 1120 with referral to the appropriate authorities for investigation and, if warranted, personnel action or prosecution. The authority to confiscate a personal PED based on reasonable suspicion for review of images, recordings, or videos is the responsibility of Code 1120 or law enforcement personnel (i.e. Naval Security Forces, Naval Criminal Investigative Service, etc.).

(2) Civilian employees failing to comply with the requirements of this instruction may be subject to disciplinary action.

(3) Military members failing to comply with the requirements of this instruction may be subject to disciplinary action per Uniform Code of Military Justice.

(4) Contractors and subcontractors found to be in violation of this instruction will have the violation reported to their Contracting Officer Representative and may be subject to criminal charges. The penalties for any contractor or subcontractor personnel who are found to be in possession of a PED with prohibited capabilities in violation of this instruction will be evaluated on a case by case basis, but at a minimum, the Standard Access Control Badge and Defense Biometric Identification System card will be confiscated. Access to the installation will be denied and an SDL will be issued per reference (e). If the violation is more severe (e.g., PED with prohibited capabilities found with pictures inside of a NAVSEA controlled space, etc.) or repeat in nature violations, contractor or subcontractor personnel may be permanently barred from access to NAVSEA controlled spaces.

(5) Visitors found to be in violation of this instruction will be reported to Code 1120 and may be subject to criminal charges.

8. Responsibilities. All PNS and PNSDETSD personnel, tenant commands, contractors, and visitors will comply with the processes set forth in this instruction and will act proactively to ensure the use of PEDs is consistent with the protection and safeguarding of controlled and classified information.

9. Records Management. Records created as a result of this instruction, regardless of media and format, must be managed per Secretary of the Navy (SECNAV) Manual SECNAV M-5210.1 of September 2019.

10. Review and Effective Date. Per Office of the Chief of Naval Operations (OPNAV) Instruction OPNAVINST 5215.17A, Code 1120 will review this instruction annually on or before the anniversary of its issuance date to ensure applicability, currency, and consistency with Federal, Department of Defense, SECNAV, and Navy policy and statutory authority using OPNAV 5215/40, Review of Instruction. This instruction will be in effect for 10 years, unless revised or canceled in the interim, and will be reissued by the 10-year anniversary date if it is still required, unless it meets one of the exceptions in OPNAVINST 5215.17A, paragraph 9. Otherwise, if the instruction is no longer required, it will be processed for cancellation as soon as the need for cancellation is known following the guidance in OPNAV Manual 5215.1 of May 2016.

11. Forms and Information Management Control

a. The Portsmouth Naval Shipyard Photography Permit is obtained by Code 1120.

b. The Visitor Portable Computer Equipment Registration is obtained from Code 1120.

c. The Webcam User Agreement is obtained from Code 1120.

Michael Craig Oberdorf
M. C. OBERDORF

Releasability and distribution:
This instruction is not cleared for public release and is available electronically only via the PNS Instructions and Notices Web site.
A, B, D, O, W

REFERENCES

(a) DoD Directive 8100.02 of 14 April 2004
(b) E.O. 12333, United States Intelligence Activities, as amended
(c) 50 U.S.C. 2406, Deputy Administrator for Naval Reactors
(d) 42 U.S.C. 7158, Naval Reactors and Military Application Programs, as amended
(e) OPNAVINST N9210.3
(f) Committee on National Security Systems Instruction No. 4009 of 6 Apr 2015
(g) NAVADMIN 216/15, Cyber Hygiene Authorization to use Personal Wearable Fitness Devices
(h) NAVSHIPYD PTSMHINST 5530.8 Security Deficiency Log Process
(i) NAVSHIPYD PTSMHINST 5510.27 Information and Personnel Security Program
(j) NAVADMIN 290/15 Use of Unclassified Navy and Marine Corps Intranet Laptops with Embedded Wireless
(k) FCC Office of Engineering and Technology Bulletin 62
(l) ALNAV 019/16, Acceptable Use of Authorized Personal Portable Electronic Devices
(m) FCC Office of Engineering and Technology Bulletin 65
(n) NAVSEA ltr 08B/15-06050 – Approval of Limited Usage of Cellular Telephones Within Naval Nuclear Propulsion Program Emergency Control Centers
(o) DoD Instruction 8510.01 of 12 March 2014
(p) NAVSEAINST 5510.2D Access and Movement Control
(q) NAVSEA 03I Letter Ser 03I/100 dated 01 November 2023
(r) Navy Security Note 08-15, Guidance on the Proper Use of DOD Mobility Classified Capability-Secret (DMCC-S) Device and Other Authorized Classified PED
(s) DON CIO Message DTG 281759Z AUG12, Processing of Electronic Storage Media for Disposal

PED FUNCTIONALITY MATRIX

PED Functionality or Device Type	Example Device	Secure Room/NWA/ Occupied Dry Dock/Shipboard	CNIA	SI	CIA / ENCLAVE
PED with Camera Disabled (camera logically disabled, removed and epoxied, or without camera capabilities)	Smart Phone (Blackberry, iPhone, Android). Smart Watch (Apple, Samsung, Fitbit, Garmin)	Prohibited	Allowed	Allowed	Allowed
PED with Camera Enabled (camera enabled)	Smart Phone (Blackberry, iPhone, Android). Smart Watch (Apple, Samsung, Fitbit, Garmin)	Prohibited	Prohibited	Allowed	Allowed
Personal Photography Device (primary function)	Digital Camera, Video Camera	Prohibited	Prohibited	Prohibited	Prohibited
Radio Frequency (RF) Receivers	Pager, Satellite Radio	Prohibited	Allowed	Allowed	Allowed
RF Transmitter.[1, 2]	RF Radio Transceiver, Walkie-Talkie	Prohibited	Allowed	Allowed	Allowed
Audio-Recording Device[5] (primary function)	Digital voice recorders	Prohibited	Prohibited	Prohibited	Prohibited
Devices without Removable Memory (Play-Only)	Moving Picture Experts Group Audio Layer 3 (MP3) players, compact disc (CD) players	Allowed	Allowed	Allowed	Allowed

PED Functionality or Device Type	Example Device	Secure Room/NWA/ Occupied Dry Dock/Shipboard	CNIA	SI	CIA / ENCLAVE
Medical Monitoring or Testing Devices[3,4]	Heart monitor, glucose tester, hearing aids, insulin pump	Allowed per Reference (a)	Allowed per Reference (a)	Allowed per Reference (a)	Allowed per Reference (a)
PWFD[6]	See Note 6	Allowed per Reference (g)	Allowed per Reference (g)	Allowed per Reference (g)	Allowed per Reference (g)

Notes:

1. An RF transmitter is defined as any RF transmitter, with the exception of single-function cellphones that are addressed separately.
2. Amateur Radio Emergency Service and Radio Amateur Civil Emergency Service members must be approved in writing, per NAVSEAINST 5239.2B, Cybersecurity Program, to carry hand-held transceivers.
3. Includes medical devices prescribed by a physician.
4. PEDs that are used in conjunction with a medical device must comply with the requirements of this instruction.
5. Primary function audio recording devices must be approved in writing by the Code 1120 Security Director prior to purchase and use.
6. PWFDs are allowed in all areas if requirements of paragraph 7.f are met. If any prohibited capabilities exist on the PWFD as indicated in paragraph 7.g.(3), then the PWFD is not allowed inside a classified space.

USE OF APPLE FACE ID ON FLANK SPEED-ENROLLED PNS ISSUED EQUIPMENT

1. Background. Face ID is an Apple, Inc authentication technology used on some of their mobile devices. User authentication (i.e., "logging in") is frequently required as part of Flank Speed's security interface. The use of Face ID is an acceptably low risk for Flank Speed-enrolled devices, given that:

a. Face ID technology provides sufficiently low false-positive rates of identification.

b. The data used by and stored for Face ID is contained within a secured enclave on the device. The data is encrypted and is not copied nor backed up off of the device.

c. Flank Speed-enrolled devices have the photography capabilities disabled, through the centrally managed security profiles. This prevents the use of the devices' cameras to capture images and video.

2. Action: PNS personnel are authorized to enable and use Face ID for authentication on Code 109-issued iOS Flank Speed enrolled devices. When using Face ID, the "Attention" settings must be enabled as follows:

a. "Require Attention for Face ID."

b. "Attention Aware Features."

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SECTION 01 20 00.00 22
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PRICE AND PAYMENT PROCEDURES (PWD ME)
11/20, CHG 4: 08/24

PART 1 GENERAL

This Section applies to Design-Bid-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EP 1110-1-8	(2021) Engineering and Design -- Construction Equipment Ownership and Operating Expense Schedule

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

Earned Value Report; G

Accounting Summary of Unit Cost Quantities; G

1.3 EARNED VALUE REPORT

1.3.1 Data Required

This Contract requires the use of a cost-loaded Network Analysis Schedule (NAS). Schedule of Prices must not be used with cost-loaded Network Analysis Schedule (NAS). Use Earned Value Report derived from cost-loaded NAS. Within 15 calendar days of Contract Award, prepare and deliver to the Contracting Officer a Schedule of Prices (construction Contract) as directed by the Contracting Officer. Provide a detailed breakdown of the Contract price, giving quantities for each of the various kinds of work, unit prices, and extended prices. Indirect activities (e.g., meetings, presentations, submittals, surveys, haul road or storage yard maintenance, dust control, security) will not be cost loaded. Evenly distribute the costs across the associated construction activities. Contractor overhead and profit including salaries for field office personnel, if applicable, must be proportionately spread over all pay items and not included as individual pay items.

1.3.2 Payment Schedule Instructions

Payments will not be made until the Earned Value Report from the Cost-Loaded NAS has been submitted to and accepted by the Contracting

Officer.

Additionally, the Earned Value Report must be separated as follows:

a. Primary Facilities Cost Breakdown:

Defined as work on the primary facilities out to the 5 foot line. Work out to the 5 foot line includes construction encompassed within a theoretical line 5 foot from the face of exterior walls and includes attendant construction, such as pad mounted HVAC cooling equipment, cooling towers, and transformers placed beyond the 5 foot line.

b. Supporting Facilities Cost Breakdown:

Defined as site work, including incidental work, outside the 5 foot line.

If as-built drawings, eOMSI, or operations and maintenance manuals are required, payments for either will not be made until complete and final submissions are approved by the Contracting Officer. The following minimum amounts must be included as Schedule of Prices line items:

Minimum Schedule of Prices values for complete final approved as-built drawings (based on awarded Contract amount: $0-$150K 4%; $150k-$700k 3%; $700k - $2.49M 2%; $2.5M + 2% (max $200k).

Minimum Schedule of Prices value for complete final approved Operation and Maintenance Manuals (based on awarded Contract amount: $0-$150K 2%; $150k-$700k 2%; $700k - $2.49M 1%; $2.5M + 1% (max $200k).

Minimum Schedule of Prices value for complete final approved eOMSI Workbook (based on awarded Contract amount: $0-$150K 2%; $150k-$700k 2%; $700k - $2.49M 1%; $2.5M + 1% (max $200k).

Minimum Schedule of Prices value for complete final approved Coordination Drawings (based on awarded Contract amount: $0-$150K 2%; $150k-$700k 2%; $700k - $2.49M 1%; $2.5M + 1% (max $200k).

Refer to 01 78 00.00 22 Closeout Submittals paragraph AS-BUILT DRAWINGS and CONTENT OF INVOICE paragraph below for monthly progress as-built requirements.

1.3.3 Schedule Requirements for HVAC TAB

The field work requirements in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC must be broken down in the Earned Value Report from the Cost-Loaded NAS and in the Construction Progress Documentation by separate line items which reflect measurable deliverables. The value for each pay item listed below will be established on a case by case basis for each Contract. The line items are as follows:

a. Approval of Design Review Report: The TABS Agency is required to conduct a review of the project plans and specifications to identify any feature, or the lack thereof, that would preclude successful testing and balancing of the project HVAC systems. Submit the resulting findings to the Government to allow correction of the design. The progress payment will not be issued until the report is reviewed and approved.

b. Approval of the pre-field engineering report: The TABS Agency submits a report which outlines the scope of field work. The report must contain details of what systems will be tested, procedures to be used, sample report forms for reporting test results and a quality control checklist of work items that must be completed before TABS field work commences.

c. Season I field work: Incremental payments are issued as the TABS field work progresses. The TABS Agency mobilizes to the project site and executes the field work as outlined in the pre-field engineering report. The HVAC water and air systems are balanced and operational data must be collected for one seasonal condition (either summer or winter depending on project timing).

d. Approval of Season I report: On completion of the Season I field work, the data is compiled into a report and submitted to the Government. The report is reviewed, and approved, after ensuring compliance with the pre-field engineering report scope of work.

e. Completion of Season I field QA check: Contract QC and Government representatives meet the TABS Agency at the jobsite to retest portions of the systems reported in the Season I report. The purpose of these tests are to validate the accuracy and completeness of the previously submitted Season I report.

f. Approval of Season II report: The TABS Agency completes all Season II field work, which is normally comprised mainly of taking heat transfer temperature readings, in the season opposite of that under which Season I performance data was compiled. Compile this data into a report and submit to the Government. On completion of submittal review to ensure compliance with the pre-field engineering report scope, progress payment is issued. Progress payment is less than that issued for the Season I report since most of the water and air balancing work effort is completed under Season I.

1.4 CONTRACT MODIFICATIONS

In conjunction with the Contract Clause DFARS 252.236-7000 Modification Proposals-Price Breakdown, and where actual ownership and operating costs of construction equipment cannot be determined from Contractor accounting records, base equipment use rates upon the applicable provisions of the EP 1110-1-8.

1.5 CONTRACTOR'S INVOICE AND CONTRACT PERFORMANCE STATEMENT

1.5.1 Content of Invoice

Requests for payment will be processed in accordance with the Contract Clause FAR 52.232-27 Prompt Payment for Construction Contracts and FAR 52.232-5 Payments Under Fixed-Price Construction Contracts. Invoices not completed in accordance with contract requirements will be returned to the Contractor for correction of the deficiencies. Include the documents listed below in the requests for payment.

a. The Contractor's invoice, on NAVFAC Form 4330 furnished by the Government, showing in summary form, the basis for arriving at the amount of the invoice. Form 4330 must include certification by

Quality Control (QC) Manager as required by the Contract.

b. Contractor's Monthly Estimate for Voucher and Contractors Certification (NAVFAC Form 4330) with Subcontractor and supplier payment certification. Other documents, including but not limited to, that need to be received prior to processing payment include the following submittals as required. These items are still required monthly even when a pay voucher is not submitted.

c. Monthly Work-hour report.

d. Updated Construction Progress Schedule and tabular reports required by the contract.

e. Contractor Safety Self Evaluation Checklist.

f. Updated submittal register.

g. Solid Waste Disposal Report.

h. Certified payrolls.

i. Updated testing logs.

j. Monthly Progress As-Builts (Provide one (1) set of up to date working as-built drawings (CADD) in the specified software and format, hard copy and electronic). Refer to 01 78 00.00 22 Closeout Submittals paragraph As-Built Drawings for additional information.

k. Accounting Summary of Unit Price Quantities.

l. Other supporting documents as requested.

1.5.2 Submission of Invoices

If DFARS Clause 252.232-7006 Wide Area WorkFlow Payment Instructions is included in the Contract, provide the documents listed in above paragraph CONTENT OF INVOICE in their entirety as attachments in Wide Area Work Flow (WAWF) for each invoice submitted. The maximum size of each WAWF attachment is two megabytes, but there are no limits on the number of attachments. If a document cannot be attached in WAWF due to system or size restriction, provide it as instructed by the Contracting Officer.

Submit/review draft (pencil requisition) invoice with Engineering Technician (ET) prior to uploading to WAWF.

Monthly invoices and supporting forms for work performed through the anniversary award date of the Contract must be submitted to the Contracting Officer within 5 calendar days of the date of invoice. For example, if Contract award date is the 7th of the month, the date of each monthly invoice must be the 7th and the invoice must be submitted by the 12th of the month.

1.5.3 Final Invoice

a. A final invoice must be accompanied by the Contractor's Final Release. If the Contractor is incorporated, the Final Release must contain the corporate seal. An officer of the corporation must sign and the corporate secretary must certify the Final Release.

b. For final invoices being submitted via WAWF, the original Contractor's Final Release Form must be provided directly to the respective Contracting Officer prior to submission of the final invoice. Once receipt of the original Final Release Form has been confirmed by the Contracting Officer, the Contractor must then submit final invoice and attach a copy of the Final Release Form in WAWF.

c. Final invoices not accompanied by the Contractor's Final Release will be considered incomplete and will be returned to the Contractor.

1.6 PAYMENTS TO THE CONTRACTOR

Payments will be made on submission of itemized requests by the Contractor which comply with the requirements of this section, and will be subject to reduction for overpayments or increase for underpayments made on previous payments to the Contractor.

1.6.1 Obligation of Government Payments

The obligation of the Government to make payments required under the provisions of this Contract will, at the discretion of the Contracting Officer, be subject to reductions and suspensions permitted under the FAR and agency regulations including the following in accordance with FAR 32.103 Progress Payments Under Construction Contracts:

a. Reasonable deductions due to defects in material or workmanship;

b. Claims which the Government may have against the Contractor under or in connection with this Contract;

c. Unless otherwise adjusted, repayment to the Government upon demand for overpayments made to the Contractor; and

d. Failure to maintain accurate "as-built" or record drawings in accordance with FAR 52.236.21.

1.6.2 Payment for Onsite and Offsite Materials

Progress payments may be made to the Contractor for materials delivered on the site, for materials stored off construction sites, or materials that are in transit to the construction sites under the following conditions:

a. FAR 52.232-5(b) Payments Under Fixed Price Construction Contracts.

b. Materials delivered on the site, but not installed, and off-site materials to be considered for progress payment must be major high cost, long lead, special order, or specialty items, not susceptible to deterioration or physical damage in storage or in transit to the construction site. Examples of materials acceptable for payment consideration include, but are not limited to, structural steel, non-magnetic steel, non-magnetic aggregate, equipment, machinery, large pipe and fittings, precast/prestressed concrete products, plastic lumber (e.g., fender piles/curbs), and high-voltage electrical cable. Materials not acceptable for payment include consumable materials such as nails, fasteners, conduits, gypsum board, glass, insulation, and wall coverings.

c. Provide the Contracting Officer with documentation (e.g., Purchase

Orders or Material Contracts) to justify payment for materials in advance of installation. When appropriate, include the Defense Priorities and Allocations System (DFAS) ratings (e.g., DX or DO) on purchase orders or material contracts to ensure timely delivery.

d. Materials to be considered for progress payment prior to installation must be specifically and separately identified in the Contractor's estimates of work submitted for the Contracting Officer's approval in accordance with Schedule of Prices requirement of this Contract. Requests for progress payment consideration for such items must be supported by documents establishing their value (e.g., paid invoices) and that the title requirements of the clause at FAR 52.232-5 Payments Under Fixed-Price Construction Contracts have been met.

e. Ensure materials are adequately insured and protected from theft and exposure. Provide insurance documentation (e.g., Builder's Risk Insurance) which covers the overall value of the materials to be paid in advance of installation.

f. Provide a written consent from the surety company with each payment request for offsite materials.

g. Materials to be considered for progress payments prior to installation must be stored within 50 miles of the Installation (Portsmouth Naval Shipyard). Other locations are subject to written approval by the Contracting Officer.

h. Materials in transit to the job site or storage site are not acceptable for payment.

i. Provide written consent allowing the Government access to the remote storage location for inspection of the stored materials.

j. Materials must be separated from other stored materials and must be clearly identified with the contract number indicating where the materials will be used.

1.7 UNIT PRICE PAYMENT ITEMS

Payment items for the work of this Contract on which the Contract unit price payments will be made are listed in the PRICE SCHEDULE. The unit price and payment made for each item listed consititutes full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, and for performing all work required for each of the unit price items.

1.7.1 Measurement/Accounting Summary of Unit Cost Quantities

a. Per FAR clause 52.236-16 Quantity Surveys:

(1) Quantity surveys shall be conducted, and the data derived from these surveys shall be used in computing the quantities of work performed and the actual construction completed and in place.

(2) The Contractor shall conduct the original and final surveys and surveys for any periods for which progress payments are requested. All these surveys shall be conducted under the direction of a representative of the Contracting Officer, unless the Contracting

Officer waives this requirement in a specific instance. The Government shall make such computations as are necessary to determine the quantities of work performed or finally in place. The Contractor shall make the computations based on the surveys for any periods for which progress payments are requested.

(3) Promptly upon completing a survey, the Contractor shall furnish the originals of all field notes and all other records relating to the survey or to the layout of the work to the Contracting Officer, who shall use them as necessary to determine the amount of progress payments. The Contractor shall retain copies of all such material furnished to the Contracting Officer.

b. Provide notification and schedule inspection dates and times with the Contracting Officer upon discovery of conditions that require the use of Unit Price items. Contracting Officer to coordinate findings on site with QC Manager and authorize application of Unit Price items prior to execution of any work related to Unit Price scope.

c. The QC Manager must assign a quantity to each repair identified during the inspection and provide an updated accounting summary for all areas inspected as the inspections progress and are completed for Contracting Officer's review.

1.7.2 Payment

Payment will be made for costs associated with Unit Cost Items, which includes performing required work and other operations incidental thereto, upon approved Unit Prices Documentation and Accounting Plan.

a. Submit an Accounting Summary of Unit Price Quantities with each monthly invoice, whether or not payment is being requested for unit-priced items. Organize this summary in table form with the column headings listed below and include updated data for every unit-priced item listed in the Price Schedule:

(1) Line Item designation as stated in the Price Schedule.

(2) Description as stated in the Price Schedule.

(3) Estimated Quantity as stated in the Price Schedule.

(4) Unit of Measure as stated in the Price Schedule.

(5) Sum of Quantities Included in Prior Invoices.

(6) Quantity Balance Prior to This Invoice. (Col 4 - Col 5).

(7) Quantity Invoiced This Period.

(8) Remaining Quantity After This Invoice (Col 6 - Col 7).

b. Prior to submitting this Summary and associated invoice, obtain Contracting Officer's concurrence of quantities to be invoiced for each unit-priced item.

c. Copy the data in column 8 into column 6 of next period's Summary.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 30 00.00 22
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ADMINISTRATIVE REQUIREMENTS (PWD ME)
11/20, CHG 4: 08/24

PART 1 GENERAL

This Section applies to Design-Bid-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Facility Turnover Planning Meeting Agenda and Red Zone (RZ) Checklist-POAM; G

NAVFAC PWD ME Internal Service Requirements List; G

Archaeological Coordination Meeting; G

View Location Map

Progress and Completion Pictures

1.3 VIEW LOCATION MAP

Submit, prior to or with the first digital photograph submittals, a sketch or drawing indicating the required photographic locations. Update as required if the locations are moved.

1.4 PROGRESS AND COMPLETION PICTURES

Photographically document site conditions prior to start of construction operations. Provide monthly, and within one month of the completion of work, digital photographs, 1600x1200x24 bit true color minimum resolution in JPEG file format showing the sequence and progress of work. Take a minimum of 20 digital photographs each week throughout the entire project from a minimum of ten different viewpoints selected by the Contractor unless otherwise directed by the Contracting Officer. Submit with the monthly invoice two sets of digital photographs, each set on a separate compact disc (CD) or data versatile disc (DVD), cumulative of all photos to date. Indicate photographs demonstrating environmental procedures. Provide photographs for each month in a separate monthly directory and name each file to indicate its location on the view location sketch. Also provide the view location sketch on the CD or DVD as a digital file. Include a date designator in file names. Photographs provided are for unrestricted use by the Government.

1.5 MINIMUM INSURANCE REQUIREMENTS

Provide the minimum insurance coverage required by FAR 28.307-2 LIABILITY, during the entire period of performance under this Contract. Provide other insurance coverage as required by State law.

1.6 SUPERVISION

Provide, at a minimimum, personnel to meet the requirements of the following supervisory roles for the project, as outlined throughout the Division 01 Specifications:

TITLE	SPECIFICATION SECTION	RESPONSIBILITY/RESTRICTIONS
Project Superintendent	01 30 00.00 22	Sole role, must be direct hire of the Prime.
Project Manager	01 30 00.00 22	Sole role, must be direct hire of the Prime.
Site Safety and Health Officer(SSHO)	01 35 26.00 22	Sole role, must be direct hire of the Prime.
Archaeological Coordinator	01 44 00.00 22	
Quality Control (QC) Manager	01 45 00.00 22	Sole role, must be direct hire of the Prime.
Quality Control (QC) Specialists	01 30 00.00 22	
Special Inspector (SI)	01 30 00.00 22	
Environmental Manager	01 57 19.00 22	May be combined with the QC Manager

Project Superintendent
Project Manager
Site Safety and Health Officer (SSHO)
Archaeological Coordinator
Quality Control (QC) Manager
Quality Control (QC) Specialists
Special Inspector (SI)
Environmental Manager

The requirements of each role are enumerated in the following paragraphs. Where the requirements for a role are enumerated in another section, a reference to that section is provided.

1.6.1 Project Superintendent

1.6.1.1 Minimum Communication Requirements

Have at least one qualified superintendent, or competent alternate, capable of reading, writing, and conversing fluently in the English language, on the job-site at all times during the performance of Contract work. In addition, if a Quality Control (QC) representative is required on the Contract, then that individual must also have fluent English communication skills.

1.6.1.2 Qualifications

Provide project superintendent with a minimum of 10 years experience in construction with at least 5 of those years as a superintendent on projects similar in size and complexity. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance. The individual must be capable of interpreting a critical path schedule and construction drawings. The qualification requirements for the alternate superintendent are the same as for the project superintendent. The Contracting Officer may request proof of the superintendent's qualifications at any point in the project if the performance of the superintendent is in question.

The Project Superintendent must be on site during working hours. The Superintendent **cannot** be the Project Manager, nor the Quality Control Manager, nor the Site Safety and Health Officer (SSHO).

The Project Superintendent must be a direct employee of the Prime Contractor, not an employee of a subcontractor.

1.6.1.3 Duties

The project superintendent is primarily responsible for managing Subcontractors and coordinating day-to-day production and schedule adherence on the project. The superintendent is required to attend Specification Review Meeting, Pre-Construction Meeting, Environmental Brief, Outage Coordination Meetings, Outage Safety Review Meeting, NAVFAC Red Zone meetings, MEP Coordination Meetings, Partnering meetings, Preparatory meetings and Quality Control meetings. The superintendent or qualified alternate must be on-site at all times during the performance of this Contract until the work is completed and accepted.

1.6.1.4 Non-Compliance Actions

The project superintendent is subject to removal by the Contracting Officer for non-compliance with requirements specified in the Contract and for failure to manage the project to insure timely completion. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor.

1.6.2 Project Manager

1.6.2.1 Requirements

The Contractor must assigned a qualified Project Manager to the project. The Project Manager must be on-site at all times during the performance of this Contract until the work is completed and accepted.

The Project Manager **cannot** be the Superintendent, nor the Quality Control Manager, nor the Site Safety and Health Officer (SSHO).

1.6.2.2 Qualifications

Approval of Project Manager by the Contracting Officer is required prior to start of construction. Provide resumes for the proposed Project Manager describing their experience with references and qualifications to the Contracting Officer for approval. Any substitutions during the construction period must be approved by the Contracting Officer. The Contracting Officer reserves the right to interview the proposed Project Manager at any time in order to verify the submitted qualifications. Any replacement personnel must have the same or better qualifications than the previous individual (PM/QCM/SSHO, etc.). The resumes for the personnel must provide a summary of projects of similar size and responsibility. The Project Manager must complete the course entitled "Construction Quality Managemetn for Contractors" prior to the start of construction.

1.6.2.3 Non-Compliance Actions

The Project Manager is subject to removal by the Contracting Officer for non-compliance with requirements specified in the Contract and for failure to manage the project to insure timely completion. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor.

1.6.3 Site Safety and Health Officer (SSHO)

A Site Safety and Health Officer (SSHO) is a Contractor employee that is responsible for overseeing and ensuring implementation of the Prime Contractor's Safety and Occupational Health (SOH) program according to the Contract, EM 385-1-1, and applicable federal, state, and local requirements.

Provide Site Safety and Health Officers (SSHOs) in accordance with the requirements of Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME)

The SSHO must be a direct employee of the Prime Contractor, not an employee of a subcontractor.

1.6.4 Archaeological Coordinator

Provide an Archaeological Coordinator in accordance with the requirements of Section 01 44 00.00 22 ARCHAEOLOGICAL MONITORING (PWD ME).

1.6.5 Quality Control (QC) Manager

The Quality Control (QC) Manager is responsible for managing and implementing the QC program on this Contract. Provide a QC Manager in accordance with the requirements of Section 01 45 00.00 22 QUALITY CONTROL (PWD ME).

The QC Manager must be a direct employee of the Prime Contractor, not an employee of a subcontractor.

1.6.6 Quality Control (QC) Specialists

Provide a separate Quality Control (QC) Specialist for each of the areas as listed in Paragraph QUALITY CONTROL (QC) SPECIALISTS in Section 01 45 00.00 22 QUALITY CONTROL (PWD ME). The QC Specialists must assist and report to the QC Manager and must have no duties other than their assigned quality control duties. The QC Specialists must be provided in accordance with the requirements of Section 01 45 00.00 22 QUALITY CONTROL (PWD ME).

1.6.7 Special Inspector (SI)

Refer to Section 01 45 35.00 22 SPECIAL INSPECTIONS (PWD ME).

1.6.8 Environmental Manager

Provide an Environmental Manager in accordance with the requirements of Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME).

1.7 SCHEDULE OF MAJOR MEETINGS

The following is a list of major project meetings required between the Contractor, NAVFAC, PNSY, and other governing bodies, to help the Contactor understand the anticipated major project meetings, schedule requirements, and attendees. This is not intended to be be an exhaustive list of all meetings required for the project, and does not include the Contractor's internal meetings to facilitate the work. Refer to other specification sections for additional meetings required for the project.

Meetings listed in the table must be attended by the Project Manager, Project Superintendent, QC Manager, SSHO, Contracting Officer, and Contracting Officer's Representative. Additional required attendees for each meeting are as shown in the table and as otherwise listed throughout specifications.

SCHEDULE OF MAJOR MEETINGS				
Meeting	Reference Specification	Schedule	Onsite or Virtual	Special Attendees
Div 01 Spec Review	01 11 00.00 22 01 30 00.00 22	Within 5 working days of Contract Award	Onsite	Officer of the Prime Contractor Management Rep of Major Subcontractors
PreConstruction	01 30 00.00 22 01 35 26.00 22	Within 45 working days of Contract Award	Onsite	Major Subcontractor (s)

Outage Coordination/ CSN1	01 14 00.00 22 01 30 00.00 22 01 35 26.00 22	Prior to Outage Submittal	Onsite	Applicable Subcontractor(s) PWD-ME DM PWD ME UEM Rep Designer of Record Applicable Stakeholders
Outage Safety Review	01 14 00.00 22 01 30 00.00 22 01 35 26.00 22	Immediately before starting related work	Onsite	Applicable Subcontractor(s) Utility Personnel
Archaeological Coordination	01 30 00.00 22	Prior to start of any ground disturbing activity or excavation	Onsite	Archaeological Coordinator Archaeologist PWD-ME CM/ET PWD-ME CRM
Environmental Brief	01 30 00.00 22 01 35 26.00 22	Within 45 working days of Contract Award	Virtual	Environmental Manager PNSY C106 Rep
MEP Coordination	01 30 00.00 22			
MEP Controls	01 30 00.00 22			
Facility Turnover (RedZone)	01 30 00.00 22 01 78 23.00 22		Onsite	
Senior Leadership Team	01 30 00.00 22	Monthly	Virtual	
Partnering	01 30 00.00 22	Within 35 days of Contract Award		
Coordination and Mutual Understanding	01 45 00.00 22	Within 10 working days of submitting QC Plan	Onsite	Flood Barrier Supplier Rep Special Inspector Environmental Manager Subcontractor(s) Alternate QC Manager
Site Waste Removal (SWR) Meeting	01 57 19.00 22 31 00 00.00 22	14 Calendar days prior to removal of waste from the project site	Onsite	Environmental Manager PWD ME CM PNSY C106 Rep

Construction Waste Management	01 74 19.00 22	Within 45 working days of Contract Award	Virtual	Environmental Manager
Excavation Meeting	31 00 00.00 22	Within 5 working days of earthwork operations	Onsite	Contractor's Geotechnical Engineer and Testing Agency, Government Geotechnical Engineer and Civil Engineer
Hazardous Materials Meeting	02 83 00.00 22 02 83 00.00 22	Prior to removal abatement work	Onsite	Contractor's Competent Person and Industrial Hygienist
Commissioning Kickoff Meeting	01 91 00.15 22	Within 60 working days after contract award	Onsite	Construction Manager, Quality Control Team, Designer of Record, and the Government Acceptance Testing Representatives and other Government team members
Regular Commissioning Coordination Meetings	01 91 00.15 22	Monthly (bi-weekly once within 30 days of the scheduled date for commissioning testing)	Virtual	Construction Manager, Quality Control Team, Designer of Record, and the Government Acceptance Testing Representatives and other Government team members
Operation and Maintenance Manual and Facility Data Workbook Coordination Meeting	01 78 23.15 22	Prior to submission of O&M Manual submittal	Virtual	The Contractor's O&M Manual and FDW preparer, Quality Control Manager, Commissioning Authority, the Government Design Manager and FDW reviewer, and discipline specific sub-contractors

O&M Manual Submittal Coordination Meeting	01 78 23.15 22	Following submission of O&M Manual progress submittal(s)	Virtual	The Contractor's O&M Manual and FDW preparer, Quality Control Manager, Commissioning Authority, the Government Design Manager and FDW reviewer, and discipline specific sub-contractors
Facility Turnover Meeting	01 78 23.15 22	Following submission of O&M Manual progress submittal(s)	Virtual	The Contractor's O&M Manual and FDW preparer, Quality Control Manager, Commissioning Authority, the Government Design Manager and FDW reviewer, and discipline specific sub-contractors
As-Built Drawing Content	01 78 00.00 22	30 calendar days prior to submittal of progress as-built drawings	Virtual	The Contractor's as-built drawings preparer, Quality Control Manager, the Government Design Manager and PNSY CADD department representative
Quality Control Plan Meeting	01 45 00.00 22	Prior to submission of QC Plan	Virtual	The Quality Control Manager and Contracting Officer
Quality Control Weekly Meeting	01 45 00.00 22	Weekly	Virtual	Project Superintendent, Project Manager, Assistant QC Manager, the QC Specialists, the Special Inspector, the Special Inspector of Record, CxC, the foremen who are performing the ongoing work, the contracting officer, and the DOR Project Manager

1.8 DIVISION 01 SPECIFICATION REVIEW MEETING

The Specification Review Meeting must take place within 5 working days after contract award and prior to the Preconstruction Meeting. The meeting must include a discussion to develop a mutual understanding of the Division 01 specification requirements. This meeting must include the Prime Contractor, the Prime and Subcontrators performing the work, the Contracting Officer, PWD ME DM, Designer of Record and the NAVFAC PWD ME UEM representative.

At the conclusion of the meeting, the Project Superintendent must certify that the Prime Contractor and its Subcontractors have read, understand and agree to the requirements of the Division 01 specifications by signing a document which will be prepared by the Contracting Officer.

1.9 PRECONSTRUCTION MEETING

Immediately after award, prior to commencing any work at the site, coordinate with the Contracting Officer a time and place to meet for the Preconstruction Meeting. **The Pre-Construction meeting must take place within 45 Calendar days from receiving the Notice to Proceed, and prior to commencement of any work at the site.** The purpose of this meeting is to discuss and develop a mutual understanding of the administrative requirements of the Contract including but not limited to: daily reporting, invoicing, value engineering, safety, Base-access, parking plan, archaeological requirements, outage requests, hot work permits, schedule requirements, quality control, earned value report, shop drawings, submittals, cybersecurity, prosecution of the work, Government acceptance, final inspections, and Contract close-out. Contractor must present and discuss their basic approach to scheduling the construction work and any required phasing. Refer to 01 35 26.00 22 Governmental Safety Requirements (PWD ME) paragraph PRECONSTRUCTION MEETING for additional information.

1.9.1 Attendees

Contractor attendees must include the Project Manager, Project Superintendent, Site Safety and Health Officer (SSHO), Archaeological Coordinator, Quality Control (QC) Manager, Environmental Manager, and major Subcontractors.

1.9.2 Submittals

This section compiles in one location the SD-01 Preconstruction Submittals that the Contractor is required to submit. The requirements of each submittal are detailed in the specification sections as listed below. This list is not meant to be comprehensive; the Contractor is responsible for confirming the complete list of SD-01 Preconstruction Submittals by reviewing the specifications, including the Submittal Register attached to Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME).

The following SD-01 Preconstruction Submittals must be submitted to the Contracting Officer fifteen (15) calendar days prior to the preconstruction meeting:

Specification	**SD #**	**SD Description**	**Submittal Description**	**Specification Paragraph**
01 31 23.13 20	01	Preconstruction Submittals	List of Contractor's Personnel; G	
01 35 26.00 22	01	Preconstruction Submittals	Accident Prevention Plan; G	
01 45 00.00 22	01	Preconstruction Submittals	Contractor Quality Control (CQC) Plan; G	

Specification	SD #	SD Description	Submittal Description	Specification Paragraph
01 45 00.00 22	01	Preconstruction Submittals	Basis of Design and Design Intent; G	

The following SD-01 Preconstruction Submittals must be submitted to the Contracting Officer at or before the preconstruction meeting:

Specification	SD #	SD Description	Submittal Description	Specification Paragraph
01 20 00.00 22	01	Preconstruction Submittals	Earned Value Report; G	
01 32 17.00 20	01	Preconstruction Submittals	Baseline NAS; G	
01 32 17.00 20	01	Preconstruction Submittals	Weekly Look Ahead Schedule; G	
01 32 17.00 20	01	Preconstruction Submittals	Outage Schedule; G	
01 33 00.00 22	01	Preconstruction Submittals	Submittal Register; G	
01 50 00.00 22	01	Preconstruction Submittals	Construction Site Plan; G	

The following SD-01 Preconstruction and SD-10 Closeout submittals must be submitted to the Contracting Officer prior to the start of construction:

Specification	SD #	SD Description	Submittal Description	Specification Paragraph
01 30 00.00 22	01	Preconstruction Submittals	Facility Turnover Planning Meeting Agenda and Red Zone (RZ) Checklist-POAM; G	
01 30 00.00 22	01	Preconstruction Submittals	NAVFAC PWD ME Internal Service Requirements List; G	
01 44 00.00 22	01	Preconstruction Submittals	Archaeological Monitoring Plan; G	
01 44 00.00 22	01	Preconstruction Submittals	Archaeologist Qualifications; G	
01 44 00.00 22	01	Preconstruction Submittals	Archaeological Coordinator Designation Letter; G	
01 50 00.00 22	01	Preconstruction Submittals	Traffic Control Plan; G	
01 50 00.00 22	01	Preconstruction Submittals	Employee Parking Plan; G	

Specification	SD #	SD Description	Submittal Description	Specification Paragraph
01 50 00.00 22	01	Preconstruction Submittals	Contractor Computer Cybersecurity Compliance Statements; G	
01 50 00.00 22	01	Preconstruction Submittals	Contractor Temporary Network Cybersecurity Compliance Statements; G	
01 57 19.00 22	01	Preconstruction Submittals	Preconstruction Survey; G	
01 57 19.00 22	01	Preconstruction Submittals	Solid Waste Management Plan; G	
01 57 19.00 22	01	Preconstruction Submittals	Regulatory Notifications; G	
01 57 19.00 22	01	Preconstruction Submittals	Environmental Management Plan; G	
01 57 19.00 22	01	Preconstruction Submittals	Dirt and Dust Control Plan	
01 57 19.00 22	10	Closeout Submittals	Contractor Hazardous Material Inventory Log; G	
01 57 19.00 22	01	Preconstruction Submittals	Storm Water Management/Erosion and Sedimentation Control Plan; G	
01 57 19.00 22	01	Preconstruction Submittals	Spill Prevention Control and Countermeasure (SPCC) Plan; G	
01 57 19.00 22	01	Preconstruction Submittals	Environmental Manager Qualifications; G	
01 74 19.00 22	01	Preconstruction Submittals	Construction Waste Management Plan; G	

Confirm, in writing, the construction start date with the Contracting Officer Representative at least two (2) working days prior to start date.

Prepare and distribute minutes of all meetings to the attendees within three (3) working days of the preconstruction meeting.

1.10 OUTAGE COORDINATION MEETING

The Outage Coordination Meeting is to be held prior to submittal of the outage request package and prior to beginning work on the utility system requiring shut-down (outage). The meeting must include a discussion to develop a mutual understanding of the administrative requirements including but not limited to outage requests. The meeting must also review any Tier 1 Shipyard Infrastructure that may be impacted by the work and review the TIER 1 INFRASTRUCTURE REQUIREMENTS specified in Section 01 14 00.00 22.

This meeting must include the Prime Contractor, the Prime and Subcontractors performing the work, the Contracting Officer, PWD ME DM, Designer of Record and the NAVFAC PWD ME UEM representative.

All parties must fully coordinate all the activities required to support the outages. Refer to 01 14 00.00 22 paragraph OUTAGE COORDINATION MEETING for additional information.

1.11 OUTAGE SAFETY REVIEW MEETING

During the Outage Safety Review meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HECP training in accordance with EM 385-1-1. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

For utility distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the Subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

1.12 ENVIRONMENTAL BRIEF

Attend an Environmental Brief prior to commencing any work on the Portsmouth Naval Shipyard. The Brief will be conducted by the Contracting Officer's Representative (COR) with PWD ME & Shipyard C106 Environmental Representatives. The Brief will be part of the Preconstruction Meeting agenda or held as a separate meeting depending on the size and scope of work and potential environmental impact, as determined by the Government.

The meeting will discuss details for all the submittals that are required in Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME) including review of the submittal requirements for the Environmental Management Plan, important notes regarding Installation Restoration (IR), Historic, and Cultural Resources, Wetlands, and Waste Generation, Concrete Wash Water/Wastewater, Hazard Materials Management including sampling, testing and disposal requirements and other applicable topics.

At the Brief, provide the following information: types, quantities, and use of hazardous materials that will be brought onto the Portsmouth Naval Shipyard; types and quantities of wastes/wastewater that may be generated during the Contract; types and quantities of oil that will be brought onto the activity; and pollution control measures for spill prevention and control, and any bulk oil storage container information including quantity and type of product stored. Discuss the results of the Preconstruction

Survey at this time.

Develop a mutual understanding relative to the details of environmental protection, including measures for protecting natural resources, historic properties, required reports, required permits, specific permit requirements, and other measures to be taken. Identify additional environmental concerns specific to the site (e.g., historic, archaeological and natural resources, IR, erosion and sediment control, spill prevention and control, soil management, testing and disposal requirements, etc.). See Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME), Paragraph 1.5.3.

The meeting must include the Project Manager, Project Superintendent, Site Safety and Health Officer (SSHO), Archaeological Coordinator, Quality Control (QC) Manager, Environmental Manager, and major Subcontractors.

1.13 ARCHAEOLOGICAL COORDINATION MEETING

Prior to the start of any ground disturbing/excavation work that is located in an Archaeological Site identified on the Drawings, the Archaeological Coordinator, Archaeologist, Quality Control (QC) Manager, PWD ME CM/ET and PWD-ME Cultural Resources Manager must meet to review the proposed approach to the work, establish a protocol for collection of artifacts, and review the proposed and review relevant documentation.

For projects with a duration exceeding one year, re-initial coordination meeting must be held on an annual basis to ensure continued alignment on archaeological protocols. In addition, a re-initial coordination meeting must be held should there be a change in key personnel (Archaeological Subcontractor; Archaeological Coordinator; QC Manager, Project Superintendent, Project Manager, etc.)

1.14 MEP COORDINATION MEETING

The Contractor must schedule in person coordination meetings with the mechanical, electrical, piping, and fire protection trades on a monthly basis at a minimum. The intent of the meetings is to review the progress of the coordination drawings and coordinate work being installed.

See Section 01 30 10.00 22 COORDINATION PROCEDURES AND CONSTRUCTION DRAWINGS (PWD ME).

Meeting minutes must be provided within 24 hours of completion of the MEP coordination meeting.

The QC organization will be responsible for leading the MEP Coordination Meetings with other trades encouraged to participate.

1.15 MEP CONTROLS MEETING

The Contractor must schedule in person coordination meeting with the mechanical and controls trades, QC Manager, DOR, and NAVFAC PWD-ME Mechanical Representative at the beginning of project to review system controls and sequencing. The intent of the meeting is to verify the design intent for the equipment controls and sequence of operations, review potential equipment selection, identify potential conflicts that may impact the intended design intent, and discuss the contractor's means of providing the required controls and sequence of operations.

Meeting minutes must be provided within 48 hours of completion of the MEP Controls Meeting.

The QC organization will be responsible for leading the MEP Controls Meeting.

1.16 FACILITY TURNOVER PLANNING MEETINGS (Red Zone Meetings)

Meet with the Government to identify strategies to ensure the project is carried to expeditious closure and turnover to the Client. Start the turnover process at the Preconstruction Meeting with a discussion of the Red Zone (RZ) process and convene at regularly scheduled RZ Meetings beginning at approximately 75 percent of construction completion or sooner. Include the following in the facility Turnover effort:

1.16.1 Red Zone Checklist

See Appendix A of this Section for an example of the Facility Turnover Planning Meeting Agenda and Red Zone (RZ) Checklist-POAM. Contracting Officer's Representative (COR) will provide the Contractor a copy of the Red Zone Checklist template in advance of the RZ turnover meeting.

At the initial Red Zone Facility Turnover meeting, NAVFAC, the Client and Contractor will modify the Red Zone Checklist template by adding or deleting critical activities applicable to the project and assign planned completion dates for each activity. This becomes the Red Zone POAM which will be utilized through to the contract completion.

Items listed on the checklists are required to remain on the checklists if they are part of the project/Contract or required by construction convention. Items not listed on the checklists, but required in the Contract or by construction convention, must be added to the checklists by the Contractor, Client and NAVFAC. The Contracting Officer may request additional activities be added to the Red Zone Checklist at any time as necessary. Checklists are applicable to all Contracts no matter what Category of Work. The Point of Contact and due date must initially be determined during the Facility Turnover Planning Meeting by the NAVFAC, Client and Contractor leads. During execution of the NRZ process, for each item on the entire list, the Construction Manager (CM) must indicate date completed and initial to indicate completion of the item. If a party fails to complete an item by the due date, this must be noted on the checklist and a new due date established and indicated.

1.16.2 Meetings

a. Upon Government acceptance of the RZ Checklist-POAM, the COR will send out the accepted RZ Checklist-POAM to all attendees. The Project Superintendent is required to lead regular Red Zone Meetings beginning at approximately 75 percent project completion, or three (3) to six (6) months prior to Beneficial Occupancy Date (BOD), whichever comes first.

b. The Contracting Officer will establish the frequency of the meetings, which is expected to increase as the project completion draws nearer. At the beginning, Red Zone meetings may be every two weeks then increase to weekly towards the final month of the project.

c. Using RZ Checklist-POAM as a Plan of Action and Milestones (POAM) and basis for discussion, review upcoming critical activities and strategies to ensure work is completed on time.

d. Durring the Red Zone Meetings, discuss with the CM/ET any upcoming activities that require Government involvement.

e. All parties will maintain their copy of the RZ Checklist-POAM documenting the actual completion dates as work is completed and update the RZ Checklist-POAM with revised planned completion dates as necessary to match progress. The CM will maintain the master RZ Checklist-POAM, periodically distributing a scanned copy of the current RZ Checklist-POAM to attendees via email after significant progress is made.

1.16.3 NAVFAC PWD ME Internal Service Requirements List

An initial, pre-edited draft of the NAVFAC PWD ME Internal Service Requirements List is included in Appendix B of this Section.

Include all information usually listed on manufacturer's name plate. The Internal Service Requirements List must include, as applicable, the following for each piece of equipment installed: description of item, location (by room number), manufacturer, model number, serial number, capacity, floor coverings, wall and ceiling surfaces; types and square footage of coverage, lighting fixtures, bathroom fixtures, windows, and HVAC filters.

Submit a preliminary Internal Service List to the COR at the initial Facility Turnover Meeting. Provide the final completed Internal Service List with all required facility system/equipment information to the COR at least ninety (90) calendar days prior to the project BOD.

1.17 SENIOR LEADERSHIP TEAM (SLT) MEETINGS

A representative of the Contractor's Senior Leadership Team must participate in monthly SLT Meetings with their Government counterparts over the Contract duration. Meetings will be held via MS Teams. The purpose of this meeting is to maintain open dialog amongst senior leaders to discuss project administration, specifically with respect to issues, impacts, and mitigations.

The SLT will establish approximately 10 Key Performance Indicators (KPIs) that will be monitored and reported by the Contractor to SLT over the course of the project. KPIs will use data customarily available to the Contractor and data provided by the Government. The Contractor will coordinate with the Government on how the KPIs are reported and presented. Reporting will generally be a tabular or graphical representations. Anticipated KPIs for tracking include, but are not limited to:

a. Percentage of preconstruction submittals as listed in paragraph SUBMITTALS submitted on time.

b. Percentage of preconstruction submittals as listed in paragraph SUBMITTALS reviewed and returned by the Government within 14 calendar days of submittal.

c. Number of Contractor submittals submitted on time.

d. Number of Contractor submittals under Government review over 14 calendar days.

e. Percentage of Contractor submittals returned by the Government marked Disapproved, Rejected, or Revise and Resubmit.

f. Number of open Contractor RFIs under Government review over 21 calendar days.

g. Number of approved Contract modifications and total value.

h. Number of accepted proposed change orders outstanding and for how long.

i. Percentage of Work in Place (WIP).

1.18 EXECUTIVE LEADERSHIP TEAM (ELT) MEETINGS

A representateive of the Contractor's Executive Leadership Team must participate in quarterly ELT Meetings with their Government counterparts over the Contract duration. Meetings will be held by MS Teams.

1.19 PARTNERING

To most effectively accomplish this Contract, the Contractor and Government must form a cohesive partnership with the common goal of drawing on the strength of each organization in an effort to achieve a successful project without safety mishaps, conforming to the Contract, within budget, and on schedule. The partnering team must consist of personnel from both the Government and Contractor including project level and corporate level leadership positions. Key Personnel from the supported command, end user (who will occupy the facility), NAVFAC, PWD Maine Design and Construction team and Subject Matter Experts (SME's), FEAD, Project Manager (PM), Design Manager (DM), Construction Manager (CM), Engineering Technician (ET), Contractor, key Subcontractors, and the Designer of Record are required to participate in the Partnering process.

1.19.1 Facilitated (Formal) Partnering

a. Within 35 calendar days after award and prior to the start of work, host a Formal Partnering session with key personnel from the project team including both Contractor and Government personnel. All costs associated with the Partnering session including the third-party independent Facilitator Consultant, meeting room and other incidental items are the responsibility of the Contractor.

b. Before the Facilitated (Formal) Partnering session, coordinate with the Facilitator all requirements for incidental items (such as audio-visual equipment, easels, flipchart paper, colored markers, note pads, pens/pencils, colored flash cards) and have these items available at the Partnering session. Provide copies of any documents required for distribution to all attendees. Participants will bear their own costs for meals, lodging and transportation associated with Partnering.

c. The Initial Partnering Session must be a duration of one day and be held at a location off base as agreed to by the Contracting Officer. Partnering session may take place concurrently with the Preconstruction Meeting.

d. Facilitator must be experienced in conducting corporate Partnering sessions and must be a third-party independent facilitating consultant

- not an employee of the Contractor. The Facilitator is responsible for leading all aspects of the Partnering session necessary to achieve the Partnering goal.

e. An outcome of the Partnering session must be an escalation matrix agreed upon by both the Government and Contractor, which identifies key Government and Contractor decision makers by name and anticipated decision durations.

f. Host follow-on Partnering Sessions at three- to six-month intervals or more frequently if needed and lasting generally a half day or less. Attendees need only be those required to resolve current issues. The same Facilitator used in the Initial Partnering session must lead the follow-on sessions unless an alternative is permitted by the Contractor Officer. All costs associated with follow-on Partnering sessions are the responsibility of the Contractor.

1.20 REQUESTS FOR INFORMATION (RFI)

Submit requests for information using Government provided request form. Refer to Section 01 45 00.00 22 Quality Control (PWD ME) for additional information.

1.21 ELECTRONIC MAIL (E-MAIL) ADDRESS

Establish and maintain electronic mail (e-mail) capability along with the capability to open various electronic attachments as text files, pdf files, and other similar formats. Within 10 days after Contract award, provide the Contracting Officer a single (only one) e-mail address for electronic communications from the Contracting Officer related to this Contract including, but not limited to, Contract documents, invoice information, request for proposals, and other correspondence. The Contracting Officer may also use email to notify the Contractor of Base access conditions when emergency conditions warrant, such as hurricanes or terrorist threats. Multiple email addresses are not allowed.

It is the Contractor's responsibility to make timely distribution of all Contracting Officer initiated e-mail with its own organization including field office(s). Promptly notify the Contracting Officer, in writing, of any changes to this email address.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

Date

01 30 00 APPENDIX A

NAVFAC Red Zone
Facility Turnover Planning Meeting

AGENDA

I. Introduction and Overview – Purpose	CM

The purpose of the Facility Turnover Planning Meeting is to address elements within the project team's purview – schedule management, assure completed facility complies with contract requirements, and other contractual issues. Each member of the project delivery team (Client, NAVFAC, and the contractor) has critical responsibilities to ensure timely completion and turnover of the new facility and each member should execute the NRZ process to achieve this end. The NRZ process provides a final re-focusing of attention to details of those key elements critical for a successful construction contract completion. In implementing NRZ processes, the NAVFAC/Contractor/Client team take a collective "snapshot" of contract status, identifying remaining actions to be accomplished, and confirm required resources needed for successful contract completion and turnover to the Client.

The Facility Turnover Planning Meeting is a collaborative effort between the Client, NAVFAC, and the contractor and results in a completed "NRZ Checklist/POAM Items" list that identifies the major items (and their due dates) that must be completed by the Contractor, the Client and the NAVFAC team to ensure timely completion of the contract.

II. Attendees

NAVFAC Echelon IV (PM); NAVFAC FEAD/ROICC Team (AROICC, CM, ET/QA, Contracting Officer); Client Team (Project Manager, Program Coordinator, User/Tenant); Contractor Team (Project Manager, Project Superintendent, CQC Manager)

III. Schedule to Completion (POAM)	Contractor
IV. Schedule of Final Outfitting and Occupancy (POAM)	Client
V. Critical feature(s) of project (POAM)	CM
VI. Transfer of Maintenance Responsibility	CM
VII. Systems training & O&M Manuals (POAM)	CQC Manager
VIII. Other Items to include on NRZ checklists	All
IX. Summary of Required Actions and Responsibility	CM

Guidelines for conducting Facility Turnover Planning Meeting are as follows:

a. Meeting is held at approximately 75% construction contract completion or three to six months prior to BOD. NAVFAC representatives will include the Project Manager, Construction Manager/AROICC (CM) and Design Manager (DM), as appropriate. The contractor representatives include applicable prime contractor staff and decision-makers from major subcontractors. Design-Build contractors will have A-E representatives attending. The Client should include representatives from Public Works Officer (PWO) staff, a Client scope and financial decision maker, a user tenant representative, a facility start-up person, and others such as SPAWAR, NMCI, telephone, and furniture contractor, etc.

b. The purpose of the meeting is to plan the remaining work, identify critical project features that still need to be completed (such as "soft" construction contract requirements as shown on the NRZ Checklist/POAM Items), and to complete the filling out of the "NRZ Checklist/POAM Items".

c. The contractor, client and NAVFAC provide a POC and due date for each item on their checklist. The team fills in the checklists by selecting items applicable to the project, selects due dates on each item, and appoints a person who has responsibility to ensure the item gets completed by the due date. The CM will be responsible to monitor the milestones.

NRZ Checklist/POAM Items

The table below provides typical NRZ checklist items for contractor, Client, and NAVFAC actions. Items listed on the checklists are required to remain on the checklists if they are part of the project/contract or required by construction convention. Items not listed on the checklists, but required in the contract or by construction convention, must be added to the checklists by the contractor, Client and NAVFAC. Checklists are applicable to all contracts no matter what Category of Work.

The Point of Contact and due date shall initially be determined during the Facility Turnover Planning Meeting by the NAVFAC, client and contractor leads. During execution of the NRZ process, for each item on the entire list, the Construction Manager (CM) shall indicate date completed and initial to indicate completion of the item. If a party fails to complete an item by the due date, this should be noted on the checklist and new due date established and indicated. The completed NRZ Checklist/POAM shall be placed in the contract file.

	A	B	C	D	E	F	G	H	I
1			Edit the document to remove Facility Delivery items that are not in the contract. To delete the rows highlight the row(s) and use Control - (minus sign). To insert additional rows use Control + (plus sign).						
2	Revision - 4/24/23		*** INFO / DIRECTIONS** **- This NRZ Checklist/POAM is a tool to track the status of critical activities required for BOD, to help ensure their timely completion and prevent delays with facility acceptance and turnover.** **- The critical activities are organized by section according to requirements (e.g. NMCI & BCO acceptance and connections)** **- Critical items missing from this list should be added as necessary to ensure the list is comprehensive. Likewise, unnecessary items should be deleted.** **- Any critical items left off the NRZ Checklist that are later identified after initial NRZ meeting is conducted should be added immediately so their progress can be tracked.** **- A copy of the latest project customized NRZ Checklist/POAM shall be maintained in the Project contract file.**						
3			**Resp.**	**Critical Activities**	**Point of Contact**	**Due Date**	**Actual Complete Date**	**CM Initials**	**Notes**
4				**Required for Facility Delivery (BOD)**					
5			KTR	Electrical Systems Testing					
6			NAVFAC	Transformer Performance Verification					
7			KTR	Generator Testing					
8			NAVFAC	Generator Performance Verification					
9			KTR	Final Electrical Connections					
10			NAVFAC	Coordinate Final Electrical Connections					Coordinate w/PW Shops
11			NAVFAC	Acceptance Testing of Critical Electrical Systems					Exceptions must be raised to FEAD Leadership with technical evaluation, for acceptance.
12			KTR	Final Gas Connections					
13			NAVFAC	Coordinate Final Gas Connections					Coordinate w/PW Shops
14			KTR	Final Water Connections					
15			KTR	Super chlorination of Potable Water Systems					
16			KTR	Plumbing/Backflow Testing					
17			NAVFAC	Coordinate Final Water Connections					Coordinate w/PW Shops
18			KTR	Critical System Start-up: (Edit & add detail accordingly)					
19									
20			KTR	Duct Air Leakage Testing (DALT)					
21			NAVFAC	DALT Field Acceptance					
22			KTR	Air Barrier Testing					
23			NAVFAC	All HVAC Commissioning/Testing Complete					Exceptions must be raised to FEAD Leadership with technical evaluation, for acceptance.
24									
25			KTR	Elevator Testing					
26			NAVFAC	Elevator Certification					Notify ML elev inspector 30 days prior to required site visit.
27									
28			KTR	Boiler/UPV Testing					
29			NAVFAC	Boiler/UPV Certification					Notify ML boiler inspector 30 days prior to required site visit.
30									
31			KTR	Crane Testing					
32			C700	Crane Certification					Notify C700 30 days prior to required testing.
33									
34			KTR	Fire Alarm/Sprinkler Testing					
35			NAVFAC	FA & FP test results to ML & DOR					days prior to acceptance testing by FPE

	A	B	C	D	E	F	G	H	I
3			**Resp.**	**Critical Activities**	**Point of Contact**	**Due Date**	**Actual Complete Date**	**CM Initials**	**Notes**
36			NAVFAC	Fire Alarm/Sprinkler Acceptance Test (exceptions must be raised to Chief Engineer).					Notify ML FPE 30 days prior to required testing. Exceptions must be raised to FEAD Leadership with technical evaluation, for acceptance.
37			NAVFAC	**All Life Safety Inspections/Testing complete**					Exceptions must be raise up through the DCBL chain so that the Command can take action under our Technical Authority.
38									
39			KTR	Keying Plan Meeting					
40			Client	Keying Plan Meeting					
41			NAVFAC	Keying plan to NAVFAC Locksmith					
42			KTR	Deliver Lockset Cores to CM/ET					
43			NAVFAC	Lockset cores & keys and installed					Locksmith
44									
45			KTR	NMCI Connections/IT Systems Testing					
46			Client	NMCI Install or other networks (C109)					
47			Client	Secure Network Installations					
48			KTR	IDS & SCIF Testing					
49			Client	IDS & SCIF Security Insp & Acceptance					
50			NAVFAC	IDS & SCIF Insp & Test					
51			KTR	Telecommunications Connections & Test					
52			Client	Telecommunication install (BCO phones)					
53			Client	Mod service contracts for Phone/Utilities/ Custodial/Grounds					
54									
55			KTR	Commissioning Functional Performance Testing (FPT)					
56			KTR	TAB (Air and Water Balancing)					
57			KTR	TAB Proportional Balancing Report					
58			NAVFAC	Proportional Balanc'g TAB Field Accept					
59			KTR	TAB Field Acceptance Testing					
60			KTR	TAB Season 1 Report*					
61			KTR	TAB Field Acceptance Testing					
62			NAVFAC	1st Season TAB Field Acceptance					
63			KTR	DDC Trend Logs / Endurance Testing					
64			KTR	ACATS (controls) Testing					
65			NAVFAC	ACATS Performance Verification					
66			KTR	Performance Verification Testing (PVT) Controls					
67									
68			NAVFAC	Roofing final acceptance					Exceptions must be raised to FEAD Leadership with technical evaluation, for acceptance.
69									
70			NAVFAC	Underwater Structure Acceptance					Exceptions must be raised to FEAD Leadership with technical evaluation, for acceptance.

	A	B	C	D	E	F	G	H	I
3			**Resp.**	**Critical Activities**	**Point of Contact**	**Due Date**	**Actual Complete Date**	**CM Initials**	**Notes**
71									
72			NAVFAC	Performance Verification Testing (PVT) Controls NAVFAC Acceptance					
73									
74			KTR	Facility-Related Control System Cybersecurity Commissioning					
75			NAVFAC	Completed Cyber Hygiene Checklist					
76			NAVFAC	Completed RMF Step 4 Validation					
77									
78			Client	GFE status/delivery schedule (GFCI, GFGI)					
79			Client	Client provided equipment KTR installed					
80			Client	Client provided equipment SELF installed					
81									
82			NAVFAC	Training Coordinated/Scheduled with FMS					
83			KTR	System Training of Navy Personnel					
84			Client	Attend Training					
85									
86			KTR	Provide As-Built Dwgs & floor plans in CAD					
87			NAVFAC	CM to Submit floor plan As-Builts to CAD Dept.					RQ'd for Fire Bill - RQ'd before BOD(A)
88			KTR	Submit sample as-built drawings for CAD review					Submit 3 drawings from ea. discipline for review
89			NAVFAC	CM to submit to DM who must review and submit Sample As-Built documents to CAD Dept.					
90									
91			KTR	Pre-Warranty Conference					
92			KTR	Contractor's Pre Final Punch List Complete					
93			NAVFAC	Obtain a copy and ensure QA Inspections are completed.					Include a completed copy for BOD Mangament sign off.
94			NAVFAC	Obtain a copy and ensure Special Inspections are completed.					Include a completed copy for BOD Mangament sign off.
95			KTR	Pre Final Inspection					
96			NAVFAC	Client walk-thru Inspection					
97			KTR	Pre-Final Inspection					
98			NAVFAC	Pre-Final Inspection					
99			KTR	Punch List					
100			KTR	Final Inspection					
101			NAVFAC	Final Inspection					
102									
103			NAVFAC	BOD/Use and Possession					
104			Client	Ribbon-cutting ceremony					
105			Client	Planned User Move-in					
106									
107			KTR	Delivery of O&M Manuals					Must be provided before BOD
108			NAVFAC	CM to submit the O&M's to DM who must review and submit to NAVFAC PWD ME Requirements FMS					
109									
110			KTR	Delivery of Product Warranties					Must be provided before BOD

	A	B	C	D	E	F	G	H	I
3			Resp.	Critical Activities	Point of Contact	Due Date	Actual Complete Date	CM Initials	Notes
111			NAVFAC	CM to submit Warranties to NAVFAC PWD ME Requirements FMS					
112									
113			KTR	As-Built Drawings					Must be provided 30 cal days before BOD
114			NAVFAC	CM to submit the As-Builts to DM who must review and submit documents to CAD Dept.					
115			KTR	Submit eOMSI Facility Data Workbook					Must be provided before BOD
116			NAVFAC	CM to submit the eOMSI Workbook to the DM who must review and submit to the NAVFAC PWD ME FOS.					
117			KTR	Submit the Internal Service Requirement List (ISRL).					Must be provided before BOD
118			NAVFAC	CM to submit the ISRL to the DM who must review and submit to the NAVFAC PWD ME FSC. Confirm ISRL submitted to FSC.					
119									
120				**Required for Contract Close-out:**					
121			NAVFAC	Warranty documentation to FMS					Must be provided before CCD
122			KTR	Site Restoration					
123			KTR	Final Landscaping					
124			KTR	Spare Parts, Extra Stock, Special Tools, etc					
125									
126			KTR	Final Demobilization and Clean-up					
127			KTR	Temp Construction Fence Removed					
128			KTR	Project Close-out Meeting					
129			NAVFAC	As-Builts to PWD ME PDC CADD Branch (Preston Gowen)					
130									
131									
132			KTR	Closeout & submit permits and inspection reports					Notice of Termination (NOT); Storm water insp rpts; Stormwater BMP Install rpts; Archaeological Monitor'g rpts and artifacts, salvaged mat. etc.
133			NAVFAC	Closeout & submit permits and inspection reports					NPDES, Notice of Termination (NOT); Storm water insp rpts; Stormwater BMP Install rpts; Archaeological Monitor'g rpts and artifacts, salvaged mat. etc.
134			KTR	Seasonal/Deferred Commissioning FPT					
135			KTR	2^{nd} Season TAB Report					
136			NAVFAC	2^{nd} Season TAB Field Accept Testing					
137			KTR	Annual Elevator Maintenance Complete					
138			NAVFAC	Annual Elevator Maintenance Complete					
139			NAVFAC	Authority to Operate delivered to client					
140			KTR	Commissioning 10-Month/Warranty Visit					
141			NAVFAC	Inform PM of BOD					

	A	B	C	D	E	F	G	H	I
3			**Resp.**	**Critical Activities**	**Point of Contact**	**Due Date**	**Actual Complete Date**	**CM Initials**	**Notes**
142			NAVFAC	BOD (A) entered into eContracts					
143			NAVFAC	BOD Letter to Contractor					Sample ltr can be found here
144			NAVFAC	Acceptance Letter to Client					Sample ltr can be found here
145			NAVFAC	Process recycled/recovered materials report					
146			NAVFAC	Contractor Evaluations (CPARS) Complete					assist PM
147			NAVFAC	Finalize Outstanding Contract Mods					

ISRL Instruction Sheet

The new ISRL is broken out in 3 areas

Section 1 is known serviceable inventory currently under contract. This information needs be obtained during the design phase of a project and conveyed in the ISRL under "current inventory-filled out by government". For a complete inventory, you will need to see the FSC department. Contact David Nedeau or Rachael Andrews. The following is an example:

Fire Suppression		
EQUIPMENT	**quantity**	**INFORMATION / LOCATION**
WET SYS	**1**	**1st floor east side mechanical room**
DRY SYS	**1**	**1st floor east side mechanical room**
PRE-ACTION	**1**	**1st floor east side mechanical room**
STAND PIPE	**2**	**one in each stairwell**
DIESEL FIRE PUMP	**1**	**Cummins 6BTA59F, 208 HP / 1st floor east side mechanical room**
ELECTRIC FIRE PUMP	**1**	**1st floor east side pump room**

Section 2 is the responsibility of the contractor. In the first block, the contractor needs to indicate what they have done with the known equipment or what they have added to the facility that was not previously there. You will see below, that all of the equipment noted by the government has either been removed, replaced, or is not part of the scope, with a small note section describing the action taken. If it is new to the facility, the contractor will indicate it on a blank line below the known listed inventory (example indicated in RED).

Fire Suppression								
EQUIPMENT	quantity	INFORMATION / LOCATION		removed	replaced	new to building	not part of scope	notes
WET SYS	1	1st floor east side mechanical room			x			like in kind
DRY SYS	1	1st floor east side mechanical room			x			replaced with wet
PRE-ACTION	1	1st floor east side mechanical room		x				removed from building
STAND PIPE	2	one in each stairwell					x	
DIESEL FIRE PUMP	1	cummins 6BTA59F, 208 HP / 1st floor east side mechanical room			x			like in kind
ELECTRIC FIRE PUMP	1	1st floor east side pump room					x	
						x		add wet system SE corner

Block 2 of the contractor required information is specific information related to the new systems installed. If an item was not part of the scope or removed from the building, then this information does not need to be filled in. If a system was replaced or is new to the building, then the subsequent data is required.

EQUIPMENT	Fire Pump Data		Fire Pump Driver Data				
	Manufacturer	GPM	Manufacturer	Model	Serial #	HP	RPM
WET SYS							
WET SYS							
DIESEL FIRE PUMP	Peerless	2500	Cummins	7BTK86E	125487	208	2500
WET SYS							

Section 3 is listed to the far right of the sheet labeled "information only". The information contained in this column, are items that need to be captured if they do not already appear under the inventory that the government has already provided. For instance, if a system or component is newly installed in a facility, refer to the list of items in the category to see if it needs to be captured on this sheet, if you do not see it listed then it does not need to be recorded.

systems to include in this report are:	
Automatic Sprinkler Systems	Gas Systems
Fire Pumps and Drivers	Chemical Systems
Air Compressors	Range Hoods
Stand Pipes	Foam Systems
Drum Drips	

*** This form needs to be filled out at the end of each phase of the contract and turned into the government no less than 90 days prior to occupancy *** : Each ISRL needs to be job specific. If it doesn't pertain to the job, delete/hide the rows that do not apply. This will prevent the contractor from filling in information not pertinent to the job.

For information or assistance filling out the ISRL, please contact the FSC department.

David Nedeau – X4850

Rachael Andrews – X4851

NAVFAC Internal Services Requirements List

Project Name (include BLDG#): **e-Project#**

****** this form needs to be filled out at the end of each phase of the contract and turned into the government no less than 90 days prior to occupancy*****

CURRENT INVENTORY - FILLED OUT BY GOVERNMENT | **FILLED OUT BY CONTRACTOR** | **INFORMATION ONLY**

Janitorial Accessories & Cleaning Services Required - ** ATTACH FLOOR PLAN CADD FILE **

Room Use (Administration / Bathroom / Kitchenette / Locker Room, etc) and Room Number	Room #	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	Room Use (Administration / Bathroom / Kitchenette / Locker Room, etc)	Room Number	Flooring Type (VCT, Resilient, Carpet, Ceramic, Concrete, Wood, other)	Square Footage (Include Shower Floors)	Planned Number of Occupants	Shower Square Footage		Quantities						(Dispensers MUST be compatible with existing Paper and Soap Products.			Consult with FSC)	equipment to include in this report are:	
													Floors	Walls	Toilets	Urinals	Sinks	Mirrors	Water Fountains	Lockers	Paper Towel Dispensers	Toilet Paper Dispensers	Soap Dispensers	Soap (Grit) Dispensers	Paper Towel Dispensers	Soap Dispensers
																									Toilet Paper Dispensers	Recessed Trash Cans
																									Flooring	Fixtures

Fire Suppression

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	Fire Pump Data		Fire Pump Driver Data					systems to include in this report are:	
									Manufacturer	GPM	Manufacturer	Model	Serial #	HP	RPM		
SAMPLE WET SYS	1	1st floor east side mechanical room		x			like in kind	WET SYS								Automatic Sprinkler Systems	Gas Systems
DRY SYS	1	1st floor east side mechanical room		x			replaced with wet	WET SYS								Fire Pumps and Drivers	Chemical Systems
PRE-ACTION	1	1st floor east side mechanical room	x				removed from building									Air Compressors	Range Hoods
STAND PIPE	2	one in each stairwell				x										Stand Pipes	Foam Systems
DIESEL FIRE PUMP	1	cummins 6BTA59F. 208 HP / 1st floor east side mechanical room		x			like in kind	DIESEL FIRE PUMP	Peerless	2500	Cummins	7BTK86E	125487	208	2500	Drum Drips	
ELECTRIC FIRE PUMP	1	1st floor east side pump room				x											
					x		add wet system SE corner	WET SYS									

Fire Alarm Systems Equipment *(not applicable @ PNSY)*

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	Control Panel			Heat and Smoke Detectors			Addressable System	Horn Strobes	systems to include in this report are:
								Manufacturer	Model	Type	Smoke or Heat	Mfgr	Model	Yes or No	Quantity	Fire Alarm System Components

Boilers, Furnaces, Heating Equipment

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	Manufacturer	Model	Serial #	MBH	Fuel	Date of Manufacture	systems to include in this report are:
															Boilers
															Furnaces

HVAC *(not applicable @ PNSY - Shops maintained)*

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	quantity	Unique ID	Manufacturer	Model	Serial #	Tonnage / HP	# of Filters	Size of Filters	Belt Specs	Refrigerant	systems to include in this report are:
																			Major HVAC Components

Boilers and Unfired Pressure Vessels (UPV) Certification

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	Boiler or UPV	Description	National Board #	USN Number	Date Inspected	systems to include in this report are:
														Boilers
														Pressure Vessels

Back Flow Preventers

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT		Location Floor #	Manufacturer	Model	Serial #	Size	Type	System installed in	systems to include in this report are:
																	Backflow Prevention Devices

Emergency Generators

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	Manufacturer	Model	Serial #	KW	HP	systems to include in this report are:
														Emergency generators

Water Cooler Filters

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	Floor	Water Cooler		Filter		systems to include in this report are:
										Mfgr	Model #	Model #	Water Sentry Y/N	Water Coolers

Gutters & Downspouts

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	Gutters				DownSpouts				Roof Drains	systems to include in this report are:
									Size	Material	Height	Length	Material	# of Drops	Size	Height	Quantity	Low Sloped Roofs

Overhead Doors

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	Door #	Width	Height	Manufacturer	Type	Fire Door Y / N	Electric / Manual	Interior / Exterior	Voltage	systems to include in this report are:
																		Overhead Doors

Motion Activated Glass Doors

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	Model Number	Serial Number	Dimension Width	Height	systems to include in this report are:
													Motion Activated Doors

Low Sloped Roof

NAVFAC Internal Services Requirements List

Project Name (include BLDG#): **e-Project#**

****** this form needs to be filled out at the end of each phase of the contract and turned into the government no less than 90 days prior to occupancy*****

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	Roofing Material		Year Installed	Eave Height	Square Footage	Roof Drawing Included?											systems to include in this report are:	
																									Low Sloped Roofs	

Elevator

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	Speed (FPM)	# of Stops	Capacity (Pounds)	Manufacturer	Installer	Date Installed	VTE Type	Rope Gripper Y/N	Siesmic Installed	Smoke Detectors Y/N	Shunt Trip Y/N	Fire Extinquisher Y/N	Machineroom Sprinklers Y/N				systems to include in this report are:	
																									VTEs	

Lightning Protection

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	quantity	Provide drawings showing Lightning Protection Equipment and locations															systems to include in this report are:	
																									Lightning Protected buildings	

Storm Water Management Soil Filters, Ponds, etc.

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	Provide Drawings Showing Desgn and Location of OWS																systems to include in this report are:	
																									Storm Water Management Systems Equipment	

Oil and Water Separators

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	Provide Drawings Showing Desgn and Location of OWS																systems to include in this report are:	
																									Oil Water Separators	

Motorized Gate / Turnstile

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	Model Number		Serial Number														systems to include in this report are:	

Other ST Recurring Services

EQUIPMENT	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes	EQUIPMENT	quantity																Include in this report are Systems and Equipment not listed above including but not limited to;	
																									UPS Systems	Automatic Transfer Switch
		Consult with FSC, FMD, and CI before filling out																							Sewage Lift Stations	Catch Basins
		the Other ST Recurring Services section																								

Other FX Recurring Services

Service	quantity	INFORMATION / LOCATION	removed	replaced	new to building	not part of scope	notes																		Services to Include but not limited to	
																									Grounds Maintenance	Refuse Removal
		Consult with FSC, FMD, and CI before filling out																							Pest Control	Snow Removal
		the Other FX Recurring Services section																								

SECTION 01 30 10.00 22
^^
COORDINATION PROCEDURES AND COORDINATION DRAWINGS (PWD ME)
01/22

PART 1 GENERAL

This Section applies to Design-Bid-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 SUMMARY

Requirements of this Section apply to, and are a component part of, each Section of the specifications.

1.2 GENERAL COORDINATION PROCEDURES

1.2.1 Coordination

Coordinate construction operations included in different Sections of the Specifications to ensure efficient and orderly installation of each part of the Work. Coordinate construction operations included in different Sections that depend on each other for proper installation, connection, and operation. Each subcontractor must participate in coordination requirements. Certain areas of responsibility are assigned to a specific subcontractor.

a. Schedule construction operations in sequence required to obtain the best results, where installation of one part of the Work depends on installation of other components, before or after its own installation.

b. Coordinate installation of different components to ensure maximum performance and accessibility for required maintenance, service, and repair.

c. Make adequate provisions to accommodate items scheduled for later installation.

1.2.2 Distribution of Memoranda

Prepare memoranda for distribution to each party involved, outlining special procedures required for coordination. Include such items as required notices, reports, and list of attendees at meetings.

a. Prepare similar memoranda for the Government and separate subcontractors if coordination of their Work is required.

1.2.3 Project Coordination Meetings

Conduct weekly Project Coordination Meetings either separately or in conjunction with weekly Quality Control Meetings to discuss project planning, coordination, and performance of on-going and future construction activities. Refer to 01 45 00.00 22 Quality Control (PWD ME) paragraph QUALITY CONTROL WEEKLY MEETINGS for additional information.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation;

submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-02 Shop Drawings

Coordination Drawings; G

1.4 COORDINATION DRAWINGS

1.4.1 General

At a minimum provide thorough and complete coordination drawings indicating as a minimum existing conditions including existing structure, walls, openings, floor to floor heights, ceiling heights, clear opening dimensions, existing utilities to remain, proposed openings to be provided by contractor through existing floors, walls, partitions, roof deck, along with proposed location, layout, and arrangement of fire suppression, fire alarm and mass notification system, plumbing, HVAC, electrical, communication, and electronic safety and security systems including, but not limited to, piping, HVAC equipment, ductwork, sprinkler system, cable tray, communications conduit trapeze runs, lighting fixtures, conduit feeders, floor poke throughs, equipment, panelboards, control panels, equipment service areas, and other items that must be shown to ensure a coordinated installation. Drawings must indicate adequate clearance for operation, maintenance, and replacement of operating equipment and devices. Coordinate with each applicable Section in this Project Specification.

Submit coordination drawings according to requirements above, in Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME), and in individual Sections, and additionally where installation is not completely indicated on Shop Drawings, where limited space availability necessitates coordination, or if coordination is required to facilitate integration of products and materials fabricated or installed by more than one entity. The intent of the contractor produced coordination drawings is to identify conflicts, and resolve them prior to on-site construction activities to the benefit of the Contractor and the Government.

Note: Electronic files of the Contract Drawings will be made available for use in the preparation of coordination drawings.

a. Content: Project-specific information, drawn accurately to a scale large enough to indicate and resolve conflicts. Do not base coordination drawings on standard printed data. Include the following information, as applicable:

    1. Use applicable Drawings as a basis for preparation of coordination drawings. Prepare sections, elevations, and details as needed to describe relationship of various systems and components.
    2. Coordinate the addition of trade-specific information to coordination drawings in a sequence that best provides for coordination of the information and resolution of conflicts between installed components before submitting for review.
    3. Indicate functional and spatial relationships of components of architectural, structural, civil, mechanical, plumbing, fire suppression, communication systems, fire alarm and mass notification system, and electrical systems.
    4. Indicate space requirements for routine maintenance and for

anticipated replacement of components during the life of the facility.
5. Show location and size of access doors required for access to concealed dampers, valves, and other controls.
6. Indicate required installation sequences.
7. Indicate dimensions shown on Drawings. Specifically note dimensions that appear to be in conflict with submitted equipment and minimum clearance requirements. Provide alternative sketches indicating proposed resolution of such conflicts. Minor dimension changes and difficult installations will not be considered changes to the Contract.

b. Timing: Contractor must provide Coordination Drawings for review and approval, complete with required building components and contract requirements, a minimum of 180 days prior or less to the ordering or installation of mechanical, electrical, fire protection, or plumbing systems.

1.4.2 Coordination Drawing Organization

Organize coordination drawings as follows:

a. Floor Plans and Reflected Ceiling Plans: Show architectural and structural elements, and mechanical, plumbing, fire suppression, communication systems, fire alarm and mass notification system, and electrical Work. Show locations of visible ceiling-mounted devices relative to acoustical ceiling grid. Supplement plan drawings with section drawings where required to adequately represent the Work.

b. Ceiling Cavity Space: Indicate existing structural beams, trusses, and other components that provide for support of ceiling and wall systems, mechanical and electrical equipment, and related Work. Locate components within ceiling cavities to accommodate layout of light fixtures and other components indicated on Drawings. Indicate areas of conflict between light fixtures and other components.

c. Mechanical and Electrical Rooms: Provide coordination drawings for mechanical and electrical rooms showing plans and elevations of mechanical, plumbing, fire suppression, communication systems, fire alarm and mass notification system, and electrical equipment.

d. Structural Penetrations: Indicate penetrations and openings required for all disciplines in existing floors and masonry walls.

e. Mechanical and Plumbing Work: Show the following:

1. Sizes and bottom elevations of ductwork, piping, and conduit runs, including insulation, bracing, flanges, and support systems.
2. Dimensions of major components, such as dampers, valves, diffusers, access doors, cleanouts, air handlers, terminal units, boiler plant equipment, control panels, and electrical distribution equipment.
3. Fire-rated enclosures around ductwork.
4. Plumbing piping locations and sizes and locations of plumbing fixtures and equipment.
5. Drawings must include existing to remain portions of mechanical and plumbing systems including ductwork, piping, and equipment.

f. Electrical Work: Show the following:

1. Runs of vertical and horizontal conduit 1-1/4 inches in diameter and larger.
2. Light fixture, exit light, smoke detector, and other fire alarm locations, including dimensions.
3. Panelboards, transformers, control panels, relays, motor controls, and disconnect switches, including dimensions.
4. Location and dimensions of pull boxes and junction boxes, dimensioned from column center lines.
5. Locations and dimensions of cable trays.
6. Drawings must include existing to remain portions of conduits and electrical systems and components.

g. Communication Systems: Show the following:

1. Locations of panels and components.
2. Locations and dimensions of cable trays.
3. Drawings must include existing to remain portions of conduits and cable trays.

h. Fire Suppression System: Show the following:

1. Locations and dimensions of risers, main and branch lines, pipe drops, and sprinklers. Drawings must include the existing-to-remain portions of the system. Drawings must include temporary piping to be installed to minimize system impairments.

i. Fire Alarm and Mass Notification System: Show the following:

1. Locations of panels, initiating devices, notification appliances, and conduit risers. Drawings must include the existing portions of the system that are to remain until the new sytem has been approved.

j. Review and Approval: Coordination drawings will be reviewed by the Government to confirm that in general the Work is being coordinated, but not for the details of the coordination, which are the Contractor's responsibility. If the Government determines that coordination drawings are not being prepared in sufficient scope or detail, or are otherwise deficient, the Government will so inform Contractor, who must make suitable modifications and resubmit. Government approval of coordination drawing submittal is required prior to the start of any associated construction activities.

1.4.3 Coordination Digital Data Files

Prepare coordination digital data files according to the following requirements:

a. File Preparation Format: Same digital data software program, version, and operating system as original Drawings (Revit 2023), or format discussed with and approved by the Contracting Officer. At Contractor's option BIM software can be used with Navisworks export provided to Government for review.

b. Digital Data Software Program: Provide separate layers or other means to distinguish between the different scope of work elements.

c. File Submittal Format: Submit or post coordination drawing updates at the time of monthly invoicing for review by the Government. Submission of Coordination Drawings for review must be provided as both PDF and Digital Version. Digital Version must be in the same software program that was used for the original drawing development.

d. All digital files will become part of the project and will become property of the Navy at the end of the project.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

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SECTION 01 31 23.13 20
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ELECTRONIC CONSTRUCTION AND FACILITY SUPPORT CONTRACT MANAGEMENT SYSTEM (eCMS)

08/23, CHG 1: 02/24

PART 1 GENERAL

1.1 CONTRACT ADMINISTRATION

Utilize the Naval Facilities Engineering Systems Command's (NAVFAC's) Electronic Construction and Facility Support Contract Management System (eCMS) for the transfer, sharing, and management of electronic technical submittals and documents. The web-based eCMS is the designated means of transferring technical documents between the Contractor and the Government. Paper media or e-mail submission, including originals or copies, of the documents identified in Table 1 are not permitted, except where eCMS is unavailable, non-functional, or specifically requested in addition to electronic submission. When specifically requested to provide documents outside of eCMS, upload all final project documentation (e.g. documents that are signed and/or adjudicated by the Government) mentioned in Table 1 into the subject eCMS document management folders that are associated with that document type. Include the identification number of the document, type of document; the name/subject or title; and for daily reports the date (day of work) with format YYYY/MM/DD in the filename. For example: For RFI’s 0011_RFI_Roof_Leaking.doc; For submittals 32a_Submittals_Light_Fixture.pdf; For Daily Reports 0132_Daily_Report_20190504.xls. Contact the Contracting Officer's Representative (COR) regarding availability of eCMS training and reference materials.

At Contractors discretion, and as approved by the Contracting Officer, alternate web-based construction management software may be used.

All government contracting specialist/officer, legal, and command communications will remain the same.

1.2 USER PRIVILEGES

The Contractor's key staff may be provided access to eCMS. Contact the Contracting Officers Representative (COR) for eCMS account access. Project roles and system roles will be established to control each user's menu, application, and software privileges, including the ability to create, edit, or delete objects. Additional project roles may be assigned for workflow. The COR makes the final decision on roles for the project. User's ability to view, edit documents may be lowered at the discretion of the COR.

Only one eCMS user account is required regardless of the number of user's projects. Notify the COR within seven calendar days if a contractor user is no longer associated with company or project so they can remove them from any open record and inactivate them from the project.

If using alternate construction management software, the Contractor must not limit or restrict COR access and must provide any program/project documentation upon request.

1.2.1 eCMS Subcontractor Users

If the contractor's user is a subcontractor, the subcontractor must be registered under the name of their company and email. For example it is common for contractors to contract Quality Control Managers. The QC Manager's account should be under their company's name and email reducing the number of eCMS accounts required.

1.2.2 Users with Multiple Roles

Users may have multiple roles associated with their account within eCMS. Roles are used in workflow. When a user is added to the project, they will be assigned the default role when the user was created. Contact the COR to change or add roles to the user for the project.

1.2.3 Loss of Privilege

Users may lose privilege to access eCMS at the discretion of the Contracting Officer. The eCMS is a collaborative system that allows flexibility of use and does not restrict all inappropriate user actions. User activities are logged into eCMS in visible and background data collection. Users found to use eCMS in an inappropriate action may have their eCMS access revoked. Examples include, but not limited to, fraudulent representations, sharing user accounts with others, and changing approved records without the consent of the COR. Depending on the severity of the infraction, the users can lose eCMS access for a period of time, permanently for the project, or lose eCMS access for any project. The contractor may appeal the suspension in writing to the contracting officer within 14 calendar days of notice. The appeal must identify the infraction, supporting information, and steps to ensure the infraction will not happen in the future.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

List of Contractor's Personnel; G

For Division 1 government approved Pre-Construction submittals, combine into a single Pre-Construction Submittal Package. Annotated with SD Type of SD-01. Pre-Construction submittal package approval date will be used as a KPI.

1.4 SYSTEM REQUIREMENTS AND CONNECTIVITY

1.4.1 General

NAVFAC eCMS requires a web-browser (platform-neutral) and Internet connection. For best results, recommend using browser in InPrivate/Incognito mode; Internet speeds greater than 40mbps when uploading files, computers with high RAM and Solid State Drives, "White List" eCMS website, Zip or Split files for better uploading. Non-NAVFAC Users are not to use VPN when using eCMS per NAVFAC IT.

The use of eCMS is required by the Contractor and all associated costs and time necessary to utilize eCMS will be borne by the Contractor with no allowance for time extensions and at no additional cost to the government.

1.4.2 Contractor Personnel List

Within 20 calendar days of contract award, provide to the Contracting Officer a list of Contractor's personnel who will have the responsibility for the transfer, sharing and management of electronic technical submittals, RFIs, daily reports, and other files and will require access to the eCMS. Project personnel roles which must be filled as applicable in the eCMS, include, at a minimum, the Contractor's Project Manager, Superintendent, Quality Control (QC) Manager, Principal, and Site Safety and Health Officer (SSHO). Notify the COR immediately of any personnel changes to the project. The Contracting Officer reserves the right to perform a security check on all potential users.

Provide the following information:

Company Name

Name (First, Last)

Email Address

Project Role (CQM, SSHO, Superintendent, CM, PM, Principal)

Existing or New eCMS User

1.5 SECURITY CLASSIFICATION

In accordance with Department of Navy guidance, all military construction contract data are unclassified, unless specified otherwise by a properly designated Original Classification Authority (OCA) and in accordance with an established Security Classification Guide (SCG). Refer to the project's OCA when questions arise about the proper classification of information.

In conformance with the Freedom of Information Act (FOIA), DoD INSTRUCTION 5200.48 CONTROLLED UNCLASSIFIED INFORMATION (CUI), and DoD requirements, any unclassified project documentation uploaded into the eCMS must be designated either "U - UNCLASSIFIED" (U) or "CUI - CONTROLLED UNCLASSIFIED INFORMATION" (CUI).NAVFAC eCMS must only be used for the transaction of unclassified information associated with construction projects. Controlled Unclassified Identification (CUI) documents may be loaded into eCMS with the appropriate markings.

1.5.1 Markings on CUI Documents

Contractor's proprietary information, or documents determined by the originator in accordance with CUI guidance, should be marked CUI. Proprietary information not marked CUI can be released under the Freedom of Information Act (FOIA). Apply the appropriate markings before any document is uploaded into eCMS. Markings are not required on Unclassified U documents.

1.6 eCMS UTILIZATION

Establish, maintain, and update data and documentation in the eCMS throughout the duration of the contract. Utilize eCMS to transfer all submittals, RFIs, daily reports, and other files required by contract to be forwarded to the government.

1.6.1 Restricted Information

Personally Identifiable Information (PII) transmittal such as credit card, driver's license, passport, social security, and payroll number are not permitted in eCMS. Name, address, and email are permitted. Pre-negotiation information such as cost estimates that require formal negotiations are not allowed. For example, proposed changes over the SAP level of $250k require formal negotiations. Cost estimates for LEAN, ULTRA LEAN, and Design Changes under the SAP level are at the discretion of the COR's or Contract Specialist/Officer's direction. The eCMS must only be used for the transaction of unclassified information associated with construction projects. Controlled Unclassified Identification (CUI) documents may be loaded into eCMS with the appropriate markings. Uploading of files directly into the Documents folder is not allowed. All documents must be uploaded using an eCMS module.

Project photos must not be uploaded to eCMS. All photos must be reviewed by Portsmouth Naval Shipyard Security prior to any public release.

1.6.2 Naming Convention for Files

Titles of files uploaded are to be descriptive of the purpose and content of the file. For example RFI_ROOF_Leak.doc or for submittals, SUB_LIGHT_FIXTURE.pdf. Titles of file to be uploaded must only contain uppercase letters, lowercase letters, numbers, hyphens (-), underscores (_) and periods (.). Use of any other characters is not allowed and may create an error. When practicable, adding the record number to the title is desired. For example RFI_XYZ12345_ROOF_Leak.doc. Uploading files with the same title will create a new revision in eCMS. Original revision is Rev 0, the first revision is Rev 1. Uploaded files are to use the default file location regardless of the module used unless directed by the COR.

Table 1 also identifies which eCMS application is to be used in the transmittal of data (these are subject to change based on the latest software configuration).

Table 1 - Project Documentation Types

SUBJECT/NAME	**REMARKS**	**eCMS APPLICATION**
As-Built Drawings	Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager	Submittals

SUBJECT/NAME	REMARKS	eCMS APPLICATION
Building Information Modeling (BIM)	Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager	Submittals
Construction Permits	Refer to rules of the issuing activity, state or jurisdiction	Submittals
Construction Schedules (Activities and Milestones)	After the schedule submittal is approved by the COR, import the schedule file into the scheduling application, and select "Approve" to establish a new schedule baseline.	Submittals and Scheduling
Construction Schedules	After the schedule submittal is approved by the COR, import the schedule file into the scheduling application, and select "Approve" to establish a new schedule baseline.	Submittals and Scheduling
Construction Schedules (3-Week Look ahead)	Import the schedule file into the scheduling application, and select "Approve" to establish a new schedule baseline	Scheduling and Meeting Minutes
Daily Production Reports	Provide weather conditions, crew size, man-hours, equipment, and materials information	Daily Report
Daily Quality Control (QC) Reports	Provide QC Phase, Definable Features of Work Identify visitors	Daily Report
Designs and Specifications	Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager	Submittals
Environmental Notice of Violation (NOV), Corrective Action Plan	Refer to rules of the issuing activity, state or jurisdiction	Submittals
Environmental Protection Plan (EPP)		Submittals
Invoice (Supporting Documentation)	Applies to supporting documentation only. Invoices are submitted in Wide-Area Workflow (WAWF)	Submittals

SUBJECT/NAME	**REMARKS**	**eCMS APPLICATION**
Jobsite Documentation, Bulletin Board, Labor Laws, SDS	Redact any PII information when loaded into eCMS	Submittals
Meeting Minutes		Meeting Minutes
Modification Documents	Provide final modification documents for the project. Upload into Modifications RFPs folder	Communications
Operations & Maintenance Support Information (OMSI/eOMSI), Facility Data Worksheet	1. Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager 2. Design reviews will be performed in existing "Dr Checks"	Submittals
Photographs	Subject to base/installation restrictions	Submittals
QCM Initial Phase Checklists		Meeting Minutes or Checklists
QCM Preparatory Phase Checklists		Meeting Minutes or Checklists
Quality Control Plans		Submittals
QC Certifications		Submittals
QC Punch List		Punch Lists
Red-Zone Checklist		Punch List or Checklists
Rework Items List		Punch Lists
Request for Information (RFI) Post-Award		RFIs
Safety Plan		Submittals
Safety - Activity Hazard Analyses (AHA)		Submittals

SUBJECT/NAME	**REMARKS**	**eCMS APPLICATION**
Safety - Mishap Reports		Daily Report
Shop Drawings	Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager	Submittals
Storm Water Pollution Prevention (Notice of Intent - Notice of Termination)	Refer to rules of the issuing activity, state or jurisdiction	Submittals
Submittals and Submittal Register		Submittals
Testing Plans, Logs, and Reports		Submittals
Training/Reference Materials		Submittals
Training Records (Personnel)	Redact any PII information if storing in eCMS	Submittals
Utility Outage/Tie-In Request/Approval		Submittals
Warranties/BOD Letter		Submittals
Quality Assurance Reports		Checklists (Government initiated)
Non-Compliance Notices		Non-Compliance Notices (Government initiated)
Other Government-prepared documents		GOV ONLY
Letters to government contracting, claims, REAs, and other contracting officer communications	eCMS is not the primary tool to use in contracting officer communications. eCMS can only store documents or letters after the submission to the contracting officer is made.	Communications

SUBJECT/NAME	REMARKS	eCMS APPLICATION
All Othere Documents	Refer to FOIA guidelines and contact the FOIA official to determine whether exemptions exist	As applicable

1.6.3 RFIs Module

Create contractor RFIs using eCMS RFIs module. The contractor must confirm the numbering convention with the COR if different than eCMS default.

If the government (GOV) Response has "No" Cost or Schedule Impact, this reply is given with the expressed understanding that it does not constitute a basis for any change in the amount or time of subject contract. Information provided in this response does not authorize work not currently included in the contract. If GOV Response is "Yes" or "Potentially" then this response may require a change to the contract. If the contractor disagrees with either the government's No Cost determination or No Schedule impact determination, or both, the contractor has 14 calendar days to notify the COR and Contracting Officer in writing.

1.6.4 Submittals Module

Create contractor submittals using eCMS Submittals module. The contractor must confirm the numbering convention with the COR if different than eCMS default.

1.6.5 Submittal Packages Module

Create submittal packages using the eCMS Submittal Packages module in lieu of or in addition to Related Objects. Submittal Packages track completion of the packaged submittals and is used in NAVFAC HQ's KPIs.

1.6.6 Communications Module

Create communications using the eCMS Communications module. The Communications module is used to create or document communications that are not a part of other eCMS modules. Use of Communications module will memorialize information into an eCMS record file. The following are Types of Communications:

Email

Memo to File

Face to Face

Telephone

Web Collaboration

Photos

Other Documents

Other

Unless directed by the COR, upload documents or files that do not have a corresponding eCMS module. Choose "Photos" Type for Photos and "Other Documents" for all other documents.

1.6.7 Issues Module

Create or respond to issues using the eCMS Issues module. Respond to CPARS issues using the Issues module.

1.6.8 Meeting Minutes Module

Create or respond to Meeting Minutes using the eCMS Meetings module.

Document required contractual meetings. Dates of meetings are used in NAVFAC KPIs. Minimum meetings in eCMS include the following:

Post Award Kickoff (PAK)

Pre-construction (Pre Con)

Initial and Preparatory Three Phases of Control

MEP Coordination

Equipment/System Controls Coordination

Partnering

Quality Control (QC)

1.6.9 Potential Change Items Module

Not used.

1.6.10 Daily Report Module

Create Daily Reports using the eCMS Daily Report Module. The contractor must confirm the numbering convention with the COR if different than eCMS default.

1.6.11 Punchlists Testing Logs (Legacy)

Punchlist Testing Logs is a legacy program that is being replaced by the Punch Lists Module. This module is to be used for reference of past projects. Use the Punch Lists Module for all future work.

1.6.12 Punch Lists Module

The eCMS Punch Lists module is useful more than just for Punchlists. The module includes the capability of batch editing, create items from Optical Character Recognition (OCR) plans, assign tasks and track completion of individual items.

Create the following using the Punch Lists module:

Rework Items List

DFOW List

Punch-Out Inspection

Pre-Final Punchlist Inspection

Final Punchlist Inspection

Testing Logs

1.6.13 FWD UltraLean COAR RFP Module

Not used.

1.6.14 Non-Compliance Notices (NCN) Module

Respond to Non-Compliance Notices listed in the Non-Compliance Notices module.

1.6.15 Checklists

Use Checklist listed in the contractor's eCMS menu and as directed by the COR. Checklists capture data and is used in dashboards and KPIs.

1.6.15.1 Partnering Team Health Survey Checklist

Contractor must use the eCMS checklist to document the partnering team health survey. Partnering Team Health Survey is in accordance with the Partnering Specification of this contract.

1.6.16 Flysheets

Use Flysheets listed in the contractor's eCMS menu, if available, and as directed by the COR. Flysheets allow the contractor to print out information from other systems and upload into eCMS. The eCMS will use OCR to capture the information as data. Flysheets capture data and used in dashboards and KPIs.

1.6.17 eCMS Outage

In the case where eCMS is unavailable for 8 hours or more, paper or email may be used in the interim to maintain project schedule.

Once the system is operational, all final records are required to be recreated using the appropriate module. Subject/title of the record to include the type of record i.e., RFI/Submittal/Daily Report/Communication/Other, the identification number(s), and the statement "Processed Outside of eCMS". Example, "RFI 001 Processed Outside of eCMS".

1.6.18 User Account Activity

NAVFAC eCMS captures user data and activities that are directly related to the user's account. The user agrees through the use of eCMS, their account activities will be captured and can be displayed on eCMS printed reports.

1.7 QUALITY ASSURANCE

Requested Government response dates on Submittals must be in accordance with the terms and conditions of the Contract unless previously agreed by the COR. Requesting response dates earlier than the required review and response time, without concurrence by the Government COR, may be cause for rejection.

Incomplete submittals will be rejected without further review and must be resubmitted. Required Government response dates for resubmittals must reflect the date of resubmittal, not the original submittal date.

All submittals and associated attachments must be transmitted to the Government via the COR. Transmittals are no longer required when using eCMS since approval status is tracked on the submittal. Transmittal forms can be attached to submittals if approved by the COR. Submittals requiring government approval are "Transmitted For" "Approval". Submittals requiring contractor approved submittals, including those designated for Contractor Quality Control approval or Information Only, are "*Transmitted For" "Information Only" in the Submittal Module. Provide and sign the QC certification or approving statement on the attachment per submittal specification section. When Submittal Packages are required, use eCMS Submittal Packages after creating individual submittals. Importing Submittals from the Submittal Register is optional. Contact the COR for the data conversion requirements.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

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SECTION 01 32 17.00 20
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COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS)
11/23, CHG 1: 08/24

PART 1 GENERAL

1.1 DEFINITIONS

The cost-loaded Network Analysis Schedule (NAS) is a tool to manage the project, both for Contractor and Government activities. The NAS is also used to report progress, evaluate time extensions, and provide the basis for progress payments.

For consistency, when scheduling software terminology is used in this section, the terms in Primavera's scheduling programs are used.

1.2 SCHEDULE REQUIREMENTS PRIOR TO THE START OF WORK

1.2.1 Preliminary Scheduling Meeting

Before preparation of the Project Baseline Schedule, and prior to the start of work, meet with the Contracting Officer to discuss the proposed schedule and the requirements of this section. Propose projected data dates for monthly update schedules for the project and incorporate each monthly update submittal into submittal register. Discuss required forms, terminology, and submittal requirements of this section and other requirements related to schedule management for this contract. Use the NAVFAC-provided Preliminary Scheduling Meeting Guideline in the completion of the intended mutual understanding.

1.2.2 Project Baseline NAS

Submit the Baseline NAS within 30 calendar days after contract award. Data date must be set to contract award date and no progress recorded for any activity. Only bonds and the required Preliminary Scheduling Meeting may be recorded as complete prior to acceptance of the Baseline NAS. The acceptance of a Baseline NAS is a condition precedent to:

a. The Contractor starting demolition work or construction stage(s) of the contract.

b. Processing Contractor's invoices(s) for any items other than bonds.

c. Review of any schedule updates. The contractor is still required to submit monthly network analysis updates. Without an accepted baseline the Government may return schedule update as received for record.

Submittal of the Baseline NAS is the Contractor's certification that the submitted schedule meets the requirements of the Contract Documents and represents the Contractor's plan on how the work will be accomplished. Provide all items listed in paragraph REQUIRED TABULAR REPORTS AND NATIVE P6 XER FILES with baseline NAS submittal.

1.3 THREE-WEEK LOOK AHEAD SCHEDULE

1.3.1 Weekly CQC Coordination and Production Meeting

Submit the first 3-week look ahead schedule 30 calendar days after contract award and continue to submit weekly until contract completion.

Transmit the Weekly Look Ahead Schedule Submittal to the Contracting Officer on the first day of the current week that the look ahead schedule covers and at least one working day prior to Quality Control meetings or as agreed to by Contracting Officer. Contractor is required to provide all attendees at meetings that require discussion of the schedule.

1.3.2 Weekly Look Ahead Schedule Requirements

Prepare and issue a 3-Week Look Ahead schedule to provide a more detailed day-to-day plan of upcoming work identified on the Project Network Analysis Schedule. Requirements include:

a. For each Look Ahead schedule activity, identify parent Baseline NAS activity number(s). The parent NAS activity is the activity in the Baseline NAS that would incorporate the Look Ahead schedule activity requirement and/or scope of work.

   (1) Contractor must provide color coding identifying the following:

      (a) Green: Activities that are being performed per the dates depicted in the Baseline NAS.

      (b) Red: Activities being performed after/longer than that shown in the Baseline NAS. For these activities, include a comments column to the right briefly stating the reason for the delay.

      (c) Yellow: 3 weeks look ahead activities that have slipped since the previous week. For these activities, include a comments column to the right briefly stating the reason for the delay.

b. Update schedule each week to show the planned work for the current and following two-week period. Also include previous week, as-built work, showing actual start and finish dates. Provide a copy of updated schedule a minimum of 2 working days prior to weekly QC meeting. See 01 45 00.00 22 Quality Control (PWD ME) paragraph QUALITY CONTROL WEEKLY MEETINGS for additional information.

c. Include upcoming outages, closures, preparatory meetings, and initial meetings, testing, and Special Inspections including and required Government QA inspections.

d. Clearly identify longest path activities on the Weekly Look Ahead Schedule. Include a key or legend that distinguishes longest path activities. Include all Longest Path activity NAS start/finish dates exceeded and/or occurring during this period.

e. Identify responsibility for each activity.

f. The detail work plans are to be bar chart type schedules, derived from but maintained separately from the Baseline NAS on an electronic spreadsheet program and printed on 11 by 17 inch sheets as directed by the Contracting Officer.

g. Activities must not exceed 5 working days in duration and have sufficient level of detail to assign crews, tools and equipment required to complete the work.

h. Lookahead activities are not limited to only onsite work. Include but not limited to submittal, offsite fabrication, and procurement activities if needed to ensure longest path is maintained. Include design activities and coordination when part of scope.

i. Upload Lookahead to eCMS as a submittal by the date and time required by Contracting Officer.

1.4 MONTHLY NETWORK ANALYSIS

Submittal of Monthly NAS is the Contractor's certification that the submitted schedule meets the requirements of the Contract Documents and represents the Contractor's plan on how the work will be accomplished. Provide all items listed in paragraph REQUIRED TABULAR REPORTS AND NATIVE P6 XER FILES with the monthly NAS submittal.

1.4.1 Monthly Network Analysis Updates

a. Regardless of whether an invoice is being submitted monthly, or no progress in the schedule occurred, an updated schedule (or with new data date) must be submitted monthly to the Government. The Monthly NAS update must be submitted within 10 calendar days of the data date.

b. Regardless of whether a baseline or other schedule is submitted and/or accepted, an updated schedule must be submitted monthly to the Government. Monthly updates submitted before the baseline being accepted will be considered as (interim and for information only) submittal.

c. Provide all items listed in paragraph REQUIRED TABULAR REPORTS AND NATIVE P6 XER FILES, with each monthly NAS update submittal.

d. Meet with Government representative(s) at monthly intervals to review and agree on the information presented in the updated project schedule. The submission of an accepted, updated schedule to the Government is a condition precedent to the processing of the Contractor's invoice.

e. Activity progress must incorporate as-built events as they occurred and correspond to records including but not limited to submittals and daily production and quality control reports.

f. Software Settings: Handle schedule calculations and Out-of-Sequence progress (if applicable) through Retained Logic, not Progress Override. Show all activity durations and float values in days. Show activity progress using Remaining Duration. Set default activity type to "Task Dependent".

g. Update schedule must reflect current Contract Completion Date and contract value in accordance with all conformed contract modifications issued prior to data date of NAS update.

1.4.2 As-Built Schedule

As a condition precedent to the release of retention and making final payment, submit an "As-Built Schedule," as the last schedule update showing all activities at 100 percent completion. This schedule must reflect the exact manner in which the project was actually constructed. Refer to 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME) paragraph AS-BUILT DRAWINGS and 01 20 00.00 22 PRICE AND PAYMENT PROCEDURES (PWD ME) paragraph CONTENT OF INVOICE for additional information.

1.4.3 Outage Schedule

The Contractor must provide a comprehensive Outage Schedule prior to the Pre-Construction meeting that includes all required Outages (Road, Rail, Utility, etc.) required to complete the work. The Schedule must include the dates the outage request packages will be submitted and must include a 15 calendar day Government Concurrence period. See Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME) for Outage Requirements. The Contractor must attend the Outage Coordination Meeting specified in Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME) to ensure the Outage Package requirements are understood and to review the Outage Schedule with the Contracting Officer's Representatives. Refer to 01 35 26.00 22 GOVERNMENT SAFETY REQUIREMENTS (PWD ME) paragraph UTILITY OUTAGE REQUIREMENTS and 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME) paragraph OUTAGE COORDINATION MEETING for additional information.

1.5 CORRESPONDENCE AND TEST REPORTS

Reference Schedule activity IDs that are being addressed in each correspondence (e.g., letters, Requests for Information (RFIs), e-mails, meeting minute items, Production and QC Daily Reports, material delivery tickets, photographs) and test report (e.g., concrete, soil compaction, weld, pressure).

1.6 ADDITIONAL SCHEDULING REQUIREMENTS

Other specification sections may include additional scheduling requirements, including systems to be inspected, tested and commissioned, and submittal procedures. Those schedule requirements must be incorporated into the Baseline NAS.

1.7 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

Baseline NAS; G

Designated Project Scheduler; G

Three-Week Look Ahead Schedule; G

Outage Schedule; G

SD-07 Certificates

Weekly Look Ahead Schedule

Monthly Network Analysis Updates; G

SD-11 Closeout Submittals

As-Built Schedule; G

1.8 SOFTWARE

Prepare and maintain project schedules using Primavera P6 software in a version compatible with Government's current version. Importing data into P6 using data conversion techniques or third party software is cause for rejection of the submitted schedule. Schedules with Performing Organizational Breakdown Structure (POBS) data is cause for rejection.

1.9 DESIGNATED PROJECT SCHEDULER

Within 30 calendar days of contract award, submit to the Contracting Officer for approval an individual who will serve as the Designated Project Scheduler. Include a copy of the candidate's resume with qualifications. The Contracting Officer may remove the Designated Project Scheduler, and require replacement, if the scheduler does not effectively fulfill their duties in accordance with the contract requirements. Payment request will not be processed without an approved Designated Project Scheduler.

1.9.1 Qualifications

The Designated Project Scheduler must have prepared and maintained at least three previous construction schedules, of similar size and complexity to this contract, using Primavera P6.

1.9.2 Duties

Duties of the Designated Project Scheduler include to be actively involved and be responsible for the following:

a. Prepare Baseline NAS.

b. Prepare monthly schedule updates.

c. Prepare tabular reports.

d. Prepare Time Impact Analysis (TIA) as necessary.

e. Provide certification that NAS and TIA submittals conform to the contract requirements.

f. Participate with the Prime Contractor and Government Representative in a monthly teleconference call,meeting at the job site in-person, and scheduled with sufficient time to support the Monthly Network Analysis Updates process, to discuss project status, schedule updates, critical activities, potential delays, and contract modifications impacting the schedule. Have a computer with P6 software available during the meeting.

1.10 NETWORK SYSTEM FORMAT

Prepare the schedule in accordance with the following Primavera P6 settings and parameters. Deviation from these settings and parameters, without prior consent of the Contracting Officer, is cause for rejection of schedule submission.

1.10.1 Schedule Activity Properties and Level of Detail

1.10.1.1 Activity Identification and Organization

a. Identify construction activities planned for the project and other activities that could impact project completion if delayed in the NAS.

b. Each activity must have a unique name.

c. Identify administrative type activity/milestones, including all pre-construction submittal and permit requirements prior to demolition or construction stage.

d. Include times for procurement, Contractor quality control and construction, acceptance testing and training in the schedule.

e. Include the Government approval time required for the submittals that require Government Approval prior to construction, as indicated in Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME).

f. Create separate activities for each Phase, Area, Floor Level and Location the activity is occurring.

g. Do not use construction category activity to represent non-work type reference (e.g., Serial Letter, Request for Information) in NAS. Place Non-work reference within the P6 activity details notebook.

Activity categories included in the schedule are specified below.

1.10.1.2 Activity Logic

a. With the exception of the Contract Award and Contract Completion Date (CCD) milestone activities, activities must not be open-ended; each activity must have at least one predecessor and at least one successor.

b. Activities must not have open start or open finish (dangling) logic.

c. Do not use lead or lag logic without Contracting Officer prior approval.

d. Minimize redundant logic ties.

e. Once an activity exists on the schedule it must not be deleted or renamed to change the scope of the activity and must not be removed from the schedule logic without approval from the Contracting Officer.

(1) While an activity cannot be deleted, where said activity is no longer applicable to the schedule, but must remain within the logic stream for historical record, change the activity original and remaining duration to zero and clearly label "(NO LONGER REQUIRED)" after the activity name. Actual finish date for activity that falls behind the data date. Redistribute

accordingly any remaining budget associated with that activity, to other remaining appropriate activity.

(2) Document any such change in the activities' "Notebook," including a date and explanation for the change.

(3) The ID number for a "NO LONGER REQUIRED" activity must not be re-used for another activity.

1.10.1.3 Longest Path Activity Baseline Limitation

For P6 settings, critical activities are defined as being on the Longest Path. Longest Path (Critical) Activities must not make up more than 30 percent of all activity within the Construction Baseline Schedule.

1.10.1.4 Assigned Calendars

All NAS activity must be assigned calendars that reflect required and anticipated non-work days.

1.10.1.5 Activity Categories

Activities must align with installation workdays/holidays, etc. Coordinate with Contracting Officer for installation schedule.

1.10.1.5.1 Pre-construction Activities

Examples of pre-construction activities include, but are not limited to, bond approval, permits, pre-construction submittals and approvals. Include pre-construction activities that are required to be completed prior to the Contractor starting the demolition or construction stage of work.

1.10.1.5.2 Procurement Activities

Examples of procurement activities include, but are not limited to: Material/equipment submittal preparation, submittal and approval of material/equipment; material/equipment fabrication and delivery, and material/equipment on-site. As a minimum, separate procurement activities must be provided for critical items, long lead items, items requiring Government approval and material/equipment procurement for which payment will be requested in advance of installation. Show each delivery with relationship tie to the Construction Activity specifically for the delivery.

1.10.1.5.3 Government Activities

Government and other agency activities that could impact progress must be clearly identified. Government activities include, but are not limited to; Government approved submittal reviews, Government conducted inspections/tests, environmental permit approvals by State regulators, utility outages, installation drills and scheduled boat/ship activities, and delivery of Government Furnished Material/Equipment.

The Government will provide a list of QA inspections and testing at the pre-construction meeting. These inspections may vary based on the Contractor's performance and are subject to change throughout the duration of the project.

(Note: These inspections are to be performed by the Government's QA Team and are separate from any special inspections required to be performed by the Contractor as part of the Contract.)

The Contractor must include the Government QA Inspection items in the Baseline NAS and highlighted in the 3-Week Look Ahead schedules. The Contractor's QC Manager must coordinate these inspections with the Contracting Officer's Representative to ensure the inspections do not cause any construction delays. Any deficiencies or non compliant work must be corrected immediately and any delays related to rework is the Contractor's responsibility. Any requests for time or costs related to this rework will be rejected.

1.10.1.5.4 Construction Quality Management (CQM) Activities

The Preparatory and Initial Phase meetings for each Definable Feature of Work identified in the Contractor's Quality Control Plan must be included in the Three-Week Look Ahead Schedule. Preparatory and Initial phase meetings are not required in the NAS, but can be represented by a start milestone linked to successor parent Construction Activity. The Follow-up Phase must be represented by the Construction Activities themselves in the NAS.

1.10.1.5.5 Construction Activities

On-site construction activities that are the responsibility of the Contractor or Contractor's subcontractors must not have a duration in excess of 20 working days without prior approval by the Contracting Officer. Contractor activities must be driven by calendars that reflect Saturdays, Sundays and all Federal Holidays as non-work days, unless otherwise defined in this contract. Federal Holidays are as defined in 5 USC 6103.

1.10.1.5.6 Turnover and Closeout Activities

Include activities or milestones for items on the NAVFAC Red Zone Checklist/POAM that are applicable to this project. As a minimum, include required Contractor testing, required Government acceptance inspections on equipment, Pre-Final Inspection, Punch List Completion, Final Inspection and Acceptance. Add an unconstrained start milestone for the initial NAVFAC Red Zone - Facility Turnover Planning Meeting at approximately 75 percent construction contract completion or six months prior to Contract Completion Date (CCD), whichever is sooner. Include a separate NAVFAC Red Zone - Facility Turnover Planning Meeting Milestone for each phase if the contract requires construction to be completed in phases.

1.10.1.5.7 Testing of HVAC - DALT, TAB, and PVT Activities

Include in the baseline schedule, activities and milestones associated with Government acceptance of Duct Air Leakage Test (DALT), Testing, Adjusting, and Balancing (TAB) and Performance Verification Test (PVT) as required and in accordance with Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC and Section 23 09 00.00 22 INSTRUMENTATION AND CONTROL FOR HVAC.

a. Identify the general area or location(s) for Government Acceptance Testing of DALT, TAB and PVT.

b. Incorporate into the baseline schedule, time periods required for advance notification of work, and Government submittal review in accordance with Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC, paragraph DALT AND TAB SUBMITTAL AND WORK SCHEDULE.

c. Include the following as schedule activities or milestones:

(1) Pre-DALT/TAB/PVT Meeting

(2) TAB Design Review Report, Government review

(3) TAB Pre-Field Engineering Report, Government review

(4) DALT Field Work

(5) DALT Field Acceptance Testing

(6) Certified Final DALT Report, Government review

(7) Control Contractors PVT Plan, Government review

(8) Equipment Suppliers PVT Plan, Government review

(9) Season I TAB Field Work

(10) Season I Certified Final TAB Report, Government review

(11) Endurance Testing, Government review

(12) PVT Field Work

(13) PVT Report, Government review

(14) Season I TAB Field Acceptance Testing

(15) Season II TAB Field Work

(16) Season II Certified Final TAB Report, Government review

(17) Season II TAB Field Acceptance Testing

(18) Post-Occupancy Endurance Testing Government review

(19) Post-Occupancy PVT Field Work

1.10.1.5.8 Commissioning Activities

Include in the baseline schedule activities and milestones associated with Commissioning.

a. Identify the general area or location(s) of systems for Commissioning Inspection and Testing

b. Incorporate into the baseline schedule time periods for Government submittal review

1.10.1.6 Contract Milestones and Constraints

1.10.1.6.1 Project Start Date Milestones

Include as the first activity on the schedule a start milestone titled, "Contract Award", which must have a Mandatory Start constraint equal to the Contract Award Date.

1.10.1.6.2 Pre-Construction Meeting Milestone

Include an unconstrained finish milestone on the schedule titled, "Pre-Construction Meeting". The Pre-Construction meeting may be a single day, or it may range over several days. The intent to cover all the Pre-Con topics, including Partnering.

1.10.1.6.3 Preconstruction Submittals Finish Milestone

Include an unconstrained finish milestone on the schedule titled, "Preconstruction Submittals". This milestone is complete when all required preconstruction submittals have been reviewed and approved by the Government.

1.10.1.6.4 Contractor Mobilization Finish Milestone

Include an unconstrained finish milestone on the schedule titled, "Contractor Mobilization".

1.10.1.6.5 NAVFAC Red Zone - Facility Turnover Planning Meeting Milestones

See paragraph TURNOVER AND CLOSEOUT ACTIVITIES above.

1.10.1.6.6 Substantial Completion Milestone

Include an unconstrained finish milestone on the schedule titled "Substantial Completion." Substantial Completion is defined as the point in time the Government would consider the project ready for beneficial occupancy wherein by mutual agreement of the Government and Contractor, Government use of the facility is allowed while construction access continues in order to complete remaining items (e.g., punch list and other close out submittals). Include a separate Substantial Completion Milestone for each phase if the contract requires construction to be completed in phases.

1.10.1.6.7 Projected Completion Milestone

Include an unconstrained finish milestone on the schedule titled "Projected Completion." Projected Completion is defined as the point in time all contract requirements are complete and verified by the Government with a successful Final Inspection in accordance with Section 01 45 00.00 22 QUALITY CONTROL. This milestone must have the Contract Completion Date (CCD) milestone as its only successor.

1.10.1.6.8 Contract Completion Date (CCD) Milestone

Last schedule entry must be an unconstrained finish milestone titled "Contract Completion (CCD: DD-MM-YY)." DD-MM-YYYY is the current contract completion date at data date, day-month-year corresponding to P6 Must Finish By Date. Baseline NAS milestone updates of Project Completion finish date for longest path must reflect calculated float as positive or negative based on CCD. Calculation of schedule updates must be such that if the finish of the "Projected Completion" milestone falls after the contract completion date, then negative float is calculated on the longest path. If the finish of the "Projected Completion" milestone falls before the contract completion date, the float calculation must reflect positive float on the longest path.

1.10.1.6.9 Additional Milestones

Provide up to 5 additional milestones as required by Contracting Officer.

1.10.1.7 Work Breakdown Structure & Activity Code

At a minimum, establish a Work Breakdown Structure (WBS) and provide activity codes identified as follows:

1.10.1.7.1 Work Breakdown Structure (WBS)

Group all activities and milestones within appropriate WBS elements/levels including, at a minimum, the following:

a. Project Milestones:

   (1) Management Milestones

   (2) Project Administrative Meetings

   (3) Permits

b. Pre-Construction Phase:

   (1) Submittals and Reviews

   (2) Procurement

   (3) Mobilization

c. Construction Phase: Create multiple elements/levels in accordance with project specific definable features of work including in WBS descending order as follows:

   (1) General Area

      (a) Type of Work Item

         1. Location

d. Project Closeout: Include activity items such as, but not limited to, Punchlist, Demobilization, O&M, As-built Drawings, Training, and As-built NAS.

e. Modifications: Create Conformed and Non-Conformed elements/levels under Modification WBS. Create multiple elements/levels as the project progresses identified by issue and Fragnet placed in Conformed for modifications issued prior data date, or Non-Conformed for issues not modified to contract prior data date.

f. Removed Activity: Activity is "removed" by remaining within logic sequence, eliminating duration and adding "(NO LONGER REQUIRED)" after

Activity Name in Activity Table.

g. Commissioning & Testing:

(1) Specific area/locations of commissioning

(2) Final Testing

(3) Training

1.10.1.7.2 Responsibility Code

Use "RESP" for Activity Code Name. All activities in the Baseline NAS must be identified with the resource for completing the task. Activities must not belong to more than one responsible party.

1.10.1.7.3 Activity Category Code

Provide user defined "CAT" codes for Project Level activity codes. Use the following codes:

a. Assign "PROC" value to Procurement type activity

b. Assign "PRE-CON" value to Pre-construction activity

c. Assign "CONS" value to Construction type activity

d. Assign "TEST" value to dedicated testing type activities

e. Assign "CX" value to dedicated Commissioning type activities

f. Assign "CLOS" value to dedicated Close Out type activity

g. Assign "OTHR" to other activity not otherwise designated

1.10.1.7.4 Drawing Code

Identify all activities in the Baseline NAS with its respective Drawing Code. The Drawing Code is the Sheet Number on the primary project drawing which indicates work to be performed. If an activity does not have an applicable Drawing Code (e.g., Mobilize), the code must be "0000".

1.10.1.7.5 Additional Activity Codes

Provide up to an additional five activity codes as required by the Contracting Officer

1.10.1.8 Adverse Weather Lost Work Days

Use the National Oceanic and Atmospheric Administration's (NOAA) Summary of Monthly Normals report to obtain the historical average number of days each month with precipitation, using a nominal 30-year, greater than 0.10 inch precipitation amount parameter, as indicated on the Station Report for the NOAA location closest to the project site as the basis for establishing a "Weather Calendar" showing the number of anticipated non-workdays for each month due to adverse weather, in addition to Saturdays, Sundays and all Federal Holidays as non-work days.

Assign the Weather Calendar to any activity that could be impacted by adverse weather. The Contracting Officer will issue a modification in accordance with the contract clauses, giving the Contractor a time only extension for the difference of days between the anticipated and actual adverse weather delay if the number of actual adverse weather delay days exceeds the number of days anticipated for the month in which the delay occurs and the adverse weather delayed activities are on the longest path to contract completion in the period when delay occurred. A lost workday due to weather conditions is defined as a day in which the Contractor cannot work at least 50 percent of the day on the impacted activity. Impacts resulting from adverse weather must be documented in Narrative Report for the month that it occurred. Weather days incurred must be noted on the daily contractor quality report identifying the specific activity id(s) from the Baseline NAS that were impacted. This same information must be included with the monthly NAS submission.

Make changes to P6 project calendars to reflect as-built conditions where work occurred where originally anticipated as non-work days, and where work did not occur (lost work day).

Weather day impacts incurred as a result of schedule delays not caused by the government, must not be applied against the weather days owned by the contract under this specification section and must be solely bourn/recovered by the contractor.

1.10.1.9 Anticipated Restricted Delays

Unless otherwise noted or defined in Section 01 14 00.00 22 WORK RESTRICTIONS, allow in the schedule ten (10) lost workdays per calendar year for instances where base access is not permitted or where work areas are temporarily not accessible for security reasons which causes a delay in the work. Use Anticipated Restricted Delays as basis for establishing a "Security Calendar" showing the number of anticipated non-workdays for each month due to anticipated restrictions, in addition to anticipated adverse weather, Saturdays, Sundays and all Federal Holidays as non-work days. Assign the Security Calendar to any activity that could be impacted by restriction delays. The Contracting Officer will issue a modification in accordance with the contract clauses, giving the Contractor a time extension for the difference of days between the anticipated and actual lost work days if the number of actual restriction delay days exceeds ten (10) days per calendar year. A lost workday due to restriction delay is defined as a day in which the Contractor cannot work at least 50 percent of the day on the impacted activity. Restricted days incurred must be noted on the daily contractor quality report identifying the specific activity id(s) from the Baseline NAS that were impacted. This same information must be included with the monthly NAS submission.

Impacts resulting from restriction delays must be documented in Narrative Report for the month that it occurred.

Make changes to P6 project calendars to reflect as-built conditions where work occurred where originally anticipated as non-work days, and where work did not occur (lost work day).

Restricted day impacts incurred as a result of schedule delays not caused by the government, must not be applied against the restricted days owned by the contract under this specification section and must be solely bourn/recovered by the contractor.

1.10.1.10 Cost Loading

The Baseline Network Analysis Schedule (NAS) must be cost-loaded and will provide the basis for progress payments. Earned Value Reports must be derived from and correspond to cost loaded NAS. Use the Critical Path Method (CPM) and the Precedence Diagram Method (PDM) to satisfy time and cost applications.

Sum of all costs assigned to activities must equal the current contract value as of the data date of the baseline or update schedule. Activity cost loading must be without front-end loading. Provide additional documentation if requested by the Contracting Officer.

1.10.1.10.1 Cost Loading Activities

a. Cost load construction activities. Define construction activities as on-site work activities that can be traced directly to the construction, alteration, demolition, repair of a project facility, infrastructure, or sitework.

b. Assign costs for materials which payment will be requested in advance of installation, including their quantities, to their respective procurement activities that were approved by the Contracting Officer. Assign labor costs, including their quantities, for material and equipment paid for after installation to their respective construction activities.

c. Costs for mobilization will not be paid as individual pay items with the exception of batch plant set-up, mobilization of dredging equipment or other similar labor-intensive situations. If payment for mobilization costs are approved, assign costs for demobilization activities. The value of demobilization activities must not be less than 40 percent of the total costs for mobilization/demobilization activities. If not approved as a separate line item, evenly disperse over construction activities the costs for mobilization and/or demobilization.

d. The value of commissioning, testing, and closeout WBS section may not be less than 10 percent of the total costs for procurement and construction activities.

e. ALL activities assigned Government responsibility will have Zero Cost. No contractor cost should be assigned to an activity designated as a Government responsibility.

f. Do not include field overhead positions as individual cost loaded activities. Evenly disperse indirect costs (e.g., meetings, presentations, submittals, surveys, haul road or storage yard maintenance, dust control, security) and profit to construction activities assigned to the Construction Phase WBS level.

1.10.1.10.2 Partial Payment

Breakdown unit of measure and cost must be defined within P6 Activity Detail Expenses for partial payment of any cost loaded activity. Lump sum cost loaded activity will not be partially paid.

1.10.2 Schedule Software Settings and Restrictions

a. Activity Constraints: Date/time constraint(s), other than those required by the contract, are not allowed unless accepted by the Contracting Officer. Identify any constraints proposed and provide an explanation for the purpose of the constraint in the Narrative Report as described in paragraph REQUIRED TABULAR REPORTS.

b. Default Progress Data Disallowed: Actual Start is date work begins on activity with intent to pursue work to substantial completion. Actual Finish is date work is substantially complete to point where successor activity can begin. Actual dates on the CPM schedule must correspond with activity dates reported on the Contractor Quality Control and Production Reports.

c. At a minimum, include the following settings and parameters in P6 Schedule preparation:

(1) General: Define or establish Calendars and Activity Codes at the "Project" level, not the "Global" level.

(2) Admin Drop-Down Menu, Admin Preferences, Time Periods Tab:

(a) Set time periods for P6 to 8.0 Hours/Day, 40.0 Hours/Week, 172.0 Hours/Month and 2000.0 Hours/Year.

(b) Use assigned calendar to specify the number of work hours for each time period: Must be checked.

(3) Admin Drop-Down Menu, Admin Preferences, Earned Value Tab:

(a) Technique for computing performance percent complete: Use "Activity percent Complete"

(b) Technique for computing Estimate to Complete: Use "PF = 1".

(c) Earned Value Calculation: Use "Budgeted values with current dates".

(4) Project Level, Dates Tab:

(a) Set "Must Finish By" date to "Contract Completion Date", and set "Must Finish By" time to 05:00pm.

(5) Project Level, Defaults Tab:

(a) Duration Type: Set to "Fixed Duration & Units".

(b) Percent Complete Type: Set to "Physical".

(c) Activity Type: Set to "Task Dependent".

(d) Calendar: Set to "Standard 5 Day Workweek". Calendar must reflect Saturday, Sunday and all Federal holidays as non-work days. Alternative calendars may be used with Contracting Officer approval.

(6) Project Level, Calculations Tab:

(a) Default Price/Unit for activities without resource or role Price/Units: Set to "$1/h".

(b) Activity percent complete based on activity steps: Must be Checked.

(c) Link Budget and At Completion for not started activities: Must be Checked.

(d) Reset Remaining Duration and Units to Original: Must be Selected.

(e) Subtract Actual from At Completion: Must be Selected.

(f) Recalculate Actual units and Cost when duration percent complete changes: Must be Checked.

(g) Update units when costs change on resource assignments: Must be Unchecked.

(h) Link Actual to Date and Actual This Period Units and Cost: Must be Checked.

(7) Project Level, Settings Tab:

(a) Define Critical Activities: Check "Longest Path".

(8) Work Breakdown Structure Level, Earned Value Tab:

(a) Technique for Computing Performance Percent Complete: "Activity percent complete" is selected.

(b) Technique for Computing Estimate to Complete (ETC): "PF = 1" is selected.

1.10.3 Required Tabular Reports and Native P6 XER Files

Include the following reports with the Baseline, Monthly Update and any other required schedule submittals:

a. Time Scaled Logic Schedule

Provide formatted 11 by 17-inch Time-scaled Logic Schedule in color and landscape-oriented with each schedule submittal. Clearly show activities on the longest path setting Gantt chart longest path activity bars to red. Group activities by WBS and sort by finish date in ascending order. Include the following information in column form for each activity and include accompanying Gantt chart:

(1) Activity ID

(2) Activity Name

(3) Original Duration

(4) Remaining duration

(5) Physical Percent Complete

(6) Start Date

(7) Finish Date

(8) Total Float

(9) Project-level Calendar

b. Previous Monthly Update Comparison Time Scaled Logic Schedule (Submit with all Monthly Update Schedule Submittals.)

Provide formatted 11 by 17-inch Time-scaled Logic Schedule in color and landscape-oriented with each monthly update schedule submittal. Clearly show activities on the current month longest path setting Gantt chart longest path activities bars to red. Show previous month activities as yellow bars and previous month milestones in yellow within Gantt chart. Sort by finish date in ascending order. Filter activities for longest path. Maintain and assign the accepted previous month update or the accepted baseline schedule for the first update submittal as the baseline and primary baseline in P6 before printing the schedule. Include the following information in column form for each activity and include accompanying Gantt chart:

(1) Activity ID

(2) Activity Name

(3) Original Duration

(4) Current Month Remaining Duration

(5) Current Month Start Date

(6) Previous Month Update Start Date (BL Project Start)

(7) Start Date Delta between Current Month and Previous Month (Variance - BL Project Start Date)

(8) Current Month Finish Date

(9) Previous Month Finish Date (BL Project Finish)

(10) Finish Date Delta between Current Month and Previous Month (Variance - BL Project Start Date)

(11) Current Month Total Float

c. P6 native XER file: Include the back-up native .xer program file compatible with the current Government version of Primavera P6. Each native schedule file must have a unique file name to include project name and data date using (yyyy-mm-dd) convention. Each native schedule must have a unique Project ID and Project Name.

d. Log Report: P6 Scheduling/Leveling Report.

e. Narrative Report: Identify and justify:

(1) Provide Project Summary Data in format below:

(a) Data Date ______

(b) Award Date: _______

(c) Original Project Duration: _____ days post Award Date

(d) Current Project Duration: ______ days post Award Date

(e) Time percent elapsed: ______ percent at data date

(f) Original CCD: _______

(g) Current CCD: _______ (thru MOD _______)

(h) Anticipated CCD: _____ (___ calendar days early/late)

(i) Original Contract Value: $____________

(j) Current Contract Value: $_________

(k) Invoiced Amount: $_________ (____ percent)

(l) Cost Growth: ____ percent

(m) Schedule Growth: ____ percent

(n) There are a total of ______ activities, ______ activities complete (____ percent), ____ activities in progress (____ percent), _____ activities not started (____ percent). Of the in progress and not started activities; ____ (____ percent) are on the longest path. The longest path has duration of _____ calendar days from data-date to anticipated project completion.

(2) Progress made in each area of the project;

(3) Longest Path;

(4) Date/time constraint(s), other than those required by the contract

(5) Listing of all changes made between the previous schedule and current updated schedule include: added or deleted activities, original and remaining durations for activities that have not started, logic (sequence constraint lag/lead), milestones, planned sequence of operations, longest path, calendars or calendar assignments, and cost loading;

(6) Any decrease in previously reported activity Earned Amount;

(7) Pending items and status thereof, including permits, changes orders, and time extensions;

(8) Status of Contract Completion Date and interim milestones;

(9) Status of Projected Completion Milestone and account of difference in calendar days between previous update Projected Completion Milestone

(10) Current and anticipated delays listing Activity Names and IDs for impacted activities(describe cause of delay and corrective

actions(s) and mitigation measures to minimize);

(11) Description of current and potential future schedule problem areas.

(12) Identification of any weather and restricted lost time as compared to anticipated weather for the month and anticipated restricted days for which the update is submitted. Impacts resulting from adverse weather must be documented in tabular form showing the calendar month (or billing period) with the days on which construction activity incurred Lost Work Days due to adverse weather. In narrative form, describe the adverse weather cause such as precipitation measurement, temperature, wind or other influencing factors, and why work was impacted. Describe the construction activity(s) that was (were) scheduled, impacted.

(13) Explain in schedule narratives the reason for any critical path delay from the previous update period. Identify any activities that drove critical path delay by their activity ID(s) and names, with an explanation as to why.

Each entry in the narrative report must cite the respective Activity ID and Activity Name, the date and reason for the change, and description of the change.

f. Earned Value Report: Derive from and correspond to P6 cost loaded schedule. List all activities having a budget amount cost loaded. Compile total earnings on the project from notice to proceed to current progress payment request. Show current budget, previous physical percent complete, to-date physical percent complete, previous earned value, to-date earned value, cost this period and cost to complete on the report for each activity.

g. Schedule Variance Control (SVC) Diagram: With each schedule submission, provide a SVC diagram showing 1) A Cash Flow Curve indicating planned project cost based on each of projected early and projected late activity finish dates and 2) one curve for Earned Value to-date. Revise Cash Flow Curves when the contract is modified, or as directed by the Contracting Officer Include a legend on report clearly indication 3 curves: early finish, late finish, and earned-value to date.

Use the following settings in Activity Usage Profile Options:

(1) In the Data section, under Display, the radio box for Cost must be selected.

(2) In the Data section, under Filter for Bars/Graphs, the checkbox for Total must be checked.

(3) In the Show Bars/Curves section:

(a) Under the By Date column, the checkboxes for Baseline, Actual and Remaining Late must be checked. The checkboxes for Budgeted and Remaining Early must be unchecked.

(b) Under the Cumulative column, the checkboxes for Baseline, Actual and Remaining Late must be checked. The checkboxes for Budgeted and Remaining Early must be unchecked.

(c) Set the color for Baseline to green.

(d) Set the color for Actual to blue.

(e) Set the color for Remaining Late to red.

(4) In the Show Earned Value Curves section, the checkboxes for Planned Value Cost, Earned Value Cost and Estimate at Completion must be unchecked.

h. Logic Diagram showing timescale from data date to 60 days after data date with filter for longest path. Leave Group By selection blank and sort by finish date in ascending order.

i. Baseline or Monthly Update Checklist as applicable completed and certified by Qualified Scheduler. Baseline NAS and Monthly Update Schedule Checklists can be found on the Whole Building Design Guide website at https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-32-17-00-20

j. Screen shot PDF of Primavera P6 Time Periods Settings referenced in paragraph SCHEDULE SOFTWARE SETTINGS AND RESTRICTIONS, list item c.(2): ADMIN DROP-DOWN MENU, ADMIN PREFERENCES, TIME PERIODS TAB

k. Screen shot PDF of Primavera P6 Earned Value Settings referenced in paragraph SCHEDULE SOFTWARE SETTINGS AND RESTRICTIONS, list item c.(3): ADMIN DROP-DOWN MENU, ADMIN PREFERENCES, EARNED VALUE TAB

l. Daily Reported Production Activity: Submit on a monthly basis, in electronic spreadsheet (format provided by the Government), summary of daily reported production activity for the reporting month in the update schedule. Use the following columns for reporting:

(1) Date

(2) Activity ID

(3) Work Description

(4) Contractor

(5) Billable Hours

m. HVAC/COMMISSIONING KTR Checklist. See Section 23 09 00.00 22 INSTRUMENTATION AND CONTROL FOR HVAC (PWD-MAINE)(J&A) and Section 23 09 13.00 22 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC for checklist requirements. Complete checklist listing all items with baseline submittal. Complete checklist as required to complete project, submitting complete checklist in intermediately following monthly update submittal.

1.11 CONTRACT MODIFICATION

1.11.1 Time Impact Analysis (TIA)

Submit a Time Impact Analysis with each cost and time proposal for a proposed change. TIA must illustrate the influence of each change or delay on the Contract Completion Date or milestones. No time extensions will be granted nor delay damages paid unless a delay occurs which consumes all available Project Float, impacts the longest path, and extends the Projected Completion beyond the Contract Completion Date.

a. Each TIA must be in both narrative and schedule form. The narrative must:

1. Define the scope and conditions of the change;

2. Provide start and finish dates of impact;

3. Successor(s) and predecessor(s) activities to impact period;

4. Responsible party;

5. Describe how the impact originated;

6. How it impacts the schedule's longest path.

b. The schedule submission must consist of three native XER files:

(1) Fragnet used to define the scope of the changed condition

(2) Most recent accepted schedule update as of the time of the impact start date. Update this schedule to show all activity progress as of the time of the impact start date. The impact start date is identified as the time when an existing activity is impeded for either starting or finishing.

(3) The impacted schedule that has the Fragnet inserted in the updated schedule and the schedule "run" so that the new completion date is determined.

c. For claimed as-built project delay, the inserted Fragnet TIA method must be modified to account for as-built events known to occur after the data date of schedule update used. Updated schedules for periods following the impact start date will be used to evaluate how the project progressed (as-built) through the finish of impact. Impact to longest path must be determined for each following update period.

d. All TIAs must include any mitigation, and must determine the apportionment of the overall delay assignable to each individual delay. Apportionment must provide identification of delay type and classification of delay by compensable and non-compensable events. The associated narrative must clearly describe analysis methodology used, and the findings in a chronological listing beginning with the earliest delay event.

(1) Identify and classify types of delay defined as follows:

(a) Force majeure delay (e.g., weather delay): Any delay event caused by something or someone other than the Government or the Contractor, or the risk of which has not been assigned solely to the Government or the Contractor. If the force majeure delay is on the longest path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, classified as a non-compensable event.

(b) A Contractor-delay: Any delay event caused by the Contractor, or the risk of which has been assigned solely to the Contractor. If the contractor-delay is on the longest path, in absence of other types of concurrent delays, Contractor is not granted extension of contract time, and classified as a non-compensable event. Where absent other types of delays, and having impact to project completion, Contractor must provide to Contracting Officer a Corrective Action Plan identifying plan to mitigate delay.

(c) A Government-delay: Any delay event caused by the Government, or the risk of which has been assigned solely to the Government. If the Government-delay is on the longest path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, and classified as a compensable event.

(2) Functional concurrency must be used to analyze concurrent delays, where: separate delay issues delay project completion, do not necessarily occur at same time, rather occur within same monthly schedule update period at minimum, or within same as-built period under review. If a combination of functionally concurrent delay types occurs, it is considered Concurrent Delay, which is defined in the following combinations:

(a) Government-delay concurrent with contractor-delay: excusable time extension, classified non-compensable event.

(b) Government-delay concurrent with force majeure delay: excusable time extension, classified non-compensable event.

(c) Contractor-delay concurrent with force majeure delay: excusable time extension, classified non-compensable event.

(3) Pacing delay reacting to another delay (parent delay) equally or more critical than paced activity must be identified prior to pacing. Contracting Officer will notify Contractor prior to pacing. Contractor must notify Contracting Officer prior to pacing. Notification must include identification of parent delay issue, estimated parent delay time period, paced activity(s) identity, and pacing reason(s). Pacing Concurrency is defined as follows:

(a) Government-delay concurrent with contractor-pacing: excusable time extension, classified compensable event.

(b) Contractor-delay concurrent with Government-pacing: inexcusable time extension, classified non-compensable event

e. Submit electronic file containing the narrative and the source schedule files used in the time impact analysis.

1.12 PROJECT FLOAT

Project Float is the length of time between the Contractor's Projected Completion Milestone and the Contract Completion Date. Project Float available in the schedule will not be for the exclusive use of either the Government or the Contractor.

The use of Resource Leveling is prohibited. Techniques used for the purpose of artificially adjusting activity durations to consume float and influence longest path are prohibited.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

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SECTION 01 33 00.00 22
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SUBMITTAL PROCEDURES (PWD ME)
08/18, CHG 4: 02/21

PART 1 GENERAL

This Section applies only to Design-Bid-Build projects at the Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 DEFINITIONS

1.1.1 Submittal Descriptions (SD)

Submittal requirements are specified in the technical sections. Examples and descriptions of submittals identified by the Submittal Description (SD) numbers and titles follow:

SD-01 Preconstruction Submittals

Submittals that are required prior to or commencing with the start of work on site. Submittals that are required prior to or at the start of construction (work) or the next major phase of the construction on a multiphase contract. No work is authorized without an approved submittal.

Preconstruction Submittals include schedules and a tabular list of locations, features, and other pertinent information regarding products, materials, equipment, or components to be used in the work.

List of required Preconstruction Submittals can be found in Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME).

SD-02 Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project.

Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials, systems or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-04 Samples

Fabricated or unfabricated physical examples of materials, equipment or workmanship that illustrate functional and aesthetic characteristics of a material or product and establish standards by which the work can be judged.

Color samples from the manufacturer's standard line (or custom color samples if specified) to be used in selecting or approving colors for the project.

Field samples and mock-ups constructed on the project site establish standards ensuring work can be judged. Includes assemblies or portions of assemblies that are to be incorporated into the project and those that will be removed at conclusion of the work.

SD-05 Design Data

Design calculations, mix designs, analyses or other data pertaining to a part of work.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accord with specified requirements. Unless specified in another section, testing must have been within three years of date of contract award for the project.

Report that includes findings of a test required to be performed on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report that includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports

Daily logs and checklists

Final acceptance test and operational test procedure

SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by responsible officials of manufacturer of product, system or material attesting that the product, system, or material meets specification requirements. Must be dated after award of project contract and clearly name the project.

Document required of Contractor, or of a manufacturer, supplier, installer or Subcontractor through Contractor. The document purpose is to further promote the orderly progression of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications.

Confined space entry permits

Text of posted operating instructions

The Contractor must maintain submitted certifications and licenses through the duration of the project. Any certification or license that expires prior to or during the execution of the affected Definable Feature of Work (DFOW) must be resubmitted to the Contracting Officer for approval.

SD-08 Manufacturer's Instructions

Preprinted material describing installation of a product, system or material, including special notices and (SDS)concerning impedances, hazards and safety precautions.

SD-09 Manufacturer's Field Reports

Documentation of the testing and verification actions taken by manufacturer's representative at the job site, in the vicinity of the job site, or on a sample taken from the job site, on a portion of the work, during or after installation, to confirm compliance with manufacturer's standards or instructions. The documentation must be signed by an authorized official of a testing laboratory or agency and state the test results; and indicate whether the material, product, or system has passed or failed the test.

Factory test reports.

SD-10 Operation and Maintenance Data

Data provided by the manufacturer, or the system provider, including manufacturer's help and product line documentation, necessary to maintain and install equipment, for operating and maintenance use by facility personnel.

Data required by operating and maintenance personnel for the safe and efficient operation, maintenance and repair of the item.

Data incorporated in an operations and maintenance manual or control system.

eOMSI submittals per Section 01 78 24.00 20 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI).

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

Submittals required for Guiding Principle Validation (GPV) or Third Party Certification (TPC).

Special requirements necessary to properly close out a construction contract. For example, Record Drawings and as-built drawings. Also, submittal requirements necessary to properly close out a major phase of construction on a multi-phase contract.

Red Zone documents per Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME).

1.1.2 Approving Authority

Office or designated person authorized to approve the submittal.

1.1.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or “S” classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals`

Submittal Register; G

1.3 PACKAGING AND SUBMISSION OF SUBMITTALS

Prepare and submit submittals required by each individual Specification Section. Each required submittal as listed in the submittal register must be packaged and submitted individually so that it can be tracked, reviewed, and returned in a concise and orderly fashion.

1.4 SUBMITTAL CLASSIFICATION

1.4.1 Government Approved (G)

Government approval is required for extensions of design, critical materials, variations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Government.

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, submittals are considered to be "shop drawings."

Submittal review period for Government Approved (G) submittals is outlined in paragraph SCHEDULING with additional clarifications in paragraph VARIATIONS.

1.4.2 Government Approved (S)

Government approval is required for (S) submittals. (S) submittals are associated with sustainability requirements. Refer to "S" Submittals for Sustainability Documentation paragraph in specification Section 01 33 29.00 22 SUSTAINABILITY REQUIREMENTS AND REPORTING (PWD ME).

1.4.3 For Information Only

Submittals not requiring Government approval will be for information only. Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

1.5 FORWARDING SUBMITTALS REQUIRING GOVERNMENT APPROVAL

As soon as practicable after award of contract, and before procurement or fabrication, provide through Government's Electronic Construction and

Facility Support Contract Management System (eCMS) submittals required in the technical sections of this specification, including shop drawings, product data and samples.

1.5.1 O&M Data

Submit data specified for a given item within 30 calendar days after the item is delivered to the contract site.

In the event the Contractor fails to deliver O&M data within the time limits specified, the Contracting Officer may withhold from progress payments 50 percent of the price of the items to which such O&M data apply.

1.5.2 Submittals Reserved for NAVFAC ATLANTIC Approval

As an exception to the standard submittal procedure for Government Approval, submit the following to NAVFAC MIDLANT Reviewers:

a. All fire protection system submittals

b. All fire alarm system submittals

c. All elevator submittals

d. Section 01 91 00.15 20 TOTAL BUILDING COMMISSIONING: SD-06 Commissioning Plan, Certificate of Readiness, and Commissioning Report submittals

e. Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC: SD-06 field test report submittals

f. 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS: SD-06 field test report submittals

g. Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC: All submittals

1.6 PREPARATION

1.6.1 Transmittal Form

Transmit each submittal, except sample installations and sample panels, using the transmittal form prescribed by the Contracting Officer. Include all information prescribed by the transmittal form and required in paragraph IDENTIFYING SUBMITTALS. Use the submittal transmittal forms to record actions regarding samples.

1.6.2 Identifying Submittals

The Contractor's Quality Control Manager must prepare, review and stamp submittals, including those provided by a subcontractor, before submittal to the Government.

Identify submittals, except sample installations and sample panels, with the following information permanently adhered to or noted on each separate component of each submittal and noted on transmittal form. Mark each copy of each submittal identically, with the following:

a. Project title and location

b. Construction contract number

c. Dates of the drawings and revisions

d. Name, address, and telephone number of Subcontractor, supplier, manufacturer, and any other Subcontractor associated with the submittal.

e. Section number of the specification by which submittal is required

f. Submittal description (SD) number of each component of submittal

g. For a resubmission, add alphabetic suffix on submittal description, for example, submittal 18 would become 18A, to indicate resubmission

h. Product identification and location in project.

1.6.3 Submittal Format

1.6.3.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.6.3.2 Format for SD-02 Shop Drawings

Provide shop drawings not less than 8 1/2 by 11 inches nor more than 30 by 42 inches, except for full-size patterns or templates. Prepare drawings to accurate size, with scale indicated, unless another form is required. Ensure drawings are suitable for reproduction and of a quality to produce clear, distinct lines and letters, with dark lines on a white background.

a. Include the nameplate data, size, and capacity on drawings. Also include applicable federal, military, industry, and technical society publication references.

b. Dimension drawings, except diagrams and schematic drawings. Prepare drawings demonstrating interface with other trades to scale. Use the same unit of measure for shop drawings as indicated on the contract drawings. Identify materials and products for work shown.

Submit an electronic copy of drawings in PDF format and native electronic format.

1.6.3.2.1 Drawing Identification

Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph IDENTIFYING SUBMITTALS.

Number drawings in a logical sequence. Each drawing is to bear the number of the submittal in a uniform location next to the title block. Place the Government contract number in the margin, immediately below the title block, for each drawing.

Reserve a blank space, no smaller than 3 by 4 inches on the right-hand side of each sheet for the Government disposition stamp.

1.6.3.3 Format of SD-03 Product Data

Present product data submittals for each section. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification section number and paragraph number to which it pertains.

1.6.3.3.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information and format as required for submission of SD-07 Certificates.

Provide product data in units used in the Contract documents. Where product data are included in preprinted catalogs with another unit, submit the dimensions in contract document units, on a separate sheet.

1.6.3.3.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.6.3.3.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. Partial submittals will not be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

1.6.3.4 Format of SD-04 Samples

1.6.3.4.1 Sample Characteristics

Furnish samples in the following sizes, unless otherwise specified or unless the manufacturer has prepackaged samples of approximately the same size as specified:

a. Sample of Equipment or Device: Full size.

b. Sample of Materials Less Than 2 by 3 inches: Built up to 8 1/2 by 11 inches.

c. Sample of Materials Exceeding 8 1/2 by 11 inches: Cut down to 8 1/2 by 11 inches and adequate to indicate color, texture, and material variations.

d. Sample of Linear Devices or Materials: 10 inch length or length to be supplied, if less than 10 inches. Examples of linear devices or materials are conduit and handrails.

e. Sample Volume of Nonsolid Materials: Pint. Examples of nonsolid materials are sand and paint.

f. Color Selection Samples: 2 by 4 inches. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified. Sizes and quantities of samples are to represent their respective standard unit.

g. Sample Panel: 4 by 4 feet.

h. Sample Installation: 100 square feet.

1.6.3.4.2 Sample Incorporation

Reusable Samples: Incorporate returned samples into work only if so specified or indicated. Incorporated samples are to be in undamaged condition at the time of use.

Recording of Sample Installation: Note and preserve the notation of any area constituting a sample installation, but remove the notation at the final clean-up of the project.

1.6.3.4.3 Comparison Sample

Samples Showing Range of Variation: Where variations in color, finish, pattern, or texture are unavoidable due to nature of the materials, submit sets of samples of not less than three units showing extremes and middle of range. Mark each unit to describe its relation to the range of the variation.

When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.

1.6.3.5 Format of SD-05 Design Data

Provide design data and certificates on 8 1/2 by 11 inch paper.

1.6.3.6 Format of SD-06 Test Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.6.3.7 Format of SD-07 Certificates

Provide design data and certificates on 8 1/2 by 11 inch paper.

1.6.3.8 Format of SD-08 Manufacturer's Instructions

Present manufacturer's instructions submittals for each section. Include the manufacturer's name, trade name, place of manufacture, and catalog model or number on product data. Also include applicable federal, military, industry, and technical-society publication references. If supplemental information is needed to clarify the manufacturer's data, submit it as specified for SD-07 Certificates.

Submit the manufacturer's instructions before installation.

1.6.3.8.1 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.6.3.9 Format of SD-09 Manufacturer's Field Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.6.3.10 Format of SD-10 Operation and Maintenance Data (O&M)

Comply with the requirements specified in Section 01 78 23.00 22 OPERATION AND MAINTENANCE DATA (PWD ME)for O&M Data format.

1.6.3.11 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.6.4 Source Drawings for Shop Drawings

1.6.4.1 Source Drawings

The entire set of source drawing files (DWG) will not be provided to the Contractor. Request the specific Drawing Number for the preparation of shop drawings. Only those drawings requested to prepare shop drawings will be provided. These drawings are provided only after award.

1.6.4.2 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim, and waives to the fullest extent permitted by law any claim or cause of action of any nature against the Government, its agents, or its subconsultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities, or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic source drawing files are not construction documents. Differences may exist between the source drawing files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic source drawing files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. The Contractor is responsible for determining if any conflict exists. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished source drawing files, the signed and sealed construction documents govern. Use of these source drawing files does not relieve the Contractor of the duty to fully comply with the contract documents, including and without limitation the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indication of ownership (seals, logos, signatures, initials and dates).

1.6.5 Electronic File Format

Provide submittals in electronic format, with the exception of material samples required for SD-04 Samples items. Compile the submittal file as a single, complete document, to include the Transmittal Form described within, and also separately include the native files which were used to create the PDF. The files should include the original digital files used to create the submittal. Name the electronic submittal file specifically according to its contents, and coordinate the file naming convention with the Contracting Officer. Electronic files must be of sufficient quality that all information is legible. Use PDF as the electronic format, unless otherwise specified or directed by the Contracting Officer. Generate PDF files from original documents with bookmarks so that the text included in the PDF file is searchable and can be copied. If documents are scanned, optical character resolution (OCR) routines are required. Index and bookmark files exceeding 30 pages to allow efficient navigation of the file. When required, the electronic file must include a valid electronic signature or a scan of a signature.

All electronic submittal documents must be transferred, shared, and managed through Government's Electronic Construction and Facility Support Contract Management System (eCMS). Refer to Section 01 31 23.13 20 ELECTRONIC CONSTRUCTION AND FACILITY SUPPORT CONTRACT MANAGEMENT SYSTEM.

1.7 QUANTITY OF SUBMITTALS

1.7.1 Number of SD-04 Samples

a. Submit two samples, or two sets of samples showing the range of variation, of each required item. One approved sample or set of samples will be retained by the approving authority and one will be returned to the Contractor.

b. Submit one sample panel or provide one sample installation where directed. Include components listed in the technical section or as directed.

c. Submit one sample installation, where directed.

d. Submit one sample of nonsolid materials.

1.8 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Provide information-only submittals to the Contracting Officer a minimum of 14 calendar days prior to the Preparatory Meeting for the associated Definable Feature of Work (DFOW). Approval of the Contracting Officer is not required on information only submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information and will return only the transmittal cover sheet to the Contractor. Normally, submittals for information only will not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.

1.9 PROJECT SUBMITTAL REGISTER

A sample Project Submittal Register showing items of equipment and materials for when submittals are required by the specifications is provided as "Appendix A - Submittal Register."

1.9.1 Submittal Management

Prepare and maintain a submittal register, as the work progresses. Do not change data that is output in columns (c), (d), (e), and (f) as delivered by Government; retain data that is output in columns (a), (g), (h), and (i) as approved. As an attachment, provide a submittal register showing items of equipment and materials for which submittals are required by the specifications. This list may not be all-inclusive and additional submittals may be required.

Column (c): Lists specification section in which submittal is required.

Column (d): Lists each submittal description (SD Number. and type, e.g., SD-02 Shop Drawings) required in each specification section.

Column (e): Lists one principal paragraph in each specification section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not consider entries in column (e) as limiting the project requirements.

Column (f): Lists the approving authority for each submittal.

Thereafter, the Contractor is to track all submittals by maintaining a complete list, including completion of all data columns and all dates on which submittals are received by and returned by the Government.

1.9.2 Preconstruction Use of Submittal Register

Submit the submittal register. Include the QC plan and the project schedule. Verify that all submittals required for the project are listed and add missing submittals. Coordinate and complete the following fields on the register submitted with the QC plan and the project schedule:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for the approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

Column (i) Contractor Material: Date that Contractor needs material delivered to Contractor control.

1.9.3 Contractor Use of Submittal Register

Update the following fields with each submittal throughout the contract.

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

1.9.4 Approving Authority Use of Submittal Register

Update the following fields:

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

1.9.5 Action Codes

1.9.5.1 Government Review Action Codes

"A" - "Approved as submitted"

"AN" - "Approved as noted"

"RR" - "Disapproved as submitted"; "Completed"

"NR" - "Not Reviewed"

"RA" - "Receipt Acknowledged"

1.9.6 Delivery of Copies

Submit an updated electronic copy of the submittal register to the Contracting Officer with each invoice request. Provide an updated Submittal Register monthly regardless of whether an invoice is submitted.

1.10 VARIATIONS

Variations from contract requirements require Contracting Officer approval pursuant to contract Clause FAR 52.236-21 Specifications and Drawings for Construction, and will be considered where advantageous to the Government.

1.10.1 Considering Variations

Discussion of variations with the Contracting Officer before submission will help ensure that functional and quality requirements are met and minimize rejections and resubmittals. For variations that include design changes or some material or product substitutions, the Government may require an evaluation and analysis by a licensed professional engineer hired by the contractor.

Specifically point out variations from contract requirements in a transmittal letter. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

Proposing Variations

When proposing variation, deliver a submittal, clearly marked as a "VARIATION" to the Contracting Officer, with documentation illustrating the nature and features of the variation including any necessary technical submittals and why the variation is desirable and beneficial to Government. If lower cost is a benefit, also include an estimate of the cost savings. In addition to documentation required for variation, include the submittals required for the item. Clearly mark the proposed variation in all documentation. Refer to Section 01 33 10.00 22 PROCUREMENT SUBSTITUTION REQUIREMENTS (PWD ME)for additional requirements.

The Contracting Officer will indicate an approval or disapproval of the variation request; and if not approved as submitted, will indicate the Government's reasons therefore. Any work done before such approval is received is performed at the Contractor's risk."

Specifically point out variations from contract requirements in a substitution request cover sheet. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

1.10.2 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.10.3 Review Schedule Extension

In addition to the normal submittal review period, a period of 30 calendar days will be allowed for the Government to consider submittals with variations.

1.11 SCHEDULING

Contracting Officer review will be completed within 21 calendar days after the date of submission, except as specified otherwise.

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals. Allow an additional 21 calendardays for review and approval of submittals for HVAC control, fire alarm, and fire protection systems.

a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. The Contractor is responsible for additional time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as for the initial submittal.

b. Submittals required by the contract documents are listed on the submittal register. If a submittal is listed in the submittal register but does not pertain to the contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not relieve the Contractor of supplying submittals required by the contract documents but that have been omitted from the register or marked "N/A."

c. Resubmit the submittal register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further resubmittal is required.

d. Except as specified otherwise, allow a review period, beginning with receipt by the approving authority, that includes at least 14 calendar days for submittals for QC manager approval and 21 working days for submittals where the Contracting Officer is the approving authority. The period of review for submittals with Contracting Officer approval begins when the Government receives the submittal from the QC organization.

e. For submittals requiring review by a Government fire protection engineer, allow a review period, beginning when the Government receives the submittal from the QC organization, of 30 calendar days for return of the submittal to the Contractor.

1.11.1 Reviewing, Certifying, and Approving Authority

The QC Manager is responsible for reviewing all submittals and certifying that they are in compliance with contract requirements. The approving authority on submittals is the QC Manager unless otherwise specified. At each "Submittal" paragraph in individual specification sections, a notation "G" following a submittal item indicates that the Contracting Officer is the approving authority for that submittal item. Provide an additional copy of the submittal to the Government Approving authority, per the requirements of paragraph FORWARDING SUBMITTALS REQUIRING GOVERNMENT APPROVAL.

1.11.2 Constraints

Conform to provisions of this section, unless explicitly stated otherwise for submittals listed or specified in this contract.

Submit complete submittals for each definable feature of the work. At the same time, submit components of definable features that are interrelated as a system.

When acceptability of a submittal is dependent on conditions, items, or materials included in separate subsequent submittals, the submittal will be returned without review.

Approval of a separate material, product, or component does not imply approval of the assembly in which the item functions.

1.11.3 QC Organization Responsibilities

a. Review submittals for conformance with project design concepts and compliance with contract documents.

b. Process submittals based on the approving authority indicated in the submittal register.

    (1) When the QC manager is the approving authority, take appropriate action on the submittal from the possible actions defined in paragraph APPROVED SUBMITTALS.

    (2) When the Contracting Officer is the approving authority or when variation has been proposed, forward the submittal to the Government, along with a certifying statement, or return the submittal marked "not reviewed" or "revise and resubmit" as appropriate. The QC organization's review of the submittal determines the appropriate action.

c. Ensure that material is clearly legible.

d. Stamp each sheet of each submittal with a QC certifying statement or an approving statement, except that data submitted in a bound volume or on one sheet printed on two sides may be stamped on the front of the first sheet only.

    (1) When the approving authority is the Contracting Officer, the QC organization will certify submittals forwarded to the Contracting Officer with the following certifying statement:

"I hereby certify that the (equipment) (material) (article) shown and marked in this submittal is that proposed to be incorporated with Contract Number _____ is in compliance with the contract drawings and specification, can be installed in the allocated spaces, and is submitted for Government approval.

Certified by Submittal Reviewer _____________________, Date _______
(Signature when applicable)

Certified by QC Manager _____________________________, Date ______"
(Signature)

(2) When approving authority is the QC manager, the QC manager will use the following approval statement when returning submittals to the Contractor as "Approved" or "Approved as Noted."

"I hereby certify that the (material) (equipment) (article) shown and marked in this submittal and proposed to be incorporated with Contract Number _____ is in compliance with the contract drawings and specification, can be installed in the allocated spaces, and is approved for use.

Certified by Submittal Reviewer ______________________, Date ______
(Signature when applicable)

Approved by QC Manager _______________________________, Date _____"
(Signature)

e. Sign the certifying statement or approval statement. The QC organization member designated in the approved QC plan is the person signing certifying statements. The use of original ink for signatures is required. Stamped signatures are not acceptable.

f. Update the submittal register as submittal actions occur, and maintain the submittal register at the project site until final acceptance of all work by the Contracting Officer.

g. Retain a copy of approved submittals and approved samples at the project site.

h. For "S" submittals, provide a copy of the approved submittal to the Government Approving authority.

1.12 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

a. Note the date on which the submittal was received from the QC manager.

b. Review submittals for approval within the scheduling period specified in paragraph SCHEDULING and only for conformance with project design concepts and compliance with contract documents.

c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS and with comments and markings appropriate for the action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals. One copy of the submittal will be retained by the Contracting Officer and one copy of the submittal will be returned to the Contractor.

1.12.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.

b. Submittals marked "approved as noted" or "approved, except as noted, resubmittal not required," authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.

c. Submittals marked "not approved," "disapproved," or "revise and resubmit" indicate incomplete submittal or noncompliance with the contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal is approved.

d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by Contractor, or is not complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.

e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

1.13 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications, give notice to the Contracting Officer as required under the FAR clause titled CHANGES. The Contractor is responsible for the dimensions and design of connection details and the construction of work. Failure to point out variations may cause the Government to require rejection and removal of such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and resubmit in accordance with the procedures above. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

All revisions made to a previous submittal must be clouded and specifically indicated in the transmittal to clearly delinate and identify what has changed. The Government has the right to reject revised submtitals that do not cloud or delineate changes from previous submittals.

1.14 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals is not to be construed as a complete check, and indicates only that the general method of construction, materials, detailing, and other information are satisfactory.

Approval or acceptance by the Government for a submittal does not relieve the Contractor of the responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained with in each submittal accurately conforms with the requirements of the contract documents.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.15 APPROVED SAMPLES

Approval of a sample is only for the characteristics or use named in such approval and is not be construed to change or modify any contract requirements. Before submitting samples, provide assurance that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.

Match the approved samples for materials and equipment incorporated in the work. If requested, approved samples, including those that may be damaged in testing, will be returned to the Contractor, at its expense, upon completion of the contract. Unapproved samples will also be returned to the Contractor at its expense, if so requested.

Failure of any materials to pass the specified tests will be sufficient cause for refusal to consider, under this contract, any further samples of the same brand or make as that material. The Government reserves the right to disapprove any material or equipment that has previously proved unsatisfactory in service.

Samples of various materials or equipment delivered on the site or in place may be taken by the Contracting Officer for testing. Samples failing to meet contract requirements will automatically void previous approvals. Replace such materials or equipment to meet contract requirements. Approval of the sample by the Contracting Officer does not relieve the Contractor of its responsibilites under the Contract.

1.16 WITHHOLDING OF PAYMENT

Payment for materials incorporated in the work will not be made if required approvals have not been obtained. No payment for materials incorporated in the work will be made unless all required DOR approvals or required Government approvals have been obtained. No payment will be made for any materials incorporated into the work for any conformance review submittals or information-only submittals found to contain errors or deviations from the Solicitation or Accepted Proposal.

1.17 CERTIFICATION OF SUBMITTAL DATA

Certify the submittal data as follows on Form ENG 4025: "I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the Contract drawings and specifications except as otherwise stated."

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

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SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
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CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION GOVT OR A/E REVWR	CONTRACTOR: SCHEDULE DATES SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	CONTRACTOR ACTION ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 11 00.00 22	SD-01 Preconstruction Submittals														
			Work Sequencing and	1.5	G												
			Preparation Plan														
			Salvage Plan	1.12	G												
			SD-11 Closeout Submittals														
			RZ Checklist	1.13	G												
		01 14 00.00 22	SD-01 Preconstruction Submittals														
			List of Contact Personnel	1.5.1.1	G												
			Outage Plan	1.5.5	G												
			Tier 1 Infrastructure Coordination	1.10	G												
			Drawings														
		01 20 00.00 22	SD-01 Preconstruction Submittals														
			Earned Value Report	1.3	G												
			Accounting Summary of Unit Cost	1.7.1	G												
			Quantities														
		01 30 00.00 22	SD-01 Preconstruction Submittals														
			Facility Turnover Planning	1.16.1	G												
			Meeting Agenda and Red Zone														
			(RZ) Checklist-POAM														
			NAVFAC PWD ME Internal	1.16.3	G												
			Service Requirements List														
			Archaeological Coordination	1.13	G												
			Meeting														
			View Location Map	1.3													
			Progress and Completion	1.4													
			Pictures														

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		01 30 10.00 22	SD-02 Shop Drawings															
			Coordination Drawings	1.4.1	G													
		01 31 23.13 20	SD-01 Preconstruction Submittals															
			List of Contractor's Personnel	1.4.2	G													
		01 32 17.00 20	SD-01 Preconstruction Submittals															
			Baseline NAS	1.2.2	G													
			Designated Project Scheduler	1.9	G													
			Three-Week Look Ahead	1.3	G													
			Schedule															
			Outage Schedule	1.4.3	G													
			SD-07 Certificates															
			Weekly Look Ahead Schedule	1.3.1														
			Monthly Network Analysis	1.4.1	G													
			Updates															
			SD-11 Closeout Submittals															
			As-Built Schedule	1.4.2	G													
		01 33 00.00 22	SD-01 Preconstruction Submittals															
			Submittal Register	1.9	G													
		01 33 10.00 22	SD-01 Preconstruction Submittals															
			Procurement Substitution	1.6	G													
			Request - Pre-Award (Bidding)															
			Period															
			Substitution Request - Post	1.7	G													
			Award															
		01 33 29.00 22	SD-01 Preconstruction Submittals															

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(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 33 29.00 22	Preliminary High Performance	1.5.3.2	G													
			and Sustainable Building Checklist															
			Sustainability Action Plan	1.4.1	G													
			Preliminary Sustainability	1.5.3.2	G													
			eNotebook															
			SD-11 Closeout Submittals															
			Final High Performance and	1.5.3.2	G													
			Sustainable Building Checklist															
			Final Sustainability eNotebook	1.5.3.2	G													
			Amended Final Sustainability	1.5.3.2	G													
			eNotebook															
			Amended Final High	1.5.3.2	G													
			Performance and Sustainable															
			Building Checklist															
			Third Party Certification		G													
			Certificate, Assessment, or															
			Validation and Compliance Report															
		01 35 26.00 22	SD-01 Preconstruction Submittals															
			Accident Prevention Plan (APP)		G													
			APP - Construction	1.9.1	G													
			Dive Operations Plan		G													
			Indoor Air Quality (IAQ)	1.18	G													
			Management Plan															
			SD-06 Test Reports															
			Monthly Exposure Reports	1.5	G													
			Notifications and Reports	1.14	G													

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(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 35 26.00 22	Accident Reports	1.14.2	G													
			LHE Inspection Reports	1.14.3	G													
			SD-07 Certificates															
			Contractor Safety Self-Evaluation	1.6	G													
			Checklist															
			Crane Operators/Riggers	1.8.7	G													
			Standard Lift Plan	1.9.3.2	G													
			Standard Lift Plan	1.14.4	G													
			Critical Lift Plan	1.9.3.3	G													
			Naval Architecture Analysis	1.9.3.4	G													
			Activity Hazard Analysis (AHA)	1.10	G													
			Confined Space Entry Permit	1.11.1	G													
			Hot Work Permit	1.11.1	G													
			Hot Work Permit	1.15	G													
			Certificate of Compliance	1.14.4	G													
			Third Party Certification of	1.14.5	G													
			Floating Cranes and															
			Barge-Mounted Mobile Cranes															
			License Certificates	1.16	G													
			Radiography Operation Planning	1.16.1	G													
			Work Sheet															
		01 44 00.00 22	SD-01 Preconstruction Submittals															
			Archaeologist Qualifications	1.4	G													
			Archaeological Coordinator	1.5														
			Archaeological Monitoring Plan	1.6	G													
			SD-06 Test Reports															

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(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 44 00.00 22	Archaeological Resource Report	3.1	G												
			Archaeological Resource Report	3.2.3	G												
			Archaeological Monitoring Report	3.2.2	G												
			- Monthly Progress Reports														
			Archaeological Monitoring Report	3.2.2	G												
			- Draft														
			Archaeological Monitoring Report	3.2.2	G												
			- Final Draft														
			Archaeological Monitoring Report	3.2.2	G												
			- Final														
			Curated Artifacts	3.3	G												
		01 45 00.00 22	SD-01 Preconstruction Submittals														
			Contractor Quality Control (CQC)	1.5.2	G												
			Plan														
			SD-06 Test Reports														
			Verification Statement	1.13.3													
			SD-07 Certificates														
			Certificate Of Readiness	1.6.5.1													
		01 45 35.00 22	SD-01 Preconstruction Submittals														
			SIOR Letter of Acceptance	3.1.1	G												
			Project Manual	3.1.1	G												
			Written NDT Practices	3.1.1													
			Written NDT Practices	3.1.3													
			SD-06 Test Reports														
			Special Inspection Daily Reports	3.1.1	G												
			Special Inspection Daily Reports	3.1.2	G												

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(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 45 35.00 22	Special Inspection Daily Reports	3.1.3	G												
			Special Inspection Daily Reports	3.1.3	G												
			Special Inspection Biweekly Reports	3.1.1	G												
			Special Inspection Biweekly Reports	3.1.2	G												
			SD-07 Certificates														
			AISC Certified Steel Fabricator	2.1	G												
			Certificate of Compliance	2.1	G												
			Special Inspector of Record		G												
			Special Inspector	1.5	G												
			Qualification Records for Nondestructive Testing Technicians	3.1.1	G												
			Qualification Records for Nondestructive Testing Technicians	3.1.3	G												
			SD-11 Closeout Submittals														
			Interim Report	3.1.1	G												
			Interim Report	3.1.3	G												
			Comprehensive Final Report	3.1.1	G												
			Final Report	3.1.3	G												
		01 50 00.00 22	SD-01 Preconstruction Submittals														
			Construction Site Plan	1.3	G												
			Traffic Control Plan	3.4.1	G												

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(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 50 00.00 22	Contractor Computer	1.6.1.4	G													
			Cybersecurity Compliance															
			Statements															
			Contractor Temporary Network	1.6.6	G													
			Cybersecurity Compliance															
			Statements															
			Employee Parking Plan	1.7	G													
			SD-03 Product Data															
			Backflow Preventers	1.4	G													
			SD-06 Test Reports															
			Backflow Preventer Tests	3.5	G													
			SD-07 Certificates															
			Backflow Tester	1.4.1	G													
			Backflow Preventers	1.4	G													
		01 57 19.00 22	SD-01 Preconstruction Submittals															
			Preconstruction Survey	1.5.1	G													
			Solid Waste Management Plan	3.4.1	G													
			Regulatory Notifications	1.5.2	G													
			Environmental Management Plan	3.1	G													
			(EMP)															
			Dirt and Dust Control Plan	3.13.1	G													
			Contractor Hazardous Material	3.8.1	G													
			Inventory Log															
			Stormwater Management/Erosion	3.2.1	G													
			and Sedimentation Control Plan															

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(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 19.00 22	Spill Prevention, Control, and Countermeasures (SPCC) Plan	3.1		G												
			Environmental Manager Qualifications	1.5.4		G												
			Employee Training Records	1.5.5		G												
			SD-06 Test Reports															
			Laboratory Analysis	3.12.4.2		G												
			Disposal Requirements	3.14.2		G												
			Erosion and Sediment Control Inspection and Corrective Action	3.2.3		G												
			Monthly Solid Waste Disposal Report	3.4.2		G												
			SD-07 Certificates															
			ECATTS Certificate Of Completion	1.4.1		G												
			Erosion And Sediment Control Inspector	1.5.5		G												
			SD-11 Closeout Submittals															
			Stormwater Management and Erosion Control Compliance Notebook	3.2.1		G												
			Waste Determination Documentation	3.5		G												
			Disposal Documentation for Hazardous and Regulated Waste	3.12.4.5		G												

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(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 19.00 22	Contractor Hazardous Material	3.8.1	G													
			Inventory Log															
			Hazardous Waste/Debris	3.12.4	G													
			Management															
			Regulatory Notifications	1.5.2	G													
			Asbestos Free Certification Form	3.12.4.3	G													
			Assembled Employee Training	1.5.5	G													
			Records															
			Sales Documentation	3.4.2	G													
		01 58 00.00 22	SD-02 Shop Drawings															
			Preliminary One Line	1.3.1.1	G													
			Preliminary Drawing Indicating	1.4.1	G													
			Layout And Text Content															
			SD-04 Samples															
			Final Rendering	1.3.1.2	G													
			Final Framed Rendering	1.3.1.3	G													
			Facility Recognition Plaque	1.3.2	G													
		01 74 19.00 22	SD-01 Preconstruction Submittals															
			Construction Waste Management	1.7	G													
			Plan															
			SD-11 Closeout Submittals															
			Final Construction Waste	1.9	G													
			Diversion Report															
		01 78 00.00 22	SD-03 Product Data															
			Warranty Management Plan	1.6.1														
			Warranty Tags	1.6.4														

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ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION / GOVT OR A/E REVWR	CONTRACTOR: SCHEDULE DATES SUBMIT	CONTRACTOR: SCHEDULE DATES APPROVAL NEEDED BY	CONTRACTOR: SCHEDULE DATES MATERIAL NEEDED BY	CONTRACTOR ACTION ACTION CODE	CONTRACTOR ACTION DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY DATE FWD TO OTHER REVIEWER	APPROVING AUTHORITY DATE RCD FROM OTH REVIEWER	APPROVING AUTHORITY ACTION CODE	APPROVING AUTHORITY DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 78 00.00 22	Final Cleaning	3.4													
			Spare Parts Data	1.5													
			SD-08 Manufacturer's Instructions														
			Instructions	1.6.1													
			Preventative Maintenance	1.8	G												
			Condition Monitoring (Predictive	1.8	G												
			Testing)														
			Inspection	1.8	G												
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.3	G												
			Manuals														
			SD-11 Closeout Submittals														
			As-Built Drawings	3.1	G												
			Record Drawings	3.2	G												
			As-Built Record of Equipment	1.7.1	G												
			and Materials														
			Warranted Equipment and	1.6.1													
			Materials														
			Certification of EPA Designated	2.1	G												
			Items														
			Certification Of USDA Designated	2.2	G												
			Items														
			Red Zone Documents	1.7.4	G												
			Facility Data Workbook (FDW),	1.7.5	G												
			Final Submission														

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		01 78 00.00 22	Post Installation Sanitary System	1.7.6	G													
			Survey															
			High Performance and		G													
			Sustainable Building (HPSB)															
			Checklist															
			List Of Installed Building															
			Equipment															
		01 78 23.00 22	SD-10 Operation and Maintenance															
			Data															
			Facility Data Workbook	1.4	G													
			Training Plan	3.1.1	G													
			Training Outline	3.1.3	G													
			Training Content	3.1.2	G													
			Operation And Maintenance	3.2.1	G													
			Manual, Progress Submittal															
			Operation And Maintenance	3.2.2	G													
			Manual, Prefinal Submittal															
			Operation And Maintenance	3.2.3	G													
			Manual, Final Submittal															
			SD-11 Closeout Submittals															
			Training Video Recording	3.1.4	G													
			Validation of Training Completion	3.1.6	G													
			Training Plan	3.1.1	G													
			Record Drawings And Utility	1.6.6.7	G													
			Systems															
		01 78 24.00 22	SD-11 Closeout Submittals															

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		01 78 24.00 22	Facility Data Workbook (FDW),	1.3.1	G													
			Progress Submittal															
			Facility Data Workbook (FDW),	1.3.2	G													
			Prefinal Submittal															
			Facility Data Workbook (FDW),	1.3.3	G													
			Final Submittal															
		01 91 19	SD-01 Preconstruction Submittals															
			Building Enclosure		G													
			Commissioning Specialist															
			Qualifications															
			Building Enclosure Testing Work	1.8	G													
			Plan															
			PRELIMINARY BECx PLAN	3.2	G													
			SD-03 Product Data															
			Thermal Imaging Camera	2.3	G													
			Test Equipment	2.1	G													
			FINAL BECx PLAN	3.3	G													
			SD-05 Design Data															
			Envelope Surface Area	3.7.2	G													
			Calculations															
			SD-06 Test Reports															
			Completed Building Envelope		G													
			Inspection Checklists															
			Pressure Test Procedures	3.7.5	G													
			Air Leakage Test Report	1.12.3	G													
			Air Leakage Test Report	3.7.5.6	G													

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		01 91 19	Diagnostic Test Report	1.12.3	G													
			Diagnostic Test Report	3.7.6.6	G													
			FINAL BECx REPORT	3.9														
			SD-07 Certificates															
			Pressure Test Agency	1.12.2.1														
			Thermographer Qualifications	1.12.2.2														
			Certificate of Readiness	1.14														
			SD-10 Operation and Maintenance															
			Data															
			Training	3.8	G													
		02 41 00	SD-01 Preconstruction Submittals															
			Demolition/Deconstruction Plan	1.2.2	G													
			Existing Conditions	1.11														
			Qualifications of Jacking	3.3	G													
			Contractor															
			Photographic Documentation	3.3.1	G													
			SD-02 Shop Drawings															
			Temporary shoring design	3.2	G													
			drawings															
			Jacking design drawings	3.3	G													
			SD-05 Design Data															
			Temporary shoring design	3.2														
			calculations															
			Jacking design calculations	3.3	G													
			SD-07 Certificates															
			Notification	1.7	G													

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						CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION			APPROVING AUTHORITY					
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		02 41 00	SD-11 Closeout Submittals														
			Receipts	3.4.4													
		02 42 91	SD-01 Preconstruction Submittals														
			Work Plan	1.2	G												
		02 82 00.00 22	SD-03 Product Data														
			Amended Water	1.2.2	G												
			Safety Data Sheets (SDS) for All	1.3.9	G												
			Materials														
			Encapsulants	2.1	G												
			Respirators	3.1.2.1	G												
			Local Exhaust Equipment	3.1.7	G												
			Pressure Differential Automatic	3.1.7	G												
			Recording Instrument														
			Vacuums	3.1.8	G												
			Glovebags	3.1.10	G												
			SD-06 Test Reports														
			Air Sampling Results	1.5.5	G												
			Pressure Differential Recordings	1.5.6	G												
			for Local Exhaust System														
			Clearance Sampling	3.2.14.5	G												
			Asbestos Disposal Quantity	3.3.3.2	G												
			Report														
			SD-07 Certificates														
			Employee Training	1.3.4	G												
			Notifications	1.3.5	G												
			Respiratory Protection Program	1.3.7	G												

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		02 82 00.00 22	Asbestos Hazard Abatement Plan	1.3.10	G												
			Asbestos Hazard Abatement Plan	1.3.10	G												
			Checklist														
			Testing Laboratory	1.3.11	G												
			Medical Certification	1.3.14	G												
			Private Qualified Person	1.5.1	G												
			Documentation														
			Competent Person	1.5.2	G												
			Worker's License	1.5.3	G												
			Contractor's License	1.5.4	G												
			Federal, State or Local Citations	1.5.7	G												
			on Previous Projects														
			Encapulants	1.2.15	G												
			Equipment Used to Contain	3.1	G												
			Airborne Asbestos Fibers														
			Water Filtration Equipment	3.1.3.3	G												
			Vacuums	3.1.8	G												
			Ventilation Systems	3.1.8	G												
			SD-11 Closeout Submittals														
			Permits	1.3.5													
			Licenses	1.3.5	G												
			Notifications	1.3.5	G												
			Respirator Program Records	1.3.7.1	G												
			Rental Equipment	1.7.1	G												
		02 83 00.00 22	SD-01 Preconstruction Submittals														
			Competent Person	1.5.1.1	G												

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		02 83 00.00 22	Training Certification	1.5.1.2	G												
			Occupational and Environmental	1.5.2.4	G												
			Assessment Data Report														
			Medical Examinations	1.5.2.5	G												
			Lead, Cadmium, and Chromium	1.5.2.9	G												
			(VI) Waste Management Plan														
			Licenses, Permits and	1.5.4	G												
			Notifications														
			Lead, Cadmium, and Chromium	1.5.2.2	G												
			(VI) Compliance Plan														
			Lead, Cadmium, and Chromium	1.5.2.3	G												
			(VI) Compliance Plan Checklist														
			Waste Characterization Sampling	1.5.2.10	G												
			Plan														
			Sample Results	3.4.1.1	G												
			SD-03 Product Data														
			Respirators	1.6.1	G												
			Vacuum Filters	1.6.4	G												
			Negative Air Pressure System	1.6.7	G												
			Materials and Equipment	2.1	G												
			Expendable Supplies	2.1.1	G												
			Local Exhaust Equipment	3.1.1.5	G												
			Pressure Differential Automatic	3.1.1.5	G												
			Recording Instrument														
			Pressure Differential Log	3.1.1.6	G												
			SD-06 Test Reports														

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		02 83 00.00 22	Sampling and Analysis	1.3.3	G													
			Occupational and Environmental	1.5.2.4	G													
			Assessment Data Report															
			Sampling Results	1.5.2.4	G													
			Pressure Differential Recordings	1.5.3	G													
			For Local Exhaust System															
			SD-07 Certificates															
			Testing Laboratory	1.5.1.3	G													
			Third Party Consultant	1.5.1.4	G													
			Qualifications															
			Notification of the	3.1.1.1	G													
			Commencement of Work															
			Impacting Lead, Cadmium, or															
			Chromium (VI)															
			Clearance Certification	3.5.1.1	G													
			SD-11 Closeout Submittals															
			Turn-In Documents or Weight	3.5.2.1	G													
			Tickets															
		02 84 16.00 22	SD-07 Certificates															
			Qualifications of CIH	1.8.1	G													
			Training Certification	1.8.1	G													
			PCB and Lamp Removal Work	1.8.2	G													
			Plan															
			PCB and Lamp Disposal Plan	1.8.3	G													
			SD-11 Closeout Submittals															
			Transporter Certification	3.5.2	G													

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		02 84 16.00 22	Certification of Decontamination	3.2.4													
			Certificate of Disposal and/or recycling	3.5.2.1													
			DD Form 1348-1	3.5.3.1													
		02 84 33.00 22	SD-01 Preconstruction Submittals														
			PCB Removal Work Plan	1.7.3	G												
			PCB Disposal Plan	1.7.4	G												
			SD-07 Certificates														
			Training certification	1.7.1													
			Qualifications of CIH	1.7.2	G												
			Notification	1.7.5													
			Transporter Certification	3.8	G												
			Certification of Decontamination	3.5.4													
			Post Cleanup Sampling	3.5.5	G												
			Certificate of Disposal	3.8.1													
		03 01 00	SD-01 Preconstruction Submittals														
			Qualifications	1.5.3	G												
			Work Plan	1.5.4	G												
			Work Plan	1.6.1	G												
			Quality Control Plan	1.5.2	G												
			SD-03 Product Data														
			Polymers	2.3	G												
			Integral Color Admixture	2.3	G												
			Miscellaneous Materials And Equipment	2.4													
			SD-04 Samples														

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		03 01 00	Mockup	1.5.5	G												
			Color Chart	1.5.5	G												
			SD-05 Design Data														
			Formwork	3.1.3	G												
			Repair Procedures	1.5.1	G												
			Mixture Proportioning	2.5	G												
			SD-06 Test Reports														
			Mixture Proportioning	2.5	G												
			Quality Control	3.1.5	G												
			Quality Control	3.2.3	G												
			Quality Control	3.3.3	G												
			Polymers	2.3	G												
			Miscellaneous Materials and	2.4													
			Equipment														
			SD-07 Certificates														
			Qualifications	1.5.3													
			Polymers	2.3													
			SD-08 Manufacturer’s Instructions														
			Equipment for Concrete	2.1													
			Preparation														
			Polymers	2.3													
			Miscellaneous Materials and	2.4													
			Equipment														
		03 30 00	SD-01 Preconstruction Submittals														
			Concrete Curing Plan	1.6.3.1	G												
			Quality Control Plan	1.6.5	G												

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		03 30 00	Quality Control Personnel	1.6.6	G													
			Certifications															
			Quality Control Organizational	1.6.6	G													
			Chart															
			Laboratory Accreditation	1.6.8	G													
			SD-02 Shop Drawings															
			Reinforcing Steel	1.6.2.1	G													
			SD-03 Product Data															
			Joint Sealants	2.4.5	G													
			Joint Filler	2.4.4	G													
			Formwork Materials	2.1	G													
			Cementitious Materials	2.3.1	G													
			Vapor Retarder	2.4.6	G													
			Concrete Curing Materials	2.4.1	G													
			Reinforcement	2.6	G													
			Liquid Chemical Floor Hardeners	2.4.3.1	G													
			and Sealers															
			Admixtures	2.3.4	G													
			Waterstops	2.2.2	G													
			Biodegradable Form Release	2.2.3	G													
			Agent															
			Pumping Concrete	1.6.3.2	G													
			Nonshrink Grout	2.4.2	G													
			SD-05 Design Data															
			Concrete Mix Design	1.6.1.1	G													
			SD-06 Test Reports															

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		03 30 00	Concrete Mix Design	1.6.1.1	G												
			Aggregates	1.6.4.3	G												
			Compressive Strength Tests	3.13.2.3	G												
			Air Content	3.13.2.4	G												
			Slump Tests	3.13.2.1	G												
			Water	2.3.2	G												
			SD-07 Certificates														
			Reinforcing Bars	2.6.1	G												
			VOC Content for Form Release	1.6.3.3													
			Agents, Curing Compounds, and														
			Concrete Penetrating Sealers														
			Safety Data Sheets	1.6.3.4													
			Field Testing Technician and	1.6.6.1	G												
			Testing Agency														
			SD-08 Manufacturer’s Instructions														
			Liquid Chemical Floor Hardeners	2.4.3.1													
			and Sealers														
			Joint Sealants	2.4.5													
			Curing Compound	2.4.1													
		04 01 00.91 22	SD-01 Preconstruction Submittals														
			Masonry Quality Control Plan	1.6.1	G												
			Project Training Program	1.6.1	G												
			Qualifications	1.6.2	G												
			SD-02 Shop Drawings														
			Photographic Documentation	1.6.5	G												
			SD-03 Product Data														

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TITLE AND LOCATION
B60: PRODUCTION FACILITY REPAIRS - SIOP

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION	GOVT OR A/E REVWR	CONTRACTOR: SCHEDULE DATES SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	CONTRACTOR ACTION ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		04 01 00.91 22	Qualifications	1.6.2	G													
			Cleaning and Restoration Methods	1.6.6	G													
			Cleaning Materials	2.1	G													
			Replacement Mortar	2.2.1.1	G													
			Replacement Mortar	2.2.1.2	G													
			Mortar	1.3.7	G													
			Replacement Masonry Units	1.6.8.2	G													
			SD-04 Samples															
			Mock-ups	1.6.8	G													
			SD-06 Test Reports															
			Testing and Matching	2.2.1.1	G													
			SD-07 Certificates															
			Repair Materials	2.2														
		04 20 00	SD-01 Preconstruction Submittals															
			Hot Weather Procedures	1.4.1	G													
			Cold Weather Procedures	1.4.2	G													
			SD-02 Shop Drawings															
			Cut CMU	3.3.2.1	G													
			Detail Drawings	3.4.1.1	G													
			SD-03 Product Data															
			Cement	2.2.2.2.1	G													
			Cementitious Materials	2.4.1.1	G													
			SD-05 Design Data															
			Masonry Compressive Strength	2.1.2	G													

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		04 20 00	Fire-Rated Concrete Masonry	2.2.2.3	G												
			Units														
			SD-06 Test Reports														
			Fire-Rated Concrete Masonry	2.2.2.3	G												
			Units														
			Field Testing of Mortar	3.6.1.1	G												
			Field Testing of Grout	3.6.1.2	G												
			SD-07 Certificates														
			Concrete Masonry Units (CMU)	2.2.2.2	G												
			Concrete Brick	2.2.2.4	G												
			SD-08 Manufacturer's Instructions														
			Admixtures for Masonry Mortar	2.4.1.3													
			Admixtures for Grout	2.4.2.2													
			SD-10 Operation and Maintenance														
			Data														
			Take-Back Program	3.8													
		05 12 00	SD-02 Shop Drawings														
			Fabrication Drawings	1.4.1	G												
			SD-03 Product Data														
			Shop Primer	2.6.2	G												
			Welding Electrodes and Rods	2.4.1	G												
			Direct Tension Indicator Washers	2.3.1.3	G												
			Non-Shrink Grout	2.4.2	G												
			Tension Control Bolts	2.3.2	G												
			SD-06 Test Reports														
			Bolts, Nuts, and Washers	2.3	G												

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		05 12 00	Weld Inspection Reports	3.7.1.2	G													
			Bolt Testing Reports	3.7.2.1	G													
			SD-07 Certificates															
			Steel	2.2	G													
			Bolts, Nuts, and Washers	2.3	G													
			Galvanizing	2.5	G													
			AISC Structural Steel Fabricator Quality Certification	1.3	G													
			Welding Procedures and Qualifications	1.4.2.1	G													
			Welding Electrodes and Rods	2.4.1	G													
			Certified Welding Inspector	3.7.1.1	G													
			NDT Technician	3.7.1.2	G													
			Welding Procedure Specifications (WPS)	3.4	G													
		05 30 00	SD-02 Shop Drawings															
			Fabrication Drawings	1.3.3	G													
			SD-03 Product Data															
			Accessories	2.2	G													
			Deck Units	2.1	G													
			Galvanizing Repair Paint	2.1.3	G													
			Touch-Up Paint	2.1.3														
			Welding Equipment	1.3.2														
			Welding Rods and Accessories	1.3.2	G													
			SD-05 Design Data															
			Deck Units	2.1	G													

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		05 30 00	SD-07 Certificates															
			Welder Qualifications	1.3.2	G													
			Welding Procedures	1.3.2	G													
			Manufacturer's Certificate	1.3.1	G													
		05 40 00	SD-02 Shop Drawings															
			Framing Components	1.5.1	G													
			SD-03 Product Data															
			Studs, Joists	2.1	G													
			SD-07 Certificates															
			Load-Bearing Cold-Formed Metal	1.4	G													
			Framing															
		05 50 13	SD-02 Shop Drawings															
			Floor Gratings	2.3	G													
			Bollards/Pipe Guards	2.4	G													
			Attic Hatch	2.7	G													
			SD-03 Product Data															
			Floor Gratings	2.3	G													
			Slotted Channel Framing	2.6	G													
			Attic Hatch	2.7	G													
		05 51 00	SD-02 Shop Drawings															
			Iron and Steel Hardware	2.1	G													
			Steel Shapes, Plates, Bars, and	2.1	G													
			Strips															
			Metal Stair System	2.2.1	G													
			SD-03 Product Data															
			Galvanized Carbon Steel Sheets	2.4.3	G													

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		05 51 00	Gray Iron Castings	2.4.4	G													
			Concrete Inserts	2.3.2	G													
			Masonry Anchorage Devices	2.3.3	G													
			Protective Coating	2.2.3	G													
			Steel Pan Stairs	2.2.2	G													
			Steel Stairs	2.3.1	G													
			SD-05 Design Data															
			Steel Stair Calculations	1.3.2	G													
			SD-07 Certificates															
			Welding Procedures	1.3.1	G													
			Welder Qualification	1.3.1	G													
			SD-08 Manufacturer’s Instructions															
			Gray Iron Castings	2.4.4														
			Protective Coating	2.2.3														
			Masonry Anchorage Devices	2.3.3	G													
		05 51 33	SD-02 Shop Drawings															
			Ladders	2.3	G													
			SD-03 Product Data															
			Ladders	2.3	G													
			SD-07 Certificates															
			Fabricator Certification for Ladder Assembly	1.3	G													
		05 52 00	SD-02 Shop Drawings															
			Fabrication Drawings	2.1	G													
			Iron and Steel Hardware	3.2	G													

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		05 52 00	Steel Shapes, Plates, Bars and	3.2	G													
			Strips															
			SD-03 Product Data															
			Structural-Steel Plates, Shapes,	2.2.1	G													
			and Bars															
			Structural-Steel Tubing	2.2.2	G													
			Masonry Anchorage Devices	2.2.4	G													
			Protective Coating	2.1.2	G													
			Steel Railings and Handrails	2.2.6	G													
			Anchorage and Fastening	3.2	G													
			Systems															
			SD-05 Design Data															
			Steel Railings and Handrail	1.3.3	G													
			Calculations															
			SD-07 Certificates															
			Welding Procedures	1.3.1	G													
			Welder Qualification	1.3.2	G													
			SD-08 Manufacturer’s Instructions															
			Installation Instructions	3.2														
		06 10 00	SD-03 Product Data															
			Adhesives	2.5.1	G													
			SD-06 Test Reports															
			Preservative-treated	1.4.3	G													
			SD-07 Certificates															
			Certificate of Grade	1.9.1	G													
			Preservative Treatment	1.7	G													

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		06 20 00	SD-02 Shop Drawings														
			Detail Drawings Indicating All	1.3	G												
			Wood Assemblies														
			SD-03 Product Data														
			Wood Products	2.1	G												
			Hardware and Accessories	2.4	G												
		06 41 16.00 10	SD-02 Shop Drawings														
			Shop Drawings	2.9	G												
			Installation	3.1													
			SD-03 Product Data														
			Wood Materials	2.1													
			High Pressure Decorative	2.3	G												
			Laminate (HPDL)														
		06 61 16	SD-02 Shop Drawings														
			Detail Drawings	1.4.2	G												
			Installation	3.1	G												
			SD-03 Product Data														
			Solid Polymer Material	2.1	G												
			Qualifications	1.4.1													
			Fabrications	2.3													
			SD-10 Operation and Maintenance														
			Data														
			Clean-up	3.2													
		07 13 53	SD-03 Product Data														
			Manufacturer's Standard Details	1.3	G												

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		07 13 53	Elastomeric Waterproofing Sheet	2.1	G												
			Material														
			Primers, Adhesives, and Mastics	1.4	G												
			Primers, Adhesives, and Mastics	2.1	G												
			SD-06 Test Reports														
			Elastomeric Waterproofing Sheet	2.1	G												
			Material														
			Field Quality Control	3.6	G												
			Protective Covering	3.7	G												
			SD-07 Certificates														
			Elastomeric Waterproofing Sheet	2.1													
			Material														
			Primers, Adhesives, and Mastics	1.4	G												
			Primers, Adhesives, and Mastics	2.1	G												
			Protective Coverings	1.4	G												
			Special Warranties	1.8	G												
			Certificates of Compliance		G												
			SD-08 Manufacturer’s Instructions														
			Primers, Adhesives, and Mastics	1.4	G												
			Primers, Adhesives, and Mastics	2.1	G												
		07 21 16	SD-03 Product Data														
			Blanket Insulation	2.1	G												
			Accessories	2.3													
			SD-08 Manufacturer’s Instructions														
			Insulation	3.3.1													
		07 22 00	SD-02 Shop Drawings														

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		07 22 00	Insulation Board Layout	1.3	G													
			SD-03 Product Data															
			Insulation	2.1	G													
			Fasteners	2.4	G													
			Moisture Control	2.3	G													
			Recycled Content For Insulation		S													
			Cover Board	2.2	G													
			SD-06 Test Reports															
			Flame Spread Rating	1.7.1														
			SD-07 Certificates															
			Installer Qualifications	1.6	G													
			Indoor Air Quality For Insulation		S													
			SD-08 Manufacturer’s Instructions															
			Fasteners	2.4														
			Insulation	2.1														
		07 31 13	SD-03 Product Data															
			Shingles	2.1.1														
			SD-04 Samples															
			Shingles	2.1.1	G													
			SD-08 Manufacturer’s Instructions															
			Application	3.3														
			SD-11 Closeout Submittals															
			Contractor's Warranty	1.5.1														
		07 42 13	SD-01 Preconstruction Submittals															
			Qualification of Manufacturer	1.5.3	G													

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		07 42 13	Qualification of Installation	1.5.4	G													
			Contractor															
			SD-02 Shop Drawings															
			Installation Drawings	1.5.1.1	G													
			SD-03 Product Data															
			Recycled Content;	2.1	G													
			Wall Panels	2.2.1	G													
			Factory Color Finish	2.2.2	G													
			Closure Materials	1.5.5	G													
			Pressure Sensitive Tape	2.5.4.4														
			Sealants and Caulking	2.5.4.1														
			Repair Paint															
			Accessories	1.5.5	G													
			Accessories	2.5	G													
			SD-04 Samples															
			Wall Panels	2.2.1	G													
			Fasteners	2.4	G													
			Metal Closure Strips	2.5.3	G													
			Color chart	2.2.2.2	G													
			Mock-up	1.5.7	G													
			SD-05 Design Data															
			Wind load design analysis	1.5.1.2	G													
			SD-06 Test Reports															
			Leakage Tests	3.7.2	G													
			Wind Load Tests	1.3.2	G													
			SD-08 Manufacturer’s Instructions															

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		07 42 13	Installation	3.3	G													
			SD-09 Manufacturer’s Field Reports															
			Manufacturer's Field Reports	3.8.1	G													
			SD-11 Closeout Submittals															
			Warranty	1.8	G													
			Maintenance Instructions	1.5.6	G													
			20 year 'No Dollar Limit' warranty for labor and material	1.8.1														
		07 42 63	SD-01 Preconstruction Submittals															
			Qualification of Manufacturer	1.5.3	G													
			Qualification of Installer	1.5.4	G													
			SD-02 Shop Drawings															
			Fabrication and Installation drawings	1.5.1														
			Wall Panel Assemblies	1.5.1	G													
			Flashing and Accessories	1.5.1	G													
			Anchorage Systems	1.5.1	G													
			SD-03 Product Data															
			sustainable acquisition	1.5.1	G													
			Manufacturer's catalog data	1.5.1														
			Factory Color Finish	1.5.1	G													
			Sub-girts and Formed Shapes	1.5.1	G													
			Closure Materials	1.5.1	G													
			Insulation	1.5.1	G													
			Pressure Sensitive Tape	1.5.1														

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		07 42 63	Sealants and Caulking	2.4.4.1														
			Accessories	1.5.1	G													
			SD-04 Samples															
			Wall Panel Assemblies	1.5.1	G													
			manufacturer's color charts and	1.5.1	G													
			chips															
			Mock-up	1.5.5	G													
			SD-05 Design Data															
			Wind Design Analysis	1.5.1	G													
			SD-06 Test Reports															
			Leakage Tests	3.7.2	G													
			Wind Load Tests	1.3.2	G													
			SD-07 Certificates															
			wall system assembly wind load	1.5.1														
			classification listings															
			SD-08 Manufacturer’s Instructions															
			Installation of Wall Panels	1.5.1	G													
			SD-11 Closeout Submittals															
			Warranty	1.8	G													
			Instructions	1.5.1														
			Safety Data Sheets	1.5.1														
			20 year 'No-Dollar-Limit' Warranty	1.5.1														
		07 52 00	SD-02 Shop Drawings															
			Roof Plan	1.4.6	G													
			SD-03 Product Data															
			Modified Bitumen Sheets	2.2	G													

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ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION	GOVT OR A/E REVWR	CONTRACTOR: SCHEDULE DATES SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	CONTRACTOR ACTION: ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY: DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		07 52 00	Cold-Applied Membrane	2.4	G													
			Adhesive															
			Primer	2.6	G													
			Pre-Manufactured Accessories	2.10														
			Warranty	1.8	G													
			SD-05 Design Data															
			Wind Uplift Calculations	1.4.5	G													
			SD-07 Certificates															
			Qualification of Manufacturer	1.4.1	G													
			Qualification of Applicator	1.4.2	G													
			Qualification of Engineer of	1.4.3	G													
			Record															
			SD-08 Manufacturer's Instructions															
			Modified Bitumen Membrane	3.3.5	G													
			Application															
			Flashing	3.3.6	G													
			Cold Adhesive Applied Modified	3.3.3.1	G													
			Bitumen Membrane															
			Cold Weather Installation	1.6	G													
			SD-11 Closeout Submittals															
			Warranty	1.8	G													
			Information Card	3.8	G													
			Instructions To Government	3.7														
			Personnel															
		07 57 14	SD-03 Product Data															

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		07 57 14	Sprayed (Bio-Based)	2.1	G												
			Polyurethane Foam Insulation														
			Protective (Ignition Barrier)	2.2	G												
			Coating														
			Primer	2.3													
			SD-04 Samples														
			Sprayed (Bio-Based)	2.1	G												
			Polyurethane Foam Insulation														
			SD-07 Certificates														
			Materials	2.1													
			Qualification of Manufacturer	1.4.1													
			Qualification of Applicator	1.4.2	G												
			SD-08 Manufacturer’s Instructions														
			Sprayed (Bio-Based)	2.1													
			Polyurethane Foam Insulation														
			Protective (Ignition Barrier)	2.2													
			Coating														
			Primer	2.3													
			Surface Area Preparation	3.3													
		07 62 13	SD-02 Shop Drawings														
			Sheet Metal	2.1.6	G												
			SD-03 Product Data														
			Contractor Quality Control	3.10													
			SD-04 Samples														
			Materials	2.1													
		07 81 00	SD-02 Shop Drawings														

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		07 81 00	Shop Drawings	1.6.4	G												
			SD-03 Product Data														
			Spray-Applied Fireproofing		G												
			Protection Materials	2.3.1	G												
			Documentation														
			Evaluation Reports	1.4.3	G												
			SD-06 Test Reports														
			Fire Resistance Rating	1.4.2	G												
			Field Tests	3.4	G												
			SD-07 Certificates														
			Installer Qualifications	1.6.1	G												
			Inspector Qualifications	1.6.2	G												
			Surface Preparation Report	3.1	G												
		07 84 00.00 22	SD-02 Shop Drawings														
			Firestopping System	2.1	G												
			SD-03 Product Data														
			Firestopping Materials	2.2	G												
			Firestopping Materials	3.1	G												
			SD-06 Test Reports														
			Inspection Forms	3.3.1	G												
			Inspection Report	3.3.2	G												
			SD-07 Certificates														
			Firestopping Inspector	1.5.2													
			Qualifications														
			Firestopping Materials	2.2													
			Firestopping Materials	3.1													

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(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		07 84 00.00 22	Firestopping Installer	1.5.1													
			Qualifications														
		07 92 00	SD-03 Product Data														
			Sealants	2.1	G												
			Primers	2.2													
			Bond Breakers	2.3													
			Backstops	2.4	G												
			SD-07 Certificates														
			Indoor Air Quality For Interior		S												
			Sealants														
		08 01 52	SD-02 Shop Drawings														
			Shop Drawings	1.4	G												
			Evaluation Survey	3.1	G												
			SD-03 Product Data														
			Hardware	2.4	G												
			Weatherstripping	3.2.10	G												
			Qualifications	1.4	G												
			SD-04 Samples														
			Hardware	2.4													
			Moldings	3.2.10													
			Weatherstripping	3.2.10													
			Mock-up (Repair)		G												
		08 11 13	SD-02 Shop Drawings														
			Doors	2.1	G												
			Doors	2.1	G												
			Frames	2.4	G												

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		08 11 13	Frames	2.4	G													
			Accessories	2.2														
			SD-03 Product Data															
			Doors	2.1	G													
			Frames	2.4	G													
			Accessories	2.2														
		08 14 00	SD-02 Shop Drawings															
			Doors	2.1	G													
			SD-03 Product Data															
			Doors	2.1	G													
			Accessories	2.2														
			Water-resistant Sealer	2.3.6														
			Warranty	1.5														
			SD-04 Samples															
			Door Finish Colors	2.3.5.2	G													
			SD-06 Test Reports															
			Cycle-Slam	2.4														
			Hinge Loading Resistance	2.4														
			SD-11 Closeout Submittals															
			Warranty	1.5														
		08 31 00	SD-03 Product Data															
			Access Doors And Panels	1.3	G													
			Hardware	1.3.1	G													
			Accessories	2.2.6	G													
			Recycled Content	2.1														
		08 33 23	SD-02 Shop Drawings															

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		08 33 23	Overhead Coiling Doors	2.2.1	G													
			Counterbalancing Mechanism	2.2.3	G													
			Manual Door Operators	2.2.4	G													
			Electric Door Operators	2.2.5	G													
			Bottom Bars	2.2.1.3	G													
			Guides	2.1.1.1	G													
			Mounting Brackets	2.2.3.1	G													
			Hood	2.2.2.2	G													
			Installation Drawings	2.1.1.1	G													
			SD-03 Product Data															
			Overhead Coiling Doors	2.2.1	G													
			Hardware	2.2.2	G													
			Counterbalancing Mechanism	2.2.3	G													
			Manual Door Operators	2.2.4	G													
			Electric Door Operators	2.2.5	G													
			Recycled content for steel curtain	2.2.1.1	S													
			slats															
			SD-05 Design Data															
			Overhead Coiling Doors	2.2.1	G													
			Hardware	2.2.2	G													
			Counterbalancing Mechanism	2.2.3	G													
			Electric Door Operators	2.2.5	G													
			SD-10 Operation and Maintenance															
			Data															
			Operation and Maintenance	1.3.2	G													
			Manuals															

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		08 33 23	SD-11 Closeout Submittals															
			Warranty	1.3.1	G													
		08 51 13	SD-02 Shop Drawings															
			Storm Windows	2.1	G													
			SD-03 Product Data															
			Hardware	2.1.5	G													
			Fasteners	2.1.4	G													
			Accessories	2.1.5	G													
			SD-04 Samples															
			Mock-Ups	1.4.2.1	G													
			SD-06 Test Reports															
			Survey	3.1	G													
		08 51 70	SD-02 Shop Drawings															
			Shop Drawings	1.5	G													
			SD-03 Product Data															
			Hardware	2.4	G													
			Weatherstripping	3.2.9	G													
			Qualifications	1.5	G													
			SD-04 Samples															
			Hardware	2.4	G													
			Weatherstripping	3.2.9	G													
			Survey	3.1.1	G													
		08 71 00	SD-02 Shop Drawings															
			Hardware Schedule	1.4	G													
			Keying System	2.3.6	G													
			SD-03 Product Data															

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		08 71 00	Hardware Items	2.3	G												
			SD-08 Manufacturer's Instructions														
			Installation	3.1													
			SD-10 Operation and Maintenance														
			Data														
			Hardware Schedule	1.4	G												
			SD-11 Closeout Submittals														
			Key Bitting	1.5.1													
		08 81 00	SD-03 Product Data														
			Glass		G												
			Glazing Accessories	1.3	G												
			Window Film		G												
			SD-08 Manufacturer's Instructions														
			Setting and Sealing Materials	2.2													
			Glass Setting	3.2													
		08 91 00	SD-02 Shop Drawings														
			Metal Louvers	1.4	G												
			Metal Louvers	2.2	G												
			SD-03 Product Data														
			Metal Louvers	1.4	G												
			Metal Louvers	2.2	G												
			SD-04 Samples														
			Metal Louver Samples	1.5	G												
		09 01 90.50	SD-03 Product Data														
			Work Plan	1.2	G												
			Materials	1.2													

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		09 01 90.50	Qualifications	1.4.4													
			SD-07 Certificates														
			Work Plan	1.2													
		09 01 90.60	SD-03 Product Data														
			Work Plan	1.2	G												
			Chemical Paint Removers	2.1													
			Qualifications	1.4.4													
			SD-04 Samples														
			Replacement Wood Flooring														
			Wood Putty														
			Stain														
			SD-07 Certificates														
			Work Plan	1.2													
		09 22 00	SD-02 Shop Drawings														
			Metal Support Systems	2.1	G												
			SD-03 Product Data														
			Metal Support Systems	2.1													
			Recycled Content for Metal Support Systems		S												
		09 29 00	SD-03 Product Data														
			Cementitious Backer Units	2.1.4	G												
			Glass Mat Water-Resistant Gypsum Tile Backing Board	2.1.3	G												
			Water-Resistant Gypsum Backing Board	2.1.2	G												
			Accessories	2.1.8													

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		09 29 00	Gypsum Board	2.1.1	G												
			Shaftwall Liner Panel	2.1.7	G												
			Recycled Content for Paper		S												
			Facing and Gypsum Cores														
			VOC Content of Joint Compound		S												
			SD-07 Certificates														
			Asbestos Free Materials	2.1	G												
			Indoor Air Quality for Gypsum	2.1.1													
			Board														
			SD-08 Manufacturer’s Instructions														
			Safety Data Sheets	2.1													
			SD-10 Operation and Maintenance														
			Data														
			Manufacturer Maintenance	2.1													
			Instructions														
		09 30 10	SD-03 Product Data														
			Porcelain Tile	2.1.1	G												
			Transition Strips	2.5.1	G												
			Mortar, Grout, and Adhesive	2.3	G												
			SD-08 Manufacturer’s Instructions														
			Manufacturer's Approved	3.8													
			Cleaning Instructions														
		09 51 00	SD-03 Product Data														
			Acoustical Units	2.2	G												
		09 65 00	SD-03 Product Data														

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		09 65 00	Resilient Flooring and	2.8	G												
			Accessories														
			Adhesives	2.4													
			Solid Vinyl Tile	2.1													
			Wall Base	2.2													
			SD-08 Manufacturer’s Instructions														
			Surface Preparation	3.2	G												
			Installation	3.1	G												
			SD-10 Operation and Maintenance														
			Data														
			Resilient Flooring and	2.8	G												
			Accessories														
		09 67 23.13	SD-02 Shop Drawings														
			Installation Drawings	2.1	G												
			SD-04 Samples														
			Hardboard Mounted Epoxy	1.5.3	G												
			Flooring														
			Mockups	1.5.1	G												
			SD-05 Design Data														
			Design Mix Data	1.2.3	G												
			SD-07 Certificates														
			Listing of Product Installations	1.5.2													
			Referenced Standards	1.5													
			Certificates														
			SD-11 Closeout Submittals														
			Warranty	1.6	G												

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		09 68 00	SD-03 Product Data														
			Carpet	2.1	G												
			Carpet	2.3	G												
			Carpet	2.5	G												
			Moldings	2.8	G												
			SD-06 Test Reports														
			Moisture and Alkalinity Tests	3.2	G												
			SD-08 Manufacturer’s Instructions														
			Surface Preparation	3.1													
			SD-10 Operation and Maintenance														
			Data														
			Cleaning and Protection	3.5	G												
			SD-11 Closeout Submittals														
			Warranty	1.5	G												
		09 90 00	SD-03 Product Data														
			Coating	2.1	G												
			Product Data Sheets	2.1													
			Sealant	3.3.5													
			SD-08 Manufacturer’s Instructions														
			Application Instructions	3.4.1													
			Mixing	2.1													
			SD-10 Operation and Maintenance														
			Data														
			Coatings	2.1	G												
		10 14 00.20	SD-02 Shop Drawings														
			Detail Drawings	1.4.1	G												

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		10 14 00.20	SD-03 Product Data														
			Installation	3.1	G												
			Warranty	1.6	G												
			SD-04 Samples														
			Software	1.3	G												
			SD-10 Operation and Maintenance														
			Data														
			Protection and Cleaning	3.1.1	G												
		10 21 13	SD-02 Shop Drawings														
			Fabrication Drawings	2.1	G												
			Installation Drawings	3.2	G												
			SD-03 Product Data														
			Cleaning and Maintenance	2.1	G												
			Instructions														
			Colors And Finishes	2.7													
			Anchoring Devices and Fasteners	2.2.1													
			Hardware and Fittings	2.2.3													
			Brackets	2.2.2													
			Door Hardware	2.2.4													
			Toilet Enclosures	2.3.1													
			Urinal Screens	2.3.2													
			Pilaster Shoes	2.5													
			Finishes	2.2.3.2	G												
			SD-07 Certificates														
			Warranty	1.5	G												

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ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION GOVT OR A/E REVWR	CONTRACTOR: SCHEDULE DATES SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	CONTRACTOR ACTION ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		10 21 13	SD-10 Operation and Maintenance														
			Data														
			Plastic Identification	2.1.1	G												
		10 26 00	SD-03 Product Data														
			Wall Panels	2.2	G												
			SD-10 Operation and Maintenance														
			Data														
			Wall Panels	2.2	G												
		10 28 13	SD-03 Product Data														
			Finishes	2.1.2	G												
			Accessory Items	2.2	G												
			SD-04 Samples														
			Finishes	2.1.2	G												
			Accessory Items	2.2													
		10 44 16	SD-02 Shop Drawings														
			Fire Extinguishers	2.1.1	G												
			Accessories	Part 2	G												
			Cabinets	Part 2	G												
			Wall Brackets	2.2.2	G												
			Schedule	1.5	G												
			SD-03 Product Data														
			Fire Extinguishers	2.1.1	G												
			Accessories	Part 2	G												
			Cabinets	Part 2	G												
			Wall Brackets	2.2.2	G												
			Replacement Parts List	3.2.1	G												

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(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		10 44 16	SD-04 Samples															
			Equipment Samples	1.3.1	G													
			SD-07 Certificates															
			Fire Extinguishers Certifications	2.1.1	G													
			Manufacturer's Warranty with	1.4	G													
			Inspection Tag															
		12 00 01.00 20	SD-01 Preconstruction Submittals															
			FF&E Schedule and Schedule	3.1.1.3	G													
			Updates															
			Contractor's Interior Designer's	3.1.1.1	G													
			Qualifications															
			SD-04 Samples															
			BVD FF&E Mock-Up	3.1.2.6	G													
			Final FF&E Mock-Up	3.1.2.9	G													
			Interior Finish Construction	3.1.2.13	G													
			Submittals															
			Post Option Award FF&E Finish	3.2.2	G													
			Submittals															
			SD-05 Design Data															
			FF&E Basis of Design Package	3.1.2.2.2	G													
			FF&E BVD Request for Quotation	3.1.2.5	G													
			(RFQ) Package															
			BVD Package	3.1.2.7	G													
			Pre-Final FF&E Package 'Over	3.1.2.10	G													
			the Shoulder'															
			Pre-Final FF&E Package	3.1.2.11	G													

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		12 00 01.00 20	Final FF&E Package	3.1.2.9	G													
			Final FF&E Package	3.1.2.12	G													
			SD-10 Operation and Maintenance															
			Data															
			Warranty and Maintenance	3.5.1	G													
			Information															
			SD-11 Closeout Submittals															
			Punch List(s) Submittal	3.3.2	G													
		14 24 23	SD-02 Shop Drawings															
			Elevator	2.1	G													
			Elevator Components	1.2.1	G													
			Elevator Components	1.2.2	G													
			Elevator Machine		G													
			Elevator Controller	1.2.1	G													
			Wiring Diagrams	1.3.4	G													
			Elevator Machine Room		G													
			Coordination Drawings															
			SD-03 Product Data															
			Elevator	2.1	G													
			Elevator Components	1.2.1	G													
			Elevator Components	1.2.2	G													
			Data Sheets	1.2.2	G													
			Elevator Microprocessor	2.5.2	G													
			Controller															
			Auxiliary Power System	1.2.3.3	G													
			Oil Heater	1.2.1	G													

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		14 24 23	Oil Cooler	1.2.1	G												
			SD-05 Design Data														
			Heat Loads	1.2.3.2													
			Reaction Loads	1.2.3.1													
			SD-07 Certificates														
			Qualified Elevator Inspector (QEI)	3.2													
			Elevator Parts and Components		G												
			Price Lists														
			Warranty	1.4	G												
			Elevator Technicians'		G												
			Qualifications														
			Endorsement Letter	1.3.1.1	G												
			Welders' Qualifications	1.2.4	G												
			Elevator Controller Certification	2.5.2.3	G												
			SD-10 Operation and Maintenance														
			Data														
			Elevator	2.1	G												
			Maintenance Control Program	1.2.5	G												
			(MCP)														
			Software and Documentation	2.5.2.2	G												
		21 13 13	SD-01 Preconstruction Submittals														
			Qualified Fire Protection Engineer	1.2.3	G												
			(QFPE)														
			Sprinkler System Designer	1.4.2.1	G												
			Sprinkler System Installer	1.4.2.2	G												
			SD-02 Shop Drawings														

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		21 13 13	Shop Drawing	1.4.1.1	G												
			SD-03 Product Data														
			Piping	2.2	G												
			Fittings	2.2.1.2	G												
			Valves	2.2.4	G												
			Relief Valves	2.7.5	G												
			Sprinklers	2.6	G												
			Pipe Hangers and Supports	2.2.3	G												
			Sprinkler Alarm Switch	2.3.1	G												
			Valve Supervisory (Tamper)	2.3.2	G												
			Switch														
			Fire Department Connection	2.5	G												
			Backflow Prevention Assembly	2.4	G												
			Air Vent	2.7.6	G												
			Hose Valve	2.4.1	G												
			Nameplates	2.1.2	G												
			SD-05 Design Data														
			Hydraulic Calculations	1.2.1.2	G												
			SD-06 Test Reports														
			Test Procedures	3.6.1	G												
			SD-07 Certificates														
			Verification of Compliant	3.6.2.1	G												
			Installation														
			Request for Government Final	3.6.2.2	G												
			Test														

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		21 13 13	SD-10 Operation and Maintenance														
			Data														
			Operating and Maintenance	3.8	G												
			(O&M) Instructions														
			Spare Parts	1.6	G												
			SD-11 Closeout Submittals														
			As-built drawings	3.8													
		21 30 00	SD-01 Preconstruction Submittals														
			Fire Pump Installation Related	1.3													
			Submittals														
			Qualified Fire Protection Engineer	1.7.1	G												
			(QFPE)														
			SD-02 Shop Drawings														
			Installation Drawings	3.3.1	G												
			As-Built Drawings	3.10.2	G												
			Piping Layout	3.3.2	G												
			Pump Room	3.3.2	G												
			SD-03 Product Data														
			Catalog Data	2.1	G												
			Spare Parts	1.6													
			Preliminary Tests	3.8.2													
			Field Tests	3.8	G												
			Manufacturer's Representative	1.7.6	G												
			Field Training	3.10.1	G												
			Navy Formal Inspection and	3.8.3													
			Tests														

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		21 30 00	SD-06 Test Reports														
			Preliminary Tests	3.8.2													
			Navy Formal Inspection and	3.8.3	G												
			Tests														
			SD-07 Certificates														
			Qualified Fire Protection Engineer	1.7.1													
			(QFPE)														
			Qualifications of Welders	1.7.2													
			Qualifications of Installer	1.7.3													
			Preliminary Test Certification	1.7.4													
			Final Test Certification	1.7.5													
			SD-10 Operation and Maintenance														
			Data														
			Operating and Maintenance	3.10.1	G												
			Instructions														
			Flow Meter	2.13													
		22 00 00	SD-02 Shop Drawings														
			Coordination Information	1.2													
			SD-03 Product Data														
			Pipe Materials	2.1.1	G												
			Pipe Fittings	2.1.2	G												
			Valves	2.3	G												
			Valve Operating Mechanisms	2.3.1	G												
			Backflow Preventers	2.7	G												
			Thermostatic Mixing Valves	2.3.5	G												
			Wall Hydrant	2.3.4	G												

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(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		22 00 00	Automatic Flow Control Valves	2.3.7	G												
			Coffee Maker Outlet Box	2.4.1	G												
			Flush Valve Water Closets	2.5.1	G												
			Flush Valve Urinals	2.5.2	G												
			Wall Hung Lavatories	2.5.4	G												
			Countertop Lavatories	2.5.3	G												
			General Purpose Sinks	2.5.5	G												
			Utility Sink	2.5.6	G												
			Three-Station Sink	2.5.7	G												
			Mop Receptors	2.5.8	G												
			Electric Water Coolers	2.5.9	G												
			Emergency Eye and Face Wash	2.5.10	G												
			Expansion Tanks	2.9	G												
			Electric Water Heater	2.10	G												
			Pumps	2.11	G												
			Trap Primer Assembly	2.15.6	G												
			Water Hammer Arresters	3.1.1.8	G												
			Hose Bibbs	2.3.3	G												
			Floor Drains	2.6.1	G												
			Floor Sinks	2.6.2	G												
			Compressed Air System	2.12	G												
			SD-06 Test Reports														
			Tests, Flushing and Disinfection	3.7	G												
			Post Installation Sanitary System	3.7.1.3	G												
			Survey														

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		22 00 00	SD-10 Operation and Maintenance															
			Data															
			Fixtures	2.5	G													
			Plumbing System	3.7.1	G													
		23 03 00.00 20	SD-02 Shop Drawings															
			ATFP Bracing Details	1.10.1	G													
			SD-03 Product Data															
			ATFP Bracing Components	1.10.1	G													
		23 05 48.00 40	SD-02 Shop Drawings															
			Installation Drawings	1.2	G													
			Outline Drawings	1.2	G													
			SD-03 Product Data															
			Equipment and Performance	1.2	G													
			Data															
			Isolators	2.2.6	G													
			SD-06 Test Reports															
			Type of Isolator	2.4	G													
			Type of Base	2.4	G													
			Allowable Deflection	2.4	G													
			Measured Deflection	2.4	G													
		23 05 93	SD-01 Preconstruction Submittals															
			Records of Existing Conditions	1.3.3	G													
			Independent TAB Agency and	1.5.1	G													
			Personnel Qualifications															
			TAB Design Review Report	1.5.4.2	G													
			SD-02 Shop Drawings															

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						CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION			APPROVING AUTHORITY					
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		23 05 93	TAB Schematic Drawings and Report Forms	1.3.3	G												
			SD-03 Product Data														
			Equipment and Performance Data	1.3	G												
			TAB Related HVAC Submittals	1.5.1.3	G												
			SD-06 Test Reports														
			Completed Pre-Final DALT Report	3.2.5	G												
			Certified Final DALT Report	3.2.8	G												
			Prerequisite HVAC Work Checkout List	1.5.4.4	G												
			Prerequisite HVAC Work Checkout List	1.5.4.4	G												
			Prerequisite HVAC Work Checkout List	1.5.4.4	G												
			Proportional Balancing	3.3.6.1	G												
			Season 1	3.3.6.2	G												
			Season 2	3.3.6.2	G												
			SD-07 Certificates														
			Independent TAB Agency and Personnel Qualifications	1.5.1	G												
			DALT and TAB Submittal and Work Schedule	1.5.4.1	G												
			TAB Pre-Field Engineering Report	1.5.4.4	G												

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(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		23 05 93	Instrument Calibration	1.5.5	G												
			Certificates														
			DALT and TAB Procedures	3.6	G												
			Summary														
			Completed Pre-Final DALT Work	3.6	G												
			Checklist														
			Advance Notice of Pre-Final	3.2.2	G												
			DALT Field Work														
			Proportional Balancing	3.3.6.1	G												
			Season 1	3.3.6.2	G												
			Season 2	3.3.6.2	G												
		23 07 00	SD-02 Shop Drawings														
			Coordination Information	1.3													
			SD-03 Product Data														
			Pipe Insulation Systems	2.3	G												
			Pipe Insulation Systems	3.2	G												
			Duct Insulation Systems	3.3	G												
			Equipment Insulation Systems	3.4	G												
			SD-08 Manufacturer’s Instructions														
			Pipe Insulation Systems	2.3													
			Pipe Insulation Systems	3.2													
			Duct Insulation Systems	3.3													
			Equipment Insulation Systems	3.4													
		23 08 00	SD-06 Test Reports														
			Pipe Flushing, Testing, And	1.5	G												
			Water Treatment Reports														

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		23 08 00	Completed Pre-Functional	3.3	G													
			Checklists															
			Seasonal Test Report	3.12	G													
			Full-Load Test Report	3.13	G													
			SD-07 Certificates															
			Certificate Of Readiness	1.6	G													
		23 09 00.00 22	SD-02 Shop Drawings															
			DDC Contractor Design Drawings	3.3	G													
			Draft As-Built Drawings	3.3	G													
			Final As-Built Drawings	3.3	G													
			SD-03 Product Data															
			Programming Software	1.8.3	G													
			Controller Application Programs	1.8.4	G													
			Configuration Software	1.8.1	G													
			Controller Configuration Settings	1.8.2	G													
			Manufacturer's Product Data	2.2	G													
			Niagara Framework Supervisory	1.8.5	G													
			Gateway Backups															
			SD-06 Test Reports															
			Existing Conditions Report	3.1.1	G													
			Pre-Construction Quality Control	1.9.1	G													
			(QC) Checklist															
			Post-Construction Quality Control	1.9.2	G													
			(QC) Checklist															
			Start-Up Testing Report	3.5.2	G													

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ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION	GOVT OR A/E REVWR	CONTRACTOR: SCHEDULE DATES SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	CONTRACTOR ACTION ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		23 09 00.00 22	Control Contractor's Performance	3.6.5	G													
			Verification Testing Plan															
			Equipment Supplier's	3.6.3.1	G													
			Performance Verification Testing															
			Plan															
			Endurance Testing Results	3.6.8.3	G													
			Performance Verification Test	3.6.9	G													
			Report															
			SD-10 Operation and Maintenance															
			Data															
			Operation and Maintenance	3.7	G													
			(O&M) Instructions															
			Training Documentation	3.9.1	G													
			SD-11 Closeout Submittals															
			Enclosure Keys	2.5	G													
			Password Summary Report	3.2.7.1	G													
			Closeout Quality Control (QC)	1.9.3	G													
			Checklist															
		23 21 13.00 20	SD-02 Shop Drawings															
			Hot water heating system	1.3.1	G													
			SD-03 Product Data															
			Finned tube radiators and	2.4.1	G													
			convectors															
			Calibrated balancing valves	2.1.6.6	G													
			Combination strainer and pump	2.1.10	G													
			suction diffuser															

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(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		23 21 13.00 20	Pumps	2.3.1	G													
			Expansion tanks	2.3.2	G													
			Air separation tanks	2.3.3	G													
			Hot water heating pipe	2.1.1	G													
			Bypass chemical feeder	2.9.1	G													
			SD-06 Test Reports															
			Hydrostatic test of piping system	3.4.1	G													
			Auxiliary equipment and accessory tests	3.4.2	G													
			SD-07 Certificates															
			Welding procedures	1.5.2.1														
			Welder's qualifications	1.5.2.2														
			SD-10 Operation and Maintenance Data															
			Finned Tube Radiators and Convectors	2.4.1	G													
			Pumps	2.3.1	G													
			Calibrated Balancing Valves	2.1.6.6	G													
			Combination Strainer and Pump Suction Diffuser	2.1.10	G													
			Expansion tanks	2.3.2	G													
			Air separation tanks	2.3.3	G													
		23 22 26.00 20	SD-03 Product Data															
			Convertors	2.3	G													
			Condensate Return Pumping Units	2.4.1	G													

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		23 22 26.00 20	Valves	2.2.7	G												
			Valve Operating Mechanism	2.2.7.7	G												
			Steam Meters	2.2.11.1	G												
			Traps	2.2.11.3	G												
			Strainers	2.2.11.4	G												
			Steam Separators	2.2.11.5	G												
			Flash Tanks	2.2.11.8	G												
			Expansion Joints	2.2.9	G												
			Instrumentation	2.2.10	G												
			SD-06 Test Reports														
			Steam Piping	2.2.1	G												
			Copper Tubing	2.2.8.2													
			Valves	2.2.7													
			Instrumentation	2.2.10													
			Pipe and Pipe System	2.2	G												
			Convertors	2.3	G												
			Condensate Return Pumping	2.4.1	G												
			Units														
			SD-07 Certificates														
			Welding Procedure	1.4.1	G												
			Welder's Performance	1.4.2	G												
			Qualification Record														
			List of Welders and Welder's	1.4.2													
			Symbols														
			SD-08 Manufacturer’s Instructions														
			Convertors	2.3	G												

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		23 22 26.00 20	Condensate Return Pumping	2.4.1	G													
			Units															
		23 23 00	SD-02 Shop Drawings															
			Refrigerant Piping System	2.3	G													
			SD-03 Product Data															
			Refrigerant Piping System	2.3	G													
			Spare Parts	1.5.2														
			Refrigerant Piping Tests	3.5														
			Verification of Dimensions	3.1														
			SD-06 Test Reports															
			Refrigerant Piping Tests	3.5	G													
			SD-07 Certificates															
			Service Organization	2.1														
			SD-10 Operation and Maintenance															
			Data															
			Maintenance	1.5	G													
			Operation and Maintenance	3.4	G													
			Manuals															
			Demonstrations	3.4	G													
		23 30 00	SD-02 Shop Drawings															
			Detail Drawings	1.5.4	G													
			SD-03 Product Data															
			Insulated Nonmetallic Flexible	2.7.1.1	G													
			Duct Runouts															
			Duct Connectors	2.7.1.1	G													
			Duct Access Doors	2.7.2	G													

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						CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION			APPROVING AUTHORITY					
ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION GOVT OR A/E REVWR	SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		23 30 00	Fire Dampers	2.7.3	G												
			Manual Balancing Dampers	2.7.4	G												
			Diffusers	2.7.5.1	G												
			Registers and Grilles	2.7.5.2	G												
			Centrifugal Fans	2.8.1.1	G												
			In-Line Centrifugal Fans	2.8.1.2	G												
			Air Handling Units	2.9	G												
			Blower-Coil Units	2.10.1	G												
			Energy Recovery Devices	2.11	G												
			Test Procedures	1.5.5													
			Indoor Air Quality for Duct	2.7.1	G												
			Sealants														
			SD-06 Test Reports														
			Performance Tests	3.10	G												
			SD-07 Certificates														
			Ozone Depleting Substances	1.5.3	G												
			Technician Certification														
			SD-08 Manufacturer’s Instructions														
			Manufacturer's Installation	3.2													
			Instructions														
			Operation and Maintenance	3.12.2													
			Training														
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.12.1	G												
			Manuals														

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		23 30 00	Fire Dampers	2.7.3	G													
			Manual Balancing Dampers	2.7.4	G													
			Centrifugal Fans	2.8.1.1	G													
			In-Line Centrifugal Fans	2.8.1.2	G													
			Air Handling Units	2.9	G													
			Energy Recovery Devices	2.11	G													
			SD-11 Closeout Submittals															
			Indoor Air Quality During	3.11	G													
			Construction															
		23 81 00	SD-03 Product Data															
			Spare Parts	3.7.1														
			System Performance Tests	3.6	G													
			Training	3.4	G													
			Inventory	1.5														
			Supplied Products	2.1	G													
			Manufacturer's Standard Catalog	2.2	G													
			Data															
			SD-06 Test Reports															
			Refrigerant Tests, Charging, and	3.5	G													
			Start-Up															
			System Performance Tests	3.6	G													
			SD-07 Certificates															
			Service Organizations	3.7.2														
			SD-10 Operation and Maintenance															
			Data															

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		23 81 00	Operation and Maintenance	3.4		G												
			Manuals															
			SD-11 Closeout Submittals															
			Ozone Depleting Substances	2.2.2.3		S												
		23 82 00.00 20	SD-03 Product Data															
			Unit Heaters	2.1		G												
			SD-10 Operation and Maintenance															
			Data															
			Unit Heaters	2.1														
		25 05 11	SD-01 Preconstruction Submittals															
			Wireless and Wired Broadcast	3.2.2.2		G												
			Communication Request															
			Device Account Lock Exception	3.3.2		G												
			Request															
			Multiple Ethernet Connection	3.2.4.1		G												
			Device Request															
			Contractor Computer	1.9.1.6		G												
			Cybersecurity Compliance															
			Statements															
			Contractor Temporary Network	1.9.6		G												
			Cybersecurity Compliance															
			Statements															
			Cybersecurity Interconnection	1.7.2		G												
			Schedule															
			Proposed STIG and SRG	1.7.1		G												
			Applicability Report															

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION GOVT OR A/E REVWR	CONTRACTOR: SCHEDULE DATES SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	CONTRACTOR ACTION ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		25 05 11	Pre-Construction Control System	1.7.4	G												
			Inventory Report														
			Contractor Personnel	1.6.1	G												
			Certifications														
			SD-02 Shop Drawings														
			Network Communication, Ports,	1.7.3	G												
			Protocols and Services Report														
			Cybersecurity Network (Riser)	1.7.7	G												
			Diagram														
			System Data Flow Diagram	1.7.5	G												
			SD-03 Product Data														
			Control System Cybersecurity	1.7.9	G												
			Documentation														
			SD-06 Test Reports														
			Wireless Communication Test	3.2.2.3	G												
			Report														
			Control System Cybersecurity	3.13.1	G												
			Testing Procedures														
			Control System Cybersecurity	3.13.3	G												
			Testing Report														
			Antivirus/Antimalware Scan	3.12.1	G												
			Results														
			SD-07 Certificates														
			Software Licenses	1.8	G												
			SD-11 Closeout Submittals														

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		25 05 11	Password Change Summary	3.4.5.5	G													
			Report															
			Enclosure Keys	3.3.7	G													
			Software and Configuration	1.7.6	G													
			Backups															
			Device Audit Record Upload	3.5.4.2	G													
			Software															
			System Maintenance Tool	3.8	G													
			Software															
			Control System Scanning Tools	3.10.2	G													
			STIG, SRG and Vendor Guide	1.7.8	G													
			Compliance Result Report															
			Final Control System Inventory	1.7.4	G													
			Report															
		25 08 10	SD-06 Test Reports															
			PVT Plan	3.1.1	G													
			PVT Phase I Report	3.1.2.1	G													
			PVT Phase II Report	3.1.2.2	G													
			SD-07 Certificates															
			Test Instrumentation Calibration	1.4	G													
			Certificates															
		25 10 10.00 22	SD-02 Shop Drawings															
			UMCS Contractor Design	3.2.2	G													
			Drawings															
			Draft As-Built Drawings	3.2.3	G													
			Final As-Built Drawings	3.2.3	G													

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		25 10 10.00 22	SD-03 Product Data															
			Product Data Sheets	2.1.5	G													
			Computer Software		G													
			Enclosure Keys		G													
			SD-05 Design Data															
			UMCS IP Network Bandwidth	3.2.1	G													
			Usage Estimate															
			SD-06 Test Reports															
			Pre-Construction QC Checklist	1.7	G													
			Post-Construction QC Checklist	1.7	G													
			Existing Conditions Report	3.1	G													
			Start-Up and Start-Up Testing	3.4	G													
			Report															
			PVT Phase I Procedures	3.5.1	G													
			PVT Phase I Report	3.5.2	G													
			PVT Phase II Report	3.5.3	G													
			SD-10 Operation and Maintenance															
			Data															
			Operation and Maintenance	1.8	G													
			(O&M) Instructions															
			Preventive Maintenance Work	3.6.6.1	G													
			Plan															
			Basic Training Documentation	3.7.1	G													
			Advanced Training	3.7.1	G													
			Documentation															

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		25 10 10.00 22	Refresher Training	3.7.1	G													
			Documentation															
			SD-11 Closeout Submittals															
			Closeout QC Checklist	1.7	G													
		26 08 00	SD-06 Test Reports															
			Acceptance Tests and	3.1	G													
			Inspections															
			SD-07 Certificates															
			Qualifications	1.4.1	G													
			Acceptance Test and Inspections	1.4.3	G													
			Procedure															
		26 20 00	SD-02 Shop Drawings															
			Panelboards	2.13	G													
			Transformers	2.15	G													
			Cable Trays	2.3	G													
			Wireways	2.27	G													
			SD-03 Product Data															
			Receptacles	2.12	G													
			Circuit Breakers	2.13.3	G													
			Switches	2.10	G													
			Transformers	2.15	G													
			Enclosed Circuit Breakers	2.14	G													
			Motor Controllers	2.17	G													
			Combination Motor Controllers	2.19.1	G													
			Manual Motor Starters	2.18	G													
			Secondary Bonding Busbar	2.22.3	G													

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		26 20 00	Surge Protective Devices	2.28	G													
			Cable Trays	2.3	G													
			SD-06 Test Reports															
			600-volt Wiring Test	3.5.2	G													
			Grounding System Test	3.5.5	G													
			Transformer Tests	3.5.3	G													
			Ground-fault Receptacle Test	3.5.4	G													
			SD-07 Certificates															
			Fuses	2.11	G													
			SD-09 Manufacturer's Field															
			Reports															
			Transformer Factory Tests	2.30.1														
			SD-10 Operation and Maintenance															
			Data															
			Electrical Systems	1.6.1	G													
		26 24 13	SD-02 Shop Drawings															
			Switchboard Drawings	1.5.2	G													
			SD-03 Product Data															
			Switchboard	2.2	G													
			SD-06 Test Reports															
			Switchboard Design Tests	2.5.2	G													
			Switchboard Production Tests	2.5.3	G													
			Acceptance Checks and Tests	3.5.1	G													
			SD-10 Operation and Maintenance															
			Data															

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(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		26 24 13	Switchboard Operation and	1.6.1	G													
			Maintenance															
			SD-11 Closeout Submittals															
			Assembled Operation and	1.6.2	G													
			Maintenance Manuals															
			Equipment Test Schedule	2.5.1	G													
			Required Settings	3.5	G													
			Service Entrance Available Fault	2.7	G													
			Current Label															
		26 27 14.00 20	SD-02 Shop Drawings															
			Installation Drawings	1.4.1	G													
			SD-03 Product Data															
			Electricity Meters	2.1	G													
			Current Transformer	2.1.2	G													
			Communications	2.2	G													
			SD-06 Test Reports															
			Acceptance Checks and Tests	3.2.1	G													
			System Functional Verification	3.2.2	G													
			Building Meter Installation Sheet	3.2.1	G													
			Meter Configuration Template	2.1.1	G													
			Meter Configuration Report	3.2.1	G													
			SD-10 Operation and Maintenance															
			Data															
			Electricity Meters and	1.5.1	G													
			Accessories															
			SD-11 Closeout Submittals															

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		26 27 14.00 20	System Functional Verification	3.2.2	G													
		26 29 23	SD-02 Shop Drawings															
			Schematic Diagrams	1.5.1	G													
			Interconnecting Diagrams	1.5.2	G													
			Installation Drawings	1.5.3	G													
			As-Built Drawings	1.5.3	G													
			SD-03 Product Data															
			Adjustable Speed Drives	2.1	G													
			Wires and Cables	2.3														
			Equipment Schedule	1.5.4														
			SD-06 Test Reports															
			ASD Test	3.3.1														
			Performance Verification Tests	3.3.2														
			Endurance Test	3.3.3														
			SD-07 Certificates															
			Testing Agency's Field	3.3.1	G													
			Supervisor															
			SD-08 Manufacturer’s Instructions															
			Installation instructions	1.5.5														
			SD-09 Manufacturer’s Field															
			Reports															
			ASD Test Plan	2.5.1	G													
			Standard Products	1.5.6														
			SD-10 Operation and Maintenance															
			Data															
			Adjustable Speed Drives	2.1														

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		26 32 15	SD-02 Shop Drawings														
			Engine-Generator Set and	2.1.1	G												
			Auxiliary Equipment														
			Auxiliary Systems	1.4.9	G												
			Detailed Drawings	1.4.8	G												
			Acceptance	3.17	G												
			SD-03 Product Data														
			Engine-Generator Set	2.1.1	G												
			Efficiencies														
			Emissions	2.1.1	G												
			Emissions	2.11	G												
			Filters	2.7.2	G												
			Special Tools	2.13	G												
			Special Tools	2.13	G												
			Remote Alarm Annunciator	1.4.9	G												
			Remote Alarm Annunciator	2.12.4	G												
			Engine-Generator Parameter	2.1.1													
			Schedule														
			Heat Exchanger	2.8.2													
			Generator	2.14													
			Manufacturer's Catalog	2.5													
			Spare Parts	1.6.2													
			Onsite Training	3.12													
			Vibration-Isolation	1.4.4													
			Posted Data and Instructions	3.16	G												
			Instructions	3.6.7.1	G												

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		26 32 15	Experience	1.4.6													
			Field Engineer	1.4.7													
			General Installation	3.2													
			Exciter	2.15													
			SD-05 Design Data														
			Performance Criteria	2.14													
			Sound Limitations	2.9	G												
			Integral Main Fuel Storage Tank	2.6.4													
			Power Factor	3.6.1.2													
			Heat Exchanger	2.8.2													
			Time-Delay on Alarms	2.18.5													
			Cooling System	2.8													
			Vibration Isolation	1.4.4													
			Battery Charger	2.2													
			Battery Charger	2.12.3.2													
			Capacity Calculations for	2.1.1	G												
			Engine-Generator Set														
			Brake Mean Effective Pressure	2.1.1	G												
			(BMEP) Calculations														
			Torsional Vibration Stress	2.1.1	G												
			Analysis Computations														
			Capacity Calculations for	2.1.1	G												
			Batteries														
			Turbocharger Load Calculations	2.1.1	G												
			SD-06 Test Reports														
			Performance Tests	3.6.9													

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		26 32 15	Factory Inspection and Tests	2.25													
			Factory Tests	2.25.2													
			Onsite Inspection and Tests	3.6	G												
			Acceptance Checks and Tests	3.9.2	G												
			Functional Acceptance Tests	3.10.2	G												
			Maintenance Procedures	3.12	G												
			Operation and Maintenance	3.12	G												
			Manuals														
			Inspections	3.6.3	G												
			Functional Acceptance Test	3.9.6	G												
			Procedure														
			SD-07 Certificates														
			Cooling System	2.8													
			Vibration Isolation	1.4.4													
			Prototype Test	2.25.2													
			Reliability and Durability	2.1.5													
			Fuel System Certification	1.4.11	G												
			Start-Up Engineer	3.8	G												
			Instructor's Qualification Resume	3.11.1	G												
			Engine Emission Limits	2.1.2.1	G												
			Sound Limitations	2.9													
			Site Visit	3.1													
			Current Balance	2.14.1													
			Materials and Equipment	2.4													
			Factory Inspection and Tests	2.25													
			Engine Tests	3.6.4	G												

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		26 32 15	Generator Tests	3.6.5	G													
			Assembled Engine-Generator Set Tests	3.6.6	G													
			SD-10 Operation and Maintenance Data															
			Preliminary Assembled Operation and Maintenance Manuals	3.9.5	G													
			Posted Data and Instructions	3.16	G													
			Training Plan	3.11.2	G													
		26 41 00	SD-01 Preconstruction Submittals															
			Lightning Protection System Installers Documentation	1.2.3	G													
			SD-02 Shop Drawings															
			Overall Lightning Protection System	1.4.1.1	G													
			Each Major Component	1.4.1.2	G													
			SD-06 Test Reports															
			Lightning Protection and Grounding System Test Plan	1.4.3	G													
			Grounding Systems Testing	3.3.2	G													
			SD-07 Certificates															
			Component UL Listed and Labeled	1.4.2	G													
			Lightning Protection System Inspection Certificate	1.4.4	G													
			Roof Manufacturer's Warranty	3.1.1.1	G													

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		26 41 00	UL Master Label Certification	3.3.3	G													
		26 51 00	SD-02 Shop Drawings															
			Luminaire Drawings	1.5.1	G													
			Occupancy/Vacancy Sensor	1.5.8	G													
			Coverage Layout															
			Lighting Control System One-Line	1.7.2	G													
			Diagram															
			Sequence of Operation for	2.5.1	G													
			Lighting Control System															
			SD-03 Product Data															
			Luminaires	2.2	G													
			Light Sources	2.3	G													
			LED Drivers	2.4	G													
			Luminaire Warranty	1.6.1	G													
			Lighting Controls Warranty	1.6.2	G													
			Local Area Controller	2.5.1.1.1	G													
			Lighting Control Panel	2.5.1.2.1	G													
			Switches	2.5.2.1	G													
			Digital Switch Timers	2.5.2.2	G													
			Wall Box Dimmers	2.5.2.3	G													
			Scene Wallstations	2.5.2.4	G													
			Occupancy/Vacancy Sensors	2.5.2.5	G													
			Photosensors	2.5.2.6	G													
			Time Clocks	2.5.2.7	G													
			Power Packs	2.5.2.5.5	G													
			Mini Inverters	2.6.4	G													

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		26 51 00	Exit Signs	2.6.1	G													
			Emergency Drivers	2.6.3	G													
			SD-05 Design Data															
			Luminaire Design Data	1.5.2	G													
			SD-06 Test Reports															
			ANSI/IES LM-79 Test Report	1.5.3														
			ANSI/IES LM-80 Test Report	1.5.4														
			ANSI/IES TM-21 Test Report	1.5.5														
			ANSI/IES TM-30 Test Report	1.5.6														
			Occupancy/Vacancy Sensor	3.2.1.1	G													
			Verification Test															
			Photosensor Verification Test	3.2.1.1	G													
			SD-07 Certificates															
			LED Driver and Dimming Switch	1.5.7	G													
			Compatibility Certificate															
			SD-10 Operation and Maintenance															
			Data															
			Lighting System	1.7.1	G													
			Lighting Control System	1.7.2	G													
			Maintenance Staff Training Plan	3.3.2.1	G													
			End-User Training Plan	3.3.2.2	G													
		27 10 00	SD-02 Shop Drawings															
			Telecommunications Drawings	1.6.1.1	G													
			Telecommunications Space	1.6.1.2	G													
			Drawings															
			SD-03 Product Data															

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		27 10 00	Telecommunications Cabling	2.3	G													
			Patch Panels	2.4.5	G													
			Telecommunications	2.5	G													
			Outlet/Connector Assemblies															
			Equipment Support Frame	2.4.2	G													
			Connector Blocks	2.4.3	G													
			SD-06 Test Reports															
			Telecommunications Cabling	3.5.1	G													
			Testing															
			SD-07 Certificates															
			Telecommunications Contractor	1.6.2.1	G													
			Key Personnel	1.6.2.2	G													
			Manufacturer Qualifications	1.6.2.3	G													
			Test Plan	1.6.3	G													
			SD-09 Manufacturer's Field															
			Reports															
			Factory Reel Tests	2.10.1	G													
			SD-10 Operation and Maintenance															
			Data															
			Telecommunications Cabling and	1.10.1	G													
			Pathway System															
			SD-11 Closeout Submittals															
			Record Documentation	1.10.2	G													
		27 32 50	SD-02 Shop Drawings															
			Nameplates	1.5.1.1.1	G													
			Wiring Diagrams	1.5.1.1.1	G													

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		27 32 50	System Layout	1.5.1.1.13	G													
			Identifying And Instructional	1.5.1.1.14	G													
			Signage															
			SD-03 Product Data															
			Technical Data And Computer	1.5.1.1.2.1	G													
			Software															
			Base / Master Station	2.4.1	G													
			Power Supply	2.4.3	G													
			Call Boxes	2.4.2	G													
			Cabling And Wiring	2.4.4	G													
			SD-06 Test Reports															
			Field Quality Control	1.5.1.2.1	G													
			Testing Procedures	1.5.1.2.2	G													
			SD-07 Certificates															
			Installer	1.5.1.3.1	G													
			Qualified Professional Engineer	1.5.1.3.2	G													
			(QPE)															
			Request For Inspection And	1.5.1.3.3	G													
			Tests															
			Testing	1.5.1.3.4	G													
			Instructor	3.5.1	G													
			SD-10 Operation and Maintenance															
			Data															
			Operation And Maintenance	1.5.1.4.1	G													
			(O&M) Instructions															
			SD-11 Closeout Submittals															

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		27 32 50	As-Built Drawings	1.5.1.4.2	G													
			Spare Parts	3.7.3	G													
		28 08 10	SD-05 Design Data															
			Test Plan	3.1	G													
			SD-06 Test Reports															
			Final Test Report	3.4	G													
			Pre-Acceptance Test Certification	3.2	G													
			SD-07 Certificates															
			Qualifications	1.4.1														
		28 10 05	SD-02 Shop Drawings															
			ESS Components	1.3.3.1	G													
			Access Control System (ACS)	2.3	G													
			Security Hardware Door	1.3.3.1	G													
			Schedule															
			Overall System Schematic	1.3.3.1	G													
			SD-03 Product Data															
			Access Control Unit	2.3.4	G													
			Access Control Devices	2.3.5	G													
			Communications Interface	2.4	G													
			Devices															
			Batteries	2.5.1	G													
			Component Enclosure	2.7	G													
			SD-05 Design Data															
			Backup Battery Capacity	1.5.1	G													
			Calculations															
			Error and Throughput Rates	2.3.2	G													

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(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		28 10 05	SD-07 Certificates															
			Contractor Qualifications	1.3.4.1	G													
			Instructor Qualifications	1.3.4.2	G													
			Contact Card Readers	2.3.5.1.1	G													
			Contactless Card Readers	2.3.5.1.2	G													
			Wire And Cable	2.4.3	G													
			Bonding, Grounding, And Shielding	3.1.8	G													
			SD-10 Operation and Maintenance Data															
			ESS Components	1.3.3.1	G													
			ESS Software, Data Package 4	1.6	G													
			ESS Training	3.6	G													
			ESS Training Outline	3.6.1	G													
			SD-11 Closeout Submittals															
			As-Built Drawings	1.7	G													
		28 31 76	SD-01 Preconstruction Submittals															
			Qualified Fire Protection Engineer (QFPE)	1.3.2	G													
			Fire alarm system designer	1.8.2.1	G													
			Supervisor	1.8.2.2	G													
			Technician	1.8.2.3	G													
			Installer	1.8.2.4	G													
			Test Technician	1.8.2.5	G													
			Fire Alarm System Site-Specific Software Acknowledgement	1.7	G													

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		28 31 76	SD-02 Shop Drawings														
			Nameplates	1.8.1.3	G												
			Instructions	2.2.4	G												
			Wiring Diagrams	1.8.1.4	G												
			System Layout	1.8.1.5	G												
			Notification Appliances	1.8.1.6	G												
			Initiating devices	1.8.1.7	G												
			Amplifiers	1.8.1.8	G												
			Battery Power	1.8.1.9	G												
			Voltage Drop Calculations	1.8.1.10	G												
			SD-03 Product Data														
			Fire Alarm and Mass Notification	2.3	G												
			Control Unit (FMCU)														
			Local Operating Console (LOC)	1.4.4	G												
			Amplifiers	1.8.1.8	G												
			Tone Generators	2.5	G												
			Digitalized voice generators	2.5	G												
			LCD Annunciator	2.6.1	G												
			Manual Stations	2.7	G												
			Smoke Detectors	2.8	G												
			Duct Smoke Detectors	2.8.2	G												
			Heat Detectors	2.9	G												
			Addressable Interface Devices	2.10	G												
			Addressable Control Modules	2.11	G												
			Isolation Modules	2.12	G												
			Notification Appliances	1.8.1.6	G												

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		28 31 76	Textual Display Sign Control	2.13.3	G												
			Panel														
			Textual Display Signs	2.13.3	G												
			Batteries	2.15.1	G												
			Battery Chargers	2.15.2	G												
			Supplemental Notification	2.15.1.1	G												
			Appliance Circuit Panels														
			Auxiliary Power Supply Panels	2.15.1.1	G												
			Surge Protective Devices	2.16	G												
			Alarm Wiring	2.16	G												
			Back Boxes and Conduit	3.3.4	G												
			Ceiling Bridges	3.2.9	G												
			Terminal Cabinets	3.3.2	G												
			Radio Transmitter and Interface	2.19.1	G												
			Panels														
			Electromagnetic Door Holders	2.20.3	G												
			Document Storage Cabinet	3.12.3	G												
			SD-06 Test Reports														
			Test Procedures	3.8.1	G												
			SD-07 Certificates														
			Verification of Compliant	3.8.2.1	G												
			Installation														
			Request for Government Final	3.8.2.2	G												
			Test														
			SD-10 Operation and Maintenance														
			Data														

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		28 31 76	Operation and Maintenance	3.10	G													
			(O&M) Instructions															
			Instruction of Government	3.11	G													
			Employees															
			SD-11 Closeout Submittals															
			As-Built Drawings	1.8.1.13														
			Spare Parts	1.10.1														
		31 00 00.00 22	SD-01 Preconstruction Submittals															
			Shoring and Sheeting Plan	3.1.1	G													
			Excavation and Trenching Plan	1.6.1	G													
			Dewatering Work Plan	1.13.2	G													
			Dewatering Work Plan	3.1.4.2	G													
			Excess Soil Disposal & Soil	1.12	G													
			Sampling Plan															
			SD-03 Product Data															
			Buried Warning and Identification	3.7.1	G													
			Tape															
			Geotextiles	2.6	G													
			SD-06 Test Reports															
			Dewatering Performance Records	3.1.4.2	G													
			Select Material Test	3.10.2.2	G													
			Material Test Report	3.11	G													
			Density Tests	3.11	G													
			Soil Sampling for Waste	1.9	G													
			Characterization															
			Fill and Backfill	3.10.2.1	G													

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		31 00 00.00 22	Borrow Materials	2.5	G												
			Moisture Content Tests	3.8	G												
		31 21 13	SD-03 Product Data														
			Radon Mitigation Systems	2.1.2	G												
			Components														
			SD-06 Test Reports														
			Post Mitigation Testing	3.3.2	G												
			SD-07 Certificates														
			Contractor Qualifications	1.8.1.1	G												
			Testing Laboratory Certification	1.8.2	G												
			Radon Mitigation System	3.3.1	G												
			Inspection														
			SD-08 Manufacturer’s Instructions														
			Radon Mitigation Systems	2.1.2													
			Components														
			SD-10 Operation and Maintenance														
			Data														
			Radon Mitigation Systems	2.1	G												
			SD-11 Closeout Submittals														
			Radon Detector Location Log	1.6	G												
		31 32 11	SD-01 Preconstruction Submittals														
			Erosion control inspector	1.5													
			qualifications														
			SD-03 Product Data														
			Catch Basin Inlet Protection	2.1	G												
			Filters														

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		31 32 11	Compost Filter Sock	3.3.2	G												
			SD-06 Test Reports														
			Maintenance and Inspection Logs	3.5													
		31 63 29	SD-02 Shop Drawings														
			Design Drawings	1.7.1	G												
			Fabrication and Installation	1.3.1	G												
			Drawings														
			SD-03 Product Data														
			Installation Plan	1.3.1	G												
			Casing	2.3													
			SD-05 Design Data														
			Design Drawings	1.7.1	G												
			Cement Grout Mix Design	2.1.1.1	G												
			Pile Load Tests	3.3.1	G												
			SD-06 Test Reports														
			Pile Load Tests	3.3.1	G												
			Grout Mix Proportions	2.1.1.1	G												
			Grout Compressive Strength	3.3.1	G												
			SD-07 Certificates														
			Designer Qualifications	1.5.5	G												
			Installer Qualifications	1.5.6	G												
			Pile Materials	1.5.3	G												
			Field Testing Technician	3.3.3	G												
			SD-11 Closeout Submittals														
			As Built Data	3.4.1	G												
		32 01 19.61	SD-03 Product Data														

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		32 01 19.61	Sealants	2.1	G												
			Manufacturer's	3.4.2													
			Recommendations														
			Application Equipment List	3.1													
			SD-08 Manufacturer’s Instructions														
			Sealants	2.1	G												
		32 05 33	SD-01 Preconstruction Submittals														
			Integrated Pest Management	2.4	G												
			Plan														
			SD-03 Product Data														
			Fertilizer	2.1	G												
			Mulches Topdressing	2.3	G												
			Organic Mulch Materials	2.3.1	G												
			SD-07 Certificates														
			Maintenance Inspection Report	3.3.1	G												
		32 12 17	SD-03 Product Data														
			Material Batch Tickets	1.3.3	G												
			SD-05 Design Data														
			Mix Design	1.3.2	G												
			Mix Design	2.4	G												
			SD-06 Test Reports														
			Pavement Courses	3.5.2.1	G												
		32 17 23	SD-01 Preconstruction Submittals														
			Qualifications	1.3.2	G												
			SD-03 Product Data														

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		32 17 23	Surface Preparation Equipment		G												
			List														
			Application Equipment List	2.1.1	G												
			Exterior Surface Preparation	3.2													
			Solventborne Paint	2.2.1	G												
			SD-08 Manufacturer’s Instructions														
			Solventborne Paint	2.2.1	G												
		32 92 19	SD-03 Product Data														
			Wood Cellulose Fiber Mulch	2.5.3	G												
			Fertilizer	2.4	G												
			SD-06 Test Reports														
			Topsoil Composition Tests	2.2.3	G												
			SD-07 Certificates														
			Seed	2.1	G												
		33 11 00	SD-01 Preconstruction Submittals														
			Connections	3.1.1	G												
			SD-03 Product Data														
			Pipe, Fittings, Joints and	2.1.1	G												
			Couplings														
			Valves	2.1.2	G												
			Valve Boxes	2.1.2.2	G												
			Pipe Restraint	2.2.1	G												
			Corporation Stops	2.2.3.1	G												
			Precast Concrete Thrust Blocks	2.2.1.2	G												
			Disinfection Procedures	3.2.2	G												
			SD-06 Test Reports														

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		33 11 00	Bacteriological Samples	3.3.1.3	G													
			Leakage Test	3.3.1.2														
			Hydrostatic Test	3.3.1.1	G													
		33 71 02	SD-03 Product Data															
			Conduit, Ducts, and Fittings	2.1	G													
			Duct Sealant	2.1.4	G													
			SD-06 Test Reports															
			Field Acceptance Checks and	3.11.1	G													
			Tests															
			SD-07 Certificates															
			Cable Installer Qualifications	1.5.1	G													
		33 82 00	SD-02 Shop Drawings															
			Telecommunications Outside	1.6.1.1	G													
			Plant															
			Telecommunications Entrance	1.6.1.2	G													
			Facility Drawings															
			SD-03 Product Data															
			Wire and Cable	2.7	G													
			Cable Splices, and Connectors	2.4	G													
			Closures	2.3	G													
			Building Protector Assemblies	2.2.1	G													
			Protector Modules	2.2.2	G													
			SD-06 Test Reports															
			Pre-installation Tests	3.5.1	G													
			Acceptance Tests	3.5.2	G													
			Outside Plant Test Plan	1.6.3	G													

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		33 82 00	SD-07 Certificates															
			Telecommunications Contractor	1.6.2.1	G													
			Key Personnel	1.6.2.2	G													
			Manufacturer's Qualifications	1.6.2.3	G													
			SD-08 Manufacturer’s Instructions															
			Building Protector Assembly	2.2.1	G													
			Installation															
			Cable Tensions	3.1.6.1	G													
			Fiber Optic Splices	3.1.7.1	G													
			SD-09 Manufacturer’s Field															
			Reports															
			Factory Reel Test Data	2.12.1	G													
			SD-10 Operation and Maintenance															
			Data															
			Telecommunications Outside	1.6.1.1	G													
			Plant (OSP)															
			SD-11 Closeout Submittals															
			Record Documentation	1.8.1	G													

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SECTION 01 33 10.00 22
^^
PROCUREMENT SUBSTITUTION REQUIREMENTS (PWD ME)
03/25

PART 1 GENERAL

This Section applies to only Design-Bid-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities. This Section applies to Pre-Award (Bidding) period and Post Award.

1.1 SUMMARY

This Section includes administrative and procedural requirements for substitutions.

A general description for each piece of equipment, material, and component scheduled on the drawings and described in the specifications include the salient physical, functional, or performance characteristics of the brand name item that an "equal" item must meet to be acceptable for award. The use of brand name manufacturer's and model numbers indicated as the basis of design were utilized in the preparation of the project's Contract Documents. "Or equals" are acceptable when the salient characteristics of the scheduled or specified equipment, component or device, which are firm requirements, are met without exception.

1.2 REFERENCES (OPA WORK AROUND)

a. FAR 11.104 - This specification only refers to "or equal" substitutions to the basis of design listed on the schedule or described in the specifications.

b. FAR 11.105 - This specification is not intended to be used for proprietary or Items Peculiar to One Manufacturer. (Indicated for Reference Only)

c. FAR 11.104 - Use of Brand Name or Equal Purchase Descriptions:

1. While the use of performance specifications is preferred to encourage offerors to propose innovative solutions, the use of brand name or equal purchase descriptions may be advantageous under certain circumstances.

2. Brand name or equal purchase descriptions must include, in addition to the brand name, a general description of those salient physical, functional, or performance characteristics of the brand name item that an "equal" item must meet to be acceptable for award. Use brand name or equal descriptions when the salient characteristics are firm requirements.

3. Note: a Justification and Approval (J&A) document is NOT required.

d. FAR 11.105 - Items Peculiar to One Manufacturer:

Agency requirements shall not be written so as to require a particular brand name, product, or a feature of a product, peculiar to one manufacturer, thereby precluding consideration of a product manufactured by another company, unless -

1. The particular brand name, product, or feature is essential to the Government's requirements, and market research indicates other companies' similar products, or products lacking the particular feature, do not meet, or cannot be modified to meet, the agency's minimum needs;

   a) The authority to contract without providing for full and open competition is supported by the required justifications and approvals (see 6.302-1); or

   b) The basis for not providing for maximum practicable competition is documented in the file (see 13.106-1(b)) or justified (see 13.501) when the acquisition is awarded using simplified acquisition procedures.

2. The documentation or justification is posted for acquisitions over $ 25,000. (See 5.102(a)(6).)

   1) (b) For multiple award schedule orders, see 8.405-6.

3. Note: a Justification and Approval (J&A) document is IS required.

1.3 DEFINITIONS

1.3.1 Substitutions

Changes in products, materials, equipment, and methods of construction from those required by the Contract Documents and proposed by the Contractor.

1.3.1.1 Substitutions for Cause

Changes proposed by the Contractor that are required due to changed Project conditions, such as unavailability of product, regulatory changes, or unavailability of required warranty terms.

1.3.1.2 Substitutions for Convenience

Changes proposed by the Contractor or the Government that are not required in order to meet other Project requirements but may offer an advantage to the Contractor or the Government.

1.3.2 Procurement Substitution Requests

Requests for changes in products, materials, equipment, and methods of construction from those indicated in the Procurement and Contracting Documents, submitted prior to receipt of bids.

1.3.3 Substitution Requests

Requests for changes in products, materials, equipment, and methods of construction from those indicated in the Contract Documents, submitted following Contract award.

1.4 PROCUREMENT SUBSTITUTIONS - PRE-AWARD (BIDDING) PERIOD

1.4.1 General

By submitting a bid, the Bidder represents that its bid is based on materials and equipment described in the Procurement and Contracting Documents, including Amendments. Bidders are encouraged to request approval of qualifying substitute materials and equipment when the Specifications Sections list materials and equipment by product or manufacturer name.

1.4.2 Procurement Substitution Requests

Procurement Substitution Requests will be received and considered by the Government when the following conditions are satisfied, as determined by the Contracting Officer; otherwise requests will be returned without action:

a. Extensive revisions to the Contract Documents are not required.

b. Proposed changes are in keeping with the general intent of the Contract Documents, including the level of quality of the Work represented by the requirements therein.

c. The request is fully documented and properly submitted.

1.4.3 Substitution Requirements

The requested substitution for convenience (not for cause) needs to meet the following requirements:

a. Substitutions for convenience which provide benefit solely to the Contractor shall be at the contractor's risk and cost.

b. Be of the same efficiency and not require additional energy cost to operate the equipment, systems, and associated components.

c. Be of roughly the same dimensions, weight, physical size, electrical characteristics, construction, etc. as the basis of design listed in the specifications and identified on the drawing schedules.

d. The performance of the proposed substitution must be equal to or greater than the basis of design listed in the specifications and identified on the drawing schedules.

e. The Contractor, not the Government, shall be responsible for all the additional costs of design, engineering, construction, project management, etc., to design, provide, furnish, and install the proposed substitution over and above the cost related to the basis of design listed in the specifications and identified on the drawing schedules.

f. The subcontractor must coordinate with the Contractor and any other associated subcontractors to address any costs related to the proposed substitution.

g. Note: There may be conditions or circumstances such as existing buildings, mechanical rooms, or confined or limited architectural spaces and areas where the proposed substitution will not be acceptable as an option to the basis of design listed in the specifications and identified on the drawing schedules.

1.5 SUBMITTALS

Government Acceptance or Approval does not remove responsibility from the Contractor for their actions or liability.

Government approval is required for submittals with a "G" classification; submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Procurement Substitution Request - Pre-Award (Bidding) Period; G
Substitution Request - Post Award; G

1.6 PROCUREMENT SUBSTITUTION REQUEST - PRE-AWARD (BIDDING) PERIOD

Submit documentation identifying product, fabrication, or installation method regarding the proposed substitution. Include Specification Section number(s) and Drawing numbers and titles.

Procurement Substitution Request must be made in writing in compliance with the following requirements:

a. Request for substitution of equipment, component, and devices may be submitted by each contractor after contract award as part of the submittal process for review and comment by the Government and Designer of Record (DOR) for the project. Request for substitutions for Proprietary Items shall only be accepted if the listed product is no longer available or cannot be procured during the project construction schedule.

b. The final determination of the acceptability of the substitution is by the Contracting Officer and the DOR. Reviews of substitution requests must be based upon the requirements in this Section in terms of the substituted products performance, efficiency and fit within the constraints of the project. The burden is on the Contractor to confirm these features and coordinate the product within the physical constraints of the project. The decision shall be final.

c. Submittal Format: Submit electronically, copies of each written Procurement Substitution Request, using the Substitution Request Cover Letter located at the end of this Section.

    1) Identify the product or the fabrication or installation method to be replaced in each request. Include related Specification Sections and drawing titles and numbers.

    2) Provide complete documentation on both the product specified and the proposed substitute, including the following information as appropriate:

        a) Point-by-point comparison of specified and proposed substitution product data, fabrication drawings, and installation procedures.
        b) Copies of current, independent third-party test data of

salient product or system characteristics.

c) Samples where applicable or when requested by Contracting Officer.

d) Detailed comparison of significant qualities of the proposed substitution with those of the Work specified. Significant qualities may include attributes such as performance, weight, size, durability, visual effect, sustainable design characteristics, warranties, and specific features and requirements indicated. Indicate deviations, if any, from the Work specified.

e) Material test reports from a qualified testing agency indicating and interpreting test results for compliance with requirements indicated.

f) Research reports, where applicable, evidencing compliance with building code in effect for Project and applicable UFC's.

g) Coordination information, including a list of changes or modifications needed to other parts of the Work and to construction performed by the Government and separate contractors, which will become necessary to accommodate the proposed substitution.

3) Provide certification by manufacturer that the substitution proposed is equal to or superior to that required by the Procurement and Contracting Documents, and that its in-place performance will be equal to or superior to the product or equipment specified in the application indicated.

4) Bidder, in submitting the Procurement Substitution Request, waives the right to additional payment or an extension of Contract Time because of the failure of the substitution to perform as represented in the Procurement Substitution Request.

1.7 SUBSTITUTION REQUEST - POST AWARD

Submit three (3) copies of each request for consideration. Identify product or fabrication or installation method to be replaced. Include Specification Section number, title, and Drawing numbers and titles.

a. Submittal Format:

1. Substitution Request Form: Use the Substitution Request Cover Sheet form located at the end of this section.

2. Documentation: Show compliance with requirements for substitutions and the following, as applicable:

a) Statement indicating why specified product, fabrication, or installation could not be provided, if applicable.

b) Coordination information, including a list of changes or revisions needed to other parts of the Work and to construction performed by the Government and separate contractors that will be necessary to accommodate the proposed substitution.

c) Detailed comparison of significant qualities of the proposed substitution with those of the Work specified. Include annotated copy of applicable Specification Section(s). Significant qualities may include attributes such as performance, weight, size, durability, visual effect, sustainable design characteristics, warranties, and specific features and requirements indicated. Indicate deviations, if any, from the Work specified.

d) Product Data, including drawings and descriptions of products and fabrication and installation procedures.

e) Samples, where applicable or requested.

f) Certificates and qualification data, where applicable or requested.

g) List of similar installations for completed projects with project names and addresses and names and addresses of Designers of Record and Owners/Government Agencies.

h) Material test reports from a qualified testing agency indicating and interpreting test results for compliance with requirements indicated.

i) Research reports evidencing compliance with applicable UFCs and building codes in effect for the Project.

j) Detailed comparison of Contractor's construction schedule using proposed substitution with products specified for the Work, including effect on the overall Contract Duration. If specified product or method of construction cannot be provided within the Contract Duration, include letter from manufacturer, on manufacturer's letterhead, stating date of receipt of purchase order, lack of availability, or delays in delivery.

k) Cost information, including a proposal of change, if any, in the Contract Sum.

l) Contractor's certification that proposed substitution complies with requirements in the Contract Documents except as indicated in the substitution request, is compatible with related materials, and is appropriate for applications indicated.

m) Contractor's waiver of rights for additional payment or time that may subsequently become necessary because of failure of the proposed substitution to produce the results required by the Contract Documents.

1.8 CONTRACTING OFFICER'S ACTION (BIDDING)

The Contracting Officer may request additional information or documentation necessary for evaluation of the Procurement Substitution Request. Contracting Officer will notify all bidders of acceptance of the proposed substitute by means of an Amendment to the Procurement and Contracting Documents.

1.8.1 Contracting Officer's Action (Post Award)

If necessary, there will be a request for additional information or documentation for evaluation within seven (7) days of receipt of a request for substitution. Notification will be provided to the Contractor of acceptance or rejection of proposed substitution within 15 days of receipt of request, or seven (7) days of receipt of additional information or documentation, whichever is later.

a. Forms of Acceptance: Change Order, Construction Change Directive (Contract Modification), or Architect's Supplemental Instructions (ASI) for minor changes in the Work.

b. Use product specified in the Contract Documents if the Contracting Officer does not issue a decision on use of a proposed substitution within the time allocated.

1.8.2 Conditions

The Contracting Officer will consider Contractor's request for substitution when the following conditions are satisfied. If the following conditions are not satisfied, the Contracting Officer will return requests without action, except to record noncompliance with these requirements:

a. Requested substitution offers the Government a substantial advantage in cost, time, energy conservation, or other considerations, after deducting additional responsibilities the Government must assume. Contractor's additional responsibilities may include compensation to the DOR for redesign and evaluation services, increased cost of other construction by the Governemnt, and similar considerations.

b. Requested substitution does not require extensive revisions to the Contract Documents.

c. Requested substitution is consistent with the Contract Documents.

d. Requested substitution will not adversely affect project cost.

e. Requested substitution will not adversely affect Contractor's construction schedule/Contract Duration.

f. Requested substitution is consistent with the Contract Documents and will produce indicated results.

g. Requested substitution will provide the specified warranty.

h. Substitution request is fully documented and properly submitted.

i. Requested substitution has received necessary approvals of authorities having jurisdiction.

j. Requested substitution is compatible with other portions of the Work.

k. Requested substitution has been coordinated with other portions of the Work.

l. Requested substitution provides specified warranty.

m. If requested substitution involves more than one (1) subcontractor, the requested substitution has been coordinated with other portions of the Work, is uniform and consistent, is compatible with other products, and is acceptable to all subcontractors involved.

1.8.3 Contracting Officer's Approval

Contracting Officer's approval of a substitution during bidding does not relieve Contractor of the responsibility to submit required shop drawings and to comply with all other requirements of the Contract Documents.

1.9 QUALITY ASSURANCE

Compatibility of Substitutions: Investigate and document compatibility of proposed substitution(s) with related products and materials. Engage a qualified testing agency to perform compatibility tests recommended by manufacturer's.

1.10 PROCEDURES

1.10.1 Coordination

Revise or adjust affected work as necessary to integrate work of the approved substitutions.

1.11 SUBSTITUTIONS

1.11.1 Substitutions for Cause

Submit requests for substitution(s) immediately on discovery of need for change, but not later than 15 days prior to time required for preparation and review of related submittals.

a. Conditions: Contracting Officer will consider Contractor's request for substitution when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

   1) Requested substitution is consistent with the Contract Documents and will produce indicated results.

   2) Substitution request is fully documented and properly submitted.

   3) Requested substitution will not adversely affect Contractor's construction schedule/Contract Duration.

   4) Requested substitution has received necessary approvals of authorities having jurisdiction.

   5) Requested substitution is compatible with other portions of the Work.

   6) Requested substitution has been coordinated with other portions of the Work.

   7) Requested substitution provides specified warranty.

   8) If requested substitution involves more than one (1) subcontractor, the requested substitution has been coordinated with other portions of the Work, is uniform and consistent, is compatible with other products, and is acceptable to all subcontractors involved.

1.12 SUBSTITUTIONS FOR CONVENIENCE

The Contracting Officer will consider requests for substitution(s) if received within 30 days after the Notice of Award. Requests received after that time may be considered or rejected at discretion of the Contracting Officer.

a. Conditions: Contracting Officer will consider Contractor's request for substitution when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1) Requested substitution offers the Government a substantial advantage in cost, time, energy conservation, or other considerations, after deducting additional responsibilities the Government must assume. The Government's additional responsibilities may include compensation to DOR for redesign and evaluation services, increased cost of other construction by the Government, and similar considerations.

2) Requested substitution does not require extensive revisions to the Contract Documents.

3) Requested substitution is consistent with the Contract Documents and will produce indicated results.

4) Substitution request is fully documented and properly submitted.

5) Requested substitution will not adversely affect Contractor's construction schedule/Contract Duration.

6) Requested substitution has received necessary approvals of authorities having jurisdiction.

7) Requested substitution is compatible with other portions of the Work.

8) Requested substitution has been coordinated with other portions of the Work.

9) Requested substitution provides specified warranty.

10) If requested substitution involves more than one (1) subcontractor, the requested substitution has been coordinated with other portions of the Work, is uniform and consistent, is compatible with other products, and is acceptable to all subcontractors involved.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not Used.

SUBSTITUTION REQUEST COVER SHEET

This cover sheet is required for all proposed substitutions including "or equals".

Contractor: ____

Subcontractor: ____

This Request Reference Number :_________________________________

Substitution Summary:

____

____

Specification Section and Paragraph Number:_____________________

Contract Drawing and Detail Reference:__________________________

Date of This Request:____

This Request Prepared By:_______________________________________

Conditions: Indicate all conditions that apply to this proposed substitution:

[ ] Substitution required because specified item is no longer available.

[ ] Substitution required because Contractor believes specified item is incorrect, inappropriate, or incompatible.

[ ] Substitution recommended because it offers the Government substantial advantage in:

[ ] Quality

[ ] Time

[ ] Cost

[ ] This is an "or equal".

Evidence: Indicate evidence attached:

[ ] Tabulated side by side comparison of specified item and proposed substitution directly comparing each feature, characteristic, and performance.

[ ] Manufacturer's product data for both specified item and proposed substitution. Origin of all information included on tabulated side by side comparison is highlighted.

[ ] Details showing how the proposed substitution interfaces with adjacent work.

[ ] Certification that warranty, if any required, will be provided as

required.

[ ] Certification that this proposed substitution is coordinated with all related and adjacent work.

[ ] Complete cost change information.

[ ] Higher cost to Government as stated in cost change information

[ ] No change in cost to Government

[ ] Lower cost to Government, credit to Government as stated in cost change information

Deviations from Contract Documents: Itemize all deviations from Contract Documents if this proposed substitution is approved.

____

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____

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The undersigned attests that the undersigned has carefully examined this entire submission and that the requirements of the Contract Documents have been met.

Should this alternate material or equipment fail to meet the expectations of the the Government within the mandated design liability period, this Contractor shall replace the material and or equipment with the original specified items at no additional cost to the Government. This Contractor shall be responsible for all collateral impacts to the project for replacement of such deviations and substitutions.

Firm: ____

By: ____

Signature: ____

Date: ____

Printed or Typed Name Title:

____

-- End of Section --

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SECTION 01 33 29.00 22
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SUSTAINABILITY REQUIREMENTS AND REPORTING (PWD ME)
02/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

COUNCIL ON ENVIRONMENTAL QUALITY (CEQ) (WHITE HOUSE)

HPSB Guiding Principles	(2016) Guiding Principles for Sustainable Federal Buildings and Determining Compliance with the Guiding Principles for Sustainable Federal Buildings

INTERNATIONAL CODE COUNCIL (ICC)

ICC IgCC	(2018) International Green Construction Code

U.S. DEPARTMENT OF AGRICULTURE (USDA)

FSRIA 9002	Farm Security and Rural Investment Act Section 9002 (USDA BioPreferred Program)

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 1-200-02	(2020; Change 2, 2022) High Performance and Sustainable Building Requirements
UFC 3-600-01	(2016; with Change 6, 2021) Fire Protection Engineering for Facilities

U.S. DEPARTMENT OF ENERGY (DOE)

Energy Star	(1992; R 2006) Energy Star Energy Efficiency Labeling System (FEMP)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 247	Comprehensive Procurement Guideline for Products Containing Recovered Materials

1.2 SUMMARY

This section includes requirements for Sustainability documentation and reporting submittals per the federally mandated High Performance and Sustainable Building (HPSB) or HPSB "Guiding Principles" (GP),[and Third Party Certification (TPC) requirements,] in accordance with UFC 1-200-02 High Performance and Sustainable Building Requirements, and other identified requirements.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

Preliminary High Performance and Sustainable Building Checklist; G

Sustainability Action Plan; G

Preliminary Sustainability eNotebook; G

SD-06 Test Reports

SD-11 Closeout Submittals

Final High Performance and Sustainable Building Checklist; G

Final Sustainability eNotebook; G

Amended Final Sustainability eNotebook; G

Amended Final High Performance and Sustainable Building Checklist; G

Third Party Certification Certificate, Assessment, or Validation and Compliance Report; G

1.4 GUIDING PRINCIPLES VALIDATION (GPV)

Provide the following sustainability activities and documentation to verify achievement of HPSB Guiding Principles Validation (GPV):

a. Analysis of each Guiding Principle Requirement and how project complies. Include final government approved narrative(s) in the HPSB Checklist submittal. Multiple checklists indicate multiple buildings that require individual HPSB Checklist tracking.

b. No changes to the HPSB Checklist are allowed without approval from the Contracting Officer, in accordance with Section 01 33 00.00 22 SUBMITTAL REQUIREMENTS (PWD ME). Immediately bring to the attention of the Contracting Officer any project changes that impact meeting the approved HPSB Guiding Principles Requirements for this project. Demonstrate the change will not increase the life-cycle cost and maintains or improves the building performance.

c. Documentation of all work required to incorporate the applicable HPSB Guiding Principles requirements indicated on the HPSB Checklist and in this contract, including all "S" submittals.

d. Sustainability Action Plan.

e. Construction related documentation for the project Sustainability eNotebook and keep updated with regularly-scheduled Construction Quality Control Meetings. Include construction related documentation containing the following components:

(1) HPSB Checklist(s)

(2) Sustainability Action Plan

(3) Documentation illustrating HPSB Guiding Principles Requirements compliance, including "S" submittals

1.4.1 Sustainability Action Plan

Include the following information in the Sustainability Action Plan:

a. Analysis of each HPSB Guiding Principles Requirement and how project will comply. Final government approved narrative(s) must be included in the HPSB Checklist submittal.

b. Name and contact information for: Contractor's Point of Contact (POC) ensuring sustainability goals are accomplished and documentation is assembled. For TPC that include on-site visit by third party representative, provide list of required attendees.

c. Indoor Air Quality plan.

1.4.2 Calculations

Provide all calculations, product data, labels and product certifications required in this specification to demonstrate compliance with the HPSB Guiding Principles Requirements.

1.5 SUSTAINABILITY SUBMITTALS

Provide HPSB Checklist and other documentation in the Sustainability eNotebook to indicate compliance with the sustainability requirements of the project.

1.5.1 High Performance Sustainable Building (HPSB) Checklist

Provide construction documentation that provides proof of, and supports compliance with, the completed HPSB Checklist.

1.5.1.1 HPSB Checklist Submittals

Submit updated HPSB Checklist with each Sustainability eNotebook submittal.

1.5.2 "S" Submittals for Sustainability Documentation

"S" submittals are the sustainability documentation requirements cited in the various sections of this contract. Submit the GPV sustainability documentation required in this section as "S" submittals in all affected UFGS Sections.

a. Highlight GPV compliance data in "S" submittal.

b. Add "S" submittals to the Sustainability eNotebook only after submittal approval, and bookmark them as required in paragraph SUSTAINABILITY ENOTEBOOK below.

c. Ensure all approved "S" submittals are included in each Sustainability eNotebook submittal.

1.5.3 Sustainability eNotebook

The Sustainability eNotebook is an electronic organizational file that serves as a repository for all required sustainability submittals. To support documentation of compliance with an approved HPSB checklist, provide and maintain a comprehensive and current Sustainability eNotebook. Include all required data in Sustainability eNotebook, to support full compliance with the HPSB Guiding Principles Requirements, including:

a. HPSB checklist

b. Sustainability Action Plan

c. Calculations

d. Labels

e. "S" submittals

f. Certifications, assessments, or validations and compliance report

1.5.3.1 Sustainability eNotebook Format

Provide Sustainability eNotebook in the form of an Adobe PDF file; bookmark each HPSB Guiding Principles Requirement and sub-bookmark at each document. Match format to HPSB Guiding Principles numbering system indicated herein. Maintain up-to-date information, such as spreadsheets, templates, with each current submittals.

Contracting Officer may deduct from the monthly progress payment accordingly if Sustainability eNotebook information is not current and on track per project goals.

1.5.3.2 Sustainability eNotebook Submittal Schedule

Provide Sustainability eNotebook Submittals at the following milestones of the project:

a. Preliminary Sustainability eNotebook

Submit preliminary Sustainability eNotebook with updated Preliminary High Performance and Sustainable Building Checklist at the first post award meeting in accordance with Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME).

b. Construction Quality Control Meetings.

Provide up-to-date GP documentation in the Sustainability eNotebook for each meeting.

c. Final Sustainability eNotebook

Submit updated Sustainability eNotebook with updated Final High Performance and Sustainable Building Checklist with TPC Checklist at Beneficial Occupancy Date (BOD). Final progress payment retainage may be held by Contracting Officer until Final Sustainability construction phase documentation is complete.

d. Amended Final Sustainability eNotebook

Amend and resubmit the Amended Final Sustainability eNotebook with Amended Final High Performance and Sustainable Building Checklist, to include post-occupancy corrections, updates, and requirements. Final progress payment retainage may be held by Contracting Officer until amended final sustainability documentation is complete. Submit the Amended Final Sustainability eNotebook Submittal on DVDs to the Contracting Officer no later than 30 days after final GP determination.

1.6 DOCUMENTATION REQUIREMENTS

a. Incorporate each of the following HPSB Guiding Principles requirements into project and provide documentation that proves compliance with each listed requirement. Items below are organized by HPSB Guiding Principles. For life-cycle cost analysis requirements, one document with all analyses is acceptable, with Contracting Officer approval.

b. For each of the following paragraphs that require the use of products listed on Government-required websites, provide documentation of the process used to select products, or process used to determine why listed products do not meet project performance requirements.

1.6.1 Commissioning (Cx)

Develop and incorporate Commissioning requirements into the documents, in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

1.6.2 Energy Efficient Products

Provide only energy-using products that are Energy Star rated or have Federal Energy Management Program (FEMP) recommended efficiency. Where Energy Star or FEMP recommendations have not been established, provide most efficient products that are life-cycle cost-effective. Provide only energy using products that meet FEMP requirements for low standby power consumption. Energy efficient products can be found at: https://www.energy.gov/eere/femp/federal-energy-management-program and http://www.energystar.gov/.

For construction submittal documentation, provide proof that product is labeled energy efficient and complies with the cited requirements.

1.6.3 Building-Level Power Metering

Provide building-level meters for electricity, natural gas, and steam where applicable.

1.6.3.1 Construction Submittal Documentation

Provide manufacturer's data validating compatibility with base-wide system and component advanced meter requirements.

1.6.4 Indoor Water Use

Provide Construction Documentation proof that fixtures are labeled EPA

WaterSense, for products available with EPA WaterSense labeling; for all other fixtures, proof they comply with EPA WaterSense efficiency requirements.

1.6.5 Indoor Water Metering

Provide building-level meters for potable water use. Provide the requirements cited in the following paragraphs:

1.6.5.1 Construction Submittal Documentation

Provide manufacturer's data validating compatibility with base-wide system and component advanced meter requirements.

1.6.6 Outdoor Water Use

Where new irrigation is required, provide only non-potable sources. Provide the requirements cited in the following paragraphs:

1.6.6.1 Construction Submittal Documentation

Provide manufacturer's data validating compatibility with base-wide system and component advanced meter requirements.

1.6.7 Outdoor Water Meters

Provide meters for outdoor systems that use potable water. Provide the requirements cited in the following paragraphs:

1.6.7.1 Construction Submittal Documentation

Provide manufacturer's data validating compatibility with base-wide system and component advanced meter requirements.

1.6.8 Moisture Control

Provide the following:

1.6.8.1 Construction Submittal Documentation

Ensure construction materials are separated and protected in accordance with other sections in this contract document, with adequate humidity controls during construction. In accordance with Section 01 78 23.00 22 OPERATION AND MAINTENANCE DATA (PWD ME), includes plan for ongoing building moisture control.

Coordinate with the moisture control requirements of Section 01 45 00.00 22 QUALITY CONTROL (PWD ME).

1.6.9 Reduce Volatile Organic Compounds (VOC) (Low-Emitting Materials)

Meet the requirements of Table 3-1 at the end of this specification.

For Construction submittal documentation, provide certifications or labels that demonstrate compliance with cited requirements, based on the attached TABLE 3-1.

1.6.10 Indoor Air Quality During Construction

Prior to construction, create indoor air quality plan. Develop and implement an IAQ construction management plan during construction and flush building air before occupancy.

For new construction and for renovation of unoccupied existing buildings, meet the requirements of ICC IgCC 1001.3.1.5 (10.3.1.4) Indoor Air Quality (IAQ) Construction Management. Coordinate with moisture control requirements in Section 01 45 00.00 22 QUALITY CONTROL (PWD ME).

Provide documentation showing that after construction ends and prior to occupancy, HVAC filters were replaced and building air was flushed out in accordance with the cited standard.

1.6.11 Recycled Content

Comply with 40 CFR 247. Refer to: https://www.epa.gov/smm/comprehensive-procurement-guideline-cpg-program for assistance identifying products cited in 40 CFR 247. Selected products must comply with non-proprietary requirements of the Federal Acquisition Regulation and must meet performance requirements.

1.6.11.1 Construction Submittal Documentation

a. Provide manufacturers' documents stating the recycled content by material, or written justification for claiming one of the exceptions allowed on the cited website.

b. Substitutions: Submit for Government approval for proposed alternative products or systems that provide equivalent performance and appearance and have greater contribution to project recycled content requirements. For all such proposed substitutions, submit with the Sustainability Action Plan accompanied by product data demonstrating equivalence.

c. In order to complete compliance with FAR 52.223-9 Estimate of Percentage of Recovered Material Content for EPA Designated Items, refer to submittal requirement for recycled/recovered material content in Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME).

1.6.12 Bio-Based Products

Provide products and materials composed of the highest percentage of bio-based materials (including rapidly renewable resources and certified sustainably harvested products), consistent with FSRIA 9002 USDA BioPreferred Program, to the maximum extent possible without jeopardizing the intended end use or detracting from the overall quality delivered to the end user and when available at a reasonable cost. Use only supplies and materials of a type and quality that conform to applicable specifications and standards.

Comply with FSRIA 9002 USDA BioPreferred Program. Refer to www.biopreferred.gov for the product categories and BioPreferred Catalog. Selected products must comply with non-proprietary requirements of the Federal Acquisition Regulation and must meet performance requirements. Provide the following documentation:

a. USDA BioPreferred label for each product; for bio-based products used on project but not listed with BioPreferred program, provide bio-based content and percentage.

b. In order to complete compliance with FAR 52.223-1 Biobased Product Certification, refer to submittal requirement for biobased products in Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME), paragraphs CERTIFICATION OF EPA DESIGNATED ITEMS and CERTIFICATION OF USDA DESIGNATED ITEMS.

1.6.13 Waste Material Management (Recycling – Construction)

Divert demolition and construction debris in accordance with Section 01 74 19.00 22 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL (PWD ME).

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 SUSTAINABILITY COORDINATION

Provide sustainability focus and coordination at all meetings to achieve sustainability goals. Coordinate meeting requirements with other UFGS Sections meeting requirements in this project. Ensure the designated [TPC accredited]sustainability professional responsible for GP [and TPC]documentation participates in these meetings to coordinate documentation completion. Review GP[and TPC] sustainability requirements, HPSB Checklist[and TPC] documentation, Sustainability Action Plan, and completeness status of Sustainability eNotebook[, and TPC status] at the following meetings:

a. Pre-Construction Conference

b. Construction Quality Control Meetings

c. Facility Turnover Meetings

Conduct review no later than 60 days before final turnover and identify any outstanding issues that affect correct completion of all documentation[and final TPC certification, assessment or validation], and actions that will achieve requirements. Conduct corrective actions prior to turnover, to ensure all requirements are achieved.

3.2 TABLE 3-1 VOLATILE ORGANIC COMPOUNDS (VOC) (LOW EMITTING MATERIALS) REQUIREMENTS

TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements

Source: ICC IgCC Chapter 8 (Materials) (Interior Applications Only)

MATERIAL CATEGORY	**EMISSIONS REQUIREMENT**		**MATERIALS WITH ADDED VOC REQUIREMENT**	**EMISSIONS REQUIREMENTS**
Adhesives and Sealants	**CDPH/EHLB/Standard method V1.1 (California Section 01350)** (Use "office" or "classroom" space limits for all applications)	or	Adhesives (carpet, resilient, wood flooring; base cove; ceramic tile; drywall and panel; primers) Sealants (acoustical; firestop; HVAC Air duct; primers) Caulks	**SCAQMD Rule 1168** (Use "other" category for HVAC duct sealant) (for firestop adhesive, UFC 3-600-01 overrides conflicting requirements)
			Aerosol adhesives	**Section 3 of Green Seal Standard GS-36** (except: cleaners, solvent cements, and primers used with plastic piping and conduit in plumbing, fire suppression, and electrical systems; HVAC air duct sealants when the application space air temp is less than 40 F (4.5 C).

TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements

Source: ICC IgCC Chapter 8 (Materials) (Interior Applications Only)

MATERIAL CATEGORY	EMISSIONS REQUIREMENT		MATERIALS WITH ADDED VOC REQUIREMENT	EMISSIONS REQUIREMENTS
Paints and Coatings	**CDPH/EHLB/Standard method V1.1 (California Section 01350)** (Use "office" or "classroom" space limits for all applications)	or	Flat and nonflat, nonflat high-gloss, specialty, basement specialty, fire-resistive, floor, low-solids, rust preventative, wood, reflective wall coatings; concrete/masonry sealers; primers; sealers; undercoaters; shellacs (clear and opaque); stains; varnishes; conjugated oil varnish; lacquer; clear brushing lacquer	**Green Seal Standard GS-11**

TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements

Source: ICC IgCC Chapter 8 (Materials) (Interior Applications Only)

MATERIAL CATEGORY	**EMISSIONS REQUIREMENT**		**MATERIALS WITH ADDED VOC REQUIREMENT**	**EMISSIONS REQUIREMENTS**
Paints and Coatings	**CDPH/EHLB/Standard method V1.1 (California Section 01350)** (Use "office" or "classroom" space limits for all applications)	or	Concrete curing compounds; dry fog, faux finishing, graphic arts (sign paints), industrial maintenance, mastic texture, metallic pigmented, multicolor, recycled coatings; pretreatment wash primers, reactive penetrating sealers; specialty primers, wood preservatives, and zinc primers	**California Air Resources Board (CARB) Suggested Control Measure for Architectural Coatings** or **SCAQMD Rule 1113r**
Paints and Coatings	**CDPH/EHLB/Standard method V1.1 (California Section 01350)** (Use "office" or "classroom" space limits for all applications)	or	High-temperature coatings; stone consolidants; swimming-pool coatings; tub- and tile-refining coatings; and waterproofing membranes	**California Air Resources Board (CARB) Suggested Control Measure for Architectural Coatings**

TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements

Source: ICC IgCC Chapter 8 (Materials) (Interior Applications Only)

MATERIAL CATEGORY	**EMISSIONS REQUIREMENT**		**MATERIALS WITH ADDED VOC REQUIREMENT**	**EMISSIONS REQUIREMENTS**
Floor Covering Materials	For carpet, all locations: **CDPH/EHLB/Standard Method V1.1 (California Section 01350)** or label for **Section 9** of **CDPH/EHLB/Standard Method V1.1 (California Section 01350)**		none	none
Insulation	**CDPH/EHLB/Standard method V1.1** (California Section 01350) (Use "office" or "classroom" space limits for all applications)		none	none

TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements

Source: ICC IgCC Chapter 8 (Materials) (Interior Applications Only)

MATERIAL CATEGORY	**EMISSIONS REQUIREMENT**		**MATERIALS WITH ADDED VOC REQUIREMENT**	**EMISSIONS REQUIREMENTS**
Composite Wood, Wood Structural Panel, and Agrifiber Products, no added urea-formaldehyde resins including laminating adhesives for composite wood and agrifiber assemblies - particleboard, medium density fiberboard (MDF), wheatboard, strawboard, panel substrates, door cores	Third-party certification (approved by CARB) of **California Air Resource Board's (CARB) regulation**, Airborne Toxic Control Measure to Reduce Formaldehyde Emissions from Composite Wood Products	or	none	**CDPH/EHLB/Standard method V1.1** (California Section 01350) (Use "office" or "classroom" space limits for all applications) (except: Structural panel components such as plywood, particle board, wafer board, and oriented strand board identified as "EXPOSURE 1," "EXTERIOR," or "HUD-APPROVED" are considered acceptable for interior use.)
Office Furniture Systems and Seating installed prior to occupancy	**ANSI/BIFMA X7.1 ANSI/BIFMA X7.1:** (95-percent of installed office furniture system workstations and seating units) **Section 7.6.2 of ANSI/BIFMA e3** (50-percent of office furniture system workstations and seating units)		none	none

TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements

Source: ICC IgCC Chapter 8 (Materials) (Interior Applications Only)

MATERIAL CATEGORY	**EMISSIONS REQUIREMENT**		**MATERIALS WITH ADDED VOC REQUIREMENT**	**EMISSIONS REQUIREMENTS**
Ceiling and Wall assemblies and systems including: acoustical treatments; ceiling panels and tiles; tackable wall panels and coverings; wall coverings; wall and ceiling paneling and planking	**CDPH/EHLB/Standard method V1.1 (California Section 01350)** (Use "office" or "classroom" space limits for all applications)		none	none

-- End of Section --

SECTION 01 35 26.00 22

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GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME)

05/24

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at the Portsmouth Naval Shipyard.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 52.2	(2017) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ANSI/ASSP)

ANSI/ASSP A10.22	(2007; R 2017) Construction and Demolition Operations - Safety Requirements for Rope-Guided and Non-Guided Workers’ Hoists
ANSI/ASSP A10.34	(2021) Protection of the Public on or Adjacent to Construction Sites
ANSI/ASSP A10.44	(2020) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations
ANSI/ASSP Z244.1	(2016; R 2020) The Control of Hazardous Energy Lockout, Tagout and Alternative Methods
ANSI/ASSP Z359.0	(2023) Z359 Committee Guidance Document for Definitions and Nomenclature Used in Z359 Fall Protection and Fall Restraint Standards
ANSI/ASSP Z359.1	(2020) The Fall Protection Code
ANSI/ASSP Z359.11	(2021) Safety Requirements for Full Body Harnesses
ANSI/ASSP Z359.12	(2019) Connecting Components for Personal Fall Arrest Systems
ANSI/ASSP Z359.13	(2013; R 2022) Personal Energy Absorbers and Energy Absorbing Lanyards
ANSI/ASSP Z359.14	(2021) Safety Requirements for Self-Retracting Devices for Personal Fall

	Arrest and Rescue Systems
ANSI/ASSP Z359.15	(2024) Safety Requirements for Single Anchor Lifelines and Fall Arresters for Personal Fall Arrest Systems
ANSI/ASSP Z359.16	(2016) Safety Requirements for Climbing Ladder Fall Arrest Systems
ANSI/ASSP Z359.18	(2017) Safety Requirements for Anchorage Connectors for Active Fall Protection Systems
ANSI/ASSP Z359.2	(2023) Minimum Requirements for a Comprehensive Managed Fall Protection Program
ANSI/ASSP Z359.3	(2019) Safety Requirements for Lanyards and Positioning Lanyards
ANSI/ASSP Z359.4	(2013; R 2022) Safety Requirements for Assisted-Rescue and Self-Rescue Systems, Subsystems and Components
ANSI/ASSP Z359.6	(2016) Specifications and Design Requirements for Active Fall Protection Systems
ANSI/ASSP Z359.7	(2019) Qualification and Verification Testing of Fall Protection Products
ANSI/ASSP Z490.1	(2024) Criteria for Accepted Practices in Safety, Health, and Environmental Training

ASME INTERNATIONAL (ASME)

ASME B30.20	(2021) Below-the-Hook Lifting Devices
ASME B30.22	(2023) Articulating Boom Cranes
ASME B30.23	(2022) Personnel Lifting Systems - Safety Standard for Cableways, Cranes, Derricks, Hoists, Hooks, Jacks, and Slings
ASME B30.26	(2015; R2020) Rigging Hardware
ASME B30.3	(2019) Tower Cranes
ASME B30.5	(2021) Mobile and Locomotive Cranes
ASME B30.7	(2021) Winches
ASME B30.8	(2020) Floating Cranes and Floating Derricks
ASME B30.9	(2021) Slings

ASTM INTERNATIONAL (ASTM)

ASTM D6245	(2012) Using Indoor Carbon Dioxide Concentrations to Evaluate Indoor Air Quality and Ventilation
ASTM D6345	(2010) Standard Guide for Selection of Methods for Active, Integrative Sampling of Volatile Organic Compounds in Air
ASTM F855	(2023) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048	(2016) Guide for Protective Grounding of Power Lines
IEEE C2	(2023) National Electrical Safety Code

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z89.1	(2014; R 2019) American National Standard for Industrial Head Protection

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA Z535.2	(2023) American National Standard for Environmental and Facility Safety Signs

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10	(2022; ERTA 1 2021) Standard for Portable Fire Extinguishers
NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations
NFPA 306	(2024) Standard for Control of Gas Hazards on Vessels
NFPA 51B	(2024) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work
NFPA 70	(2023) National Electrical Code
NFPA 70E	(2024) Standard for Electrical Safety in the Workplace

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

ANSI/SMACNA 008	(2007) IAQ Guidelines for Occupied Buildings Under Construction, 2nd Edition

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-222	(2023) Structural Standard for Antenna Supporting Structures and Antennas and Small Wind Turbine Support Structures
TIA-1019	(2012; R 2016) Standard for Installation, Alteration and Maintenance of Antenna Supporting Structures and Antennas

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 20	Standards for Protection Against Radiation
29 CFR 26	(2020) Administrative Review Board Rules of Practice and Procedure
29 CFR 1910	Occupational Safety and Health Standards
29 CFR 1910.146	Permit-required Confined Spaces
29 CFR 1910.147	Control of Hazardous Energy (Lock Out/Tag Out)
29 CFR 1910.333	Selection and Use of Work Practices
29 CFR 1915	Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment
29 CFR 1915.89	Control of Hazardous Energy (Lockout/Tags-Plus)
29 CFR 1919	Gear Certification
29 CFR 1926	Safety and Health Regulations for Construction
29 CFR 1926.1400	Cranes and Derricks in Construction
29 CFR 1926.16	Rules of Construction
29 CFR 1926.450	Scaffolds
29 CFR 1926.500	Fall Protection
29 CFR 1926.552	Material Hoists, Personal Hoists, and Elevators
29 CFR 1926.553	Base-Mounted Drum Hoists
49 CFR 173	Shippers - General Requirements for Shipments and Packagings

CPL 2.100	(1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146
ATTACHMENT A	"CONTRACTOR CRANE, LHE,CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS"
ATTACHMENT B	"NAVFAC DIG SAFE UTILITY LOCATE REQUEST FORM"
ATTACHMENT C	"CONTRACTOR DIG-SAFE CERTIFICATION FORM"
ATTACHMENT D	"CONTRACTOR UTILITY LOCATE - PRE-EXCAVATION SAFETY CHECKLIST"
ATTACHMENT E	"CONTRACTOR DAILY CHECKLIST FOR TRENCHING/EXCAVATION"
ATTACHMENT F	"UTILITY STRIKE REVIEW CHECKLIST"

The attachments are included at the end of this Section. If the attachments are missing, notify the Contracting Officer.

1.2 DEFINITIONS

The following definitions are for the convenience of the reader. If there is a referenced document in the text of this specification section, that is the document that should define terms for that paragraph. If further clarification is needed, contact the Contracting Officer.

1.2.1 Site Safety and Health Officer (SSHO)

A Contractor Employee that is responsible for overseeing and ensuring implementation of the prime Contractor's Safety and Occupational Health (SOH) program according to the Contract, EM 385-1-1, applicable federal, state, and local requirements.

1.2.1.1 Level One SSHO

A designated employee with full-time SOH responsibility that meets and follows the requirements of EM 385-1-1.

1.2.1.2 Level Two SSHO

A designated employee with Level Two SSHO responsibility that meets and follows the requirements of EM 385-1-1. Level Two SSHOs cannot be assigned to projects that have a residual Risk Assessment Code (RAC) of high or extremely high.

1.2.1.3 Level Three SSHO

A designated Qualified Person or Competent Person with SOH responsibility that meets and follows the requirements of EM 385-1-1. Level 3 SSHOs cannot be assigned to projects that have a residual RAC of high or extremely high.

1.2.1.4 Alternate SSHO

An employee that meets the definition of the contract-required level SSHO, but is not the primary SSHO.

1.2.2 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge, and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are unsanitary, hazardous, or dangerous to personnel, and who has authorization to take prompt corrective measures to eliminate them.

1.2.3 Competent Person, Confined Space

The CP, Confined Space, is a person meeting the competent person requirements as defined in EM 385-1-1, with thorough knowledge of OSHA's Confined Space Standard, 29 CFR 1910.146 and 29 CFR 1926 Subpart AA, and designated in writing to be responsible for the immediate supervision, implementation, and monitoring of the confined space program, who through training, knowledge, and experience in confined space entry is capable of identifying, evaluating, and addressing existing and potential confined space hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.4 Competent Person, Cranes and Rigging

The CP, Cranes and Rigging, as defined in EM 385-1-1, is a person meeting the competent person, who has been designated in writing to be responsible for the immediate supervision, implementation, and monitoring of the Crane and Rigging Program, who through training, knowledge, and experience in crane and rigging is capable of identifying, evaluating, and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.5 Competent Person, Excavation/Trenching

A CP, Excavation/Trenching, is a person meeting the competent person requirements as defined in EM 385-1-1 and 29 CFR 1926, who has been designated in writing to be responsible for the immediate supervision, implementation, and monitoring of the excavation/trenching program, who through training, knowledge, and experience in excavation/trenching is capable of identifying, evaluating, and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.6 Competent Person, Fall Protection

The CP, Fall Protection, is a person meeting the competent person requirements as defined in EM 385-1-1 and in accordance with ANSI/ASSP Z359.0, who has been designated in writing by the employer to be responsible for immediate supervising, implementing, and monitoring of the fall protection program, who through training, knowledge, and experience in fall protection and rescue systems and equipment, is capable of identifying, evaluating, and addressing existing and potential fall hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.7 Competent Person, Scaffolding

The CP, Scaffolding is a person meeting the competent person requirements in EM 385-1-1, and designated in writing by the employer to be responsible for immediate supervising, implementing, and monitoring of the scaffolding program. The CP for Scaffolding has enough training, knowledge, and experience in scaffolding to correctly identify, evaluate and address existing and potential hazards and also has the authority to take prompt corrective measures with regard to these hazards. CP qualifications must be documented and include experience on the specific scaffolding systems/types being used, assessment of the base material that the scaffold will be erected upon, load calculations for materials and personnel, and erection and dismantling. The CP for scaffolding must have a documented, minimum of 8-hours of scaffold training to include training on the specific type of scaffold being used (e.g. mast-climbing, adjustable, tubular frame), in accordance with EM 385-1-1.

1.2.8 Competent Person (CP) Trainer

A competent person trainer is a person who is qualified in the material presented, and who possesses a working knowledge of applicable technical regulations, standards, equipment and systems related to the subject matter on which they are training Competent Persons. A competent person trainer must be familiar with the typical hazards and the equipment used in the industry they are instructing. The training provided by the competent person trainer must be appropriate to that specific industry. The competent person trainer must evaluate the knowledge and skills of the competent persons as part of the training process.

1.2.9 High Risk Activities

High Risk Activities are activities that involve work at heights, crane and rigging, excavations and trenching, scaffolding, electrical work, and confined space entry.

1.2.10 High Visibility Accident

A High Visibility Accident is any mishap which may generate publicity or high visibility.

1.2.11 Load Handling Equipment (LHE)

LHE is a term used to describe cranes, hoists and all other hoisting equipment (hoisting equipment means equipment, including crane, derricks, hoists and power operated equipment used with rigging to raise, lower or horizontally move a load).

1.2.12 Medical Treatment

Medical Treatment is treatment administered by a physician or by registered professional personnel under the standing orders of a physician. Medical treatment does not include first aid treatment even through provided by a physician or registered personnel.

1.2.13 Near Miss

A Near Miss is a mishap resulting in no personal injury and zero property damage, but given a shift in time or position, damage or injury may have occurred (e.g., a worker falls off a scaffold and is not injured; a crane swings around to move the load and narrowly misses a parked vehicle).

1.2.14 Operating Envelope

The Operating Envelope is the area surrounding any crane or load handling equipment. Inside this "envelope" is the crane, the operator, riggers and crane walkers, other personnel involved in the operation, rigging gear between the hook, the load, the crane's supporting structure (i.e. ground or rail), the load's rigging path, the lift and rigging procedure.

1.2.15 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.2.16 Qualified Person, Fall Protection (QP for FP)

A QP for FP is a person meeting the requirements of EM 385-1-1 and ANSI/ASSP Z359.0, with a recognized degree or professional certificate and with extensive knowledge, training, and experience in the fall protection and rescue field who is capable of designing, analyzing, and evaluating and specifying fall protection and rescue systems.

1.2.17 Recordable Injuries or Illnesses

Recordable Injuries or Illnesses are any work-related injury or illness that results in:

a. Death, regardless of the time between the injury and death, or the length of the illness;

b. Days away from work (any time lost after day of injury/illness onset);

c. Restricted work;

d. Transfer to another job;

e. Medical treatment beyond first aid;

f. Loss of consciousness; or

g. A significant injury or illness diagnosed by a physician or other licensed health care professional, even if it did not result in (a) through (f) above.

1.2.18 USACE Property and Equipment

Interpret "USACE" property and equipment specified in USACE EM 385-1-1 as Government property and equipment.

1.2.19 Load Handling Equipment (LHE) Accident or Load Handling Equipment Mishap

A LHE accident occurs when any one or more of the eight elements in the operating envelope fails to perform correctly during operation, including operation during maintenance or testing resulting in personnel injury or death; material or equipment damage; dropped load; derailment; two-blocking; overload; or collision, including unplanned contact between the load, crane, or other objects. A dropped load, derailment, two-blocking, overload, and collision are considered accidents, even though no material damage or injury occurs. A component failure (e.g., motor burnout, gear tooth failure, or bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, or roll over). Document an LHE mishap or accident using the NAVFAC prescribed Navy Crane Center (NCC) accident form.

1.3 SUBMITTALS

Government Acceptance or Approval does not remove responsibility from the Contractors for their actions or liability.

Government approval is required for submittals with a "G" classification; submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G

APP - Construction; G

Dive Operations Plan; G

Indoor Air Quality (IAQ) Management Plan; G

SD-06 Test Reports

Monthly Exposure Reports; G

Notifications and Reports; G

Accident Reports; G

LHE Inspection Reports; G

SD-07 Certificates

Contractor Safety Self-Evaluation Checklist; G

Crane Operators/Riggers; G

Standard Lift Plan; G

Critical Lift Plan; G

Naval Architecture Analysis; G

Activity Hazard Analysis (AHA); G

Confined Space Entry Permit; G

Hot Work Permit; G

Certificate of Compliance; G

Third Party Certification of Floating Cranes and Barge-Mounted Mobile Cranes; G

License Certificates; G

Radiography Operation Planning Work Sheet; G

1.4 PUBLIC HEALTH EMERGENCIES

In the event of a declared public health emergency, follow safety precautions as required by the Occupational Safety and Health Administration (OSHA) www.osha.gov, the Centers for Disease Control and Prevention (CDC) www.cdc.gov, and as required by federal, state and local requirements.

1.5 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report by the fifth of each month. This report is a compilation of employee-hours worked each month for all site workers, both Prime and Subcontractor. Failure to submit the report may result in retention of up to 10 percent of the progress payment.

1.6 CONTRACTOR SAFETY SELF-EVALUATION CHECKLIST

Contracting Officer will provide a "Contractor Safety Self-Evaluation checklist" to the Contractor at the preconstruction meeting. Complete the checklist monthly and submit with each request for payment. This submission is required monthly even when a progress payment is not requested. An acceptable score of 90 or greater is required. Failure to submit the completed safety self-evaluation checklist or achieve a score of at least 90 may result in retention of up to 10 percent of the voucher. The Contractor Safety Self-Evaluation checklist can be found on the Whole Building Design Guide website at www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-35-26

1.7 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1, and applicable Federal, State, and local laws, ordinances, criteria, rules, and regulations at the date of the solicitation for this Contract. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this Specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.7.1 Subcontractor Safety Requirements

For this Contract, neither Contractor nor any Subcontractor may enter into Contract with any Subcontractor that fails to meet the following requirements. The term Subcontractor in this and the following paragraphs means any entity holding a Contract with the Contractor or with a Subcontractor at any tier.

1.7.1.1 Experience Modification Rate (EMR)

Subcontractors on this Contract must have an effective EMR less than or equal to 1.10, as computed by the National Council on Compensation

Insurance (NCCI) or if not available, as computed by the state agency's rating bureau in the state where the Subcontractor is registered, when entering into a subcontract agreement with the Prime Contractor or a Subcontractor at any tier. The Prime Contractor may submit a written request for additional consideration to the Contracting Officer where the specified acceptable EMR range cannot be achieved. Relaxation of the EMR range will only be considered for approval on a case-by-case basis for special conditions and must not be anticipated as tacit approval. Contractor's Site Safety and Health Officer (SSHO) must collect and maintain the certified EMR ratings for all Subcontractors on the project and make them available to the Government at the Government's request.

1.7.1.2 OSHA Days Away from Work, Restricted Duty, or Job Transfer (DART) Rate

Subcontractors on this Contract must have a DART rate, calculated from the most recent, complete calendar year, less than or equal to 3.4 when entering into a subcontract agreement with the Prime Contractor or a Subcontractor at any tier. The OSHA Dart Rate is calculated using the following formula:

(N/EH) x 200,000

where:

N = number of injuries and illnesses with days away, restricted work, or job transfer

EH = total hours worked by all employees during most recent, complete calendar year

200,000 = base for 100 full-time equivalent workers (working 40 hours per week, 50 weeks per year)

The Prime Contractor may submit a written request for additional consideration to the Contracting Officer where the specified acceptable OSHA Dart rate range cannot be achieved for a particular Subcontractor. Relaxation of the OSHA DART rate range will only be considered for approval on a case-by-case basis for special conditions and must not be anticipated as tacit approval. Contractor's Site Safety and Health Officer (SSHO) must collect and maintain self-certified OSHA DART rates for all Subcontractors on the project and make them available to the Government at the Government's request.

1.8 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

The Quality Control Manager cannot be the SSHO on the project.

1.8.1 Site Safety and Health Officer (SSHO)

1.8.1.1 Qualifications of SSHO

All SSHOs must meet the training and experience requirements identified in the EM 385-1-1 and this Contract.

The SSHO must ensure that the requirements of 29 CFR 1926.16 are met for the project. Provide a Safety oversight team that includes a minimum of one (1) person at each project site to function as the Site Safety and Health Officer (SSHO). The SSHO or an Alternate SSHO meeting the requirements of the contract-required level SSHO but is not the primary SSHO must be at the project site at all times to implement and administer the Contractor's safety program and Government-accepted Accident Prevention Plan. The SSHO and Alternate SSHO must have the required training, experience, and qualifications in accordance with EM 385-1-1, and all associated sub-paragraphs. Note the "Project Site" is the specific area of construction within the limits of work as defined by the contract documents. The SSHO shall be a direct employee of the prime contractor, not an employee of a subcontractor.

1.8.1.2 Duties of SSHO

All SSHOs must carry out the roles and responsibilities as identified in this Contract and the EM 385-1-1. All SSHOs will be designated on an ENG Form 6282, provided by the Contracting Officer. Superintendent, QC Manager, and SSHO are subject to dismissal if their required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

The SSHO must:

a. Conduct daily safety and health inspections and maintain a written log which includes area/operation inspected, date of inspection, identified hazards, recommended corrective actions, and estimated and actual dates of corrections. Attach safety inspection logs to the Contractor's daily production report.

b. Conduct mishap investigations and complete required accident reports. Report mishaps and near misses.

c. Use and maintain OSHA's Form 300 to log work-related injuries and illnesses occurring on the project site for Prime Contractors and Subcontractors, and make available to the Contracting Officer upon request. Post and maintain the Form 300 on the site Safety Bulletin Board.

d. Maintain applicable safety reference material on the job site.

e. Attend the pre-construction meeting, pre-work meetings including preparatory meetings, and periodic in-progress meetings.

f. Review the APP and AHAs for compliance with EM 385-1-1, and approve, sign, implement, and enforce them.

g. Establish a Safety and Occupational Health (SOH) Deficiency Tracking System that lists and monitors outstanding deficiencies until resolution.

h. Ensure Subcontractor compliance with safety and health requirements.

i. Maintain a list of hazardous chemicals on site and their associated Safety Data Sheets (SDS).

j. Maintain a weekly list of high hazard activities involving energy, equipment, excavation, entry into confined space, and elevation, and be prepared to discuss details during QC Meetings.

k. Provide and keep a record of site safety orientation and indoctrination for Contractor employees, Subcontractor employees, and site visitors.

Superintendent, QC Manager, and SSHO are subject to dismissal if the above or any other required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out. The dismissal of the Superintendent, QC Manager, and/or SSHO will not be cause for claims for a Contract modification(s) for an extension to the Contract duration or for additional compensation.

1.8.2 Roles and Responsibilities of Prime Contractor and SSHO

The Prime Contractor and SSHO must ensure that the requirements of all applicable OSHA and EM 385-1-1 are met for the project. The Prime Contractor must ensure an SSHO or an equally qualified Alternate SSHO(s) is at the worksite at all times to implement and administer the Contractor's safety program and Government accepted Accident Prevention Plan. If the required SSHO has to temporarily (that is, up to 24 hours / 1 day) leave the site of work due to unforeseen or emergency situations, a Level One, Two, or Three SSHO may be used in the interim and must be on the site of work at all times when work is being performed.

If the SSHO must be off-site for a period longer than 24 hours / 1 day, a qualified alternate SSHO that meets the contract requirements must be onsite.

a. Prime contractor must ensure all SSHOs:

(1) Are designated on an ENG Form 6282.

(2) Meet minimum training and experience requirements identified in EM 385-1-1.

(3) Execute roles and responsibilities identified in EM 385-1-1.

The primary SSHO may not be absent from the site for more than two weeks at one time and not more than 30 work days in a calendar year.

1.8.3 Additional Requirements

The SSHO must be a direct employee of the Prime Contractor, not an employee of a subcontractor.

The Level One SSHO must not serve as the Quality Control Manager. The One SSHO must not serve as the Superintendent.

The SSHO must have completed a 40-hour contract safety awareness course based on the content and principles of EM 385-1-1, and instructed in accordance with the guidelines of ANSI/ASSP Z490.1, by a trainer meeting the qualifications of paragraph QUALIFIED TRAINER REQUIREMENTS. If the SSHO does not have a current certification, certification must be obtained within 60 days, maximum, of Contract award.

1.8.4 Contract Site Safety And Health Officer(s)(SSHOs) Minimum Requirements

Provide a minimum of one Level One SSHO that meets the requirements of EM 385-1-1 for this project.

1.8.5 Competent Person (CP) Qualifications

Provide Competent Persons in accordance with EM 385-1-1 and herein. Competent Persons for high risk activities include confined space, cranes and rigging, excavation/trenching, fall protection, and electrical work. The CP for these activities must be designated in writing, and meet the requirements for the specific activity (i.e. competent person, fall protection).

The Competent Person identified in the Contractor's Safety and Health Program and accepted Accident Prevention Plan, must be on-site at all times when the work that presents the hazards associated with their professional expertise is being performed. Provide the credentials of the Competent Persons(s) to the the Contracting Officer for information in consultation with the Portsmouth Naval Shipyard Safety Office.

1.8.5.1 Competent Person (CP) for Confined Space Entry

Provide a CP for Confined Space Entry who meets the requirements of EM 385-1-1 and herein. The CP for Confined Space Entry must supervise the entry into each confined space in accordance with EM 385-1-1.

If the work involves operations that handle combustible or hazardous materials, this person must have the ability to understand and follow through on the air sampling, Personal Protective Equipment (PPE), and instructions of a Marine Chemist, Coast Guard authorized persons, or Certified Industrial Hygienist. Confined space and enclosed space work must comply with NFPA 306, OSHA 29 CFR 1915, Subpart B, "Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment," or as applicable, 29 CFR 1910.147 for general industry, and 29 CFR 26 for construction.

1.8.5.2 Competent Person for Scaffolding

Provide a Competent Person for Scaffolding who meets the requirements of EM 385-1-1 and herein.

1.8.5.3 Competent Person for Fall Protection

Provide a Competent Person for Fall Protection who meets the requirements of EM 385-1-1 and herein.

1.8.6 Qualified Trainer Requirements

Individuals qualified to instruct the 40 hour Contract safety awareness course, or portions thereof, must meet the definition of a Competent Person Trainer as defined in the EM 385-1-1, and, at a minimum, possess a working knowledge of the following subject areas: EM 385-1-1, Electrical Standards, Lockout/Tagout, Fall Protection, Confined Space Entry for Construction, Excavation, Trenching and Soil Mechanics, and Scaffolds in accordance with 29 CFR 1926.450, Subpart L.

Instructors are required to:

a. Prepare class presentations that cover construction-related safety requirements.

b. Ensure that all attendees attend all sessions by using a class roster signed daily by each attendee. Maintain copies of the roster for at least five (5) years. This is a certification class and must be attended 100 percent. In cases of emergency where an attendee cannot make it to a session, the attendee can make it up in another class session for the same subject.

c. Update training course materials whenever an update of the EM 385-1-1 becomes available.

d. Provide a written exam of at least 50 questions. Students are required to answer 80 percent correctly to pass.

e. Request, review, and incorporate student feedback into a continuous course improvement program.

1.8.7 Crane Operators/Riggers

Provide Operators, Signal Persons, and Riggers meeting the requirements in EM 385-1-1 for Riggers, Crane Operators, and Signal Persons. In addition, for mobile cranes with Original Equipment Manufacturer (OEM) rated capacities of 50,000 pounds or greater, designate crane operators qualified by a source that qualifies crane operators (i.e., union, a Government agency, or an organization that tests and qualifies crane operators). Provide proof of current qualification.

1.8.8 Meetings

1.8.8.1 Preconstruction Meeting

a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction meeting. This includes the Project Superintendent, Site Safety and Occupational Health Officer, Quality Control Manager, and any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program, and procedures associated with it).

b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures, and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays. The development of the APP and Project Schedule shall commence after receipt of the Notice to Proceed from the Contracting Officer.

c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

d. See Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME) paragraph PRECONSTRUCTION MEETING for additional information.

1.8.8.2 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors on the project location. The SSHO, supervisors, foremen, or Collateral Duty Safety Officer's (CDSOs) must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.9 ACCIDENT PREVENTION PLAN (APP)

1.9.1 APP - Construction

Submit the Accident Prevention Plan (APP) for review and acceptance by the Government at least 15 calendar days prior to the start, after being given Notice to Proceed. A competent person must prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of EM 385-1-1, ENG Form 6293, and as specified and supplemented herein. Cover all paragraph and subparagraph elements in EM 385-1-1. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and occupational health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential Subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of Subcontractors. The Government considers the Prime Contractor to be the "controlling employer" for all worksite safety and health of the Subcontractors. Contractor's are responsible for informing their Subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one trade/craft from interfering with or creating hazardous working conditions for other trades/crafts, and inspecting Subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed in accordance with the APP and ENG Form 6293 Accident Prevention Plan Worksheet, by an officer of the firm (Prime Contractor senior person), the individual preparing the APP, the on-site superintendent, the designated SSHO, the Contractor Quality Control Manager, and any designated Certified Safety Professional (CSP) or Certified Health Physicist (CIH). The SSHO must provide and maintain the APP and a log of signatures by each Subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Submit the APP to the Contracting Officer within 30 calendar days of Contract award and not less than 10 calendar days prior to the date of the preconstruction meeting for acceptance. Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the

APP and attachments will be enforced as part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, Project Superintendent, SSHO, and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e. imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within twenty-four (24) hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ANSI/ASSP A10.34), and the environment.

1.9.2 Names and Qualifications

Provide plans in accordance with the requirements outlined in EM 385-1-1, including the following:

a. Names and qualifications (resumes including education, training, experience, and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.

b. As a minimum, designate and submit qualifications of Competent Persons (CP) for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use, and maintenance. Designate and submit qualifications for additional CPs as applicable to the work performed under this Contract.

1.9.3 Plans

Provide plans in the APP in accordance with the requirements outlined in EM 385-1-1, including, but not limited to, the following:

1.9.3.1 Confined Space Entry Plan

Develop a confined or enclosed space entry plan in accordance with EM 385-1-1, applicable OSHA standards 29 CFR 1910, 29 CFR 1915, and 29 CFR 1926, OSHA Directive CPL 2.100, and any other Federal, State, and local regulatory requirements identified in this Contract. Identify the qualified person's name and qualifications, training, and experience. Delineate the qualified person's authority to direct work stoppage in the event of hazardous conditions. Include procedure for rescue by Contractor personnel and the coordination with emergency responders. (If there is no confined space work, include a statement that no confined space work exists and none will be created.)

1.9.3.2 Standard Lift Plan (SLP)

Plan all lifts to avoid situations where the operator cannot maintain safe control of the lift. Prepare a written SLP in accordance with EM 385-1-1 using ENG Form 6203 for every lift or series of lifts (if duty cycle or routine lifts are being performed). The SLP must be developed, reviewed, and accepted by all personnel involved in the lift in conjunction with the associated AHA. Signature on the AHA constitutes acceptance of the plan. Maintain the SLP on the LHE for the current lift(s) being made. Maintain historical SLPs for a minimum of three (3) months. Keep a copy of the SLP on the LHE for the specific current approved lift(s) being made. Keep a copy of every SLP for a minimum of three (3) months. Use the ENG Form 6203 (Crane Standard Lift Plan), included as an appendix to this Section.

1.9.3.3 Critical Lift Plan - Crane or Load Handling Equipment

Provide a Critical Lift Plan as required by EM 385-1-1. Before any critical lift, a CP or QP must prepare a Critical Lift Plan and submit it to the Contracting Officer or Contracting Officer's Representative for acceptance. Use the ENG Form 6213 (Load-Handling Equipment Crane Operation Critical Lift Plan). In addition, Critical Lift Plans are required for the following:

a. Lifts over 50 percent of the capacity of barge mounted mobile crane's hoist.

b. When working around energized power lines where the work will get closer than the minimum clearance distance in EM 385-1-1.

c. For lifts with anticipated binding conditions.

d. When erecting cranes.

1.9.3.3.1 Critical Lift Plan Planning and Schedule

Critical lifts require detailed planning and additional or unusual safety precautions. Develop and submit a critical lift plan to the Contracting Officer 30 calendar days prior to critical lift. Comply with load testing requirements in accordance with EM 385-1-1.

1.9.3.3.2 Lifts of Personnel

In addition to the requirements of EM 385-1-1, for lifts of personnel, demonstrate compliance with the requirements of 29 CFR 1926.1400 and EM 385-1-1.

1.9.3.4 Barge Mounted Mobile Crane Lift Plan

Provide a Naval Architecture Analysis and include an LHE Manufacturer's Floating Service Load Chart in accordance with EM 385-1-1.

1.9.3.5 Multi-Purpose Machines, Material Handling Equipment, and Construction Equipment Lift Plan

Multi-purpose machines, material handling equipment, and construction equipment used to lift loads that are suspended by rigging gear, require proof of authorization from the machine OEM that the machine is capable of making lifts of loads suspended by rigging equipment. Written approval from a qualified registered professional engineer, after a safety analysis is performed, is allowed in lieu of the OEM's approval. Demonstrate that the operator is properly trained and that the equipment is properly configured to make such lifts and is equipped with a load chart.

1.9.3.6 Fall Protection and Prevention (FP&P) Plan

The plan must be in accordance with the requirements of EM 385-1-1 and ANSI/ASSP Z359.2, be site specific, and address all fall hazards in the work place and during different phases of construction. Address how to protect and prevent workers from falling to lower levels when they are exposed to fall hazards above 6 feet. A competent person or qualified person for fall protection must prepare and sign the plan documentation. Include fall protection and prevention systems, equipment and methods employed for every phase of work, roles and responsibilities, assisted rescue, self-rescue and evacuation procedures, training requirements, and monitoring methods. Review and revise, as necessary, the Fall Protection and Prevention Plan documentation as conditions change, but at a minimum every six (6) months, for lengthy projects, reflecting any changes during the course of construction due to changes in personnel, equipment, systems, or work habits. Keep and maintain the accepted Fall Protection and Prevention Plan documentation at the job site for the duration of the project. Include the Fall Protection and Prevention Plan documentation in the Accident Prevention Plan (APP). A Fall Protection and Prevention Plan is required when employees are working at heights or exposed to fall hazards. The plan must be developed by a CP or QP and updated as conditions change, but at least every 6 months.

1.9.3.7 Rescue and Evacuation Plan

Provide a Rescue and Evacuation Plan in accordance with EM 385-1-1 and ANSI/ASSP Z359.2, and include in the FP&P Plan and as part of the APP. Include a detailed discussion of the following: methods of rescue; methods of self-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility.

1.9.3.8 Hazardous Energy Control Program (HECP)

Develop a HECP in accordance with EM 385-1-1, 29 CFR 1910.147, 29 CFR 1910.333, 29 CFR 1915.89, ANSI/ASSP Z244.1, and ANSI/ASSP A10.44. Submit this HECP as part of the Accident Prevention Plan (APP). Conduct a preparatory meeting and inspection with all effected personnel to coordinate all HECP activities. Document this meeting and inspection in accordance with EM 385-1-1. Ensure that each employee is familiar with and complies with these procedures.

1.9.3.9 Excavation and Trenching Plan

Identify the safety and health aspects of excavation and provide and prepare the plan in accordance with EM 385-1-1 and Section 31 00 00.00 22 EARTHWORK.

1.9.3.10 Lead, Cadmium, and Chromium Compliance Plan

Identify the safety and health aspects of work involving lead, cadmium, and chromium, including occupant protection and prepare in accordance with Section 02 83 00.00 22 MANAGEMENT OF LEAD, CADMIUM, AND CHROMIUM (VI) DURING RENOVATION, DEMOLITION, CONSTRUCTION, REMOVAL, AND ABATEMENT (PNS PROJECTS).

1.9.3.11 Asbestos Hazard Abatement Plan

Identify the safety and health aspects of asbestos work, and prepare in accordance with Section 02 82 00.00 22 ASBESTOS REMEDIATION (PNS PROJECTS).

1.9.3.12 Site Safety and Health Plan

Identify the safety and health aspects, and prepare in accordance with Section 01 35 29.13 HEALTH, SAFETY, AND EMERGENCY RESPONSE PROCEDURES FOR CONTAMINATED SITES.

1.9.3.13 Polychlorinated Biphenyls (PCB) Plan

Identify the safety and health aspects of Polychlorinated Biphenyls work, and prepare in accordance with Sections 02 84 33 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCBs) and 02 61 23 REMOVAL AND DISPOSAL OF PCB CONTAMINATED SOILS.

1.9.3.14 Lighting Lamps and Ballasts Containing PCBs and Mercury Plan

Identify the safety and health aspects of Polychlorinated Biphenyls and Mercury work, and prepare in accordance with Section 02 84 16.00 22 HANDLING OF LIGHTING BALLASTS AND LAMPS CONTAINING PCBs AND MERCURY (PWD ME).

1.9.3.15 Site Demolition Plan

Identify the safety and health aspects, and prepare in accordance with Section 02 41 00 DEMOLITION AND DECONSTRUCTION and referenced sources. Include engineering survey as applicable.

1.10 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or Subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, Subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager, and the Subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1 or as directed by the Contracting Officer. Submit the AHA for review at least 15 working days prior to the start of each activity task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel, and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.10.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.10.2 AHA Signature Log

Each employee performing work as part of an activity, task, or DFOW must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.11 DISPLAY OF SAFETY INFORMATION

1.11.1 Safety Bulletin Board

Prior to commencement of work, erect a safety bulletin board at the job site. Where size, duration, or logistics of project do not facilitate a bulletin board, an alternative method, acceptable to the Contracting Officer, that is accessible and includes all mandatory information for employee and visitor review, may be deemed as meeting the requirement for a bulletin board. Include and maintain information on safety bulletin board as required by EM 385-1-1. Additional items required to be posted include:

a. Confined space entry permit.

b. Hot work permit.

1.11.2 Safety and Occupational Health (SOH) Deficiency Tracking System

Establish a SOH deficiency tracking system that lists and monitors the status of SOH deficiencies in chronological order. Use the tracking system to evaluate the effectiveness of the APP. A monthly evaluation of the data must be discussed in the QC or SOH meeting with everyone on the project. The list must be posted on the project bulletin board and must be updated daily, and provide the following information:

a. Date deficiency identified;

b. Description of deficiency;

c. Name of person responsible for correcting deficiency;

d. Projected resolution date; and

e. Date actually resolved.

1.12 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.13 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. Government has no responsibility to provide emergency medical treatment.

1.14 NOTIFICATIONS AND REPORTS

1.14.1 Accident Notification

Notify the Contracting Officer as soon as practical, but no more than twenty-four (24) hours, after any mishaps, including recordable accidents, incidents, and near misses, as defined in EM 385-1-1 Accident Reporting Timeline, any report of injury, illness, or any property damage. For LHE or rigging mishaps, notify the Contracting Officer as soon as practical but not more than four (4) hours after the accident.

The Contractor is responsible for obtaining appropriate medical and emergency assistance and for notifying fire, law enforcement, and regulatory agencies. Immediate reporting is required for electrical mishaps, to include Arc Flash; shock; uncontrolled release of hazardous energy (includes electrical and non-electrical); load handling equipment or rigging; fall from height (any level other than same surface); and underwater diving (if applicable). These mishaps must be investigated in depth to identify all causes and to recommend hazard control measures.

Table
Accident Reporting Required Timeline

Accident Type	**Notify KO or COR**	**Complete Final Accident Report on ENG 3394 and provide to KO or COR**
Fatality, in-patient hospitalization, amputation, eye loss, or property damage over $600,000.	Immediately, no later than (NLT) 8 Hours	Within 7 Days
All other accidents and near misses	Immediately, no later than (NLT) 24 Hours	Within 7 Days

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any accident or near miss.

1.14.2 Accident Reports

a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1, to establish the root cause(s) of the accident. Complete the applicable NAVFAC Contractor Incident Reporting System (CIRS), and electronically submit via the NAVFAC Enterprise Safety Applications Management System (ESAMS). Complete and submit an accident investigation report in ESAMS within 7 days for accidents as defined by EM 385-1-1. Complete the investigation report within 30 days. Accidents must include a written report submitted as an attachment in ESAMS using the following outline:

(1) Summary description to include:

(a) process

(b) findings

(c) outcomes

(2) Root Cause

(3) Direct Factors

(4) Indirect and Contributing Factors

(5) Corrective Actions

(6) Recommendations

All accidents are reportable, regardless of whether or not it is recordable.

b. Near Misses: Complete the applicable documentation in NAVFAC Contractor Incident Reporting System (CIRS), and electronically submit via the NAVFAC Enterprise Safety Applications Management System (ESAMS). The Contracting Officer will provide the Contractor with the required forms. Near miss reports are considered positive and proactive Contractor safety management actions.

c. Conduct an accident investigation for any load handling equipment accident (including rigging gear accidents) to establish the root cause(s) of the accident. Complete the Load Handling Equipment (LHE) Accident Report (Crane and Rigging Accident Report) form and provide the report to the Contracting Officer within 30 calendar days of the accident. Do not proceed with crane operations until cause(s) is determined and corrective actions have been implemented to the satisfaction of the Contracting Officer. The Contracting Officer will provide a blank copy of the accident report form.

1.14.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.14.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

Provide a FORM 16-1 Certificate of Compliance for LHE entering the Portsmouth Naval Shipyard under this Contract and in accordance with EM 385-1-1. Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1 and using ENG Form 6203 Standard Pre-Lift Crane Plan/Checklist for each lift planned. Submit SLP to the Contracting Officer for approval within 15 calendar days in advance of planned lift.

Provide a Critical Lift Plan as required by EM 385-1-1, using NAVFAC MidLant Critical Lift Plan Form. Critical lifts require detailed planning and additional or unusual safety precautions. Develop and submit a critical lift plan to the Contracting Officer 30 calendar days prior to critical lift. Comply with load testing requirements in accordance with

EM 385-1-1.

1.14.5 Third Party Certification of Floating Cranes and Barge-Mounted Mobile Cranes

Floating cranes and barge-mounted mobile cranes used to perform work under the terms of this Contract must be certified in accordance with 29 CFR 1919 by an OSHA accredited person prior to submitting the required Lift Plan. Include proof of certification with the initial Lift Plan submission.

1.15 HOT WORK PERMIT

1.15.1 Permit and Personnel Requirements

Submit and obtain a written permit no later than five (5) working days prior to performing "Hot Work" (i.e. welding or cutting) or operating other flame-producing/spark producing devices, from the Portsmouth Naval Shipyard Fire Department. A permit is required from the Explosives Safety Office for work in and around where explosives are processed, stored, or handled. CONTRACTORS ARE REQUIRED TO MEET ALL CRITERIA BEFORE A PERMIT IS ISSUED. Provide at least two (2) 20 pound 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must be current inspection tagged, and contain an approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" performed at the Portsmouth Naval Shipyard. The Fire Watch must be trained in accordance with NFPA 51B and remain on-site for a minimum of one (1) hour after completion of the task or as specified on the hot work permit.

When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency Portsmouth Naval Shipyard Fire Department phone number. REPORT ANY FIRE, NO MATTER HOW SMALL, TO THE RESPONSIBLE PORTSMOUTH NAVAL SHIPYARD FIRE DEPARTMENT AND THE CONTRACTING OFFICER IMMEDIATELY.

a. <u>Duration of Hot Work</u>

At the discretion of the Portsmouth Naval Shipyard Fire Inspectors, a hot work permit may be written for up to one work week, (Monday through Friday), provided work is being performed at one specific location and will not hinder any life safety code.

Hot work permits must be issued for five (5) work days with a time period not to exceed twelve (12) hours. Request for weekend permits must be submitted at least two (2) working days in advance of the work to be performed and will only be valid for the weekend requested.

b. <u>Hot Work Within a Confined Space</u>

Hot work permits within a confined space will be issued on a daily basis with a time period not to exceed a twelve (12) hour shift. All requests for hot work permits within a confined space must be received a minimum of two (2) complete working days prior to entry. A duplicate copy of the air monitoring results will be furnished to the fire inspector at time of issuance.

1.15.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or a Certified Industrial Hygienist for "HOT WORK" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in EM 385-1-1.

1.16 RADIATION SAFETY REQUIREMENTS

Submit License Certificates, employee training records, and Leak Test Reports for radiation materials and equipment to the Contracting Officer and Radiation Safety Office (RSO), and Contracting Oversight Technician (COT) for all specialized and licensed material and equipment proposed for use on the construction project. Maintain on-site records whenever licensed radiological materials or ionizing equipment are on Government property.

Protect workers from radiation exposure in accordance with 10 CFR 20, ensuring any personnel exposures are maintained As Low As Reasonably Achievable.

1.16.1 Radiography Operation Planning Work Sheet

Submit a Gamma and X-Ray Radiography Operation Planning Work Sheet to the Contracting Officer 14 days prior to commencement of operations involving radioactive materials or radiation generating devices for all equipment to be used on Portsmouth Naval Shipyard.

The Contracting Officer and COT will review the submitted worksheet and provide questions and comments.

Contractors must use primary dosimeters process by a National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratory.

1.16.2 Site Access and Security

Coordinate site access and security requirements with the Contracting Officer and COT for all radiological materials and equipment containing ionizing radiation that are proposed for use on a Government facility. For gamma radiography materials and equipment, a Government escort is required for any travels on the Portsmouth Naval Shipyard. The Navy COT or Government authorized representative will meet the Contractor at a designated location outside the Portsmouth Naval Shipyard, ensure safety of the materials being transported, and will escort the Contractor for gamma sources onto the Portsmouth Naval Shipyard, to the job site, and off the Portsmouth Naval Shipyard.

Provide a copy of all calibration records, and utilization records to the COT for radiological operations performed on the site.

1.16.3 Loss or Release and Unplanned Personnel Exposure

Loss or release of radioactive materials, and unplanned personnel exposures must be reported immediately to the Contracting Officer, RSO, and Portsmouth Naval Shipyard Security Department Emergency Number.

1.16.4 Site Demarcation and Barricade

Properly demark and barricade an area surrounding radiological operations to preclude personnel entrance, in accordance with EM 385-1-1, Nuclear Regulatory Commission, and Applicable State regulations and license requirements, and in accordance with requirements established in the accepted Radiography Operation Planning Work Sheet.

Do not close or obstruct streets, walks, and other facilities occupied and used by the Government without written permission from the Contracting Officer.

1.16.5 Security of Material and Equipment

Properly secure the radiological material and ionizing radiation equipment at all times, including keeping the devices in a properly marked and locked container, and secondarily locking the container to a secure point in a Contractor's vehicle or other approved storage location during transportation and while not in use. While in use, maintain a continuous visual observation on the radiological material and ionizing radiation equipment. In instances where radiography is scheduled near or adjacent to buildings or areas having limited access or one-way doors, make no assumptions as to building occupancy. Where necessary, the Contracting Officer will direct the Contractor to conduct an actual building entry, search, and alert. Where removal of personnel from such a building cannot be accomplished and it is otherwise safe to proceed with the radiography, position a fully instructed employee inside the building or area to prevent exiting while external radiographic operations are in process.

1.16.6 Transportation of Material

Comply with 49 CFR 173 for Transportation of Regulated Amounts of Radioactive Material. Notify Local Fire authorities and the site Radiation Safety Officer (RSO) of any Radioactive Material use.

1.16.7 Schedule for Exposure or Unshielding

Actual exposure of the radiographic film or unshielding the source must be scheduled only upon final approval from the local COTS or RSO Representative and must not be initiated until after 5 p.m. on weekdays.

1.16.8 Transmitter Requirements

Adhere to the Portsmouth Naval Shipyard policy concerning the use of transmitters, such as radios and cell phones. Obey Emissions control (EMCON) restrictions.

1.17 CONFINED SPACE ENTRY REQUIREMENTS.

Confined space entry must comply with EM 385-1-1, 29 CFR 1926 Subpart AA, 29 CFR 1910, 29 CFR 1910.146, and Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used. Contractors entering and working in confined spaces while performing shipyard industry work are required to follow the requirements of 29 CFR 1915 Subpart B.

1.17.1 Entry Procedures

Prohibit entry into a confined space by personnel for any purpose, including hot work, until the qualified person has conducted appropriate tests to ensure the confined or enclosed space is safe for the work intended and that all potential hazards are controlled or eliminated and documented. Comply with EM 385-1-1 for entry procedures. Hazards pertaining to the space must be reviewed with each employee during review of the AHA.

1.17.2 Forced Air Ventilation

Forced air ventilation is required for all confined space entry operations and the minimum air exchange requirements must be maintained to ensure exposure to any hazardous atmosphere is kept below its action level.

1.17.3 Sewer Wet Wells

Sewer wet wells require continuous atmosphere monitoring with audible alarm for toxic gas detection.

1.17.4 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue from a confined space.

1.18 CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN

Submit an IAQ Management Plan within 15 calendar days after Contract award and not less than 10 calendar days before the preconstruction meeting. Revise and resubmit Plan as required by the Contracting Officer. Make copies of the final plan available to all workers on site. Include provisions in the Plan to meet the requirements specified below and to ensure safe, healthy air for construction workers and building occupants.

1.18.1 Requirements During Construction

Provide for evaluation of indoor Carbon Dioxide concentrations in accordance with ASTM D6245. Provide for evaluation of volatile organic compounds (VOCs) in indoor air in accordance with ASTM D6345. Use filters with a Minimum Efficiency Reporting Value (MERV) of 8 in permanently installed air handlers during construction.

1.18.1.1 Control Measures

Meet or exceed the requirements of ANSI/SMACNA 008 to help minimize contamination of the building from construction activities. The five requirements of this manual which must be adhered to are described below:

a. HVAC protection: Isolate return side of HVAC system from surrounding environment to prevent construction dust and debris from entering the duct work and spaces.

b. Source control: Use low emitting paints and other finishes, sealants, adhesives, and other materials as specified. When available, cleaning products must have a low VOC content and be non-toxic to minimize building contamination. Utilize cleaning techniques that minimize dust generation. Cycle equipment off when not needed. Prohibit idling motor vehicles where emissions could be drawn into building. Designate receiving/storage areas for incoming material that minimize IAQ impacts.

c. Pathway interruption: When pollutants are generated use strategies such as 100 percent outside air ventilation or erection of physical barriers between work and non-work areas to prevent contamination.

d. Housekeeping: Clean frequently to remove construction dust and debris. Promptly clean up spills. Remove accumulated water and keep work areas dry to discourage the growth of mold and bacteria. Take extra measures when hazardous materials are involved.

e. Scheduling: Control the sequence of construction to minimize the absorption of VOCs by other building materials.

1.18.1.2 Moisture Contamination

a. Remove accumulated water and keep work dry.

b. Use dehumidification to remove moist, humid air from a work area.

c. Do not use combustion heaters or generators inside the building.

d. Protect porous materials from exposure to moisture.

e. Remove and replace items which remain damp for more than a few hours.

1.18.2 Requirements After Construction

After construction ends and prior to occupancy, conduct a building flush-out or test the indoor air contaminant levels. Flush-out must be a minimum 2 weeks with MERV-13 filtration media as determined by ASHRAE 52.2 at 100 percent outside air. Air contamination testing must be consistent with EPA's current Compendium of Methods for the Determination of Air Pollutants in Indoor Air. After building flush-out or testing and prior to occupancy, replace filtration media. Filtration media must have a MERV of 13 as determined by ASHRAE 52.2.

1.19 HIGH NOISE LEVEL PROTECTION

Operations that involve the use of equipment with output of high noise levels (i.e. jackhammers, air compressors, and explosive-actuated devices) must be coordinated with the Contracting Officer. Use of any such equipment outside normal working hours must be approved in writing by the Contracting Officer prior to commencement of work.

1.20 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

a. Secure outside equipment and materials and place materials that could be damaged in protected areas.

b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.

c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS

2.1 CONFINED SPACE SIGNAGE

Provide permanent signs integral to or securely attached to access covers for new permit-required confined spaces. Signs for confined spaces must comply with NEMA Z535.2. Signs wording: "DANGER -- PERMIT-REQUIRED CONFINED SPACE, DO NOT ENTER" in bold letters a minimum of one inch in height and constructed to be clearly legible with all paint removed. The signal word "DANGER" must be red and readable from 5 feet.

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHAs, Federal and State OSHA regulations, and other related submittals and Portsmouth Naval Shipyard fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory minimum PPE includes:

a. Head Protection Hat That Meets ANSI/ISEA Z89.1

b. Long Pants

c. Appropriate Safety Footwear

d. Appropriate Class Reflective Vests

e. Gloves

f. Safety Glasses

g. Hearing Protection

h. Fall Protection

i. Reflective Foul Weather Gear When Needed

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones, or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.1.2 Hazardous Material Use

Each hazardous material must receive approval from the Contracting Officer or their designated representative prior to being brought onto the job site or prior to any other use in connection with this Contract. Allow a minimum of 10 working days for processing of the request for use of a hazardous material.

3.1.3 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury, polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. Notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on Portsmouth Naval Shipyard.

3.1.4 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e. 29 CFR Part 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during demolition, repair, renovation, or construction operations, stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If the material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If the material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification pursuant to FAR 52.243-4 Changes and FAR 52.236-2 Differing Site Conditions.

3.2 UTILITY OUTAGE REQUIREMENTS

See Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME) Paragraph entitled FACILITY/UTILITY/SERVICE OUTAGE & INTERRUPTIONS PROCESS for Outage requirements.

3.3 UTILITY OUTAGE SAFETY AND COORDINATION REVIEW MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage safety review meeting in accordance with EM 385-1-1, Hazardous Energy Control Program (HECP). This meeting must include the Prime Contractor, the Prime and Subcontractors performing the work, the Contracting Officer, and the NAVFAC PWD ME UEM representative. All parties must fully coordinate HEC activities with one another. During the safety review and coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HECP training in accordance with EM 385-1-1. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the Subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1. The meeting will occur immediately preceding the start of work and following the completion of the outage safety and coordination review meeting. Both the safety preparatory inspection coordination meeting and the outage coordination safety review meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1, 29 CFR 1910.333, 29 CFR 1915.89, ANSI/ASSP A10.44, NFPA 70E, and paragraph entitled HAZARDOUS ENERGY CONTROL PROGRAM (HECP) herein.

Contracting Officer will, upon request, apply lockout/tag-out tags and take other actions that, because of experience and knowledge, are known to be necessary to make the particular equipment safe to work on.

No person, regardless of position or authority, must operate any switch, valve, or equipment that has an official lockout/tag-out tag attached to it, nor can such tag be removed except as provided in this Section.

No person must work on any equipment that requires a lockout/tag-out tag unless he/she, his/her immediate supervisor, project leader, or a subordinate has in his/her possession the stubs of the required lockout/tag-out tags.

When work is to be performed on electrical circuits, only qualified personnel are to perform work on electrical circuits.

A supervisor who is required to enter an area protected by a lockout/tag-out tag will be considered a member of the protected group provided he/she notifies the holder of the tag stub each time he/she enters and departs from the protected area.

Portsmouth Naval Shipyard and NAVFAC Personnel use a red lock and a red tag to indicate personnel are working on the systems. Use of a red lock and a red tag is highly encouraged to maintain continuity throughout the Portsmouth Naval Shipyard. The use of another colored locks and tags (blue for Portsmouth Naval Shipyard workers and Yellow for NAVFAC personnel) indicate that the system is out of service for some reason.

Identification markings on building light and power distribution circuits must not be relied on for established safe work conditions.

Before clearance will be given on any equipment other than electrical (generally referred to as mechanical apparatus), the apparatus, valves, or systems must be secured in a passive condition with the appropriate vents, pins, and locks.

Pressurized or vacuum systems must be vented to relieve differential pressure completely.

Vent valves must be tagged open during the course of the work.

Where dangerous gas or fluid systems are involved, or in areas where the environment may be oxygen deficient, system or areas must be purged, ventilated, or otherwise made safe prior to entry.

3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the Subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1. The meeting will occur immediately preceding the start of work and following the completion of the outage safety and coordination review meeting. Both the safety preparatory inspection coordination meeting and the outage coordination safety review meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECP. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1.

3.4.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECP. In accordance with EM 385-1-1, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal notification to the Government (by completing the Government form if provided by Contracting Officer's Representative), confirming that steps of de-energization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.4.4 Tag Placement

Lockout/tag-out tags must be completed in accordance with the regulations printed on the back thereof and attached to any device which, if operated, could cause an unsafe condition to exist.

If more than one group is to work on any circuit or equipment, the employee in charge of each group must have a separate set of lockout/tag-out tags completed and properly attached.

When it is required that certain equipment be tagged, the Government will review the characteristics of the various systems involved that affect the safety of the operations and the work to be performed; take the necessary actions, including voltage and pressure checks, grounding, and venting, to make the system and equipment safe to work on; and apply such lockout/tag-out tags to those switches, valves, vents, or other mechanical devices needed to preserve the safety provided. This operation is referred to as "Providing Safety Clearance."

3.4.5 Tag Removal

When any individual or group has completed its part of the work and is clear of the circuits or equipment, the supervisor, project leader, or individual for whom the equipment was tagged must turn in his/her signed lockout/tag-out tag stub to the Contracting Officer. That group's or individual's lockout/tag-out tags on equipment may then be removed on authorization by the Contracting Officer.

3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment, and rescue and evacuation procedures in accordance with ANSI/ASSP Z359.2 and EM 385-1-1.

3.5.1 Training

Institute a fall protection training program. As part of the Fall Protection Program, provide training for each employee who might be exposed to fall hazards and using personal fall protection equipment. Provide training by a competent person for fall protection in accordance with EM 385-1-1. Document training and practical application of the competent person in accordance with EM 385-1-1 and ANSI/ASSP Z359.2 in the AHA.

3.5.2 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1, 29 CFR 1926.500 Subpart M, ANSI/ASSP Z359.0, ANSI/ASSP Z359.1, ANSI/ASSP Z359.2, ANSI/ASSP Z359.3, ANSI/ASSP Z359.4, ANSI/ASSP Z359.6, ANSI/ASSP Z359.7, ANSI/ASSP Z359.11, ANSI/ASSP Z359.12, ANSI/ASSP Z359.13, ANSI/ASSP Z359.14, ANSI/ASSP Z359.15, ANSI/ASSP Z359.16, and ANSI/ASSP Z359.18.

3.5.2.1 Additional Personal Fall Protection Measures

In addition to the required fall protection systems, other protective measures such as safety skiffs, personal floatation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1. Personal fall protection systems and equipment are required when working from an articulating or extendable boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

3.5.2.2 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabiners must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 3,600 lbs in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 6 feet, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1.

3.5.3 Fall Protection for Roofing Work

Implement fall protection controls based on the type of roof being constructed and work being performed. Evaluate the roof area to be accessed for its structural integrity including weight-bearing capabilities for the projected loading.

a. Low Sloped Roofs:

   (1) For work within 6 feet of an edge, on a roof having a slope less than or equal to 4:12 (vertical to horizontal), protect personnel from falling by the use of conventional fall protection systems (personal fall arrest/restraint systems, guardrails, or safety nets) in accordance with EM 385-1-1 and 29 CFR 1926.500. A safety monitoring system is not adequate fall protection and is not authorized.

   (2) For work greater than 6 feet from the unprotected roof edge, in addition to the use of conventional fall protection systems the use of a warning line system is also permitted, in accordance with 29 CFR 1926.500 and EM 385-1-1.

b. Steep-Sloped Roofs: Work on a roof having a slope greater than 4:12 (vertical to horizontal) requires a personal fall arrest system, guardrails with toe-boards, or safety nets. This requirement also applies to residential or housing type construction.

3.5.4 Horizontal Lifelines (HLL)

Provide HLL in accordance with EM 385-1-1. Commercially manufactured horizontal lifelines (HLL) must be designed, installed, certified, and used, under the supervision of a qualified person, for fall protection as part of a complete fall arrest system which maintains a safety factor of 2 (29 CFR 1926.500). The competent person for fall protection may (if deemed appropriate by the qualified person) supervise the assembly, disassembly, use and inspection of the HLL system under the direction of the qualified person. Locally manufactured HLLs are not acceptable unless they are custom designed for limited or site specific applications by a Registered Professional Engineer who is qualified in designing HLL systems.

3.5.5 Guardrails and Safety Nets

Design, install, and use guardrails and safety nets in accordance with EM 385-1-1 and 29 CFR 1926 Subpart M.

3.5.6 Rescue and Evacuation Plan and Procedures

When personal fall arrest systems are used, ensure that the mishap victim can self-rescue or can be rescued promptly should a fall occur. Prepare a Rescue and Evacuation Plan and include a detailed discussion of the following: methods of rescue; methods of self-rescue or assisted-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility. Include the Rescue and Evacuation Plan within the Activity Hazard Analysis (AHA) for the phase of work, in the Fall Protection and Prevention (FP&P) Plan, and the Accident Prevention Plan (APP). The plan must be in accordance with the requirements of EM 385-1-1, ANSI/ASSP Z359.2, and ANSI/ASSP Z359.4.

3.6 SHIPYARD REQUIREMENTS

All personnel who enter the Controlled Industrial Area (CIA) must wear mandatory personal protective equipment (PPE) at all times and comply with PPE postings of shops both inside and outside the CIA.

3.7 WORK PLATFORMS

3.7.1 Scaffolding

Provide employees with a safe means of access to the work area on the scaffold. Climbing of any scaffold braces or supports not specifically designed for access is prohibited. Comply with the following requirements:

a. Scaffold platforms greater than 20 feet in height must be accessed by use of a scaffold stair system.

b. Ladders commonly provided by scaffold system manufacturers are prohibited for accessing scaffold platforms greater than 20 feet maximum in height.

c. An adequate gate is required.

d. Employees performing scaffold erection and dismantling must be qualified.

e. Scaffold must be capable of supporting at least four times the maximum intended load, and provide appropriate fall protection as delineated in the accepted fall protection and prevention plan.

f. Stationary scaffolds must be attached to structural building components to safeguard against tipping forward or backward.

g. Special care must be given to ensure scaffold systems are not overloaded.

h. Side brackets used to extend scaffold platforms on self-supported scaffold systems for the storage of material are prohibited. The first tie-in must be at the height equal to 4 times the width of the smallest dimension of the scaffold base.

i. Scaffolding other than suspended types must bear on base plates upon wood mudsills (2 in x 10 in x 8 in minimum) or other adequate firm foundation.

j. Scaffold or work platform erectors must have fall protection during the erection and dismantling of scaffolding or work platforms that are more than six feet.

k. Delineate fall protection requirements when working above six feet or above dangerous operations in the Fall Protection and Prevention (FP&P) Plan and Activity Hazard Analysis (AHA) for the phase of work.

3.7.2 Elevated Aerial Work Platforms (AWPs)

Workers must be anchored to the basket or bucket in accordance with manufacturer's specifications and instructions (anchoring to the boom may only be used when allowed by the manufacturer and permitted by the CP). Lanyards used must be sufficiently short to prohibit a worker from climbing out of the basket. The climbing of rails is prohibited. Lanyards with built-in shock absorbers are acceptable. Self-retracting devices are not acceptable. Tying off to an adjacent pole or structure is not permitted unless a safe device for 100 percent tie-off is used for the transfer.

Use of AWPs must be operated, inspected, and maintained as specified in the operating manual for the equipment and delineated in the AHA. Operators of AWPs must be designated as qualified operators by the Prime Contractor. Maintain proof of qualifications on site for review and include in the AHA.

3.8 EQUIPMENT

3.8.1 Material Handling Equipment (MHE)

a. Material handling equipment such as forklifts must not be modified with work platform attachments for supporting employees unless specifically delineated in the manufacturer's printed operating instructions. Material handling equipment fitted with personnel work platform attachments are prohibited from traveling or positioning while personnel are working on the platform.

b. The use of hooks on equipment for lifting of material must be in accordance with manufacturer's printed instructions. Material Handling Equipment Operators must be trained in accordance with OSHA 29 CFR 1910, Subpart N.

c. Operators of forklifts or power industrial trucks must be licensed in accordance with OSHA.

3.8.2 Load Handling Equipment (LHE)

The following requirements apply. In exception, these requirements do not apply to commercial truck mounted and articulating boom cranes used solely to deliver material and supplies (not prefabricated components, structural steel, or components of a systems-engineered metal building) where the lift consists of moving materials and supplies from a truck or trailer to the ground; to cranes installed on mechanics trucks that are used solely in the repair of shore-based equipment; to crane that enter the Portsmouth Naval Shipyard but are not used for lifting; nor to other machines not used to lift loads suspended by rigging equipment. However, LHE accidents occurring during such operations must be reported.

a. Equip cranes and derricks as specified in EM 385-1-1.

b. Notify the Contracting Officer 15 working days in advance of any LHE entering the Portsmouth Naval Shipyard, in accordance with EM 385-1-1, so that necessary quality assurance spot checks can be coordinated. Prior to cranes entering Federal activities, a Crane Access Permit must be obtained from the Contracting Officer. A copy of the permitting process will be provided at the Preconstruction Meeting. Contractor's operator must remain with the crane during the spot check. Rigging gear must comply with OSHA ASME B30.9 Standards safety standards.

c. Comply with the LHE manufacturer's specifications and limitations for erection and operation of cranes and hoists used in support of the work. Perform erection under the supervision of a designated person (as defined in ASME B30.5). Perform all testing in accordance with the manufacturer's recommended procedures.

d. Comply with ASME B30.5 for mobile and locomotive cranes, ASME B30.22 for articulating boom cranes, ASME B30.3 for construction tower cranes, ASME B30.8 for floating cranes and floating derricks, ASME B30.9 for slings, ASME B30.20 for below the hook lifting devices, and ASME B30.26 for rigging hardware.

e. Under no circumstance make a lift at or above 90 percent of the cranes rated capacity in any configuration.

f. When operating in the vicinity of overhead transmission lines, operators and riggers must be alert to this special hazard and follow the requirements of EM 385-1-1 and ASME B30.5 or ASME B30.22 as applicable.

g. Do not use crane suspended personnel work platforms (baskets) unless the Contractor proves that using any other access to the work location would provide a greater hazard to the workers or is impossible. Do not lift personnel with a line hoist or friction crane. Additionally, submit a specific AHA for this work to the Contracting Officer. Ensure the activity and AHA are thoroughly reviewed by all involved personnel.

h. Inspect, maintain, and recharge portable fire extinguishers as specified in NFPA 10, Standard for Portable Fire Extinguishers.

i. All employees must keep clear of loads about to be lifted and of suspended loads, except for employees required to handle the load.

j. Use cribbing when performing lifts on outriggers.

k. The crane hook/block must be positioned directly over the load. Side loading of the crane is prohibited.

l. A physical barricade must be positioned to prevent personnel access where accessible areas of the LHE's rotating superstructure poses a risk of striking, pinching, or crushing personnel.

m. Maintain inspection records in accordance by EM 385-1-1, including shift, monthly, and annual inspections, the signature of the person performing the inspection, and the serial number or other identifier of the LHE that was inspected. Records must be available for review by the Contracting Officer.

n. Maintain written reports of operational and load testing in accordance with EM 385-1-1, listing the load test procedures used along with any repairs or alterations performed on the LHE. Reports must be available for review by the Contracting Officer.

o. Certify that all LHE operators have been trained in proper use of all safety devices (e.g. anti-two block devices).

p. Take steps to ensure that wind speed does not contribute to loss of control of the load during lifting operations. At wind speeds greater than 20 mph, the operator, the rigger, and the lift supervisor must cease all crane operations, evaluate conditions and determine if the lift may proceed. Base the determination to proceed or not on wind calculations per the manufacturer and a reduction in LHE rated capacity if applicable. Include this maximum wind speed determination as part of the activity hazard analysis plan for that operation.

q. Follow FAA guidelines when required based on project location.

3.8.3 Machinery and Mechanized Equipment

a. Proof of qualifications for operator must be kept on the project site for review.

b. Manufacture specifications or owner's manual for the equipment must be on-site and reviewed for additional safety precautions or requirements that are sometimes not identified by OSHA or USACE EM 385-1-1. Incorporate such additional safety precautions or requirements into the AHAs.

3.8.4 Base Mounted Drum Hoists

a. Operation of base mounted drum hoists must be in accordance with EM 385-1-1 and ANSI/ASSP A10.22.

b. Rigging gear must be in accordance with applicable ASME/OSHA standards.

c. When used on telecommunication towers, base mounted drum hoists must be in accordance with TIA-1019, TIA-222, ASME B30.7, 29 CFR 1926.552, and 29 CFR 1926.553.

d. When used to hoist personnel, the AHA must include a written standard operating procedure. Operators must have a physical examination in accordance with EM 385-1-1, and trained, at a minimum, in accordance with EM 385-1-1. The base mounted drum hoist must also comply with OSHA Instruction CPL 02-01-056 and ASME B30.23.

e. Material and personnel must not be hoisted simultaneously.

f. Personnel cage must be marked with the capacity (in number of persons) and load limit in pounds.

g. Construction equipment must not be used for hoisting material or personnel or with trolley/tag lines. Construction equipment may be used for towing and assisting with anchoring guy lines.

3.8.5 Use of Explosives

Use of explosives or blasting is not permitted at Portsmouth Naval Shipyard.

3.9 EXCAVATIONS AND UTILITY LOCATING

Soil classification must be performed by a competent person in accordance with 29 CFR 1926 and EM 385-1-1.

3.9.1 Utility Locations

a. General

Excavation or ground penetrating work is defined as any operation in which earth, rock or other material below ground is moved or otherwise displaced,by means of power and hand tools, power equipment which includes grading,trenching, digging, boring, auguring, tunneling, scraping and cable or pipe driving except tilling of soil, gardening or displacement of earth, rock or other material for agricultural purposes. Removal of bituminous concrete pavement or concrete is not considered excavation. Ground penetrating work may include but is not limited to installing fence posts, probes, borings, piles, sign posts, stakes or anchor rods of any kind that penetrates the soil more than 3-inches. The “Excavator” is defined as the person directly responsible for performing the excavation or ground penetrating work.

b. Underground Utilities Location

The Prime Contractor/Excavator must fully comply with the State of Maine “DIG SAFE “law (Title 23, MRSA 3360-A). Existing underground utilities shown on the plans are based on PNS Total Map and are shown in their approximate locations only.

The Prime Contractor/Excavator must pre-mark the excavation area in “White Paint Only”. (Field notes may be done in Pink paint). The Excavator must notify “DIG SAFE” (1-888-344-7233) at least within 14calendar days, but no more than 30 calendar days prior to the commencement of the excavation or ground penetrating activity. The Excavator must notify Unitil within 14 calendar days, but no more than 30 calendar days prior to the commencement of excavation or ground penetrating activity when the activity is within twenty (20) feet of a natural gas line.

The Prime Contractor / Excavator must prepare a NAVFAC DIG Safe Utility Locate Request Form (Attachment B) at least within 14 calendar days prior to the commencement of the excavation or ground penetrating activity and submit the Form to the CM/ET. (The NAVFAC Dig Safe Utility Locate Request Form Attachment B is attached at the end of this Section.)

The Government will locate and mark the underground utilities within 14 calendar days of receiving the Dig Safe Notification. Unitil will locate and mark the underground natural gas utilities.

Excavation or ground penetrating activities cannot commence until the utilities have been marked in the field and the NAVFAC Dig Safe Utility Locate Request Form has been returned indicating the PWD ME Dig Safe review process has been completed and excavation has been approved by the Contracting Officer and the PWD ME Dig Safe Coordinator.

If the excavation or ground penetrating activities do not commence within 27 days of Dig Safe notification or the excavation work is expanded outside the location originally specified in the notification, the Excavator must renotify Dig Safe, the Contracting Officer and the PWD ME Dig Safe Coordinator.

The Contractor must maintain the utility markings throughout the excavation period at the Contractor’s expense.

The Contractor must contact the PWD ME Dig Safe Coordinator (DSC) if there are any questions regarding the underground utilities or the Dig Safe process.

c. Third Party Utility Locate

The Contractor must provide the services of a third party, qualified independent utility locating company/person(s) (Cannot be the Government’s Utility Locating Firm) to positively identify underground utilities in the work area. The third party independent locating firm is in addition to the PWD ME Dig Safe Process.

The Third Party review must be completed in conjunction with the PWD ME Dig Safe marking process. Once the Third Party Locate Company has completed their review of the excavation area and the review of the Government markings are confirmed, the Third Party Locate Company and Contractor must sign the Contractor Dig-Safe Certification Form (Attachment C) and submit the form to the Contracting Officer/DSC prior to commencing excavation. If the Third Party Locate Company finds any discrepancies with the Government's utility markings, the Contractor must notify the Contracting Officer/DSC immediately and work with the DSC to resolve the discrepancy **PRIOR TO THE START OF EXCAVATION.**

3.9.2 Utility Location Verification

Physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within three (3) feet of the underground system. When any adjacent construction work is expected to come within twenty (20) feet of a natural gas utility, the Contractor must notify Unitil in addition to notifying "DIG SAFE".

3.9.3 Utilities Within and Under Concrete (roofs, walls, floors,etc) Bituminous Asphalt, and Other Impervious Surfaces (Including Dry Dock and Berthing Walls)

Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. This will require utility locating **both before and after** concrete slab removal when excavation continues below bottom of slab. Whenever Contract work involves chipping, saw cutting, drilling or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with Portsmouth Naval Shipyard UEM department in addition to location and depth verification by a third party, independent, private locating company.

The third party, independent, private locating company must locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings does not alleviate the Contractor from meeting this requirement. Any markings made during the utility investigation must be maintained by the Contractor throughout the Contract.

3.9.4 Excavating with Hazardous or Tier 1 Utilities Within the Excavation Area (as identified by DSC or plans)

a. The Contractor must employ supplemental daily Dig Safe procedures to include an additional checkoff on Contractor Daily Activity Plan to confirm all utilities have been clearly marked and reviewed with the SSHO. The SSHO and the Excavator must complete and sign CONTRACTOR UTILITY LOCATE – PRE-EXCAVATION SAFETY CHECKLIST (Attachment D). This checklist must be reviewed and signed by the PWD ME ET prior to the commencing any excavation/trenching work.

b. The Contractor must provide a Dig Safe laminated utility color coding system posted in or near all heavy digging equipment for easy reference to type of utility.

c. The SSHO must complete a pre-excavation walk as part of the morning procedure to help ensure all known utilities are identified and markings are refreshed with the appropriate color-coded paint. The Excavation/Trenching Competent person must complete the Contractor Daily Checklist for Trenching/Excavation included in Attachment E. The Daily Checklist must be completed prior commencing excavation/trenching work and must be submitted with the CM/ET daily.

d. Contractor must provide additional danger signage, to mark areas of known live underground utilities. Signage must comply with NEMA Z535.2.

e. Contractor must ensure a ‘spotter’ accompanies the equipment operator during excavation work.

f. Contractor to provide Construction CM/ET notification no later then 7 working days prior to holding the preparatory and initial pre-excavation/demo safety review meeting. The CM/ET, as well as the PWD ME DSC, ***WILL*** attend this meeting to ensure safety and preservation of Tier 1 utilities is discussed with adequate focus.

g. The Contractor must confirm and identify the closest utility isolation points and develop mitigation strategies with the utility owner (Coordinate with the PWD ME DSC) to ensure the safe excavation adjacent to these utilities. Utility Outages to isolate utility systems may need to be considered in circumstances where the excavation work cannot be completed safely.

h. All construction activities (including excavation adjacent to Tier 1 Infrastructure) and outages directly impacting or within three (3) feet of Tier 1 infrastructure, or within twenty (20) feet of natural gas utilities, are required to have a Tier 1 Infrastructure Risk Analysis and Risk Mitigation plan (Tier 1 Risk Mitigation Plan). The Contract Documents identify the Tier 1 Infrastructure that could be impacted by the work associated with this project.

The CI Risk Mitigation Plan (See Example below)is the Contractor's responsibility. General information related to Project Specific Outage Phasing and Sequencing may be provided in the Contract Documents. See Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME), Paragraph entitled FACILITY/UTILITY/SERVICE OUTAGE & INTERRUPTIONS PROCESS for Outage Requirements.

The Contractor's proposed means and methods related to any work that may impact Tier 1 infrastructure are subject to Government approval. The anticipated phasing and sequencing for completing this work may be outlined in the Drawings. Any additional work and costs associated with any changes to the phasing and sequencing based on the Contractor's means and methods are the responsibility of the Contractor and will not be a basis for a claim for additional cost and time. The Contracting Officer will require approval of each CI Risk Mitigation Plan prior to work, and when associated with an outage, it must be submitted along with the Outage Request.

DEFINITION:

a. "Tier 1 Infrastructure" are Shipyard facilities (utilities and structures) that require additional work restrictions and procedures when working on or in the vicinity of. Coordinate with the Contracting Officer for a list of Tier 1 Infrastructure associated with the scope of the project.

TIER 1 INFRASTRUCTURE RISK ANALYSIS (RA) AND RISK MITIGATION PLAN (RMP)

(Example additional information may be required to address the RA & RMP requirements)

1. Work: (describe in enough detail to allow someone not involved to have a good general understanding of scope):

2. Risks to Tier 1 Infrastructure:

    a. Describe what could go wrong with the work and the Impact to Shipyard operations (Air, Water, Electrical Power, Telecom/IT, Rail or Structures).

        i. Risk Assessment: Evaluate the risk of what could go wrong. (Probability of failure/Severity of Impact. See RISK DETERMINATION TABLE - in the Outage Form.)

        ii. Mitigated Risk Assessment: Evaluate the reduced risk of what could go wrong based on steps in Work Control Plan and Oversight outlined below.

    b. What could go wrong #2:

        i. Risk Assessment #2: Evaluate the risk of what could go wrong. (Probability of failure/Severity of Impact)

ii. Mitigated Risk Assessment #2: Evaluate the reduced risk of what could go wrong based on steps in Work Control Plan and Oversight outlined below. State why or how the risk is reduced.

3. Work Control Plan:

a. Describe specific steps and actions (how, when, and by who) to mitigate the risks to Tier 1 infrastructure that were identified above. (examples only shown below)

i. Specific utilities/services in the area will be secured prior to work, or, potholing/GPR or other utility/service locating has been performed to identify specific utility/service location and depth, which is included in Preparatory Phase meeting agenda.

ii. Protective timber matting installed over crane and train rail to prevent damage to rails supporting train and crane movements.

iii. Noise/Dust/Vibration. Managed per contract requirements, and to be briefed during Initial Phase meeting.

iv. Detailed work plan.

v. Specific selection of equipment/tools for work.

vi. Foreman/safety pre-job reviews.

vii. Foreman pre-job work brief with crew.

4. Oversight: Describe specific oversight work leading up to and for the execution of work (how, when, and by who) to mitigate the risks to Tier 1 infrastructure that were identified above. (examples only shown below)

a. Daily Contractor Production Reports.

b. Daily Contractor Quality Control (QC) Reports.

c. QC Manager field visits (include periodicity).

d. Periodic project Coordination Meeting (include parties involved)

i. If there is a utility strike, the Contractor must stop all work and notify the Contracting Officer's Representatives immediately. The Contractor must complete the Underground Utility Strike Review Checklist Attachment F with 2 working days of the Utility Strike. See requirements in Paragraphs herein entitled MISHAP NOTIFICATIONS and ACCIDENT REPORTS for additional requirements.

3.10 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1.

3.10.1 Electrical Work

As described in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Cables must be verified as de-energized by using a spiking tool, or equivalent insulation displacement voltage verification method to confirm de-energized state. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all medium and high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety footwear, insulating gloves, and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and 29 CFR 1910.147.

3.10.2 Qualifications

Electrical work must be performed by QP personnel with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National, and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with State and Local requirements applicable to where work is being performed.

3.10.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

3.10.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

Check grounding circuits to ensure that the circuit between the ground and a grounded power conductor has a resistance low enough to permit sufficient current flow to allow the fuse or circuit breaker to interrupt the current.

3.10.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification, and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

3.10.6 Portable Extension Cords

Size portable extension cords in accordance with manufacturer ratings for the tool to be powered and protected from damage. Immediately remove from service all damaged extension cords. Portable extension cords must meet the requirements of EM 385-1-1, NFPA 70E, and OSHA electrical standards.

-- End of Section --

CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

1 CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

1.1 The following is a list of requirements that contractors shall comply with for all contracts that may result in the use of a category 1 or 4 crane, multi-purpose machines, forklifts, construction equipment, and rigging gear when used on Navy property to lift suspended loads. Non-compliance with the requirements of this instruction may result in denial of access, stopping of operations, or removal from Navy property.

1.2 References:

1.2.1 NOT USED

1.2.2 American Society of Mechanical Engineers (ASME) B30.3 (tower cranes), B30.5 (mobile cranes), B30.8 (floating cranes), B30.9 (slings), B30.20 (below the hook lifting devices), B30.22 (articulating booms), B30.26 (rigging hardware); ANSI/ITSDF B56.6 (rough terrain forklifts); Safety Standards for Cableways, Cranes, Derricks, Hoists, Hooks, Jacks, and Slings

1.2.3 CFR, Title 29, Chapter XVII, Part 1917, Marine Terminals.

1.2.4 CFR, Title 29, Chapter XVII, Part 1926, Safety and Health Regulations for Construction

1.2.5 CFR, Title 29, Chapter XVII, Part 1915, Occupational Safety and Health Standards for Shipyard Employment

1.2.6 OPNAVINST 5100.23, Navy Safety and Occupational Health Program Manual

1.2.7 EM 385-1-1, Safety and Health Requirements Manual, U.S. Army Corps of Engineers

1.2.8 NAVFAC Guide Specification NFGS-01525D, Safety Requirements

1.3 These requirements are solely intended to provide for the protection of Government property and personnel and are not intended to, and do not, in any manner whatsoever, relieve the contractor of its responsibility, including, without limitation, its responsibility for the protection of its equipment and personnel.

1.4 Notification Requirement: Contractor shall notify the Contracting Officer 7 calendar days in advance of the intent of bringing a non-Navy owned crane onto Navy property or of any multi-purpose machines, material handling equipment, or construction equipment that may be used in a crane-like application to lift suspended loads. The contractor shall also specify when crane entry onto Navy property is scheduled during back shift, weekend, or holiday hours of operation. All entries shall be through a prearranged entry point. The following documentation shall be provided along with notification: a copy of Form 16-1 and objective evidence of operator qualifications for cranes with rated capacities of 2,000 lbs. or greater. Failure to schedule or provide necessary documentation may result in the crane being denied access to the facility.

1.5 The contractor shall comply with applicable reference 1.2.2 standards (e.g., B30.3 for construction tower cranes, B30.5 for mobile cranes, B30.8 for floating cranes, B30.9 for slings, B30.20 for below the hook lifting devices, and B30.22 for articulating boom cranes), B30.26 for rigging hardware, and ANSI/ITSDF B56.6 for rough terrain forklifts). For barge mounted mobile cranes, require a third party certification from an OSHA accredited organization (or from a state accredited organization for those states with OSHA approved state plans), a load indicating device, a wind-indicating device, and a marine type list and trim indicator readable in one-half degree increments. Third party certification is not required for barge-mounted mobile cranes at naval activities in foreign countries.

1.6 Certification of Compliance (reference 1.2.1): The contractor shall complete a certificate of compliance that the crane (or other machine if used to lift suspended loads) and rigging gear meet applicable OSHA and ANSI/ASME regulations (with the contractor citing which OSHA regulations are applicable, e.g., cranes/multi-purpose machines used in cargo transfer shall comply with reference 1.2.3; cranes/multi-purpose machines used in construction, demolition, or maintenance shall comply with reference 1.2.4; cranes/multi-purpose machines used in ship repair shall comply with reference 1.2.5; slings shall comply with ASME B30.9; rigging hardware shall comply with ASME B30.26). For cranes (or other machine if used to lift suspended loads) and rigging equipment at naval activities in foreign countries, the contractor shall certify that the crane and rigging gear conform to the appropriate host country safety standards. The contractor shall also certify that all of its crane (or other machine) operators working on the naval activity have been trained not to bypass safety devices (e.g., anti-two block devices) during lifting operations, and that its operators, riggers, and company officials are aware of the actions required in the event of an accident as specified in the contract. Require that the certifications be posted on the crane. When a crane on Navy property is not authorized for use, the Certification of Compliance shall state, "Operation of this Crane is NOT Authorized."

1.7 The contractor shall certify (reference 1.2.1) that the crane or machine operator is qualified and trained for the operation of the crane to be used. For mobile and commercial truck mounted cranes with OEM rated capacities of greater than 2,000 pounds, the crane operator shall be designated as qualified by a source that qualifies crane operators (i.e., a union, a government agency, or an organization that tests and qualifies crane operators). Proof of current qualification shall be provided.

1.8 For multi-purpose machines, material handling equipment and construction equipment used to lift loads suspended by rigging equipment, the contractor shall have proof or authorization from the machine OEM that the machine is capable of making lifts of loads suspended by rigging equipment. The contractor shall demonstrate that the equipment is properly configured to make such lifts and is equipped with a load chart.

1.9 All hooks used on cranes, hoists, other machines, and rigging gear shall have self-closing latches or the throat opening shall be "moused" (secured with wire, rope, heavy tape, etc.) or otherwise secured to prevent the attached item from coming free of the hook under a slack condition. The following exceptions apply and shall be approved by the contractor's technical organization: items where the hook throat is fully obstructed and not available for manual securing and lifts where securing the hook throat increases the danger to personnel such as forge shop, dip tank, or underwater work.

1.10 Loading Limitations:

CAUTION: Piers and waterfront areas such as along dry docks and quay walls may have load restrictions.

1.10.1 The contractor shall notify the Contracting Officer prior to moving a crane on a pier, dry dock, or other waterfront area. Provide the Contracting Officer with the crane make, model, and configuration in which it is to be used.

1.10.2 The contractor shall comply with crane access routes and load limitations issued with the contract.

1.10.3 Allowable Surface Loads. Loads transferred to soils and pavements shall be minimized to a desired maximum of 3000 pounds per square foot, by placement of cribbing or steel pads under rubber-tired crane outriggers and trailer stanchions/sand shoes, or by placement of mats under treads of crawler cranes. Visually inspect areas adjacent to cribbing or plates and report any unusual bituminous pavement surface conditions, irregularities, or cracking to the Contracting Officer.

1.10.3.1 Outriggers of rubber-tired cranes shall be landed on two layers of timbers of appropriate thickness, oriented at right angles to each other, or landed on properly designed steel pads. Treads of crawler cranes shall run on appropriate mats. Use and design of cribbing, plates and mats shall be in a manner consistent with general construction industry standards.

1.10.3.2 Position loads that will remain on trailers detached from tractors to attain a distribution of 65 percent to rear axles and 35 percent to front support stanchions/sand shoes. For example, assuming an 83000 pound maximum gross weight and a soil bearing pressure of 3000 pounds per square foot, the required support under each sand shoe would be 2.5 feet x 2.5 feet. Accordingly, two tiers of timber cribbing at right angles, each 2.5 feet x 2.5 feet x 4 inches, or a properly designed 2.5 feet x 2.5 feet steel pad would be utilized under each trailer stanchion/ sand shoe.

1.11 Prior to making any critical lift, the contractor shall provide a critical lift plan for each of the following lifts: lifts over 75 percent of the capacity of the crane, hoist, or other machine (50 percent of the capacity of a barge mounted mobile crane's hoists) at any radius of lift; lifts involving more than one crane, hoist, or other machine; lifts of personnel (lifts of personnel suspended by rigging equipment from multi-purpose machines, material handling equipment, or construction equipment shall not be permitted); lifts made in the vicinity of overhead power lines; erection of cranes; and lifts involving non-routine rigging or operation, sensitive equipment, or unusual safety risks. The plan shall include the following as applicable:

1.11.1 The size and weight of the load to be lifted, including crane (or other machine) and rigging equipment that add to the weight. The OEM's maximum load capacities for the entire range of the lift shall also be provided.

1.11.2 The lift geometry, including the crane (or other machine) position, boom length and angle, height of lift, and radius for the entire range of the lift. Applies to both single and tandem crane/machine lifts.

1.11.3 A rigging plan, showing the lift points, rigging equipment, and rigging procedures.

1.11.4 The environmental conditions under which lift operations are to be stopped.

1.11.5 For lifts of personnel, the plan shall demonstrate compliance with the requirements of reference

1.11.6 For barge mounted mobile cranes, barge stability calculations identifying crane placement/footprint; barge list and trim based on anticipated loading; and load charts based on calculated list and trim specific to the barge the crane is mounted on. The amount of list and trim shall be within the crane manufacturer's requirements.

1.11.7 For lifts in the vicinity of overhead power lines (i.e., if any part of the crane or other machine, including the fully extended boom of a telescoping boom crane or machine, or the load could approach the distances noted in figure 10-3 of reference 1.2.1 during a proposed operation), the plan shall demonstrate compliance to 29 CFR 1926.550(a)(15).

1.12 The following additional documentation is required for contractor provided tower cranes (those cranes defined by ASME B30.3).

1.12.1 Foundation design and requirements

1.12.2 Installation instructions1.12.4

1.12.3 Assembly and disassembly instructions including climbing/jumping instructions if applicable

1.12.4 Operating manual, limitations, and precautions

1.12.5 Periodic inspection and maintenance requirements

1.13 Crane and Rigging Gear Accident Reporting and Record Keeping: Contractor's operating cranes on Navy property shall report all WHE accidents that occur incidental to an operation, project, or facility as prescribed by paragraphs (1.10.1) through (1.10.3) requirements below. Contractors shall report directly to their respective Contracting Officer. There are two general categories of accidents as defined below. Crane accidents are those that occur during operation of a crane. Rigging gear accidents are those that occur when gear is used by itself in weight handling operation i.e., without a crane.

1.13.1 Crane Accident: For the purpose of this definition, it is assumed there is an "operating envelope" around any crane, and inside the envelope are the following elements:

- The crane
- The operator
- The rigger(s) and crane walker
- Other personnel involved in the operation (supervisor, mechanic, tag line handler, engineer, etc.)
- The rigging gear between the hook and the load
- The load
- The crane's supporting structure (ground, rail, etc.)
- The lift procedure

1.13.1.1 Definition: A crane accident occurs when any one or more of the six elements in the operating envelope fails to perform correctly during operation, including operation during maintenance, or testing resulting in the following:

- Personnel injury or death. Minor injuries that are inherent in any industrial operation, including strains and repetitive motion related injuries, shall be reported by the normal personnel injury reporting process in lieu of these requirements.
- Material or equipment damage
- Dropped load
- Derailment
- Two-blocking
- Overload (This includes load tests when the test load tolerance is exceeded.)
- Collision, including unplanned contact between the load, crane, and/or other objects.

A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, roll over, etc.). *[Bullets]* 3, 4, 5, 6, and 7 are considered crane accidents even though no material damage or injury occurs.

Exception: If a crane is used as an anchor point for a portable hoist/rigging gear, rigging gear accident as defined in paragraph 1.10.2 below is not considered a crane accident if the crane is not being operated (no functions are in motion) at the time of the rigging gear accident, unless the accident results in an overload or damage to the crane, in which case it shall be reported as a crane accident.

1.13.2 Rigging Gear Accidents: For the purpose of this definition, it is assumed there is an "operating envelope" around any weight handling operation, and inside the envelope are the following:

- Rigging gear and miscellaneous equipment
- The user of the gear or equipment
- Other personnel involved in the operation (supervisor, mechanic, tag line handler, engineer, etc.)
- The load
- The gear or equipment's supporting structure
- The load's rigging path

- The rigging procedure

1.13.2.1 Definition. A rigging gear accident occurs when any one or more of the five elements in the operating envelope fails to perform correctly during weight handling operations resulting in the following:

- Personnel injury or death. Minor injuries that are inherent in any industrial operation, including strains and repetitive motion related injuries, shall be reported by the normal personnel injury reporting process of the activity in lieu of these requirements.
- Material or equipment damage that requires the damaged item to be repaired because it can no longer perform its intended function. This does not include superficial damage such as scratched paint, damaged lagging, or normal wear on rigging gear.
- Dropped load.
- Two-blocking of cranes and powered hoists.
- Overload. (This includes load tests when the test load tolerance is exceeded.)

A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped load, damaged load, etc.). *[Bullets]* 3, 4, and 5 are considered accidents even though no material damage or injury occurs.

1.13.3 The contractor shall notify the Contracting Officer as soon as practical, but not later than four hours, after any WHE accident. The contractor shall secure the accident site and protect evidence until released by the Contracting Officer. The contractor shall conduct an accident investigation to establish the root cause(s) of the accident. Crane operations shall not proceed until cause is determined and corrective actions have been implemented to the satisfaction of the Contracting Officer. The contractor shall provide the Contracting Officer within 30 days of any accident a Crane and Rigging Gear Accident Report using the form provided in reference1.2.1 consisting of a summary of circumstances, an explanation of causes(s), photographs if available, and corrective actions taken. These notifications and reporting requirements are in addition to those promulgated by reference 1.2.6 and related claimant instructions.

1.14 Each contractor shall perform the following actions prior to conducting crane operations on Navy property:

1.14.1 <u>Inspection Requirements</u>: It shall be the sole responsibility of the contractor to assure the Contracting Officer and/or designated Navy personnel that the crane and associated rigging gear are in good working order and safe for use.

1.14.1.1 <u>Crane Inspection</u>: Perform pre-operational inspection of the crane in the presence of a representative of the Contracting Office of the crane prior to starting work on Navy property. Inspection shall meet all applicable reference 1.2.2, reference 1.2.7 (for NAVFAC construction contracts), and OSHA requirements.

1.14.1.2 <u>Wire Rope Inspection</u>: Perform a Wire Rope Inspection in the presence of a representative of the contracting office to applicable reference 1.2.2, reference 1.2.7 (for NAVFAC construction contracts), and OSHA requirements.

1.14.1.3 <u>Rigging Gear Inspection</u>: Perform a Rigging Gear Inspection in the presence of a representative of the contracting office to applicable reference 1.2.2, reference 1.2.7 (for NAVFAC construction contracts), and OSHA requirements.

SECTION 01 35 26 – ATTACHMENT A

CONTRACTOR CRANE ENTRY CHECKLIST

1	Crane Company:	**Date of Entry:**			
	Crane Manufacturer/Crane Model/Crane Number:	**Time of Entry:**			
2	Date of Annual Inspection Expiration				
3	Date of Quadrennial Inspection Expiration				
4	Name & phone number of Contracting Official (or designated local representative)	**Contracting Official**			
		Phone Number			
5	Does the package include a routine or critical lift plan?			**Yes** ☐	**No** ☐
6	Location of lift site?				
7	Duration crane will be continuously on the job site (hrs, days, weeks…)				
8	Does plan include certification from contractor that the crane complies with ASME B30 standard [B30.5 (mobile cranes), B30.8 (floating cranes), B30.22 (articulating boom cranes), or B30.3 (construction tower cranes)] as applicable?			**Yes** ☐	**No** ☐
9	Does plan include a certificate of compliance?			**Yes** ☐	**No** ☐
10	Which OSHA regulations does the certificate of compliance indicate? (For cranes used in cargo transfer, 29 CFR 1917 applies; for cranes used in construction, demolition, or maintenance, 29 CFR 1926 applies; for cranes used in shipbuilding, ship repair, or ship breaking, 29 CFR 1915 applies).				
11	Does plan include valid medical certificate and proof of operator qualification from a source that qualifies crane operators (union, governmental agency, or an organization that tests and qualifies crane operators)? Verify qualification for each back-up operator (if provided) on the certificate of compliance.		**Yes** ☐	**No** ☐	**N/A** ☐
12	Does the plan designate a qualified Rigger-in-Charge			**Yes** ☐	**No** ☐
13	What is the weight of the heaviest load to be lifted?		**lbs.**		
14	What is the weight of the rigging gear?		**lbs.**		
15	What are the crane components (and their weights) that add to the weight of the load (hook, jib, etc.)?	**Main Block**	**lbs.**		
		Aux. Block	**lbs.**		
		Jib (Stowed)	**lbs.**		
		Jib (Erected)	**lbs.**		
		Other	**lbs.**		
16	What is the maximum total crane lift (sum of 13, 14 & 15 above)?	**TOTAL**	**lbs.**		
17	What is the capacity of the crane as configured		**lbs.**		
18	What percentage of crane capacity does this lift represent?		**%**		

19	What is the main boom length? If a jib will be utilized, indicate the length and offset.	**MAIN**	**JIB**	**OFFSET**	
20	What are the minimum and maximum load radii?	**Min**		**Max**	
21	Does the plan include the manufacturer's load chart for entire range of lift(s)?			**Yes** ☐	**No** ☐
22	Does plan include ground loading and outrigger reaction data to determine cribbing requirements, or a Waterfront Operational Permit?		**Yes** ☐	**No** ☐	**N/A** ☐
23	For crawler crane, does the plan indicate area restrictions for operation?		**Yes** ☐	**No** ☐	**N/A** ☐
24	For floating crane, does plan include maximum allowable list?		**Yes** ☐	**No** ☐	**N/A** ☐
25	For mobile crane mounted on barge, is crane equipped with load indicating device? Wind indicating device? Marine type list and trim indicator (readable in one-half degree increments)?		**Yes** ☐	**No** ☐	**N/A** ☐
26	For mobile crane mounted on barge, does plan include revised load chart?		**Yes** ☐	**No** ☐	**N/A** ☐
27	What are the environmental conditions under which crane operations are to be stopped?				
28	Will the crane perform critical lifts? **(If no, skip items 29 –49.)**			**Yes** ☐	**No** ☐
29	What circumstances require this lift to be classified as a critical lift? (Blind lift, 75% of chart, non-routine rigging, etc.)				
30	What are the exact dimensions of the load? (L x W x H)				
31	Does the plan indicate the crane position? (Overhead view)			**Yes** ☐	**No** ☐
32	What is the maximum lift height of the lift?				
33	What is the minimum boom angle?				
34	What is the maximum boom angle?				
35	What is the name of the operator?				
36	Indicate name(s) of backup operator (if required).				
37	Does the plan show lift points?			**Yes** ☐	**No** ☐
38	Does the plan describe the rigging procedures?			**Yes** ☐	**No** ☐
39	Does the plan indicate rigging hardware requirements?			**Yes** ☐	**No** ☐

40	For personnel lifts, does the plan demonstrate compliance with 29 CFR 1926.550?	**Yes** ☐	**No** ☐	**N/A** ☐
41	Does EM 385-1-1 govern this lift?	**Yes** ☐	**No** ☐	**N/A** ☐
42	What are the coordination and communication requirements for the lift (e.g., radio and hand signals)?			
43	For tandem or tailing crane lifts, does the plan indicate the make and model of the crane, the line, boom, and swing speeds, and the requirement for an equalizer beam?	**Yes** ☐	**No** ☐	**N/A** ☐
44	For floating cranes, refer to questions 20-22?			
45	What is the name of the lift supervisor?			
56	Does the plan indicate the qualifications of the lift supervisor?		**Yes** ☐	**No** ☐
47	What are the names of the riggers?			
48	Does the plan indicate the qualifications of the riggers?		**Yes** ☐	**No** ☐
49	Did all involved personnel (Operator, Riggers, Lift Supervisor, etc.) sign the critical lift plan?		**Yes** ☐	**No** ☐

Name	Organization	**Signature**	**Date**	**Phone**
Contracting Official:				
wed By				

<table>
<tr><td colspan="2"><u>FORM 16-1</u>

<u>Certificate of Compliance for LHE and Rigging</u></td></tr>
<tr><td colspan="2">This certificate shall be signed by an official of the company that provides LHE/cranes and rigging gear for any application under this contract.</td></tr>
<tr><td>Contracting Officer's Point of Contact:
(Government Designated Representative)</td><td>Phone #:</td></tr>
<tr><td>Prime Contractor/Phone #:</td><td>Contract Number:</td></tr>
<tr><td>SSHO/QC:</td><td>Phone #:</td></tr>
<tr><td colspan="2">LHE Manufacturer/Type/Capacity:</td></tr>
<tr><td colspan="2">LHE Operator(s) Name(s):</td></tr>
<tr><td colspan="2">I certify that:
1. The above noted LHE and all rigging gear conform to the EM 385-1-1, applicable OSHA regulations (host country regulations in foreign countries) and applicable ASME standards.
2. The operator(s) noted above has been trained, qualified and designated in accordance with the requirements in Section 16, EM 385-1-1 for the operation of the above noted LHE.
3. The operator(s) noted above has been trained not to bypass safety devices during LHE operations.
4. The operator(s), rigger(s) and company official (staff) are aware that immediate notification to the GDA of any incident or accident involving this equipment is required.</td></tr>
<tr><td>Company Official Signature:</td><td>Date:</td></tr>
<tr><td colspan="2">Company Official Name/Title:</td></tr>
<tr><td colspan="2">Post on Crane/LHE.
(In Cab and Contractor's Office for each LHE onto USACE Project/Property)</td></tr>
</table>

FORM 16-2
Standard Pre-Lift Crane Plan/Checklist

DATE: ___/___/______ Job Number: ______________ Location: _____________________

TIME: ___________ Completed By (Competent Person):_____________________

NOTE: Applies to Cranes, Derricks, Hoists and Power-Operated equipment that can be used to hoist, lower and/or horizontally move a suspended load (includes excavators, forklifts, Rough Terrain equipment, etc., when used with rigging).

Crane Considerations		Yes	No
1	Are the lifts within the crane's rated capacities? (based on boom height, radius)		
2	Boom deflections considered?		
3	Have all potential crane boom obstructions been identified?		
4	Have Environmental Considerations been addressed? (Wind, Weather-Lightning)		
5	Have electrical hazards been addressed (Overhead / Underground) - Clearance distances established? - Is a spotter required? - Public Utility contact required?		
6	Crane swing radius properly barricaded and personnel advised of hazards?		

Comments:

Load		Yes	No
1	Weights and Centers of Gravity (COG) have been Determined?		
2	Anything Inside / Outside the loads that could shift during the lift?		
3	Determine if the rigging needs protection from the loads?		
4	All anchor bolts, hold downs, or fasteners have been removed?		
5	Potential for binding – are load cells required to verify the loads are free?		
6	Attachment points rated to take load weight?		
7	Are the loads structurally capable of being lifted? (bending & twisting issues)		
8	Is a critical lift plan required per the EM section 16.H?		

Comments:

FORM 16-2 (cont'd)
Standard Pre-Lift Crane Plan/Checklist

Rigging		Yes	No
1	All rigging has been inspected by a Qualified Rigger?		
2	Have sling angles been calculated?		
3	Are shackles correctly sized for the sling eyes?		
4	Are softeners needed?		

Comments:

Personnel		Yes	No
1	The roles, responsibilities and qualifications for personnel have been defined? (Operator, Lift Supervisor, Rigger, Signal Person)		
2	A Pre-Lift meeting has been conducted?		
3	Personnel trained per the EM?		

Comments:

Area Preparation		Yes	No
1	The locations for the load landings has been selected and prepared?		
2	Blocking and or Cribbing is available to set the loads on?		
3	Travel paths have been determined and cordoned off?		
4	Other personnel in the area have been notified of the lifts?		
5	Have ground bearing support questions been addressed?		

Comments:

Crane Operator: ______________________________ Date: ____________

Riggers: ____________________________________ Date: ____________

Signal Person: _______________________________ Date: ____________

Others: _____________________________________ Date: ____________

NAVFAC MIDLANT

CRITICAL LIFT PLAN

Date:	**Qualified Person:**
Location:	

Contractor is required to fill sections A.thru F. prior to submitting plan.
Sections G.- H to be completed by PWD Maine ET

A. TOTAL LOAD

1. Load Weight ______ lbs
2. Wt. of Aux. Block ______ lbs
3. Wt. of Main Block ______ lbs
4. Wt. of Lifiting Beam ______ lbs
5. Wt. of Sling/Shackles ______ lbs
6. Wt. of Jib/Ext. (erected/stowed) ______ lbs
7. Wt. of Hoist Rope ______ lbs
8. Other: ______ lbs

TOTAL WEIGHT ______

Note: Source of load weight (Drawings, Calcs, etc.) must be attached on Page 2.

E. CRANE PLACEMENT ***(Mobile Cranes Only)***

1. Maximum Bearing Pressure ______ PSF
   Note: Bearing Pressure Calculations must be attached on Page 3.
2. Ground Conditions Suitable for Load? ____ YES / NO
   Note: Ground Condition Calculations must be attached on Page 3.
3. High Voltage or Electrical Hazards? ____ YES / NO
   Note: If Elelctrical Hazards are present they must be shown on Page 4.
4. Obstructions to Lift or Swing? ____ YES / NO
   Note: If Obstructions are present they must be shown on Page 4.
5. Travel with Load Required? ____ YES / NO
6. Other? ______

B. CRANE

1. Type of Crane Mobile Hydraulic Truck
2. Maximum Crane Capacity ______ lbs.
3. Radius (Maximum) ______ ft.
4. Radius (Minimum) ______ ft.
5. Boom Length (Maximum) ______ ft.
6. Boom Length (Minimum) ______ ft.
7. Crane Capacity (Max Radius) ______ lbs.
8. Crane Capacity (Min Radius) ______ lbs.
9. Boom Angle (Maximum) ______ deg.
10. Boom Angle (Minimum) ______ deg.
11. Gross Load of Crane ______ lbs.
12. Lift is ______ % of the Crane's rated capacity
13. If Jib/Ext. is to be used:
    Length ______ ft.
    Offset ______ ft.
14. Rated Capacity of Jib/Ext. lbs

F. OPERATOR QUALIFICATIONS

1. Certified Operator? ____ YES / NO
2. Option? ____
3. Certified for Type, Class & Capacity? ____ YES / NO
4. Designated in writing by emp______

G. COMPLETED IN FIELD:	***(YES)***	***NO***	***PWD ET***
1. Crane Inspected			
2. Rigging Inspected			
3. Crane Set-up			
4. Overhead Hazard Check			
5. Swing Check			
6. Counterweight Check			
7. Operator Qualifications			
8. Signal Person Qualifications			
9. Rigger Qualifications			
10. Load Chart in Crane			
11. Load Test			
12. Tag Lines			
13. Wind Conditions			
14. Traffic Hazard Check			
15. Site Control			

C. HOIST ROPE	***Main***	***Aux 1***	***Aux 2***
1. # of Parts			
2. Rope Diamter			
3. Capacity			

D. RIGGING

1. Hitch Type(s) ______
2. No. of Slings: ______ Size: ______
3. Sling Type: ______
4. Sling Assembly Capacity: ______ lbs.
5. Shackle Size(s): ______
6. Shackle Rated Capacity(s) ______ lbs.

H. SIGNATURES IN FIELD:

1. Crane Operator ______
2. Rigger ______
3. Signal Person ______
4. Lift Supervisor ______
5. PWD ET ______
6. Other ______

Rev.
Jan-15

U.S. Army Corps of Engineers

CRITICAL LIFT PLAN

For use of this form, see EM 385-1-1, Section 16. Proponent agency is Crane HHWG.

SITE PLAN

Show here or attach site plan and sequencing

NAVFAC
Naval Facilities Engineering Systems Command
PWD Maine / OICC Maine

P2 CONTRACTOR CRANE / MULTI PURPOSE MACHINE & RIGGING OPERATION CHECKLIST	
Prime Contractor: Phone Number:	Crane Operator's Name:
Crane Supplier: Phone Number:	Competent Riggers Name:
Crane Manufacturer: Model No.	Crane Capacity: Crane Serial No:
Inspection Due Date:	APP Rep: Print____________________________________ Sign_____________________________________

CONTRACTOR CRANE OR RIGGING OPERATION CHECKLIST				
		YES	NO	N/A
1	Is the Certificate of Compliance, P-1, in the operator's cab (or in the contractor's on-site office for rigging operations) with the current operator's name listed?			
2	Is the crane/machine transited to and from the job site correctly? Are the OEM instructions for travel being followed?			
3	Does the operator know the weight of the load to be lifted?			
4	Is the load to be lifted within the crane/machine manufacturer's rated capacity in its present configuration?			
5	Are outriggers/stabilizers required and, if so, are they properly extended and down?			
6	If outrigger/stabilizers are used, and the wheels are not off the ground is this the correct setup in accordance with the OEM?			
7	Is the crane/machine level and on firm ground, or if the ground is not firm are adequate supporting materials provided?			
8	If supporting materials are provided, is the entire surface of the outrigger/stabilizer pad supported and is the supporting material of sufficient strength to safely support the loaded outrigger/stabilizer pad?			
9	If outriggers/stabilizers are not used, is the crane/machine rated for on-rubber lifts by the OEM's load chart?			
10	Is the swing radius of the crane counterweight clear of people and obstructions and are accessible areas within the swing area barricaded to prevent injury or damage?			

Figure P-2 (1 of 3)

		YES	NO	N/A
11	Is the Anti-two block safety device functional? When block is reconfigured has the device been retested to insure correct operation.			
12	Has the hook been centered over the load in such a manner to minimize swing?			
13	Is the load well secured and balanced in the sling or lifting device after it is lifted a few inches for verification?			
14	Is the lift and rotation path clear of obstructions?			
15	If rotation of the load being lifted is hazardous, is a tagline or other restraint being used?			
16	Are personnel prevented from standing or passing under a suspended load?			
17	Is the operator paying full attention to the signal person?			
18	Are proper signals being used? Is the operator responding properly to the signals? Are radios used for blind lifts?			
19	Are empty hooks lashed or otherwise secured during travel to prevent swinging?			
20	Does the operator remain at the controls while the load is suspended?			
21	Does the operator ensure that side loading is prohibited?			
22	Are personnel prevented from riding on a load?			
23	Are start and stop motions in a smooth fluid motion (no sudden acceleration or deceleration)?			
24	Is the lift a critical lift?			
25	If so, is a lift plan provided and understood and check-off sheets initialed and signed off?			
26	If overhead power lines are in the vicinity, is a critical lift plan provided addressing the requirements of 29 CFR 1926.1407-1411?			
27	If pick and carry operations are allowed and performed, are OEM directions followed (e.g. rotation lock engaged, boom centered over front or rear, etc.)?			
28	When the crane/machine is left unattended, is it in a safe condition?			
29	Is rigging gear undamaged and acceptable for the application?			
30	Does rigging gear meet applicable ASME or host nation standards (e.g., ASME B30.9 for slings, B30.10 for hooks, B30.26 for rigging hardware such as shackles, safety hoist rings, and eyebolts, B30.20 for below the hook lifting devices)?			

Figure P-2 (2 of 3)

		YES	NO	N/A
31	Was the rigging gear inspected prior to use?			
32	Is sling protection used to protect slings (especially synthetic slings) and equipment from damage due to abrasion and sharp corners and edges?			
33	Is the rigging gear used in accordance with its working load limit? Is the working load limit marked on the rigging gear?			
34	Are positive latching devices (or "moussing") used on crane and rigging hooks?			
35	If a mobile crane is used on a barge, are all rules of 29 CFR 1926.1437(construction) being followed?			
36	If a mobile crane is used on a barge are the outriggers/stabilizers blocked or are the crawlers traveling in a defined space as allowed by 29 CFR 1926.1437 (construction)?			
37	For floating cranes, are rules of 29 CFR 1915 (ship repair) or 29 CFR 1926.1437 (construction) being followed?			
38	If a multi-purpose machine, forklift, or construction equipment is being used, is there proof from the OEM (or qualified PE) that the machine is approved for suspended load lifting and is there a load chart?			
39	If a personnel lift is being performed with a crane or base mounted hoist, are all requirements of NAVFAC P-307, paragraph 11.1.g (5) for a crane or 11.1.m for a base mounted hoist being followed?			

Contractor:	Subcontractor:
Location:	Date:

Notes:

Location of Lift:____

Material being lifted and Weight of lift:____

Signature of Government Representative:

Figure P-2 (3 of 3)

NAVFAC DIG SAFE UTILITY LOCATE REQUEST FORM

A Utility Locate Request Form is required for ANY Excavation (i.e. ground penetrating or concrete slab cutting, coring or drilling) either inside or outside of a building, which will penetrate more than 3", on the Shipyard. This Form shall be submitted to NAVFAC DSC at least fourteen (14) calendar days prior to Excavation.

INFORMATION THE EXCAVATOR WILL NEED TO OBTAIN A DIG SAFE TICKET (when the Excavator contacts DIG SAFE they will obtain a Dig Safe Ticket as proof of notification.)

Caller Details:

Caller Name ____________________Title_______________

Phone #_____________________ Fax #____________________ Alt # ________________

Email address ____________________ Business Hours _______ to _______

Company Name____

Address____

City____ State ___________ Zip ______

Location Details (where excavation is planned)

(Check one) State: MA______ME______NH______RI______VT______

City/Town ____

(Optional) Latitude ___________________ Longitude ___________________

Address/Intersection____

Nearest Cross Street ____

Additional Information ____

Type of Work____

Excavation Planned Depth (feet) ____________

Area (i.e. St to building, in the St., sidewalk area, right side of building)

____

Pre-marked? Yes _______ No _________

Excavation planned Start Month _____Day _______Year _____ Time (military) _____

Excavator Doing Work (if not same as above) __________________________________

Form Revision Date 4/04/23 **Attachment B cont'd**

NAVFAC DIG SAFE UTILITY LOCATE REQUEST FORM

A Utility Locate Request Form is required for ANY Excavation (i.e. ground penetrating or concrete slab cutting, coring or drilling) either inside or outside of a building, which will penetrate more than 3", on the Shipyard. Submit this Form to PWD-ME DSC at least fourteen (14) calendar days prior to Excavation. See Note on ticket expiration.

Part I – Completed by the Excavator.

Today's Date: ____/____/_____ DIG SAFE Ticket #:_________________

Ticket Expiration Date:(60 days from date obtained): _____/_____/_____
Utility Companies Dig Safe will notify: ____________________________________

Requested by: ___________________________Phone #:______________________

Code # / Company: _____________________ E-mail:_________________________

Contract #:________________________Project Title: ___________________________

Shipyard POC: ______________________Phone #________________________

Excavation Location: ______________________Area Pre-Marked YES____, NO______

Type of work: ____

Depth: (ft): _______ Anticipated Excavation Date: ___/____/___Time: (military) ________

Attach a map or the contract drawings showing the excavation/ground penetrating area.

NAVFAC Construction Contractors: Complete Part 1 and Submit Form to NAVFAC CM/ET
AE Firms/Sub consultants: Complete Part 1 and Submit Form to PWD-ME DM
Shipyard Personnel: Complete Parts 1 & 2 and Submit Form to PWD-ME DSC

Part 2 – To be completed by the NAVFAC CM/ET/DM (If Shipyard work, Part 2 must be completed by Requestor)

Date: ___/___/____ Name: ____________________Phone #__________________ Locate Priority: Routine (3 Business days) ____; Emergency (<3 Business days)____
________ (DSC Approval Req if <3 Business days)

Part 1 Reviewed and Complete: YES______, NO______ Initial: ________

Submit Completed Form to DSC. The DSC will review the Request and forward to the PWD-ME FSC Utility Locate Contract Rep who will review and forward to the Utility Locating Company for action.

Part 3 – To be completed by FSC Utility Locating Company

Approved by FSC Utility Locate Contractor: Initials _________

Date Utilities marked in the field: _____/____/_____ Name: _____________________

Comments: ____

Utility Plan Discrepancies Noted: YES____, NO____ (PWD-ME FSC PAR & DSC must be Notified if discrepancies are found: Notification made: YES____ to: ________________

Comments: ____

Submit to PWD-ME FSC Contract Rep who will forward to the DSC. If Plan Discrepancies are noted, provide a scaled markup plan showing the correct utility locations.

Form Revision Date 4/04/23 **Attachment B cont'd**

NAVFAC DIG SAFE UTILITY LOCATE REQUEST FORM

Part 4 – To be completed by PWD-ME DSC

Date__/___/____ Logged Into Database: YES ___NO ____ #________

Any Hazardous/Mission Critical Utilities in Excavation Area:

YES__ NO__ *(if YES, ensure Excavator is briefed to include additional controls of Appendix B.)*

Comments: ____

Any unknown utilities, anomalies (including increased risk of passive detection modes), or discrepancies noted by FSC Utility Locate in Part 3 have been discussed with Project Team and adequate risk mitigations are in place.

DSC or DSC Representative Approval for excavation. (*See note below).

Signature/Date: ____________________________________

DSC return a copy of the completed Form to NAVFAC CM/ET/DM or Shipyard Personnel Requesting the Utility Locate.

NAVFAC CM/ET/DM/Requestor ensure the Excavator is aware if any Plan Discrepancies were found, establish risk mitigations, and to use extreme caution when excavating.

If Plan Discrepancies are noted, the PWD DSC shall coordinate any Utility Plan Updates with PWD-ME GIS lead.

Note:

If work has not begun within 30 days from the Dig Safe Ticket approval date or if the excavation area expands beyond the original limits, the Excavator must renew the Dig Safe ticket by notifying DIG SAFE (1-888-344-7233) and the Contracting Officer. If the proposed excavation is not completed within 60 calendar days after obtaining the Dig Safe ticket, the ticket expires. Tickets should be renewed by calling DIG SAFE (as needed) at least 72 hours prior to expiration so there is always an active ticket. The utility locators (not 3rd Party) must be notified to mark out each time a ticket is renewed.

**DSC approval does not waive the construction contractor's responsibility to perform a 3rd Party Locate (if specified by contract) prior to beginning excavation. The excavator is always responsible to ensure utilities are not damaged during excavation. DSC approval does not diminish or change that responsibility.*

Form Revision 4/4/23
Attachment C

Public Works Department, Maine

Reset Form

CONTRACTOR DIG-SAFE CERTIFICATION FORM

References:

1. Government Safety Requirements (PWD ME) Section 01 35 26.00 22

Project Title: ____
Prime Contractor:____
Contract #: ____
Initial Dig-Safe Request #:__________________________________

A. Third Party Utility Location Certification

In accordance with the Specification Section 01 35 26.00 22 Government Safety Requirements para 3.7.1, The Contractor must provide the services of a third party qualified independent utility locating company/person(s) (cannot be the Government's Utility Locating Firm) to positively identify underground utilities in the work area. The utility locating company shall indicate the utility type, size, and depth of all utilities within the proposed area of excavation including, but not limited to, voice/data, electrical, gas, water, air, sanitary sewer, and storm drainage. The third party independent locating firm is in addition to the PWD ME Dig Safe Process.

By signing below, the Contractor certifies that the Third Party Utility Locate Company has complete a review of the excavation area.

Third Party Utility Locate Company:__________________________________

Date Utility Locate was completed:__________________________________

Print Name of the Utility Locator:__________________________________

Utility Locator Signature/Date:____________________________________

Any Discrepancies Found with the Marked Utilities (Yes/No): Yes_____**No**____

(If Yes, notify the Contracting Officer Immediately)

Prime Contractor Signature/Date:__________________________________

Public Works Department, Maine

CONTRACTOR DIG-SAFE CERTIFICATION FORM

B. Prime Contractor Utility Location Certification

Prime Contractor has performed a review of the following documents to ensure a comprehensive understanding of the sub-surface area to be excavated.

- Government's Dig-Safe report and associated drawings.
- Independent 3rd party Dig-Safe report and associated drawings.
- Project Contract drawings
- Government Total Map of approximate existing utility locations.

Prior to proceeding with excavation the prime contractor has the responsibility to access associated documentation of all items listed above and to mitigate the risk of a potential utility strike.

The final risk assessment shall be identified in the Contractor AHA and reviewed with the Contracting Officers Rep. Also, any hazardous/mission critical utilities identified by the DSC or plans, are covered in the Contractors AHA's and pre-job Safety Briefs prior to commencing excavation.

By signing below, the Contractor certifies the above documentation has been thoroughly reviewed and evaluated for inconsistencies, anomalies, and stated inadequacies, and the prime contractor has risk mitigation actions planned to ensure excavation does not damage underground utilities.

Prime Contractor Signature/date; ______________________________________

Reset Form

Public Works Department, Maine

Form Revision Date 4/04/23 **Attachment D**

CONTRACTOR UTILITY LOCATE – PRE-EXCAVATION SAFETY CHECKLIST

Project Title: ____

Prime Contractor: ____

Contract No.: ____

Excavator: ____

Initial ____ **Gov't and Utility (if listed in Part 1 Appendix A) Dig Safe Markings Completed & Visible in the field.**

Initial ____ **NAVFAC Dig Safe Utility Locate Request Form returned from DSC and any Noted Discrepancies are Resolved and risks mitigated.**

Initial ____ **Contractor's Independent Third Party Utility Locate Completed and any discrepancies are resolved with NAVFAC (Attach Third Party Certification Form).**

Are Hazardous or Mission Critical Utilities identified in the excavation area (as identified by DSC or plans)?: Yes ___ or No: ___
If Yes complete the following:

☐ The Contractor must supplement daily Dig Safe procedures to include an additional checkoff on Contractor Daily Activity Plan ensuring all utilities have been clearly marked and reviewed with the SSHO.

☐ The Contractor must have a Dig Safe laminated utility color coding system posted in or near all heavy digging equipment for easy reference to type of utility.

☐ The SSHO must complete a pre-excavation walk as part of the morning procedure to ensure all known utilities are identified and markings are refreshed with the appropriate color-coded paint.

☐ Contractor must provide additional danger signage, to mark areas of known live underground utilities.

☐ Contractor must ensure a 'spotter' accompanies the equipment operator during excavation work.

☐ Contractor must provide Construction CM/ET notification no later than 7 working days prior to the preparatory and initial pre-excavation/ demo safety review meeting. (The CM/ET and the DSC will attend this meeting to ensure safety and preservation of Critical Utilities is discussed with adequate focus.)

☐ The Contractor must confirm & identify the closest utility isolation points & develop mitigation strategies with the utility owner (Coordinate with the DSC) to ensure the safe excavation adjacent to these utilities. Utility Outages to isolate utility systems may need to be considered in circumstances where the excavation work cannot be completed safely.

***NOTE: The Hazardous or Mission Critical Utility requirements listed above are in addition to, and do not replace other requirments. ALL UTILITIES require hand excavation using wood or fiberglass tools when work is expected to come within 3' of the utility, per contract (Div 1 Specs, General Notes, Excavation specs).**

Prime Contractor SSHO Signature/Date: ______________________________

Excavator Signature/Date: ______________________________

NAVFAC ET Signature/Date: ______________________________

NAVFAC
Naval Facilities Engineering Command

Public Works Department, Maine

Daily Worksite Checklist for Trenching/Excavation Sites

Project:		Date:	Weather:	Soil Type:
Trench Depth:	Length:	Width:	Type of Protective System:	

Signature of Competent Person:

Yes	No	N/A	Excavation
			Trench box extends at least 18 inches above the vertical wall of the excavation and to within 2 feet of the bottom of the trench (or less if soil collapsing behind or below trench box).
			Trench box installed in accordance with manufacturers specific instructions and use limitations.
			Trench box inspected for damage or defects and pins and spreaders are securely installed.
			If other soil protective systems are used, they are installed in accordance with manufacturers Instructions OR are approved by a Registered Professional Engineer.
			All employees at worksite trained in trenching safety procedures.
			Surface encumbrances such as utility poles, heavy equipment supported or removed.
			Heavy equipment safety zone at least 1½ times depth of trench for if not supported.
			Employees protected from loose rock or soil.
			Spoils, materials, and equipment set back a minimum of 2' from edge of excavation.
			Walkways and bridges over excavations 6' or more in depth are at least 20 inches wide and are equipped with required guardrails.
			Ladders placed no more than 25 feet apart.
			Employees prohibited from working or walking under suspended loads.
			Employees prohibited from working on faces of sloped or benched excavations above other employees.
			Warning system established and used when mobile equipment is operating near edge of excavation.
			Barriers provided if trench opening is not readily apparent.
			Barriers, fences available to secure area if left overnight.

Yes	No	N/A	Personal Protective Equipment
			Hard hats worn by all employees.
			Work boots or safety shoes worn by all employees.
			Eye protection worn by all employees (if applicable).
			Hearing protection worn by all employees (if applicable).
			Warning vests, or other highly visible PPE provided and worn by all employees exposed to vehicular traffic.

Daily Worksite Checklist for Trenching/Excavation Sites – pg 2

Yes	No	N/A	Utilities
			Utility companies contacted and/or utilities located.
			Exact location of utilities marked when near excavation.
			Underground installations protected, supported, or removed when excavation is open.

Yes	No	N/A	Wet Conditions
			Precautions taken to protect employees from accumulation of water.
			Water removal equipment monitored by Competent Person.
			Surface water controlled or diverted.
			Inspection made after each rainstorm.

Yes	No	N/A	Hazardous Atmosphere
			Atmosphere tested when there is a possibility of oxygen deficiency or build-up of hazardous gases. If yes, atmosphere will be tested every ___________________.
			Oxygen content is between 19.5% and 21%.
			Flammable gas build-up to 20% of lower explosive limit (LEL).
			Toxic Levels of gases are below limits set on gas monitor.
			Ventilation blowing into space and air intake placed away from vehicle exhaust.
			Program Manager contacted if atmosphere is above established limits. Source of contaminant to be determined and eliminated prior to entry or Program Manager will establish special procedures for entry.

Emergency Procedures for Trench Cave-Ins

- **GET ALL OTHER EMPLOYEES OUT OF THE TRENCH!!**
- **CALL 438-2333**
- **NOTIFY COMPETENT PERSON**

- **Note time**
- **Note location of trapped worker(s)**
- **Leave all victims hand tools in place**
- **Shut down all heavy equipment**
- **Stop nearby traffic that may cause vibration**
- **Keep everyone back from trench at least 50 feet**
- **Gather information for rescue team**
- **WAIT for rescue team. Do not attempt to rescue.**

Note-Do not attempt to dig the person out using hand tools or heavy equipment. This could cause the trench to collapse further and could cause further injuries!!!!

Form Reset

Rev. 4/03/23

Cover Sheet for
Contractor Dig Safe Utility Locate Request Form
(PWD Maine & OICC Maine & other)

Ktr		*Check blocks when action completed.*
☐		**Prime Contractor Name:**
☐		**Contract Title:**
☐		**Requestor Name:**
☐		**Location of Excavation:**
Ktr	**ET**	***Attach Items as Noted for DSC for Review:***
☐	☐	**Dig Safe Ticket #:**
☐	☐	**Date of Excavation Start:**
☐	☐	**Renewal Date:** **(If work is planned to exceed 60 days from when State issued Ticket#)**
☐	☐	**If job includes digging <20' from a gas line, Unitil has been called by Prime Ktr.**
☐	☐	**If Potholing is required by contract, notified DSC with anticipated dates.** *Potholing is N/A.* ☐
☐	☐	**Ensure General Ktr has latest Total Map & Project Plans prior to mapping:**
☐	☐	**Navfac Dig Safe Utility Locate Request Form Attached:** **(Call in data page and Parts #1 thru #4)**
☐	☐	**Confirm white paint is marked and all items cleared out of the way:**
☐	☐	**Ensure Genral Ktr has latest Total Map & Project Plans prior to mapping:**
☐	☐	**Contractor Dig Safe Certification Form Completed and Attached:**
☐	☐	**Contractor Utility Locate Pre-Excavation Safety Check List Completed and Attached:**
☐	☐	**Anomalies, Dig Safe Notes, Risks, ORM, Mitigations to be discussed at Preparatory Mtg:**
☐	☐	**Anomalies, Dig Safe Notes, Risks, ORM, Mitigations to be discussed at Initial Phase Mtg:**
☐	☐	**Inside bldgs. for borings:** **(as needed based on design, and for PWD ME shop)**

ET Validation all steps are completed. Sign/date_______________________________

Public Works Department, Maine

PWD -ME Dig Safe Coordinator
gary.r.gagnon4.civ@us.navy.mil
207-994-9208

All fields are mandatory and must be filled in.
Prime Contractor to complete and submit within 2 working days of the utility strike. For detailed instructions, refer to Dig Safe Strike Checklist Instructions.

UNDERGROUND UTILITY STRIKE REVIEW CHECKLIST

1. General Information:

a. Project & Contract#: ____
b. Excavation Site: ______________________________________
c. Utility Impacted: _____________________________________
d. Depth of Utility Impacted (ft.): _______________
e. Date Utility was impacted:____/____/____, Time Utility was impacted:__________
f. Describe how and why damage occurred, what equipment was being used, what went wrong: (attach separate sheet as needed)
____
____
____
g. Prime Contractor company name: ____________________________________
h. Excavator company name: ____
i. Project ET: ____
j. PWD Locate Request Form Revision date used:___/___/____ (Attach a completed copy)
k. Is this identified as a Hazardous/Mission Critical utility on Part 4 of the Dig Safe Request Form? Yes: ______, No:______

2. Utility Locate Information:

a. Dig Safe Locate Ticket;

i. Original Dig Safe Ticket #: ___________ and date pulled: ____________
ii. Latest date Dig Safe Ticket was renewed: ____________ N/A: ______
iii. Date of latest FSC Utility Locate markings in excavation area: ________________
iv. Was Prime Contractor maintaining visible markings? Yes:_____, No:_____
v. Was the impacted utility currently marked where it was struck? Yes:_____
1. If not, describe situation:____
____
vi. Did FSC Utility Locate KTR flag anomalies or discrepancies on Field notes that apply to this utility? (Attach FSC Locate sketch and field notes) Yes:______, No:_____
1. Was DSC involved in resolution of anomalies or discrepancies? Yes:____, or why not?____
2. Was there a post FSC Locate Tailgate discussion? Yes:______, or why not?
____

3. Did Prime Contractor attempt to mitigate utility impact risk from anomalies or discrepancies? Yes:______, No:______
Describe how, or why not (include separate sheet if needed): ______________
____
____

vii. Did PWD ME Dig Safe Coordinator approve excavation on Utility Locate Request Form Part 4: Yes: ________, or why not?______________________________

b. 3rd Party Utility Locate Information:

i. 3rd Party Locate Company: ____
ii. Date of the 3rd Party Locate:____/_____/______
iii. Did the 3rd Party flag anomalies or discrepancies on field notes that apply to this utility? (Attach 3rd Party Locate sketch and field notes) Yes:_____, No:______

1. Was DSC involved in resolution of anomalies or discrepancies? Yes: ____, or why not?____
2. Was there a post 3rd Party Tailgate discussion? Yes: ______, or why not?
____

Did Prime Contractor attempt to mitigate utility impact risk from anomalies or discrepancies? Yes: ______, No: ______
Describe how, or why not (include separate sheet if needed): ______________
____
_ ____

iv. Did Prime Contractor complete the Contractor Dig Safe Certification Form prior to excavation? (attach completed copy): Yes:_____, or why not?
____

3. Preparatory & Initial Meetings Information:

a. Date of the Preparatory Meeting: ____/____/_____
b. Did PWD DSC attend the Preparatory meeting? Yes: _____or why not? ______________
c. Were Haz/Mission Critical Utilities reviewed at the meeting? Yes: _____, No:________
d. Did PWD UEM personnel attend the meeting? Yes: _______, No: ________
e. Were the Utility Markings in the field reviewed? Yes: ______, No: _______
f. Were FSC and 3rd Party Identified discrepancies or anomalies noted and discussed? Yes: _____, If No, why not? _____, ____
g. Date of the Initial Meeting: __________________
h. Were Utility Risk Mitigation Strategies reviewed (attach minutes): Yes: ______, if No, why not? _____, ____
i. Were Utility Outages part of the Risk Mitigation Plan? Yes: _______, No: ________
j. Was excavation performed with expected precaution? Yes: _______, No: ________

4. Utility Mapping Review (DSC input):

a. Did the FSC Dig Safe KTR have the latest Total Map? (Explain): Yes: ____, No: ____

____

____

b. Were inaccuracies between the field conditions & the Total Map found? Explain; Yes: _____, No: ______, ____

____

c. If Total Map inaccuracies were identified for this area, has CAD updated for most current As-Built Info? Yes: ______, No: _______

d. Was digging within 3’ of marked utility? Yes: _______, No: ________

5. Items to submit along with this Review Checklist:

a. _______A copy of the Dig Safe Ticket.

b. _______Diagram/sketch of the area with specific location of the strike.

c. _______Appropriate Contractor’s Daily Work Notes.

d. _______FSC Dig Safe Locate Report (sketch and field notes).

e. _______3rd Party Locate Report (sketch and field notes).

f. _______Statement from Equipment Operator.

g. _______Preparatory Meeting Minutes.

h. _______Initial Meeting Minutes.

6. Project Specific Comments:

____

____

____

____

7. What could be done better?

a.

b.

c.

8. Lessons Learned:

a.

b.

c.

ATTACHMENTS FOR DETAILS AND EXPLAINATIONS AS NEEDED

The Color Coding System

Color coded paint or flags are used to identify the type of underground facilities.

RED	*Electric Power Lines, Cables, Conduit and Lighting Cables*
YELLOW	*Gas, Oil, Steam, Petroleum or Gaseous Materials*
ORANGE	*Communication, Alarm or Signal Lines, Cables or Conduit*
BLUE	*Potable Water*
GREEN	*Sewers and Drain Lines*
PURPLE	*Non-Potable Water: Reclaimed Water, Irrigatiion and Slurry Lines*
PINK	*Temporary Survey Markings/ Unknown Utilities*

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SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS
05/24

PART 1 GENERAL

1.1 REFERENCES

Various publications are referenced in other sections of the specifications to establish requirements for the work. These references are identified in each section by document number, date and title. The document number used in the citation is the number assigned by the standards producing organization (e.g., ASTM B564 Standard Specification for Nickel Alloy Forgings). However, when the standards producing organization has not assigned a number to a document, an identifying number has been assigned for reference purposes.

1.2 ORDERING INFORMATION

The addresses of the standards publishing organizations whose documents are referenced in other sections of these specifications are listed below, and if the source of the publications is different from the address of the sponsoring organization, that information is also provided.

ACOUSTICAL SOCIETY OF AMERICA (ASA)
1305 Walt Whitman Road, Suite 110
Melville, NY 11747-4300
Ph: 516-576-2360
Fax: 631-923-2875
E-mail: asa@acousticalsociety.org
Internet: https://acousticalsociety.org/

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)
30 West University Drive
Arlington Heights, IL 60004-1893
Ph: 847-394-0150
Fax: 847-253-0088
E-mail: communications@amca.org
Internet: http://www.amca.org

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)
2311 Wilson Blvd, Suite 400
Arlington, VA 22201
Ph: 703-524-8800
Internet: http://www.ahrinet.org

ALUMINUM ASSOCIATION (AA)
1400 Crystal Drive
Suite 430
Arlington, VA 22202
Ph: 703-358-2960

Internet: https://www.aluminum.org/

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)
1900 E Golf Rd, Suite 1250
Schaumburg, IL 60173

Ph: 847-303-5664
E-mail: customerservice@FGIAonline.org
Internet: https://fgiaonline.org/

AMERICAN ASSOCIATION OF RADON SCIENTISTS AND TECHNOLOGISTS (AARST)
527 N Justice Street
Hendersonville, NC 28739
Ph: 828-348-0185

E-mail: info@aarst.org
Internet: http://aarst-nrpp.com/wp/

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)
555 12th Street NW, Suite 1000
Washington, DC 20004
Ph: 202-624-5800
Fax: 202-624-5806
E-Mail: info@aashto.org
Internet: https://www.transportation.org/

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)
1 Davis Drive
P.O. Box 12215
Research Triangle Park, NC 27709-2215
Ph: 919-549-8141

E-Mail: ordering@aatcc.org
Internet: https://www.aatcc.org/

AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)
1001 N. Fairfax Street Suite 500
Alexandria, VA 22314
Ph: 703-842-0030
E-mail: info@americanbearings.org
Internet: https://www.americanbearings.org/

AMERICAN CONCRETE INSTITUTE (ACI)
38800 Country Club Drive
Farmington Hills, MI 48331-3439
Ph: 248-848-3800
Fax: 248-848-3701
Internet: https://www.concrete.org/

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)
3640 Park 42 Drive
Cincinnati, OH 45241
Ph: 513-742-2020
Fax: 513-742-3355
Email: customerservice@acgih.org
Internet: https://www.acgih.org/

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)
130 East Randolph, Suite 2000
Chicago, IL 60601
Ph: 312-670-2400
Fax: 312-670-5403
Steel Solutions Center: 866-275-2472
E-mail: solutions@aisc.org

Internet: https://www.aisc.org/

AMERICAN IRON AND STEEL INSTITUTE (AISI)
25 Massachusetts Avenue, NW Suite 800
Washington, DC 20001
Ph: 202-452-7100
Internet: https://www.steel.org/

AMERICAN LADDER INSTITUTE (ALI)
1300 Summer Avenue
Clevelan, OH 44115
Ph: 216-241-7333
Fax: 216-241-0105
E-mail: info@americanladderinstitute.org
Internet: https://www.americanladderinstitute.org

AMERICAN LUMBER STANDARDS COMMITTEE (ALSC)
7470 New Technology Way, Suite F
Frederick, MD 21703
Ph: 301-972-1700
Fax: 301-540-8004
E-mail: alsc@alsc.org
Internet: http://www.alsc.org

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
1899 L Street, NW,11th Floor
Washington, DC 20036
Ph: 202-293-8020
Fax: 202-293-9287
E-mail: info@ansi.org
Internet: https://www.ansi.org/

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)
1201 Dublin Road Suite G04

Columbus, OH 43215-1045
Ph: 800-222-2768 or 614-274-6003

E-mail: customersupport@asnt.org
Internet: https://www.asnt.org/

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)
1801 Alexander Bell Drive
Reston, VA 20191
Ph: 800-548-2723; 703-295-6300
Email: customercare@asce.org
Internet: https://www.asce.org/

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)
180 Technology Parkway NW
Peachtree Corners, GA 30092
Ph: 404-636-8400 or 800-527-4723
Fax: 404-321-5478
E-mail: ashrae@ashrae.org
Internet: https://www.ashrae.org/

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Two Park Avenue

New York, NY 10016-5990
Ph: 800-843-2763
Fax: 973-882-1717
E-mail: customercare@asme.org
Internet: https://www.asme.org/

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)
520 N. Northwest Highway
Park Ridge, IL 60068
Ph: 847-699-2929
E-mail: customerservice@assp.org
Internet: https://www.assp.org/

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)
18927 Hickory Creek Drive, Suite 220
Mokena, IL 60448
Ph: 708-995-3019
Fax: 708-479-6139
Internet: http://www.asse-plumbing.org

AMERICAN WATER WORKS ASSOCIATION (AWWA)
6666 W. Quincy Avenue
Denver, CO 80235 USA
Ph: 303-794-7711 or 800-926-7337
Fax: 303-347-0804
Internet: https://www.awwa.org/

AMERICAN WELDING SOCIETY (AWS)
8669 NW 36 Street, #130
Miami, FL 33166-6672
Ph: 800-443-9353
Email: customercare@aws.org
Internet: https://www.aws.org/

AMERICAN WOOD COUNCIL (AWC)
50 Catoctin Circle SE, Suite 201
Leesburg, VA 20176
Ph: 800-890-7732 or 202-463-2766
Fax: 412-741-0609
E-mail: info@awc.org
Internet: https://www.awc.org/

AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)
P.O. Box 361784
Birmingham, AL 35236-1784
Ph: 205-733-4077
Fax: 205-733-4075
Internet: http://www.awpa.com

APA - THE ENGINEERED WOOD ASSOCIATION (APA)
7011 South 19th St.
Tacoma, WA 98466-5333
Ph: 253-565-6600
Fax: 253-565-7265
Internet: https://www.apawood.org/

ASPHALT ROOFING MANUFACTURERS ASSOCIATION (ARMA)
2331 Rock Spring Road
Forest Hill, MD 21050

Ph: 443-640-1075
Fax: 443-640-1031
Email: info@asphaltroofing.org
Internet: https://asphaltroofing.org/

ASSOCIATED AIR BALANCE COUNCIL (AABC)
1015 18th St. NW Suite 603
Washington, DC 20036
Ph: 202-737-0202
Fax: 202-315-0285
E-mail: info@aabc.com
Internet: https://www.aabc.com/

ASSOCIATION OF THE WALL AND CEILING INDUSTRY (AWCI)
513 West Broad Street, Suite 210
Falls Church, VA 22046
Ph: 703-538-1600
Fax: 703-534-8307
Internet: https://www.awci.org/

ASTM INTERNATIONAL (ASTM)
100 Barr Harbor Drive, P.O. Box C700
West Conshohocken, PA 19428-2959
Ph: 610-832-9500
Fax: 610-832-9555
E-mail: service@astm.org
Internet: https://www.astm.org/

BUILDERS HARDWARE MANUFACTURERS ASSOCIATION (BHMA)
529 14th Street NW Suite 1280
Washington, DC 20045
Ph: 212-297-2122
Fax: 212-370-9047
Internet: https://www.buildershardware.com/

CALIFORNIA AIR RESOURCES BOARD (CARB)
1001 I Street
Sacramento, CA 95814
Ph: 800-242-4450
Email: helpline@arb.ca.gov
Internet: https://ww2.arb.ca.gov/

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)
PO Box 997377, MS 0500
Sacramento, CA 95899-7377
Ph: 916-558-1784
Internet: https://www.cdph.ca.gov/

CARPET AND RUG INSTITUTE (CRI)
P.O. Box 2048
Dalton, GA 30722-2048
Ph: 706-278-3176
Fax: 706-278-8835
Internet: https://carpet-rug.org/

CAST IRON SOIL PIPE INSTITUTE (CISPI)
2401 Fieldcrest Drive

Mundelein, IL 60060
Ph: 224-864-2910
Internet: https://www.cispi.org/

CENTERS FOR DISEASE CONTROL AND PREVENTION (CDC)
1600 Clifton Road
Atlanta, GA 30329-4027
Ph: 800-232-4636
TTY: 888-232-6348
Internet: https://www.cdc.gov

COMPOSITE PANEL ASSOCIATION (CPA)
19465 Deerfield Avenue, Suite 306
Leesburg, VA 20176
Ph: 703-724-1128
Fax: 703-724-1588
Internet: https://www.compositepanel.org/

COMPRESSED GAS ASSOCIATION (CGA)
8484 Westpark Drive Suite 220
McLean, Virginia 22102
Ph: 703-788-2700
Fax: 703-961-1831
E-mail: cga@cganet.com
Internet: https://www.cganet.com/

CONCRETE REINFORCING STEEL INSTITUTE (CRSI)
933 North Plum Grove Road
Schaumburg, IL 60173-4758
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Email: crsi@crsi.org
Fax: 847-517-1206
Internet: http://www.crsi.org/

CONSUMER ELECTRONICS ASSOCIATION (CEA)
1919 South Eads St.
Arlington, VA 22202
Ph: 703-907-7600
E-mail: CTA@CTA.tech
Internet: https://www.cta.tech/

COPPER DEVELOPMENT ASSOCIATION (CDA)
Internet: https://www.copper.org/

COUNCIL ON ENVIRONMENTAL QUALITY (CEQ) (WHITE HOUSE)
722 Jackson Place
Washington DC 20506
Internet: https://www.whitehouse.gov/administration/eop/ceq

CSA GROUP (CSA)
178 Rexdale Blvd.
Toronto, ON, Canada M9W 1R3
Ph: 416-747-4044
Fax: 416-747-2510
E-mail: member@csagroup.org
Internet: https://www.csagroup.org/

DOOR AND ACCESS SYSTEM MANUFACTURERS ASSOCIATION (DASMA)
1300 Sumner Avenue

Cleveland, OH 44115-2851
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Fax: 216-241-0105
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Internet: https://www.dasma.com/

ELECTRICAL GENERATING SYSTEMS ASSOCIATION (EGSA)
P.O. Box 73206
Washington, DC 20056
Ph: 561-750-5575
Fax: 561-395-8557
Email: e-mail@egsa.org
Internet: http://www.egsa.org

ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)
310 Maxwell Road, Suite 200
Alpharetta, GA 30009
Ph: 678-393-9990
Fax: 678-393-9998
E-mail: emikoski@ecianow.org
Internet: https://www.ecianow.org

EUROPEAN COMMITTEE FOR STANDARDIZATION (CEN/CENELEC)
CEN-CENELEC Management Centre
Rue de la Science 23
B - 1040 Brussels, Belgium
Ph: 32-2-550-08-11
Fax: 32-2-550-08-19
Internet: https://www.cencenelec.eu/

EUROPEAN UNION (EU)
European Commission
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1000 Bruxelle
Belgium
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Internet: https://commission.europa.eu/index_en

FLUID CONTROLS INSTITUTE (FCI)
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E-mail: fci@fluidcontrolsinstitute.org
Internet: https://fluidcontrolsinstitute.org/

FM GLOBAL (FM)
270 Central Avenue
Johnston, RI 02919-4949
Ph: 401-275-3000
Fax: 401-275-3029
Internet: https://www.fmglobal.com/

GLASS ASSOCIATION OF NORTH AMERICA (GANA)
National Glass Association
1945 Old Gallows Rd., Suite 750
Vienna, VA 22182
Ph: 866-342-5642
Ph: 703-442-4890

Fax: 703-442-0630
Internet: https://www.glass.org/

GREEN SEAL (GS)
601 13th St NW 12th Floor
Washington, DC 20005
Ph: 202-872-6400
Fax: 202-872-4324
E-mail: greenseal@greenseal.org
Internet: https://www.greenseal.org/

GYPSUM ASSOCIATION (GA)
962 Wayne Ave., Suite 620
Silver Spring, MD 20910
Ph: 301-277-8686
Fax: 301-277-8747
E-mail: info@gypsum.org
Internet: https://www.gypsum.org/

HYDRAULIC INSTITUTE (HI)
300 Interpace Parkway
Parsippany, NJ 07054-4405
Ph: 973-267-9700
Fax: 973-267-9055
Internet: http://www.pumps.org

ICC EVALUATION SERVICE, INC. (ICC-ES)
3060 Saturn Street, Suite 100
Brea, CA 92821
Ph: 800-423-6587
Fax: 562-695-4694
E-mail: es@icc-es.org
Internet: https://icc-es.org/

ILLUMINATING ENGINEERING SOCIETY (IES)
120 Wall Street, Floor 17
New York, NY 10005-4001
Ph: 212-248-5000
Fax: 212-248-5018
E-mail: membership@ies.org
Internet: https://www.ies.org/

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
445 and 501 Hoes Lane
Piscataway, NJ 08854-4141
Ph: 732-981-0060 or 800-701-4333
Fax: 732-981-9667
E-mail: onlinesupport@ieee.org
Internet: https://www.ieee.org/

INSULATED CABLE ENGINEERS ASSOCIATION (ICEA)
P.O. Box 493
Miamitown, OH 45041-9998
E-mail: info@icea.net
Internet: https://www.icea.net/

INSULATING GLASS MANUFACTURERS ALLIANCE (IGMA)
27 N. Wacker Dr. Suite 365
Chicago, IL 60606-2800

Ph: 613-233-1510
Fax: 613-482-9436
E-mail: enquiries@igmaonline.org
Internet: https://fgiaonline.org//

INTELLIGENCE COMMUNITY STANDARD (ICS)
Homeland Security Digital Library
Ph: 831-272-2437
E-mail: hsdl@nps.edu
Internet: https://www.hsdl.org/c/

INTERNATIONAL CODE COUNCIL (ICC)
200 Massachusetts Ave, NW Suite 250
Washington, DC 20001
Ph: 800-786-4452 or 888-422-7233
Fax: 202-783-2348
E-mail: order@iccsafe.org
Internet: https://www.iccsafe.org/

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)
3050 Old Centre Ave. Suite 101
Portage, MI 49024
Ph: 269-488-6382 or 1-888-300-6382
Fax: 269-488-6383
Internet: https://www.netaworld.org/

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)
3, rue de Varembe, 1st floor
P.O. Box 131
CH-1211 Geneva 20, Switzerland
Ph: 41-22-919-02-11
Fax: 41-22-919-03-00
E-mail: info@iec.ch
Internet: https://www.iec.ch/

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)
ISO Central Secretariat
BIBC II
Chemin de Blandonnet 8
CP 401 - 1214 Vernier, Geneva
Switzerland
Ph: 41-22-749-01-11
E-mail: central@iso.ch
Internet: https://www.iso.org

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)
1101 Wilson Blvd, Suite 1425
Arlington, VA 22209-1762
Ph: 703-525-1695
Fax: 703-528-2148
Internet: https://safetyequipment.org/

INTERNET ENGINEERING TASK FORCE (IETF)
c/o Association Management Solutions, LLC (AMS)
5177 Brandin Court
Fremont, California 94538
Ph: 510-492-4080
Fax: 510-492-4001
E-mail: ietf-info@ietf.org

Internet: https://www.ietf.org/

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)
127 Park Street, NE
Vienna, VA 22180-4602
Ph: 703-281-6613
E-mail: info@msshq.org
Internet: http://msshq.org

MASTER PAINTERS INSTITUTE (MPI)
2800 Ingleton Avenue
Burnaby, BC CANADA V5C 6G7
Ph: 1-888-674-8937
Fax: 1-888-211-8708
E-mail: info@paintinfo.com or techservices@mpi.net
Internet: http://www.mpi.net/

METAL BUILDING MANUFACTURERS ASSOCIATION (MBMA)
1300 Sumner Avenue
Cleveland, OH 44115-2851
Ph: 216-241-7333
Fax: 216-241-0105
Internet: https://www.mbma.com/

METAL FRAMING MANUFACTURERS ASSOCIATION (MFMA)
330 N. Wabash Avenue
Chicago, IL 60611
Ph: 312-644-6610
E-mail: MFMAstats@smithbucklin.com
Internet: http://www.metalframingmfg.org/

MIDWEST INSULATION CONTRACTORS ASSOCIATION (MICA)
7250 Poe Avenue, Suite 410
Dayton, OH 45414
Ph: 888-294-0084
Fax: 937-278-0317
Email: info@micainsulation.org
Internet: https://www.micainsulation.org/

MODBUS ORGANIZATION, INC (MODBUS)
800 Federal St.
Andover, MA 01810
Ph: 508-435-7170
Fax: 508-435-7172
Internet: http://www.modbus.org

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)
800 Roosevelt Road, Bldg C, Suite 312
Glen Ellyn, IL 60137
Ph: 630-942-6591
Fax: 630-790-3095
E-mail: info@naamm.org
Internet: http://www.naamm.org

NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)
1201 Pennsylvania Ave. NW, Suite 1200
Washington, DC 20004
Ph: 202-991-6300

Fax: 202-217-4171
Internet: https://www.necanet.org/

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)
1300 North 17th Street, Suite 900
Arlington, VA 22209
Ph: 703-841-3200
Email: communications@nema.org
Internet: https://www.nema.org

NATIONAL ELEVATOR INDUSTRY, INC. (NEII)
5537 SW Urish Road
Topeka, KS 66610
Ph: 703-589-9985
E-Mail: info@neii.org
Internet: https://nationalelevatorindustry.org/

NATIONAL ENVIRONMENTAL BALANCING BUREAU (NEBB)
8575 Grovemont Circle
Gaithersburg, MD 20877
Ph: 301-977-3698
Fax: 301-977-9589
Internet: http://www.nebb.org

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
1 Batterymarch Park
Quincy, MA 02169-7471
Ph: 800-344-3555
Fax: 800-593-6372
Internet: https://www.nfpa.org

NATIONAL INSTITUTE FOR CERTIFICATION IN ENGINEERING TECHNOLOGIES (NICET)
1420 King Street
Alexandria, VA 22314-2794
Ph: 888-476-4238 (1-888 IS-NICET)
E-mail: tech@nicet.org
Internet: https://www.nicet.org/

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)
Patriots Plaza 1
395 E Street, SW, Suite 9200
Washington, DC 20201
Ph: 800-232-4636
Fax: 513-533-8347
Internet: https://www.cdc.gov/niosh/

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)
100 Bureau Drive
Gaithersburg, MD 20899
Ph: 301-975-2000
Internet: https://www.nist.gov/

NATIONAL PARK SERVICE (NPS)
National Park Service
1849 C Street, NW
Washington, DC 20240
Ph: 202-208-6843
Internet: https://www.nps.gov/

NATIONAL ROOFING CONTRACTORS ASSOCIATION (NRCA)
10255 West Higgins Road, Suite 600
Rosemont, IL 60018-5607
Ph: 847-299-9070
Fax: 847-299-1183
Internet: http://www.nrca.net

NORTHEASTERN LUMBER MANUFACTURERS ASSOCIATION (NELMA)
272 Tuttle Road
Cumberland Center, ME 04021
Ph: 207-829-6901
Fax: 207-829-4293
E-mail: info@nelma.org
Internet: https://www.nelma.org/

NSF INTERNATIONAL (NSF)
789 North Dixboro Road
P.O. Box 130140
Ann Arbor, MI 48105
Ph: 734-769-8010 or 800-NSF-MARK
Fax: 734-769-0109
E-mail: info@nsf.org
Internet: http://www.nsf.org

OPC FOUNDATION (OPC)
16101 N. 82nd Street
Suite 3B
Scottsdale, AZ 85260-1868
Ph: 480-483-6644
Fax: 480-483-7202
Email: office@opcfoundation.org
Internet: https://opcfoundation.org/

PLUMBING AND DRAINAGE INSTITUTE (PDI)
800 Turnpike Street, Suite 300
North Andover, MA 01845
Ph: 978-557-0720 or 800-589-8956
E-Mail: pdi@PDIonline.org
Internet: http://www.pdionline.org

SCIENTIFIC CERTIFICATION SYSTEMS (SCS)
2000 Powell Street, Suite 600
Emeryville, CA 94608
Ph: 510-452-8000
Fax: 510-452-8001
E-mail: info@SCSglobalservices.com
Internet: https://www.scsglobalservices.com/

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)
4201 Lafayette Center Drive
Chantilly, VA 20151-1219
Ph: 703-803-2980
Fax: 703-803-3732
Internet: https://www.smacna.org/

SINGLE PLY ROOFING INDUSTRY (SPRI)
465 Waverley Oaks Road, Suite 421
Waltham, MA 02452
Ph: 781-647-7026
Fax: 781-647-7222
E-mail: info@spri.org
Internet: https://www.spri.org/

SOCIETY FOR PROTECTIVE COATINGS (SSPC)
800 Trumbull Drive
Pittsburgh, PA 15205
Ph: 877-281-7772 or 412-281-2331
Fax: 412-444-3591
E-mail: customerservice@sspc.org
Internet: http://www.sspc.org

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)
400 Commonwealth Drive
Warrendale, PA 15096
Ph: 877-606-7323 or 724-776-4841
Fax: 724-776-0790
E-mail: customerservice@sae.org
Internet: https://www.sae.org/

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)
21865 Copley Drive
Diamond Bar, CA 91765
Ph: 909-396-2000
E-mail: webinquiry@aqmd.gov
Internet: http://www.aqmd.gov

SOUTHERN PINE INSPECTION BUREAU (SPIB)
P.O. Box 10915
Pensacola, FL 32524-0915
Ph: 850-434-2611 or 800-995-7742
Fax: 850-434-1290
E-mail: spib@spib.org
Internet: https://www.spib.org/

SPRAY POLYURETHANE FOAM ALLIANCE (SPFA)
3927 Old Lee Hwy. #101B
Fairfax, VA 22030
Ph: 800-523-6154
Fax: 703-222-5816
Internet: http://www.sprayfoam.org

STEEL DECK INSTITUTE (SDI)
P.O. Box 426
Glenshaw, PA 15116
Ph: 412-487-3325
Fax: 412-487-3326
Internet: https://www.sdi.org/

STEEL DOOR INSTITUTE (SDI/DOOR)
30200 Detroit Road
Westlake, OH 44145
Ph: 440-899-0010
Fax: 440-892-1404
E-mail: info@steeldoor.org

Internet: https://www.steeldoor.org/

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)
15 Technology Parkway South, Suite 115
Peachtree Corners, GA 30092
Ph: 800-332-8686 or 770-446-1400
Fax: 770-446-6947
E-mail: memberconnection@tappi.org
Internet: http://www.tappi.org

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)
1320 North Courthouse Rosd, Suite 200
Arlington, VA 22201
Ph: 703-907-7700
Fax: 703-907-7727
E-mail: marketing@tiaonline.org
Internet: https://www.tiaonline.org/

THE MASONRY SOCIETY (TMS)
105 South Sunset Street, Suite Q
Longmont, CO 80501-6172
Ph: 303-939-9700
Fax: 303-541-9215
E-mail: info@masonrysociety.org
https://masonrysociety.org/

TILE COUNCIL OF NORTH AMERICA (TCNA)
100 Clemson Research Boulevard
Anderson, SC 29625
Ph: 864-646-8453
Fax: 864-646-2821
E-mail: info@tileusa.com
Internet: https://www.tcnatile.com/

TRIDIUM, INC (TRIDIUM)
3951 Westerre Parkway, Suite 350
Richmond, VA 23233
Ph: 804-747-4771
Fax: 804-747-5204
E-mail: support@tridium.com
Internet: https://www.tridium.com/

U.S. AIR FORCE (USAF)
E-mail: usaf.pentagon.saf-aa.mbx.AFDPO-PPL@mail.mil
Internet: https://www.e-publishing.af.mil/

U.S. ARMY CORPS OF ENGINEERS (USACE)
CRD-C DOCUMENTS available on Internet:
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Internet: http://www.publications.usace.army.mil/
or
https://www.hnc.usace.army.mil/Missions/Engineering-Directorate/TECHINFO/

U.S. DEFENSE LOGISTICS AGENCY (DLA)
Andrew T. McNamara Building
8725 John J. Kingman Road

Fort Belvoir, VA 22060-6221
Ph: 877-352-2255
E-mail: dlacontactcenter@dla.mil
Internet: http://www.dla.mil

U.S. DEPARTMENT OF AGRICULTURE (USDA)
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Gastonia, NC 28054-2193
Ph: 704-810-8884
E-mail: PA@usda.gov
Internet: https://www.ams.usda.gov/
Order Other Publications from:
USDA Rural Development
Rural Utilities Service
STOP 1510, Rm 5135
1400 Independence Avenue SW
Washington, DC 20250-1510
Phone: (202) 720-9540
Internet:
https://www.rd.usda.gov/about-rd/agencies/rural-utilities-service

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Washington, DC 20301-1400
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E-mail: customerservice@ntis.gov
Internet: https://www.ntis.gov/
Obtain Military Specifications, Standards and Related Publications from:
Acquisition Streamlining and Standardization Information System (ASSIST)
Department of Defense Single Stock Point (DODSSP)
Document Automation and Production Service (DAPS)
Building 4/D
700 Robbins Avenue
Philadelphia, PA 19111-5094
Ph: 215-697-6396 - for account/password issues
Internet: https://assist.dla.mil/online/start/; account registration required
Obtain Unified Facilities Criteria (UFC) from:
Whole Building Design Guide (WBDG)
National Institute of Building Sciences (NIBS)
1090 Vermont Avenue NW, Suite 700
Washington, DC 20005
Ph: 202-289-7800
Fax: 202-289-1092
Internet:
https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc

U.S. DEPARTMENT OF ENERGY (DOE)
1000 Independence Avenue Southwest
Washington, D.C. 20585
Ph: 202-586-5000

Fax: 202-586-4403
E-mail: The.Secretary@hq.doe.gov
Internet: https://www.energy.gov/

U.S. DEPARTMENT OF HOUSING AND URBAN DEVELOPMENT (HUD)
HUD User
P.O. Box 23268
Washington, DC 20026-3268
Ph: 800-245-2691 or 202-708-3178
TDD: 800-927-7589
Fax: 202-708-9981
E-mail: helpdesk@huduser.gov
Internet: https://www.huduser.gov

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)
1200 Pennsylvania Avenue, N.W.
Washington, DC 20004
Ph: 202-564-4700
Internet: https://www.epa.gov
--- Some EPA documents are available only from:
National Technical Information Service (NTIS)
5301 Shawnee Road
Alexandria, VA 22312
Ph: 703-605-6060 or 1-800-363-2068
Fax: 703-605-6880
TDD: 703-487-4639
E-mail: info@ntis.gov
Internet: https://www.ntis.gov/

U.S. FEDERAL AVIATION ADMINISTRATION (FAA)
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TTY: 888-835-5322
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Bookstore: 202-512-0132
Internet: https://www.gpo.gov/

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)
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Washington, DC 20590
Ph: 202-366-4000
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Internet: https://highways.dot.gov/
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Internet: https://www.gpo.gov/

U.S. GENERAL SERVICES ADMINISTRATION (GSA)
General Services Administration
1800 F Street, NW
Washington, DC 20405
Ph: 1-844-472-4111
Internet: https://www.gsaelibrary.gsa.gov/ElibMain/home.do
Obtain documents from:
Acquisition Streamlining and Standardization Information System (ASSIST)
Internet: https://assist.dla.mil/online/start/; account registration required

U. S. GREEN BUILDING COUNCIL (USGBC)
2101 L St NW, Suite 600
Washington, DC 20037
Ph: 202-828-7422
Internet: https://new.usgbc.org/

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)
8601 Adelphi Road
College Park, MD 20740-6001
Ph: 866-272-6272
Internet: https://www.archives.gov/
Order documents from:
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732 N. Capitol Street, NW
Washington, DC 20401
Ph: 202-512-1800 or 866-512-1800
Bookstore: 202-512-0132
Internet: https://www.gpo.gov/

UL SOLUTIONS (UL)
333 Pfingsten Road
Northbrook, IL 60062
Ph: 877-854-3577 or 360-817-5500
E-mail: CustomerExperienceCenter@ul.com
Internet: https://www.ul.com/
UL Directories available through IHS at https://accuristech.com/

WEST COAST LUMBER INSPECTION BUREAU (WCLIB)
1010 South 336th Street #210
Federal Way, WA 98003
Ph: 253-835-3344

Internet: https://www.plib.org/

WESTERN WOOD PRODUCTS ASSOCIATION (WWPA)
2 Centerpointe Drive STE 360
Lake Oswego, OR 97035
Ph: 503-224-3930
E-mail: info@wwpa.org
Internet: http://www.wwpa.org

WINDOW AND DOOR MANUFACTURERS ASSOCIATION (WDMA)
2001 K Streeet NW
3rd Floor North
Washington, DC 20006
Ph: 202-367-1157
or
330 N Wabash Avenue, Suite 2000
Chicago, IL 60611
Ph: 312-321-6802
E-mail: membersupport@wdma.com
Internet: https://www.wdma.com/

WOODWORK INSTITUTE (WI)
1455 Response Road, Suite 110
Sacremento, CA 95815
Ph: 916-372-9943
Fax: 916-372-9950
E-mail: info@woodinst.com
Internet: https://woodworkinstitute.com

PART 2 PRODUCTS

Not used

PART 3 EXECUTION

Not used

-- End of Section --

SECTION 01 44 00.00 22

ARCHAEOLOGICAL MONITORING (PWD ME)
10/23

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at the Portsmouth Naval Shipyard and at PWD ME AOR Facilities.

1.1 SUMMARY

Perform archaeological monitoring for all forms of soil disturbing activities and excavations within archaeologically sensitive portions of the project area **as identified in project drawings**. Work must include but is not limited to: background research and review of previous surveys, monitoring during excavation, installation of components, regrading, and other work indicated or specified. Results of monitoring efforts must be summarized in Archaeological Monitoring Reports , including progress reports, a draft monitoring report, a final draft monitoring report, and a final monitoring report. Artifacts collected during monitoring efforts must be catalogued and curated.

The Government will provide available existing documentation pertaining to the project area, which will include previous investigations, plans, photographs, and research.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 79	(2022) Curation of Federally Owned orAdministered Archeological Collections
36 CFR 61	(2017) Procedures for State, Tribal, and Local Government Historic Preservation Programs

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

Archaeologist Qualifications; G

Archaeological Coordinator Designation Letter

Archaeological Monitoring Plan; G

SD-06 Test Reports

Archaeological Resource Report; G

Archaeological Monitoring Report - Monthly Progress Reports; G

Archaeological Monitoring Report - Draft; G

Archaeological Monitoring Report - Final Draft; G

Archaeological Monitoring Report - Final; G

Curated Artifacts (if collected); G

1.4 QUALITY ASSURANCE

The Contractor must submit Archaeologist Qualifications prior to submission of the Archaeological Monitoring Plan. The Archaeologist Qualifications must be approved by the Government, in accordance with the procedures of Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME), prior to the Archaeological Coordination Meeting and submission of the Archaeological Monitoring Plan. The Archaeologist must meet the Secretary of the Interior's Professional Qualification Standards for Archaeologist as outlined in 36 CFR 61 as required by the National Historic Preservation Act (Section 112, a, 1, A), OPNAVIST 5090.1E (Section 13-3.6), and the Principal Investigator must be an approved Archaeologist on the Maine Historic Preservation Commission's Level II Approved List. Field staff do not need to be named on the Level II list if they are supervised by an archaeologist with those qualifications.

The Archaeologist must specialize in historic archaeology and have experience monitoring construction work in urban or industrial archaeological settings. The Archaeologist must have a demonstrated track record of success in carrying projects to completion, including the submission of all final project documentation and curated artifacts.

1.5 COORDINATION

The Contractor must designate in writing, with an Archaeological Coordinator Designation Letter, the Archaeological Coordinator responsible for coordination with the archaeological subcontractor. This individual must have a thorough understanding of the project schedule. The Contractor must notify Contractoring Officer of any changes in the Archaeological Coordinator in writing.

The Archeological Contractor is responsible for obtaining permission and clearance from the Contracting Officer to enter areas of the required fieldwork.

1.5.1 ARCHAEOLOGICAL COORDINATION MEETING

Prior to the start of any ground disturbing/excavation work and prior to submission of the Archaeological Monitoring Plan, the Archaeological Coordinator, project archaeologist, Contractor's QC Manager, PWD ME CM/ET and PWD-ME Cultural Resources Manager must meet at an Archaelogical Coordination Meeting to review the proposed scope of work, establish a protocol for artifact collection, and review the relevant documentation.

The Contractor is responsible for recording minutes of all meetings and furnishing a copy of the minutes to the Contracting Officer within 3 calendar days of the meeting.

For projects with planned excavation activities, exceeding, or spanning, more than one year, re-initial archaeological coordination meetings must be held on an annual basis to ensure continued alignment on archaeological protocols. In addition, a re-initial archaeological coordination meeting must be held should there be a change in key personnel (Archaeological Subcontractor; Archaeological Coordinator; QC Manager, Project Superintendent, Project Manager, etc.)

1.6 ARCHAEOLOGICAL MONITORING PLAN

The Contractor must submit and receive approval of an Archaeological Monitoring Plan prior to the start of construction. The monitoring plan must be informed by the discussions held at the Archaeological Coordination Meeting and must include:

a. Identification of the areas to be monitored. Provide supporting graphics.

b. A description of anticipated or potential resources within the project area, including anticipated locations. Provide supporting graphics.

c. A description of field methods appropriate to the project including but not limited to mapping/GPS data collection methods, photography, night-work protocols, dewatering protocols, working within Installation Restoration (IR) Sites, human remains discovery protocols, discovery of a National Register-eligible archaeological site, and procedures for encountering potential UXO.

d. Project-specific archeological resource discovery protocols (anticipated and unanticipated) as approved through consultation and as discussed at the Archaeological Coordination Meeting. Include protocols for chain of communication, work stoppage, assessment, documentation, and resumption of work.

e. Artifact collection/discard methods.

f. Reporting requirements.

g. Curation requirements and anticipated repository.

h. Archaeologist Qualifications, including experience related to urban or industrial archaeological settings. These qualifications must have been approved prior to submission of the Archaeological Monitoring Plan, as outlined in Paragraph QUALITY ASSURANCE.

i. Points of contact.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 INSPECTION AND MONITORING

a. The Archaeologist must observe excavation activities for archaeological resources and must be authorized to collect archaeological material as warranted for analysis. The Archaeologist must be present on the project site according to a schedule agreed upon by the Contracting Officer until the Contracting Officer agrees that the project construction activities could have no effects on significant archaeological resources.

b. The Archaeologist must advise all project Contractors and subcontractors to be on the alert for evidence of the presence of the expected resource(s), of how to identify the evidence of the expected resource(s), and of the appropriate protocol in the event of an archaeological resource discovery. The archaeologist must be present at preparatory and initial meetings relating to definable features of work that include excavation activities. GPS points, photographs, sketch maps, and measured drawings must be provided, as applicable, to document resources encountered during the course of excavations.

c. If an intact archaeological resource is encountered, all soil disturbing activities in the vicinity of the resource must cease. The Archaeologist/Archaeological Coordinator must be empowered to halt excavation in the area and must immediately notify the Contracting Officer of the encountered archaeological resource. The Archaeologist must, after making a reasonable effort to assess the identity, integrity, and significance of the encountered archaeological resource, present the findings of this assessment to the Contracting Officer. Following verbal communication of this information, the assessment must be formalized in an Archaeological Resource Report.

d. If suspected human remains (including but not limited to: bone, bone fragments, teeth, mummified flesh, burials, and creamations) are encountered, stop work in the surrounding area immediately and notify the Contracting Officer and PWD-ME ENV. The remains must be treated with respect and not be disturbed or photographed. In coordination with the Contracting Officer and PWD-ME ENV, provide the materials and labor necessary to secure the area to prevent damage and protect the discovery.

e. In the event that potential munitions of explosive concern are identified (i.e., cannonballs, mortars, bullets, etc.), the Archaeologist must immediately halt excavation in the area. The Project Superintendent or SSHO must immediately notify Security Dispatch, and then notify the Contracting Officer. The Contractor must vacate the area to a safe distance. The Contractor must not resume work in the area until directed by the Government.

f. The Contractor must be responsible for the safety of all work conducted and must provide a work area that is safe during archaeological investigations. This includes providing appropriate shoring, access, and egress routes, dewatering, and any needed equipment. No one will be expected to enter the excavation until it has been deemed safe. It will be the responsibility of the Contractor to ensure that the archaeological monitors have the ability to enter the excavation as needed.

g. The Contractor may be performing work at night. When working at night, the Contractor must provide adequate and ample light to allow the monitoring archaeologists the ability to successfully monitor the area for archaeological resources. Lighting provided must be sufficient to illustrate changes in soil type/color, stratigraphy, and to identify/document artifacts and features. If adequate lighting cannot be provided, or if the sensitivity of resources warrant, nighttime work cannot occur within those areas. Refer to Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME) and the plans for the locations where nighttime work is not permitted.

h. The Archaeologist must document the resource(s) encountered with photographs (including scale), mapping, measured drawings, collection of GPS points, written description of the resource; and collect artifacts, as appropriate for further analysis.

3.2 ARCHAEOLOGICAL MONITORING REPORTS

Record findings of the archaeological monitoring efforts in written reports at required intervals, as outlined herein, and with sufficient information. Reports must always include a combination of narratives and graphics. The Contractor is required to prepare and submit all data, reports, graphics, maps, charts, etc. in an electronic format consistent with the platform currently used by the PWD ME Cultural Resources GIS database. Details and requirements will be provided to the Contractor by the Contracting Officer upon award of this Contract. The report must include a recommendation for the future treatment of archaeological resources encountered during construction that were left in place.

3.2.1 Overall Report Requirements:

Upon the completion of ground disturbing work, the archaeological monitoring effort must be documented in a comprehensive monitoring report. The report must adhere to the industry standards and the Maine SHPO's Archaeological Survey Guidelines and the State Historic Preservation Officer's Standards for Archaeological Work in Maine. The report must summarize the monitoring work, resources encountered, and document disposition. For resources left in place at the conclusion of construction, the report must include recommendations for future treatment. The report must include graphics including photographs, measured drawings, and maps (created by GPS points and showing the location(s) of any resources documented, including those resources left in situ).

Contracting Officer review of all monitoring report submissions will be completed within 45 calendar days after the date of submission.

3.2.2 Report Iterations:

a. Archaeological Monitoring Report - Monthly Progress Reports: Provide a brief monthly progress report consisting of a table indicating the dates of archaeological monitoring activities, the names of the archaeological monitors on site each day, and a summary of resources encountered/not encountered. The dates must be keyed to an accompanying map showing the areas monitored.

b. Archaeological Monitoring Report - Draft: Provide one (1) hard copy and one (1) electronic copy of a draft report (99 percent complete) for review by the Government. Once the draft report has been reviewed by the Government, the Contractor must be notified of any recommended changes in a timely manner by the Contracting Officer. Production of the final draft report must be within thirty (30) business days of receipt of the Government comments.

c. Archaeological Monitoring Report - Final Draft: Provide two (2) hard copies and one (1) electronic copy of the final draft report for review by the Government and Maine SHPO. The final draft report must be completed in accordance with current State of Maine archaeological survey requirements. Draft GIS data must also be provided at this time.

d. Archaeological Monitoring Report - Final: Once the final draft report has been reviewed by the SHPO, the Contractor must be notified of any recommended changes in a timely manner by the Contracting Officer. Production of three (3) hard copies and three (3) electronic copies of the final reports must be within thirty (40) business days of receipt of the Government comments.

The Contractor must submit all informal documents regarding archaeological field site reconnaissance and archaeological site mapping (field notes, etc.) directly to the Contracting Officer. Upon request or after acceptance of the final report, all field notes, photographs, artifacts, and related data will be turned over to the Contracting Officer.

The Contractor must submit all documentation in editable electronic format (Microsoft Word, Excel, ESRI ArcINFO, etc.) and in PDF upon completion of work. Three (3) hard copies and three (3) electronic copies of the report must also be submitted upon completion and acceptance of work. The electronic copies must be submitted on DVDs or CDs.

3.2.3 ARCHAEOLOGICAL RESOURCE REPORT

If a potentially significant resource is encountered, summarize the results of the archaeological investigations in an Archaeological Resource Report. The Archaeologist, in coordination with the Contractor, must produce the Resource Report to include a map and written summary of the discovery to support PWD ME coordination with SHPO. The Report must include, at a minimum, a physical description of the resource and its context; the depth at which it was encountered; potential resource identification; a description of the proposed impacts to the resource; and recommended follow-on steps, including options to avoid or minimize impacts on the potential site. The Resource Report must be submitted to the Contracting Officer no later than 2 business days after the potentially significant resource is encountered.

3.3 ARTIFACT CURATION

Artifacts recovered during the monitoring effort must be cleaned, recorded, and curated in accordance with 36 CFR 79. The Archaeologist must identify a repository for permanent curation and pay initial curation fees. If no suitable repository is identified, curated artifacts must be provided to the Government with the Archaeological Monitoring Report - Final submission.

-- End of Section --

SECTION 01 45 00.00 22
^^
QUALITY CONTROL (PWD ME)
08/23, CHG 1: 05/24

PART 1 GENERAL

This Section applies to only Design-Bid-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 52.2	(2017) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

ASTM INTERNATIONAL (ASTM)

ASTM C1077	(2017) Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation
ASTM D3666	(2016) Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials
ASTM D3740	(2019) Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
ASTM D6245	(2012) Using Indoor Carbon Dioxide Concentrations to Evaluate Indoor Air Quality and Ventilation
ASTM D6345	(2010) Standard Guide for Selection of Methods for Active, Integrative Sampling of Volatile Organic Compounds in Air
ASTM E329	(2021) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection
ASTM E543	(2021) Standard Specification for Agencies Performing Non-Destructive Testing

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

U.S. GREEN BUILDING COUNCIL (USGBC)

LEED GBDC	(2009) LEED Reference Guide for Green Building Design and Construction

1.2 PAYMENT

Separate payment will not be made for providing and maintaining an effective Quality Control program. Include all associated costs in the applicable Schedule item.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

Contractor Quality Control (CQC) Plan; G

SD-06 Test Reports

Verification Statement

SD-07 Certificates

Certificate Of Readiness

1.4 GENERAL REQUIREMENTS

Establish and maintain an effective quality control (QC) system that complies with FAR 52.246-12 Inspection of Construction. QC is comprised of plans, procedures, and organization necessary to produce an end product that complies with the Contract requirements. The QC system covers all construction operations, both onsite and offsite, and must be keyed to the proposed construction sequence. The Quality Control Manager, Superintendent, Project Manager, Site Safety and Health Officer (SSHO), and all on-site supervisors are responsible for the quality of work and are subject to removal by the Contracting Officer for non-compliance with the quality requirements specified in the Contract. The Quality Control Manager must maintain a physical presence at the work site at all times and is the primary individual responsible for all quality control.

1.5 QUALITY CONTROL (QC) PROGRAM REQUIREMENTS

Establish and maintain a QC program as described in this Section. This QC program is a key element in meeting the objectives of NAVFAC Commissioning. The QC program consists of a QC Organization, QC Plan, QC Plan Meeting(s), a Coordination and Mutual Understanding Meeting, QC meetings, three phases of control, submittal review and approval, historic coordination drawings review and approval, testing, completion inspections, QC certifications, independent Special Inspections in accordance with Section 01 45 35.00 22 SPECIAL INSPECTIONS (PWD ME), and documentation necessary to provide materials, equipment, workmanship, fabrication, construction and operations which comply with the requirements of this Contract. The QC program must cover on-site and off-site work and be keyed to the work sequence. No construction work or testing may be performed unless the Contractor has approved site supervision representative(s) (QC Manager, SSHO, and Superintendent) within visual control of any work being performed. Contractor may be required to provide multiple site supervision personnel to meet this requirement.

The QC Manager must report to an officer of the firm and not be subordinate to the Project Superintendent, or the Project Manager. The QC Manager must be a direct employee of the Prime Contractor, not an employee of a subcontractor. The QC Manager, Project Superintendent, and Project Manager must work together effectively. Although the QC Manager is the primary individual responsible for quality control, all individuals will be held responsible for the quality of work on the job.

1.5.1 Meetings

1.5.1.1 Quality Control Plan Meeting

Prior to submission of the QC Plan, the Contractor must schedule Division 01 Specification Review Meeting with the Contracting Officer to discuss the QC Plan requirements of this Contract.

This meeting must discuss and develop a mutual understanding of the QC Plan requirements prior to plan development and submission and to agree on the Contractor's list of Definable Feature of Work (DFOW).

1.5.1.2 Coordination and Mutual Understanding Meeting

Before start of construction, prior to acceptance by the Government of the CQC Plan, and within 21 calendar days of Contract Award, meet with the Contracting Officer and discuss the Contractor's quality control system. During the meeting, a mutual understanding of the system details must be developed, including the forms for recording the CQC operations, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's management and control with the Government's Quality Assurance. Also review the Special Inspection requirements outlined in Section 01 45 35.00 22 SPECIAL INSPECTIONS (PWD ME). Minutes of the meeting will be prepared by the QC Manager and signed by the Contractor, and the Government. Provide a copy of the signed minutes to all attendees. At a minimum the Coordination and Mutual Understanding Meeting must be repeated when a new QC Manager is appointed. There can be other occasions when subsequent conferences will be called by either party to reconfirm mutual understandings or address deficiencies in the CQC system or procedures which can require corrective action by the Contractor.

1.5.1.2.1 Purpose

The purpose of this meeting is to develop a mutual understanding of the QC details, including documentation, administration for on-site and off-site work, design intent, Cx in accordance with Section 01 91 00.15 20 TOTAL BUILDING COMMISSIONING, environmental requirements and procedures, coordination of activities to be performed, Special Inspections, and the coordination of the Contractor's management, production, and QC personnel. At the meeting, the Contractor must explain in detail how three phases of control will be implemented for each DFOW, as well as how each DFOW will be affected by each management plan or requirement as listed below:

a. Waste Management Plan.

b. Procedures for noise and acoustics management.

c. Environmental Management Plan.

d. Environmental regulatory requirements.

e. Cx Plan requirements in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

f. Special Inspections.

g. Indoor Air Quality (IAQ) Management Plan

1.5.1.2.2 Coordination of Activities

Coordinate activities included in various sections to assure efficient and orderly installation of each component. Coordinate operations included under different sections that are dependent on each other for proper installation and operation. Schedule construction operations with consideration for indoor air quality as specified in the IAQ Management Plan. Coordinate prefunctional tests and startup testing with CxC. Coordinate special inspections.

1.5.1.2.3 Attendees

As a minimum, the Contractor's personnel required to attend include an officer of the firm, the Project Manager, Project Superintendent, QC Manager, Alternate QC Manager, Assistant QC Manager, QC Specialists, Special Inspector, Commissioning Provider (CxC), Environmental Manager, and subcontractor representatives as approved by the Contracting Officer. Each subcontractor who will be assigned QC responsibilities must have a principal of the firm at the meeting. Minutes of the meeting will be prepared by the QC Manager and signed by the Contracting Officer. Provide a copy of the signed minutes to all attendees and include a copy in the QC Plan.

1.5.1.3 Quality Control (QC Meetings)

After the start of construction, conduct weekly QC meetings led by the QC Manager at the work site with the Project Superintendent, Project Manager, Assistant QC Manager, the QC Specialists, the Special Inspector, CxC, and the foremen who are performing the work of the DFOWs. The QC Manager must prepare an agenda and provide a copy to the Contracting Officer no later than two (2) working days before the meeting. The QC Manager must prepare the minutes of the meeting and provide a copy to the Contracting Officer within two (2) calendar days after the meeting. The Contracting Officer may attend these meetings. As a minimum, accomplish the following at each meeting:

a. Review the minutes of the previous meeting.

b. Review the schedule and the status of work and deficiencies/rework. Review the most current approved schedule (in accordance with schedule specification) and the status of work and deficiencies/rework.

c. Review the status of submittals and Request for Information (RFIs).

d. Review the work to be accomplished in the next 3 weeks as defined by the schedule section paragraph WEEKLY LOOK AHEAD in Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS) and all documentation required for that work.

e. Review Testing Plan and Log including status of tests performed since last QC Meeting.

f. Resolve QC and production problems. Discuss status of pending change orders.

g. Address items that may require revising the QC Plan.

h. Review Accident Prevention Plan (APP) and effectiveness of the safety program.

i. Review environmental requirements and procedures.

j. Review Environmental Management Plan.

k. Review Waste Management Plan.

l. Review the status of training completion.

m. Review Cx Plan and progress. Review Issues Log and resolution.

n. Review IAQ Management Plan

o. Review and confirm status of accounting of unit price items specified in PRICE SCHEDULE document.

p. Review Non-Compliance Notices and any actions required to address non-complaint work/actions

q. Review Special Inspections (Section 01 45 35.00 22 SPECIAL INSPECTIONS) (PWD ME)and any deficiencies identified by the Special Inspectors. Review actions taken or planned actions to correct any deficiencies noted.

1.5.2 Contractor Quality Control (CQC) Plan

Submit no later than 45 calendar days after Contract Award, the CQC Plan proposed to implement the requirements FAR 52.246-12 Inspection of Construction. Construction will be permitted to begin only after acceptance of the CQC Plan and other Contract requirements.

1.5.2.1 Content of Contractor Quality Control (CQC) Plan

Provide a CQC Plan, prior to start of construction that includes a table of contents, with major sections identified, pages numbered sequentially, and that documents the proposed methods and responsibilities for accomplishing quality control during the construction of the project. The CQC Plan must at a minimum include the following sections:

a. A description of the quality control organization and acknowledgment that the CQC staff will implement the three phase control system for all aspects of the work specified.

b. An organizational chart showing the quality control organization with individual names and job titles and lines of authority up to an executive of the company at the home office.

c. NAMES AND QUALIFICATIONS: Names and qualifications, in resume format, (including position titles and durations for qualifying experiences) for each person in the QC organization. Include the Construction Quality Management (CQM) for Contractors course certifications for the QC personnel as required by the paragraphs entitled CONSTRUCTION QUALITY MANAGEMENT TRAINING.

d. DUTIES, RESPONSIBILITY AND AUTHORITY OF QC PERSONNEL: Duties, responsibilities, and authorities of each person in the QC organization.

e. OUTSIDE ORGANIZATIONS: A listing of outside organizations, such as architectural and consulting engineering firms, that will be employed by the Contractor and a description of the services these firms will provide. Example: The fire protection engineer who designs the sprinkler system.

f. APPOINTMENT LETTERS: Letters signed by an officer of the firm appointing the QC Manager and Alternate QC Manager, CxC, and stating that they are responsible for implementing and managing the QC program as described in this Contract. Include in this letter the responsibility of the QC Manager and Alternate QC Manager to implement and manage the three phases of control, and their authority to stop work which is not in compliance with the Contract. Letters of direction are to be issued by the QC Manager to all other QC Specialists outlining their duties, authorities, and responsibilities. Include copies of the letters in the QC Plan.

g. SUBMITTAL PROCEDURES AND INITIAL SUBMITTAL REGISTER: Procedures for reviewing, approving, scheduling, and managing submittals, including those of subcontractors, offsite fabricators, suppliers, and purchasing agents. Provide the name(s) of the person(s) in the QC organization authorized to review and certify submittals prior to approval. Provide the initial submittal of the Submittal Register as specified in Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME).

h. TESTING LABORATORY INFORMATION: Testing laboratory information required by the paragraph ACCREDITATION REQUIREMENTS, as applicable.

i. TESTING PLAN AND LOG: A Testing Plan and Log that includes the tests required, associated feature of work required, referenced by the specification paragraph number requiring the test, the frequency, and the person responsible for each test. Use Government forms to log and track tests.

j. Procedures to complete construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected. This phase is performed prior to beginning work on each definable feature of work, after all required plans, documents, materials are approved, and after copies are at the work site.

k. Reporting procedures, included proposed reporting formats.

l. Procedures for submitting and reviewing design changes/variations prior to submission to the Contracting Officer.

m. LIST OF DEFINABLE FEATURES: A Definable Feature of Work (DFOW) is a task that is separate and distinct from other tasks and has control requirements and work crews unique to that task. A DFOW is identified by different trades or disciplines, or it is work by the same trade in a different environment. A DFOW is by definition any item or activity on the construction schedule, and the schedule specification provides direction regarding how the DFOWs are to be structured. Include in the list of DFOWs for all activities on the Construction Schedule. Although each section of the specifications can generally be considered a definable feature of work, there are frequently more than one definable features under a particular section. Identify the specification section number and schedule activity ID for each DFOW listed. The DFOW list will be reviewed in coordination with the construction scheduled and agreed upon during the Coordination of Mutual Understanding Meeting.

n. PROCEDURES FOR PERFORMING AND TRACKING THE THREE PHASES OF CONTROL: Identify procedures used to ensure the three phases of control to manage the quality on this project. For each Definable Feature of Work (DFOW), a Preparatory and Initial phase checklist must be filled out during the Preparatory and Initial phase meetings. Conduct the Preparatory and Initial Phases and meetings with a view towards obtaining quality construction by planning ahead and identifying potential problems for each DFOW.

o. Coordinate scheduled work with Special Inspections required by Section 01 45 35.00 22 SPECIAL INSPECTIONS (PWD ME), the Statement of Special Inspections, and the Special Inspections Project Manual. Where the applicable code issued by the International Code Council (ICC) calls for inspection by the Building Official, the Contractor must include the inspections in the Quality Control Plan and must perform the inspections required by the applicable ICC. The Contractor must perform these inspections using independent qualified inspectors. Include the Special Inspection Project Manual requirements in the QC Plan.

p. PROCEDURES FOR COMPLETION INSPECTION: Procedures for identifying and documenting the completion inspection process. Include in these procedures the responsible party for punch out inspection, pre-final inspection, and final acceptance inspection.

q. TRAINING PROCEDURES AND TRAINING LOG: Procedures for coordinating and documenting the training of personnel required by the Contract.

r. ORGANIZATION AND PERSONNEL CERTIFICATION LOG: Procedures for coordinating, tracking and documenting all certifications required for entities such as subcontractors, testing laboratories, suppliers, and personnel. The QC Manager will ensure that certifications are current, appropriate for the work being performed, and will not lapse during any period of the Contract that the work is being performed.

s. DAILY CONTRACTOR QUALITY CONTROL REPORT FORM: Template that includes fields for the following:

    1) Date.
    2) Sequential report number.
    3) Weather and temperature.

4) Number of personnel on site by trade or by Subcontract.
5) Kind and number of major equipment on site.
6) Tests performed and their results, if known.
7) Materials and equipment delivered to the site and their conditions.
8) Names, affiliations, and positions of visitors to the site with brief explanations of the reasons for visits.
9) Brief description of each work activity, noting items that were completed that day.
10) Items of work that need attention at a later date, and why.
11) Any accidents and injuries.
12) Items of concern with respect to maintenance of quality.
13) Accounting update of unit price items specified in the Price Schedule utilizing the numbering convention indicated in the Unit Prices Form (e.g., 0001g).
14) Any other items of significance.

1.5.3 Acceptance of the Quality Control (QC) Plan

The Contracting Officer's acceptance of the Contractor QC Plan is required prior to the start of construction. The Contracting Officer reserves the right to require changes in the QC Plan and operations as necessary, including removal or addition of personnel, to ensure the specified quality of work. The Contracting Officer reserves the right to interview any member of the QC organization at any time in order to verify the submitted qualifications. All QC organization personnel are subject to acceptance by the Contracting Officer. The Contracting Officer may require the removal of any individual for non-compliance with quality requirements specified in the Contract. The removal and replacement of QC organization personnel will not be cause for claim of additional compensation or extensions of the Contract Completion Date (CCD).

1.5.4 Preliminary Construction Work Authorized Prior to Acceptance

The only construction work that is authorized to proceed prior to the acceptance of the QC Plan is mobilization of storage and office trailers, temporary utilities, and surveying with specific prior approval of the Contracting Officer.

1.5.5 Notification of Changes

Notify the Contracting Officer, in writing, of any proposed changes in the QC Plan or changes to the QC organization personnel, in accordance with the submittal requirements in Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME). Proposed changes are subject to acceptance by the Contracting Officer.

1.5.6 Special Inspections

Perform all required Special Inspections per Section 01 45 35.00 22 SPECIAL INSPECTIONS (PWD ME), the statement of Special Inspections and the Schedule of Special Inspections.

1.6 QUAILTY CONTROL (QC) ORGANIZATION

1.6.1 Quality Control (QC) Manager

1.6.1.1 Duties

Provide a QC Manager to implement and manage the QC program. The

Contractor shall have approved site supervision representative(s) (QC Manager, SSHO, and Superintendent) within visual control of any work being performed. Contractor may be required to provide multiple site supervision personnel to meet this requirement. The QC Manager must be a direct employee of the Prime Contractor, not an employee of a subcontractor. The QC Manager must not perform the duties of Project Superintendent, nor the duties of Project Manager, or SSHO. The only duties and responsibilities of the QC Manager are to manage and implement the QC program on this Contract. The QC Manager must attend the partnering meetings, QC Plan Meetings, Coordination and Mutual Understanding Meeting, conduct the QC meetings, perform the three phases of control except for those phases of control designated to be performed by QC Specialists, perform submittal review and approval, ensure testing is performed, and provide QC certifications and documentation required in this Contract. The QC Manager is responsible for managing and coordinating the three phases of control and documentation performed by the QC Specialists, testing laboratory personnel, and any other inspection and testing personnel required by this Contract. The QC Manager is the manager of all QC activities and must not be the Special Inspector. The QC manager is responsible for notifying the Special Inspector of activities which require their review. The QC manager is responsible for coordinating the Special Inspection activities, see paragraph CONTRACTOR'S QUALITY CONTROL (QC) MANAGER, in Section 01 45 35.00 22 SPECIAL INSPECTIONS (PWD ME). The QC manager is responsible for the quality control for Controlled Area perimeter construction.

1.6.1.2 Qualifications

The QC Manager must be an individual with a minimum of 15 years combined experience in the following positions: Project Superintendent, QC Manager, Project Manager, Project Engineer or Construction Manager on similar size and type construction contracts which included the major trades that are part of this Contract. The individual must have at least 4 years experience as a QC Manager. The individual must be familiar with the requirements of EM 385-1-1, and have experience in the areas of hazard identification, safety compliance, and sustainability.

The QC Manager and all members of the QC organization must be capable of reading, writing, and conversing fluently in the English language.

1.6.1.3 Construction Quality Management Training

In addition to the above experience and education requirements, the QC Manager and all members of the QC team must have completed the CQM for Contractors course. If the QC Manager does not have a current certification, obtain the CQM for Contractors course certification within 90 days of award. This course is periodically offered by the Naval Facilities Engineering Systems Command and the Army Corps of Engineers. Contact the Contracting Officer for information on the next scheduled class.

The Construction Quality Management Training certificate expires after 5 years. If the QC Manager's certificate has expired, retake the course to remain current.

The QC Managers for the work sites listed above are each responsible for implementing and managing the QC Program at the designated work site. They must perform the three phases of control, perform submittal review, ensure testing is performed, and prepare QC certifications and documentation required by this Contract. The qualification requirements for the QC Managers at each site are listed in the paragraph QUALIFICATIONS.

1.6.2 Organizational Changes

Maintain the QC staff with personnel as required by the specification section at all times. When it is necessary to make changes to the QC staff, revise the CQC Plan to reflect the changes and submit the changes to the Contracting Officer for acceptance.

1.6.3 Alternate Quality Control (QC) Manager Duties and Qualifications

Designate an alternate for the QC Manager at the work site to serve in the event of the designated QC Manager's absence. The period of absence may not exceed 2 weeks at one time, and not more than 30 workdays during a calendar year. The qualification requirements for the Alternate QC Manager must be the same as for the QC Manager.

1.6.4 Assistant Quality Control (QC) Manager Duties and Qualifications

Provide a full-time assistant to the QC Manager at the work site to perform the three phases of control, perform submittal review, ensure testing is performed, and prepare QC certifications and documentation required by this Contract. The Assistant QC Manager must be on the work site during supplemental work shifts beyond the regular shift and perform the duties of the QC Manager during such supplemental shift work. The qualification requirements for the Assistant QC Manager must be an individual with a minimum of 5 years combined experience in the following positions: Project Superintendent, QC Manager, Project Manager, Project Engineer, or Construction Manager on similar size and type construction contracts which include the major trades that are part of this Contract. The individual must have at least 2 years experience as a QC Manager. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance.

The Assistant QC Manager must be capable of reading, writing, and conversing fluently in the English language.

1.6.5 Commissioning

Refer to Section 01 91 00.15 BUILDING COMMISSIONING for project commissioning requirements.

Commissioning (Cx) is a systematic, quality-focused process for delivery of a project focusing on verifying and documenting all commissioned systems and assemblies are installed, tested, and operating as they were planned and designed to meet the project requirements. The Quality Control requirements outlined in this specification section are key in supporting the objectives of the Cx process, specifically coordinating testing, documenting, and verifying proper system operation. Properly executed the Quality Control support of Cx ensures timely execution of necessary tasks to deliver the fully commissioned and operating systems in coordination with the overall construction and project schedule.

Provide Cx in addition to the quality control requirements of this section and not as a substitute for quality control requirements. The QC Manager is responsible for carrying out the three phases of control while ensuring the functional performance and integrated systems tests are coordinated with the Cx provider as required for each system to be commissioned.

1.6.5.1 Certificate Of Readiness

The QC Manager must issue a Certificate of Readiness for Government approval for each system to be commissioned. Schedule Functional Performance Tests for each system only after the Certificate of Readiness has been approved by the Government for the system. The Certificate of Readiness certifies that all required inspections have been completed and deficiencies that were identified through any prior review, inspection, or test activity have been corrected before the start of Functional Performance Tests. Refer to Cx requirements in Section 01 91 00.15 20 TOTAL BUILDING COMMISSIONING for a list of systems to be commissioned and detailed requirements for the Cx provider.

1.6.6 Quality Control (QC) Specialists

Provide a separate QC Specialist at the work site (work sites associated with construction activities for the project including off-site fabrication shops) for each of the areas as listed in the Matrix listed below, who must assist and report to the QC Manager and who must have no duties other than their assigned quality control duties. These individuals or specialized technical companies must be a first tier Subcontractor. QC Specialists must be physically present at the work site with frequency as indicated in the Experience Matrix below, to participate in the QC meetings, perform the three phases of control, including participation in Preparatory and Initial Phase meetings, and to perform and document Follow-up inspections as an extension of the QC Manager for each definable feature of work in their area of responsibility. QC Specialists must assist and be present for training events, and Critical System Acceptance inspections by the Government. Qualification, experience, Area of Responsibility, and frequency of QC surveillance are provided in the Matrix listed herein.

Experience Matrix		
1. Area	**2-1. Qualification** **2-2. Experience**	**3-1. Area of Responsibility** **3-2. Frequency**
Testing, Adjusting and Balancing (TAB) Personnel	2.1 TAB Team Field Leader must be a member of AABC or an experienced technician of the firm certified by the NEBB. 2-2. 3 years experience immediately preceding this Contract	3-1. Area of Responsibility: Testing, Adjusting, and Balancing of HVAC Systems. 3-2. Frequency of QC surveillance and inspection will be weekly during installation and including final inspection. Frequency of QC surveillance and inspection of off-site fabrication will be weekly during fabrication activities.
Structural	2-1. Graduate Civil Engineer (with Structural Track or Focus) or Construction Manager with 2-2. 2 years experience observing structural features of work in the field.	3-1. Area of Responsibility: Scope depicted on Structural Drawings 3-2. Frequency of QC surveillance and inspection will be periodic during installation and including final inspection.
Fire Protection QC Specialist (QFPE)	Note: See paragraph FIRE PROTECTION QC SPECIALIST (QFPE)	Note: See paragraph FIRE PROTECTION QC SPECIALIST (QFPE)

Experience Matrix		
1. Area	**2-1. Qualification** **2-2. Experience**	**3-1. Area of Responsibility** **3-2. Frequency**
Mechanical	2-1. Mechanical Engineer with a professional engineer registration, and 2-2. 10 years related experience or 10 years experience in mechanical systems design.	3-1. Area of Responsibility: Installation of mechanical systems and off-site duct work fabrication. 3-2. Frequency of QC surveillance and inspection will be bi-weekly during installation and including final inspection. Frequency of QC surveillance and inspection of off-site fabrication will be weekly during fabrication activies.
Architectural	2-1. Project Architect in any state, and 2-2. 5 years experience	3-1. Area of Responsibility: Architectural. 3-2. Frequency of QC surveillance and inspection will be bi-weekly during installation and including final inspection.
Concrete, Pavements and Soils	2-1. Materials Technician 2-2. 2 years experience for the appropriate area.	3-1. Area of Responsibility: Including, but not limited to, reinforcing inspection, concrete testing, soil compaction testing, and pavement testing. 3-2. Frequency of QC surveillance and inspection will be daily while work is being performed and including final inspection.

Experience Matrix		
1. Area	**2-1. Qualification** **2-2. Experience**	**3-1. Area of Responsibility** **3-2. Frequency**
Coating Inspector	2-1. Coating Inspector must be employed and qualified to SSPC QP 5, Level III, by the selected coating inspection company.	3-1. Area of Responsibility: Interior and Exterior Coatings of Welded Steel Water Tanks. 3-2. Frequency of QC surveillance and inspection will be as specified in Section 09 97 13.16 INTERIOR COATING OF WELDED STEEL WATER TANKS and Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES
Protective Coatings Specialist (PCS)	Note: See paragraph QUALIFICATIONS OF CERTIFIED PROTECTIVE COATINGS SPECIALIST (PCS) in Section 09 97 13.16 INTERIOR COATING OF WELDED STEEL WATER TANKS and Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES.	3-1. Area of Responsibility: Interior and Exterior Coatings of Welded Steel Water Tanks. 3-2. Frequency of QC surveillance and inspection will be as specified in Section 09 97 13.16 INTERIOR COATING OF WELDED STEEL WATER TANKS and Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES
Geotechnical	2-1. Graduate Geotechnical Engineer with a professional engineer registration in the state of Maine, with 2-2. Specialized experience in geotechnical engineering.	Note: See Section 31 00 00.00 22 EARTHWORK for areas of responsibility and frequency of QC surveillance and inspection requirements.

Experience Matrix		
1. Area	**2-1. Qualification** **2-2. Experience**	**3-1. Area of Responsibility** **3-2. Frequency**
Elevator	2-1. Elevator Manufacturer's Technical Representative. 2-2. 5 years minimum with elevator system used.	3-1 Area of responsibility: Preparation, installation, and testing of elevator systems. 3-2 Frequency of QC surveillance and inspection will be minimum of three times a week during installation and full time during testing.
Environmental	2-1. Environmental Engineer, with 2-2. 5 years experience in the testing, handling, and disposal of contaminated soil and groundwater.	3-1. Area of Responsiblity: Oversight of the excavation, handling, characterization, transportation, and disposal of contaminated soil and groundwater. Ensure compliance with PNSY Soil Management SOPs and applicable state and federal requirements for the handling of contaminated materials. 3-2. Frequency of QC surveilance and inspection will be: -weekly during excavation in non-IR areas of the project; -full-time during excavation in IR areas of the project; and -full-time during the collection of soil and groundwater samples.

Experience Matrix		
1. Area	**2-1. Qualification** **2-2. Experience**	**3-1. Area of Responsibility** **3-2. Frequency**
Electrical	2-2. 5 years of experience supervising electrical features of work in the field with a construction company.	3-1. Area of Responsibility: Medium voltage electrical equipment and supporting low voltage power and communication systems. 3-2. Frequency of QC surveillance and inspection will be weekly during installation and including final inspection.
Roofing Manufacturer's Technical Representative	2-1. Preparation, installation and testing of roofing systems, accessories, and roof mounted items. 2-2. 5 years experience minimum	3-1. Area of Responsibility: 3-2. Frequency of QC surveillance and inspection will be monthly during installation and including final inspection.
Building Envelope QC Specialist	2-1. Roofing Manufacturer's Technical Representative 2-2. 5 years minimum with roofing system used.	3-1. Area of Responsiblity: Installation and testing of roofing and exterior below grade water proofing. 3-2. Frequency of QC surveillance and inspection will be once a week during installation, once a week during flashing installation, and full-time during roof testing.

1.6.7 Fire Protection QC Specialist

Provide a Fire Protection Quality Control Specialist (QFPE) within the QC organization to perform quality control related activities as specified herein on fire protection and life safety systems installed under this Contract. Refer also to QC requirements in 21 13 13 WET PIPE SPRINKLER

SYSTEMS, FIRE PROTECTION and 28 31 76 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, ADDRESSABLE.

1.6.7.1 Qualifications

The QFPE must have the following qualifications:

a. Be a registered Professional Engineer (P.E.) licensed by a Licensing Board in the United States, the District of Columbia, Guam or Puerto Rico, having passed the National Council of Examiners for Engineering and Surveying (NCEES) written examination specifically in the discipline of Fire Protection Engineering.

b. Have a minimum of 5 years of Fire Protection Engineering experience on projects of similar relevance and complexity to the fire protection work specified under this Contract.

c. Other than the contractual obligations with the Prime Contractor, the QFPE must have no other business relationship (i.e., employee, owner, partner, operating officer, distributor, salesman, technical representative, family relationship, or financial investment) with the Prime Contractor or subcontractors.

d. Be employed by an independent engineering firm or company. The firm may identify multiple, to a maximum of five, licensed Fire Protection Engineers for the performance of the duties under the Contractor but must submit the names and qualifications for Government approval for all individuals identified prior to them performing any work under this Contract.These individuals may not be substituted without prior approval from the Contracting Officer.

1.6.7.2 Responsibilities

QFPE duties and responsibilities:

a. Assist in the development of the QC Plan including the Testing Plan and Log and executing the three phases of control for work involving the installation and testing of fire protection and life safety systems as an extension of the QC Manager.

b. Participate in project QC Meetings. Participate in Preparatory and Initial Phase meetings and perform and Follow-up inspections for work involving the installation and testing of fire protection and life safety systems.

c. Review and certify that all submittals pertaining to fire and life safety systems are complete and accurate prior to submission to the Government for approval. The Government FPE retains the role as the Authority Having Jurisdiction (AHJ). The QFPE is responsible for the delegated design of the sprinkler system, including fire pumps, and the fire alarm and mass notification system. The QFPE specialist is responsible for ensuring fire protection and life safety submittals are complete and accurate and all corrections have been made prior to submission to the Government. The Government reserves the right to reject any submittal that has not been reviewed first by the QFPE and so marked, in writing, attesting to such review and completeness of the submittal.

d. The Government reserves the right to reject any submittal or construction that is not in compliance to Contract. Government reviews do not relieve the Contractor responsibility for providing adequate quality control measures and do not constitute or imply acceptance of Contract variation.

e. Perform construction surveillance in accordance with the Schedule of Fire Protection System Inspections. Construction surveillance includes but is not limited to performing periodic on-site inspections during construction at specified milestones, performing a pre-final inspection of installed systems and witnessing functional testing; and participating and documenting in an on-site final acceptance inspection of fire protection and life safety systems with the Government FPE. :

f. Document inspection results on a QFPE report prepared each day inspections are performed. The report must include a description of the visual inspection or observation performed, a written summary of findings, a conclusion on compliance with the Contract documents, and signature of the QFPE Specialist. In person inspection must be documented via photo (.jpeg). Photographic documentation must include before and after conditions and physical measurements. Forward the QFPE daily report to the QC Manager who must include the report with the submission of their daily QC Report to the Government each day. Every site visit by the QFPE must be documented on a QFPE daily report.

1.6.7.3 Schedule of Fire Protection System Inspections

A schedule, prepared by the Fire Protection DOR, which lists each of the required visual inspetions and observations required by the QFPE. The schedule is included at the end of this UFGS section.

1.6.8 Special Inspector

The Special Inspector (SI) must be an independent third party hired directly by the Prime Contractor. The SI and SIOR must not be a company employee of the Contractor or any subcontractor performing the work to be inspected. The qualifications of the SI and SIOR are defined in Section 01 45 35.00 22 SPECIAL INSPECTIONS (PWD ME).

1.7 SUBMITTAL AND DELIVERABLES REVIEW AND APPROVAL

Procedures for submission, review and approval of submittals are described in Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME). Procedures must include field verification of relevant dimensions and component characteristics by the QC organization prior to submittal being sent to the Contracting Officer. The CQC organization is responsible for certifying that all submittals and deliverables are in compliance with the Contract. When Section 01 91 00.15 BUILDING COMMISSIONING are included in the Contract, the submittals required by those sections have to be coordinated with Section 01 33 00 SUBMITTAL PROCEDURES to ensure adequate time is allowed for each type of submittal required.

1.8 REQUESTS FOR INFORMATION (RFI)

If clarification is required regarding work scope or if unforseen conditions/ discrepancies arise, the Contractor must submit Requests for Information (RFI) to the Contracting Officer for review and response. Submit RFI's using Government provided request form. Contractor must provide a proposed course of action as well as indicate if there is cost/time impact associated with the potential changes. Contractor Officer review will be completed within 21 days after date of submission, except as specified otherwise.

1.9 THREE PHASES OF CONTROL

CQC enables the Contractor to ensure that the construction, including that of subcontractors and suppliers, complies with the requirements of the Contract. At least three phases of control must be conducted by the QC Manager to adequately cover both on-site and off-site work for each definable feature of the construction work as follows:

1.9.1 Preparatory Phase

Document the results of the preparatory phase actions by separate minutes prepared by the QC Manager and attached to the daily CQC report. Instruct to applicable workers as to the acceptable level of workmanship required to meet Contract specifications.

Notify the Contracting Officer at least five (5) working days in advance of each preparatory phase.

The meeting must be conducted by the QC Manager and attended by the QC Specialists, the Project Superintendent, the SSHO, the CxC, the Special Inspector, and the foreman responsible for the DFOW or as approved by the Contracting Officer. When the DFOW will be accomplished by a subcontractor, that subcontractor's foreman must attend the preparatory phase meeting. Document the results of the preparatory phase actions in the daily Contractor Quality Control Report and in the Preparatory Phase Checklist (Use Attachment A for the PWD ME Preparatory Phase Checklist). This phase is performed prior to beginning work on each DFOW. (Note: Preparatory Meeting shall only be held if the shop drawings or submittals have been approved, hard copies are printed for the meeting, the APP and appropriate AHA related to the DFOW have been submitted and the appropriate personnel as stated above are present for the meeting). Perform the following prior to beginning work on each DFOW:

a. Review each paragraph of the applicable specification sections, reference codes, and standards. Make available during the preparatory inspection a copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field. Maintain and make available in the field for use by Government personnel until final acceptance of the work.

b. Review the Contract drawings.

c. Verify that field measurements are as indicated on construction or shop drawings or both before confirming product orders, to minimize waste due to excessive materials.

d. Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved. Verify receipt of approved factory test results, when required.

e. Review the testing plan and ensure that provisions have been made to provide the required QC testing.

f. Examine the work area to ensure that the required preliminary work has been completed and complies with the Contract and ensure any deficiencies/rework items in the preliminary work have been corrected and confirmed by the Contracting Officer

g. Review coordination of product/material delivery to designated prepared areas to execute the work.

h. Examine the required materials, equipment, and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data and are properly stored.

i. Check to assure that all materials and equipment have been tested, submitted, and approved.

j. Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.

k. Discuss specific controls to be used, construction methods, construction tolerances, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential problems for each DFOW. Ensure any portion of the plan requiring separate Contracting Officer acceptance has been approved.

l. Review the Cx requirements in accordance with Section 01 91 00.15 BUILDING COMMISSIONING and ensure all preliminary work items have been completed and documented.

m. Review Special Inspections required by Section 01 45 35.00 22 SPECIAL INSPECTIONS (PWD ME), the Statement of Special Inspections and the Schedule of Special Inspections.

n. If work includes demolition work, review demolition work plan to ensure compliance with EM 385-1-1.

o. If work includes excavation, at a minimum the Dig Safe Coordinator / CM / ET shall be invited to the prep/initial phase meetings and will review Utility Locating Requirements. (Note: Special attention is required if Hazardous or Mission Critical Utilities are within the excavation area.) If Hazardous and Mission Critical Utilities (as identified in the Scope of Work & Drawings) are located within the excavation area, the PWD ME Dig Safe Coordinator (DSC), PWD ME CM/ET, PWD ME Design Manager (DM) and the assigned PWD ME Project Civil Engineer must be present at the meeting to review requirements to ensure safety and preservation of critical utilities is discussed with adequate focus. Contractor to provide Construction CM/ET notification no later then 7 working days prior to holding the preparatory and initial pre-excavation/demo safety review meeting. The CM/ET, as well as the PWD ME DSC, attend this meeting to ensure safety and preservation of critical utilities is discussed with adequate focus.

p. If work includes excavation, review Soil Management Requirements (Note: Special attention is required if the work is within an IR site or Hazardous Soils are within the excavation area. Include Code 106.3 Representatives at this meeting to discuss Soil Management processes and procedures.)

q. If the work is subject to any environmental (EV) permitting, Archaeological Monitoring, or SHPO approvals, review the terms and conditions of the applicable permits as well as any supplemental controls/ approvals that are required to complete the work. (Note: Include PWD ME EV Representatives at this meeting to review the project specific requirements).

r. If work includes the removal and disposal of Hazardous materials, review the appropriate work plans to ensure the work plans have been approved and any comments are understood by the work execution team. (Note: Include Code 106.3 Representatives at this meeting to discuss Hazardous Materials handling and disposal processes and procedures.)

1.9.2 Initial Phase

Notify the Contracting Officer at least 2 working days in advance of each initial phase. When construction crews are ready to start work on a DFOW, conduct the initial phase with the QC Specialists, the Project Superintendent, the Special Inspector, and the foreman responsible for that DFOW. Observe the initial segment of the DFOW to ensure that the work complies with Contract requirements. Document the results of the initial phase in the daily CQC Report and in the Initial Phase Checklist (See Attachment B for the Initial Phase Checklist). Repeat the initial phase for each new crew to work on-site, or when acceptable levels of specified quality are not being met. Indicate the exact location of initial phase for definable feature of work for future references and comparison with follow-up phases. Perform the following for each DFOW:

a. Check work to ensure that it is in full compliance with the Contract requirements. Review minutes of the preparatory meeting.

b. Verify adequacy of controls to ensure full Contract compliance. Verify required control inspection and testing comply with the Contract.

c. Establish level of workmanship and verify that it meets the minimum acceptable workmanship standards. Compare with required sample panels as appropriate.

d. Resolve any workmanship issues.

e. Resolve conflicts.

f. Ensure that testing is performed by the approved laboratory.

g. Check work procedures for compliance with the APP and the appropriate AHA to ensure that applicable safety requirements are met.

h. Review project specific work plans (i.e. Cx, HAZMAT Abatement, Stormwater Management) to ensure all preparatory work items have been completed and documented.

i. Coordinate scheduled work with special inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections and the Schedule of Special Inspections.

1.9.3 Follow-Up Phase

Perform the following for on-going DFOW daily, or more frequently as necessary, until the completion of each DFOW. The Final Follow-up for any DFOW will clearly note in the daily report the DFOW is completed, and all deficiencies/rework items have been completed in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Each DFOW that has completed the Initial Phase and has not completed the Final Follow-up must be included on each daily report. If no work was performed on that DFOW for the period of that daily report, it must be so noted. Document all Follow-Up activities for DFOWs in the daily CQC Report:

a. Ensure the work including control testing complies with Contract requirements until completion of that particular work feature. Record checks in the CQC documentation.

b. Maintain the quality of workmanship required.

c. Ensure that testing is performed by the approved laboratory.

d. Ensure that deficiencies/rework items are being corrected. Conduct final follow-up checks and correct all deficiencies prior to the start of additional features of work which may be affected by the deficient work.

e. Do not build upon nor conceal non-conforming work.

f. Assure manufacturers' representatives have performed necessary inspections if required and perform safety inspections.

g. Review the Cx requirements in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

h. Coordinate scheduled work with Special Inspections required by Section 01 45 35.00 22 SPECIAL INSPECTIONS (PWD ME), the Statement of Special Inspections and
the Schedule of Special Inspections.

1.9.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phases on the same DFOW if the quality of on-going work is unacceptable, if there are changes in the applicable QC organization, if there are changes in the on-site production supervision or work crew, if work on a DFOW has not started within 45 days of the initial preparatory meeting or has resumed after 45 days of of inactivity, or if other problems develop.

1.9.5 Notification of Three Phases of Control for Off-Site Work

Notify the Contracting Officer at least2 weeks prior to the start of the preparatory and initial phases.

1.9.6 Deficiency/Rework Items List

The QC Manager must maintain a list of work that does not comply with the Contract, identifying what items need to be corrected, the activity ID number associated with the item, the date the item was originally discovered, the date the item will be corrected by, and the date the item was corrected.

The QC Manager reviews the list at each weekly QC Meeting:

a. There is no requirement to report a deficiency/rework item that is corrected the same day it is discovered.

b. No successor task may be advanced beyond the preparatory phase meeting until all deficiencies/rework items have been cleared by the QC Manager and concurred with by the Contracting Officer. This must be confirmed as part of the Preparatory Phase activities.

c. Attach a copy of the "Deficiency/Rework Items List" to the last daily CQC Report of each month.

d. The Contractor is responsible for including those items identified by the Contracting Officer.

e. All deficiencies/rework items must be confirmed as corrected by the QC Manager, and concurred by the Contracting Officer, prior to commencement of any completion inspections per paragraph COMPLETION INSPECTIONS unless specifically exempted by the Contracting Officer.

f. Non-Compliance with these requirements is grounds for removal in accordance with paragraph ACCEPTANCE OF THE QUALITY CONTROL (QC) PLAN.

g. All delays, concurrent or related to failure to manage, monitor, control, and correct deficiencies/rework items are entirely the responsibility of the Contractor and can not be made the subject, or any component of any request for additional time or compensation.

1.10 TESTING

Perform specified or required tests to verify that control measures are adequate to provide a product which conforms to Contract requirements. Upon request, furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and acceptance tests when specified. Procure the services of an U.S. Army Corps of Engineers approved testing laboratory or establish an approved testing laboratory at the project site or within 10 miles. Perform the following activities and record and provide the following data:

a. Verify that testing procedures comply with Contract requirements.

b. Verify that facilities and testing equipment are available and comply with testing standards.

c. Check test instrument calibration data against certified standards.

d. Verify that recording forms and test identification control number system, including all test documentation requirements, have been prepared.

e. Record results of all tests taken, both passing and failing on the CQC report for the date taken. Specification paragraph reference, location where tests were taken, and the sequential control number identifying the test. If approved by the Contracting Officer, actual test reports are submitted later with a reference to the test number and date taken. Provide an information copy of tests performed by an offsite or commercial test facility directly to the Contracting Officer. Failure to submit timely test reports as stated results in nonpayment for related work performed and disapproval of the test facility for this Contract.

1.10.1 Accreditation Requirements

Construction materials testing laboratories must be accredited by a laboratory accreditation authority and must submit a copy of the Certificate of Accreditation and Scope of Accreditation. The laboratory's scope of accreditation must include the appropriate ASTM standards (ASTM E329, ASTM C1077, ASTM D3666, ASTM D3740, ASTM E543) listed in the technical Sections of the specifications. Laboratories engaged in Hazardous Materials Testing must meet the requirements of OSHA and EPA. The policy applies to the specific laboratory performing the actual testing, not just the Corporate Office.

1.10.2 Laboratory Accreditation Authorities

Laboratory Accreditation Authorities include the National Voluntary Laboratory Accreditation Program (NVLAP) administered by the National Institute of Standards and Technology at https://www.nist.gov/nvlap, the American Association of State Highway and Transportation Officials (AASHTO) Accreditation Program at http://www.aashtoresource.org/aap/overview, International Accreditation Services, Inc. (IAS) at http://www.iasonline.org, U. S. Army Corps of Engineers Materials Testing Center (MTC) at http://www.erdc.usace.army.mil/Media/FactSheets/FactSheetArticleView/tabid/9254/Article/476661/materials-testing-center.aspx, the American Association for Laboratory Accreditation (A2LA) program at http://www.a2la.org.

1.10.3 Capability Check

The Contracting Officer retains the right to check laboratory equipment in the proposed laboratory and the laboratory technician's testing procedures, techniques, and other items pertinent to testing, for compliance with the standards set forth in this Contract. Laboratories utilized for testing soils, concrete, asphalt, and steel must meet criteria detailed in ASTM D3740 and ASTM E329.

1.10.4 Test Results

Cite applicable Contract requirements, tests or analytical procedures used. Provide actual results and include a statement that the item tested or analyzed conforms or fails to conform to specified requirements. If the item fails to conform, notify the Contracting Officer immediately. Conspicuously stamp the cover sheet for each report in large red letters "CONFORMS" or "DOES NOT CONFORM" to the specification requirements, whichever is applicable. Test results must be signed by a testing laboratory representative authorized to sign certified test reports. Furnish the signed reports, certifications, and other documentation to the Contracting Officer via the QC Manager. Furnish a summary report of field tests at the end of each month, per the paragraph entitled DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER.

1.10.5 Test Reports and Monthly Summary Report of Tests

Furnish the signed reports, certifications, and a summary report of field tests at the end of each month to the Contracting Officer. Attach a copy of the summary report to the last daily CQC Report of each month. Provide a copy of the signed test reports and certifications to the Operation and Maintenance Support Information (OMSI) preparer for inclusion into the OMSI documentation, in accordance with Sections 01 78 23.00 22 OPERATION

AND MAINTENANCE DATA (PWD ME) and 01 78 24.00 20 FACILITY DATA WORKBOOK (FDW).

1.11 Quality Control (QC) CERTIFICATIONS

1.11.1 Contractor Quality Control (CQC) Report Certification

Contain the following statement within the CQC Report: "On behalf of the Contractor, I certify that this report is complete and correct and equipment and material used, and work performed during this reporting period is in compliance with the Contract drawings and specifications to the best of my knowledge, except as noted in this report."

1.11.2 Invoice Certification

Furnish a certificate to the Contracting Officer with each payment request, signed by the QC Manager, attesting that as-built drawings are current, coordinated and attesting that the work for which payment is requested, including stored material, complies with Contract requirements.

1.11.3 Completion Certification

Upon completion of work under this Contract, the QC Manager must furnish a certificate to the Contracting Officer attesting that "the work has been completed, inspected, tested and is in compliance with the Contract." Provide a copy of this final QC Certification for completion to the preparer of the Operation & Maintenance (O&M) documentation.

1.12 COMPLETION INSPECTIONS

1.12.1 Punch-Out Inspection

Near the completion of all work or any increment thereof, established by a completion time stated in the Contract Clause entitled "Commencement, Prosecution, and Completion of Work," or stated elsewhere in the specifications, the QC Manager and the CxC (if applicable) must conduct an inspection of the work and develop a "punch list" of items which do not conform to the approved drawings, specifications, and Contract. Include in the punch list any remaining items on the "Deficiency/Rework Items List", that were not corrected prior to the Punch-Out Inspection as approved by the Contracting Officer in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Include within the punch list the estimated date by which the deficiencies will be corrected. Provide a copy of the punch list to the Contracting Officer.

The QC Manager, or staff, must make follow-on inspections to ascertain that all deficiencies have been corrected. All punch list items must be confirmed as corrected by the QC Manager and concurred by the Contracting Officer. Once this is accomplished, notify the Government that the facility is ready for the Government "Pre-Final Inspection".

1.12.2 Pre-Final Inspection

The Government and QC Manager will perform this inspection to verify that the facility is complete and ready to be occupied. A Government "Pre-Final Punch List" will be documented by the QC Manager as a result of this inspection. The QC Manager will ensure that all items on this list are corrected and concurred by the Contracting Officer prior to notifying the Government that a "Final" inspection with the Client can be scheduled. All items noted on the "Pre-Final" inspection must be corrected and concurred in a timely manner and be accomplished before the Contract completion date for the work, or any increment thereof, if the project is divided into increments by separate completion dates unless exceptions are directed by the Contracting Officer.

1.12.3 Final Acceptance Inspection

Notify the Contracting Officer at least 14 calendar days prior to the date a final acceptance inspection can be held. State within the notice that all items previously identified on the pre-final punch list will be corrected and acceptable, along with any other unfinished Contract work, by the date of the final acceptance inspection. The Contractor must be represented by the QC Manager, the Project Superintendent, the CxC, and others deemed necessary. Attendees for the Government will include the Contracting Officer, other Government QA personnel, and personnel representing the Client. Failure of the Contractor to have all Contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance with the Contract Clause entitled "Inspection of Construction."

1.13 DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER

1.13.1 Construction Documentation

Reports are required for each day shift and night shift that work is performed and must be attached to the Contractor Quality Control Report prepared for the same day/night. Maintain current and complete records of on-site and off-site QC program operations and activities. Account for each calendar day, including daytime work and nighttime work, throughout the life of the Contract.

The Project Superintendent and the QC Manager must prepare and sign the Contractor Production and CQC Reports, respectively. Every space on the forms must be filled in. Use N/A if nothing can be reported in one of the spaces. The reporting of work must be identified by terminology consistent with the construction schedule. In the "Remarks" sections of the reports, enter pertinent information including directions received, problems encountered during construction, work progress and delays, conflicts or errors in the drawings or specifications, field changes, safety hazards encountered, instructions given and corrective actions taken, delays encountered, a record of visitors to the work site, quality control problem areas, deviations from the QC Plan, construction deficiencies encountered, and meetings held. For each entry in the report(s), identify the Schedule Activity No. that is associated with the entered remark.

1.13.2 Quality Control Activities

CQC and Contractor Production reports will be prepared daily/nightly to maintain current records providing factual evidence that required quality control activities and tests have been performed. Include in these records the work of subcontractors and suppliers on an acceptable form that includes, as a minimum, the following information:

a. The name and area of responsibility of the Contractors and any subcontractors.

b. Operating plant/equipment with hours worked, idle, or down for repair.

c. Work performed each day/night, giving location, description, and by whom. When a Network Analysis Schedule (NAS) is used, identify each item of work performed each day by NAS activity number.

d. Control phase activities performed. Preparatory, and Initial phase Checklists associated with the DFOW referenced to the construction schedule. Follow-up phase activities identified to the DFOW. If testing or specific QC Specialist activities are associated with the Follow-up phase activities for a specific DFOW note this and include those reports.

e. Test and control activities performed with results and references to specifications and drawings requirements. Identify the control phase (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST.

f. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications and drawings requirements.

g. Submittals and deliverables reviewed, with Contract reference, by whom, and action taken.

h. Offsite surveillance activities, including actions taken.

i. Job safety evaluations stating what was checked, results, and instructions or corrective actions.

j. Instructions given/received and conflicts in plans and specifications.

1.13.3 Verification Statement

Indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. Cover both conforming and deficient features and include a statement that equipment and materials incorporated in the work and workmanship comply with the Contract.

For work during regular working hours, furnish the original and one copy of these records in report form to the Government by 10:00 AM the next working day after the date covered by the report. For work outside of regular hours (e.g., nighttime work), furnish the original and one copy of these records in report form to the Government by 10:00 PM the next working day after the date covered by the report. As a minimum, prepare and submit one report for every 7 days of no work and on the last day of a no work period. All calendar days need to be accounted for throughout the life of the Contract. The first report following a day/night of no work will be for that day/night only. Reports need to be signed and dated by the QC Manager. Include copies of test reports and copies of reports prepared by all subordinate quality control personnel within the QC Manager Report.

1.13.4 Reports from the Quality Control (QC) Specialist(s)

Document inspection results on a QC specialist report prepared each day/night work is performed in their area of responsibility. The report must include a description of the visual inspection or observation performed, a written summary of findings, a conclusion on compliance with the Contract documents, and signature of the QC Specialist. In person inspections must be documented with Video/photographs. Video/photographic documentation of deficiencies must include before and after conditions and physical measurements, as necessary. Forward the QC daily report to the QC Manager who must include the report with the submission of their daily QC Report to the Government each day. Every site visit by the QC Specialist must be documented on a QC Specialist daily report.

1.13.5 Quality Control Validation

Establish and maintain the following in an electronic folder. Divide folder into a series of tabbed sections as shown below. Ensure folder is updated at each required progress meeting. This information must be readily available to the Contracting Officer during all business hours.

a. CQC Meeting minutes in accordance with paragraph QUALITY CONTROL (QC) MEETINGS.

b. All completed Preparatory and Initial Phase Checklists, arranged by specification section, further sorted by DFOW referenced to the construction schedule. Submit each individual Phase Checklist the day the phase event occurs as part of the CQC daily report.

c. All milestone inspections, arranged by Activity Number referenced to the construction schedule.

d. An up-to-date copy of the Testing Plan and Log with supporting field test reports, arranged by specification section referenced to the DFOW to which individual reports results are associated. Individual field test reports will be submitted within 2 working days after the test is performed in accordance with the paragraph QUALITY CONTROL ACTIVITIES. Monthly Summary Report of Tests: Submit the report as an electronic attachment to the CQC Report at the end of each month.

e. Copies of all Contract modifications, arranged in numerical order. Also include documentation that modified work was accomplished.

f. An up-to-date copy of the DEFICIENCY/REWORK ITEMS LIST.

g. Cx documentation in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

h. Special Inspection reports.

i. Upon commencement of Completion Inspections of the entire project or any defined portion, maintain up-to-date copies of all punch lists issued by the QC staff to the Contractor and subcontractors and all punch lists issued by the Government in accordance with the paragraph COMPLETION INSPECTIONS.

1.13.6 Testing Plan and Log

As tests are performed, the CxC and the QC Manager will record on the "Testing Plan and Log" the date the test was performed and the date the test results were forwarded to the Contracting Officer. Attach a copy of the updated "Testing Plan and Log" to the last daily CQC Report of each month. Provide a copy of the final "Testing Plan and Log" to the preparer of the Operation & Maintenance (O&M) documentation.

1.13.7 As-Built Drawings

The QC Manager is required to ensure the as-built drawings, required by Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME) are kept current on a daily basis and marked to show deviations which have been made from the Contract drawings. The as-built drawings document commences with the QC Manager ensuring all amendments, or changes to the Contract prior to Contract award are accurately noted in the initial document set creating the accurate baseline of the contract prior to any work starting. Ensure each deviation has been identified with the appropriate modifying documentation (e.g., PC No., Modification No., Request for Information No.). The QC Manager or QC Specialist assigned to an area of responsibility must initial each revision. Upon completion of work, the QC Manager will furnish a certificate attesting to the accuracy of the as-built drawings prior to submission to the Contracting Officer.

1.14 NOTIFICATION ON NON-COMPLIANCE

The Contracting Officer will notify the Contractor of any detected non-compliance with the Contract. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, must be deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of a claim for extension of time for excess costs or damages by the Contractor.

1.15 CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN

Submit an IAQ Management Plan within 15 calendar days after Contract Award and not less than 10 calendar days before the preconstruction meeting. Revise and resubmit the Plan as required by the Contracting Officer. Make copies of the final plan available to all workers on site. Include provisions in the Plan to meet the requirements specified below and to ensure safe, healthy air for construction workers and building occupants.

1.15.1 Requirements During Construction

Provide for evaluation of indoor Carbon Dioxide concentrations in accordance with ASTM D6245. Provide for evaluation of volatile organic compounds (VOCs) in indoor air in accordance with ASTM D6345. Use filters with a Minimum Efficiency Reporting Value (MERV) of 8 in permanently installed air handlers during construction.

1.15.2 Control Measures

Meet or exceed the requirements of ANSI/SMACNA 008, Chapter 3, to help minimize contamination of the building from construction activities. The five requirements of this manual which must be adhered to are described below:

a. HVAC Protection: Isolate return side of HVAC system from surrounding environment to prevent construction dust and debris from entering the duct work and spaces.

b. Source Control: Use low emitting paints and other finishes, sealants, adhesives, and other materials as specified. When available, cleaning products must have a low VOC content and be non-toxic to minimize building contamination. Utilize cleaning techniques that minimize dust generation. Cycle equipment off when not needed. Prohibit idling motor vehicles where emissions could be drawn into building. Designate receiving/storage areas for incoming material that minimize IAQ impacts.

c. Pathway Unterruption: When pollutants are generated use strategies such as 100 percent outside air ventilation or erection of physical barriers between work and non-work areas to prevent contamination.

d. Housekeeping: Clean frequently to remove construction dust and debris. Promptly clean up spills. Remove accumulated water and keep work areas dry to discourage the growth of mold and bacteria. Take extra measures when hazardous materials are involved.

e. Scheduling: Control the sequence of construction to minimize the absorption of VOCs by other building materials.

1.15.3 Moisture Contamination

a. Remove accumulated water and keep work dry.

b. Use dehumidification to remove moist, humid air from a work area.

c. Do not use combustion heaters or generators inside the building.

d. Protect porous materials from exposure to moisture.

e. Remove and replace items which remain damp for more than a few hours.

1.15.4 Requirements for Construction

After construction ends and prior to occupancy, conduct a building flush-out or test the indoor air contaminant levels. Flush-out must be a minimum of two (2) weeks with MERV-13 filtration media as determined by ASHRAE 52.2 at 100 percent outside air, or in accordance with LEED GBDC. Air contamination testing must be consistent with EPA's current Compendium of Methods for the Determination of Air Pollutants in Indoor Air, and with the LEED GBDC. After building flush-out or testing and prior to occupancy, replace filtration media. Filtration media must have a MERV of 13 as determined by ASHRAE 52.2.

1.15.5 Contractor QC Self-Evaluation Checklist

Checklist" to the Contractor at the pre-construction meeting. Complete the checklist monthly and submit with each request for payment voucher. An acceptable score of 90 or greater is required. Failure to submit the completed safety self-evaluation checklist or achieve a score of at least 90 may result in retention of up to 10 percent of the voucher. The Government will randomly perform QC inspections following the provided checklist. If the Government inspection score is less than 90 the Government will withhold 10 percent from the next invoice.

1.16 DELIVERY, STORAGE, AND HANDLING

Designate receiving/storage areas for incoming material to be delivered according to installation schedule and to be placed convenient to work area in order to minimize waste due to excessive materials handling and misapplication. Store and handle materials in a manner as to prevent loss from weather and other damage. Keep materials, products, and accessories covered and off the ground, and store in a dry, secure area. Prevent contact with material that may cause corrosion, discoloration, or staining. Protect all materials and installations from damage by the activities of other trades.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

PREPARATORY PHASE CHECKLIST

SPEC SECTION		DATE	
CONTRACT NO	DEFINABLE FEATURE OF WORK (DFOW)	SCHEDULE ACT NO.	INDEX #

PERSONNEL PRESENT GOVERNMENT REP NOTIFIED HOURS IN ADVANCE: ◯ YES ◯ NO Add Del

NAME	POSITION	COMPANY/GOVERNMENT

SUBMITTALS

REVIEW SUBMITTALS AND/OR SUBMITTAL REGISTER. HAVE ALL SUBMITTALS BEEN SUBMITTED AND APPROVED? (IF NO, WHAT ITEMS HAVE NOT BEEN SUBMITTED OR APPROVED) ◯ YES ◯ NO

ARE ALL MATERIALS ON HAND? ◯ YES ◯ NO
IF NO, WHAT ITEMS ARE MISSING.

CHECK APPROVED SUBMITTALS AGAINST DELIVERED MATERIAL. (THIS SHOULD BE DONE AS MATERIAL ARRIVES.)
COMMENTS:

MATERIAL STORAGE

ARE MATERIALS STORED PROPERLY? ◯ YES ◯ NO
IF NO, WHAT ACTION IS TAKEN?

SPECIFICATIONS

REVIEW EACH PARAGRAPH OF SPECIFICATIONS.

DISCUSS PROCEDURE FOR ACCOMPLISHING THE WORK.

CLARIFY ANY DIFFERENCES.

PRELIMINARY WORK

ENSURE PRELIMINARY WORK IS CORRECT AND PERMITS ARE ON FILE. IF NOT, WHAT ACTION IS TAKEN?

TESTING

IDENTIFY TEST TO BE PERFORMED, FREQUENCY, AND BY WHOM.

WHEN REQUIRED?

WHERE REQUIRED?

REVIEW TESTING PLAN.

HAS TEST FACILITIES BEEN APPROVED?

SAFETY

ACTIVITY HAZARD ANALYSIS APPROVED? ◯ YES ◯ NO
REVIEW APPLICABLE PORTION OF EM 385-1-1.

MEETING COMMENTS

NAVY/ROICC COMMENTS DURING MEETING.

OTHER ITEMS

OTHER ITEMS OR REMARKS.

QC MANAGER DATE

Reset Form

Note: Preparatory Meeting shall only be held if the shop drawings or submittal have been approved, the APP and appropriate AHA related to the DFOW have been submitted and the appropriate personnel (QC Manager, QC Specialists, Project Sup, CxC, Special Inspector, Special Inspector of Record, Foreman responsible for the DFOW, subcontractor foreman) are present for the meeting.

PREPARATORY PHASE CHECKLIST Part 2, DIV 1 Compliance		
Date	Contract No.	Project Title

☐ Identify any changed conditions or modifications that may impact the execution of the work.

☐ Examine the work area to ensure that the required preliminary work has been completed.

☐ Examine the required materials, equipment, and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data.

☐ Review each paragraph of the applicable specification Sections.

☐ Review the Contract drawings

☐ Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved. Verify receipt of approved factory test results, when required.

☐ Discuss specific controls used and construction methods, constructions tolerances, layout/survey controls (horizontal and vertical controls) necessary, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential programs for each DFOW.

☐ Review the testing plan and ensure that provisions have been made to provide the required QC testing and applicable required A/E (DOR) Quality Assurance Inspections.

☐ Review the listing of Government QA inspections that may be implemented as part of the execution of the work.

☐ Review any Special Inspections (Section 01 45 35) required as part of the execution of the work.

☐ Discuss the QC documentation required to be collected as part of the work. (e.g. Special inspector reports within 24 hours, material testing reports as soon as available.)

☐ If work includes demolition work, review demolition work plan to ensure compliance with EM 385-1-1 Section 23 Demolition, Renovation and Re-Occupancy.

☐ If work includes excavation, review Utility Locating Requirements. (Note: special attention is required if Hazardous or Mission Critical Utilities are within the excavation area. (If Hazardous and Mission Critical Utilities (as identified in the Scope of Work & Drawings) are located within the excavation area, the PWD ME Dig Safe Coordinator, PWD ME CM/ET, PWD ME Design Manager (DM) and the assigned PWD-ME Project Civil Engineer must be present at the meeting to review requirements to ensure safety and preservation of critical utilities as discussed with adequate focus.)

☐ If work includes excavation, review Soil Management Requirements (Note: Special attention is required if the work is within an IR site or hazardous soils are within the excavation area. Include Code 106.3 Representatives at this meeting to discuss Soil Management process and procedures.)

☐ If the work is subject to any environmental (EV) permitting, Archaeological Monitoring, or SHPO approvals, review the terms and conditions of the applicable permits as well as any supplemental controls/approvals that are required to complete the work. (Note: Include PWD-ME EV Representatives at this meeting to review the project specific requirements.

☐ If work includes the removal and disposal of Hazardous materials, review the appropriate work plans to ensure the work plans have been approved and any comments are understood by the work execution team. (Note: Include Code 106.3 Representatives at this meeting to discuss Hazardous Materials handling and disposal processes and procedures.

☐ Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.

☐ Review the processes/strategies that will be implemented to address any issues if the work does not go as planned or if any unforeseen conditions are encountered or if any changes arise that may impact the successful execution of the work.

____

QC Manager

INITIAL PHASE CHECKLIST

INITIAL PHASE CHECKLIST		SPEC SECTION	DATE
CONTRACT NO	DEFINABLE FEATURE OF WORK (DFOW)	SCHEDULE ACT NO.	INDEX #

PERSONNEL PRESENT

GOVERNMENT REP NOTIFIED HOURS IN ADVANCE: YES NO

Add
Del

NAME	POSITION	COMPANY/GOVERNMENT

PROCEDURE COMPLIANCE

IDENTIFY FULL COMPLIANCE WITH PROCEDURES IDENTIFIED AT PREPARATORY. COORDINATE PLANS, SPECIFICATIONS AND SUBMITTALS

COMMENTS:

PRELIMINARY WORK

ENSURE PRELIMINARY WORK IS COMPLETE AND CORRECT. IF NOT, WHAT ACTION IS TAKEN?

WORKMANSHIP

ESTABLISH LEVEL OF WORKMANSHIP
WHERE IS WORK LOCATED?

IS SAMPLE PANEL REQUIRED? YES NO
WILL THE INITIAL WORK BE CONSIDERED A SAMPLE? YES NO
(IF YES, PROVIDE THE LOCATION WHERE THE INITIAL WORK CONSIDERED AS A SAMPLE IS LOCATED AND THE PRECAUTIONS TAKEN TO PRESERVE THE SAMPLE

RESOLUTION

RESOLVE ANY DIFFERENCES.

COMMENTS:

CHECK SAFETY

REVIEW JOB CONDITIONS USING EM 385-1-1, ACTIVITY HAZARD ANALYSIS AND MSDS

COMMENTS:

OTHER

OTHER ITEMS OR REMARKS

QC MANAGER DATE

SCHEDULE OF FIRE PROTECTION SYSTEM INSPECTIONS BY FPQC SPECIALIST

1. INSTRUCTIONS for FPQC Specialist:

Perform surveillance of fire protection system installation during construction and during pre-final and final acceptance inspections. Surveillance includes either visual inspections or observations performed by the FPQC as indicated in the following Schedule of Fire Protection System Inspections. Check boxes in the schedule indicate the inspections or observations that are applicable to this project, which must be performed and documented by the FPQC.

Surveillance of fire protection system inspections is divided into three phases:

Surveillance During Construction – Includes periodic on-site visits by the FPQC during construction

Pre-Final Acceptance Inspection – FPQC must be onsite during pre-final acceptance inspection

Final Acceptance inspection – FPQC must be onsite during final acceptance inspection

2. INSPECTION TYPE

VISUAL INSPECTION:

Perform on-site visual inspections as indicated in the Schedule of Fire Protection System Inspections as construction progresses to confirm work conforms with the contract requirements including but not limited to verifying proper installation of equipment and materials in accordance with applicable codes. Where work will be covered up by finished walls and ceilings, inspections must be scheduled and performed when work is still visible.

OBSERVATION:

Observe component and system testing in person on-site as indicated in the Schedule of Fire Protection System Inspections. Confirm testing is completed in accordance with applicable codes and that testing was successful.

DOCUMENTATION:

Document all inspections and observations in the FPQC Report in accordance with UFGS 01 45 00. Document any deficiencies and all work observed that has been successfully performed in accordance with the contract documents. Record deficiencies in the rework list.

DESIGNER NOTES (delete this box after reviewing): **Rev 2-14-23**

1. Fire Protection DOR must edit this schedule based on the project scope.
2. Indicate inspection and observation tasks applicable to your project that are to be performed by the FPQC by checking boxes next to each task required. Either delete unchecked sections or leave them in the schedule unchecked.
3. This schedule does NOT need to be printed in color.
4. When finished editing, delete this note box and save this schedule as a PDF and insert into the project specifications immediately following UFGS Section 01 45 00.

SCHEDULE OF FIRE PROTECTION SYSTEM INSPECTIONS BY FPQC SPECIALIST

PERFORM SURVEILLANCE DURING CONSTRUCTION

~ FIRE SUPPRESSION SYSTEM ~

PERFORM TASK IF CHECKED	TASK	INSPECTION TYPE	DESCRIPTION
☐	1. Verify proper installation of underground service main and service; laterals, thrust blocks, tie-rods and connection to above ground piping.	**VISUAL INSPECTION**	✓ Visually inspect materials ✓ Visually inspect installation
☐	2. Verify proper orientation of fire hydrants and fire department access	**VISUAL INSPECTION**	✓ Visually inspect installation and orientation
☐	3. Verify proper installation of interior and exterior attachments, coatings, and vortex plate prior to filling fire protection water tank	**VISUAL INSPECTION**	✓ Visually inspect fire protection water tank installation
☐	4. Hydrostatic/leak test for underground piping	**OBSERVE**	✓ Verify proper test method ✓ Confirm test was successful
☐	5. Hydrostatic/leak test for above ground piping prior to ceiling installation	**OBSERVE**	✓ Verify proper test method ✓ Confirm test was successful
☐	6. Underground system flush prior to connection to the riser	**OBSERVE**	✓ Verify system was properly flushed
☐	7. Forward flow test for Backflow preventer	**OBSERVE**	✓ Verify proper test method ✓ Confirm test was successful
☐	8. Suction pipe flush prior to fire pump startup	**OBSERVE**	✓ Verify system was properly flushed
☐	9. Hydrostatic test fire pump suction and discharge piping	**OBSERVE**	✓ Verify proper test method ✓ Confirm test was successful
☐	10. Fire pump startup	**OBSERVE**	✓ Confirm successful startup
☐	11. Verify proper installation of sprinkler piping prior to wall or ceiling installation	**VISUAL INSPECTION**	✓ Inspect pipe hangars, bracing, sprinkler heads ✓ Inspect for sprinkler obstructions; damaged, painted or covered heads ✓ Verify proper location of control valves, drains, vents, backflow, test header ✓ Verify proper component mounting heights
☐	12. Document inspections and observations in FPQC Report	**DOCUMENT**	✓ Submit report for each day inspections or observations occur ✓ Annotate deficiencies and satisfactory work observed ✓ Record rework items in FPQC report

SCHEDULE OF FIRE PROTECTION SYSTEM INSPECTIONS BY FPQC SPECIALIST

~ SPECIAL FIRE SUPPRESSION SYSTEM ~

PERFORM TASK IF CHECKED	TASK	INSPECTION TYPE	DESCRIPTION
☐	1. Verify proper installation of gaseous fire suppression system	**VISUAL INSPECTION**	✓ Inspect gas cylinders, initiating controls, shutdowns, aborts, and visual warning
☐	2. Verify proper installation of IR hangar detector	**VISUAL INSPECTION**	✓ Inspect detector and nozzle layout ✓ Inspect for obstructions, abort stations ✓ Inspect riser piping
☐	3. Verify proper installation of elevator fire protection	**VISUAL INSPECTION**	✓ Inspect elevator recall functions ✓ Inspect two-way communication
☐	4. Verify proper installation of fire suppression in secure spaces	**VISUAL INSPECTION**	✓ Inspect location of sprinkler piping and sprinklers
☐	5. Document inspections and observations in FPQC Report	**DOCUMENT**	✓ Submit report for each day inspections or observations occur ✓ Document deficiencies and satisfactory work observed in FPQC Report ✓ Record deficiencies in rework list

~ PASSIVE FIRE PROTECTION SYSTEMS ~

PERFORM TASK IF CHECKED	TASK	INSPECTION TYPE	DESCRIPTION
☐	1. Verify proper construction of fire rated assemblies	**VISUAL INSPECTION**	✓ Inspect partitions for proper fire rated construction per design ✓ Inspect doors and door hardware for proper fire rating ✓ Inspect glass for proper fire rating
☐	2. Verify proper installation of fireproofing	**VISUAL INSPECTION**	✓ Inspect for use of proper fireproofing materials ✓ Inspect for cracking, spalling or delaminating ✓ Inspect for proper application thickness
☐	3. Verify proper installation of firestopping	**VISUAL INSPECTION**	✓ Inspect to confirm firestopping system is UL tested or an engineering judgement has been obtained and submitted ✓ Inspect for use of proper firestopping materials ✓ Inspect penetrations through fire rated walls, partitions and ceilings ✓ Inspect construction joints and gaps
☐	4. Document inspections and observations in FPQC Report	**DOCUMENT**	✓ Submit report for each day inspections or observations occur

SCHEDULE OF FIRE PROTECTION SYSTEM INSPECTIONS BY FPQC SPECIALIST

			✓ Document deficiencies and satisfactory work observed in FPQC Report ✓ Record deficiencies in rework list
~ FIRE ALARM SYSTEMS ~			
PERFORM TASK IF CHECKED	TASK	INSPECTION TYPE	DESCRIPTION
☐	1. Verify proper installation of fire alarm system after conduit is installed and wiring pulled, but before devices are installed	**VISUAL INSPECTION**	✓ Inspect conduit, wiring, conduit fill, wire type ✓ Inspect installation heights of back boxes
☐	2. Document inspections and observations in FPQC Report	**DOCUMENT**	✓ Submit report for each day inspections or observations occur ✓ Document deficiencies and satisfactory work observed in FPQC Report ✓ Record deficiencies in rework list

~ END OF SURVEILLANCE DURING CONSTRUCTION SECTION

PERFORM DURING PRE-FINAL ACCEPTANCE INSPECTION

PERFORM TASK IF CHECKED	TASK	INSPECTION TYPE	DESCRIPTION
☐	1. Inspect all fire protection systems and all life safety features	**VISUAL INSPECTION**	✓ Verify proper installation of fire protection and life safety systems
☐	2. Witness functional testing of all fire protection and life safety systems	**OBSERVE**	✓ Verify proper testing is performed on fire protection and life safety systems
☐	3. Identify deficiencies that require correction	**VISUAL INSPECTION**	✓ Document deficiencies in FPQC Report and record in rework list
☐	4. Verify rework items have been corrected	**VISUAL INSPECTION**	✓ Record re-inspection results on FPQC Report
☐	5. Document results of pre-final inspection and photograph unsatisfactory conditions	**DOCUMENT**	✓ Document results of pre-final inspection in FPQC Report
☐	6. Certify installation and operation of all fire protection and life safety systems conform to the contract documents	**DOCUMENT**	✓ Provide separate letter certifying conformance with contract requirements

~ END OF PRE-FINAL INSPECTION SECTION

PERFORM DURING FINAL ACCEPTANCE INSPECTION

PERFORM TASK IF CHECKED	TASK	INSPECTION TYPE	DESCRIPTION

SCHEDULE OF FIRE PROTECTION SYSTEM INSPECTIONS BY FPQC SPECIALIST

☐	1. Verify proper installation and operation of all fire protection systems and all life safety features	**VISUAL INSPECTION**	✓
☐	2. Note any deficiencies identified by Government FPE on deficiency list	**DOCUMENT**	✓ Document deficiencies identified by Government FPE on deficiency list
☐	3. Verify all deficiencies have been corrected	**VISUAL INSPECTION**	✓ Inspect rework to confirm all deficiencies have been corrected
☐	4. Certify that all deficiencies have been corrected	**DOCUMENT**	✓ Document inspection results on FPQC Report

~ END OF FINAL ACCEPTANCE INSPECTION SECTION

SECTION 01 45 35.00 22
^^
SPECIAL INSPECTIONS (PWD ME)
05/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2024) International Building Code

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01	(2023; with Change 1, 2023) Structural Engineering

1.2 GENERAL REQUIREMENTS

Perform Special Inspections in accordance with the Statement of Special Inspections, Schedule of Special Inspections, Chapter 17 of ICC IBC, and UFC 3-301-01. The Statement of Special Inspections and Schedule of Special Inspections are included as an attachment to this specification. Special Inspections are to be performed by an independent third party and are intended to ensure that the work of the Prime Contractor is in accordance with the Contract Documents and applicable building codes. Special inspections do not take the place of the three phases of control inspections performed by the Contractor's QC Manager or any testing and inspections required by other sections of the specifications.

The Contractor must provide notification to the Contracting Officer 14 days prior to completing any construction elements requiring Special Inspections.

1.3 DEFINITIONS

1.3.1 Continuous Special Inspections

Continuous Special Inspections is the constant monitoring of specific tasks by a special inspector. These inspections must be carried out continuously over the duration of the particular tasks.

1.3.2 Perform

Perform these tasks for each weld, fastener or bolted connection, and noted required verification.

1.3.3 Observe

Observe these items randomly during the course of each work day to insure that applicable requirements are being met. Observe these Special Inspections items on a periodic daily basis. Operations need not be delayed pending these inspections.

1.3.4 Special Inspector (SI)

A qualified person retained by the Contractor and approved by the Contracting Officer as having the competence necessary to inspect a particular type of construction requiring Special Inspections. The SI must be an independent third party hired directly by the Prime Contractor.

1.3.5 Associate Special Inspector (ASI)

A qualified person who assists the SI in performing Special Inspections but who must perform inspection under the direct supervision of the SI and who cannot perform inspections without the SI on site.

1.3.6 Third Party

A Special inspector must not be an employee of the Contractor or of any Sub-Contractor performing the work to be inspected.

1.3.7 Special Inspector of Record (SIOR)

A licensed professional engineer in responsible charge of supervision of all special inspectors for the project and approved by the Contracting Officer. The SIOR must be an independent third-party entity hired directly by the Prime Contractor.

1.3.8 Contracting Officer

The Government official having overall authority for administrative contracting actions. Certain contracting actions may be delegated to the Contracting Officer's Representative (COR).

1.3.9 Quality Control (QC) Manager

An individual retained by the Prime Contractor and qualified in accordance with the Section 01 45 00.00 22 QUALITY CONTROL (PWD ME) having the overall responsibility for the Contractor's QC organization.

1.3.10 Structural Engineer of Record (SER)

A registered design professional responsible for the overall design and review of submittal documents prepared by others. The SER is registered or licensed to practice their respective design profession as defined by the statutory requirements of the professional registration laws in the state in which the design professional works. The SER is also referred to as the Engineer of Record (EOR) in design documents.

1.3.11 Statement of Special Inspections (SSI)

A document developed by the SER identifying the materials, systems, components and work required to have Special Inspections. This statement is included at the end of this specification.

1.3.12 Schedule of Special Inspections (SSI)

A schedule which lists each of the required Special Inspections, the extent to which each Special Inspection is to be performed, and the required frequency for each in accordance with ICC IBC Chapter 17. This schedule is included at the end of this specification.

1.3.13 Definable Feature of Work (DFOW)

An inspection group that is separate and distinct from other inspection groups, having inspection requirements or inspectors that are unique.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

SIOR Letter of Acceptance; G

Special Inspections Project Manual; G

Special Inspections Agency's Written NDT Practices with method and evidence of regular equipment calibration where applicable.

SD-06 Test Reports

Special Inspection Daily Reports; G

Special Inspection Biweekly Reports; G

SD-07 Certificates

AISC Certified Steel Fabricator; G

Certificate of Compliance; G

Special Inspector of Record Qualifications; G

Special Inspector Qualifications; G

Qualification Records for Nondestructive Testing Technicians; G

SD-11 Closeout Submittals

Interim Report of Special Inspections from each DFOW; G

Comprehensive Final Report of Special Inspections; G

Final Report of Structural Observations where required; G

1.5 SPECIAL INSPECTOR QUALIFICATIONS

Submit qualifications for each special inspector according to the following certification requirements. The Government will consider alternate equivalent certification upon request. Without written acceptance deviations to the following qualifications are not approved.

1.5.1 Steel Construction and High Strength Bolting

1.5.1.1 Special Inspector

a. ICC Structural Steel and Bolting Special Inspector certificate with one year of related experience, or

b. Registered Professional Engineer with 3 years of related experience

1.5.1.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.2 Welding Structural Steel

1.5.2.1 Special Inspector

AWS Certified Welding Inspector.

1.5.2.2 Associate Special Inspector

AWS Certified Associate Welding Inspector

1.5.3 Nondestructive Testing of Welds

1.5.3.1 Special Inspector

NDT Level II Certificate pluse one year of related experience.

1.5.3.2 Associate Special Inspector

NDT Level I Certificate plus one year of related experience.

1.5.3.3 Inspection Agency NDT Administrator

The individual administering the Standard Written Practice must have an NDT Level III Certificate.

1.5.4 Cold Formed Steel Framing

1.5.4.1 Special Inspector

a. ICC Structural Steel and Bolting Special Inspector certificate with one year of related experience, or

b. ICC Commercial Building Inspector with one year of experience, or

c. Registered Professional Engineer with 3 years related experience

1.5.4.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.5 Concrete Construction

1.5.5.1 Special Inspector

a. ICC Reinforced Concrete Special Inspector Certificate with one year of related experience, or

b. ACI Concrete Construction Special Inspector, or

c. Registered Professional Engineer with 3 years of related experience

1.5.5.2 Associate Special Inspector

a. ACI Concrete Construction Special Inspector-in-Training, or

b. Engineer-In-Training with one year of related experience

1.5.6 Masonry Construction

1.5.6.1 Special Inspector

a. ICC Structural Masonry Special Inspector Certificate with one year of related experience, or

b. Registered Professional Engineer with 3 years of related experience

1.5.6.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.7 Verification of Site Soil Condition, Fill Placement and Load-Bearing Requirements

1.5.7.1 Special Inspector

a. ICC Soils Special Inspector Certificate with one year of related experience, or

b. National Institute for Certification in Engineering Technologies (NICET) Soils Technician Level II Certificate in Construction Material Testing, or

c. Geologist-In-Training with 3 years of related experience, or

d. Registered Professional Engineer with 3 years of related experience

1.5.7.2 Associate Special Inspector

a. NICET Soils Technician Level I Certificate in Construction Material Testing with one year of related experience, or

b. Engineer-In-Training with one year of related experience

1.5.8 Deep Foundations

1.5.8.1 Special Inspector

a. Registered Professional Engineer with 3 years of related experience

1.5.8.2 Associate Special Inspector

a. NICET Soils Technician Level I Certificate in Construction Material Testing with one year of related experience, or

b. NICET Geotechnical Engineering Technician Level I Construction or

Generalist Certificate with one year of related experience, or

c. Engineer-In-Training with one year of related experience

1.5.9 Intumescent Fire-Resistive Coatings

1.5.9.1 Special Inspector

a. ICC Spray-applied Fireproofing Special Inspector Certificate, or

b. ICC Fire Inspector I Certificate with one year of related experience, or

c. Registered Professional Engineer or Architect with related experience

1.5.9.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.10 Fire-Resistant Penetrations and Joints

Refer to Section 07 84 00.00 22 FIRESTOPPING for firestopping installation and inspection requirements.

PART 2 PRODUCTS

2.1 SPECIAL INSPECTION OF FABRICATED ITEMS

Special Inspection of fabricator's work performed in the fabricator's shop is required to be inspected in accordance with the Statement of Special Inspections and the Schedule of Special Inspections unless the fabricator is certified by the approved agency to perform such work without Special Inspections. Submit the following certification to the Contracting Officer for information to allow work performed in the fabricator's shop to not be subjected to Special Inspections.

AISC Certified Steel Fabricator.

At the completion of fabrication, submit a certificate of compliance, to be included with the comprehensive final report of Special Inspections, stating that the materials supplied and work performed by the fabricator are in accordance with the construction documents.

PART 3 EXECUTION

3.1 RESPONSIBILITIES

3.1.1 Special Inspector of Record

a. Supervise all Special Inspectors required by the Contract Documents and the IBC.

b. Submit an SIOR Letter of Acceptance to the Contracting Officer attesting to acceptance of the duties of SIOR, signed and sealed by the SIOR.

c. Verify the qualifications of all of the Special Inspectors.

d. Verify the qualifications of fabricators.

e. Submit Special Inspection agency's written NDT practices for the monitoring and control of the agency's operations to include the following:

(1) The agency's procedures for the selection and administration of inspection personnel, describing the training, experience and examination requirements for qualification and certification of inspection personnel.

(2) The agency's inspection procedures, including general inspection, material controls, and visual welding inspection.

f. Submit Qualification Records for Nondestructive Testing Technicians for nondestructive testing (NDT) technicians designated for the project.

g. Submit NDT procedures and equipment calibration records for NDT to be performed and equipment to be used for the project.

h. Prepare a Special Inspections Project Manual, which must cover the following:

(1) Roles and responsibilities of the following individuals during Special Inspections: SIOR, SI, ASI, General Contractor's QC Manager and SER.

(2) Organizational chart or communication plan, indicating lines of communication.

(3) Contractor's internal plan for scheduling inspections. Address items such as timeliness of inspection requests, whom to contact for inspection requests, and availability of alternate inspectors.

(4) Indicate the Government reporting requirements.

(5) Propose forms or templates to be used by SI and SIOR to document inspections.

(6) Indicate procedures for tracking nonconforming work and verification that corrective work is complete.

(7) Indicate how the SIOR and SI will participate in weekly QC meetings.

(8) Indicate how Special Inspections of shop fabricated items will be handled when the fabricator's shop is not certified in accordance with paragraph SPECIAL INSPECTION OF FABRICATED ITEMS.

(9) Include a section in the manual that covers each specific item requiring Special Inspections that is indicated on the Schedule of Special Inspections. Provide names and qualifications of each special inspector who will be performing the Special Inspections for each specific item. Provide details on how the Special Inspections are to be carried out for each item so that the expectations are clear for the General Contractor and the Subcontractor performing the work.

Make a copy of the Special Inspections Project Manual available on the job site during construction. Submit a copy of the Special Inspections Project Manual for approval.

i. Attend coordination and mutual understanding meeting where the information in the Special Inspections Project Manual will be reviewed to verify that all parties have a clear understanding of the Special Inspection provisions and the individual duties and responsibilities of each party.

j. Maintain a 3-ring binder for the Special Inspector's daily and special inspection biweekly reports and the Special Inspections Project Manual. This binder must be located in a conspicuous place in the project trailer/office to allow review by the Contracting Officer and the SER.

k. Submit a copy of the Special Inspector's special inspection daily reports to the QC Manager. The QC Manager must submit copies of the Special Inspector's daily reports to the Contracting Officer with the Contractor's Daily Quality Control Reports.

l. Discrepancies that are observed during Special Inspections must be reported to the QC Manager for correction. If discrepancies are not corrected before the Special Inspector leaves the site, the observed discrepancies must be documented in the daily report.

m. Submit a biweekly Special Inspections report until all work requiring Special Inspections is complete. A report is required for each biweekly period in which Special Inspection activity occurs; the report must include the following:

   (1) A brief summary of the work performed during the reporting time frame.

   (2) Changes from and discrepancies with the drawings, specifications that were observed during the reporting period.

   (3) Discrepancies which were resolved or corrected.

   (4) A list of nonconforming items requiring resolution.

   (5) All applicable test results including nondestructive testing reports.

n. At the completion of each Definable Feature of Work (DFOW) requiring Special Inspections, submit an interim report that documents the Special Inspections completed for that DFOW, including corrections of all discrepancies noted in the daily reports. Interim reports of Special Inspections must be signed and dated by the SIOR.

o. At the completion of the project submit a comprehensive final report of Special Inspections that documents the Special Inspections completed for the project including corrections of all discrepancies noted in the daily reports. The comprehensive final report of Special Inspections must be signed, dated and sealed by the SIOR.

3.1.2 Quality Control Manager

a. Supervise all Special Inspectors required by the Contract Documents and the IBC.

b. Verify the qualifications of all the Special Inspectors.

c. Verify the qualifications of all fabricators.

d. Maintain a 3-ring binder for the Special Inspector's special inspection daily reports and special inspection biweekly reports. This binder must be located in a conspicuous place in the project trailer/office to allow review by the Contracting Officer and the SER. The QC Manager must submit copies of the Special Inspector's daily reports to the Contracting Officer with the Contractor's daily quality control (qc) reports.

e. Maintain a rework items list that includes discrepancies noted on the Special Inspector's daily report.

3.1.3 Special Inspectors

a. Inspect all elements of the project which the special inspector is qualified to inspect and which are identified in the Schedule of Special Inspections.

b. Attend preparatory phase meetings related to the Definable Feature of Work (DFOW) which the Special Inspector is qualified to inspect.

c. Submit Special Inspection agency's written NDT practices for the monitoring and control of the agency's operations, to include the following:

   (1) The agency's procedures for the selection and administration of inspection personnel, describing the training, experience and examination requirements for qualification and certification of inspection personnel.

   (2) The agency's inspection procedures, including general inspection, material controls, and visual welding inspection.

d. Submit Qualification Records for Nondestructive Testing Technicians for nondestructive testing (NDT) technicians designated for the project.

e. Submit NDT procedures and equipment calibration records for NDT to be performed and equipment to be used for the project.

f. Submit a copy of the special inspection daily reports to the QC Manager.

g. Report discrepancies that are observed during Special Inspections to the QC Manager for correction. If discrepancies are not corrected before the special inspector leaves the site the observed discrepancies must be documented in the daily report.

h. Submit a biweekly Special Inspection Report until all inspections are complete. A report is required for each biweekly period in which Special Inspections activity occurs, and must include the following:

   (1) A brief summary of the work performed during the reporting time frame.

(2) Changes from and discrepancies with the drawings, specifications and mechanical or electrical component certification that were observed during the reporting period.

(3) Discrepancies which were resolved or corrected.

(4) A list of nonconforming items requiring resolution.

(5) All applicable test result, including nondestructive testing reports.

i. At the completion of each DFOW requiring Special Inspections, submit an interim report of Special Inspections that documents the Special Inspections completed for that DFOW. Identify the inspector responsible for each item inspected and corrections of all discrepancies noted in the daily reports. The interim report of Special Inspections must be signed and dated and it must indicate the certification of the Special Inspector qualifying them to conduct the inspection.

j. At the completion of the project, submit a comprehensive Final Report of Special Inspections that documents the Special Inspections completed for the project and corrections of all discrepancies noted in the daily reports. The comprehensive final report of Special Inspections must be signed and dated and it must indicate the certification of the Special Inspector qualifying them to conduct the inspection.

k. Submit special inspection daily reports to the SIOR.

3.2 DEFECTIVE WORK

Check work as it progresses, but failure to detect any defective work or materials must in no way prevent later rejection if defective work or materials are discovered, nor obligate the Contracting Officer to accept the defective work or materials. The Contractor's failure to comply with the Special Inspections specified herein will be grounds for the Contracting Officer taking any or all of the following actions including issuing a Stop Work Order, the removal of the QC Manager, and a marginal or unsatisfactory CPARS rating.

-- End of Section --

Project: B60 Production Facility Repairs - SIOP
Location: Portsmouth Naval Shipyard, Kittery, Maine
Project #: 1740579
Date: 9/17/2025

STATEMENT OF SPECIAL INSPECTIONS

Project Seismic Design Category: C
Project Risk Category: II
Project Design Wind Speed (mph): 115
Number of Stories: 2
Structure Height Above Grade (ft): 48
Hazardous Occupancy or attached to such? No Group H Occupancies (2021 IBC, Section 415)

Special Inspector of Record (SIOR)

A Special Inspector of Record (SIOR) **IS NOT** required (per UFGS 01 45 35, Section 1.3.8)

Lateral Force Resisting System (LFRS)

2021 IBC 1704.3.2 and 1704.3.3

Following is a listing of critical main wind/seismic force resisting systems for this structure. Carefully inspect these elements as part of the roles and responsiblities of the Special Inspector (reference the Schedule of Special Inspections for inspection checklists).

Vertical LFRS Elements	Notes
Mezzanines: Steel Braced Frames (R=3)	Diagonal steel braces in each principle direction.
Horizontal LFRS Elements	**Notes**
Mezzanines: Steel Deck/Concrete Slab Rigid Floor Diaphragm	Composite steel deck/concrete steel slab

Project: B60 Production Facility Repairs - SIOP
Location: Portsmouth Naval Shipyard, Kittery, Maine
Project #: 1740579
Date: 9/17/2025

Designated Seismic Systems (DSS)

(2021 IBC 1705.13.3) (ASCE 7-16, 13.2.2, C13.2.2) (UFC 3-301-01, 2.5.3)

Non-structural 'Designated Seismic Systems' (DSS) must remain operable and contain hazardous substances following a design earthquake. Accordingly, all Designated Seismic Systems must be listed below and must be certified by the manufacturer to remain both operable and/or to contain hazardous substances after a design

ELECTRICAL Designated Seismic Systems (DSS) Requiring a Certificate of Compliance
N/A

If additional space is required, append an additional sheet listing the remaining DSS

MECHANICAL/PLUMBING Designated Seismic Systems (DSS) Requiring a Certificate of Compliance
N/A

If additional space is required, append an additional sheet listing the remaining DSS

OTHER Designated Seismic Systems (DSS) Requiring a Certificate of Compliance
N/A

Final Walk Down Inspection and Report

(UFC 3 301 01 SECTION 2-5.4)

Final Walk Down Inspection of non-structural Designated Seismic Systems does not apply to this project (no Designated Seismic Systems)

SCHEDULE OF SPECIAL INSPECTIONS

Reference UFGS 01 45 35 for all requirements not noted as part of this schedule.

INSPECTION DEFINITIONS:

PERFORM: Perform these tasks for each weld, fastener or bolted connection, and noted verification.

OBSERVE: Observe these items randomly during the course of each work day to insure that applicable requirements are being met. Operations need not be delayed pending these inspections at contractor's risk.

DOCUMENT: Document, with a report, that the work has been performed in accordance with the contract documents. This is in addition to any other reports required in the Special Inspections guide specification.

CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

A. STRUCTURAL - STEEL – WELDING SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

STEEL INSPECTION PRIOR TO WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-10: Table C-N5.4-1		
TASK	INSPECTION TYPE[1]	DESCRIPTION
1. Verify that the welding procedures specification (WPS) is available	**PERFORM**	
2. Verify manufacturer certifications for welding consumables are available	**PERFORM**	
3. Verify material identification	**PERFORM**	Type and grade.
4. Welder Identification System	**PERFORM**	The fabricator or erector, as applicable, shall maintain a system by which a welder who has welded a joint or member can be identified. Stamps, if used, shall be the low-stress type.
5. Fit-up of groove welds (including joint geometry)	OBSERVE	✓ Joint preparation ✓ Dimensions (alignment, root opening, root face, bevel) ✓ Cleanliness (condition of steel surfaces) ✓ Tacking (tack weld quality and location) ✓ Backing type and fit (if applicable)
6. Configuration and finish of access holes	OBSERVE	
7. Fit-up of fillet welds	OBSERVE	✓ Dimensions (alignment, gaps at root) ✓ Cleanliness (condition of steel surfaces) ✓ Tacking (tack weld quality and location)
STEEL INSPECTION DURING WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-10: Table C-N5.4-2		
TASK	INSPECTION TYPE	DESCRIPTION
8. Use of qualified welders	**PERFORM**	Welding by welders, welding operators, and tack welders who are qualified in conformance with requirements.
9. Control and handling of welding consumables	OBSERVE	✓ Packaging ✓ Electrode atmospheric exposure control
10. No welding over cracked tack welds	OBSERVE	
11. Environmental conditions	OBSERVE	✓ Wind speed within limits ✓ Precipitation and temperature
12. Welding Procedures Specification followed	OBSERVE	✓ Settings on welding equipment ✓ Travel speed ✓ Selected welding materials ✓ Shielding gas type/flow rate ✓ Preheat applied ✓ Interpass temperature maintained (min./max.) ✓ Proper position (F, V, H, OH) ✓ Intermix of filler metals avoided
13. Welding techniques	OBSERVE	✓ Interpass and final cleaning ✓ Each pass within profile limitations ✓ Each pass meets quality requirements

[1] **PERFORM**: Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

A. STRUCTURAL - STEEL – WELDING SECTION (CONTINUED)

STEEL INSPECTION AFTER WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 2021 1705.2.1, AISC 360-10: Table C-N5.4-3		
TASK	INSPECTION TYPE [1]	DESCRIPTION
14. Welds cleaned	OBSERVE	
15. Size, length, and location of all welds	**PERFORM**	Size, length, and location of all welds conform to the requirements of the detail drawings.
16. Welds meet visual acceptance criteria	**PERFORM AND DOCUMENT**	✓ Crack prohibition ✓ Weld/base-metal fusion ✓ Crater cross section ✓ Weld profiles ✓ Weld size ✓ Undercut ✓ Porosity
17. Repair activities	**PERFORM AND DOCUMENT**	
18. Document acceptance or rejection of welded joint or member	**PERFORM**	

END SECTION

B. STRUCTURAL - STEEL – BOLTING SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

STEEL INSPECTION TASKS PRIOR TO BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-10: Table C-N5.6-1		
TASK	INSPECTION TYPE [2]	DESCRIPTION
1. Manufacture's certifications available for fastener materials	**PERFORM**	
2. Fasteners marked in accordance with ASTM requirements	OBSERVE	
3. Proper fasteners selected for joint detail (grade, type, bolt length if threads are to be excluded from shear plane)	OBSERVE	
4. Proper bolting procedure selected for joint detail	OBSERVE	
5. Connecting elements, including appropriate faying surface condition and hole preparation, if specified, meet applicable requirements	OBSERVE	
6. Proper storage provided for bolts, nuts, washers, and other fastener components	OBSERVE	
STEEL INSPECTION TASKS DURING BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-10: Table C-N5.6-2		
TASK	INSPECTION TYPE [1]	DESCRIPTION

[1] **PERFORM**: Perform these tasks for each weld, fastener or bolted connection, and required verification.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

[2] **PERFORM**: Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

7. Fastener assemblies of suitable condition, placed in all holes and washers (if required) are positioned as required	OBSERVE	
8. Joint brought to the snug-tight condition prior to pretensioning operation	OBSERVE	
9. Fastener component not turned by the wrench prevented from rotating	OBSERVE	
10. Bolts are pretensioned in accordance with RCSC Specification, progressing systematically from the most rigid point toward the free edges	OBSERVE	
STEEL INSPECTION TASKS AFTER BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-10: Table C-N5.6-3		
TASK	INSPECTION TYPE [1]	DESCRIPTION
11. Document acceptance or rejection of all bolted connections	**DOCUMENT**	

END SECTION

C. STRUCTURAL - STEEL - NON DESTRUCTIVE TESTING SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

NONDESTRUCTIVE TESTING OF WELDED JOINTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-10: Section N5.5		
TASK	INSPECTION TYPE [1]	DESCRIPTION
1. Use of qualified nondestructive testing personnel	**PERFORM**	Visual weld inspection and nondestructive testing (NDT) shall be conducted by personnel qualified in accordance with AWS D1.8 clause 7.2

END SECTION

D. STRUCTURAL - STEEL - OTHER INSPECTIONS

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

OTHER STEEL INSPECTIONS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 341-10: Tables J8-1 & J10-1		
TASK	INSPECTION TYPE [2]	DESCRIPTION
1. Anchor rods and other embedments supporting structural steel	**PERFORM**	Verify the diameter, grade, type, and length of the anchor rod or embedded item, and the extent or depth of embedment prior to placement of concrete.
2. Fabricated steel or erected steel frame	OBSERVE	Verify compliance with the details shown on the construction documents, such as braces, stiffeners, member locations and proper application of joint details at each connection.

END SECTION

[1] **PERFORM**: Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

[2] **PERFORM**: Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

E. STRUCTURAL - COLD-FORMED METAL DECK - PLACEMENT SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

METAL DECK INSPECTION PRIOR TO DECK PLACEMENT – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2011, Appendix 1, Table 1.1		
TASK	INSPECTION TYPE [1]	DESCRIPTION
1. Verify compliance of materials (deck and all deck accessories) with construction documents, including profiles, material properties, and base metal thickness	**PERFORM**	
2. Document acceptance or rejection of deck and deck accessories	**DOCUMENT**	
METAL DECK INSPECTION DURING DECK PLACEMENT – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2011, Appendix 1, Table 1.2		
TASK	INSPECTION TYPE [1]	DESCRIPTION
3. Verify compliance of deck and all deck accessories installation with construction documents	**PERFORM**	
4. Verify deck materials are represented by the mill certifications that comply with the construction documents	**PERFORM**	
5. Document acceptance or rejection of installation of deck and deck accessories	**DOCUMENT**	
METAL DECK INSPECTION AFTER DECK PLACEMENT – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2011, Appendix 1, Table 1.3		
TASK	INSPECTION TYPE [1]	DESCRIPTION
6. Welding procedure specification (WPS) available	**PERFORM**	
7. Manufactures certifications for welding consumables available	OBSERVE	
8. Material identification (type/grade)	OBSERVE	
9. Check welding equipment	OBSERVE	

END SECTION

[1] **PERFORM**: Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

F. STRUCTURAL - COLD-FORMED METAL DECK – WELDING SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

METAL DECK INSPECTION DURING WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2011, Appendix 1, Table 1.4		
TASK	INSPECTION TYPE [1]	DESCRIPTION
1. Use of qualified welders	OBSERVE	
2. Control and handling of welding consumables	OBSERVE	
3. Environmental conditions (wind speed, moisture, temperature)	OBSERVE	
4. WPS followed	OBSERVE	
METAL DECK INSPECTION AFTER WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2011, Appendix 1, Table 1.5		
TASK	INSPECTION TYPE [1]	DESCRIPTION
5. Verify size and location of welds, including support, sidelap, and perimeter welds.	**PERFORM**	
6. Welds meet visual acceptance criteria	**PERFORM**	
7. Verify repair activities	**PERFORM**	
8. Document acceptance or rejection of welds	**DOCUMENT**	

END SECTION

G. STRUCTURAL CONCRETE CONSTRUCTION

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

CONCRETE CONSTRUCTION, INCLUDING COMPOSITE DECK – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2021 IBC TABLE 1705.3 (ACI 318-19 REFERENCES NOTED IN IBC TABLE)		
TASK	INSPECTION TYPE [2]	DESCRIPTION
1. Inspect reinforcement and verify placement.	PERIODIC	Verify prior to placing concrete that reinforcement is of specified type, grade and size; that it is free of oil, dirt and unacceptable rust; that it is located and spaced properly; that specified cover has been provided; that hooks, bends, ties, stirrups and supplemental reinforcement are placed correctly; that specified lap lengths, stagger and offsets are provided; and that all mechanical connections are installed per the manufacturer's instructions and/or evaluation report.

[1] **PERFORM**: Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

[2] **PERIODIC**: Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.
CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

CONCRETE CONSTRUCTION, INCLUDING COMPOSITE DECK – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2021 IBC TABLE 1705.3 (ACI 318-19 REFERENCES NOTED IN IBC TABLE)		
TASK	INSPECTION TYPE [1]	DESCRIPTION
2. Inspect cast-in-place anchors, post-installed drilled anchors, and adhesive anchors.	PERIODIC	✓ Verify prior to placing concrete that cast in place anchors and post installed drilled anchors have proper embedment, spacing and edge distance. ✓ Inspection requirements for adhesive anchors are different from those for other post-installed anchors and require assessment and qualification under the provisions of ACI 355.4 which may require proof loading.
3. Verify use of required mix design	PERIODIC	Verify that all mixes used comply with the approved construction documents
4. Prior to concrete placement, fabricate specimens for strength tests, perform slump and air content tests, and determine the temperature of the concrete	CONTINUOUS	At the time fresh concrete is sampled to fabricate specimens for strength test, verify these tests are performed by qualified technicians.
5. Inspect concrete placement for proper application techniques	CONTINUOUS	Verify proper application techniques are used during concrete conveyance and that depositing avoids segregation or contamination. Verify that concrete is properly consolidated.
6. Verify maintenance of specified curing temperature and technique	PERIODIC	Inspect curing and cold weather, or hot weather protection procedures.
7. Inspect formwork for shape, location and dimensions of the concrete member being formed.	PERIODIC	Formwork fabrication and installation shall result in final structure that conforms to shapes, lines, and dimensions of the members as required by the construction documents.

END SECTION

H. GEOTECHNICAL - SOILS INSPECTION SECTION

THIS SECTION APPLICABLE IF BOX IS CHECKED: ☒

SOILS INSPECTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.6		
TASK	INSPECTION TYPE [1]	DESCRIPTION
1. Materials below shallow foundations are adequate to achieve the design bearing capacity	OBSERVE	
2. Excavations are extended to proper depth and have reached proper material	OBSERVE	
3. Verify use of proper materials, densities and lift thicknesses during placement and compaction of compacted fill	**CONTINUOUS**	
4. Prior to placement of compacted fill, inspect subgrade and verify that site has been prepared properly	OBSERVE	During fill placement, the special inspector shall verify that proper materials and procedures are used in accordance with the provisions of the approved geotechnical report

END SECTION

GEOTECHNICAL - MICROPILE FOUNDATIONS SECTION

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒

DRIVEN DEEP FOUNDATION ELEMENTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.7 and TABLE 1705.7		
TASK	INSPECTION TYPE	DESCRIPTION
1. Verify element materials, sizes and lengths comply with requirements	**CONTINUOUS**	
2. Inspect driving operations and maintain complete and accurate records for each element	**CONTINUOUS**	
3. Verify placement locations and plumbness, verify use of required grout mix, inspect grout and reinforcement placement for proper application techniques, record tip and butt elevations and document any damage to foundation elements.	**CONTINUOUS**	
4. For concrete elements, perform additional special inspections in accordance with the concrete section in this schedule.		

END SECTION

[1] **OBSERVE**: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

FIRE PROTECTION – FIRE RESISTANT PENETRATIONS AND JOINTS SECTION

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒

FIRE RESISTANT PENETRATIONS AND JOINTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2021 IBC SECTION 1705.18		
TASK	INSPECTION TYPE [1]	DESCRIPTION
1. Inspections of penetration firestop systems conducted in accordance with Section 07 84 00.00 22 and ASTM E 2174.	PERIODIC	Inspect firestopping in accordance with referenced sections, and document inspection results to be submitted. The inspector must inspect the applications initially to ensure adequate preparations and periodically during the work to assure that the completed work has been accomplished according to the manufacturer's written instructions and the specified requirements.
2. Inspections of fire-resistant joint systems conducted in accordance with Section 07 84 00.00 22 and ASTM E 2393.	PERIODIC	

END SECTION

[1] **PERIODIC**: Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

SECTION 01 50 00.00 22
^^
TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS (PWD ME)
11/20, CHG 3: 08/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C511	(2017; R 2021) Reduced-Pressure Principle Backflow Prevention Assembly

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-530-01	(2023) Interior and Exterior Lighting Systems

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023) National Electrical Code
NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

U.S. FEDERAL AVIATION ADMINISTRATION (FAA)

FAA AC 70/7460-1	(2016; Rev L; Change 2) Obstruction Marking and Lighting

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)

MUTCD	(2009; Rev 2012) Manual on Uniform Traffic Control Devices

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

Construction Site Plan; G

Traffic Control Plan; G

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

Employee Parking Plan; G

SD-03 Product Data

Backflow Preventers; G

SD-06 Test Reports

Backflow Preventer Tests; G

SD-07 Certificates

Backflow Tester Certification; G

Backflow Preventers Certificate of Full Approval; G

1.3 CONSTRUCTION SITE PLAN

Prior to the start of work, submit for Government approval a site construction plan showing the locations and dimensions of temporary facilities (including layouts and details for equipment and material, and onsite and offsite storage areas), access and haul routes, avenues of ingress/egress to the fenced area and details of the temporary construction safety fencing/barrier systems that comply with EM 385-1-1.

Identify any areas where vehicle track pads will be installed or which may have to be graveled to prevent the tracking of mud outside the project site limits. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, dewatering system storage tanks, and infiltration pitsworker parking areas.

Prior to the start of Construction, furnish and erect temporary project safety fencing at the work site in accordance with the plans and the following requirements:

a. Temporary project fencing (or a substitute acceptable to the Contracting Officer (GDA) and delineated in the APP) must be provided on all projects. See also EM 385-1-1.

b. Fencing must extend from grade to a minimum of 6 feet above grade and must have a maximum mesh size of 2 inches. Fencing must remain rigid/taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection.

c. Signs warning of the presence of construction hazards and requiring unauthorized persons to keep out of the construction area must be posted on the fencing. At a minimum, signs must be posted every 150 feet. Fenced sides of projects that are less than 150 feet must, at minimum, have at least one warning sign.

d. Depending upon the nature and location of the project site, the Contractor may request not to install temporary fencing in some sections of the project site. The SSHO must submit a risk analysis (AHA) to the Contracting Officer for review. This must be based on a risk analysis of public exposure and other project specific considerations, and must be included in the applicable AHA.

e. Access gates to the job site must remain closed when not actively in use to prevent nonessential personnel from entering the construction area.

If the Contracting Officer approves the request and has determined fencing is not required, install signs and other acceptable barrier systems, warning of construction hazards, and must be conspicuously posted.

If at any time it is determined the risk to the public changes based on the work, take immediate action to address any risk to the public.

1.4 BACKFLOW PREVENTERS CERTIFICATE

1.4.1 Backflow Tester Certificate

Prior to testing, submit to the Contracting Officer certification issued by the State or local regulatory agency attesting that the backflow tester has successfully completed a certification course sponsored by the regulatory agency. Tester must not be affiliated with a company participating in other phases of this Contract.

1.4.2 Backflow Prevention Training Certificate

Submit a certificate recognized by the State or local authority that states the Contractor has completed at least 10 hours of training in backflow preventer installations. The certificate must be current.

1.5 DOD CONDITION OF READINESS (COR)

DOD will set the Condition of Readiness (COR) based on the weather forecast for sustained winds 50 knots (58 mph) or greater. Contact the Contracting Officer for the current COR setting.

Monitor weather conditions a minimum of twice a day and take appropriate actions according to the approved Emergency Plan in the accepted Accident Prevention Plan, EM 385-1-1 Emergency Plan requirements and the instructions below.

Unless otherwise directed by the Contracting Officer, comply with:

a. Condition FOUR (Sustained winds of 58 mph or greater expected within 72 hours): Normal daily jobsite cleanup and good housekeeping practices. Collect and store in piles or containers scrap lumber, waste material, and rubbish for removal and disposal at the close of each work day. Maintain the construction site including storage areas, free of accumulation of debris. Stack form lumber in neat piles less than 3.3 feet high. Remove all debris, trash, or objects that could become missile hazards. Review requirements pertaining to "Condition THREE" and continue action as necessary to attain "Condition FOUR" readiness. Contact Contracting Officer for weather and COR updates and completion of required actions.

b. Condition THREE (Sustained winds of 58 mph or greater expected within 48 hours): Maintain "Condition FOUR" requirements and commence securing operations necessary for "Condition ONE" which cannot be completed within 18 hours. Cease all routine activities which might interfere with securing operations. Commence securing and stow all gear and portable equipment. Make preparations for securing buildings. Reinforce or remove formwork and scaffolding. Secure machinery, tools, equipment, materials, or remove from the jobsite. Expend every effort to clear all missile hazards and loose equipment from general Portsmouth Naval Shipyard areas. Contact Contracting Officer for weather and COR updates and completion of required actions. Review requirements pertaining to "Condition TWO" and continue action as necessary to attain "Condition THREE" readiness.

c. Condition TWO (Sustained winds of 58 mph or greater expected within 24 hours): Secure the jobsite, and leave Government premises.

d. Condition ONE. (Sustained winds of 58 mph or greater expected within 12 hours): Contractor access to the jobsite and Government premises is prohibited.

1.6 CYBERSECURITY DURING CONSTRUCTION

Meet the following requirements throughout the construction process.

1.6.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. When used, Contractor computers must meet the following requirements along with the requirements of paragraph SECURITY RESPONSIBILITIES (PNSY) of Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME):

Note: Any Computers (including tablet, laptop, or other computers) must not be introduced into nor removed from the Portsmouth Naval Shipyard. Obtain approvals from the PNSY Security Officer via the Contracting Officer. Cameras, video equipment, or similar photographic equipment installed in computers must be disabled (e.g., removed and the void filled with epoxy). Proof of computer Portsmouth Naval Shipyard Security Officer approval must be with the tablet, laptop, or other computers at all times.

1.6.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.6.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. All computers used on this project must be scanned using the installed software at least once per day.

1.6.1.3 Passwords and Passphrases

The passwords and passphrases for all computers must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.6.1.4 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-50-0 Each Statement must be signed by a cybersecurity representative for the relevant company.

1.6.2 Temporary IP Networks

Temporary Contractor-installed IP networks may be used during construction. When used, temporary contractor-installed IP networks must meet the following requirements:

1.6.2.1 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than IP networks provided under this project or Government furnished IP networks provided for this purpose. Any and all network access from outside the project site is prohibited.

1.6.3 Government Access to Network

Government personnel, as defined, prescribed, and identified by the Contracting Officer, must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification. If there is a Government agency responsible for network access, identify that agency.

1.6.4 Temporary Wireless IP Networks

1.6.4.1 Contractor Provided Wi-Fi Network

Provide standalone Wi-Fi network for authorized Government personnel. Network access must be available throughout the jobsite and for the entire Contract period. The Government will use the Wi-Fi service for coordination in the field and related activities. Do not attach or connect the temporary Wi-Fi network to Government networks or systems unrelated to construction activities.

Password information must adhere to requirements outlined in Paragraph PASSWORDS AND PASSPHRASES. Provide password and Wi-Fi network name (SSID) to the Contracting Officer for Government personnel access. Wi-Fi performance must be capable of handling bandwidth to support Government and Contractor activities, including drawing and model access onsite. Provide a point of contact for reporting Wi-Fi outages.

1.6.4.2 Temporary Wireless IP Networks Security

In addition to the other requirements on temporary IP networks, temporary wireless IP (WiFi) networks must not interfere with existing wireless network and must use WPA2 security. Network names (SSID) for wireless networks must be changed from their default values.

1.6.5 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.6.6 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-50-0 Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

1.7 EMPLOYEE PARKING PLAN

Parking on the Portsmouth Naval Shipyard is limited and is not guaranteed. Contractor's employees & Subcontractor's employees must park privately owned vehicles in the Jamaica Island Parking Lot only. Parking is not allowed in any other parking lot or area on the Portsmouth Naval Shipyard. Employee parking must not interfere with existing and established parking requirements of the Portsmouth Naval Shipyard. The Contractor and subcontractors must not violate any Town of Kittery Parking regulations. If a privately owned vehicle is found parked outside the Jamaica Island parking lot or found violating any Shipyard or Town of Kittery Parking Regulations, the vehicle will be subject to towing, a non-compliance notice will be issued and may reflect poorly on the Prime Contractor's performance rating (CPARS) and noted in their final performance evaluation. There may not be sufficient on-yard parking available for all company-owned vehicles. The Contractor may need to make arrangement for off-shipyard parking of company-owned vehicles. Limited parking will not be the basis for a claim for additional time or costs. The Employee Parking Plan must include the anticipated number of Contractor and Subcontractor company-owned vehicles that will be required to support the construction effort. The plan must identify how the Contractor will ensure company-owned vehicles will park at the Jamaica Island Parking Lot only when not in active use, and include proposed off-yard parking and shuttling that may be required for company-owned vehicles. There is a Maine DOT Park & Ride located in Kittery https://www.kitteryme.gov/home/news/maine-department-transportation-launches-new-park-

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE

2.1.1 Bulletin Board

Prior to the commencement of work activities, provide a clear weatherproof covered bulletin board not less than 36 by 48 inches in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the Contract, Wage Rate Information poster, Safety and Health Information as required by EM 385-1-1 and other information approved by the Contracting Officer. Coordinate requirements herein with 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, and in location as approved by the

Contracting Officer.

2.1.2 Project Identification Signs

The requirements for the signs, their content, and location are as indicated and as specified in Section 01 58 00.00 22 PROJECT IDENTIFICATION (PWD ME). Erect signs within 15 days after receipt of the notice to proceed. Correct the data required by the safety sign daily, with light colored metallic or non-metallic numerals.

2.1.3 Warning Signs

Post temporary signs, tags, and labels to give workers and the public adequate warning and caution of construction hazards according to the EM 385-1-1. Attach signs to the perimeter fencing every 150 feet warning the public of the presence of construction hazards. Signs must require unauthorized persons to keep out of the construction site. Correct the data required by safety signs daily. Post signs at all points of entry designating the construction site as a hard hat area.

2.2 TEMPORARY TRAFFIC CONTROL

2.2.1 Haul Roads (Not Applicable)

2.2.2 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Barricades are required whenever safe public access to paved areas such as roads, parking areas or sidewalks is prevented by construction activities or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night.

2.3 FENCING

Provide fencing along the construction site and at all open excavations and tunnels to control access by unauthorized personnel. Safety fencing must be highly visible to be seen by pedestrians and vehicular traffic. All fencing must meet the requirements of EM 385-1-1. Remove the fence upon completion and acceptance of the work. Fencing must be installed to be able to restrain a force of at least 250 pounds against it unless otherwise specified.

2.3.1 Polyethylene Mesh Safety Fencing <u>(Not Permitted)</u>

2.3.2 Chain Link Panel Fencing

Temporary panel fencing must be galvanized steel chain link panels 6 feet high. Multiple fencing panels may be linked together at the bases to form long spans as needed. Each panel base must be weighted down using sand bags or other suitable materials in order for the fencing to withstand anticipated winds while remaining upright. Fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection. Completely remove fencing and posts at the completion of construction and restore surfaces disturbed or damaged to its original condition. Equip fence with a lockable gate. Gate must remain locked when construction personnel are not present.

2.3.3 Post-Driven Chain Link Fencing (Not Permitted)

2.4 TEMPORARY SECURITY LIGHTING

Provide temporary security lighting if any existing permanent security lighting is disturbed during the execution of the work. Any CIA lighting outages must be approved through Code 1122 and temporary lighting must be provided that is equal to or greater than the current foot-candle rating of the area. Refer to UFC 3-530-01, Table 6-2, for minimum lighting criteria footcandle ratings. The clear zone at the Portsmouth Naval Shipyard is 10 feet. Allow 5 working days for review of proposed temporary lighting by Code 1122.

2.5 TEMPORARY WIRING

Provide temporary wiring in accordance with EM 385-1-1, NFPA 241 and NFPA 70. Include monthly inspection and testing of all equipment and apparatus.

2.6 BACKFLOW PREVENTERS

Certificate of Full Approval from FCCCHR List, University of Southern California, attesting that the design, size and make of each backflow preventer has satisfactorily passed the complete sequence of performance testing and evaluation for the respective level of approval. Certificate of Provisional Approval is not acceptable.

Reduced pressure principle type conforming to the applicable requirements AWWA C511. Provide backflow preventers complete with 150 pound flanged cast iron, mounted gate valve on water lines 2 inches and smaller, strainer, and 304 stainless steel or bronze, internal parts.

Backflow Preventers must be reduced pressure principle type conforming to the applicable requirements AWWA C511. Provide backflow preventers complete with 150 pound flanged mounted gate valve, stainless steel or bronze, internal parts. The particular make, model/design, and size of backflow preventers to be installed must be included in the latest edition of the List of Approved Backflow Prevention Assemblies issued by the FCCCHR List and be accompanied by a Certificate of Full Approval from FCCCHR List. After installation conduct Backflow Preventer Tests and provide test reports verifying that the installation meets the FCCCHR Manual Standards.

PART 3 EXECUTION

3.1 EMPLOYEE PARKING

Refer to paragraph Employee Parking Plan for requirements.

3.2 AVAILABILITY AND USE OF UTILITY SERVICES

3.2.1 Temporary Utilities

Provide temporary utilities required for construction. Materials may be new or used, must be adequate for the required usage, not create unsafe conditions, and not violate applicable codes and standards.

3.2.2 Government Provided Utility Services

a. Reasonable amounts of the following utilities will be made available without charge.

Utility Services		
Electricity		
Potable Water		
Sea Water		
Compressed Air		
Steam		
Sanitary Sewer		

b. The Contractor must coordinate directly with Unitil, the natural gas provider for PNSY, for use of natural gas.

c. The point at which the Government will deliver such utilities or services and the quantity available must be coordinated with the Contracting Officer. Pay all costs incurred in connecting, converting, and transferring the utilities to the work. Make connections, including providing backflow-preventing devices on connections to domestic water lines; providing meters; and providing transformers; and make disconnections. Provide the backflow device as specified above and NAVFAC PWD ME shop personnel (at the Portsmouth Naval Shipyard) will install the backflow preventer. Do not operate any Portsmouth Naval Shipyard water system control valves. Notify the Contracting Officer a minimum of 15 calendar days prior to the desired date of the installation of the backflow device. Under no circumstances will taps to Shipyard fire hydrants be allowed for obtaining domestic water.

3.2.3 Sanitation

Provide and maintain within the construction area minimum field-type sanitary facilities in accordance with EM 385-1-1. Locate the facilities behind the construction fence or out of the public view. Clean units and empty wastes at least once a week or more frequently into a municipal, district, or Portsmouth Naval Shipyard sanitary sewage system, or remove waste to a commercial facility. Obtain approval from the system owner prior to discharge into a municipal, district, or Portsmouth Naval Shipyard sanitary sewer system. Penalties or fines associated with improper discharge will be the responsibility of the Contractor. Coordinate with the Contracting Officer and follow Portsmouth Naval Shipyard regulations and procedures when discharging into the Shipyard sanitary sewer system. Maintain these conveniences at all times without nuisance. Include provisions for pest control and elimination of odors. Government toilet facilities will not be available to Contractor's personnel.

3.2.4 Telephone

Make arrangements and pay all costs for telephone facilities desired.

3.2.5 Obstruction Lighting of Cranes

Provide a minimum of two (2) aviation red or high intensity white obstruction lights on temporary structures (including cranes) over 100 feet above ground level. Light construction and installation must comply with FAA AC 70/7460-1. Lights must be operational during periods of reduced visibility, darkness, and as directed by the Contracting Officer.

3.2.6 Fire Protection

Provide temporary fire protection equipment for the protection of personnel and property during construction. Remove debris and flammable materials daily to minimize potential hazards.

3.3 STATION OPERATION AFFECT ON CONTRACTOR OPERATIONS

3.3.1 Restricted Access Areas

The Government will monitor work in areas indicated. Notify Contracting Officer at least 14 calendar days prior to starting work in these areas.

3.4 TRAFFIC PROVISIONS

3.4.1 Maintenance of Traffic

a. Conduct operations in a manner that will not close a thoroughfare or interfere with traffic on railways or road except with written permission of the Contracting Officer at least 15 calendar days prior to the proposed modification date, and provide to the Contracting Officer a Traffic Control Plan for Government approval detailing the proposed controls to traffic movement for approval. The plan must be in accordance with State and local regulations and the MUTCD, Part VI. Contractor may move oversized and slow-moving vehicles to the worksite provided requirements of the State of Maine Department of Transportation have been met.

b. Conduct work so as to minimize obstruction of traffic, and maintain traffic on at least half of the roadway width at all times. Obtain approval from the Contracting Officer prior to starting any activity that will obstruct vehicle or pedstrian traffic.

c. Provide, erect, and maintain, at Contractor's expense, lights, barriers, signals, passageways, detours, Life Safety Signage, overhead protection, and other items that may be required by the authority having jurisdiction.

d. Provide cones, signs, barricades, lights, or other traffic control devices and personnel required to control traffic. Do not use foil-backed material for temporary pavement marking because of its potential to conduct electricity during accidents involving downed power lines.

e. The area adjacent to the project site is an active Shipyard work site. All work must be coordinated to avoid impacting Portsmouth Naval Shipyard operations. Work must not impact Portsmouth Naval Shipyard Fire Department access in any way unless otherwise approved by the Contracting Officer and Portsmouth Naval Shipyard Fire Department via the outage request process.

3.4.2 Protection of Traffic

Maintain and protect traffic on all affected roads during the construction period except as otherwise specifically directed by the Contracting Officer. Measures for the protection and diversion of traffic, including the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment the work, and the erection and maintenance of adequate warning, danger, and direction signs, will be as required by the State and local authorities having jurisdiction. Provide self-illuminated (lighted) barricades during hours of darkness. All personnel working in roadways will wear brightly-colored vests or other high visibility apparel in accordance with EM 385-1-1. Protect the traveling public from damage to person and property. Minimize the interference with public traffic on roads selected for hauling material to and from the site. Investigate the adequacy of existing roads and their allowable load limit. Contractor is responsible for the repair of damage to roads caused by construction operations.

3.4.3 Rush Hour Restrictions

Do not interfere with the peak traffic flows preceding and during normal operations without notification to and approval by the Contracting Officer.

3.4.4 Dust Control

Dust control methods and procedures must be approved by the Contracting Officer. Coordinate dust control methods with 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME).

3.5 REDUCED PRESSURE BACKFLOW PREVENTERS

Provide an approved reduced pressure backflow prevention assembly at each location where the Contractor taps into the Government potable water supply.

Perform backflow preventer tests using test equipment, procedures, and certification forms conforming to those outlined in the latest edition of the Manual of Cross-Connection Control published by the FCCCHR Manual. Test and tag each reduced pressure backflow preventer upon initial installation (prior to continued water use) and monthly thereafter. Tag must contain the following information: make, model, serial number, dates of tests, results, maintenance performed, and signature of tester. Record test results on certification forms conforming to requirements cited earlier in this paragraph.

3.6 CONTRACTOR'S TEMPORARY FACILITIES

Contractor is responsible for security of their property. Provide adequate outside security lighting at the temporary facilities. Trailers must be anchored to resist high winds and meet applicable state or local standards for anchoring mobile trailers. Coordinate anchoring with EM 385-1-1. The Contract Clause entitled "FAR 52.236-10, Operations and Storage Areas" and the following apply:

3.6.1 Administrative Field Offices

Provide and maintain administrative field office facilities within the construction area at the designated site. Government office and warehouse facilities will not be available to the Contractor's personnel.

In the event a new building is constructed for the temporary project field office, it must be a minimum 12 feet in width, 16 feet in length and have a minimum of 7 feet headroom. Equip the building with approved electrical wiring, at least one double convenience outlet and the required switches and fuses to provide 110-120 volt power. Provide a work table with stool, desk with chair, two additional chairs, and one legal size file cabinet that can be locked. The building must be waterproof, supplied with a heater, have a minimum of two doors, electric lights, a telephone, a battery-operated smoke detector alarm, a sufficient number of adjustable windows for adequate light and ventilation, and a supply of approved drinking water. Provide approved sanitary facilities. Screen the windows and doors and provide the doors with deadbolt type locking devices or a padlock and heavy-duty hasp bolted to the door. Door hinge pins must be non-removable. Arrange the windows to open and to be securely fastened from the inside. Protect glass panels in windows by bars or heavy mesh screens to prevent easy access. In warm weather, provide air conditioning capable of maintaining the office at 50 percent relative humidity and a room temperature 20 degrees F below the outside temperature when the outside temperature is 95 degrees F. Unless otherwise directed by the Contracting Officer, remove the building from the site upon completion and acceptance of the work.

3.6.2 Quality Control Manager Records and Field Office

Provide on the jobsite an office with useful floor area for the exclusive use of the QC Manager. Provide a weathertight structure with adequate heating and cooling, toilet facilities, lighting, ventilation, a 4 by 8 foot plan table, a standard size office desk and chair, computer station, and working communications facilities. Provide either a 1,500 watt radiant heater and a window-mounted air conditioner rated at 9,000 Btus minimum or a window-mounted heat pump of the same minimum heating and cooling ratings. Provide a door with a cylinder lock and windows with locking hardware. Make utility connections. Locate as directed. File quality control records in the office and make available at all times to the Government. After completion of the work, remove the entire structure from the site.

3.6.3 Storage Area

Construct a temporary 6 foot high chain link fence around trailers and materials. Fence posts may be driven, in lieu of concrete bases, where soil conditions permit. Do not place or store trailers, materials, or equipment outside the fenced area unless such trailers, materials, or equipment are assigned a separate and distinct storage area by the Contracting Officer away from the vicinity of the construction site but within the Portsmouth Naval Shipyard boundaries. Trailers, equipment, or materials must not be open to public view with the exception of those items which are in support of ongoing work on the current day. Do not stockpile materials outside the fence in preparation for the next day's work. Park mobile equipment, such as tractors, wheeled lifting equipment, cranes, trucks, and like equipment within the fenced area at the end of each work day.

Keep fencing in a state of good repair and proper alignment. If the Contractor elects to traverse grassed or unpaved areas which are not established roadways with construction equipment or other vehicles, cover the grassed or unpaved areas with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways; gravel gradation must be at the Contractor's discretion. Mow and maintain grass located within the boundaries of the construction site for the duration of the project. Grass and vegetation along fences, structures, under trailers, and in areas not accessible to mowers must be edged or trimmed neatly.

3.6.4 Supplemental Storage Area

Upon request, and pending availability, the Contracting Officer will designate another or supplemental area for the use and storage of trailers, equipment, and materials. This area may not be in close proximity of the construction site but will be within the Portsmouth Naval Shipyard boundaries. Maintain the area in a clean and orderly fashion and secured if needed to protect supplies and equipment. Utilities will not be provided to this area by the Government.

3.6.5 Appearance of Trailers

a. Trailers must be roadworthy and comply with all appropriate state and local vehicle requirements. Trailers which are rusted, have peeling paint or are otherwise in need of repair will not be allowed on Portsmouth Naval Shipyard property. Trailers must present a clean and neat exterior appearance and be in a state of good repair.

b. Maintain the temporary facilities. Failure to do so will be sufficient reason to require their removal at the Contractor's expense.

3.6.6 Trailers or Storage Buildings

a. Trailers or storage buildings will be permitted, where space is available, subject to the approval of the Contracting Officer.

b. Mount a sign not smaller than 13.78 by 13.78 inches on the trailer or building that shows the company name, business phone number, emergency phone number and conforms to the following requirements and sketch:

Graphic panel	Aluminum, painted blue
Copy	Screen painted or vinyl die-cut, white
Typeface	Univers 65 u/lc
See Sketch No. 01500 (graphic).	

3.6.7 Safety Systems

Protect the integrity of installed safety systems or personnel safety devices. Obtain prior approval from the Contracting Officer if entrance into systems serving safety devices is required. If it is temporarily necessary to remove or disable personnel safety devices in order to accomplish Contract requirements, provide alternative means of protection prior to removing or disabling any permanently installed safety devices or equipment and obtain approval from the Contracting Officer.

3.6.8 Special Storage Requirements

The following special storage requirements apply:

3.6.8.1 Laydown Space

Parking and laydown space on the site is limited to the area shown on the plans. Manage the on-site work including equipment, storage trailers, material, material deliveries to allow the work to be completed within the specified Contract duration. This may require the Contractor to locate suitable storage off-site and multiple equipment mobilizations to allow the work to be completed. Equipment or materials not used to complete the work must be removed from the site. If additional offsite storage, additional mobilization or demobilizations are required, all these costs must be included in the base bid.

Failure to plan the work based on the space limitations must not be the basis for any claim nor an equitable price or Contract time adjustment.

3.6.8.2 Storage in Existing Buildings

The Contractor will be working in existing building; the storage of material will be allowed where indicated.

3.6.9 Weather Protection of Temporary Facilities and Stored Materials

Take necessary precautions to ensure that roof openings and other critical openings in the temporary facilities are monitored carefully. Take immediate actions required to seal off such openings when rain or other detrimental weather is imminent, and at the end of each workday. Ensure that the openings are completely sealed off to protect materials and equipment in the temporary facilities from damage.

3.6.9.1 Building and Site Storm Protection

When a warning of gale force winds is issued, take precautions to minimize danger to persons, and protect the work and nearby Government property. Precautions must include, but are not limited to, closing openings; removing loose materials, tools and equipment from exposed locations; and removing or securing scaffolding and other temporary work. Close openings in the work when storms of lesser intensity pose a threat to the work or any nearby Government property.

3.7 TEMPORARY PARTITIONS

Provide "No Dust" temporary partitions of wood or metal frame, heavy duty plastic sheathing and negative pressure HEPA filtered ventilation. Provide access door and vestibules as required to prevent dust from escaping the enclosure. Refer to Section 02 41 00 DEMOLITION AND DECONSTRUCTION for additional information and requirements if included as part of the specifications.

3.8 PLANT COMMUNICATIONS

Whenever the individual elements of the plant are located so that operation by normal voice between these elements is not satisfactory, install a satisfactory means of communication, such as telephone or other suitable devices and make available for use by Government personnel.

3.9 TEMPORARY PROJECT SAFETY FENCING

As soon as practicable, but not later than 15 days after the date established for commencement of work on a definable feature of work (DFOW) or in a work area, furnish and erect temporary project safety fencing at the work site. Maintain the safety fencing during the life of the work in that area and, upon completion and acceptance of the work, remove from the work site.

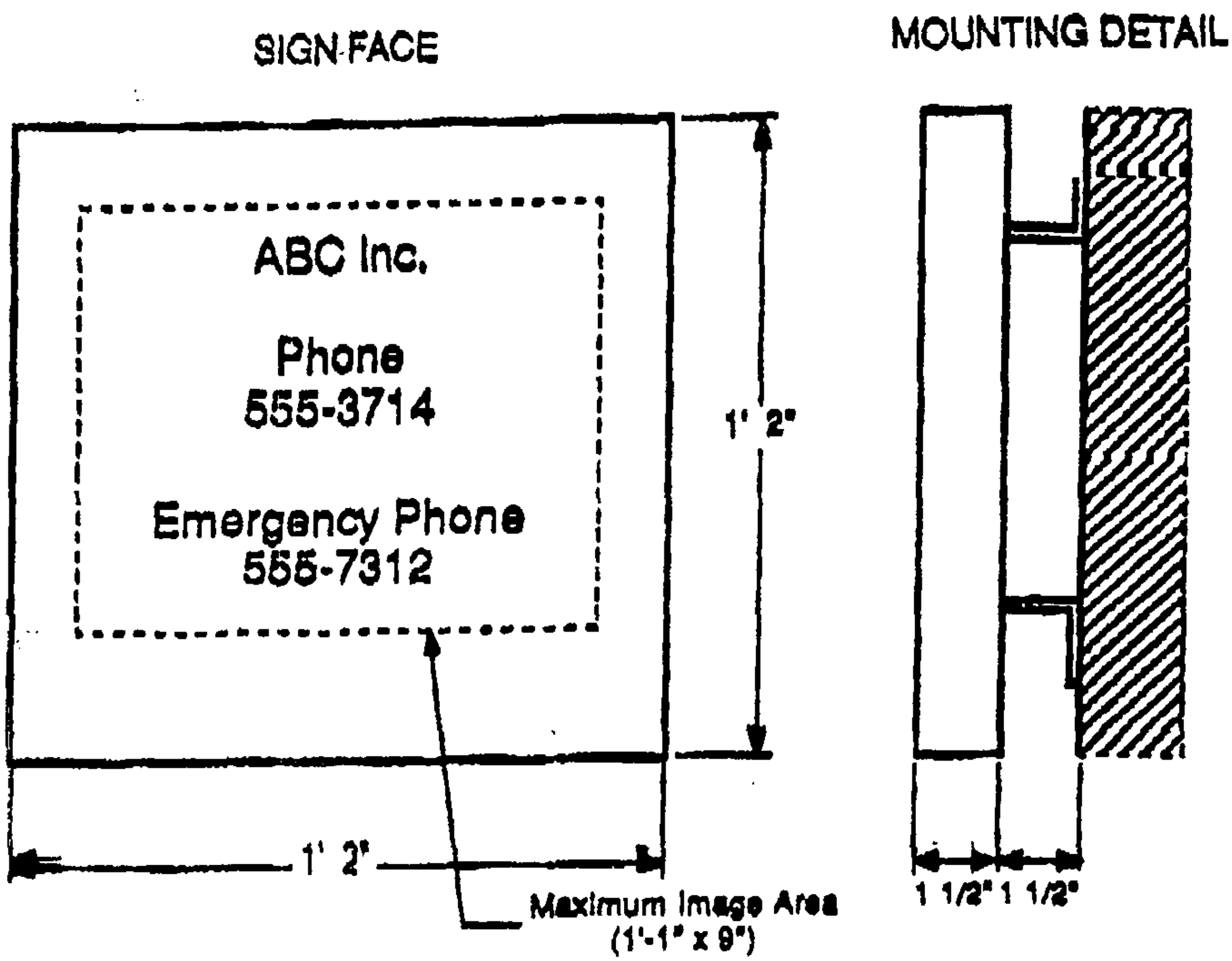
3.10 CLEANUP

Remove construction debris, waste materials, packaging material and the like from the work site daily. Any dirt or mud which is tracked onto paved or surfaced roadways must be cleaned away. Store all salvageable materials resulting from demolition activities within the fenced area described above or at the supplemental storage area. Neatly stack stored materials not in trailers, whether new or salvaged.

3.11 RESTORATION OF STORAGE AREA

Upon completion of the project remove the bulletin board, signs, barricades, haul roads, and all other temporary products from the site. After removal of trailers, materials, and equipment from within the fenced area, remove the fence. Restore areas used during the performance of the Contract to the original or better condition. Remove gravel used to traverse grassed areas and restore the area to its original condition, including top soil and seeding as necessary.

-- End of Section --



Sign requirements:

Graphic panel: Aluminum, painted blue
Copy: Screen painted or vinyl die-cut, white
Typeface: Univers 65 u/lc

SECTION 01 57 19.00 22
^^
TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME)
08/22

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard only.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only. Note: This is not an all-inclusive list of publications and other references may be applicable.

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA SW-846	(Third Edition; Update IV) Test Methods for Evaluating Solid Waste: Physical/Chemical Methods

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.120	Hazardous Waste Operations and Emergency Response
40 CFR 50	National Primary and Secondary Ambient Air Quality Standards
40 CFR 82	Protection of Stratospheric Ozone
40 CFR 112	Oil Pollution Prevention
40 CFR 241	Guidelines for Disposal of Solid Waste
40 CFR 243	Guidelines for the Storage and Collection of Residential, Commercial, and Institutional Solid Waste
40 CFR 258	Subtitle D Landfill Requirements
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 261.7	Residues of Hazardous Waste in Empty Containers
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of

	Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 266	Standards for the Management of Specific Hazardous Wastes and Specific Types of Hazardous Waste Management Facilities
40 CFR 268	Land Disposal Restrictions
40 CFR 270	EPA Administered Permit Programs: The Hazardous Waste Permit Program
40 CFR 272	Approved State Hazardous Waste Management Programs
40 CFR 273	Standards for Universal Waste Management
40 CFR 279	Standards for the Management of Used Oil
40 CFR 280	Technical Standards and Corrective Action Requirements for Owners and Operators of Underground Storage Tanks (UST)
40 CFR 300	National Oil and Hazardous Substances Pollution Contingency Plan
40 CFR 300.125	National Oil and Hazardous Substances Pollution Contingency Plan - Notification and Communications
40 CFR 302	Designation, Reportable Quantities, and Notification
40 CFR 355	Emergency Planning and Notification
40 CFR 372-SUBPART D	Specific Toxic Chemical Listings
40 CFR 761	Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions
49 CFR 171	General Information, Regulations, and Definitions
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
49 CFR 178	Specifications for Packagings
40 CFR 302	Designation, Reportable Quantities, and

Notification

STATE OF MAINE REGULATIONS

The following STATE OF MAINE REGULATIONS are available on the Internet at: http://www.maine.gov/dep/permits.htm

STATE OF MAINE Statutes are available on the internet at https://legislature.maine.gov/statutes/38/title38ch0sec0.html

STATE OF MAINE RULES are available on the internet at https://www.maine.gov/sos/rulemaking/agency-rules/department-environmental-protection-
MAINE DEP AIR BUREAU CHAPTER 101 Visible Emissions Regulations; http://www.maine.gov/dep/air/rules/index.html

MAINE DEP AIR BUREAU CHAPTER 151 Architectural and Industrial Maintenance (AIM) COATINGS; http://www.maine.gov/dep/air/rules/index.html

MAINE DEP AIR BUREAU CHAPTER 159 Control of Volatile Organic Compounds from Adhesives and Sealants; http://www.maine.gov/dep/air/rules/index.html

MAINE DEP 38 MSRA 420-C	Erosion and Sedimentation Control Law and Rules
MAINE DEP 38 MSRA 420-D	Stormwater Management
MAINE 38 MRSA 439-B	Contractors Certified in Erosion Control (Latest Edition)
MAINE DEP MSRA 481-490	Site Location of Development
MAINE 38 MSRA 850	Identification of Hazardous Waste
MAINE 38 MSRA 851	Standards for Generators of Hazardous Waste
MAINE 38 MSRA 852	Land Disposal Restrictions
MAINE 38 MSRA 853	Licensing of Transporters of Hazardous Waste
MAINE 38 MSRA 857	Hazardous Waste Manifest Requirements
MAINE 38 MSRA 858	Universal Waste Rules
MAINE 38 MSRA 860	Waste Oil Management Rules
MAINE 88 MRSA 480A-480Z	Natural Resources Protection Act
CODE OF MAINE RULE 06-096	Chapter 500 Stormwater Management
CODE OF MAINE RULE 06-096	Chapter 573 Snow Dumps: Best Management Practices for Pollution Prevention
MAINE DEP	Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition.
MEPDES Permit #MER042000	General Permit for the Discharge of Stormwater from State or Federally Owned Municipal Separate Storm Sewer Systems

PORTSMOUTH NAVAL SHIPYARD INSTRUCTIONS

NAVSHIPY PTSMH INST 5090.8 Environmental Sampling Manual

NAVSHIPY PTSMH INST 5090.30 Hazardous Waste Generator Standards

1.2 DEFINITIONS

1.2.1 Class I and II Ozone Depleting Substance (ODS)

Class I ODS is defined in Section 602(a) of The Clean Air Act. A list of Class I ODS can be found on the EPA website at the following weblink. https://www.epa.gov/ozone-layer-protection/ozone-depleting-substances.

Class II ODS is defined in Section 602(s) of The Clean Air Act. A list of Class II ODS can be found on the EPA website at the following weblink. https://www.epa.gov/ozone-layer-protection/ozone-depleting-substances.

1.2.2 Chemical Wastes

This includes salts, acids, alkalies, herbicides, pesticides, and organic chemicals.

1.2.3 Contractor Generated Hazardous Waste

Contractor generated hazardous waste is materials that, if abandoned or disposed of, may meet the definition of a hazardous waste. These waste streams would typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e., methyl ethyl ketone, toluene), waste thinners, excess paints, excess solvents, waste solvents, excess pesticides, and contaminated pesticide equipment rinse water.

1.2.4 Electronics Waste

Electronics waste is discarded electronic devices intended for salvage, recycling, and disposal.

1.2.5 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally or historically.

1.2.6 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

1.2.7 Garbage

Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.

1.2.8 Hazardous Debris

As defined in paragraph SOLID WASTE , debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) in accordance with 40 CFR 261. Hazardous debris also includes debris that exhibits a characteristic in accordance with 40 CFR 261.

1.2.9 Hazardous Materials

Hazardous material is any material that: Is defined in 49 CFR 171, and listed in 49 CFR 172, regulated as a hazardous material in accordance with 49 CFR 173, or requires a Safety Data Sheet (SDS) per 29 CFR 1910.120; or during end use, treatment, handling, packaging, storage, transportation, or disposal meets or has components that meet or have potential to meet the definition of a hazardous waste as defined by 40 CFR 261 Subparts A, B, C, or D.

Designation of a material by this definition, when separately regulated or controlled by other sections or directives, does not eliminate the need for adherence to that hazard-specific guidance which takes precedence over this instruction for "control" purposes. Such material includes ammunition, weapons, explosive actuated devices, propellants, pyrotechnics, chemical and biological warfare materials, medical and pharmaceutical supplies, medical waste and infectious materials, bulk fuels, radioactive materials, and other materials such as asbestos, mercury, and polychlorinated biphenyls (PCBs).

1.2.10 Hazardous Waste

Hazardous Waste is any material that meets the definition of a solid waste and exhibits a hazardous characteristic (ignitability, corrosivity, reactivity, or toxicity) as specified in 40 CFR 261, Subpart C, or contains a listed hazardous waste as identified in 40 CFR 261, Subpart D or as identified by Maine Department of Environmental Protection (MEDEP) MAINE 38 MSRA 850-860. Hazardous waste controls also apply to Universal Wastes.

1.2.11 Land Application

Land Application means spreading or spraying discharge water at a rate that allows the water to percolate into the soil. No sheeting action, soil erosion, discharge into storm sewers, discharge into defined drainage areas, or discharge into the "waters of the United States" must occur. Comply with federal, state, and local laws and regulations.

1.2.12 Municipal Separate Storm Sewer System (MS4) Permit

MS4 permits are those held by municipalities or installations to obtain NPDES permit coverage for their stormwater discharges.

1.2.13 National Pollutant Discharge Elimination System (NPDES)

The NPDES permit program controls water pollution by regulating point sources that discharge pollutants into waters of the United States.

1.2.14 Non-Emergency Spill Event

A non-emergency spill event is a discharge of a known material or any hazardous substance that does not pose an immediate threat to human health or the environment, can be cleaned up as part of normal housekeeping by the personnel who discovered the spill, and is not released on the soil or into any waterway inlet (for example, storm drain) or outside Navy property boundaries.

1.2.15 Oily Waste

Oily waste are those materials which are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from that POLs. Oily wastes also means materials, including wastewaters, centrifuge solids, filter residues or sludges, bottom sediments, tank bottoms, and sorbents which have come into contact with and have been contaminated by POLs.

All oily waste must be collected and turned in to Building 357 for disposal. A hazardous waste determination form may be required prior to disposal. All waste characterization sampling must be conducted in accordance with Paragraph 3.13.4.2 herein entitled SAMPLING AND ANALYSIS OF HAZARDOUS WASTE. Oily waste must be minimized through good housekeeping practices and employee education.

1.2.16 Ozone Depleting Substance (ODS)

Chlorofluorocarbons (CFCs), halons or chlorinated hydrocarbons (such as carbon tetrachloride and methyl chloroform), and hydrochlorofluorocarbon (HCFCs) which have been linked to depletion of the earth's ozone layer are all substances collectively known as ozone depleting substances or ODSs. Class I or Class II ODS substances are defined and listed in the Clean Air Act Section 602 and 40 CFR 82.

1.2.17 Regulated Waste

Those solid wastes that have specific additional Federal, State, or local controls for handling, storage, or disposal.

1.2.18 Reportable Release

A reportable release means any spilling, leaking, pumping, pouring, emitting, emptying, discharging, injecting, escaping, leaching, dumping, or disposing into the environment of a known or unknown material or hazardous substance that poses an immediate threat to human health or the environment to the air, soil, or water. Reportable releases are: a sheen of oil on the water; a violation of the Installation's or project's water permit (NPDES permit(s)); A sewage spill that threatens human health or the environment; a Comprehensive Environmental Response, Compensation, and Liability Act reportable quantity for hazardous/toxic substances (40 CFR 302); an air or hazardous substance release that is a threat to human health or the environment or released outside the facility boundaries.

1.2.19 Sediment

Sediment is soil and other debris that have eroded and have been transported by runoff water or wind.

1.2.20 Solid Waste

Solid waste is a solid, liquid, semi-solid, or contained gaseous waste. A solid waste can be a hazardous waste, non-hazardous waste, or non-Resource Conservation and Recovery Act (RCRA) regulated waste. Types of solid waste typically generated at construction sites may include:

1.2.20.1 Debris

Debris is non-hazardous solid material generated during the construction, demolition, or renovation of a structure which exceeds 2.5 inch particle size that is: a manufactured object; plant or animal matter; or natural geologic material (for example, cobbles and boulders), broken or removed concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials may be reinforced with or contain ferrous wire, rods, accessories, and weldments. A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

1.2.20.2 Green Waste

Green waste is the vegetative matter from landscaping, land clearing and grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps, and plant roots. Marketable trees, grasses, and plants that are indicated to remain, be re-located, or be re-used are not included.

1.2.20.3 Material Not Regulated as Solid Waste

Material not regulated as solid waste is nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

1.2.20.4 Non-Hazardous Waste

Non-hazardous waste is waste that is excluded from, or does not meet, hazardous waste criteria in accordance with 40 CFR 261.

1.2.20.5 Recyclables

Recyclables are materials, equipment, and assemblies such as doors, windows, door and window frames, plumbing fixtures, glazing, and mirrors that are recovered and sold as recyclable, and structural components. It also includes commercial-grade refrigeration equipment with Freon removed, household appliances where the basic material content is metal, clean polyethylene terephthalate bottles, cooking oil, used fuel oil, textiles, high-grade paper products and corrugated cardboard, stackable pallets in good condition, clean crating material, and clean rubber/vehicle tires. Paint cans that meet the definition of empty containers in accordance with 40 CFR 261.7 may be included as recyclable if sold to a scrap metal company.

1.2.20.6 Surplus Soil

Surplus soil is existing soil that is in excess of what is required for this work, including aggregates intended, but not used, for on-site mixing of concrete, mortars, and paving. Contaminated soil meeting the definition of hazardous material or hazardous waste is not included and must be managed in accordance with paragraph HAZARDOUS MATERIAL MANAGEMENT.

1.2.20.7 Scrap Metal

This includes scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the definition of hazardous material or hazardous waste is not included. Scrap metal coated with PCB contaminated paint > or = to 50 ppm is considered a hazardous waste and requires Code 106.3 coordination for recycling/disposal.

1.2.20.8 Wood

Wood is dimension and non-dimension lumber, plywood, chipboard, and hardboard. Treated and/or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included. Treated wood includes, but is not limited to, lumber, utility poles, crossties, and other wood products with chemical treatment.

1.2.20.9 Paint Cans

Metal cans that are RCRA empty of paints, solvents, thinners, and adhesives. NOTE: Aerosol (paint) cans are Hazardous Wastes and must not be disposed of as solid waste or be considered in any definition of "empty", "paint", or "metal" cans.

1.2.20.10 Special Waste

Special Waste: "Special waste" means any solid waste generated by sources other than household and typical commercial establishments that do not meet the definition of a hazardous waste but exists in such an unusual quantity or in such a chemical or physical state, or any combination thereof, that may disrupt or impair effective waste management or threaten the public health, human safety, or the environment and requires special handling, transportation, and disposal procedures. Special waste includes, but is not limited to:

(1) Ash;

(2) Industrial and industrial process waste;

(3) Sludge and dewatered septage;

(4) Debris from nonhazardous chemical spills and cleanup of those spills;

(5) Contaminated soils and dredge materials;

(6) Asbestos and asbestos-containing waste;

(7) Sand blast grit and non-liquid paint waste;

(8) PCB and PCB contaminated waste;

(9) High and low pH waste;

(10) Spent filter media residue;

(11) Shredder residue; and

(12) Railroad ties with or without creosote.

1.2.21 Surface Discharge

Surface discharge means discharge of water into drainage ditches, storm sewers, or creeks meeting the definition of "waters of the United States". Surface discharges from construction sites are discrete, identifiable sources and require a permit from the governing agency. Comply with federal, state, and local laws and regulations.

1.2.22 Wastewater

Wastewater is the used water and solids that flow through a sanitary sewer to a treatment plant.

1.2.23 Stormwater

Stormwater is any precipitation in an urban or suburban area that does not evaporate or soak into the ground, but instead collects and flows into storm drains, rivers, and streams.

1.2.24 Waters of the United States

Waters of the United States means Federally jurisdictional waters, including wetlands, that are subject to regulation under Section 404 of the Clean Water Act or navigable waters, as defined under the Rivers and Harbors Act.

1.2.25 Wetlands

Wetlands are those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions.

1.2.26 Sewage

Liquid waste designated by the Government as "domestic sanitary sewage" and normally discharged through domestic sanitary sewage systems. Liquids designated as "sewage" include human body waste, and wastewater from sinks, showers, laundries, dishwashers, and garbage disposals when these liquids use only chemicals approved by the Government for discharge into the sanitary sewer.

1.2.27 Spill Event

A spill is any release of oil or hazardous substances to the water or ground that is not controlled or permitted. This includes any spilling, leaking, pumping, emitting, discharging, injecting, escaping, leaching, disposing, or dumping of liquid or solid material that is not authorized in writing by the Contracting Officer.

1.2.28 Universal Waste

The universal waste regulations streamline collection requirements for certain hazardous wastes in the following categories: batteries, pesticides, and mercury-containing equipment (e.g., thermostats) and lamps (e.g., fluorescent bulbs). The rule is designed to reduce hazardous waste in the municipal solid waste (MSW) stream by making it easier for universal waste handlers to collect these items and send them for recycling or proper disposal. These regulations can be found at 40 CFR 273.

1.2.29 Waste Hazardous Material (WHM)

Any waste material which because of its quantity, concentration, or physical, chemical, or infectious characteristics may pose a substantial hazard to human health or the environment and which has been so designated. Used oil not containing any hazardous waste, as defined above, falls under this definition.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

Preconstruction Survey; G

Solid Waste Management Plan; G

Regulatory Notifications; G

Environmental Management Plan (EMP); G

Dirt and Dust Control Plan; G

Contractor Hazardous Material Inventory Log; G

Stormwater Management/Erosion and Sedimentation Control Plan; G

Spill Prevention, Control, and Countermeasures (SPCC) Plan; G

Environmental Manager Qualifications; G

Employee Training Records; G

SD-06 Test Reports

Laboratory Analysis; G

Disposal Requirements; G

Erosion and Sediment Control Inspection and Corrective Action; G

Monthly Solid Waste Disposal Report; G

SD-07 Certificates

ECATTS Certificate Of Completion; G

Erosion And Sediment Control Inspector; G

SD-11 Closeout Submittals

Some of the records listed below are also required as part of other submittals. For the "Records" submittal, maintain on-site a separate three-ring Environmental Records binder and submit at the completion of the project. Make separate parts to the binder corresponding to each of the applicable sub items listed below.

Stormwater Management and Erosion Control Compliance Notebook; G

Waste Determination Documentation; G

Disposal Documentation for Hazardous and Regulated Waste; G

Contractor Hazardous Material Inventory Log; G

Hazardous Waste/Debris Management; G

Regulatory Notifications; G

Asbestos Free Certification Form; G

Assembled Employee Training Records; G

Sales Documentation; G

1.4 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the Contract, environmental protection as defined. Plan for and provide environmental protective measures to control pollution that develops during construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Protect the environmental features within the project boundaries and those affected outside the limits of permanent work during the entire duration of this Contract. Comply with Federal, State, and local regulations pertaining to the environment, including water, air, solid waste, hazardous waste and substances, oily substances, and noise pollution.

Tests and procedures assessing whether construction operations comply with Applicable Environmental Laws may be required. Analytical work must be performed by qualified laboratories; and where required by law, the laboratories must be certified.

1.4.1 Training in Environmental Compliance Assessment Training and Tracking System (ECATTS)

The Environmental Manager is responsible for environmental compliance on projects. The Environmental Manager must complete ECATTS training modules (installation specific or general) prior to starting respective portions of on-site work under this Contract. If personnel changes occur for any of these positions after starting work, replacement personnel must complete ECATTS training within 14 days of assignment to the project.

Submit an ECATTS certificate of completion for personnel who have completed the required "Environmental Compliance Assessment Training and Tracking System (ECATTS)" training. This training is web-based and can be accessed from any computer with Internet access using the following instructions.

Register for NAVFAC Environmental Compliance Assessment, Training and Tracking System, by logging on to https://environmentaltraining.ecatts.com. Obtain the password for registration from the Contracting Officer.

This training has been structured to allow Contractor personnel to receive credit under this contract and also to carry forward credit to future contracts. Ensure the Environmental Manager review their training plans for new modules or updated training requirements prior to beginning work. Some training modules are tailored for specific State regulatory requirements; therefore, working in multiple states will require personnel to retake modules tailored to the State where the Contract work is being performed.

ECATTS is available for use by all personnel associated with this project. These other personnel are encouraged (but not required) to take the training and may do so at their discretion.

1.4.2 Conformance with the Environmental Management System

Perform work under this contract consistent with the policy and objectives identified in the installation's Environmental Management System (EMS). Perform work in a manner that conforms to objectives and targets of the environmental programs and operational controls identified by the EMS. Support Government personnel when environmental compliance and EMS audits are conducted by escorting auditors at the Project site, answering questions, and providing proof of records being maintained. Provide monitoring and measurement information as necessary to address environmental performance relative to environmental, energy, and transportation management goals. In the event an EMS nonconformance or environmental noncompliance associated with the contracted services, tasks, or actions occurs, take corrective and preventative actions. In addition, employees must be aware of their roles and responsibilities under the installation EMS and of how these EMS roles and responsibilities affect work performed under the contract.

Coordinate with the installation's EMS coordinator to identify training needs associated with environmental aspects and the EMS, and arrange training or take other action to meet these needs. Provide training documentation to the Contracting Officer. The Installation Environmental Office will retain associated environmental compliance records. Make EMS Awareness training completion certificates available to Government auditors during EMS audits and include the certificates in the Employee Training Records. See paragraph EMPLOYEE TRAINING RECORDS.

1.5 QUALITY ASSURANCE

1.5.1 Preconstruction Survey and Protection of Features

This paragraph supplements the Contract Clause PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS. Prior to start of any onsite construction activities, perform a Preconstruction Survey of the project site with the Contracting Officer, and when requested,take photographs showing existing environmental conditions in and adjacent to the site. Submit a report for the record with a copy provided to the Contracting Officer. Obtain a camera pass from Portsmouth Naval Shipyard Security (via Contracting Officer) prior to use of a camera at Portsmouth Naval Shipyard. Digital cameras can only be used. Digital cameras shall not have bluetooth or wifi capabilities. All computer discs must be turned over to Portsmouth Naval Shipyard Security (via Contracting Officer) for review and clearance prior to use. Include in the report a plan describing the features requiring protection under the provisions of the Contract Clauses, which are not specifically identified on the drawings as environmental features requiring protection along with the condition of trees, shrubs and grassed areas immediately adjacent to the site of work and adjacent to the Contractor's assigned storage area and access route(s), as applicable. The Contractor and Contracting Officer will sign this survey report upon mutual agreement regarding its accuracy and completeness. Protect those environmental features included in the survey report and any indicated on the drawings, regardless of interference that their preservation may cause to the work under the Contract.

1.5.2 Regulatory Notifications

Provide regulatory notification requirements in accordance with Federal, State, and local regulations. In cases where the Government will also provide public notification (such as stormwater permitting), coordinate with the Contracting Officer. Submit copies of all regulatory notifications to the Contracting Officer at least 30 calendar days prior to commencement of work activities. Typically, regulatory notifications must be provided for the following (this listing is not all inclusive): demolition, renovation, NPDES defined site work, construction, removal or use of a permitted air emissions source, and remediation of controlled substances (asbestos, hazardous waste, lead paint).

1.5.3 Environmental Brief

Attend an Environmental Brief prior to commencing any work on the Portsmouth Naval Shipyard. The Brief will be conducted by the Contracting Officer with PWD-ME EV staff & Shipyard C106 Environmental Representatives. The Brief will be part of the Preconstruction Meeting agenda or held as a separate meeting depending on the size and scope of work and potential environmental impact, as determined by the Government.

The meeting will discuss details for all the submittals that are required in accordance with this section including review of the submittal requirements for the Environmental Management Plan, important notes regarding Installation Restoration (IR), Historic, and Cultural Resources, Wetlands, Waste Generation, Concrete Wash Water/Wastewater, Hazard Materials Management including sampling, testing and disposal requirements and other applicable topics.

At the Brief, provide the following information: types, quantities, and use of hazardous materials that will be brought onto the Portsmouth Naval Shipyard; types and quantities of wastes/wastewater that may be generated during the Contract; types and quantities of oil that will be brought onto the Portsmouth Naval Shipyard; pollution control measures for spill prevention and control, and any bulk oil storage container information including quantity and type of product stored. Discuss the results of the Preconstruction Survey at this time.

Develop a mutual understanding relative to the details of environmental protection, including measures for protecting natural resources, historic properties, required reports, required permits, specific permit requirements, and other measures to be taken. Identify additional environmental concerns specific to the site (e.g., historic, archaeological and natural resources, IR, erosion and sediment control, spill prevention and control, soil management, testing and disposal requirements, etc.).

1.5.4 Environmental Manager

Appoint in writing an Environmental Manager for the project site. The Environmental Manager is directly responsible for coordinating contractor compliance with Federal, State, local, and Portsmouth Naval Shipyard requirements. The Environmental Manager cannot perform the duties of the Project Superintendent, nor the SSHO, nor the Quality Control Manager, nor the Project Manager. The Environmental Manager must ensure compliance with Hazardous Waste Program requirements (including hazardous waste handling, storage, daily turn-in, manifesting, and disposal as applicable); implement the EMP; ensure environmental permits are obtained, maintained, and closed out; ensure compliance with Stormwater Program Management requirements; ensure compliance with Hazardous Materials (storage, handling, and reporting) requirements; and coordinate any remediation of regulated substances (lead, asbestos, PCB containing building materials, PCB Universal Waste/e-Waste, etc.). The person in this position must be trained to adequately accomplish the following duties: ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to containers; ensure all personnel are trained in 40 CFR requirements in accordance with their position requirements; coordinate removal or turn-in of waste containers; implement, inspect, and maintain erosion and sediment controls as required by State law; and maintain the Environmental Records binder and required documentation, ensure compliance with all Spill Prevention, Control, and Countermeasures (SPCC) requirements, including but not limited to, the proper storage of tanks and containers and their secondary containment, inspections, spill procedures, etc. including environmental permits compliance and close-out. Submit Environmental Manager Qualifications to the Contracting Officer.

1.5.5 Employee Training Records

Prepare and maintain employee training records throughout the term of the contract meeting applicable 40 CFR requirements. Provide Employee Training Records in the Environmental Records Binder. Ensure every employee completes a program of classroom instruction or on-the-job training that teaches them to perform their duties in a way that ensures compliance with Federal, State, and local regulatory requirements for RCRA Large Quantity Generator. Provide a Position Description for each employee, by subcontractor, based on the Davis-Bacon Wage Rate designation or other equivalent method, evaluating the employee's association with hazardous and regulated wastes. This Position Description will include training requirements as defined in 40 CFR 265 for a Large Quantity Generator facility. Submit these Assembled Employee Training Records to the Contracting Officer at the conclusion of the project, unless otherwise directed.

Train personnel to meet EPA and state requirements. Conduct environmental protection/pollution control meetings for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and avoiding pollution; familiarization with statutory and contractual pollution standards; installation and care of devices, vegetative covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental protection/pollution control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of archaeological sites, artifacts, waters of the United States, and endangered species and their habitat that are known to be in the area. Provide a copy of the Erosion and Sediment Control Inspector Certification as required by Maine.

1.5.6 Non-Compliance Notifications

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with federal, state or local environmental laws or regulations, permits, and other elements of the Contractor's EPP. After receipt of such notice, inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. FAR 52.242-14 Suspension of Work provides that a suspension, delay, or interruption of work due to the fault or negligence of the Contractor allows for no adjustments to the contract for time extensions or equitable adjustments. In addition to a suspension of work, the Contracting Officer may use additional authorities under the contract or law.

1.6 SOLID WASTE DISPOSAL PLAN

Provide a Solid Waste Disposal Plan to Portsmouth Naval Shipyard Environmental Division (Code 106.3) and NAVFAC Environmental (PWD-ME EV) Staff in accordance with Paragraph entitled SOLID WASTE MANAGEMENT PLAN in Part 3 of this Section.

1.7 SITE WASTE REMOVAL (SWR) MEETING

A SWR meeting must be held with OICC PNSY, PWD-ME EV staff, and Code 106.3 fourteen (14) calendar days prior to the removal of wastes from a project site to ensure all pertinent requirements of the Standard Operating Procedure for Site Waste Removal(SWR)(Attachment A) have been met. No transport of wastes will be allowed until full concurrence is provided by OICC PNSY, PWD-ME EV staff, and Code 106.3 following the satisfactory completion of the SWR.

1.8 ENVIRONMENTAL RECORDS BINDER

Maintain on-site a separate three-ring Environmental Records Binder and submit at the completion of the project. Make separate parts within the binder that correspond to each submittal listed under paragraph CLOSEOUT SUBMITTALS in this section.

1.9 FACLITY HAZARDOUS WATER GENERATOR STATUS

Portsmouth Naval Shipyard is designated as a Large Quantity Generator. Meet the regulatory requirements of this generator designation for any work conducted within the boundaries of the Portsmouth Naval Shipyard. Comply with provisions of federal, state, and local regulatory requirements applicable to this generator status regarding training and storage, handling, and disposal of construction derived wastes.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 ENVIRONMENTAL MANAGEMENT PLAN (EMP)

Prior to initiating any work on site, the Contractor will meet with the Contracting Officer, Shipyard Code 106.3 Representatives, and PWD-ME EV Staff at the Environmental Brief to discuss the proposed Environmental Management Plan (EMP) and develop a mutual understanding relative to the details of environmental protection required to be addressed in the EMP, including measures for protecting natural resources and other measures to be taken. See paragraph ENVIRONMENTAL BRIEF for additional details on the meeting.

The EMP must be submitted in the following format and include all sections specified below, unless a separate plan submittal is accomplished for the EMP section as identified in a.(1)(b). All sections not applicable to the work shall be marked N/A.

a. Description of the Environmental Management Plan

(1) General overview and purpose

(a) A brief description of the work to be performed to include each type of work activity that will potentially impact the environment (e.g. soil excavation, hazardous waste, abatement (PCB, Lead, Asbestos), concrete slurry/wastewater, emergency generator installation, paint blasting, 55 gallon and greater oil/fuel storage etc.).

(b) A list of other plans required by environmental permit or being submitted separately (e.g. Solid Waste Management Plan, Hazardous Waste Management Plan, Storm Water Pollution Prevention Plan (SWPPP), Spill Prevention Controls and Countermeasures (SPCC) plan, etc).

(c) A copy of any standard or project specific operating procedures that will be used to effectively manage and protect the environment on the project site.

(d) Communication and training procedures that will be used to convey environmental management requirements to employees and Subcontractors.

(e) Emergency contact information for senior construction project contractor personnel and those responsible for environmental management and spill response (office phone number, cell phone number, and e-mail address).

(2) General site information including a site plan showing haul routes, construction material laydown and storage areas, tracking pads, construction trailer locations, sanitary facilities, excavation and soil stockpile locations, dewatering and infiltration areas, location map showing selected erosion and sediment controls, and all other construction facilities required for the work.

(3) A letter signed by an officer of the firm appointing the Environmental Manager and stating that person is responsible for managing and implementing the Environmental Program as described in this contract. Include in this letter the Environmental Manager's authority to direct the removal and replacement of non-conforming work.

b. Management of Natural Resources

(1) Land resources

(2) Tree protection

(3) Replacement of damaged landscape features

(4) Temporary construction

(5) Stream crossings

(6) Fish and wildlife resources

(7) Wetland areas

c. Protection of Historical and Archaeological Resources

(1) Objectives

(2) Methods

(a) Identify any historic properties affected by the project and areas of archaeological monitoring, treatment of unanticipated archaeological resources, and outline mitigation requirements per this contract.

d. Dirt and Dust Control

Additional information/requirements for dirt and dust control, including those for the Dirt and Dust Control Plan, are found in paragraph DUST CONTROL of this specification.

(1) Describe how dust will be controlled in construction zones, along haul routes, and at construction entrances/exits.

(2) List of control measures, operational practices, and BMPs to be employed for dirt and dust control. Selected BMPs must adhere to the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition. Examples include construction entrance/exit design, dust suppressants, stabilization areas, using covers on dump trucks, and sweeping and vacuuming.

e. Stormwater Management and Control

Additional information/requirements for stormwater management and control are found in paragraphs EROSION AND SEDIMENT CONTROL MEASURES and SOLID WASTE MANAGEMENT PLAN of this specification and may be necessary based on the work to be performed. This plan section must include:

(1) A statement that dumpsters and rolloffs will be equipped with a secure lid or covering. Watertight dumpsters and rolloffs are preferred. All dumpsters and rolloffs must be kept closed at all times, except when being loaded with trash or construction debris.

(2) Erodible soil areas.

(3) Temporary measures for runoff control.

(4) Best Management Practices (BMPs) selected for use from the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition.

(5) A location map of selected erosion and sediment controls and manufacture data sheets of products to be used (e.g. erosion control sock, silt sacks, filter fabric etc.).

(6) A statement that erosion and sediment control inspections will be performed for all projects with ground disturbances greater than or equal to 1 cubic yard and comply with erosion and sediment control practices within a shoreland zone. Inspections must be completed at least once per week as well as BEFORE and AFTER a storm event that has the potential to create runoff but at a minimum for every storm resulting in 0.25 inches of precipitation or greater.

(7) The Erosion and Sediment Control Inspector(s) name(s) and current certificate for Maine Certification in Erosion Control Practices.

(8) A statement that a certified individual will be responsible for management of erosion and sediment control practices at the site each day earth-moving activities occur.

(9) Erosion and sediment control BMP maintenance and corrective action procedures, see paragraph EROSION AND SEDIMENT CONTROL INSPECTION AND CORRECTIVE ACTION.

(10) Include a draft (or blank) inspection form that will be used for Erosion and Sediment Control Inspections.

(11) Concrete wastewater/slurry containment and turn-in procedures. All concrete wastewater/slurry for disposal must be turned in to Building 357 by the end of the shift generated.

f. Protection of the Environment from Waste Derived from Contract Operations

(1) Control and disposal of solid and sanitary waste.

(2) Control and disposal procedures for hazardous materials and hazardous waste management. Additional information for requirements below are found in paragraph HAZARDOUS WASTE/DEBRIS MANAGEMENT of this specification and may be necessary based on the work to be performed.

All hazardous waste generation must comply with the procedures in NAVSHIPY PTSMH INST 5090.30 Hazardous Waste Generator Standards. PTSMH INST 5090.30 will be provided by the Contracting Officer.

At a minimum, this section must include the following:

(a) A list and projected amount of potentially hazardous waste, universal waste or otherwise regulated waste streams that may be generated. Hazardous waste, universal waste, and other regulated waste (e.g., used oil, concrete wastewater/slurry, paint and paint related debris, used aerosol cans, etc.) must be turned in to Building 357, Hazardous Waste Storage Facility (HWSF) by the end of the shift generated, unless otherwise approved by Code 106.3.

(b) Procedures to be employed to complete a written waste determination for wastes which are to be generated. Written waste determinations and procedures will be as requested by the Government;

(c) Sampling/analysis plan, including laboratory method(s) that will be used for waste determinations and copies of relevant laboratory certifications (See paragraph CONTROL AND MANAGEMENT OF HAZARDOUS WASTES of this specification for details);

(d) Management procedures for daily accumulation, storage (if approved by Code 106.3), labeling, transportation, turn-in and disposal of hazardous materials and hazardous waste. All waste turned in to Building 357 must be in containers provided by HWSF personnel and have the contents clearly identified. Disposal or treatment of hazardous waste by the Contractor is not allowed unless specifically noted in contract documents or approved by Code 106.3;

(e) Management procedures and regulatory documentation ensuring disposal of hazardous waste complies with Land Disposal Restrictions, as applicable (40 CFR 268);

(f) Management procedures for recyclable hazardous materials such as lead-acid batteries, used oil, and similar;

(g) Used oil management procedures in accordance with 40 CFR 279 and PTSMH INST 5090.30. All used oil must be turned in to Building 357 by the end of the shift generated;

(h) Pollution prevention/hazardous waste minimization procedures;

(i) Plans for the disposal of hazardous and other regulated waste by permitted facilities (as approved by Code 106.3); and

(j) Procedures to be employed to ensure all required employee training records are maintained.

g. Prevention of Releases to the Environment

This plan section must include:

(1) Procedures to prevent releases to the environment and emergency response procedures and notifications to make in the event of a release to the environment. See paragraph RELEASES/SPILLS OF OIL AND HAZARDOUS SUBSTANCES of this specification for additional detail on items to include in this plan section.

(2) A Spill Prevention, Control, and Countermeasures (SPCC) Plan is

required if work is anticipated to extend beyond 6 months, AND will use bulk oil storage containers 55 gallons or greater, in accordance with 40 CFR 112. All SPCC plans must be approved by Code 106.3. Plans need not be certified by a Professional Engineer but must clearly demonstrate proper management of all tanks and containers on site. The SPCC plan must include:

(a) Type of tank or container, quantity stored, type of product stored, and location.

(b) Secondary containment required for tanks/containers 55 gallons or greater; double-wall tanks preferred.

(c) Tank inspection forms (industry standard forms are acceptable, but the use of Portsmouth Naval Shipyard inspection forms is preferred). Records must be kept for 3 years or for the duration of the project. Tanks must be inspected monthly.

i) Bulk storage containers, to include those on equipment, 55 gallons or greater require monthly inspection.

ii) Inspection procedures and an inspection sheet for the release of retained stormwater from secondary containment used for bulk storage tanks and containers.

(d) Where spill kits are located and a description of the spill kit contents for the type of spill anticipated.

(e) If transferring fuel: how often, what type of fuel, and where. Coordinate with the Contracting Officer's Representative (COR) and Code 106.3 prior to transferring fuel over water.

(f) Who to notify in case of ANY spill (Portsmouth Naval Shipyard Fire Department: 207-438-2333, Contracting Officer's Representative, NRC, MEDEP, etc.). All regulatory notification must be coordinated with Code 106.3.

(g) How to clean up a spill safely and a statement that all spill cleanup waste will be brought to Building 357 by the end of the shift generated.

h. Air Emissions/Air Permitting

(1) Installation of a permanent emergency generator for use by the government requires coordination with Code 106.3 for air permitting determination prior to the installation of any portion of the generator or generator support infrastructure.

(2) Provide details of painting and paint blasting operations to include the following, as applicable:

(a) Spray painting and blasting must be performed within containment.

(b) Tarps used to contain the material must prevent visible emissions from exceeding 20% opacity.

(c) Unless otherwise approved, debris collection systems must be suspended from the structure and extend a minimum of 10 feet wider in all directions than the area in which work is done. Collection systems must be positioned in a manner such that no loss of paint residue occurs, emptied at least once a day and must not remain in place overnight without being emptied.

(d) Painting and Blasting operations must not be performed when the direction and velocity of the wind is such that residue falls outside the limits of the collector.

(e) Spray painting dry fall paint will not be allowed if wind speeds exceed 30 feet/s (20mph), unless approved by Code 106.3.

i. Regulatory Notification and Permits

List what notifications must be made and what permits must be obtained for the work. Demonstrate that those permits were obtained or applied for by including copies of all applicable environmental permits. The EMP will not be approved until the permits have been obtained.

3.1.1 Environmental Management Plan Review

Within 30 calendar days after the Contract award date, submit the proposed Environmental Management Plan for further discussion, review, and approval.

Commencement of work must not begin until the Environmental Management Plan has been approved by the Contracting Officer, Code 106.3, and PWD-ME EV Staff.

3.1.2 Licenses, State and Federal Permits

Copies of the approved permit(s) are available from the Contracting Officer. Maintain copies of all permits at the project site. Comply with all terms and conditions of the approved permits.

Where required by the State regulatory authority, the inspections and certifications will be provided through the services of a Professional Engineer (PE), registered (licensed) in the State of Maine. Where a PE is not required, the individual must be otherwise qualified by other current State licensure, specific training, and prior experience (minimum 5 years). As a part of the quality control plan, which is required to be submitted for approval by Section 01 45 00.00 22 QUALITY CONTROL (PWD ME), provide a sub item containing the name, appropriate professional registration or license number, address, and telephone number of the professionals or other qualified persons who will be performing the inspections and certifications for each permit.

a. The following permits and consultations will be obtained by the Government:

(1) ME DEP Air Quality License

3.2 PROTECTION OF NATURAL RESOURCES

Preserve the natural resources within the project boundaries and outside the limits of permanent work and as specified in the permits issued for the work. Restore to an equivalent or improved condition upon completion of work. Confine construction activities to within the limits of the work indicated or specified.

Do not disturb fish and wildlife. Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as indicated or specified.

Except in areas to be cleared, do not remove, cut, deface, injure, or destroy trees or shrubs without the Contracting Officer's permission. Do not fasten or attach ropes, cables, or guys to existing nearby trees for anchorages unless authorized by the Contracting Officer. Where such use of attached ropes, cables, or guys is authorized, the Contractor will be responsible for any resultant damage.

Protect existing trees which are to remain to ensure they are not injured, bruised, defaced, or otherwise damaged by construction operations. Remove displaced rocks from uncleared areas. By approved excavation, remove trees with 30 percent or more of their root systems destroyed. Remove trees and other landscape features scarred or damaged by equipment operations, and replace with equivalent, undamaged trees and landscape features. Obtain Contracting Officer's approval before replacement. Tree wound paint must not be used for tree cuts or stumps.

In the event that tree protection is no longer feasible, or it is identified in the field that further tree removal is required the Contractor shall reach out to the PWD-ME EV Natural Resources Manager, via the Contracting Officer, prior to any tree or vegetation clearing. Additional tree clearing without approval is not permitted. Further mitigation and instruction is provided in the TREE AND SHRUBBERY PRESERVATION AND REPLACEMENT POLICY FOR NAVFAC PWD-ME INSTALLATIONS (Attachment B).

3.2.1 Erosion and Sediment Control Measures

The State of Maine Erosion and Sediment Control Law requires persons undertaking activity involving filling, displacing, or exposing soil or other earthen materials to take measures to prevent unreasonable erosion of soil or sediment beyond the project site or into a protected natural resource.

At the Portsmouth Naval Shipyard, the Piscataqua River, Upper Meade Pond, and Lower Meade Pond are protected natural resources under State Law. Erosion control measures must be in place before the activity begins, maintained and must remain in place and functional until the site is permanently stabilized.

Temporary and permanent erosion control measures must meet, at a minimum, the Best Management Practices (BMPs) presented in the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition. Other techniques may be employed if approved by Code 106.3 and PWD-ME EV and demonstrated to the Contracting Officer that the practice will achieve the required result of no release of sediment per State law.

Site work including any filling, excavation, landscaping, and/or other earthwork in excess of one cubic yard of disturbance, must comply with State of Maine requirements for certification in erosion and sediment control practices within a shoreland zone. A certified individual must be responsible for management of erosion and sediment control practices at the site each day earth moving activities occur. A certified individual is required to visit the site every day to ensure proper erosion and sediment control practices are followed. As an alternative, the

Contractor may choose to contract with a certified individual to supervise the work in shoreland areas.

The shipyard is entirely within the federal coastal zone as administered under the Coastal Zone Management Act. The shoreland zone is any area within 250 feet of mean high water.

a. Stormwater Management/Erosion and Sedimentation Control Plan

(1) Submit a Stormwater Management/Erosion and Sedimentation Control Plan to the Contracting Officer, for review and approval by Code 106.3 and PWD-ME Environmental, for all earthwork in excess of one cubic yard of disturbance. The Plan must demonstrate effective selection, implementation and maintenance of Best Management Practices (BMPs) demonstrating compliance with the Portsmouth Naval Shipyard's Maine Pollutant Discharge Elimination System Municipal Storm-Separate Sewer System Permit (MS4).

CMR 06-096 Chapter 500 Stormwater Management and terms and conditions specified in other approved permits (e.g., Maine Construction General Permit (MCGP), U.S. Army Corps of Engineers (ACOE)) issued for the work.

Provide details of chosen temporary erosion and sediment controls to be employed specific to the work site to include a site plan showing the locations of controls. Ensure proposed controls comply with MEDEP approved plans and State regulations.

Submit Stormwater Management and Erosion Control Compliance Notebook at project completion or as directed by the Contracting Officer.

The Plan must:

(a) Identify potential sources of pollution which may be reasonably expected to affect the quality of stormwater discharge from the site.

(b) Describe and ensure implementation of practices which will be used to reduce the pollutants in stormwater discharge at the manufacturing, storage and lay down, and construction sites.

(c) Describe and ensure full compliance with all MEDEP General Permits (MS4/MCGP) and any other regulatory permits (e.g., ACOE, USEPA) specific to the project.

(d) Describe and ensure compliance with MEDEP over winter stabilizationand construction requirements, as applicable.

(e) Identify inspections and maintenance schedules for Best Management Practices (BMPs) demonstrating compliance with the State of Maine standards. Maintenance procedures must address regular cleaning of drainage structures and repair of temporary erosion control structures, as well as a final cleaning of all drainage structures and removal and reclamation of temporary erosion and sediment control BMP's upon completion of the project.

(f) Select applicable best management practices from the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition. Submit a site plan showing locations of controls and provide manufacturer's product sheets for each control to be used.

(g) Include documentation that the individual responsible for management of erosion and sediment control practices at the site iscertified in accordance with the State of Maine DEP regulations.

(h) Control of Manufactured Concrete Product Waste Plan. At a minimum, must include a description of concrete manufacturing activities and the management of concrete wastewater/wash water as described in paragraph herein entitled CONCRETE WASH WATER.

3.2.2 Stockpiles

Assumed Non-Hazardous soil stockpiles require erosion and sediment controls around the downhill leading edge of the stockpile, or if a leading edge cannot be found, around the entire stockpile perimeter; a MEDEP approved erosion control cover on top of the stockpile when inactive 7 or more days, when stockpile operations are complete and while awaiting soil characterization test results; while awaiting test results, placarded as Pending Characterization and based on those sample results placarded as either Non-Hazardous or Hazardous; immediately cover with 6 mil plastic sheeting if soil characterization test results return as Hazardous.

Known Hazardous (IR Soil) or suspect contaminated soil stockpiles require 6 mil plastic sheeting underneath; erosion control sock around the entire stockpile perimeter; a 6 mil plastic sheeting cover unless actively being worked or while obtaining soil characterization test samples; at all times placarded as Hazardous for IR soil stockpiles. Suspect contaminated soil stockpiles must be placarded as Pending Characterization while awaiting soil characterization test results and based on those results placarded as either Non-Hazardous or Hazardous. IR soils stockpiles must be located within the boundaries of the IR site from which the soil originated. When working in or near IR sites, the contractor shall clearly pre-mark the IR site boundary to ensure soil remains segregated within their respective zones.

3.2.3 Erosion and Sediment Control Inspection and Corrective Action

Inspection reports must be kept on file at the project site and submitted electronically to the Contracting Officer upon request. The State of Maine requires inspections of disturbed and impervious areas, erosion and sediment control measures, areas used for storage that are exposed to precipitation, and locations where vehicles enter or exit the site.

Inspections must be performed at least once per week as well as BEFORE and AFTER a storm event. A storm event is any precipitation event with the potential to create runoff but at a minimum for every storm resulting in 0.25 inches of precipitation or greater. Inspection reports must document compliance with State of Maine requirements. If erosion and sediment control BMPs need to be repaired, the repair work should be initiated upon discovery of the problem but no later than the end of the next workday. If additional BMPs or significant repair of BMPs are necessary, implementation must be completed within 7 calendar days and prior to any storm event (rainfall). For each BMP requiring maintenance, BMP needing replacement, and location needing additional BMPs, note in the inspection report the corrective action taken and when it was taken.

3.2.4 Burnoff

Burnoff of the ground cover is not permitted.

3.3 PROTECTION OF CULTURAL RESOURCES

3.3.1 Archaeological Resources

Existing archaeological resources within the work area are shown on the drawings. Protect these resources and be responsible for their preservation during the life of the Contract. Refer to Section 01 44 00.00 22 ARCHAEOLOGICAL MONITORING (PWD ME) for requirements on archaeological monitoring, inspection, and reporting. The Government retains ownership over archaeological resources.

3.3.2 Historical Resources

Existing historical resources within the work area are shown on the drawings. Protect these resources and be responsible for their preservation during the life of the Contract.

3.4 SOLID WASTE MANAGEMENT

3.4.1 SOLID WASTE MANAGEMENT PLAN

Provide a written Solid Waste Management Plan (SWMP) to the Contracting Officer, of intended licensed disposal sites for Government approval and for submission to State regulatory agencies. At a minimum, the SWMP must contain, but not be limited to, the following wastes: stumps and grubbings, excess soil, construction debris, demolition debris, household solid waste, special waste, and industrial solid waste. The submission must contain the name of the disposal facility, address, facility phone number, and the waste type and quantity to be disposed of at the facility.

If waste from the site is taken to a transfer station, identify the facility or facilities at which the waste is ultimately disposed. Government approval for the facility must be obtained prior to transporting wastes off Government property.

Provide the Contracting Officer with written notification of the quantity of anticipated solid waste or debris that is anticipated or estimated to be generated by construction. Include in the report the locations where various types of waste will be disposed or recycled. Include letters of acceptance from the receiving location or as applicable; submit one copy of the receiving location State and local Solid Waste Management Permit or license showing such agency's approval of the disposal plan before transporting wastes off Government property.

3.4.2 Monthly Solid Waste Disposal Report

Monthly, submit a solid waste disposal report to the Contracting Officer. For each waste, the report will state the classification (using the definitions provided in this Section), amount, location, and name of the business receiving the solid waste.

Provide copies of the waste handling facilities' weight tickets, receipts, bills of sale, and other sales documentation. In lieu of sales documentation, a statement indicating the disposal location for the solid waste that is signed by an employee authorized to legally obligate or bind the firm may be submitted. The sales documentation must include the receiver's tax identification number and business, EPA, or State registration number, along with the receiver's delivery and business addresses and telephone numbers. For each solid waste retained for the Contractor's own use, submit the information previously described in this paragraph on the solid waste disposal report. Prices paid or received do not have to be reported to the Contracting Officer unless required by other provisions or specifications of this Contract or public law.

3.4.3 Control and Management of Solid Wastes

Pick up solid wastes, and place into containers that are regularly emptied. Containers must be kept covered except when being loaded with trash and debris. Watertight covered dumpsters and roll-offs are preferred. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Recycling is encouraged and can be coordinated with the Contracting Officer and the Portsmouth Naval Shipyard Recycling Coordinator. Remove all solid waste (including non-hazardous debris) from Government property and dispose off-site at an approved landfill. Solid waste disposal off-site must comply with the most stringent local, State, and Federal requirements including 40 CFR 241, 40 CFR 243, and 40 CFR 258. Discharges from dumpsters and roll-offs are prohibited.

Manage hazardous material used in construction including, but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, and used rags, as per environmental law and Portsmouth Naval Shipyard requirements. Note: Scrap metal coated with PCB contaminated paint > or = to 50 ppm is considered a hazardous waste and requires Code 106.3 coordination for recycling/disposal.

3.4.3.1 Dumpsters and Roll-offs

Dry weather discharges from dumpsters and roll-offs are prohibited. Equip dumpsters and roll-offs with a secure lid or covering. Keep covers closed at all times, except when being loaded with trash and debris, or provide secondary containment to ensure that discharges have a control. Locate dumpsters and roll-offs behind the construction fence or out of the public view and away from storm drains. Empty site dumpsters and roll-offs at least once a week or as needed to keep the site free of debris and trash. If necessary, provide 55-gallon trash containers to collect debris in the construction site area. Locate the trash containers behind the construction fence or out of the public view. Empty trash containers at least once a day. For large demolitions, dumpsters and roll-offs must use a cover and watertight containers are preferred; debris must not be higher than the sides before emptying. Water must not be collected in watertight roll-offs or dumpsters. Any water collected in roll-offs, dumpsters, or secondary containment could become contaminated with oil or other contaminants and may require waste characterization by the Contractor prior to release and/or removal from PNS.

3.5 WASTE DETERMINATION DOCUMENTATION

Complete a Waste Determination form (provided by the Contracting Officer at the preconstruction meeting) for Contractor-derived wastes to be generated, as required by Code 106.3. Base the waste determination upon either a constituent listing from the manufacturer used in conjunction with consideration of the process by which the waste was generated, EPA approved analytical data, and/or laboratory analysis (Safety Data Sheets (SDS) by themselves are not adequate). Attach support documentation to the Waste Determination form. As a minimum, provide a Waste Determination form for the following waste (this listing is not all inclusive): oil- and latex-based painting and caulking products, solvents, adhesives, aerosols, petroleum products, and containers of the original materials.

3.6 POLLUTION PREVENTION/HAZARDOUS WASTE MINIMIZATION

Minimize the use of hazardous materials and the generation of waste. Include procedures for pollution prevention/ hazardous waste minimization in the Hazardous Waste Management Section of the Environmental Management Plan. Obtain a copy of the Portsmouth Naval Shipyard's Pollution Prevention/Hazardous Waste Minimization Plan for reference material when preparing this part of the EMP. If no written plan exists, obtain information by contacting the Contracting Officer. Describe the anticipated types of the hazardous materials to be used in the construction when requesting information.

3.7 WHM/HW MATERIALS PROHIBITION

Do not improperly dispose of hazardous materials/hazardous waste on Government property. No hazardous material will be brought onto Government property that does not directly relate to requirements for the performance of this Contract.

Incidental materials used to support the contract including, but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, rags, clothing, empty containers, etc. may be considered hazardous wastes and must be turned in to Building 357 for disposal by the Government as described in the paragraph entitled HAZARDOUS MATERIAL MANAGEMENT of this Section. The list is illustrative rather than inclusive. Universal wastes must be managed with controls similar to those for hazardous waste.

Do not discharge any materials to sanitary sewer, storm drain, or to the Piscataqua River or conduct waste treatment or disposal on Government property without written approval by the Contracting Officer and Code 106.3.

3.8 HAZARDOUS MATERIAL MANAGEMENT

Hazardous materials for disposal may be considered hazardous wastes and must be turned in to Building 357 for hazardous waste determination by the end of the shift generated.

Include hazardous material control procedures in the Environmental Management Plan as described in paragraph ENVIRONMENTAL MANAGEMENT PLAN (EMP) herein and as applicable in the Accident Prevention Plan (APP) in accordance with Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). Address procedures and proper handling of hazardous materials, including the appropriate transportation requirements. All hazardous material must be in active use or properly stored. Hazardous materials must not be left unattended or uncontrolled while on Portsmouth Naval Shipyard.

Submit an SDS and estimated quantities to be used for each hazardous material to the Contracting Officer prior to bringing the material on the Portsmouth Naval Shipyard. Typical materials requiring SDS and quantity reporting include, but are not limited to, oil- and latex-based painting and caulking products, solvents, adhesives, aerosol, and petroleum products.Use hazardous materials in a manner that minimizes the amount of hazardous waste generated. Containers of hazardous materials must have National Fire Protection Association labels or their equivalent. Certify that hazardous materials removed from the site are hazardous materials and do not meet the definition of hazardous waste in accordance with 40 CFR 261 and state and Portsmouth Naval Shipyard requirements.

3.8.1 Contractor Hazardous Material Inventory Log

Submit the "Contractor Hazardous Material Inventory Log" (found at: https://www.wbdg.org/dod/ufgs/ufgs-forms-graphics-tables), which provides information required by EPCRA Sections 312 and 313 along with corresponding Safety Data Sheets (SDS), to the Contracting Officer at the start and at the end of construction (30 days from final acceptance), and update no later than January 31 of each calendar year during the life of the Contract. Keep copies of the SDSs for hazardous materials onsite. At the end of the project, provide the Contracting Officer with copies of the SDSs, and the maximum quantity of each material that was present at the site at any one time, the dates the material was present, the amount of each material that was used during the project, and how the material was used. The Contracting Officer may request documentation for any spills or releases, environmental reports, or off-site transfers.

3.9 PETROLEUM PRODUCTS AND REFUELING

Conduct the fueling and lubricating of equipment and motor vehicles in a manner that protects against spills and evaporation, away from storm drains and surface waters. Manage used oil generated on site in accordance with 40 CFR 279. Determine if any used oil generated while onsite exhibits a characteristic of hazardous waste. All used oil will be turned in to Building 357 and disposed of by the Government.

3.9.1 Oil Storage Including Fuel Tanks

Provide secondary containment and overfill protection for oil and hazardous substance storage tanks. Prevent oil or hazardous substances from entering the ground, drainage areas, storm drains, or navigable waters. In accordance with 40 CFR 112, surround all temporary fuel oil or petroleum storage tanks with a temporary berm or containment of sufficient size and strength to contain the contents of the tanks, plus 10 percent freeboard for precipitation. Construct the berm to be impervious to oil for 72 hours and so that no discharge will permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Use drip pans during oil transfer operations; adequate absorbent material must be onsite to clean up any spills and prevent releases to the environment. Cover tanks and drip pans during inclement weather. Provide procedures and equipment to prevent overfilling of tanks.

Monitor and remove any rainwater that accumulates in open containment dikes, berms, or spill pallets. Inspect the accumulated rainwater prior to draining from a containment dike, berm or spill pallet to the environment, to determine there is no oil sheen present.

3.9.2 Inadvertent Discovery of Petroleum Contaminated Soil or Hazardous Wastes

If petroleum contaminated soil, or suspected hazardous waste is found during construction that was not identified in the Contract Documents, immediately notify the Contracting Officer. Do not disturb this material until authorized by the Contracting Officer.

3.10 FUEL TANKS

Petroleum products and lubricants required to sustain up to 30 days of construction activity may be kept onsite. Storage and refilling practices must comply with 40 CFR 112. Secondary containment must be provided and be no less than 110 percent of the tank volume plus five inches of free-board. If a secondary berm is used for containment then the berm must be impervious to oil for 72 hours and be constructed so that any discharge will not permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Drip pans are required. Cover tanks and drip pans during inclement weather to prevent spills, as necessary.

3.11 RELEASES/SPILLS OF OIL AND HAZARDOUS SUBSTANCES

Exercise due diligence to prevent, contain, and respond to **ALL** spills of hazardous material, hazardous substances, hazardous waste, sewage, regulated gas, petroleum, lubrication oil, and other substances regulated in accordance with 40 CFR 300. In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the spill/release. In the event of **ANY** releases of oil and hazardous substances, chemicals, or gases; immediately (within 15 minutes) notify the Portsmouth Naval Shipyard Fire Department 207-438-2333, the Portsmouth Naval Shipyard's Command Duty Officer, Contracting Officer's Representative, and the Contracting Officer. If the response is inadequate, the Navy may respond. If this should occur, reimbursement to the Government for spill response assistance and analysis will be required. Contractor will provide SDSs, Waste Profile Sheets, and/or other technical data on the material spilled to emergency response personnel.

The Contractor is responsible for verbal and written notifications as required by the Federal 40 CFR 300.125 and 40 CFR 355, State, and local regulations and Navy Instructions. These notifications will be done in coordination with Code 106.3. Spill response must be in accordance with 40 CFR 300 and applicable State and local regulations. Contain and clean up these spills without cost to the Government. If Government assistance is requested or required, reimburse the Government for such assistance. Provide copies of the written notification and documentation that a verbal notification was made within 20 days.

Maintain spill cleanup equipment and materials at the work site. Clean up all hazardous and non-hazardous (WHM) waste spills. Reimburse the Government for all material, equipment, and clothing generated during any spill cleanup. Reimburse the Government for all costs incurred including sample analysis materials, equipment, and labor if the Government must initiate its own spill cleanup procedures, for spills, when:

a. Spill cleanup procedure has not begun within one hour of spill discovery/occurrence, or

b. If, in the Government's judgment, the spill cleanup is not adequately abating a life-threatening situation and/or is a threat to any body of water or environmentally sensitive areas.

c. All discharged material and other spill cleanup wastes (e.g., absorbent, disposable clothing, brushes, rags, brooms, containers, decontamination water etc.), must be collected, containerized, properly marked and brought to Building 357 by the end of the shift generated. Decontamination water must be collected and turned in separately from other generated wastes or as directed by Code 106.3.

3.12 CONTROL AND MANAGEMENT OF HAZARDOUS WASTES

At the time of the preconstruction meeting, the Contractor will be briefed and provided written information regarding hazardous waste management. The Government will provide technical and oversight guidance in all aspects of hazardous waste management.

3.12.1 General

All hazardous wastes generated within the confines of the Portsmouth Naval Shipyard must be disposed of by the Government. Accordingly, all hazardous wastes generated to accomplish the requirements of this Contract must be considered Government-generated, and must be disposed of by the Government. Do not bring hazardous wastes onto Government property. Hazardous wastes must be handled in compliance with 40 CFR 260-268, 273, 279 and State of Maine MEDEP Regulations Chapter 850 to 860. For hazardous waste spills, immediately notify the Portsmouth Naval Shipyard Fire Department 207-438-2333, the Portsmouth Naval Shipyard's Command Duty Officer, Code 106.3: 207-438-4477, Contracting Officer's Representative and the Contracting Officer.

3.12.2 Containers

Use only Government-furnished, Government-labeled containers for the packaging of hazardous soils and hazardous wastes. Containers are requested and picked up at Building 357 following approval of the Management Plan required above.

a. Segregate hazardous and non-hazardous soils/wastes. Hazardous soils/wastes must be placed into containers provided by the Government. All containers of hazardous waste must have a Code 106.3 tracking number affixed to the container, as applicable.

b. Hazardous soils must be turned over to the Government, at Building 357, as coordinated with Code 106.3. All other containers of waste hazardous materials and hazardous waste, full or partially full, must be turned in to Building 357 by the end of the shift generated.

While in control of hazardous soils and hazardous wastes, the soils/wastes must be handled in accordance with Portsmouth Naval Shipyard requirements.

c. Prior to Government acceptance of the containers, provide the certification required by the "Submittals" paragraph of this Section, and such additional information regarding contents of the containers. Submittal of a Waste Determination form may be required for proper waste characterization as requested by Code 106.3.

3.12.3 Facility Hazardous Waste Generator Status

Portsmouth Naval Shipyard is designated as a Large Quantity Generator. All work conducted within the boundaries of the Portsmouth Naval Shipyard must meet the regulatory requirements of this generator designation. Comply with all provisions of Federal, State, and local regulatory requirements applicable to this generator status regarding training and storage, handling, and disposal of all construction derived wastes.

3.12.4 Hazardous Waste/Debris Management

Identify construction activities that will generate hazardous waste or debris and universal wastes. Provide a documented waste determination for resultant waste streams as requested by Code 106.3 or other Government personnel. Hazardous waste or debris must be identified, labeled, handled, stored, and disposed of in accordance with all Federal, State, and local regulations including 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, 40 CFR 265, 40 CFR 266, 40 CFR 268, Maine Department of Environmental Protection (MEDEP) MAINE 38 MSRA 850-860, and NAVSHIPY PTSMH INST 5090.30, latest edition.

Manage hazardous wastes and universal wastes in accordance with the approved Hazardous Waste Management Section of the Environmental Management Plan. Store hazardous wastes and universal wastes in Government-provided containers in accordance with 49 CFR 173 and 49 CFR 178. Hazardous waste generated within the confines of Government facilities is identified as being generated by the Government.

Prior to removal of any hazardous waste from Government property, hazardous waste manifests must be signed by Portsmouth Naval Shipyard personnel from the Portsmouth Naval Shipyard Environmental Office. Provide the Contracting Officer with a copy of waste determination documentation for any solid waste streams that have any potential to be hazardous waste or contain any chemical constituents listed in 40 CFR 372-SUBPART D.

3.12.4.1 Waste Storage/Satellite Accumulation/90 Day Storage Areas

Accumulate hazardous waste at satellite accumulation points and in compliance with 40 CFR 262 and applicable state or local regulations.

Individual waste streams will be limited to 55 gallons of accumulation (one quart for acutely hazardous wastes). If the Contractor expects to generate hazardous waste at a rate and quantity that makes satellite accumulation impractical, the Contractor may request a temporary 90-day or 180-day, as appropriate, accumulation point be established. Regulated waste and Hazardous Waste Accumulation Areas(HWAAs) are approved by Code 106.3.

Submit a request in writing to the Contracting Officer providing the following information (Attach Site Plan to the Request):

Contract Number	____________	Contractor	______________
Haz/Waste or Regulated Waste POC	____________	Phone Number	______________
Type of Waste	____________	Source of Waste	______________
Emergency POC	____________	Phone Number	______________
Location of the Site:	____________		

Attach a Waste Determination form for the expected waste streams. Allow ten (10) working days for processing this request. Additional compliance requirements (e.g., training and contingency planning) that may be required are the responsibility of the Contractor. Barricade the designated area where waste is being stored and post a sign identifying as follows:

"DANGER - UNAUTHORIZED PERSONNEL KEEP OUT"

3.12.4.2 Sampling and Analysis of Hazardous Waste

a. Waste Sampling

Sample waste in accordance with EPA SW-846 and NAVSHIPY PTSMH INST 5090.8, latest revision. Each sampled drum or container must be clearly marked with the Contractor's identification number and cross referenced to the chemical analysis performed. All sampling events will require a Code 106.3 reviewed and approved sampling plan and availability for Code 106.3 oversight.

b. Laboratory Analysis

Follow the analytical procedure and methods in accordance with 40 CFR 261. Provide all analytical results and reports performed to the Contracting Officer and Code 106.3 Environmental Sampling Project Manager for review and approval.

All laboratory analysis for hazardous waste identification must be performed by a laboratory compliant with OPNAVINST 5090.1 Chapter 7-3.3. Proof of compliance must be made available upon request. All analyses provided by laboratories that are not compliant with the stated requirements will be rejected.

c. Analysis Type

Identify waste material/hazardous waste by analyzing for properties that are reasonably suspected of the waste. Soil and other materials may require specific analysis for acceptance to a disposal facility - contact Code 106.3 personnel at the Hazardous Waste Storage Facility, Building 357, before choosing parameters.

d. Typical Hazardous Waste and PCB Sampling Plan Requirements

i. Confirmation that the sampling plan will be reviewed and approved by the Navy (Code 106.3) prior to conducting any sampling. Code 106.3 review period is fourteen (14) calendar days from receipt of the plan.

ii. The timing of waste characterization sampling and analysis. PCB sampling must be conducted prior to the accomplishment of any work. To the extent feasible, all other waste sampling and analysis should be completed prior to the start of work. Waste sampling after the start of work requires Code 106.3 approval.

iii. A project summary with a brief description of the project, a description of the anticipated waste and how the waste will be generated, a map or diagram with sample locations and type, an estimate of the amount of waste to be generated, and the name and location of the disposal facility the waste will go to if waste is to be disposed of off-site, as approved by Code 106.3.

iv. A statement in the plan that the sample collection, management, and analysis will be performed per all applicable local, state, and federal statutes and regulations, project specifications, Navy and PNSY requirements to include NAVSHIPYD PTSMHINST 5090.8 (latest revision) and other PNSY guidance as applicable.

v. The proposed date(s) for sampling and expected duration of the sampling event(s). All hazardous waste and PCB sampling requires Code 106.3 oversight with seven (7) calendar day notification prior to a sampling event.

vi. A statement that all materials used in the collection of the samples (gloves, wipes, trash, etc.) and hazardous debris generated by the sampling process will be containerized in a bag and delivered to Building 357 for disposal by the end of the shift generated.

vii. A narrative of how the sample(s) will be taken (e.g. discrete, composite, grab), a description of the decontamination process for reusable equipment between consecutive samples, documentation that the Environmental Field Technician (EFT) conducting the waste sampling has training in proper collection and decontamination procedures, and a statement giving rationale for why the sample(s) will be representative.

viii. A list of all equipment to be used for sampling. Examples include:

1. Individual sample jars and vials;
2. Field log;
3. Site diagram;
4. Chain-of-Custody (COC) forms;
5. Nitrile gloves;
6. Stainless steel or plastic soil scoops;
7. Metal chisels of assorted sizes;
8. Razor knives and replacement blades;
9. Water and spray bottle;
10. Trash bags;
11. Disposable cleaning wipes;
12. Paper towels;
13. Duct tape;
14. Flashlight;
15. Personnel lifts; and
16. Other equipment as necessary.

ix. A listing of the constituents to be analyzed for, identifying the analytical tests methods to be performed with the corresponding EPA test methods numbers.

x. The name and location of the laboratory the samples will be sent to and confirmation that the selected lab has Maine Department of Environmental Protection certification, National Environmental Laboratories Accreditation Conference (NELAC) certification and certifications for all corresponding EPA test methods, as applicable.

xi. Confirmation that the samples will be submitted, under strict chain-of-custody (COC) requirements, to the certified laboratory.

xii. Confirmation that the sampling results report will be transmitted to the Contracting Officer and Code 106.3 for each material sampled. The report must include quality control data, be reported down to the Method Detection Limit (MDL) and include:

1. Laboratory analytical results
2. Field log book entries
3. Copy of the Chain of Custody (COC)
4. Pertinent diagrams and maps
5. All other supporting data

3.12.4.3 Asbestos Certification

Items, components, or materials disturbed by or included in work under this Contract may involve asbestos. Other materials in the general area around where work will be performed may contain asbestos. All thermal insulation, in all work areas, should be considered to be asbestos unless positively identified by conspicuous tags or previous laboratory analysis certifying them as asbestos free.

Inadvertent discovery of non-disclosed asbestos that will result in an abatement action requires a change in scope before proceeding. Upon discovery of asbestos containing material not identified in the Contract documents, immediately stop all work that would generate further damage to the material, evacuate the asbestos exposed area, and notify the Contracting Officer for resolution of the situation prior to resuming normal work activities in the affected area. Do not remove or perform work on any asbestos containing materials without the prior approval of the Contracting Officer. Do not engage in any activity, which would remove or damage such materials or cause the generation of fibers from such materials.

Asbestos containing waste must be managed and disposed of in accordance with applicable environmental law. Asbestos containing waste must be manifested and the manifest provided to the Contracting Officer. Disposal of asbestos-containing waste must be coordinated with PWD-ME EV and Code 106.3.

Provide the attached Asbestos Free Certification Form (Attachment C) prior to the Government taking beneficial occupancy certifying that all materials, including those supplied by third parties, are asbestos free.

3.12.4.4 Polychlorinated Biphenyls (PCBs)

Polychlorinated Biphenyls (PCBs) were manufactured between 1950 and 1979 and widely used in building construction and renovation materials (e.g., paints, varnishes, caulks, rubber/felt gaskets, adhesives, siding, mastics, roofing, tar paper, fireproofing etc.). Work may include the disturbance of items, components, or materials containing PCBs as identified in project documents. Suspect PCB containing materials must be analytically tested prior to disturbance and managed per 40 CFR 761 and applicable environmental law. If adequate PCB testing for industrial hygiene and waste disposal was not provided in the contract documents, the Contractor shall be responsible for PCB suspect materials testing prior to the start of work.

- Sampling of suspect PCB material requires a Code 106.3 reviewed and approved sampling plan and Code 106.3 oversight.

- Disposal of all PCB containing waste(s) $\geq$ 1 ppm must be coordinated with B357/Code 106.3 personnel.

Inadvertent discovery of undisclosed PCBs resulting in a new or expanded abatement action requires a change in scope review before proceeding. Upon discovery of PCB containing material not identified in the Contract documents, immediately stop all work that could impact PCB containing materials and notify the Contracting Officer for resolution of the situation prior to resuming normal work activities in the affected area.

3.12.4.5 Hazardous Waste Disposal

Control of stored waste, packaging, sampling, analysis, and disposal will be determined by the details in the Contract. The requirements for jobs in the following paragraphs will be used as the guidelines for disposal of any hazardous waste generated and disposed of by the Contractor. All hazardous waste disposed of by the Contractor must be approved by and coordinated with Code 106.3 prior to transporting wastes off PNS property.

a. Responsibilities for Disposal

Responsibilities include any generation of WHM/HW requiring disposal of solid waste or liquid.

(1) Agree to provide all service necessary for the final treatment/disposal of the hazardous material/waste in accordance with all local, State, and Federal laws and regulations, and the terms and conditions of the Contract within sixty (60) days after the materials have been generated. These services must include all necessary personnel, labor, transportation, packaging, detailed analysis (if required for disposal, and/or transportation, including manifesting or completing waste profile sheets, equipment, and the compilation of all documentation is required).

(2) Contain all waste in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, 40 CFR 265, 40 CFR 266, 40 CFR 268, 40 CFR 270, 40 CFR 272, 40 CFR 273, 40 CFR 279, 40 CFR 280, and 40 CFR 761.

(3) Obtaining a representative sample of the material generated for each job done to provide waste stream determination.

(4) Analyzing for each sample taken and providing analytical results to the Contracting Officer. Provide two copies of the results.

(5) Determine the DOT proper shipping names for all waste (each container requiring disposal) and will demonstrate how this determination is developed and supported by the sampling and analysis requirements contained herein to the Contracting Officer.

Disposal Turn-In Requirements

For any waste hazardous materials or hazardous waste generated which requires disposal, the following conditions must be met in order to be acceptable for disposal:

a. Drums compatible with waste contents and drums meet DOT requirements for 49 CFR 173 for transportation of materials.

b. Drums banded to wooden pallets. No more than three (3) 55-gallon drums to a pallet, or two (2) 85-gallon over packs.

c. Band using 1-1/4 inch minimum band on upper third of drum.

d. Contents of drum identified on the outside of the drum, as well as the volume of material, name of material manufacturer, and other vendor information as available.

e. Always have three (3) to five (5) inches of empty space above volume of material. This space is called 'outage'.

f. Provide disposal documentation for hazardous and regulated waste.

3.12.5 Class I ODS Prohibition

Class I ODS must not be used in the performance of this Contract, nor be provided as part of the equipment. This prohibition will be considered to prevail over any other provision, specification, drawing, or referenced documents. Class I and II ODSs are Government property and must be returned to the Government for appropriate management. Coordinate with the Installation Environmental Office to determine the appropriate location for turn in of all reclaimed refrigerant. Regulations related to the protection of stratosphere ozone may be found in 40 CFR 82.

Heating and air conditioning technicians must be certified through an EPA-approved program. Copies of certifications must be maintained at the employees' place of business and be carried as a wallet card by the technician, as provided by environmental law. Accidental venting of a refrigerant is a release and must be reported to the Contracting Officer.

3.12.6 Universal Waste/e-Waste Management

Universal waste including, but not limited to, some mercury containing building products such as florescent lamps, mercury vapor lamps, high pressure sodium lamps, CRTs, batteries, aerosol paint containers, electrical equipment containing PCBs, and consumed electronic devices, must be managed in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 273, MAINE DEP 38 MSRA 858, and applicable environmental law.

3.13 DUST CONTROL

Dust control must meet the requirements of the BMPs in the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition. Keep dust down at all times, including during nonworking periods. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates that would exceed 40 CFR 50, state, and local air pollution standards or that would cause a hazard or a nuisance. Sprinkle or treat, with dust suppressants, the soil at the site, haul roads, and other areas disturbed by operations. Dry power brooming will not be permitted. Instead, use vacuuming or sweeping. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Runoff from cutting water is prohibited from entering storm drains or leaving the work site. Do not unnecessarily shake bags of cement, concrete mortar, or plaster.

When temporary dust control measures are employed, repetitive treatment must be applied as needed to accomplish control.

Visible emissions from a fugitive emission source (including stockpiles and roadways) must not exceed an opacity of 20 percent, except for no more than five (5) minutes in any 1-hour period.

Dust suppression during demolition work may include using a manned water hose. Runoff from the site is prohibited.

3.13.1 Dirt and Dust Control Plan

Submit truck and material haul routes along with a plan for controlling dirt, debris, and dust on Portsmouth Naval Shipyard roadways. As a minimum, identify in the plan the Subcontractor and equipment for cleaning along the haul route and measures to reduce the dirt, dust, and debris from roadways.

3.14 ABRASIVE AND/OR WET BLASTING AND WATER CUTTING

3.14.1 Blasting Operations

a. Abrasive Blasting

The use of silica sand is prohibited in sandblasting.

Provide tarpaulin drop cloths and windscreens to enclose abrasive blasting operations to confine and collect dust, abrasive, agent, paint chips, and other debris.

Abrasive blasting must take place in containments with emissions vented through bag house filters and emissions must be limited to 10% opacity on a six minute block average. The bag houses must be used to control PM emission and operated properly at all times abrasive blasting is being performed.

b. Wet Blasting and Water Cutting

The use of wet blasting requires the capture and proper disposal of all wastes, including the blasting water, associated with the process. Provide enclosed containments to confine and collect wastewater and debris.

3.14.2 Disposal Requirements

Submit analytical results of the wastes and/or debris generated from blasting operations per paragraph entitled LABORATORY ANALYSIS of this Section. Hazardous waste generated from blasting operations must be managed in accordance with paragraph entitled HAZARDOUS WASTE/DEBRIS MANAGEMENT of this Section and the approved EMP. Concrete wash water and oily waste generated from blasting operations must be disposed of in accordance with the policy outlined in these specifications.

3.15 SPRAY PAINTING OPERATIONS

Spray painting operations must take place in containment. Emissions from spray painting must vent through air filters and are limited to 10%

opacity on a six-minute block average. The air filters are used to control particulate emissions.

3.16 NOISE

Make the maximum use of low-noise emission products, as certified by the EPA and/or sound deadening enclosures to limit noise within the limits of work. Blasting or the use of explosives will not be permitted. Confine any operations that may generate excessive noise to the period between 7 a.m. and 5 p.m., Monday through Friday, exclusive of holidays, unless otherwise specified or approved by the Contracting Officer.

The maximum permissible sound pressure levels, as measured at the limits of the Navy Property boundary, must not exceed the maximum noise levels as specified in the Town of Kittery's Ordinance and all applicable OSHA Regulations. The Kittery Ordinance specifies the maximum permissible sound level limit at the Navy Property Boundary to be 65 dBA from 0700 to 2100 and 60 dBA from 2100 to 0700.

For any construction or demolition operations that may generate noise levels in excess of 80 dBA in adjacent facility work areas or abuttering Non-Gov't owned properties adjacent to the work area, the Contractor must provide a Noise mitigation plan to the Contractor Offcer at least 15 calendar days prior to commencing any work that would create noise the would exceed the specified maximum noise levels. The Noise Mitigation plan may include the use low-noise emission equipment, sound deadening enclosures or other mitogation measures to ensure compliance with the Kittery Noise Ordinance levels or the adjacent work areas are not impacted.

Provide all noise monitoring equipment necessary to record and document noise levels associated with any operations generating excessive noise to ensure these noise levels specified are not exceeded. Any work activities that generates noise that exceeds the specified maximum noise levels must stop immediately. The Contractor must take corrective actions to ensure the noise levels are mitigated. Any delays in the work, are the Contractors responsibility and will not be justifcation for a time extension.

3.17 MERCURY MATERIALS

Mercury is prohibited in the construction of this facility, unless specified otherwise, and with the exception of mercury vapor lamps and fluorescent lamps. Dumping of mercury-containing materials and devices such as mercury vapor lamps, fluorescent lamps, and mercury switches, in rubbish containers is prohibited. Remove without breaking, pack to prevent breakage, and transport out of the activity in an unbroken condition for disposal as directed. Immediately (within 15 minutes) notify instances of breakage or mercury spillage to the Portsmouth Naval Shipyard Fire Department 207-438-2333, the Portsmouth Naval Shipyard's Command Duty Officer, Code 106.3: 207-438-4477, Contracting Officer's Representative and the Contracting Officer. Clean the mercury spill area(s) to the satisfaction of the Contracting Officer. Cleanup of a mercury spill must not be recycled and must be managed as a hazardous waste for disposal.

3.18 CONCRETE WASH WATER

Concrete wash water is water, pressure washing water, or storm water that has come into contact with cement, uncured concrete, concrete dust, or other material of a similar nature generated during construction activities including, but not limited to, washing down ready-mix trucks, mixers, wheelbarrows, pre casting equipment, forms, manufactured cast concrete sections, tools, concrete areas; masonry cutting operations; cleaning up of split mortar or block fill; and hosing away excess materials.

Water or stormwater that has come into contact with pre casting equipment, forms, tools, etc. which have been subjected to oil based form release agents will be considered an oily waste if a visual inspection indicates any signs of oily residue. Oily wastes must be collected and disposed of in accordance with Portsmouth Naval Shipyard policy.

Concrete wash water must be collected and either placed in a concrete wash out structure for settling solids and evaporating liquid or turned-in as wastewater to Building 357 by the end of the shift generated. As approved by Code 106.3, large volumes of concrete wastewater may be disposed of at a facility designed and approved for the disposal or recycling of concrete wastewater. Under no circumstances will clean water be added to concrete wash water/wastewater for dilution purposes or any other reason. Containment wash out structures must be watertight, designed to promote evaporation, and provide adequate freeboard to contain the wash water, solids, and rainfall to prevent overflow. Washout must only occur in designated areas that have been approved by the Contracting Officer's Representative.

Inspect all concrete wash out structures daily to determine filled capacity. Remove all materials from containment wash out structures when 75% fill capacity has been reached. Concrete wash out structure operations must be conducted to reduce the amount of concrete wastewater generated for disposal. Remove liquids or cover structures before predicted rainstorms to prevent overflows and infiltration of rainwater. Inspect structures for holes and tears daily and repair to maintain watertight conditions.

Hardened solids can be removed from containment structures and recycled, reused, or disposed of per regulatory requirements. Liquids remaining in the containment structure must be vacuumed and turned-in to Building 357 or, as approved by Code 106.3, disposed of at a facility designated for disposal/recycling of concrete wastewater.

3.18.1 Pollution Prevention

Store dry and wet concrete supplies under cover away from drainage areas. Concrete wash water must not be released to the storm drain system, sewer system, roadways, other uncontained impervious surfaces, or to natural waterways including the Piscataqua River and its tributaries. Take all precautions necessary to prevent rainwater or stormwater runoff from coming into contact with concrete wash water. Divert clean stormwater and roof runoff from contact with concrete wash water. Take all measures necessary to minimize the volume of concrete wash water generated. Protect all waterways, catch basins, and storm drain structures from potential discharges of concrete wash water. Collect and control concrete wash water separately from wastewater determined to be oily waste.

3.18.2 On-Shipyard Disposal

ALL CONCRETE WASTEWATER MUST BE COLLECTED AND TURNED-IN TO BUILDING 357 BY

THE END OF THE SHIFT GENERATED. CONCRETE WASH WATER THAT IS COLLECTED AND DEPOSITED IN WASH OUT STRUCTURES MUST BE TURNED IN TO BUILDING 357 AT OR BEFORE THE CONCRETE WASH OUT STRUCTURE REACHES 75% CAPACITY.

3.18.3 Off-Shipyard Disposal

Concrete wastewater may be disposed of off-site as approved by Code 106.3. The Contractor must submit for approval the disposal facility designated for concrete wash water/wastewater and provide careful oversight to prevent improper dumping of concrete wash water. Any clean up resulting from improper control of concrete wash water will be at no expense to the Government.

3.19 DISPOSAL OF CHLORINATED WATER AND DECHLORINATION REQUIREMENTS

Chlorinated water created during disinfection procedures must not be directly discharged to storm drains or sanitary sewers without prior dechlorination and approval by Code 106.3. Chlorinated water must be neutralized by the controlled addition of a reducing chemical such as sodium thiosulfate, sodium bisulfate, sodium sulfite, sulfur dioxide, or ascorbic acid (commonly known as Vitamin C), as approved by Code 106.3. Dechlorination must be sufficiently effective to reduce total residual chlorine concentration to 1 mg/L or below.

3.20 SNOW STORAGE AND DISPOSAL

Site and operate snow storage and disposal areas to prevent the discharge of snow directly into surface waters and minimize discharges of pollutants from snow maintenance activities. As necessary, the Contractor must employ applicable snow dump BMPs from CMR 06-096 Chapter 573 Snow Dumps: Best Management Practices for Pollution Prevention for snow storage areas.

-- End of Section --

Public Works Department, Maine

ATTACHMENT A

ASBESTOS FREE CERTIFICATION FORM

References:

1. NAVFAC P-502 Asbestos Program Management

A. Asbestos Certification

The Contractor shall not use any asbestos containing material (ACM) at any time during the Work. Contractor shall certify that all material/equipment installed in any portion of the Work is asbestos free. The Government may request that the Contractor provide Safety Data Sheets, manufacturer certifications, laboratory analytical results, and/or other documentation to support the Contractor's certification that all suspect material/equipment is asbestos free. If any material/equipment is found to contain asbestos, the Contractor shall pay for the lawful and proper removal and disposal of product(s), and re-install acceptable material/equipment all at its sole expense.

By signing below, the Contractor certifies that all material/equipment installed in any portion of the Work is asbestos free:

Project Title:

Contract No.:

Company:

Date:

Print Name:

Contractor Signature:

Public Works Department, Maine

Rev. 0
12/17/2021

ATTACHMENT B

STANDARD OPERATING PROCEDURE FOR SITE WASTE REMOVAL (SWR) PROCESS

Applicability: This SOP applies to any special waste and excavated soil removed from Portsmouth Naval Shipyard (PNSY) including dredged materials. Herein the term "waste" represents both landside and dredged excavated materials, and bulk special waste that exceeds the storage capacity of the Hazardous Waste Facility.

Overview: Soils removed from PNSY shall be characterized and disposed in accordance with the regulatory requirements applicable to the waste. NAVFAC PWD-Maine, as the Contracting Officer Representative (COR), owns the overall responsibility for ensuring this removal and disposal process is followed and shall ensure Contractor compliance. The NAVFAC PWD-Maine COR (e.g. CM or ET) shall also ensure required documentation is complete and stored in the project file.

A critical step in this process is the scheduling and execution of a Site Waste Removal (SWR) meeting held at the project site. Wastes cannot be removed from a project site before this meeting is held and documented by a signed Site Waste Removal Form *[Enclosure (1)]*. Three parties must sign the form to complete the process – the NAVFAC COR, the C106 representative, and the contractor.

NOTE: Waste characterization must be completed in accordance with C106 requirements before the SWR meeting can be scheduled. The Contractor shall prepare and submit a Sampling and Analysis Plan (SAP) to C106.3 for approval prior to implementation in accordance with the Contract Documents. The SAP must demonstrate compliance with the sampling requirements of the proposed disposal facility. Following C106 approval of the SAP, waste sampling can be performed by C106 or by the contractor with direct oversight by C106. Next, the analytical results (complete laboratory report) are submitted to C106 for review. C106 will provide direction on the disposal requirements for the waste (e.g. non-hazardous or hazardous). The sampling plan must include a site diagram clearly showing the source location of the waste and the stored waste stockpile location. This diagram must be reviewed at the SWR meeting.

SWR Process:

1. Contractor requests SWR meeting via Microsoft Outlook sent to project CM or ET and the C106 representative a minimum of three (3) business days for non-hazardous wastes and ten (10) business days for hazardous wastes prior to the desired SWR meeting date. The invite shall include a proposed hauling schedule. The meeting must be held at the removal site or area where the waste is stored and will be hauled from.
2. Prior to the meeting, the Contractor shall inspect the site and confirm the field conditions (e.g. source of excavated soils, location and characterization of stockpiled wastes designated for disposal, etc.) match the information presented on the site diagram previously provided to C106 in the waste sampling plan. Wastes shall not be removed from PNSY if the sampling plan documentation does not match the field conditions found at the removal site.
3. Hold on-site SWR meeting. Contractor, NAVFAC CM/ET, and C106 representative shall be present for discussion and signing of the Initial SWR Form *[Enclosure (1)]*. If one of these three parties is not in attendance at the SWR it must be rescheduled.

STANDARD OPERATING PROCEDURE FOR
SITE WASTE REMOVAL (SWR) PROCESS

SWR Process-Continued:

4. The SWR meeting discussion shall *review* the following:
    a. Source of the waste.
    b. Location of removal/stock pile areas designated for removal.
    c. Analytical characterization results of the samples taken for the waste designated for disposal demonstrating non-hazardous or hazardous nature of the waste.
    d. Estimated tonnage of waste to be removed.
    e. Anticipated calendar time frame and duration of waste removal activities.
    f. Phases of waste removal that will require additional C106 review and *SWR* signoff. ‘Phases’ of waste removal will be established for each specific project. The *SWR* meeting group will establish how phases are defined for that particular project. Phase examples include:
        i. One stockpile of soil on site or in one bin from excavation activity.
        ii. One barge load of dredge spoils from dredge material excavation.
        iii. One single area or multiple areas for excavation/demolition within one project site as illustrated on a site diagram for live loading projects. **LIVE LOADING REQUIRES A DAILY SWR MEETING.**
5. SWR group will complete a site walk to identify the waste source area(s) to ensure in-field conditions match the site diagram.
6. NAVFAC will sign one DRR Form for each day of removal activity. A copy of the DRR form will be given to the first truck hauling waste for the activity, and the *DRR* form will be delivered with the truck load to B-357 as proof of DRR form approval for the day. The DRR forms will include the hauler name and truck numbers as well as the information provided in the Initial SWR Form signed by C106.
7. If suspected contaminated wastes (identified by odor or visual examination) are discovered during removal, removal activities shall immediately cease *and contact* C106 to evaluate the situation.
8. The contractor will fill out the descriptive fields of the Initial SWR form. C106 representative and NAVFAC CM/ET will review the information and provide signatures if the information is complete and everyone is in concurrence with the waste removal requirements.
9. The contractor will scan both the Initial SWR Form and corresponding site diagram and send to the NAVFAC CM/ET via email.
10. NAVFAC CM/ET will send the Initial SWR form and corresponding site diagram to the C106 SWR group email **PORT_PTNH_PNS_SWR.**
11. The Contractor *w*ill email subsequent DRR forms for daily waste removal activities to Code 106 at **PORT_PTNH_PNS_SWR.**
12. On the day of hauling, the NAVFAC CM/ET shall double check that the correct stockpile was loaded for disposal, sign the DRR form, and submit to the Contractor to file.
13. The NAVFAC CM/ET will continue daily surveillance, site condition reviews, and execution of DRR forms for each shipping day. The NAVFAC CM/ET will also review the Contractor Daily Production Reports to ensure all loads are recorded so they can be traced back to disposal manifests.

ENCLOSURE 1

☐INITIAL SITE WASTE REMOVAL (SWR) ☐DAILY REMOVAL RECORD (DRR)

WASTE ACTIVITY DESCRIPTION		
1. Contract No:		
2. Contract title:		
3. Prime contractor company name:		
4. Hauling contractor company name:		
5. Anticipated hauling date range: Click or tap to enter a date. through Click or tap to enter a date.		
6. Waste type: ☐ Non-hazardous soils ☐ Hazardous soils ☐ Special waste ____________________ ☐ Other ____________________		
7a. Waste source: 7b. Current waste location:		
8. Anticipated total hauling amount (tons):		
9. PNSY approved waste disposal facility:		
WASTE REMOVAL APPROVALS – The signatures below verify that an onsite visual inspection of the material has been performed, and that this form is a record of the facts agreed upon.		
10. Prime contractor signature:		
11. NAVFAC CM/ET signature:		
12. Code 106.3 signature:		
DAILY REMOVAL RECORD - to be filled each morning of waste removal		
13. Hauling date:		
14. Anticipated hauling amount:	tons	truckloads
15. Hauling contractor Name & truck ID number(s):		
16. Prime contractor signature:		
17. NAVFAC CM/ET signature:		

WASTE REMOVAL PASS ATTRIBUTE DESCRIPTIONS

1. *Contract No*. The unique NAVFAC contract identification number applied to the project requesting waste removal.

2. *Contract title*. The unique contract title/description applied to the project requesting waste removal.

3. *Prime contractor company name*. The project's prime contractor company name.

4. *Hauling contractor company name*. The subcontractor hired to haul waste from the project site to the PNSY approved waste disposal company.

5. *Anticipated hauling date(s)*. Hauling date range agreed upon at SWR meeting for duration of waste removal in the approved phase of removal.

6. W*aste description*. Description of waste in terms of hazardous/non-hazardous designation and any other characteristic determined at the SWR meeting.

7a. W*aste source*. Detailed written description of the source of the waste to be removed. Include removal areas, buildings, streets, land marks, coordinates, and other information as available.

7b. *Current waste location*. Detailed written description of the waste location if displaced from the original source during removal. Include removal areas, bin designations, stockpile locations, buildings, streets, land marks, coordinates and other information as available.

8. *Anticipated total hauling amount (tons)*. Total amount of waste anticipated for removal in the approved phase of removal.

9. *PNSY approved waste disposal facility*. Name of the PNSY approved waste disposal facility.

10. *Prime contractor signature*. Written signature of project's prime contractor.

11. *NAVFAC CM/ET signature*. Written signature of NAVFAC CM or ET in attendance at the SWR meeting.

12. *Code 106.3 signature*. Written signature of Code 106.3 representative in attendance at the SWR meeting.

13. *Hauling date*. Date of the waste removal.

14. *Anticipated hauling amount*. Entire hauling amount (total tons) anticipated for said hauling date.

15. *Hauling contractor truck ID number(s)*. The identification number of the contractor's truck.

16. *Prime contractor signature*. Written signature of prime contractor.

17. *NAVFAC CM/ET signature*. Written signature of project's NAVFAC CM or ET overseeing daily removal of waste from the site.

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SECTION 01 58 00.00 22
^^
PROJECT IDENTIFICATION (PWD ME)
08/19, CHG 5: 08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)

AWPA U1	(2024) Use Category System: User Specification for Treated Wood

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-02 Shop Drawings

Preliminary One Line Drawings Of Project Rendering; G

Preliminary Drawing Indicating Layout And Text Content; G

SD-04 Samples

Final Rendering Sample; G

Final Framed Rendering and Copies; G

Facility Recognition Plaque; G

1.3 QUALITY CONTROL

1.3.1 Rendering

Provide the project rendering in accordance with the following drawing stages as required in the SUBMITTALS paragraph. The following submittal data is required to properly identify the appropriate view and approve the final rendering of the facility. The final painted rendering will be used to produce the image for the signboard and framed photographic copies provided to the Contracting Officer.

1.3.1.1 Preliminary One Line Drawings

Provide three different views of the facility in a preliminary single line drawing (black and white) format. These three views will represent the best angles at which to view the proposed facility showing the best design features and the three dimensional character of the facility.

1.3.1.2 Final Rendering Sample

Provide a photographic copy (8 by 10 inches minimum size) of final rendering for approval of color, landscaping, and foreground/background development prior to final submittal.

1.3.1.3 Final Framed Rendering and Copies

Provide final full color rendering of the proposed facility as specified.

1.3.2 Facility Recognition Plaque

Submit full size drawing of Facility Recognition Plaque for approval. Confirm the content (message), location and mounting with Contracting Officer prior to fabrication. The final names on the plaque will be determined at the end of the project duration to assure that current participants can be identified and recognized on the plaque.

1.4 PROJECT IDENTIFICATION SIGN

Prior to initiating any work on site, provide one project identification sign at the location indicated. Construct the sign in accordance with project sign detail, which can be downloaded at: http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables. Maintain sign throughout the life of the project. Upon completion of the project, remove the sign from the site. The Government will temporarily supply a copy of the rendering to use in the production of the final signboard artwork.

1.4.1 Project Identification Signboard

Provide a project identification signboard in accordance with attached Plates 2, 3, 4, and 5. Provide a preliminary drawing indicating layout and text content. Erect a signboard at a conspicuous location on the job site where directed by the Contracting Officer.

a. The field of the sign consists of a 4 by 8 foot sheet of grade B-B medium density overlaid exterior plywood.

b. Lumber is B or better Southern pine, pressure-preservative treated in accordance with AWPA U1. Nails are aluminum or galvanized steel.

c. Give one coat of exterior alkyd primer and two coats of exterior alkyd enamel paint to the entire signboard and supports. Perform the lettering and sign work by a skilled sign painter using paint known in the trade as bulletin colors. The colors, lettering sizes, and lettering styles are as indicated. Where preservative-treated lumber is required, utilize only cured pressure-treated wood which has had the chemicals leached from the surface of the wood prior to painting.

d. Use spray applied automotive quality high gloss acrylic white enamel paint as background for the NAVFAC logo. NAVFAC logo is an applied 2 mil film sticker/decal with either transparent or white background or paint the logo by stencil onto the sign. The weather resistant sticker/decal film is rated for a minimum of 2-year exterior vertical exposure. Mount the self-adhering sticker to the sign with pressure sensitive, permanent acrylic adhesive. Shop cut sticker/decal to rectangular shape and provide pull-off backing sheet on adhesive side of design sticker for shipping.

e. Sign paint colors (manufacturer's numbers/types listed below for color identification only)

(1) Blue = To match dark blue color in the NAVFAC logo.

(2) White = To match Brilliant White color in the NAVFAC logo.

f. NAVFAC logo must retain proportions and design integrity. NAVFAC logos in electronic format may be obtained from the WBDG at the following link:
http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables.
Use the following to choose color values for the paint to be used:

(1) Dark Blue = equivalent to CMYK values 100, 72, 0, 8.

(2) Light Blue = equivalent to CMYK values 69, 34, 0, 0.

(3) Cyan = equivalent to CMYK values 100, 9, 0, 6.

(4) Yellow = equivalent to CMYK values 0.9,94, 0.

g. Final signboard artwork (rendering) may be either mounted under plexiglass as indicated in attached Plates 2 and 5, or may be electrostatically printed on 4 mil self-adhering, weather resistant, glossy vinyl film and mounted to signboard. Provide film that is capable of full color reproduction of the building rendering and cover it with an ultra-violet protection film. Laminate the 2 mil satin gloss clear protection film to the white 4 mil vinyl image film. Utilize pressure sensitive "controltac" adhesive to attach rendering to signboard and smooth out surface with hand pressure tools in accordance with manufacturer's recommendations. Shop cut sticker to size required and provide pull-off backing sheet on adhesive side of film for shipping. Provide the rendering on film that is rated for a minimum of 2-years exterior vertical exposure.

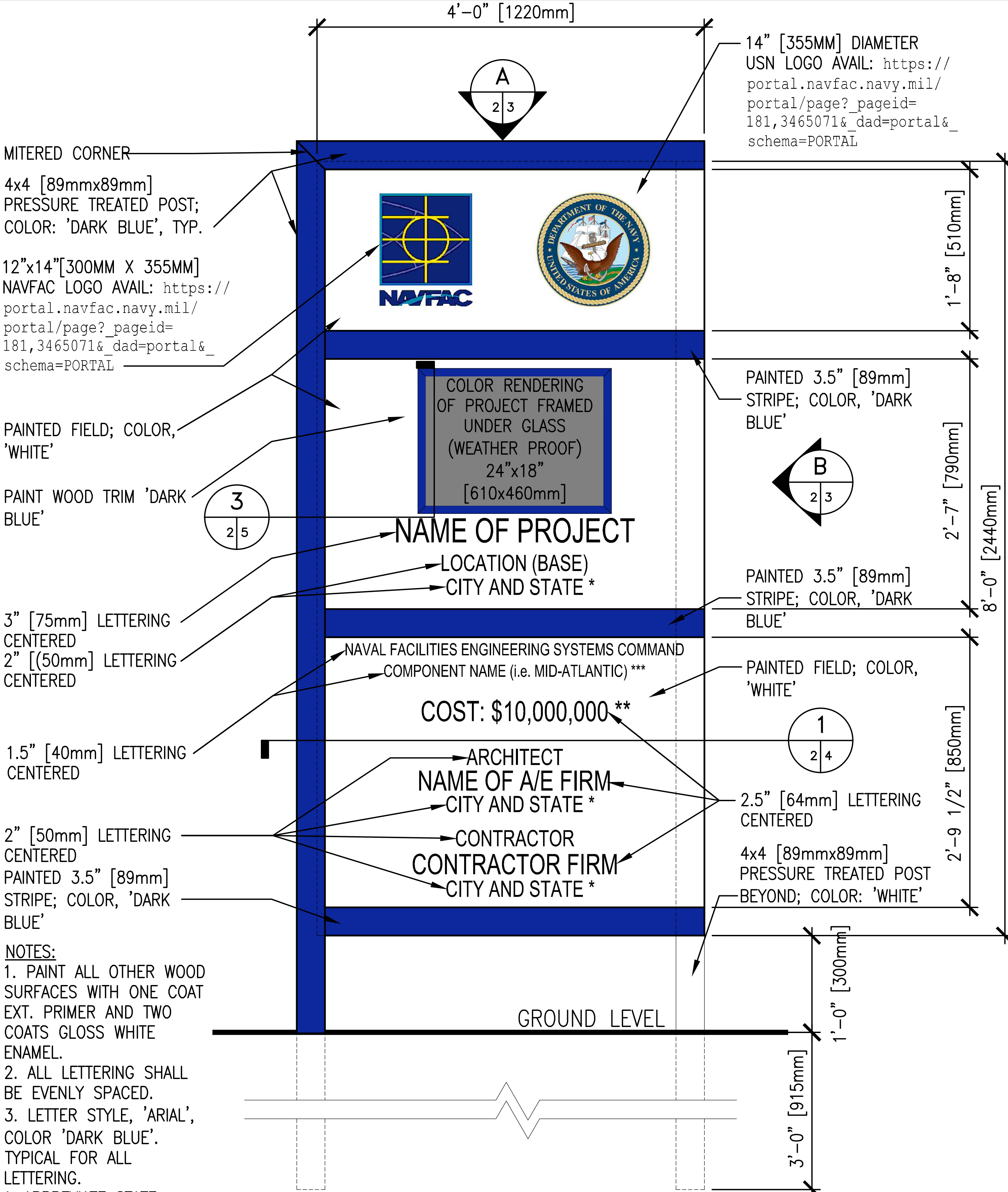
PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --



NOTES:

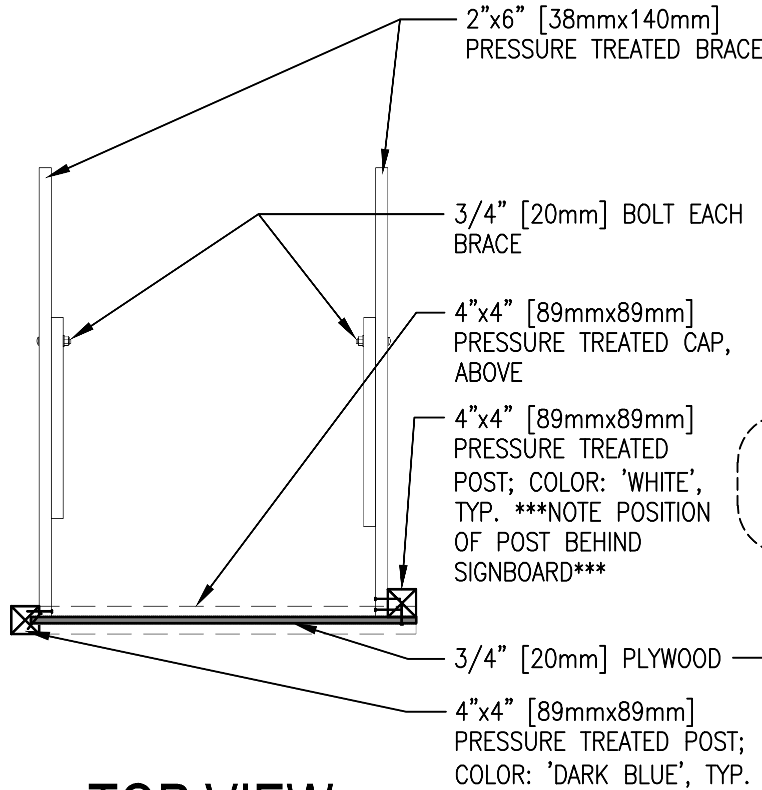
1. PAINT ALL OTHER WOOD SURFACES WITH ONE COAT EXT. PRIMER AND TWO COATS GLOSS WHITE ENAMEL.
2. ALL LETTERING SHALL BE EVENLY SPACED.
3. LETTER STYLE, 'ARIAL', COLOR 'DARK BLUE'. TYPICAL FOR ALL LETTERING.

*–ABBREVIATE STATE
**–CONFIRM USE WITH CONTRACTING OFFICER
***–VERIFY NAME WITH CONTRACTING OFFICER.

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD

SCALE: 3/4" = 1'-0"

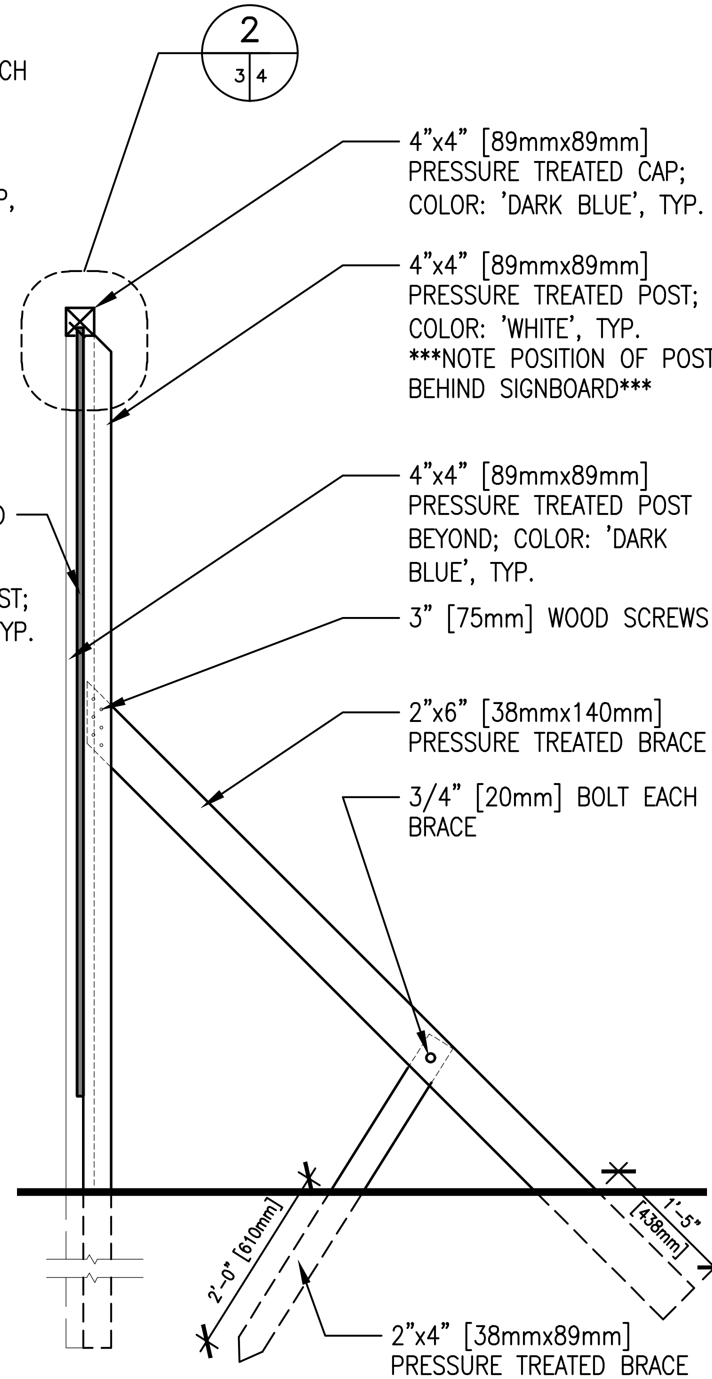
PLATE 2USN



A
1,2 | 3

TOP VIEW
SCALE: 1/2" = 1'-0"

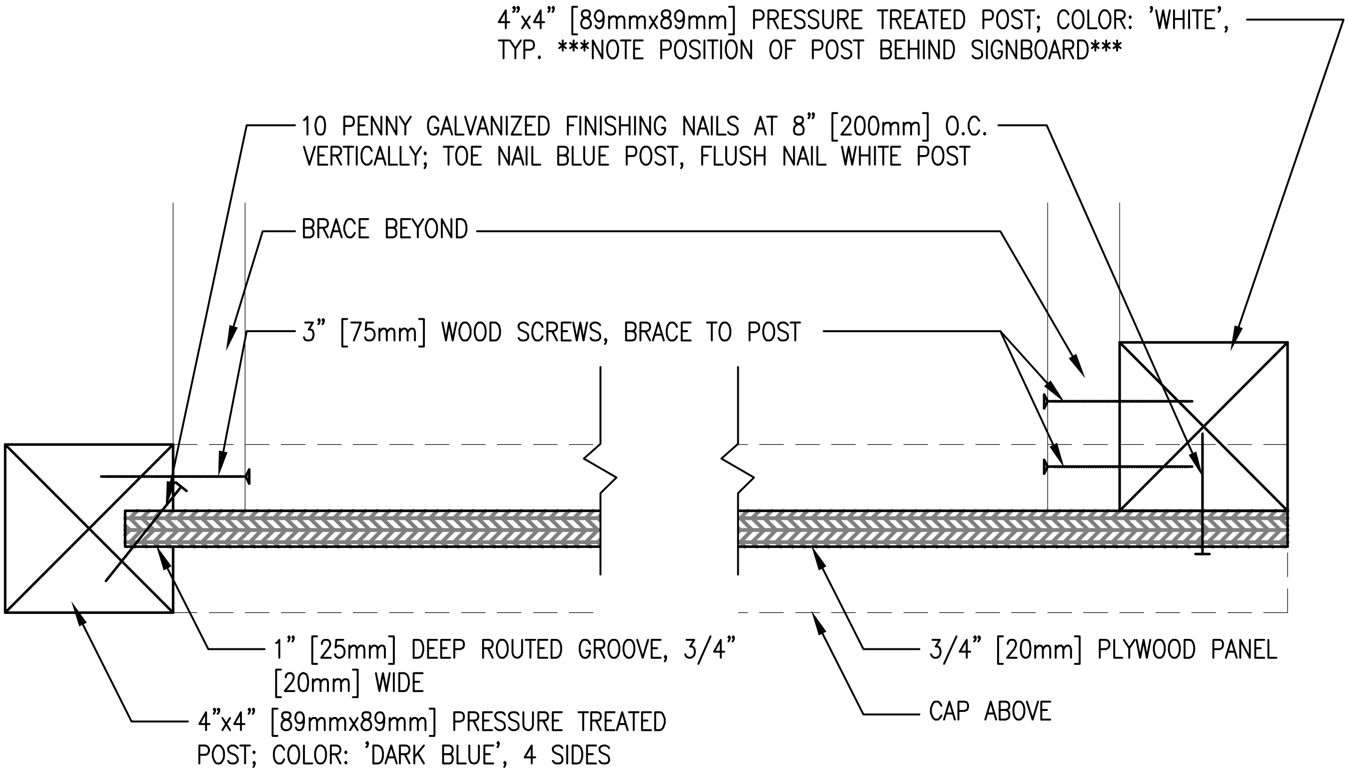
NOTES:
1. POSTS AND BRACES SHALL BE PRESSURE TREATED.
2. ALL FASTENERS SHALL BE ZINC COATED.
3. BRACING IS REQUIRED IN ALL SOIL CONDITIONS AND HIGH WIND ENVIRONMENTS.



B
1,2 | 3

SIDE VIEW
SCALE: 1/2" = 1'-0"

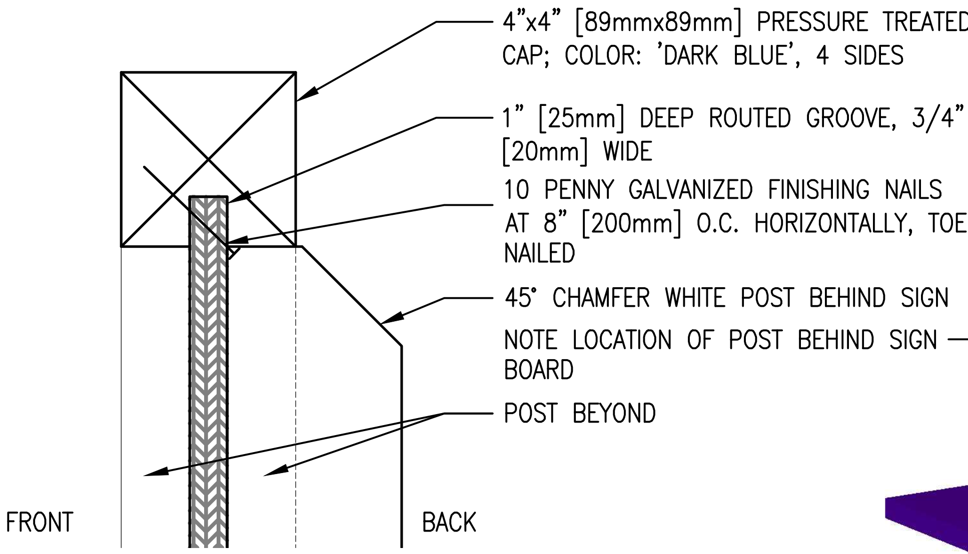
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD
SUPPORT DETAILS

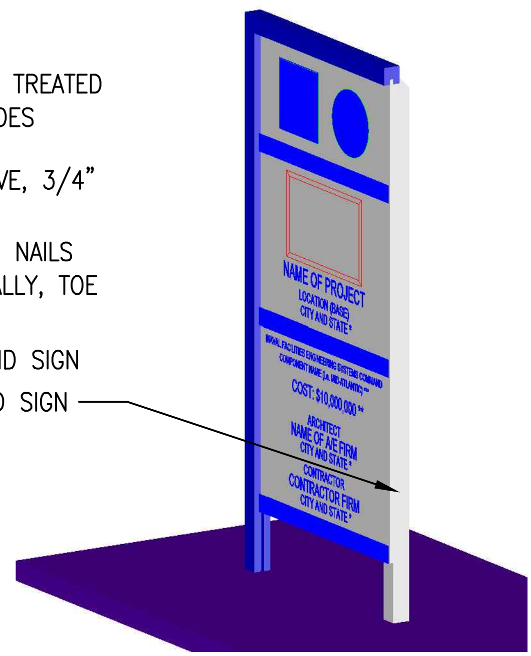


1 PLAN SECTION

1,2 | 4

SCALE: 3" = 1'-0"





2 SECTION AT TOP

1,2 | 4

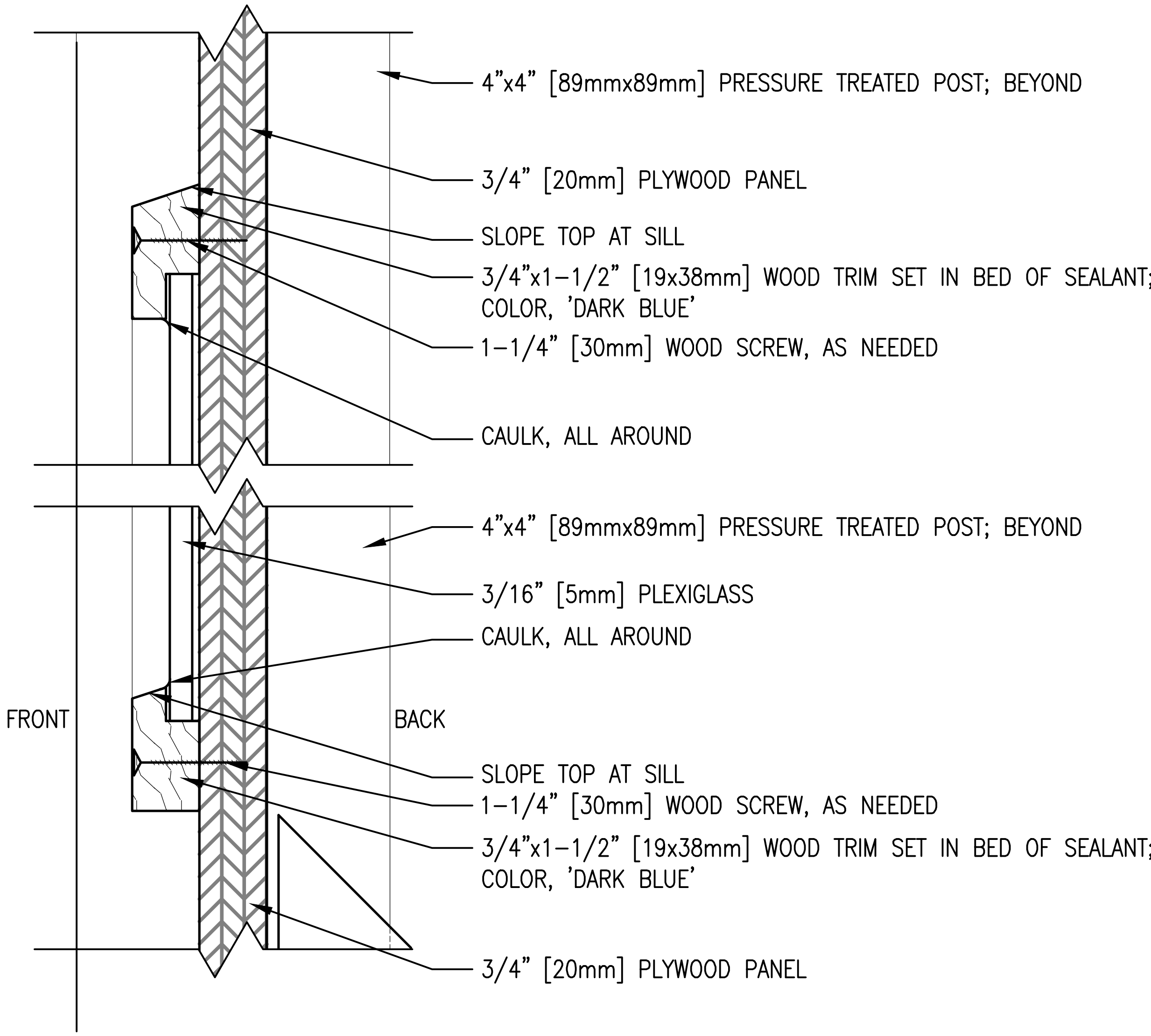
SCALE: 3" = 1'-0"

3 ISO VIEW

2 | 4

SCALE: NONE

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD SECTIONS



3 / 2 | 5

SECTION AT RENDERING FRAME

SCALE: 6" = 1'-0"

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD SECTION

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SECTION 01 74 19.00 22
^^
CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL (PWD ME)
02/19, CHG 3: 11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 273	Standards for Universal Waste Management
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
49 CFR 178	Specifications for Packagings

1.2 DEFINITIONS

1.2.1 Co-mingle

The practice of placing unrelated materials together in a single container, usually for benefits of convenience and speed.

1.2.2 Construction Waste

Waste generated by construction activities, such as scrap materials, damaged or spoiled materials, temporary and expendable construction materials, and other waste generated by the workforce during construction activities.

1.2.3 Demolition Debris/Waste

Waste generated from demolition activities, including minor incidental demolition waste materials generated as a result of Intentional dismantling of all or portions of a building, to include clearing of building contents that have been destroyed or damaged.

1.2.4 Disposal

Depositing waste in a solid waste disposal facility, usually a managed landfill or incinerator, regulated in the US under the Resource Conservation and Recovery Act (RCRA).

1.2.5 Diversion

The practice of diverting waste from disposal in a landfill or incinerator, by means of eliminating or minimizing waste, or reuse of materials.

1.2.6 Final Construction Waste Diversion Report

A written assertion by a material recovery facility operator identifying constituent materials diverted from disposal, usually including summary tabulations of materials, weight in short-ton.

1.2.7 Recycling

The series of activities, including collection, separation, and processing, by which products or other materials are diverted from the solid waste stream for use in the form of raw materials in the manufacture of new products sold or distributed in commerce, or the reuse of such materials as substitutes for goods made of virgin materials, other than fuel.

1.2.8 Reuse

The use of a product or materials again for the same purpose, in its original form or with little enhancement or change.

1.2.9 Salvage

Usable, salable items derived from buildings undergoing demolition or deconstruction, parts from vehicles, machinery, other equipment, or other components.

1.2.10 Source Separation

The practice of administering and implementing a management strategy to identify and segregate unrelated waste at the first opportunity.

1.3 CONSTRUCTION WASTE (INCLUDES DEMOLITION DEBRIS/WASTE)

Divert a minimum of 60 percent by weight of the project construction waste and demolition debris/waste from the landfill or incinerator. Follow applicable industry standards in the management of waste. Apply sound environmental principles in the management of waste. (1) Practice efficient waste management when sizing, cutting, and installing products and materials and (2) use all reasonable means to divert construction waste and demolition debris/waste from landfills and incinerators and to facilitate the recycling or reuse of excess construction materials.

1.4 CONSTRUCTION WASTE MANAGEMENT

Implement a Construction Waste Management Program for the project. Take a pro-active, responsible role in the management of construction construction waste, recycling process, disposal of demolition debris/waste, and require all subcontractors, vendors, and suppliers to participate in the Construction Waste Management Program. Establish a process for clear tracking, and documentation of construction waste and demolition debris/waste.

1.4.1 Implementation of Construction Waste Management Program

Develop and document how the Construction Waste Management Program will be implemented in a Construction Waste Management Plan. Submit a Construction Waste Management Plan to the Contracting Officer for approval. Construction waste and demolition debris/waste materials include un-used construction materials not incorporated in the final work, as well as demolition debris/waste materials from demolition activities or deconstruction activities. In the management of waste, consider the availability of viable markets, the condition of materials, the ability to provide material in suitable condition and in a quantity acceptable to available markets, and time constraints imposed by internal project completion mandates.

1.4.2 Oversight

The Environmental Manager, as specified in Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME), is responsible for overseeing and documenting results from executing the Construction Waste Management Plan for the project and is responsible for all waste being disposed of in accordance with the Contract Documents and state and Federal regulations. If any waste materials are removed from the Portsmouth Naval Shipyard or Facility not in compliance with the requirements stated in the Contract Documents, the person assigned as the Environmental Manager is subject to removal by the Contracting Officer for non-compliance with requirements specified in the Contract. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor. The Contractor will be responsible to take all required actions to address and correct any non-compliance as well as paying any fines as a result of improper handling or disposal.

1.4.3 Special Programs

Implement special programs involving rebates or similar incentives related to recycling of construction waste and demolition debris/waste materials. Retain revenue or savings from salvaged or recycling, unless otherwise directed. Ensure firms and facilities used for recycling, reuse, and disposal are permitted for the intended use to the extent required by federal, state, and local regulations.

1.4.4 Special Instructions

Provide on-site instruction of appropriate separation, handling, recycling, salvage, reuse, and return methods to be used by all parties at the appropriate stages of the projects. Designation of single source separating or commingling will be clearly marked on the containers.

1.4.5 Waste Streams

Delineate waste streams and characterization, including estimated material types and quantities of waste, in the Construction Waste Management Plan. Manage all waste streams associated with the project. Typical waste streams are listed below. Include additional waste steams not listed:

a. Land Clearing Debris

b. Asphalt

c. Masonry and CMU

d. Concrete

e. Metals (Includes, but is not limited to, banding, stud trim, ductwork, piping, rebar, roofing, other trim, steel, iron, galvanized, stainless steel, aluminum, copper, zinc, bronze.)

f. Wood (nails and staples allowed)

g. Glass

h. Paper

i. Plastics (PET, HDPE,PVC,LDPE,PP,PS, Other)

j. Gypsum

k. Non-hazardous paint and paint cans

l. Carpet

m. Ceiling Tiles

n. Insulation

o. Beverage Containers

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

Construction Waste Management Plan; G

SD-11 Closeout Submittals

Final Construction Waste Diversion Report; G

1.6 MEETINGS

Conduct Construction Waste Management meetings. After award of the Contract and prior to commencement of work, schedule and conduct a meeting with the Contracting Officer to discuss the proposed Construction Waste Management Plan and to develop a mutual understanding relative to the management of the Construction Waste Management Program and how waste diversion requirements will be met.

The requirements of this meeting may be fulfilled during the coordination and mutual Understanding meeting outlined in Section 01 45 00.00 22 QUALITY CONTROL (PWD ME). At a minimum, discuss and document waste management goals at following meetings:

a. Preconstruction meeting.

b. Regular Quality Control meetings.

c. Work safety meeting (if applicable).

1.7 CONSTRUCTION WASTE MANAGEMENT PLAN

Submit Construction Waste Management Plan within 15 days after contract award. Revise and resubmit Construction Waste Management Plan as necessary, in order for construction to begin. Execute demolition or deconstruction activities in accordance with Section 02 41 00 DEMOLITION AND DECONSTRUCTION. Manage demolition debris/waste or deconstruction materials in accordance with the approved construction waste management plan.

An approved Construction Waste Management Plan will not relieve the Contractor of responsibility for compliance with applicable environmental regulations or meeting project cumulative waste diversion requirement. Ensure all subcontractors receive a copy of the approved Construction Waste Management Plan. The plan demonstrates how to meet the project waste diversion requirement. Also, include the following in the plan:

a. Identify the names of individuals responsible for waste management and waste management tracking, along with roles and responsibilities on the project.

b. Actions that will be taken to reduce solid waste generation, including coordination with subcontractors to ensure awareness and participation.

c. Description of the regular meetings to be held to address waste management.

d. Description of the specific approaches to be used in recycling/reuse of the various materials generated, including the areas on site and equipment to be used for processing, sorting, and temporary storage of materials.

e. Name of landfill and incinerator to be used.

f. Identification of local and regional re-use programs, including non-profit organizations such as schools, local housing agencies, and organization that accept used materials such as material exchange networks and resale stores. Include the name, location, phone number for each re-use facility identified, and provide a copy of the permit or license for each facility.

g. List of specific materials, by type and quantity, that will be salvaged for resale, salvaged and reused on the current project, salvaged and stored for reuse on a future project, or recycled. Identify the recycling facilities by name, address, and phone number.

h. Identification of materials that cannot be recycled or reused with an explanation or justification, to be approved by the Contracting Officer.

i. Description of the means by which materials identified in item (g) above will be protected from contamination.

j. Description of the means of transportation of the recyclable materials (whether materials will be site-separated and self-hauled to designated centers, or whether mixed materials will be collected by a waste hauler and removed from the site).

k. Copy of training plan for subcontractors and other services to prevent contamination by co-mingling materials identified for diversion and waste materials.

l. Identification of at least 5 construction or demolition material streams for diversion.

m. Facilities or subcontractors offering construction waste transport on-site or off-site must ensure that proper shipping orders, bill of lading, manifests, or other shipping documents containing waste diversion information meet requirements of 40 CFR 273 Universal Waste Management, 49 CFR 173 Shippers - General Requirements for Shipments and Packagings, and 49 CFR 178 Specifications for Packaging. Individuals signing manifests or other shipping documents should meet the minimum training requirements.

n. List each supplier who deliver construction materials, in bulk, or package products in returnable containers or returnable packaging, or have take-back programs. List each program and the applicable material to actively monitor and track to assist in meeting waste diversion requirements on the project.

Distribute copies of the waste management plan to each subcontractor, Environmental Manager, and the Contracting Officer.

1.8 RECORDS (DOCUMENTATION)

1.8.1 General

Maintain records to document the types and quantities of waste generated and diverted though re-use, recycling and sale to third parties; through disposal to a landfill or incinerator facility. Provide explanations for materials not recycled, reused or sold. Collect and retain manifests, weight tickets, sales receipts, and invoices specifically identifying diverted project waste materials or disposed materials.

1.8.2 Accumulated

Maintain a running record of materials generated and diverted from landfill disposal, including accumulated diversion rates for the project. Make records available to the Contracting Officer during construction or incidental demolition activities. Provide a copy of the diversion records to the Contracting Officer upon completion of the construction, incidental demolitions or minor deconstruction activities.

1.9 FINAL CONSTRUCTION WASTE DIVERSION REPORT

A Final Construction Waste Diversion Report is required at the end of the project. Provide Final Construction Waste Diversion Report 60 days prior to the Beneficial Occupancy Date (BOD). The final Construction Waste Diversion Report must be included in the Sustainability eNotebook in accordance with Section 01 33 29.00 22 SUSTAINABILITY REQUIREMENTS AND REPORTING (PWD ME).

1.10 COLLECTION

Collect, store, protect, and handle reusable and recyclable materials at the site in a manner which prevents contamination, and provides protection from the elements to preserve their usefulness and monetary value. Provide receptacles and storage areas designated specifically for recyclable and reusable materials and label them clearly and appropriately to prevent contamination from other waste materials. Keep receptacles or storage areas neat and clean.

Train subcontractors and other service providers to either separate waste streams or use the co-mingling method as described in the Construction Waste Management Plan. Handle hazardous waste and hazardous materials in accordance with applicable regulations and coordinate with Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME). Separate materials by one of the following methods described herein:

1.10.1 Source Separation Method

Separate waste products and materials that are recyclable from trash and sort as described below into appropriately marked separate containers and then transport to the respective recycling facility for further processing. Deliver materials in accordance with recycling or reuse facility requirements (e.g., free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process). Separate materials into the category types as defined in the Construction Waste Management Plan.

1.10.2 Other Methods

Other methods proposed by the Contractor may be used when approved by the Contracting Officer.

1.11 DISPOSAL

Control accumulation of waste materials and trash. Recycle or dispose of collected materials off-site at intervals approved by the Contracting Officer and in compliance with waste management procedures as described in the waste management plan. Except as otherwise specified in other sections of the specifications, dispose of in accordance with the following:

1.11.1 Reuse

Give first consideration to reusing construction and demolition materials as a disposition strategy. Recover for reuse materials, products, and components as described in the approved Construction Waste Management Plan. Coordinate with the Contracting Officer to identify onsite reuse opportunities or material sales or donation available through Government resale or donation programs. Sale of recovered materials is not allowed on the Portsmouth Naval Shipyard.

1.11.2 Recycle

Recycle non-hazardous construction and demolition/debris materials that are not suitable for reuse. Track rejection of contaminated recyclable materials by the recycling facility. Rejected recyclables materials will not be counted as a percentage of diversion calculation. Recycle all fluorescent lamps, HID lamps, mercury (Hg) -containing thermostats and ampoules, and PCBs-containing ballasts and electrical components as directed by the Contracting Officer. Do not crush lamps on site as this creates a hazardous waste stream with additional handling requirements.

1.11.3 Waste

Dispose by landfill or incineration only those waste materials with no practical use, economic benefit, or recycling opportunity.

1.12 ADDITIONAL REPORTING REQUIRMENTS

Provide monthly cost and revenue data to the NAVFAC Midlant Integrated Solid Waste Management office. Submit report by e-mail to: IntegratedSolidWasteManagement@navy.mil no later than the 3rd of each month. Data must be reported on an Excel document provided by the Contracting Officer. Comply with the requirements specified in Appendix 01 74 19-1, CONSTRUCTION AND DEMOLITION SOLID WASTE REPORT.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used. -- End of Section --

CONSTRUCTION AND DEMOLITION SOLID WASTE REPORT

SITE: **Month:** ____________

Contractor's Company Name: **Contract #**

Contractor's POC and Telephone or Email Address:

Project Description:

SECTION 1	Tons	Cost	Revenue	Remarks
Recycled (tons)				
Concrete(incl: brick & block)				
Wood				
Metal				
Asphalt				
Green waste(clearing debris)				
Dirt				
Sand				
Gravel/Rock				
Mixed				
Misc				
Subtotal - Recycled	**0.00**	**$ -**	**$ -**	
SECTION 2				
Landfilled (tons)				
Concrete(incl: brick & block)				
Wood				
Metal				
Asphalt				
Green Waste(clearing debris)				
General C&D				
Dirt				
Sand				
Gravel/Rock				
Mixed				
Misc				
Subtotal - Landfilled	**0.00**	**$ -**	**$ -**	
Solid Waste (tons)				
Total Solid Waste	**0.00**	**$ -**	**$ -**	

REPORTING DEADLINE IS NO LATER THAN THE 3RD OF EACH MONTH

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SECTION 01 78 00.00 22
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CLOSEOUT SUBMITTALS (PWD ME)
05/19, CHG 1: 08/21

PART 1 GENERAL

This Section applies to only Design-Bid-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E1971	(2005; R 2011) Standard Guide for Stewardship for the Cleaning of Commercial and Institutional Buildings

GREEN SEAL (GS)

GS-37	(2017) Cleaning Products for Industrial and Institutional Use

U.S. DEPARTMENT OF DEFENSE (DOD)

FC 1-300-09N	(2024) Navy and Marine Corps Design

1.2 DEFINITIONS

1.2.1 As-Built Drawings

As-built drawings are the marked-up drawings, maintained by the Contractor on-site, that depict actual conditions and deviations from the Contract Documents. These deviations and additions may result from coordination required by, but not limited to: contract modifications; official responses to submitted Requests for Information (RFI's); direction from the Contracting Officer; design that is the responsibility of the Contractor, and differing site conditions. Maintain the as-builts throughout construction as red-lined hard copies on site. These files serve as the basis for the creation of the record drawings. Monthly progress as-builts are required with contractor's invoice, refer to 01 20 00.00 22 PRICE AND PAYMENT PROCEDURES (PWD ME) Paragraph Content of Invoice for additional information.

1.2.2 Record Drawings

The record drawings are the final compilation of actual conditions reflected in the as-built drawings.

1.3 SOURCE DRAWING FILES

Request the full set of electronic drawings, in the source format, for Record Drawing preparation, after award and at least 30 days prior to required use.

1.3.1 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction drawings and data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim and waives to the fullest extent permitted by law, any claim or cause of action of any nature against the Government, its agents or sub consultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic CAD drawing files are not construction documents. Differences may exist between the CAD files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic CAD files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished Source drawing files, the signed and sealed construction documents govern. The Contractor is responsible for determining if any conflict exists. Use of these Source Drawing files does not relieve the Contractor of duty to fully comply with the contract documents, including and without limitation, the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction drawings and data related to this contract, remove all previous indicia of ownership (seals, logos, signatures, initials and dates).

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-03 Product Data

- Warranty Management Plan
- Warranty Tags
- Final Cleaning
- Spare Parts Data

SD-08 Manufacturer's Instructions

- Posted Instructions
- Preventative Maintenance; G
- Condition Monitoring (Predictive Testing); G

Inspection; G

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

SD-11 Closeout Submittals

As-Built Drawings; G

Record Drawings; G,

As-Built Record of Equipment and Materials; G

Warranted Equipment and Materials

Certification of EPA Designated Items; G

Certification Of USDA Designated Items; G

Red Zone Documents; G

Facility Data Workbook (FDW), Final Submission; G

Post Installation Sanitary System Survey; G

High Performance and Sustainable Building (HPSB) Checklist; G

List Of Installed Building Equipment

1.5 SPARE PARTS DATA

Submit two copies of the Spare Parts Data list.

a. Indicate manufacturer's name, part number, and stock level required for test and balance, pre-commissioning, maintenance and repair activities. List those items that may be standard to the normal maintenance of the system.

1.6 WARRANTY MANAGEMENT

1.6.1 Warranty Management Plan

Develop a warranty management plan which contains information relevant to FAR 52.246-21 Warranty of Construction. At least 30 days before the planned pre-warranty conference, submit one set of the warranty management plan. Include within the warranty management plan all required actions and documents to assure that the Government receives all warranties to which it is entitled. The plan narrative must contain sufficient detail to render it suitable for use by future maintenance and repair personnel, whether tradesmen, or of engineering background, not necessarily familiar with this contract. The term "status" as indicated below must include due date and whether item has been submitted or was accomplished. Submit warranty information, made available during the construction phase, to the Contracting Officer for approval prior to each monthly pay estimate. Assemble approved information in a binder and turn over to the Government upon acceptance of the work. The construction warranty period must begin on the date of project acceptance and continue for the full product warranty period. Conduct a joint 4 month and 9 month warranty inspection, measured from time of acceptance; with the Contractor, Contracting Officer and the Customer Representative. The warranty management plan must include, but is not limited to, the following:

a. Roles and responsibilities of personnel associated with the warranty process, including points of contact and telephone numbers within the organizations of the Contractors, subcontractors, manufacturers or suppliers involved.

b. For each warranty, the name, address, telephone number, and e-mail of each of the guarantor's representatives nearest to the project location.

c. A list and status of delivery of Certificates of Warranty for extended warranty items, including roofs, HVAC balancing, pumps, motors, transformers, and for commissioned systems, such as fire protection and alarm systems, sprinkler systems, and lightning protection systems.

d. Warranted Equipment and Materials list for each warranted equipment, item, feature of construction or system indicating:

(1) Name of item.
(2) Model and serial numbers.
(3) Location where installed.
(4) Name and phone numbers of manufacturers or suppliers.
(5) Names, addresses and telephone numbers of sources of spare parts.
(6) Warranties and terms of warranty. Include one-year overall warranty of construction, including the starting date of warranty of construction. Items which have warranties longer than one year must be indicated with separate warranty expiration dates.
(7) Cross-reference to warranty certificates as applicable.
(8) Starting point and duration of warranty period.
(9) Summary of maintenance procedures required to continue the warranty in force.
(10) Cross-reference to specific pertinent Operation and Maintenance manuals.
(11) Organization, names and phone numbers of persons to call for warranty service.
(12) Typical response time and repair time expected for various warranted equipment.

e. The plans for attendance at the 4 and 9 month post-construction warranty inspections conducted by the Government.

f. Procedure and status of tagging of equipment covered by warranties longer than one year.

g. Copies of instructions to be posted near selected pieces of equipment where operation is critical for warranty or safety reasons.

1.6.2 Performance Bond

The Performance Bond must remain effective throughout the construction and warranty period.

a. In the event the Contractor fails to commence and diligently pursue any construction warranty work required, the Contracting Officer will have the work performed by others, and after completion of the work, will charge the remaining construction warranty funds of expenses incurred by the Government while performing the work, including, but not limited to administrative expenses.

b. In the event sufficient funds are not available to cover the construction warranty work performed by the Government at the Contractor's expense, the Contracting Officer will have the right to recoup expenses from the bonding company.

c. Following oral or written notification of required construction warranty repair work, respond in a timely manner. Written verification will follow oral instructions. Failure to respond will be cause for the Contracting Officer to proceed against the Contractor.

1.6.3 Pre-Warranty Conference

Prior to contract completion, and at a time designated by the Contracting Officer, meet with the Contracting Officer to develop a mutual understanding with respect to the requirements of this section. At this meeting, establish and review communication procedures for Contractor notification of construction warranty defects, priorities with respect to the type of defect, reasonable time required for Contractor response, and other details deemed necessary by the Contracting Officer for the execution of the construction warranty In connection with these requirements and at the time of the Contractor's quality control completion inspection, furnish the name, telephone number and address of a licensed and bonded company which is authorized to initiate and pursue construction warranty work action on behalf of the Contractor. This point of contact must be located within the local service area of the warranted construction, be continuously available, and be responsive to Government inquiry on warranty work action and status. This requirement does not relieve the Contractor of any of its responsibilities in connection with other portions of this provision.

1.6.4 Warranty Tags

At the time of installation, tag each warranted item with a durable, oil and water resistant tag approved by the Contracting Officer. Attach each tag with a copper wire and spray with a silicone waterproof coating. Also, submit two record copies of the warranty tags showing the layout and design. The date of acceptance and the QC signature must remain blank until the project is accepted for beneficial occupancy. Show the following information on the tag.

Type of product/material	
Model number	
Serial number	
Contract number	
Warranty period from/to	
Inspector's signature	

Construction Contractor	
Address	
Telephone number	
Warranty contact	
Address	
Telephone number	
Warranty response time priority code	
WARNING - PROJECT PERSONNEL TO PERFORM ONLY OPERATIONAL MAINTENANCE DURING THE WARRANTY PERIOD.	

1.7 PROJECT CLOSEOUT DOCUMENTS

1.7.1 As-Built Record of Equipment and Materials

Furnish one (1) copy of preliminary record of equipment and materials used on the project 15 working days prior to final inspection. This preliminary submittal will be reviewed and returned 5 working days after final inspection with Government comments. Submit two (2) sets of final record of equipment and materials 10 working days after final inspection. Key the designations to the related area depicted on the Contract Drawings. List the following data:

RECORD OF DESIGNATED EQUIPMENT AND MATERIALS DATA				
Description	Specification Section	Manufacturer and Catalog, Model, and Serial Number	Composition and Size	Where Used

1.7.2 Construction Contract Specifications

Furnish final record (as-built) construction Contract Specifications, including modifications thereto, 30 calendar days after transfer of the completed facility.

1.7.3 Real Property Equipment

Furnish a list of installed equipment furnished under this Contract. Include all information usually listed on a manufacturer's name plate. In the "EQUIPMENT-IN-PLACE LIST" include, as applicable, the following for each piece of equipment installed: description of item, location (by room number), model number, serial number, capacity, name and address of manufacturer, name and address of equipment supplier, condition, spare parts list, manufacturer's catalog, and warranty. Furnish a draft list at time of transfer. Furnish the final list 30 calendar days after transfer of the completed facility.

1.7.4 Red Zone Documents

Submit Red Zone Documents in accordance with Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME).

1.7.5 Facility Data Workbook (FDW), Final Submission

Submit Facility Data Workbook (FDW), Final Submission in accordance with Section 01 78 25.00 20.

1.7.6 Post Installation Sanitary System Survey

Submit Post Installation Sanitary System Survey CD in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE.

1.8 PREVENTATIVE MAINTENANCE

Submit Preventative Maintenance and Condition Monitoring (Predictive Testing) and Inspection schedules with instructions that state when systems should be retested.

Define the anticipated length of each test, test apparatus, number of personnel identified by responsibility, and a testing validation procedure permitting the record operation capability requirements within the schedule. Provide a signoff blank for the Contractor and the Contracting Officer for each test feature; e.g., gpm, rpm, psi. Include a remarks column for the testing validation procedure referencing operating limits of time, pressure, temperature, volume, voltage, current, acceleration, velocity, alignment, calibration, adjustments, cleaning, or special system notes. Delineate procedures for preventative maintenance, inspection, adjustment, lubrication and cleaning necessary to minimize corrective maintenance and repair.

Repair requirements must inform operators how to check out, troubleshoot, repair, and replace components of the system. Include electrical and mechanical schematics and diagrams and diagnostic techniques necessary to enable operation and troubleshooting of the system after acceptance.

1.9 COMMISSIONING

1.9.1 Building Commissioning

All Contract requirements for building commissioning must be completed prior to Contract completion.

1.9.2 HVAC Commissioning

All Contract requirements of Commissioning of HVAC Systems must be fully completed, including all testing concurrent with Building Commissioning. All Contract requirements of Testing, Adjusting, and Balancing of HVAC Systems must be fully completed, including testing and inspection, prior to HVAC commissioning, except as noted otherwise in the Section for Testing, Adjusting, and Balancing for HVAC. All Contract requirements of Space Temperature Control Systems, Direct Digital Control for HVAC and Other Control Systems, or BACnet Direct Digital Control for HVAC and Other

Building Control Systems must be fully completed, including all testing, prior to HVAC commissioning. The time required to complete all work and testing as prescribed in the specifications is included in the allotted calendar days for completion.

1.9.3 Final Closeout Documents

Submit Final Commissioning Report submittal per Section 01 91 00.15 BUILDING COMMISSIONING.

PART 2 PRODUCTS

2.1 CERTIFICATION OF EPA DESIGNATED ITEMS

Submit the Certification of EPA Designated Items as required by FAR 52.223-9 Estimate of Percentage of Recovered Material Content for EPA Designated Items and FAR 52-223-17 Affirmative Procurement of EPA designated items in Service and Construction Contracts. Include on the certification form the following information: project name, project number, Contractor name, license number, Contractor address, and certification. The certification will read as follows and be signed and dated by the Contractor. "I hereby certify the information provided herein is accurate and that the requisition/procurement of all materials listed on this form comply with current EPA standards for recycled/recovered materials content. The following exemptions may apply to the non-procurement of recycled/recovered content materials:

a. The product does not meet appropriate performance standards;

b. The product is not available within a reasonable time frame;

c. The product is not available competitively (from two or more sources);

d. The product is only available at an unreasonable price (compared with a comparable non-recycled content product)."

Record each product used in the project that has a requirement or option of containing recycled content in accordance with SECTION 01 33 29.00 22 SUSTAINABILITY REQUIREMENTS AND REPORTING (PWD ME), noting total price, total value of post-industrial recycled content, total value of post-consumer recycled content, exemptions (a, b, c, or d, as indicated), and comments. Recycled content values may be determined by weight or volume percent, but must be consistent throughout.

2.2 CERTIFICATION OF USDA DESIGNATED ITEMS

Submit the Certification of USDA Designated Items as required by FAR 52-223-1 Bio-based Product Certifications and FAR 52.223-2 Affirmative Procurement of Biobased Products Under Service and Construction Contracts. Include on the certification form the following information: project name, project number, Contractor name, license number, Contractor address, and certification. The certification will read as follows and be signed and dated by the Contractor. "I hereby certify the information provided herein is accurate and that the requisition/procurement of all materials listed on this form comply with current USDA standards for biobased materials content. The following exemptions may apply to the non-procurement of biobased content materials:

a. The product does not meet appropriate performance standards;

b. The product is not available within a reasonable time frame;

c. The product is not available competitively (from two or more sources);

d. The product is only available at an unreasonable price (compared with a comparable bio-based content product)."

Record each product used in the project that has a requirement or option of containing biobased content in accordance with SECTION 01 33 29.00 22 SUSTAINABILITY REQUIREMENTS AND REPORTING (PWD ME), noting total price, total value of post-industrial recycled content, total value of post-consumer recycled content, total value of biobased content, exemptions (a, b, c, or d, as indicated), and comments. Biobased content values may be determined by weight or volume percent, but must be consistent throughout.

PART 3 EXECUTION

3.1 AS-BUILT DRAWINGS

Provide and maintain two black line print copies of the PDF contract drawings for As-Built Drawings. Maintain the as-builts throughout construction as red-lined hard copies on site and or red-lined PDF files. Submit Final As-Built Drawings 30 days prior to Beneficial Occupancy Date (BOD).

Monthly Progress As-Builts: Provide one (1) set of working as-built drawings (CADD) in the specified software and format, hard copy and electronic, to the Contracting Officer with the monthly progress invoice. Progress as-builts must be approved during pay request process. Refer to 01 20 00.00 22 PRICE AND PAYMENT PROCEDURES (PWD ME) Paragraph Content of Invoice for additional information.

3.1.1 Markup Guidelines

Refer to Attachment: 01 78 00.00 22_Portsmouth Naval Shipyard As-Built Standards.

Make comments and markup the drawings complete without reference to letters, memos, or materials that are not part of the As-Built drawing. Show what was changed, how it was changed, where item(s) were relocated and change related details. These working as-built markup prints must be neat, legible and accurate as follows:

a. Use base colors of red, green, and blue. Color code for changes as follows:

    (1) Special (Blue) - Items requiring special information, coordination, or special detailing or detailing notes.

    (2) Deletions (Red) - Over-strike deleted graphic items (lines), lettering in notes and leaders.

    (3) Additions (Green) - Added items, lettering in notes and leaders.

b. Provide a legend if colors other than the "base" colors of red, green, and blue are used.

c. Add and denote any additional equipment or material facilities, service lines, incorporated under As-Built Revisions if not already shown in legend.

d. Use frequent written explanations on markup drawings to describe changes. Do not totally rely on graphic means to convey the revision.

e. Use legible lettering and precise and clear digital values when marking prints. Clarify ambiguities concerning the nature and application of change involved.

f. Wherever a revision is made, also make changes to related section views, details, legend, profiles, plans and elevation views, schedules, notes and call out designations, and mark accordingly to avoid conflicting data on all other sheets.

g. For deletions, cross out all features, data and captions that relate to that revision.

h. For changes on small-scale drawings and in restricted areas, provide large-scale inserts, with leaders to the applicable location.

i. Indicate one of the following when attaching a print or sketch to a markup print:

   1) Add an entire drawing to contract drawings

   2) Change the contract drawing to show changes on the drawing.

   3) Provided for reference only to further detail the initial design.

j. Incorporate all shop and fabrication drawings into the markup drawings.

3.1.2 As-Built Drawings Content/Meeting

Schedule a meeting with PNSY CADD Department 30 calendar days prior to submission of progress as-built drawings to review as-built drawing standards.

Show on the as-built drawings, but not limited to, the following information:

a. The actual location, kinds and sizes of all sub-surface utility lines. In order that the location of these lines and appurtenances may be determined in the event the surface openings or indicators become covered over or obscured, show by offset dimensions to two permanently fixed surface features the end of each run including each change in direction on the record drawings. Locate valves, splice boxes and similar appurtenances by dimensioning along the utility run from a reference point. Also record the average depth below the surface of each run.

b. The location and dimensions of any changes within the building structure.

c. Layout and schematic drawings of electrical circuits and piping.

d. Correct grade, elevations, cross section, or alignment of roads, earthwork, structures or utilities if any changes were made from contract plans.

e. Changes in details of design or additional information obtained from working drawings specified to be prepared or furnished by the Contractor; including but not limited to shop drawings, fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment, and foundations.

f. The topography, invert elevations and grades of drainage installed or affected as part of the project construction.

g. Changes or Revisions which result from the final inspection.

h. Where contract drawings or specifications present options, show only the option selected for construction on the working as-built markup drawings.

i. If borrow material for this project is from sources on Government property, or if Government property is used as a spoil area, furnish a contour map of the final borrow pit/spoil area elevations.

j. Systems designed or enhanced by the Contractor, such as HVAC controls, fire alarm, fire sprinkler, and irrigation systems.

k. Changes in location of equipment and architectural features.

l. Modifications and compliance with FC 1-300-09N procedures.

m. Actual location of anchors, construction and control joints, etc., in concrete.

n. Unusual or uncharted obstructions that are encountered in the contract work area during construction.

o. Location, extent, thickness, and size of stone protection particularly where it will be normally submerged by water.

3.2 RECORD DRAWINGS

Prepare and provide Record Drawings and Source Documents in accordance with FC 1-300-09N. Provide four copies of Record Drawings and Documents on separate CDs or DVDs 30 days after BOD.

3.3 OPERATION AND MAINTENANCE MANUALS

Provide project operation and maintenance manuals as specified in Section 01 78 23.00 22 OPERATION AND MAINTENANCE DATA (PWD ME). Provide electronic copies of the Operation and Maintenance Manual files and one hard copy of the Operation and Maintenance Manuals. Submit to the Contracting Officer for approval within 30 calendar days of the Beneficial Occupancy Date (BOD). Update and resubmit files for final approval at BOD.

3.4 CLEANUP

Provide final cleaning in accordance with ASTM E1971 and submit twocopies of the listing of completed final clean-up items. Leave premises "broom clean." Comply with GS-37 for general purpose cleaning and bathroom cleaning. Use only nonhazardous cleaning materials, including natural cleaning materials, in the final cleanup. Clean interior and exterior glass surfaces exposed to view; remove temporary labels, stains and foreign substances; polish transparent and glossy surfaces; vacuum carpeted and soft surfaces. Clean equipment and fixtures to a sanitary condition. Replace filters of operating equipment and comply with the Indoor Air Quality (IAQ) Management Plan. Clean debris from roofs, gutters, downspouts and drainage systems. Sweep paved areas and rake clean landscaped areas. Remove waste and surplus materials, rubbish and construction facilities from the site. Recycle, salvage, and return construction and demolition waste from project in accordance with Section 01 57 19.00 TEMPORARY ENVIRONMENTAL CONTROLS, PORTSMOUTH NAVAL SHIPYARD (PWD-ME) and 01 74 19.00 22 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL (PWD ME).

3.5 REAL PROPERTY RECORD (NOT APPLICABLE)

-- End of Section --

Last updated June 18, 2014

NAVFAC MID-ATLANTIC

Public Works Department
Maine

PORTSMOUTH NAVAL SHIPYARD

AS-BUILT CADD STANDARDS

June 2014

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STANDARDS FOR EDITING EXISTING CONTRACT DRAWINGS TO REFLECT AS-BUILT CONDITIONS

Following are the *Standards* for editing existing contract drawings to reflect the construction as-built conditions for **NAVFAC MID-ATLANTIC; Public Works Department Maine (PWD-ME)**. The purpose of these *Standards* is to provide a uniform system of drawing formats. This system will be used in retrieving information from the drawings in the future. Also, these *Standards* were established to eliminate various recurring problems encountered when transferring electronic files from non-government sources. All drawings modifications found to be generated outside of these standards will be returned to the drawing provider for correction at no additional cost to the ***GOVERNMENT.***

As-built Drawings:

Drawing Sets:

The contractor shall maintain at the jobsite one set of full-size hard copy prints of the contract drawings, accurately marked in red with adequate dimensions, to show all variations between the construction actually provided and that indicated or specified in the contract documents, including buried or concealed construction. **Special attention shall be given to recording the horizontal and vertical location of all buried utilities that differ from the contract drawings**. Existing utility lines and features revealed during the course of construction shall also be accurately located and dimensioned, using permanent existing features and station coordinates as a reference. Acceptable features include; building corners, centers of utility manhole covers, fire hydrants, etc. Existing topographic features that differ from those shown on the contract drawings shall also be accurately located and recorded. Where a choice of materials or methods is permitted herein or where variations in scope or character of work from that of the original contract are authorized, the drawings shall be marked to define the construction provided. The representations of such changes shall conform to standard drafting practices and shall include such supplementary notes, legends, and details as necessary to clearly portray the as-built construction. These drawings shall be available for review by the Contracting Officer at all times. Upon completion of the work, the marked up prints shall be certified as correct, signed by the Contractor, and delivered to the Contracting Officer for review by **PWD-ME**. After acceptance of the as-builts, the contractor will add the as-built information to the original contract drawing files and plot new Mylar reflecting the as-built conditions.

Requests for partial payments will not be approved if the marked prints are not kept current. Request for final payment will not be approved until Contracting Officer receives delivery of the original electronic contract drawing CADD files, modified to reflect the as-built conditions and new Mylar plotted to reflect the as-built conditions.

Computer Aided Drafting Design (CADD) As-built Drawings:

File formats:

Although all methods of CADD drawing are permitted, the final product of all computer-aided drawings shall be made compatible with the ***AutoCAD*** currently in use by **PWD-ME** Drawings created using non-AutoCAD programs that do not support AutoCAD's DWG file format, shall be transferred to the Drawing Interchange Format (DXF).

If the drawing provider's CADD software does not support the DXF format, it is the responsibility of the drawing provider to contact the Contracting Officer who will in turn contact the **PWD-ME** to make special arrangements for file transfer.

Each drawing file must not depend on any other drawing file to completely represent the finished product *(Mylar)*. All drawings found to use dependent external reference drawings, **will be returned to the drawing provider for correction**.

Each drawing file must not depend on any ***"THIRD PARTY"*** utility or software to represent the finished Mylar. Before using any third party software to create a finished product, the drawing provider shall verify that no additional cost or effort is required by the **PWD-ME** to completely represent the finished product *(Mylar).*

Drawing Sheets:

Electronic files of the contract drawings shall be provided to the Contractor following award of the contract. The contractor will add the accepted as-built conditions to the original electronic file of the drawing sheet. If additional sheets need to be added to the contract drawing package because of insufficient sheet space, it is the responsibility of the Contractor to ensure the PWD-ME CADD Drawing Standards are obtained and followed. Additional drawing numbers will be assigned by **PWD-ME** personnel prior to submittal of as-built drawings.

Basic Drawing Standards:

Electronic As-builts:

The contractor shall modify the original contract drawing files to reflect the construction contract as-built conditions reviewed and accepted by **the Contracting Officer.** Each as-built condition added to a drawing file shall be encapsulated by a closed polygon or "revision cloud. A revision symbol shall be placed outside the "revision cloud" with the appropriate letter designating the revision sequence. The contractor shall annotate in the "revision block" of each drawing file modified as to the type of revisions made to the drawing file. The contract drawings are to be edited to reflect the as-built conditions only. No part of the original drawings shall be deleted, erased or rendered illegible. Parts of the contract drawing found to be in error or modified during construction, shall be over struck using methods described not to obscure the original drawing, and annotations will be added adjacent that clearly explain the modification, including accurate dimensions locating the feature.

Drawing Sheet Management:

Where construction projects encompass multiple facilities (buildings) or systems (utilities), all plans, details, legends, notes etc., for an individual facility or system must appear on the drawing sheets for each facility or system. Standardized details must appear on the drawing sheets for each facility or system where they apply. Cross-referencing details between facilities or systems is not permitted.

Drawing Units:

All drawings of buildings, structures, floor plans, etc. are to be drawn full size, with one drawing unit equal to one inch. All site plans, location plans, etc. are to be drawn full size, with one drawing unit equal to one foot. The original drawing's origin shall not be modified.

Entity Management:

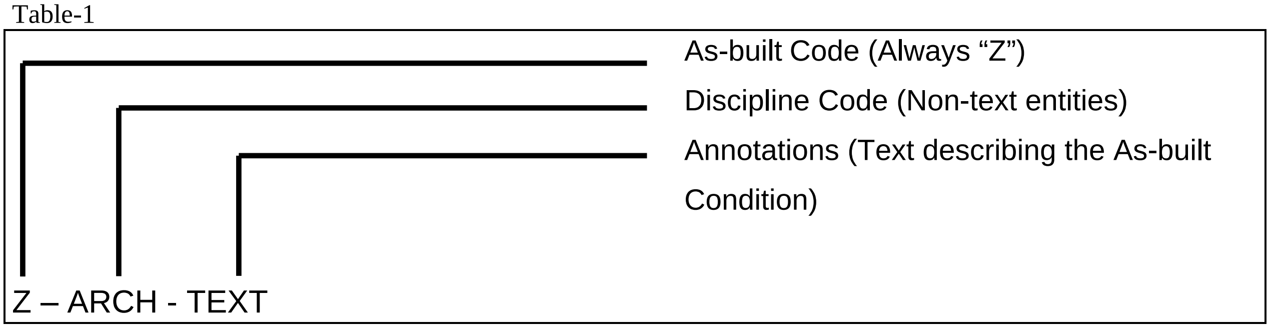
All additions to the original contract drawings will be made on additional layers as designated below. The minimum text height for the "D" size sheet is 1/8in plotted and 3/32in plotted for the "B" size sheet. All as-built modification layers are to be ***CYAN*** in, including all annotations, dimensions, leaders and callouts.

Entities discovered during construction but not represented on the original construction drawings shall be assigned to layers as described below. This information shall be added to the drawing in such a way not to obscure any original drawing entity when plotted.

Level/layer Management:

All drawing entities shall be grouped together by layer. Drawing entities shall be assigned to layers by discipline code for that entity, see table page D4. This includes non-text entities such as strikeouts, lines, circles, symbols, etc. See Table-1 for typical layer-naming.

Table-1



Identify entities by discipline. For instance: the example below shows layering configuration for all architectural as-built entities ***EXCLUDING TEXT*** that deviate from the original design:

Z-ARCH.

Annotations, dimensions, leaders and callouts for those architectural entities shall be assigned to layer:

Z-ARCH-TEXT.

All annotations shall be assigned to layers by discipline and designated as ***TEXT***.

Each layer name will follow the discipline code for that entity. See Table-2 for Discipline Descriptions:

Table-2

Discipline Code:	***Discipline:***	***Description:***
ARCH	Architectural	Architectural design, building's interior/exterior, walls, doors, windows etc.
CIVIL	Civil	Site work, external to buildings and structures. Typically surface work.
ELEC	Electrical	Electrical work pertaining to buildings/structures internal/external and distribution.
FP	Fire Protection	Fire Protection systems pertaining to buildings/structures internal/external
GEN	General	Typically drawing information, general notes, symbols, etc....
MECH	Mechanical	Mechanical work, HVAC components, compressed air, etc. pertaining to buildings/structures internal/external
PLUMB	Plumbing	Plumbing, piping and fixtures, etc.. pertaining to buildings/structures internal/external
UTIL	Utilities	Utility distribution systems, duct work, manholes and piping/cabling underground/direct buried
STRUC	Structural	All structural work, building framing, conc. duct reinforcement, misc. steel work, etc.
TELE	Telecommunications	Phone lines, cable TV, computer data lines, etc. pertaining to buildings/structures internal/external

Electronic Deliverables:

Submit all as-built documentation and drawing files to the Contracting Officer via recordable CD in the file formats discussed in this and other sections. Electronic transfer of files via E-mail or other methods is not permitted.

Standards Deviation:

Contact the Contracting Officer if questions arise about these standards or these drawing standards cannot be followed. Otherwise, all drawing files will be returned to the drawing provider for correction at no additional cost to the **PWD-ME.**

Last updated June 18, 2014

NAVFAC MID-ATLANTIC

Public Works Department
Maine

PORTSMOUTH NAVAL SHIPYARD

CADD STANDARDS

June 2014

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1. CADD STANDARDS

All design documents and deliverables prepared by In-House forces and/or contracted A/E firms including IDIQ, Design Bid-Build and Design-Build process shall conform to the following NAVFAC MIDLANT and PWD-ME CADD standards areas not addressed on this standard shall follow:

- UFC 1-300-09N

U.S. DEPARTMENT OF DEFENSE (DOD) *[UFCs available on* http://dod.wbdg.org/

These are NAVFAC MIDLANT PWD-MAINE specific guidelines that supersede all others for contract documents utilized for execution. Refer to the U.S. National CAD Standard Version 3.0 for any items not covered in this or the referenced NAVFAC UFCs.

1. A. General: (for both Design-Build and Design-Bid-Build)

For drawing templates, pen tables, sample symbology, and instructions on sheet arrangement & procedures, see attached NAVFAC MIDLANT PWD-MAINE; Sheet Border Use and Procedures. Provide *.PDF format of all drawing sheets at each submittal phase. In addition, provide *.PDF and *.DWG format of all drawing sheets at the final design submission. CADD files shall be made available upon request, in addition to the full size, stamped Mylar and half-size hard copies or as directed by the specific contract specifications.

1. B. Design Drawings

Prepare, organize, and present design drawings in accordance with the requirements of UFC 1-300-09N.

All drawings and their associated PDFs will maintain a “PRELIMINARY Not for Construction” stamp across the signature areas of the title block, until the actual final design submittal. The stamp shall be in translucent lettering across the Project Title area of the drawing title block and shall be displayed on layer “G-ANNO-TTLB-PRLM”. The “G-ANNO-TTLB-PRLM” shall be off for final submittal.

Design revisions shall be tracked by number and date in the revision block of each drawing sheet during the design process. Design revisions shall be assigned to layer “G-ANNO-TTLB-PRLM” and shall be frozen for final submittal.

1. C. Drafting Conventions

1. C.1. Fonts

The standard text height for a plotted D-size drawing shall be **1/8”** for typical text, and **1/4”** for titles, and **1”** maximum for project titles on cover sheets. The standard text height for a plotted B-size drawing shall be **3/32”** for typical text, and **1/8”** for titles, and **1/2”** maximum for project titles on cover sheets. For typical text, use the “**RomanS”** font. For titles, use the “**Swis 721**

BT" font. Both fonts shall have a width ratio of no less than **0.8**.

ROMANS – 1/8" TALL

Swis 721 BT – 1/4" TALL (up to 1")

These are the only fonts that should be used within the drawing area, with one exception: on civil site plans, the Design Manager (EIC/AIC) may use text ("RomanS" font, 1/8" height {Leroy 100}, 0.8 width factor, with an oblique angle of 12 degrees) to annotate existing site features only. Typical (or Normal) text as defined above must be used for all other annotations.

1. C. 2. Symbols

Approved symbology in use by PWD-ME is provided on the PWD-ME title sheet. On the PWD-ME title and standard sheet, borders and plot styles shall delivered at each project design kickoff MTG.

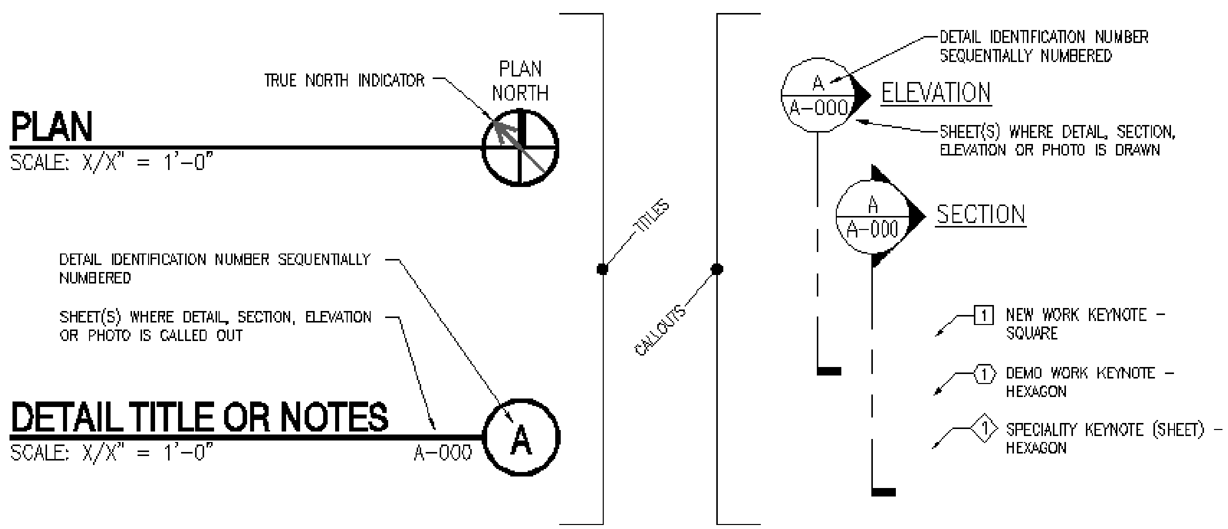

Figure 3-1 PWD-ME Approved Symbols

Note that detail, section, elevation and photograph callouts utilize two-part bubbles which indicate a detail identification number and identify sheet where the detail, section, elevation or photo is shown. Additional sheets(s) on which the detail is called out is (are) displayed to the bottom left of the bubble.

1. C. 3. Line Weights / Pen Tables

NAVFAC's Electronic Documents and Deliverables Working Group (EDDWG) developed a comprehensive pen table that utilized the National Cad Standard 255-pen table as a basis, but added much needed thinner lines and additional grayscales (shaded or screened pens) that had not made the transition from the original US Coast Guard standard. The NAVFAC-specific pen table is provided as filename **NavFacStnd.ctb**.

The pen tables can be thought of in groups of 20. After pen 19, pens 20 thru 39 are in the "Rust" range when displayed on screen. The first 10 (pens 20 – 29) look "Rust" and print shades of "Rust"; while the second 10 (pens 30 – 39) look "Rust" but print Black. Typically, design drawings for contract documents utilize the pens that print Black (currently colored pens are used for renderings and some other instances, but not normally on contract drawings).

Since these pen tables are established to be legible when printed at full-size (22"x34"), the corresponding text height is 1/8". To be consistent, the related B-size (11"x17") & A-size sheet (8-1/2"x11")—normally used for sketches, uses fonts and line weights that are ½ the size of the large format (D-size) documents—otherwise the fonts are too big and the line weights are too bold. The associated pen table for B-size & A-size documents is **NavFacStnd-Sketch.ctb**

1. D. NAVFAC Logo

A NAVFAC Publications/Logo Guide will be published at a later date, but in the interim, all deliverables shall have the NAVFAC Brand Logo on the title block area of the drawing border (as displayed in the Templates). The logo should not be removed, deleted, or turned off and should appear on all PDFs to be utilized for the electronic bid solicitations. If contract drawings are produced by an A/E firm and that firm desires to have their logo on the drawings then the A/E must apply that logo to all sheets consistently in the same location. Typically that is in the lower- or upper-right hand corner of the drawform, adjacent to the title block.

1. E. Units of Measure

Unless otherwise directed, for all projects the default shall be English units of measurement (U.S. Customary System of units). All drawings, specifications, cost estimates, and design calculations shall reference this system of units.

1. F. Datums and Coordinate Systems

VERTICAL DATUM: All elevations, grades, or profiles shall be represented in U.S. feet and referenced to NAVD88 vertical datum and will be provided as required.

HORIZONTAL DATUM: All Portsmouth Naval Shipyard mapping, utility information, site drawings and surveys are represented in U.S. feet and referenced to the state plane coordinate system, NAD83 Maine west, zone 1802. All electronic mapping and utility information shall be maintained to this system.

1. G. Drawing Numbers/File Naming

In addition to NAVFAC drawings numbers discussed in UFC 1-300-09N, Chapter 4-3, "Drawing Numbers" the Government Project Manager will provide PWD (Public Works Department) Drawing Numbers for each drawing sheet, similar to NAVFAC drawing numbers. The PWD Drawing Number to be requested by the drawing provider at the 100% design stage. The CADD files shall be named by the PWD drawing number in accordance with the following guidance: The CADD file for each ANSI D (22” x 34”) size drawing sheet shall be named by the corresponding PWD Number, minus the design year designator. Where the assigned PWD Number is: 59-08-501 the corresponding CADD file shall be named 059-501.DWG. Where a single drawing file is used to create multiple drawing sheets via layout tabs, the drawing file shall be named by the PWD Number of the first drawing sheet of the drawing set. Drawing sets shall be arranged by discipline.

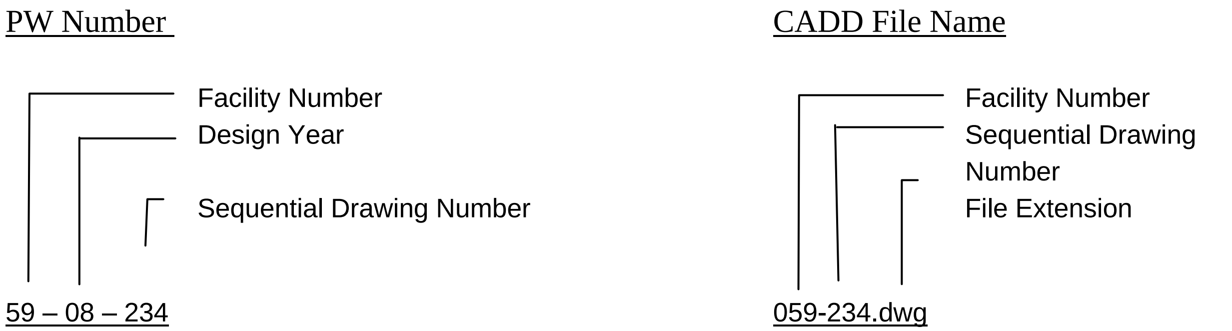

Figure 7-1 DRAWING NUMBERS/FILE NAMING

Files for sketches, drawings other than ANSI D (22” x 34”) size shall be named by construction contract number and sheet number of the first sheet of the sheet set.

1. G. 1. File Naming Convention for PDF Files:

ADOBE PDF file naming shall follow CADD file naming methods.

1. H. Electronic Submittals

Submit all CADD files for the final drawings in Autodesk Drawing (*.DWG) compatible with current PWD-ME CADD format. Drawing files shall be full files, uncompressed and unzipped. All submitted CADD files shall maintain their original and complete data structure as produced in their native Autodesk application (i.e. Architectural Desktop, Land Development Desktop, Civil 3D, Building Systems...). All files utilizing AutoCAD’s “Xref” feature shall be delivered with all external files “bound” to the drawing file. There shall be no external references in the final drawing file.

Multiple layout tabs are allowed, the tabs shall be annotated by PWD number for each sheet as described in section PWD-ME 7.0 Drawing Numbers/File Naming. The electronic file(s) shall be named by the PWD number of the first sheet of the sheet set. When required, the drawing file shall be split by sheet discipline. Corresponding PDF files shall be grouped into one file and named by the project title.

1. I. Record Documents …1.10.1 Record Drawings (As-Built drawings, **G)**

The Quality Control Manager shall deliver the marked-up As-Built drawings to the Contractor's Designer of Record who shall incorporate all as-built modifications.

The as-built modifications shall be accomplished by electronic drafting methods on the original CADD (*.DWG) design drawing files to create a complete set of record drawings. For each Record drawing, provide a CADD drawing identical to signed [A/E- or] Contractor-originated *.PDF drawings, that incorporates modifications to the as-built conditions. In addition, copy initials and dates from the Contracting Officer approved *.PDF documents to the title block of the Record CADD (*.DWG) drawings. The Record electronic files shall use the file name of the original signed CADD drawing file name with the suffix "-RD" before the file extension, ".DWG" (example; 059-501-RD.dwg – see Drawing File Naming). The original design, RFP, reference, or definitive drawings are not required for inclusion in the Record set of drawings.

After all as-built conditions are recorded on the CADD (*.DWG) files, produce a PDF file of each individual record drawing in conformance with UFC 1-300-09N. Generate PDF drawing files using a PDF page size that corresponds to the original document sheet size (NAVFAC utilizes an ANSI D-size, 22"x34" frame and a 0" border all around) and a PDF print resolution that results in clear detail of all drawing features.

Provide one set of signed and stamped record drawings, plotted on Mylar that fully represent the PDF file. Provide all project final submittals including as-builts in accordance with UFC 1-300-09N Chapter 7-5.2. Each CD-ROM shall be marked with Project Name, Construction Contract Number, Project Number, Specification Number, and Record Drawing date.

1. I. 1. Source Documents

In addition to the drawings provide the specifications, design analysis, reports, survey data, calculations, and any other contract documents utilized in creating the design package (drawings, specifications, and cost estimate) on the CD-ROM disk(s) as specified in preceding paragraph.

1. J. Forwarding Submittals

The [A/E- or] Contractor shall provide all submittals, that lend themselves to paper format (i.e. 8-1/2"x11" documents, 11"x17" or tabloid sheets, or 22"x34" drawings), in electronic *.PDF (Portable Document Format) to the Contracting Officer. The appropriate forwarding e-mail address(es) will be provided at the Pre-Construction meeting. Provision of electronic submittals with appropriate electronic transmittal may alleviate some of the paper copies required herein but must be verified at Pre-Construction meeting. In such instances a minimum of one hard copy shall be forwarded to the Contracting Officer.

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SECTION 01 78 23.00 22
^^
OPERATION AND MAINTENANCE DATA (PWD ME)
05/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE GUIDELINE 1.4	(2019) Preparing Systems Manuals for Facilities

ASTM INTERNATIONAL (ASTM)

ASTM E1971	(2005; R 2011) Standard Guide for Stewardship for the Cleaning of Commercial and Institutional Buildings
ASTM E2166	(2016; R 2023) Standard Practice for Organizing and Managing Building Data

U.S. DEPARTMENT OF DEFENSE (DOD)

FC 1-300-09N	(2024) Navy and Marine Corps Design

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-10 Operation and Maintenance Data

Facility Data Workbook; G

Training Plan; G

Training Outline; G

Training Content; G

Operation And Maintenance Manual, Progress Submittal; G

Operation And Maintenance Manual, Prefinal Submittal; G

Operation And Maintenance Manual, Final Submittal; G

SD-11 Closeout Submittals

Training Video Recording; G

Validation of Training Completion; G

Training Plan; G

Record Drawings And Utility Systems; G

1.3 MEETINGS

To assure that Operation and Maintenance Manual and Facility Data Workbook requirements are being met through the duration of the project, organize the following meetings and discuss the subsequent topics:

1.3.1 Preconstruction Meeting

At a minimum, discuss the following at the Preconstruction Meeting. Refer to Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS Paragraph Preconstruction Meeting::

a. The requirement for Operation and Maintenance Manuals and Facility Data deliverables under this contract including coordination meetings

b. Processes and method of gathering Facility Data information during construction

c. Primary roles and responsibilities associated with the development and delivery of the Operation and Maintenance Manuals and Facility Data deliverables, and

d. Identify and agree upon a date and attendance list for the meetings described below:

1.3.2 Operation and Maintenance Manual and Facility Data Workbook Coordination Meeting

Facilitate a meeting after the Preconstruction Meeting prior to the submission of the Operation and Maintenance Manual Progress Submittal. Meeting attendance must include the Contractor's Operation and Maintenance Manual and Facility Data Workbook Preparer, Quality Control Manager, the Commissioning Authority (CxA), the Government's Design Manager (DM), Contracting Officer's Representative, and Government's facility data reviewer. Include any Mechanical, Electrical, and Fire Protection Sub-Contractors.

The purpose of this meeting is to reach a mutual understanding of the scope of work concerning the contract requirements for Operation and Maintenance Manual and coordinate the efforts necessary by both the Government and Contractor to ensure an accurate collection, preparation and timely Government review of Operation and Maintenance Manual.

1.3.3 O&M Manual Submittal Coordination Meeting

Facilitate a meeting following submission and Government review of each design or progress submittal of the Operation and Maintenance Manual and Facility Data Workbook.

a. Include personnel from the Coordination meeting and any additional personnel identified.

b. The purpose of this meeting is to demonstrate ongoing compliance with the requirements identified in this specification. Discuss Government review comments and unresolved items preventing completion and Government approval of the Operation and Maintenance Manuals and Facility Data Workbook (FDW).

c. The applicable deliverables, along with Government remarks associated with review of these submittals serve as the primary guide and agenda for this meeting.

1.3.4 Facility Turnover (Red Zone) Meeting

Include Operation and Maintenance Manual in NAVFAC Red Zone (NRZ) facility turnover meetings as specified in Section 01 30 00.00 22, ADMINISTRATIVE REQUIREMENTS (PWD ME).

1.4 FACILITY DATA WORKBOOK

Develop an editable, electronic spreadsheet based on the equipment in the Operation and Maintenance Manuals that contains the information required to start a preventive maintenance program as outlined in 01 78 24.00 20 FACILITY DATA WORKBOOK (FDW). As a minimum, provide Facility Data Workbook as a list of system equipment, location installed, warranty expiration date, manufacturer, model, and serial number.

1.5 OPERATION AND MAINTENANCE MANUAL MEDIA

Assemble Operation and Maintenance Manual into an electronically bookmarked file using the most current version of Adobe Acrobat or similar software capable of producing PDF file format. Provide compact disks (CD) or data digital versatile disk (DVD) as appropriate, so that each one contains operation, maintenance and record files, project record documents, and training videos. Include a complete bookmarked operation and maintenance directory.

1.5.1 CD or DVD Label and Disk Holder or Case

Provide the following information on the disk label and disk holder or case:

a. Building Number

b. Project Title

c. Activity and Location

d. Construction Contract Number

e. Prepared For: (Contracting Agency)

f. Prepared By: (Name, title, phone number and email address)

g. Include the disk content on the disk label

h. Date

i. Virus scanning program used

1.5.2 O&M Manual Tabbed Hard Copy

Provide a hard copy of the O&M manual upon completion of the project. Provide tabs for each section and subsection for ease of navigation by the user.

1.6 O&M MANUAL CONTENT

Organize the bookmarked Operation and Maintenance Manual into the following Parts in accordance with ASHRAE GUIDELINE 1.4, and as modified and detailed below. Word template for O&M Manual is available at: https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-78-23.

1.6.1 Part 1: Executive Summary

Provide a summary of the information found in the O&M manual including the purpose of the manual and a description of the manual's organization.

1.6.2 Part 2: Facility Design and Construction

1.6.2.1 General Facility and Systems Description

Provide an overview of the intent for design and use of the facility. Provide a PDF of the Record Drawings prepared in accordance with FC 1-300-09N and 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME) and bookmarked using the sheet title and sheet number. Include uncluttered 11 by 17 inches floor plans with room numbers, type or function of space, and overall facility dimensions on the floor plans. Do not include items such as construction instructions, references, or frame numbers.

Detail the overall dimensions of the facility, number of floors, foundation type, expected number of occupants, and facility Category Code list and generally describe all the facility systems and any special building features (for example, HVAC Controls, Sprinkler Systems, Cranes, Elevators, and Generators). Include photographs marked up and labeled to show key operating components and the overall facility appearance.

1.6.2.2 Basis of Design

Provide a copy of the contract Basis of Design.

1.6.2.3 Contract Documents, RFP, Amendments, and Modifications

Provide the contract construction documents complete, to include specifications, drawings, Request for Proposal, amendments, and modifications.

1.6.2.4 Room Inventory of Real Property and Finishes

Provide a list of installed equipment furnished under this contract. Include all information usually listed on manufacturer's name plate. Include, as applicable, the following information for each piece of equipment installed: description of item, all dimensions, location by room number, model number, serial number, capacity, name and address of manufacturer, name and address of equipment supplier, condition, spare parts list, manufacturer's catalog, and warranty. Real property includes, but is not limited to, floor coverings, wall surfaces, ceiling surfaces, windows, roofing, HVAC filters, plumbing fixtures, and lighting fixtures.

Submit the final list 30 days after transfer of the completed facility.

Include spatial data defining actual net square footage and data of each room. Also include the room finish schedule including room names and numbers. Include schedules in the construction drawings in the room inventory. Add a column to each schedule to record what was provided by the contractor during construction. Provide a PDF of room inventory. Key the designations to the related area depicted on the contract drawings. List the following data:

RECORD OF DESIGNATED EQUIPMENT AND MATERIALS DATA				
Description	Specification Section	Manufacturer and Catalog, Model, and Serial Number	Composition and Size	Where Used

1.6.3 Part 3: Facilities, Systems, and Assemblies Information

1.6.3.1 Organization

Bookmark information in this section using the current version of ASTM E2166 Uniformat II, UFGS numbers, and document type as outlined in the example below. Bookmark/tab each item to the third level for easy navigation of the manual.

Example as shown in Table below:

PARTS AND SUBPART NUMBERING
3.1 B20 EXTERIOR CLOSURE (System)
3.1.1 B2030 EXTERIOR DOORS (Subsystem)
3.1.1.1 B2030110 GLAZED DOORS (Component)
3.1.1.1.1 Applicable specifications List in UFGS Format
3.1.1.1.2 Manufacturer's Operations and Maintenance Data
3.1.1.1.3 Approved Submittal
3.1.1.1.4 Coordination/Shop Drawings
3.1.1.1.5 Sequence of Operation for Operating Equipment
3.1.1.1.6 Testing Equipment Information and Performance Data
3.1.1.1.7 Routine Maintenance Requirements
3.1.1.1.8 Repair Procedures
3.1.1.1.9 Emergency Procedures & Locations of Applicable Controls
3.1.1.1.10 Warranties

PARTS AND SUBPART NUMBERING
3.1.1.1.11 Record Drawings and Utility Systems
3.1.1.1.12 Contractor / Supplies Listing and Contact Information

1.6.3.2 Related Specifications

Reference each specification related to the subsystem in this section, and locate the actual specification section in Part 2 of the O&M Manual. List specifications in table format as shown in the below example.

UFGS Number	Specification Title	Page Spec Begins in Part 2

1.6.3.3 Manufacturer's Operations and Maintenance Data

Provide a copy of all manufacturer specifications and cutsheets. Provide text-searchable, high-quality document files from the manufacturer's online or electronic documentation. Color documents are preferred. Provide documents specific to the product(s) installed under this Contract. Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Provide Uniformat II Level 3 identification for D20, D30, D40 installed equipment. When possible, do not submit document files containing multiple product catalogs from the same manufacturer, or product data from multiple manufacturers in the same files. Provide documents directly from the manufacturer whenever possible. Do not provide scanned copies of hardcopy documents. Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Include special hardware requirements, such as requirement to use high-strength bolts and nuts. Identify parts by make, model, serial number, and source of supply to allow reordering without further identification. Provide clear and legible illustrations, drawings, and exploded views to enable easy identification of the items. When illustrations omit the part numbers and description, both the illustrations and separate listing must show the index, reference, or key number that will cross-reference the illustrated part to the listed part. Group the parts shown in the listings by components, assemblies, and subassemblies in accordance with the manufacturer's standard practice. Parts data may cover more than one model or series of equipment, components, assemblies, subassemblies, attachments, or accessories, such as typically shown in a master part catalog.

1.6.3.4 Approved Submittals and Certificates

Provide a copy of all submittals documented with the required approval as applicable for each UFGS specification listed in the table outlined in applicable specifications. Include copies of SD-07 Certificates submittals documented with the required approval, SD-08 Manufacturer's Instructions submittals documented with the required approval, and SD-10 Operation and Maintenance Data submittals documents with the required approval.

1.6.3.5 Approved Coordination/Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work. Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project. Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

1.6.4 Sequence of Operation for Operating Equipment

Provide record one-line diagrams for each floor, delineating mechanical equipment location within the building. Provide specific instructions, procedures, and illustrations for the following phases of operation for the installed model and features of each system:

1.6.4.1 Safety Precautions and Hazards

List personnel hazards and equipment or product safety precautions for operating conditions. List all residual hazards identified in the Activity Hazard Analysis provided under Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). Provide recommended safeguards for each identified hazard. Specify if any certifications or licenses are required to operate the equipment.

1.6.4.2 Operator Prestart

Provide procedures required to install, set up, and prepare each system for use.

1.6.4.3 Startup, Shutdown, and Post-Shutdown Procedures

Provide narrative description for Startup, Shutdown and Post-shutdown operating procedures including the control sequence for each procedure.

1.6.4.4 Normal Operations

Provide Control Diagrams with data to explain operation and control of systems and specific equipment. Provide narrative description of Normal Operating Procedures.

1.6.4.5 Emergency Operations

Provide Emergency Procedures for equipment malfunctions to permit a short period of continued operation or to shut down the equipment to prevent further damage to systems and equipment. Provide Emergency Shutdown Instructions for fire, explosion, spills, or other foreseeable contingencies. Provide guidance and procedures for emergency operation of utility systems including required valve positions, valve locations and zones or portions of systems controlled.

1.6.4.6 Operator Service Requirements

Provide instructions for services to be performed by the operator such as lubrication, adjustment, inspection, and recording gauge readings.

1.6.4.7 Environmental Conditions

Provide a list of Environmental Conditions (temperature, humidity, and other relevant data) that are best suited for the operation of each product, component or system. Describe conditions under which the item equipment should not be allowed to run.

1.6.4.8 Operating Log

Provide forms, sample logs, and instructions for maintaining necessary operating records.

1.6.4.9 Additional Requirements for Equipment Control Systems

Provide Data Package 5 and the following for all control systems:

a. Provide a narrative description on how to perform and apply functions, features, modes, and other operations, including unoccupied operation, seasonal changeover, manual operation, and alarms. Include detailed technical manual for programming and customizing control loops and algorithms.

b. Submit complete controls equipment schedules, full as-built sequence of operations, wiring and logic diagrams, Input/Output Tables, equipment schedules, copies of checkout tests and calibrations performed by the Contractor (not Cx tests), and all associates information.

c. Full points list. Provide a listing of rooms with the following information for each room:

(1) Floor

(2) Room number

(3) Room name

(4) Air handler unit ID

(5) Reference drawing number

(6) Air terminal unit tag ID

(7) Heating or cooling valve tag ID

(8) Minimum cfm

(9) Maximum cfm

d. Full print out of all schedules and set points after testing and acceptance of the system.

e. Full as-built print out of software program.

f. Marking of system sensors and thermostats on the as-built floor plan and mechanical drawings with their control system designations.

1.6.4.10 Testing Equipment Information and Performance Data

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.6.5 Routine Maintenance Requirements

1.6.5.1 Preventive Maintenance Plan, Schedule, and Procedures

Provide manufacturer's schedule for routine preventive maintenance, inspections, condition monitoring (predictive tests) and adjustments required to ensure proper and economical operation and to minimize repairs. Provide instructions stating when the systems should be retested. Provide manufacturer's projection of preventive maintenance work-hours on a daily, weekly, monthly, and annual basis including requirements by type of activity. For periodic calibrations, provide manufacturer's specified frequency and procedures for each separate operation.

a. Define the anticipated time required to perform each test (work-hours), test apparatus, number of personnel identified by responsibility, and a testing validation procedure permitting the record operation capability requirements within the schedule. Provide a remarks column for the testing validation procedure referencing operating limits of time, pressure, temperature, volume, voltage, current, acceleration, velocity, alignment, calibration, adjustments, cleaning, or special system notes. Delineate procedures for preventive maintenance, inspection, adjustment, lubrication and cleaning necessary to minimize repairs.

b. Repair requirements must inform operators how to check out, troubleshoot, repair, and replace components of the system. Include electrical and mechanical schematics and diagrams and diagnostic techniques necessary to enable operation and troubleshooting of the system after acceptance.

1.6.5.2 Lubrication Data

Include the following preventive maintenance lubrication data, in addition to instructions for lubrication required under paragraph OPERATOR SERVICE REQUIREMENTS:

a. A table showing recommended lubricants for specific temperature ranges and applications.

b. Charts with a schematic diagram of the equipment showing lubrication points, recommended types and grades of lubricants, and capacities. Provide procedural instructions for Oil Sampling for all equipment.

c. A Lubrication Schedule showing service interval frequency.

1.6.6 Repair Procedures

Provide instructions and a list of tools required to repair or restore the product or equipment to proper condition or operating standards. Provide manufacturer's recommended procedures and instructions for correcting problems and making repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials. Specify if any certifications or licenses are required to repair the equipment.

1.6.6.1 Troubleshooting Guides and Diagnostic Techniques

Provide step-by-step procedures to promptly isolate the cause of typical malfunctions. Describe clearly why the checkout is performed and what conditions are to be sought. Identify tests or inspections and test equipment required to determine whether parts and equipment may be reused or require replacement.

1.6.6.2 Wiring Diagrams and Control Diagrams

Provide point-to-point drawings of wiring and control circuits including factory-field interfaces. Provide a complete and accurate depiction of the actual job specific wiring and control work. On diagrams, number electrical and electronic wiring and pneumatic control tubing and the terminals for each type, identically to actual installation configuration and numbering.

1.6.6.3 Removal and Replacement Instructions

Provide step-by-step procedures and a list of required specialty tools and supplies for removal, replacement, disassembly, and assembly of components, assemblies, subassemblies, accessories, and attachments. Provide tolerances, dimensions, settings and adjustments required. Use a combination of text and illustrations.

1.6.6.4 Repair Work-Hours

Provide manufacturer's projection of repair work-hours including requirements by type of craft. Identify, and tabulate separately, repair that requires the equipment manufacturer to complete or to participate.

1.6.6.5 Warranty Information

List and explain the various warranties and clearly identify the servicing and technical precautions prescribed by the manufacturers or contract documents in order to keep warranties in force. Identify if replacement of a subassembly, attachment or accessory requires the entire assembly to be replaced. Include warranty information for primary components of the system. Provide copies of warranties required by Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME).

1.6.6.6 Extended Warranty Information

List all warranties for products, equipment, components, and sub-components whose duration exceeds one year. For each warranty listed, indicate the applicable specification section, duration, start date, end date, and the point of contact for warranty fulfillment. Also, list or reference the specific operation and maintenance procedures that must be performed to keep the warranty valid. Provide copies of warranties required by Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME).

1.6.6.7 Record Drawings and Utility Systems

The record drawings are the final compilation of actual conditions reflected in the as-built drawings. Provide record drawings as outlined in 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME).

Using Record Source Drawings, show and document details of the actual installation of the utility systems, annotate and highlight the Operation and Maintenance information. Provide the following drawings at a large enough scale to differentiate designated isolation units from surrounding valves and switches.

1.6.6.8 Personnel Training Requirements

Provide information available from the manufacturers that is needed for use in training designated personnel to properly operate and maintain the equipment and systems.

1.6.6.9 Contractor / Supplier Listing and Contact Information

Provide a list that includes the name, address, telephone number, email and website of the General Contractor and each Subcontractor who installed the product or equipment, or system. For each item, also provide the name address and telephone number of the manufacturer's representative and service organization that can provide replacements most convenient to the project site. Provide the name, address, and telephone number of the product, equipment, and system manufacturers.

1.6.7 Part 4: Facility Operations

1.6.7.1 Completed Facility Operating Plan

Provide a plan that documents that procedures for the operation of systems and assemblies in the facility. The systems that should be included in the Operating Plan include, but are not limited to:

a. Electrical systems and equipment

b. Mechanical systems and equipment

c. Fire Protection systems and equipment

d. Control Systems and equipment

e. Architectural and Structural systems, fixtures, structures, and equipment

f. Vertical transportation such as elevators and escalators

1.6.7.2 Testing Equipment and Special Tool Information

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

1.6.7.3 Testing and Performance Data

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.6.7.4 Approved Field Test Reports and Manufacturer's Field Reports

Compile and provide approved Field Test Reports (SD-06) and Manufacturer's Field Reports (SD-09) submittals.

1.6.7.5 Maintenance Plans, Procedures, Checklists, Records, and Spare Parts Inventory

1.6.7.5.1 Maintenance Schedules

Include recommended maintenance schedules for systems and equipment.

1.6.7.5.2 Ongoing Commissioning Operational and Maintenance Record Keeping

Include ongoing commissioning and optimization procedures and documentation to monitor and improve the performance of facility systems.

1.6.7.5.3 Janitorial and Cleaning Plans and Procedures

Include a copy of facility cleaning and janitorial plan with procedures and intended chemicals and equipment.

Provide environmentally friendly cleaning recommendations in accordance with ASTM E1971.

1.6.7.6 Utility Record Drawings

1.6.7.6.1 Utility Schematic Diagrams

Provide a one-line schematic diagram for each utility system such as power, water, wastewater, and gas/fuel. Schematic diagram must show from the point where the utility line is connected to the mainline up to the 5 foot connection point to the facility. Indicate location or area designation for route of transmission or distribution lines; locations of duct banks, manholes/handholes or poles; isolation units such as valves and switches; and utility facilities such as pump stations, lift stations, and substations.

1.6.7.6.2 Enlarged Connection and Cutoff Plans

Provide enlarged floor plans and provide information between the 5 foot utilities connection point and where utilities connect to facility distribution. Enlarge floor plans/elevations of the rooms where the utility enters the building and indicate on these plans the locations of the main interiors and exterior connection and cutoff points for the utilities. Also enlarge floor plans/elevations of the rooms where equipment is located. Include enough information to enable someone unfamiliar with the facility to locate the connection and cutoff points. Indicate designations such as room number, panel number, circuit breaker, or valve number of each utility and equipment connection and cutoff point, and what that connection and cutoff point controls.

1.6.7.6.2.1 Description of Utility Metering and Monitoring Systems

Provide in narrative format a description of the utility metering and monitoring systems. Include locations, function, and related systems.

1.6.7.6.2.2 Procedures for Tracking Utility Use and Reporting

Procedures for usage reporting and tracking in support of establishing and monitoring utility budgets and costs, and in developing annual energy reports.

1.6.7.6.2.3 One-Line Diagrams and Meter Location of System

Provide one-line diagrams and design drawings that highlight meter locations on the site.

1.6.7.6.3 Spare Parts and Supply Lists

Provide lists of spare parts and supplies required for repair to ensure continued service or operation without unreasonable delays. Special consideration is required for facilities at remote locations. List spare parts and supplies that have a long lead-time to obtain.

1.6.8 Part 5: Training

Provide a copy of training plans used for each type of equipment along with training materials used, arranged in specification sequence. Provide a copy of training records, sign-in sheets, and agendas. Include training and documentation on the updating and continued use of the O&M Manual.

1.6.9 Part 6: Cx Project Report and TAB Report

Provide the final Cx Plan and complete Cx reports with evaluation and testing forms and records for each building system. Include relevant commissioned system assemblies test reports including installers checklists of assemblies. Provide all Cx Progress Reports, issues and resolutions logs with resolution or status of each item, and a list of any open items and seasonal or additional testing required.

1.6.10 Part 7: Regulatory Requirements

Provide information describing regulatory and policies compliance requirements or provide a reference to where it is stored.

1.6.11 Part 8: Permits

Provide information requiring frequently asked questions and associated answers or provide a reference to where it is stored.

1.6.12 Part 9: Operations and Maintenance Manual Approval

Provide a signed document stating that the project O&M Manual has been reviewed and confirming agreement with the approach it presents. Include contact information for the signer for coordination of any future changes.

1.7 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES

Provide the O&M data packages specified in individual technical sections. O&M Data Packages are one of the components of the O&M Manual. The information required in each type of data package follows:

1.7.1 Package Quality

Documents must be fully legible. Operation and Maintenance data must be consistent with the manufacturer's standard brochures, schematics, printed instructions, general operating procedures, and safety precautions.

1.7.2 Data Package 1

a. Safety precautions and hazards

b. Cleaning recommendations

c. Maintenance and repair procedures

d. Warranty information

e. Extended warranty information

f. Contractor information

g. Spare parts and supply list

1.7.3 Data Package 2

a. Safety precautions and hazards

b. Normal operations

c. Environmental conditions

d. Lubrication data

e. Preventive maintenance plan, schedule, and procedures

f. Cleaning recommendations

g. Maintenance and repair procedures

h. Removal and replacement instructions

i. Spare parts and supply list

j. Parts identification

k. Warranty information

l. Extended warranty information

m. Contractor information

1.7.4 Data Package 3

a. Safety precautions and hazards

b. Operator prestart

c. Startup, shutdown, and post-shutdown procedures

d. Normal operations

e. Emergency operations

f. Environmental conditions

g. Operating log

h. Lubrication data

i. Preventive maintenance plan, schedule, and procedures

j. Cleaning recommendations

k. Troubleshooting guides and diagnostic techniques

l. Wiring diagrams and control diagrams

m. Maintenance and repair procedures

n. Removal and replacement instructions

o. Spare parts and supply list

p. Product submittal data

q. O&M submittal data

r. Parts identification

s. Warranty information

t. Extended warranty information

u. Testing equipment and special tool information

v. Testing and performance data

w. Contractor information

x. Field test reports

1.7.5 Data Package 4

a. Safety precautions and hazards

b. Operator prestart

c. Startup, shutdown, and post-shutdown procedures

d. Normal operations

e. Emergency operations

f. Operator service requirements

g. Environmental conditions

h. Operating log

i. Lubrication data

j. Preventive maintenance plan, schedule, and procedures

k. Cleaning recommendations

l. Troubleshooting guides and diagnostic techniques

m. Wiring diagrams and control diagrams

n. Repair procedures

o. Removal and replacement instructions

p. Spare parts and supply list

q. Repair work-hours

r. Product submittal data

s. O&M submittal data

t. Parts identification

u. Warranty information

v. Extended warranty information

w. Personnel training requirements

x. Testing equipment and special tool information

y. Testing and performance data

z. Contractor information

aa. Field test reports

1.7.6 Data Package 5

a. Safety precautions and hazards

b. Operator prestart

c. Start-up, shutdown, and post-shutdown procedures

d. Normal operations

e. Environmental conditions

f. Preventive maintenance plan, schedule, and procedures

g. Troubleshooting guides and diagnostic techniques

h. Wiring and control diagrams

i. Maintenance and repair procedures

j. Removal and replacement instructions

k. Spare parts and supply list

l. Product submittal data

m. Manufacturer's instructions

n. O&M submittal data

o. Parts identification

p. Testing equipment and special tool information

q. Warranty information

r. Extended warranty information

s. Testing and performance data

t. Contractor information

u. Field test reports

v. Additional requirements for HVAC control systems

1.7.7 Changes to Submittals

Provide manufacturer-originated changes or revisions to submitted data if a component of an item is so affected subsequent to acceptance of the O&M Data. Submit changes, additions, or revisions required by the Contracting Officer for final acceptance of submitted data within 30 calendar days of the notification of this change requirement.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 TRAINING

Prior to acceptance of the facility by the Contracting Officer for Beneficial Occupancy, provide comprehensive training for the systems and equipment specified in the technical specifications. The training must be targeted for the Facilities Management Specialist, building maintenance personnel, and applicable building occupants. Instructors must be well-versed in the particular systems that they are presenting. Address aspects of the Operation and Maintenance Manual submitted in accordance with Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME). Address aspects of the FDW, as submitted in Section 01 78 24.00 20 Facility Data Workbook (FDW). Training must include classroom or field lectures based on the system operating requirements. The location of classroom training requires approval by the Contracting Officer.

3.1.1 Training Plan

Submit a written training plan to the Contracting Officer for approval at least 60 calendar days prior to the scheduled training. Training plan must be approved by the Quality Control Manager (QC) prior to forwarding to the Contracting Officer. Also, coordinate the training schedule with the Contracting Officer and QC. Include within the plan the following elements:

a. Equipment included in training

b. Intended audience

c. Location of training

d. Dates of training

e. Objectives

f. Outline of the information to be presented and subjects covered including description

g. Start and finish times and duration of training on each subject

h. Methods (e.g. classroom lecture, video, site walk-through, actual operational demonstrations, written handouts)

i. Instructor names and instructor qualifications for each subject

j. List of texts and other materials to be furnished by the Contractor that are required to support training

k. Description of proposed software to be used for video recording of training sessions.

3.1.2 Training Content

The core of this training must be based on manufacturer's recommendations and the operation and maintenance information. The QC is responsible for overseeing and approving the content and adequacy of the training. Spend 95 percent of the instruction time during the presentation on the OPERATION AND MAINTENANCE DATA. Include the following for each system training presentation:

a. Start-up, normal operation, shutdown, unoccupied operation, seasonal changeover, manual operation, controls set-up and programming, troubleshooting, and alarms.

b. Relevant health and safety issues.

c. Discussion of how the feature or system is environmentally responsive. Advise adjustments and optimizing methods for energy conservation.

d. Design intent.

e. Use of O&M Manual Files.

f. Review of control drawings and schematics.

g. Interactions with other systems.

h. Special maintenance and replacement sources.

i. Tenant interaction issues.

j. Summary/presentation of associated FDW information submitted in Section 01 78 24.00 20 FACILITY DATA WORKBOOK (FDW).

3.1.3 Training Outline

Provide the Operation and Maintenance Manual Files (Bookmarked PDF) and a written course outline listing the major and minor topics to be discussed by the instructor on each day of the course to each trainee in the course. Provide the course outline 14 calendar days prior to the training.

3.1.4 Training Video Recording

Record classroom training session(s) on video. Provide to the Contracting Officer two copies of the training session(s) in DVD video recording format. Capture within the recording, in video and audio, the instructors' training presentations including question and answer periods with the attendees. The recording camera(s) must be attended by a person during the recording sessions to assure proper size of exhibits and projections during the recording are visible and readable when viewed as training.

3.1.5 Unresolved Questions from Attendees

If, at the end of the training course, there are questions from attendees that remain unresolved, the instructor must send the answers, in writing, to the Contracting Officer for transmittal to the attendees, and the training video must be modified to include the appropriate clarifications.

3.1.6 Validation of Training Completion

Ensure that each attendee at each training session signs a class roster daily to confirm Government participation in the training. At the completion of training, submit a signed validation letter that includes a sample record of training for reporting what systems were included in the training, who provided the training, when and where the training was performed, and copies of the signed class rosters. Provide two copies of the validation to the Contracting Officer, and one copy to the Operation and Maintenance Manual Preparer for inclusion into the Manual's documentation.

3.1.7 Quality Control Coordination

Coordinate this training with the QCin accordance with Section 01 45 00.00 22 QUALITY CONTROL (PWD ME).

3.2 SUBMITTAL SCHEDULING

3.2.1 Operation and Maintenance Manual, Progress Submittal

Submit the Progress submittal when construction is approximately 50 percent complete, to the Contracting Officer for approval. Provide Operation and Maintenance Manual Files (Bookmarked PDF). Include the elements and portions of system construction completed up to this point. The purpose of this submittal is to verify progress is in accordance with contract requirements as discussed during the Operation and Maintenance

Manual Coordination Meeting.

3.2.2 Operation and Maintenance Manual, Prefinal Submittal

Submit the 100 percent submittal of the Operation and Maintenance Prefinal Submittal to the Contracting Officer for approval within 60 calendar days of the Beneficial Occupancy Date (BOD). This submittal must provide a complete, working document that can be used to operate and maintain the facility. Any portion of the submittal that is incomplete or inaccurate requires the entire submittal to be returned for correction. Any discrepancies discovered during the Government's review of Operation and Maintenance Progress submittal must be corrected prior to the Prefinal submission. The Prefinal Submittal must include Operation and Maintenance Manual Files (Bookmarked PDF).

3.2.3 Operation and Maintenance Manual, Final Submittal

Submit completed Operation and Maintenance Manual Files (Bookmarked PDF). The Final submittal is due at BOD. Any discrepancies discovered during the Government's review of the Prefinal submittal, including the Field Verification, must be corrected prior to the Final submission.

-- End of Section --

SECTION 01 78 24.00 22
^^
FACILITY DATA WORKBOOK (FDW)(PWD ME)
05/23, CHG 1: 11/23

PART 1 GENERAL

1.1 DEFINITIONS AND ABBREVIATIONS

1.1.1 eOMSI

Electronic Operation and Maintenance Support Information

1.1.2 Facility Data Workbook (FDW)

A Microsoft Excel file containing required facility information populated by the Contractor.

1.1.3 Systems

The words "system", "systems", and "equipment", when used in this document refer to as-built systems and equipment.

1.1.4 KTR

An abbreviation for "Contractor."

1.2 OPERATION AND MAINTENANCE MANUAL AND FACILITY DATA WORKBOOK MEETINGS

1.2.1 Pre-Construction Meeting

Meeting requirements are identified in Section 01 78 23.00 22 OPERATION AND MAINTENANCE DATA (PWD ME).

1.3 SUBMITTAL SCHEDULING

1.3.1 Facility Data Workbook (FDW), Progress Submittal

Submit the Progress submittal when construction is approximately 50 percent complete, to the Contracting Officer for approval. Provide Facility Data Workbook (Excel). Include the elements and portions of system construction completed up to this point.

The purpose of this submittal is to verify progress is in accordance with contract requirements as discussed during the Operation and Maintenance Manual and Facility Data Workbook Meetings. Field verify a portion of the eOMSI information in accordance with paragraph FIELD VERIFICATION.

1.3.2 Facility Data Workbook (FDW), Prefinal Submittal

Submit the 100 percent submittal of the FDW Prefinal Submittal to the Contracting Officer for approval within 60 calendar days of the Beneficial Occupancy Date (BOD). This submittal must provide a complete, working document that can be used to operate and maintain the facility. Any portion of the submittal that is incomplete or inaccurate requires the entire submittal to be returned for correction. Any discrepancies discovered during the Government's review of FDW Progress submittal must be corrected prior to the Prefinal submission.

1.3.3 Facility Data Workbook (FDW), Final Submittal

Submit completed Facility Data Workbook (Excel). The Final submittal is due at BOD. Any discrepancies discovered during the Government's review of the Prefinal eOMSI submittal, including the Field Verification, must be corrected prior to the Final eOMSI submission.

1.4 UNITS OF MEASURE

Provide FDW utilizing the units of measure used in the Government generated contract documents.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-11 Closeout Submittals

Facility Data Workbook (FDW), Progress Submittal; G

Facility Data Workbook (FDW), Prefinal Submittal; G

Facility Data Workbook (FDW), Final Submittal; G

PART 2 PRODUCTS

2.1 eOMSI FACILITY DATA WORKBOOK

An initial, pre-edited draft of the Model & Facility Data Matrix tab within the Facility Data Workbook is attached to this Section. The Government will provide this Facility Data Workbook electronically to the Contractor upon award. Add, delete, and update Mastersystems, Systems, and Subsystems that may have changed during construction, or any items that may have been omitted or missed during design, at no additional cost to the Government. Complete the KTR Facility Data File tab based on the selection of Mastersystems, Systems, and Subsystems installed. The following tabs are included in the eOMSI Facility Data File Workbook and serve the purpose stated:

a. Instructions Tab: Instructions for completing Model & Facility Data Matrix Tab and KTR Facility Data File Tab. If a discrepancy exists between what is required in this Section and the Workbook, the instructions within the workbook take precedence.

b. Model & Facility Data Matrix Tab: - The Matrix lists Required Facility Asset Fields for each SYSTEM and SUBSYSTEM. The Designer of Record selects SYSTEMS and SUBSYSTEMS that are within the project scope, which the Contractor needs to include and populate in KTR Facility Data File tab. The "Required Facility Asset Field Position Numbers," one through thirty-five, are pre-populated, and are not editable.

c. Required Facility Asset Fields Tab: Defines the 35 Required Facility Asset Field Position Numbers used in Model and Facility Data Matrix and KTR Facility Data File tabs.

d. KTR Sample Facility Data File Tab: Sample KTR eOMSI facility data file. This tab provides an example of the mandatory fields of equipment installed by the Contractor, and populated in the KTR eOMSI Facility Data File Tab, along with their descriptions.

e. KTR Facility Data File Tab: Required eOMSI facility data file deliverable provided to the Government. Provide a separate and unique new row for each facility component or piece of equipment installed. Coordinate with the Government's Contracting Officer's Representative and NAVFAC PW FMD for specific facility component naming convention.

PART 3 EXECUTION

3.1 FIELD VERIFICATION

Field verify Facility Data Workbook information with Contractor and Government personnel. Include the following personnel in this meeting: Contractor's Facility Data Workbook Preparer and Quality Control Manager, Commissioning Authority, and the Government's Contracting Officer's Representative and NAVFAC PW FMD. Request, and provide, an eOMSI Field Verification Meeting no sooner than 14 calendar days after submission of the Progress FDW submittal, and another, no sooner than 14 calendar days after submission of the Prefinal FDW submittal. During this meeting, the Government and Contractor will verify that the Facility Data Workbook is complete and accurate.

Field verify that at least 5 Subsystems under each of the Mastersystems are accurate, for a total of 25 Subsystems. For each of these items, verify that the required facility asset field, as defined in the "Model & Facility Data Matrix" tab, contains the specified data and it is accurate (i.e., item description, manufacturer, model no., serial no.). 100 percent accuracy of eOMSI information is required for successful field verification. If data discrepancies are discovered amongst the 25 Subsystems verified, resubmit an updated eOMSI FDW, and request a make-up field verification meeting. At the make-up field verification meeting 25 new Subsystems and their associated required facility asset fields will be field verified; the 25 new Subsystems must be 100 percent accurate. Any discrepancies discovered must be corrected prior to next eOMSI Facility Data Workbook Submittal.

(1) D10 - CONVEYING
(2) D20 - PLUMBING
(3) D30 - HVAC
(4) D40 - FIRE PROTECTION
(5) D50 - ELECTRICAL

-- End of Section --

KTR FACILITY DATA FILE

Each facility component or piece of equipment will be a new row. Refer to Model & Facility Data Matrix for guidance on which fields are appl

USE UPPER CASE TO FILL OUT DATA FILE

Position	1	2	3	4	5	6	7	8	9	10
Name	**Asset Num**	**Description**	**Long Description**	**MASTER SYSTEM**	**SYSTEM**	**SUBSYSTEM**	**Building Number**	**Asset Quantity**	**Replacement Cost**	**Contract Number**
Explanation	Asset tag number for MaximoTo be filled in by NAVFAC FOS	Primary Asset Name (100 Character Limit) To be completed by DOR	Additional Relevant Information (e.g. size, capacity, limits, etc…) (1000 Character Limit) To be completed by KTR		Reference values from Model & Facility Data Matrix tab (SYSTEM) Selected By DOR	Reference values from Model & Facility Data Matrix tab (SUBSYSTEM) Selected By DOR	MAXIMO building number provided by DOR (Any questions ask FOS)	To be completed by DOR (refer to Model & Facility Data Matrix, Field) Can be edited by KTR for changes during project	Install cost (material and labor) from schedule of values, bid proposal, etc. To be completed by KTR	To be completed by GVT
EXAMPLE	K000000	CONDENSING UNIT, AIR COOLED	ACCU-3A SERVES HP-24 / HP-25	D30 - HVAC	D3030 - COOLING GENERATING SYSTEMS	D3030221 - CONDENSING UNIT, AIR COOLED	PNSY-59	1	$5,000	N40085-16-C-9329

icable to specific components & equipment.

11	12	13	14	15	16	17	18	19
Task/ Delivery Order Number	Warranty Expiration Date	Installation Date	CLASSIFICATIONID	Room Number	Manufacturer	Model	Serial #	PAY ITEM
To be completed by GVT	MM/DD/YYYY To be completed by KTR	MM/DD/YYYY To be completed by KTR	Reference values from Model & Facility Data Matrix tab (SUBITEM)	Location for installed equipment/asset. Include floor & room number. (include equipment tag e.g AHU_1, FANN001) To be completed by KTR	To be completed by KTR	To be completed by KTR	To be completed by KTR	To be completed by KTR
	2/1/2019	2/1/2018	D3030221	OUTDOOR PAD 1ST SET FROM STAIRS, ACCU-3B	FUJITSU	AOUA120TLCVG	TNN000034	

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SECTION 01 91 00.15 22

TOTAL BUILDING COMMISSIONING
08/24

PART 1 GENERAL

1.1 SUMMARY

The Government has hired an independent Commissioning Firm.

This Section specifies the scope of work for the entire Commissioning Team which includes the scope of work for the Lead Commissioning Specialist, Commissioning Specialists, Designers of Record, Installing Contractors, and Equipment Suppliers. Refer to paragraph COMMISSIONING TEAM for descriptions of Commissioning Team members and assigned responsibilities.

1.2 COMMISSIONING LIMITATIONS

Commissioning Requirements specified in this Section are prohibited from fulfilling requirements specified in other Sections. Requirements specified in other Sections are prohibited from fulfilling requirements specified in this Section.

1.3 SYSTEMS TO BE COMMISSIONED

Commissioning must be performed for equipment, systems, and assemblies listed under each systems category in the following paragraphs.

Specified equipment, system, and assembly counts are totals that include all commissioned buildings and all construction phases in this project. Refer to items entitled "Sample Strategy" throughout this Section that must be applied to these counts.

1.3.1 Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category

a. Primary Systems.

(1) Up to one steam supply system featuring one steam supply service and one condensate return system consisting of steam condensate receiver and one condensate pump, steam-to-water heat exchanger and steam pressure reducing valve.

(2) Up to one hot water distribution system featuring up to two variable speed primary pumps, associated air separator, expansion tank, and controls.

(3) Up to one Building Automation System.

b. Air Handling Systems.

(1) Up to one rooftop DOAS air handling units each featuring a variable-air-volume supply air fan, a DX cooling coil, a hot water coil, air-side economizer, and packaged controls with remote VFD condensing unit.

(2) Up to one dedicated outside air systems (DOAS) each featuring a variable-air-volume supply air fan, a variable-air-volume exhaust

air fan, a DX cooling coil, a hot water coil, air-side energy recovery, and packaged controls with remote VFD Condensing Unit.

(3) Up to three blower coil units each featuring a direct expansion cooling coil, a hot water coil and remote condensing unit.

(4) Up to four split system air-conditioning units each featuring packaged controls.

c. Terminal Devices.

(1) Up to eight unit heaters (four of 8 are cabinet unit heaters) each featuring a hydronic heating coil and remote thermostat.

(2) Up to one electric unit heaters.

(3) Up to two exhaust air fans.

(4) Up to two gravity ventilators with motorized dampers.

1.3.2 Plumbing Systems Category

a. Up to two reduced pressure backflow preventers.

b. Up to three domestic heating hot water systems each featuring up to two electric water heaters each with packaged controls, up to one recirculating pump, and up to one temperature mixing valve.

c. Up to four emergency eyewash stations.

d. Up to one power plant compressed air distribution system featuring packaged controls and approximately 40 drops.

1.3.3 Utility Systems Category

a. Up to one set of whole-building energy utility metering systems.

b. Up to one whole-building water utility metering system.

1.3.4 Electrical Type I Systems Category

a. Power - all generator paralleling and loading systems.

b. Power - all generator transfer switches.

c. Power - all grounding systems.

d. Power - all industrial control systems.

e. Power - all main distribution panels.

f. Power - all panelboards.

g. Power - all step-down transformers.

h. Power - all switchgear.

i. Power - all transient voltage surge protective devices.

1.3.5 Electrical Type II Systems Category

a. Energy monitoring - all individual energy monitoring of total electrical energy, HVAC systems, interior lighting, exterior lighting, and receptacle circuits.

b. Lighting - all daylighting control systems.

c. Lighting - all emergency egress systems, control devices, and control systems.

d. Lighting - all exit systems, control devices, and control systems.

e. Lighting - all exterior systems, control devices, and control systems.

f. Lighting - all interior systems, control devices, and control systems.

g. Power systems - all automatic receptacles controls.

1.3.6 Building Envelope Systems, Components, and Assemblies Category

a. All building envelope systems including curtain walls, exterior penetrations, exterior air barrier assemblies, exterior windows, exterior doors, and skylights.

b. All roofing including low sloped roofs and steep sloped roofs.

1.4 COMPLEX SYSTEMS

Complex systems are defined as the following systems categories as listed under paragraph SYSTEMS TO BE COMMISSIONED with appropriate qualifiers and are subject to more stringent requirements as specified in this Section.

a. All equipment, systems, and assemblies listed under item entitled "Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category."

b. All equipment, systems, and assemblies listed under item entitled "Plumbing Systems Category" featuring on-board factory packaged controls such as domestic heating hot water systems, domestic solar heating hot water systems, and service water pressure-booster pump systems. Refer to the following paragraph for definition of equipment featuring on-board factory packaged controls.

c. All equipment, systems, and assemblies listed under item entitled "Utility Systems Category."

Equipment featuring on-board factory packaged controls is defined as equipment that is a complex system as defined in paragraph COMPLEX SYSTEMS which features a controller issuing sophisticated logic-based commands to its internal components to accomplish the intent of Plans and Specifications. This equipment may function as either a stand-alone piece of commissioned equipment or may function as part of a commissioned system. This controller may be equipment mounted or may be remote mounted and may or may not interface with a building control system. Examples of equipment meeting and not meeting this definition are provided in the following paragraphs.

a. Equipment meeting this definition includes chillers, boilers provided with or without master controllers, packaged pumping systems, dedicated outside air systems, packaged rooftop units, computer room air conditioners, and air terminal units.

b. Equipment not meeting this definition includes unitary air-conditioners, packaged terminal air-conditioners, and heat pumps.

c. Equipment not meeting this definition, but requiring demonstration of minimum safe motor speed either prior to or during Functional Performance Testing includes variable frequency drives.

1.5 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 90.1 - IP	(2019; Errata 1 2019; Errata 2-5 2020; Addenda BY-CP 2020; Addenda AF-DB 2020; Addenda A-G 2020; Addenda F-Y 2021; Errata 6-8 2021; Interpretation 1-4 2020; Interpretation 5-8 2021 Addenda AS-AQ2022) Energy Standard for Buildings Except Low-Rise Residential Buildings

1.6 SUBMITTALS

Commissioning Submittals are not required to be submitted by the Installing Contractor. Exchange of information between the Commissioning Team specified in this Section must still occur.

1.7 COMMISSIONING TEAM

The Commissioning Team is defined as the entity that must perform all Commissioning Activities specified in this Section for systems listed under paragraph SYSTEMS TO BE COMMISSIONED. The Commissioning Team is composed of the Lead Commissioning Specialist, Commissioning Specialists, Designers of Record, Installing Contractors, Equipment Suppliers, and the Contracting Officer's Technical Representative.

1.7.1 Lead Commissioning Specialist

The Lead Commissioning Specialist (CxC) is defined as the entity that must perform all Commissioning Responsibilities assigned in this Section that includes, but is not limited to, coordinating all aspects of the Commissioning Process; managing all Commissioning Specialists; and submitting all Commissioning Plans, schedules, reports, and documentation directly to the Contracting Officer's Technical Representative as modified in this Section.

1.7.2 Commissioning Specialists

Commissioning Specialists are defined as entities that must perform all Commissioning Responsibilities assigned in this Section that includes, but is not limited to, performing reviews, witnessing that all system components have been installed, that each control device operates, that each equipment piece operates, that complete systems operate, and that interactive systems operate in accordance with Plans and Specifications.

Commissioning Specialists must perform assigned responsibilities for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

1.7.3 Designers of Record

Designers of Record are defined as entities that must act as design professionals responsible for design of their respective system that is subject to Commissioning. Designers of Record must perform all Commissioning Responsibilities assigned in this Section.

1.7.4 Installing Contractors

Installing Contractors are defined as the General Contractor's Superintendent or Project Manager; Quality Control Representative; and Mechanical, Controls, Plumbing, Electrical, Test and Balance, Fire Protection, and Building Envelope Sub-Contractors who provide, install, or construct systems in whole or in part for this contract. The General Contractor must provide a Lead Commissioning Representative to manage all Sub-Contractor Commissioning Representatives and Equipment Suppliers to accomplish all Commissioning Responsibilities assigned in this Section. Each Sub-Contractor must provide a Commissioning Representative to coordinate and perform all Commissioning Responsibilities for commissioned equipment, systems, and assemblies provided as part of their respective contract as directed by the General Contractor. Submission of Commissioning related certifications or qualifications for the Lead Commissioning Representative and Installing Contractor Representatives is not required.

Installing Contractors must perform all Commissioning Responsibilities assigned in this Section that includes, but is not limited to, demonstrating performance of all commissioned equipment, systems, and assemblies provided as part of their respective contract and must provide all necessary materials, services, and labor required for successful demonstration as modified in this Section.

1.7.5 Equipment Suppliers

Equipment Suppliers (ES) are defined as factory authorized start-up technicians or factory representatives knowledgeable with start-up, demonstration, and testing of commissioned equipment featuring on-board factory packaged controls provided as part of their respective contract. Each Equipment Supplier must coordinate and perform all Commissioning Responsibilities assigned in this Section as directed by the General Contractor. Submission of Commissioning related certifications or qualifications for Equipment Suppliers is not required. Refer to paragraph COMPLEX SYSTEMS for definition of equipment featuring on-board factory packaged controls's.

Equipment Suppliers must perform all Commissioning Responsibilities assigned in this Section that includes, but is not limited to, demonstrating performance of all commissioned equipment provided as part of their respective contract and must provide all necessary materials, services, and labor required for successful demonstration of all commissioned equipment provided as part of their respective contract as modified in this Section.

1.7.6 Installing Contractors and Equipment Suppliers Coordination

Installing Contractors and Equipment Suppliers must coordinate and share demonstration and testing responsibilities specified in this Section for commissioned equipment featuring on-board factory packaged controls. Installing Contractors and Equipment Suppliers must simultaneously perform all aspects of demonstration and testing for these types of systems. Refer to paragraph COMPLEX SYSTEMS for definition of equipment featuring on-board factory packaged controls.

1.7.7 Contracting Officer's Technical Representative

The Contracting Officer's Technical Representative (COTR) represents the Government for all elements of the Commissioning Process. The COTR will serve as the single point of contact with the Lead Commissioning Specialist, will receive all Commissioning Submittals, may attend meetings, and may, but is not required to, witness field inspections and testing specified in this Section.

1.8 COMMUNICATIONS

1.8.1 Communications with the Contracting Officer's Technical Representative

Communication with the Contracting Officer's Technical Representative must be through the Lead Commissioning Specialist who acts as the single direct point of contact for all aspects of the Commissioning Process.

1.8.2 Communications between the Commissioning Team

Communications between the Commissioning Team including, but not limited to, verbal communications, schedule information, and exchange of information specified in this Section, must occur directly between the Commissioning Team without requiring interaction of the Contracting Officer's Technical Representative.

1.9 NUMBER OF BUILDINGS AND CONSTRUCTION PHASING

1.9.1 Number Of Buildings

This project includes one commissioned building.

a. Commissioning Design Reviews specified in this section must be performed as a single review for each submission inclusive of all commissioned buildings in this project. Refer to item entitled "Commissioning Design Reviews" for additional requirements.

b. Construction Phase Site Observations specified in this Section are inclusive of all commissioned buildings in this project. Refer to item entitled "Construction Phase Site Observations" for additional requirements.

c. Construction Phase Functional Performance Testing must be performed during a single continuous occurrence inclusive of all commissioned buildings for each systems category until completion of testing of all systems listed under respective systems category. Refer to item entitled "Construction Phase Functional Performance Testing" for additional requirements.

d. Post-Construction Phase Site Observations specified are inclusive of all commissioned buildings in this project. Refer to item entitled "Post-Construction Phase Site Observations" for additional requirements.

1.9.2 Construction Phasing

This project's construction is not phased.

1.10 ASHRAE STANDARD 90.1 COMMISSIONING COMPLIANCE

ASHRAE Standard 90.1 Commissioning compliance is required for this project.

1.10.1 General Requirements

All Commissioning Activities referred to as "Building Systems Verification and Testing" and "Building Commissioning" in ASHRAE 90.1 - IP as modified in this Section, Commissioning Activities listed in references contained in ASHRAE 90.1 - IP's paragraphs as modified in this Section, and Commissioning Activities specified in this Section must be performed. All Commissioning Activities that are referred to as "optional activities," "guidance," "should activities," "recommendations," "suggested practices," "informative information," or listed as similar designations are mandatory regardless of any qualifiers.

1.10.2 Commissioning Team Responsibilities

The Commissioning Team must, at a minimum, perform all Commissioning Activities specified in the preceding item entitled "General Requirements" and must perform all Commissioning Activities specified in this Section. This Section takes precedence should there be a conflict between requirements listed in this Section and requirements listed in ASHRAE 90.1 - IP or listed in any of its references.

1.11 SUSTAINABILITY THIRD PARTY CERTIFICATION (TPC) COMMISSIONING COMPLIANCE

Sustainability Third Party Certification (TPC) is not required for this project.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 DESIGN PHASE

3.1.1 Design Phase ASHRAE Standard 90.1 Commissioning Compliance

3.1.1.1 General Requirements

Activities must include, but not be limited to, Design Phase Commissioning Activities referred to as "Building Systems Verification and Testing" and "Building Commissioning" in ASHRAE 90.1 - IP as modified in this Section, Design Phase Commissioning Activities listed in references contained in

ASHRAE 90.1 - IP as modified in this Section, and Design Phase Commissioning Activities specified in this Section.

3.1.1.2 Lead Commissioning Specialist and Commissioning Specialists Responsibilities

The Lead Commissioning Specialist and Commissioning Specialists must, at a minimum, perform all activities specified under item entitled "Design Phase ASHRAE Standard 90.1 Commissioning Compliance."

3.1.2 Design Phase Commissioning Coordination Meeting

3.1.2.1 General Requirements

A Design Phase Commissioning Coordination meeting must be conducted no later than 30 calendar days prior to the first Plans and Specifications submission or no later than 30 calendar days after approval of Commissioning Firm certifications and all Commissioning Specialist certifications, whichever occurs last. Meeting topics must include, but not be limited to, the Commissioning Process, contract requirements, lines of communication, roles and responsibilities, schedules, documentation requirements, and logistics as specified in this Section. Meeting must be conducted by conference call.

3.1.2.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, conduct meeting and prepare meeting minutes complete with Attendance Roster.

3.1.2.3 Designers of Record and Others Responsibilities

Designers of Record attend this meeting. The Contracting Officer's Technical Representative, a User Representative, and a Public Works Department Representative may also attend this meeting.

3.1.3 Design Phase Commissioning Plan

3.1.3.1 General Requirements

The Design Phase Commissioning Plan must be prepared specific to this construction and contents must include, but not be limited to, contents specified in the following paragraphs.

a. Plan purpose.

b. Overview of Commissioning Activities.

c. Commissioning scope.

d. Commissioning Process outline.

e. Detailed description of Commissioning Activities.

f. List of all commissioned equipment, systems, and assemblies.

g. Description of performance criteria.

h. Building information.

i. Contact information for the Lead Commissioning Specialist, Commissioning Specialists, Designers of Record, and Contracting Officer's Technical Representative.

j. Criteria listing including Unified Facilities Criteria and Building Codes and Standards.

k. Description of Commissioning Team members.

l. Roles and responsibilities.

m. Lines of communication.

n. Documentation requirements.

o. Description of Commissioning Plan distribution during the Design Phase and Construction Phase.

p. Description of Issues and Resolution Logs distribution during the Design Phase and Construction Phase.

q. Description of Commissioning testing document distribution during the Design Phase and Construction Phase.

r. Description of Commissioning report distribution during the Design Phase and Construction Phase.

s. Owner's Project Requirements.

t. Description of the Basis of Design.

u. Description of Plans and Specifications reviews performed by Commissioning Specialists.

v. Description of Plans and Specifications reviews performed by the Contracting Officer's Technical Representative.

w. Examples and descriptions of Site Observation Reports and Issues And Resolution Log.

x. Listing and description of Design Phase and Construction Phase Commissioning meetings.

y. Identification of Commissioning Activities for incorporation into the Project Construction Schedule.

z. Listing of submittals required by Commissioning Specialists.

aa. Examples and description of the development of Pre-Functional Checklists and Functional Performance Test Checklists.

ab. Description of Pre-Functional Checks and Functional Performance Testing complete with sample strategies.

ac. Description of Ductwork Air Leak Test (DALT) field acceptance testing specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

ad. Description of Performance Verification Test (PVT) field acceptance testing specified in Section 23 09 00.00 22 INSTRUMENTATION AND CONTROL FOR HVAC.

ae. Description of Test and Balance (TAB) field acceptance testing specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

af. Identification of intervals at which a summary of Functional Performance Testing results will be provided to the Contracting Officer's Technical Representative for concurrence or acceptance.

ag. Listing of Operation and Maintenance Manual requirements.

ah. Description of Systems Manuals.

ai. Description of training requirements.

aj. Description of Final Commissioning Report.

3.1.3.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, prepare and submit the **Design Phase Commissioning Plan** no later than 45 calendar days after approval of Commissioning Specialists certifications.

3.1.4 Design Phase Issues And Resolution Log

3.1.4.1 General Requirements

The Design Phase Issues and Resolution Log must be prepared and maintained to track all Commissioning Issues identified by any member of the Commissioning Team. The Design Phase Issues and Resolution Log must be prepared in an electronic format that can be used throughout the Design Phase such that it can be provided to the Government upon request. Design Phase Issues and Resolution Log contents must include, but not be limited to, contents specified in the following paragraphs.

a. Project title and location.

b. Lead Commissioning Specialist's contact information.

c. Issue number.

d. Date issue was discovered.

e. Name of Commissioning Team member identifying the issue.

f. Commissioning Team member responsible for resolving the issue.

g. Current status.

h. Issue description.

i. Pictures of item if available and appropriate.

j. Effects of issue on project or building operation.

k. Possible cause of issue.

l. Proposed resolution.

m. Final resolution and final resolution date.

n. Approvals of issue resolution, including approver's name.

3.1.4.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must prepare and maintain the **Design Phase Issues And Resolution Log** and must submit one electronic copy no later than seven calendar days after the addition of an issue or a change in status of an issue.

3.1.5 Commissioning Design Reviews

3.1.5.1 General Requirements

Commissioning Design Reviews must be performed using Government provided software in accordance with ASHRAE 90.1 - IP and must include, but not be limited to, review of each submission of Plans and Specifications specified in the following paragraphs, except for the last submission. Each review must include back-checks of deficiencies identified in preceding reviews. The last submission must include only a back-check of deficiencies identified in preceding reviews, but not a complete Commissioning Design Review.

a. First submission: Final Plans and Specifications.

Commissioning Design Reviews must verify Commissioning Aspects of the Plans and Specifications, must detail compliance with the Owner's Project Requirements (OPR) and must detail compliance with provisions of ASHRAE 90.1 - IP. The OPR is for reference only and does not constitute part of this contract. Commissioning Design Reviews must identify deficiencies of commissioned equipment, systems, and assemblies that include, but not be limited to, deficiencies related to the following paragraphs as applicable to this project.

a. Adverse impacts on proper operation and maintenance.

b. Adequate accessibility for proper testing.

c. Proper specification of performance and testing criteria.

d. Proper specification of relevant control sensor and control device locations.

e. Completely, clearly, and correctly developed control sequences.

3.1.5.2 Commissioning Design Review Reports

A Commissioning Design Review Report must be prepared for each Commissioning Design Review. Each deficiency must be listed individually and must be separated into categories corresponding to systems categories listed under paragraph SYSTEMS TO BE COMMISSIONED. Commissioning Design Review Report contents must include, but not be limited to, contents specified in the following paragraphs.

a. Project title, project number, and date of review.

b. Executive summary.

c. List of Plans and Specifications reviewed.

d. Deficiency location.

e. Clear description of deficiencies complete with proposed corrective actions.

f. Relevant observations from similar applications.

g. Clarifying questions related to constructability and maintainability if necessary.

h. Compliance with provisions of ASHRAE 90.1 - IP.

Updated Commissioning Design Review Reports must be prepared to include resolution of outstanding issues reached during Commissioning Design Review meetings.

3.1.5.3 Commissioning Design Review Meetings

A Commissioning Design Review meeting must be conducted no later than 14 calendar days after preparation of each Commissioning Design Review Report to discuss proposed corrective actions and to resolve outstanding issues. The Contracting Officer's Technical Representative may attend these meetings. Meeting must be conducted by conference call.

3.1.5.4 Lead Commissioning Specialist Responsibilities

Lead Commissioning Specialist activities must include, but not be limited to, activities specified in the following paragraphs.

a. Prepare and provide Commissioning Design Review Reports to Designers of Record.

b. Conduct Commissioning Design Review meetings.

c. Update Commissioning Design Review Reports.

d. Prepare and submit **Updated Commissioning Design Review Reports** no later than 14 calendar days after each Commissioning Design Review meeting.

3.1.5.5 Commissioning Specialists Responsibilities

Commissioning Specialists activities must include, but not be limited to, performing Commissioning Design Reviews and attending Commissioning Design Review meetings for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.1.5.6 Designers of Record Responsibilities

Designers of Record activities must include, but not be limited to, providing written responses to Commissioning Design Review Reports, attending Commissioning Design Review meetings, implementing revisions to

Plans and Specifications resulting from resolution of Commissioning Design Reports, and related activities assigned in other Sections.

3.2 CONSTRUCTION PHASE

3.2.1 Construction Phase ASHRAE Standard 90.1 Commissioning Compliance

3.2.1.1 General Requirements

Activities must include, but not be limited to, Construction Phase Commissioning Activities referred to as "Building Systems Verification and Testing" and "Building Commissioning" in ASHRAE 90.1 - IP as modified in this Section, Construction Phase Commissioning Activities listed in references contained in ASHRAE 90.1 - IP as modified in this Section, and Construction Phase Commissioning Activities specified in this Section.

3.2.1.2 Lead Commissioning Specialist and Commissioning Specialists Responsibilities

The Lead Commissioning Specialist and Commissioning Specialists must, at a minimum, perform all activities specified under item entitled "Construction Phase ASHRAE Standard 90.1 Commissioning Compliance."

3.2.2 Construction Phase Sequencing

3.2.2.1 Heating, Ventilation, and Air-Conditioning (HVAC) Systems

Heating, ventilation, and air-conditioning (HVAC) systems activities specified in the following paragraphs must be performed as applicable prior to performing Functional Performance Testing of systems listed under item entitled "Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED.

a. All equipment, systems, and assemblies have been completed, cleaned, flushed, disinfected, calibrated, tested, and operate in accordance with Plans and Specifications.

b. All Ductwork Air Leakage Testing (DALT) field work, Test and Balance (TAB) field work, Pre-Final TAB Reports, and related prerequisites have been completed, submitted, and approved in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

c. All inspections, measurements, and testing of building envelope and roofing systems specified in Sections 07 42 13 METAL WALL PANELS and 07 42 63 METAL WALL PANEL ASSEMBLIES have been completed, submitted, and approved.

d. All pressure testing, infrared thermography testing, and fog testing of building envelope and roofing systems specified in Section 01 91 19 BUILDING ENCLOSURE COMMISSIONING have been completed, submitted, and approved.

e. All Pre-Functional Checks have been completed, submitted, and approved.

f. Certified TAB Report specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC has been prepared, submitted, and approved.

g. Certified Certificate of Readiness has been completed, submitted, and approved.

All completed Final Functional Performance Test Checklists for commissioned equipment, systems, and assemblies listed under item entitled "Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED must be submitted and approved prior to the Government performing any Performance Verification Testing specified in
Section 23 09 00.00 22 INSTRUMENTATION AND CONTROL FOR HVAC.

3.2.2.2 Electrical Systems

Electrical systems activities specified in the following paragraphs must be performed as applicable prior to performing Functional Performance Testing of systems listed under item entitled "Electrical Type II Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED.

a. All electrical, power generation, and lighting equipment, systems, and assemblies have been completed, calibrated, tested, and operate in accordance with Plans and Specifications.

b. Building Envelope has been enclosed according to contract documents and final construction has been completed.

c. All ceiling tiles, floor coverings, and window coverings have been put in place.

d. All lamps have completed a minimum 100-hour burn-in period.

e. All furniture has been put in place.

f. Certified Certificate of Readiness has been completed, submitted, and approved.

3.2.3 Construction Phase Scheduling

3.2.3.1 General Requirements

Commissioning Activities and other related activities for inclusion into the Project Construction Schedule must include, but not be limited to, activities specified in the following paragraphs.

a. Perform Construction Phase Commissioning Coordination meeting.

b. Submit and approve completed Pre-Functional Checklists.

c. Submit and approve proposed Functional Performance Test Checklists.

d. Submit and approve final Functional Performance Test Checklists.

e. Perform Construction Phase Site Observations.

f. Perform inspections, measurements, and testing of building envelope and roofing systems specified in Sections 07 42 13 METAL WALL PANELS and 07 42 63 METAL WALL PANEL ASSEMBLIES.

g. Perform pressure testing, infrared thermography testing, and fog testing of building envelope and roofing systems specified in Section 01 91 19 BUILDING ENCLOSURE COMMISSIONING.

h. Submit and approve Certified TAB Report specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING HVAC.

i. Submit and approve Certified Certificate of Readiness.

j. Witness Construction Phase Functional Performance Testing.

k. Correct deficiencies discovered during Construction Phase Functional Performance Testing.

l. Re-witness Construction Phase Functional Performance Testing.

m. Government Witness of Performance Verification Testing.

n. Correct deficiencies discovered during Performance Verification Testing.

o. Government Re-witness of Performance Verification Testing.

p. Provide all Systems Manuals contents.

q. Submit and approve completed Systems Manuals.

r. Perform training.

s. Submit and approve Final Commissioning Report.

t. Perform Post-Construction Phase Site Observations.

u. Witness Post-Construction Phase Functional Performance Testing if all requirements and activities for respective system could not be fully or successfully completed during the Construction Phase.

v. Correct deficiencies discovered during Post-Construction Phase Functional Performance Testing.

w. Re-witness Post-Construction Phase Functional Performance Testing.

3.2.3.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must request a current Project Construction Schedule from Installing Contractors no later than at the Construction Phase Commissioning Coordination Meeting and must prepare and provide to Installing Contractors a list of Commissioning Activities for inclusion into the Project Construction Schedule. The Lead Commissioning Specialist must associate each Commissioning Activity to a single construction activity such that if the date of the construction activity changes, the associated Commissioning Activity changes accordingly. List of Commissioning Activities for inclusion into the Project Construction Schedule is not required to be submitted to the Government.

3.2.3.3 Installing Contractors Responsibilities

Installing Contractors must provide a current Project Construction Schedule to the Lead Commissioning Specialist upon receiving request. Installing Contractors must incorporate the list of Commissioning Activities into the Project Construction Schedule and must link each Commissioning Activity to the associated construction activity identified by the Lead Commissioning Specialist such that if the date of the construction activity changes, the associated Commissioning Activity changes accordingly.

Installing Contractors must provide updated Project Construction Schedules to the Lead Commissioning Specialist at least on a monthly basis.

3.2.4 Construction Phase Commissioning Coordination Meeting

3.2.4.1 General Requirements

A Construction Phase Commissioning Coordination meeting must be conducted no later than 90 calendar days after award of construction contract to discuss topics that include, but not be limited to, the Commissioning Process, contract requirements, lines of communication, roles and responsibilities, Commissioning Activities for inclusion into the Project Construction Schedule, documentation requirements, Inspections and Testing, and logistics specified in this Section. Meeting must be conducted by conference call.

3.2.4.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, conduct meeting and prepare meeting minutes complete with Attendance Roster.

3.2.4.3 Installing Contractors and Others Responsibilities

Installing Contractors must attend this meeting. The Contracting Officer's Technical Representative, a User Representative, and a Public Works Department Representative may attend this meeting.

3.2.5 Construction Phase Commissioning Progress Meetings

3.2.5.1 General Requirements

Construction Phase Commissioning Progress meetings must be conducted to discuss topics that include, but not be limited to, unresolved Commissioning Issues, status of items specified to be provided to the Lead Commissioning Specialist, and upcoming Commissioning Activities. Number of Construction Phase Commissioning Progress meetings must be conducted for each systems category as specified in the following paragraphs. Meetings must be conducted by conference call.

a. Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category: Six meetings.

b. Plumbing Systems Category: Two meetings.

c. Utility Systems Category: No meetings.

d. Electrical Type I Systems Category: No meetings.

e. Electrical Type II Systems Category: Two meetings.

f. Building Envelope Systems, Components, and Assemblies Category: Two meetings.

Construction Phase Commissioning Progress meetings are prohibited from being combined with conducting Construction Phase Commissioning Coordination meeting, conducting Site Observations, or witnessing

Functional Performance Testing.

3.2.5.2 Commissioning Specialists Responsibilities

Each Commissioning Specialist must, at a minimum, conduct meetings and prepare meeting minutes complete with Attendance Roster for systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for systems categories assigned to other Commissioning Specialists.

3.2.5.3 Installing Contractors and Others Responsibilities

Installing Contractors must attend these meetings. The Contracting Officer's Technical Representative, a User Representative, and a Public Works Department Representative may attend these meetings.

3.2.6 Construction Phase Interim Commissioning Plan

3.2.6.1 General Requirements

The Construction Phase Interim Commissioning Plan must be prepared and must be an update to the Design Phase Commissioning Plan and contents must include, but not be limited to, contents specified in the following paragraphs.

a. All contents specified for the Design Phase Commissioning Plan.

b. Basis of Design.

c. Contact information for the entire Commissioning Team.

3.2.6.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, prepare and submit the **Construction Phase Interim Commissioning Plan** no later than 30 calendar days after the Construction Phase Commissioning Coordination meeting and no later than 14 calendar days prior to performing construction of the building envelope.

3.2.7 Construction Phase Final Commissioning Plan

3.2.7.1 General Requirements

The Construction Phase Final Commissioning Plan must be prepared and must be an update to the Construction Phase Interim Commissioning Plan and contents must include, but not be limited to, contents specified in the following paragraphs.

a. All contents specified for the Construction Phase Interim Commissioning Plan.

b. All Pre-Functional Checklists.

c. All approved final Functional Performance Test Checklists.

3.2.7.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, perform activities specified in the following paragraphs.

a. Prepare and submit the **Construction Phase Final Commissioning Plan** no later than 30 calendar days prior to performing any Functional Performance Testing.

3.2.8 Construction Phase Issues And Resolution Log

3.2.8.1 General Requirements

The Construction Phase Issues and Resolution Log must be prepared and maintained to track all Commissioning Issues identified by any member of the Commissioning Team. The Construction Phase Issues and Resolution Log must be prepared in an electronic format that can be used throughout the Construction Phase such that it can be provided to the Government upon request. Construction Phase Issues and Resolution Log contents must include, but not be limited to, contents specified in the following paragraphs.

a. Project title and location.

b. Lead Commissioning Specialist's contact information.

c. Issue number.

d. Date issue was discovered.

e. Name of Commissioning Team member identifying the issue.

f. Commissioning Team member responsible for resolving the issue.

g. Current status.

h. Issue description.

i. Pictures of item if available and appropriate.

j. Effects of issue on project or building operation.

k. Possible cause of issue.

l. Proposed resolution.

m. Final resolution and final resolution date.

n. Approvals of issue resolution, including approver's name.

3.2.8.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must prepare and maintain the **Construction Phase Issues And Resolution Log** and must submit one electronic copy no later than seven calendar days after the addition of an issue or a change in status of an issue.

3.2.9 Pre-Functional Checklists

3.2.9.1 General Requirements

Pre-Functional Checklists must be prepared for all commissioned equipment, systems, and assemblies that are subject to functional performance testing and testing required according to UFGS 01 91 19 BUILDING ENCLOSURE COMMISSIONING. Pre-Functional Checklist contents must include, but not be limited to, contents for physical inspections that demonstrate construction, installation and start-up of equipment, systems, and assemblies is complete. Refer to item entitled "Pre-Functional Checklists" in item entitled "Inspections and Testing" for more information.

Download example Pre-Functional Checklists for Section 01 91 00.15 TOTAL BUILDING COMMISSIONING at the following location: http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables. Checklists must contain the same level of detail shown in examples, but are not required to match the format of examples.

3.2.9.2 Lead Commissioning Specialist Responsibilities

Lead Commissioning Specialist must provide blank Pre-Functional Checklists to Installing Contractors. Blank Pre-Functional Checklists are not required to be submitted to the Government.

3.2.9.3 Commissioning Specialists Responsibilities

Commissioning Specialists must prepare Pre-Functional Checklists for all equipment, systems, and assemblies listed under systems categories assigned to them in this Section that are subject to functional performance testing and testing required according to UFGS 01 91 19 BUILDING ENCLOSURE COMMISSIONING. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.2.10 Functional Performance Test (FPT) Checklists

Functional Performance Test (FPT) Checklists must be prepared for all commissioned equipment, systems, and assemblies demonstrating performance is in accordance with Plans and Specifications as specified in the following paragraphs. A single final FPT Checklist must be prepared for each equipment piece, system, and assembly except that a single final FPT Checklist may be prepared for use with those that have identical features as modified in the following paragraphs. FPT Checklist contents must include, but not be limited to, step-by-step actions, expected results, and confirmation of compliance with required provisions of ASHRAE 90.1 - IP as modified in the following paragraphs. Refer to item entitled "Construction Phase Functional Performance Testing" for more information.

3.2.10.1 Complex Systems Requirements

FPT Checklists must be prepared for all complex systems as defined in paragraph COMPLEX SYSTEMS. A single FPT Checklist must be prepared for each system except that a single FPT Checklist may be prepared for use with multiple systems that have identical control sequences such as unitary air-conditioners, packaged terminal air-conditioners, and heat pumps.

Contents must include, but not be limited to, contents specified in the following paragraphs and shown in example FPT Checklists for Section 01 91 00.15 TOTAL BUILDING COMMISSIONING available for download at the following location: http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables. FPT Checklists must contain the same level of detail specified below and as shown in examples, but are not required to match the format of examples.

a. Notable system features including information about such attributes as system sizing and controls to facilitate understanding of system operation.

b. Conclusions and recommendations based on control system feature, point-to-point, actuator, and system operation observations. Conclusions must clearly indicate if system does or does not perform in accordance with requirements of Plans and Specifications. Recommendation must clearly indicate that the system should or should not be accepted by the Government.

c. Test conditions including date, beginning and ending time, and beginning and ending outdoor air conditions.

d. Identification of the equipment involved in the test.

e. Attendees present throughout the entire system test.

f. Control system feature identification including control point description, embedded/visible type, adjustable/monitoring type, actual value, and setpoint value/alarm range.

g. Point-to-point observations including demonstrating system flow meters and sensors have been calibrated and are correctly displayed on the Operator workstation.

h. Actuator operation observations demonstrating actuator responses to commands from the control system.

i. Variable frequency drive observations demonstrating minimum safe motor speed.

j. System operation observations for system-based testing demonstrating each control sequence, operation mode, and alarm condition resulting from control setpoint manipulation. System operation observations must include, but not be limited to, observations specified in the following paragraphs.

    (1) Introduction identifying testing methodology.

    (2) As-found conditions prior to system manipulation.

    (3) Clear list of test items (step numbers).

    (4) Design control sequences or interlocks segmented by unique functions.

    (5) Intended test procedures following each segmented sequence or interlock identifying the system manipulation required to initiate system response.

    (6) Expected system responses.

    (7) Space for comments for each test item.

(8) Pass or fail indication for each test.

Performance Verification Tests prepared by the Controls Contractor as specified in Section 23 09 00.00 22 INSTRUMENTATION AND CONTROL FOR HVAC are prohibited from fulfilling requirements of this Section.

3.2.10.2 Plumbing Systems Requirements

FPT Checklists must be prepared for all equipment, systems, and assemblies listed under item entitled "Plumbing Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED.

3.2.10.3 Electrical Type I Systems Requirements

FPT Checklists must not be prepared for equipment, systems, and assemblies listed under item entitled "Electrical Type I Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED.

3.2.10.4 Electrical Type II Systems Requirements

FPT Checklists must be prepared for all equipment, systems, and assemblies listed under item entitled "Electrical Type II Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED. FPT Checklists must be system-based and demonstrate end-objective performance in accordance with Plans and Specifications. FPT Checklists must not repeat fundamental performance testing such as resistance measurements required in other Sections.

3.2.10.5 Building Envelope Systems, Components, and Assemblies Requirements

FPT Checklists must not be prepared for equipment, systems, and assemblies listed under item entitled "Building Envelope Systems, Components, and Assemblies Category" in paragraph SYSTEMS TO BE COMMISSIONED.

3.2.10.6 Proposed FPT Checklist Meeting and Draft FPT Checklist Meeting

For all complex systems as defined in paragraph COMPLEX SYSTEMS, a Proposed FPT Checklist Meeting and a Draft FPT Checklist Meeting must be conducted for detailed discussions of procedures intended to demonstrate performance is in accordance with Plans and Specifications. Meetings must be conducted by conference call.

3.2.10.7 Lead Commissioning Specialist and Commissioning Specialists Responsibilities

For all equipment, systems, and assemblies other than complex systems as defined in paragraph COMPLEX SYSTEMS, the Lead and Commissioning Specialist must perform respective responsibilities specified in the following paragraphs.

a. The Lead Commissioning Specialist must submit **Final Functional Performance Test Checklists** for each systems category listed under paragraph SYSTEMS TO BE COMMISSIONED for which FPT checklists are required to be prepared. FPT checklists must be submitted as a single submittal for each systems category no later than 30 calendar days prior to testing of any system in the respective systems category.

b. Commissioning Specialists must prepare Final Functional Performance Test Checklists for equipment, systems, and assemblies listed under systems categories assigned to them in this Section for which FPT checklists are required to be prepared. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

For all complex systems as defined in paragraph COMPLEX SYSTEMS, the Lead Commissioning Specialist and Commissioning Specialists must prepare and submit Final Functional Performance Test Checklists in the sequence specified in the following paragraphs. Commissioning Specialists must perform respective responsibilities for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing respective responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

a. Commissioning Specialists must prepare Proposed Functional Performance Test Checklists.

b. The Lead Commissioning Specialist must submit **Proposed Functional Performance Test Checklists** for all complex systems as a single submittal no later than 60 calendar days prior to testing of any complex system.

c. Commissioning Specialists must conduct the Proposed Functional Performance Test Checklist Meeting attended by the Contracting Officer's Technical Representative.

d. Commissioning Specialists must prepare Draft Functional Performance Test Checklists to incorporate input from the Contracting Officer's Technical Representative.

e. The Lead Commissioning Specialist must provide Draft Functional Performance Test Checklists to Installing Contractors and Equipment Suppliers no later than seven calendar days prior to the Draft Functional Performance Test Checklist Meeting.

f. Commissioning Specialists must conduct the Draft Functional Performance Test Checklist Meeting attended by Installing Contractors and Equipment Suppliers. The Contracting Officer's Technical Representative may attend this meeting.

g. Commissioning Specialists must prepare Final Functional Performance Test Checklists to incorporate input from Installing Contractors.

h. The Lead Commissioning Specialist must submit Final Functional Performance Test Checklists for all complex systems as a single submittal no later than seven calendar days after meeting to discuss Draft Functional Performance Test Checklists and no later than 30 calendar days prior to testing of any complex system.

3.2.10.8 Installing Contractors and Equipment Suppliers Responsibilities

Installing Contractors and Equipment Suppliers must attend Draft Functional Performance Test Checklist Meeting.

3.2.11 Construction Submittals

3.2.11.1 General Requirements

Construction Submittals must be reviewed for all commissioned equipment, systems, and assemblies to the extent necessary to verify that equipment, systems, and assemblies are in compliance with Commissioning Aspects of Plans and Specifications as modified in this Section. One review for each commissioned equipment, system, and assembly must be performed coincident with Designers of Record and review must include, but not be limited to, review of shop drawings and equipment performance data. Review of resubmittals is prohibited.

3.2.11.2 Construction Submittal Review Reports

A Construction Submittal Review Report must be prepared for all commissioned equipment, systems, and assemblies and contents must include, but not be limited to, contents specified in the following paragraphs.

a. Project title, project number, and date of review.

b. Overview complete goals of the Commissioning-focused submittal review process.

c. Summary of deficiencies.

d. Ongoing record of all Construction Submittals received and reviewed, list of unresolved deficiencies, and record of communications made to resolve deficiencies.

e. Descriptive outline of reviewed construction submittals complete with the Installing Contractor's submittal designation.

f. Clear description of deficiencies complete with supporting documentation.

3.2.11.3 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist activities must include, but not be limited to, activities specified in the following paragraphs.

a. Provide list of shop drawings and equipment performance data that must be reviewed to Installing Contractors.

b. Prepare and provide Construction Submittal review comments to Designers of Record.

c. Prepare and submit **Construction Submittal Review Reports** no later than seven calendar days after completion of each Construction Submittal review.

3.2.11.4 Commissioning Specialists Responsibilities

Commissioning Specialists must perform reviews of Construction Submittals for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.2.11.5 Designers of Record Responsibilities

Designers of Record activities must include, but not be limited to, receiving and considering for incorporation comments listed in Construction Submittal Review Reports provided by the Lead Commissioning Specialist.

3.2.11.6 Installing Contractors Responsibilities

Installing Contractors must provide all requested Construction Submittals and requested supplemental information to the Lead Commissioning Specialist.

3.2.12 Construction Phase Site Observations

3.2.12.1 General Requirements

Construction Phase Site Observations must be performed as specified in the following paragraphs to verify and document Commissioning Aspects of equipment, systems, and assemblies as specified in this Section. Two eight-hour Construction Phase Site Observations must be conducted for each systems category that is not specified in the following paragraphs. Specified number of observations are totals that include all commissioned buildings and all construction phases in this project.

Construction Phase Site Observations are prohibited from being combined with witnessing Functional Performance Testing, but may be combined with Pre-Functional Checklist Verification unless indicated otherwise. Purposes of Site Observations specified in the following paragraphs for each systems category may be combined into a single Site Observation.

3.2.12.2 Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category

Site Observations must be performed for the purposes specified in the following paragraphs.

a. Two eight-hour Site Observations to identify issues that may result in unsuccessful functional performance testing and to verify and document Commissioning Aspects of commissioned equipment, systems, and assemblies listed under this systems category.

b. One eight-hour Site Observation to verify and document HVAC pipe pressure testing.

c. One eight-hour Site Observation to verify and document HVAC pump installation.

d. One eight-hour Site Observation to verify and document compliance with provisions of ASHRAE 90.1 - IP.

3.2.12.3 Plumbing Systems Category

Site Observations must be performed for the purposes specified in the following paragraphs.

a. One eight-hour Site Observation to identify issues that may result in unsuccessful functional performance testing and to verify and document Commissioning Aspects of commissioned equipment, systems, and assemblies listed under this systems category.

b. One eight-hour Site Observation to verify and document Plumbing pipe pressure testing.

c. One eight-hour Site Observation to verify and document Plumbing pump installation and compliance with provisions of ASHRAE 90.1 - IP.

3.2.12.4 Utility Systems Category

Site Observations must be performed for the purposes specified in the following paragraphs.

a. No Site Observation dedicated to this systems category. Verification and documentation of Commissioning Aspects of commissioned equipment, systems, and assemblies listed under this systems category may be combined with Site Observations for other systems categories.

b. No Site Observation dedicated to this systems category to verify and document compliance with provisions of ASHRAE 90.1 - IP. Verification and documentation of compliance with provisions of ASHRAE 90.1 - IP of commissioned equipment, systems, and assemblies listed under this systems category may be combined with Site Observations for other systems categories.

3.2.12.5 Electrical Type I Systems Category

Site Observations must be performed for the purposes specified in the following paragraphs.

a. Two eight-hour Site Observations to verify and document Commissioning Aspects and to witness testing performed by independent testing organizations of commissioned equipment, systems, and assemblies listed under this systems category and listed by the Lead Commissioning Specialist to require on-site verification and documentation by Commissioning Specialists. Independent testing organizations typically use tests readily available from testing associations such as the InterNational Electrical Testing Association (NETA).

b. No Site Observation dedicated to this systems category to verify and document compliance with provisions of ASHRAE 90.1 - IP. Verification and documentation of compliance with provisions of ASHRAE 90.1 - IP of commissioned equipment, systems, and assemblies listed under this systems category may be combined with Site Observations for other systems categories.

3.2.12.6 Electrical Type II Systems Category

Site Observations must be performed for the purposes specified in the following paragraphs.

a. Two eight-hour Site Observations to identify issues that may result in unsuccessful functional performance testing and to verify and document Commissioning Aspects of commissioned equipment, systems, and assemblies listed under this systems category.

b. One eight-hour Site Observation to verify and document compliance with provisions of ASHRAE 90.1 - IP.

3.2.12.7 Building Envelope Systems, Components, and Assemblies Category

Site Observations must be performed for the purposes specified in the following paragraphs.

a. One eight-hour Site Observation to verify, document, and witness inspections, measurements, and testing of building envelope and roofing systems performed by independent testing agencies and specified in Sections 07 42 13 METAL WALL PANELS and 07 42 63 METAL WALL PANEL ASSEMBLIES.

b. One eight-hour Site Observation to verify, document, and witness pressure testing, infrared thermography testing, and fog testing of building envelope and roofing systems performed by independent testing agencies and specified in Section 01 91 19 BUILDING ENCLOSURE COMMISSIONING.

c. Two eight-hour Site Observation to verify and document that continuous air barrier materials and assemblies are in compliance with requirements of ASHRAE 90.1 - IP while the continuous air barrier is still accessible for inspection and repair.

3.2.12.8 Summary Meetings

A Summary Meeting must be conducted at the conclusion of each Site Observation for the purpose of discussing issues discovered during each Site Observation. Meeting must be conducted on-site with all attendees present.

3.2.12.9 Site Observation Reports

A Site Observation Report must be prepared for each Site Observation for the purpose of documenting issues discovered during each Site Observation.

Site Observation Reports for all equipment, systems, and assemblies listed under item entitled "Electrical Type I Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED must include results of testing performed by independent testing organizations that were witnessed by Commissioning Specialists and test results that were not witnessed by Commissioning Specialists.

3.2.12.10 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must provide list of equipment, systems, and assemblies listed under item entitled "Electrical Type I Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED that require on-site verification and documentation by Commissioning Specialists to Installing Contractors.

The Lead Commissioning Specialist must submit **Construction Phase Site Observation Reports** complete with Summary Meeting Attendance Roster no later than seven calendar days after Site Observation.

3.2.12.11 Commissioning Specialists Responsibilities

Commissioning Specialist must perform responsibilities specified in the following paragraphs.

a. Each Commissioning Specialist must be responsible to perform Site Observations, conduct Summary Meetings, and prepare Construction Phase Site Observation Reports for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.2.12.12 Installing Contractors Responsibilities

Installing Contractors must provide Commissioning Specialists access to construction and systems for Site Observations and must attend Summary Meetings.

Installing Contractors must schedule testing to occur in the presence of Commissioning Specialists as specified in the following paragraphs.

a. All equipment, systems, and assemblies listed under item entitled "Electrical Type I Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED and listed by the Lead Commissioning Specialist to require on-site verification and documentation by Commissioning Specialists. Results of testing of equipment, systems, and assemblies listed under item entitled "Electrical Type I Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED must be provided to the Lead Commissioning Specialist that were witnessed by Commissioning Specialists and that were not witnessed by Commissioning Specialists.

b. Pressure testing, infrared thermography testing, and fog testing specified in Section 01 91 19 BUILDING ENCLOSURE COMMISSIONING and inspections, measurements, and testing specified in Sections 07 42 13 METAL WALL PANELS and 07 42 63 METAL WALL PANEL ASSEMBLIES for equipment, systems, and assemblies listed under item entitled "Building Envelope Systems, Components, and Assemblies Category" in paragraph SYSTEMS TO BE COMMISSIONED. Results of all tests, inspections, and measurements must be provided to the Lead Commissioning Specialist that were witnessed by Commissioning Specialists and that were not witnessed by Commissioning Specialists.

3.2.13 Pre-Functional Checks

Pre-Functional Checks must be performed prior to performing Functional Performance Testing of respective system and activities must include, but not be limited to, activities specified in the following paragraphs.

a. Perform Pre-Functional Checklist Verification.

3.2.13.1 Pre-Functional Checklist Verification

3.2.13.1.1 General Requirements

Pre-Functional Checklist Verification must be performed using Pre-Functional Checklists from the approved Construction Phase Final Commissioning Plan by conducting on-site spot-checks to verify accuracy after receipt of completed Pre-Functional Checklists from Installing Contractors. Verification may be combined with Construction Phase Site Observations.

3.2.13.1.2 Number of Trips

The number of trips to the site for the purpose of Pre-Functional Checklist Verification for each systems category must be up to that specified in the following paragraphs. It is encouraged that these trips be combined with Construction Phase Site Observations.

a. Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category: Two eight-hour trips.

b. Plumbing Systems Category: Two eight-hour trips.

c. Utility Systems Category: No trip dedicated to this systems category. Verification of commissioned equipment, systems, and assemblies listed under this systems category may be combined with trips for other systems categories.

d. Electrical Type I Systems Category: One eight-hour trip.

e. Electrical Type II Systems Category: Two eight-hour trips.

f. Building Envelope Systems, Components, and Assemblies Category: One eight-hour trip.

3.2.13.1.3 Sample Strategy

20-Percent sample spot-check for all equipment, systems, and assemblies for which Pre-Functional Checklists have been completed. Sample spot-check must include at least one complete item such as one complete unitary air-conditioner.

3.2.13.1.4 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must provide blank Pre-Functional Checklists to Installing Contractors.

The Lead Commissioning Specialist must submit all **completed Pre-Functional Checklists** and all associated equipment start-up reports for each systems category listed under paragraph SYSTEMS TO BE COMMISSIONED as a single submittal no later than seven calendar days after receipt of all completed checklists for equipment, systems, and assemblies listed under the respective systems category from Installing Contractors.

3.2.13.1.5 Commissioning Specialists Responsibilities

Commissioning Specialists must perform Pre-Functional Checklist Verifications and then must indicate acceptance of all Pre-Functional Checklist items for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.2.13.1.6 Installing Contractors Responsibilities

Installing Contractors must complete one Pre-Functional Checklist for each piece of equipment and system. Initial each Pre-Functional Checklist item for their area of responsibility indicating the item's construction, installation, and start-up as applicable has been verified to be complete and is ready for testing.

Installing Contractors must provide completed and initialed Pre-Functional Checklists and all associated equipment start-up reports to the Lead Commissioning Specialist as a single package for each systems category no later than seven calendar days after completion of inspection of all checklist items for each piece of equipment and system.

3.2.14 Certified Certificates of Readiness and Certificates of Readiness

3.2.14.1 General Requirements

A single Certified Certificate of Readiness must be prepared for each equipment, system, and assembly subject to Functional Performance Testing and testing required according to UFGS 01 91 19 BUILDING ENCLOSURE COMMISSIONING by reviewing, signing, and dating Certificates of Readiness to indicate each equipment, system, and assembly is complete and ready for testing. Testing of any equipment, system, or assembly is prohibited until after its Certified Certificate of Readiness has been submitted and approved. Government approval of Certified Certificates of Readiness does not constitute the Government's responsibility for equipment, systems, or assemblies being complete and ready for testing.

A single Certificate of Readiness must be prepared for each equipment, system, and assembly subject to Functional Performance Testing. Certificate of Readiness contents must include, but not be limited to, contents specified in the following paragraphs to the extent applicable to the specific system being tested.

a. Cover sheet with spaces for signatures and dates by the Lead Commissioning Specialist and by Installing Contractors to indicate each system is complete and ready for testing.

b. Equipment and system start-up reports.

c. Control system start-up reports.

d. Diagnostic test reports.

e. Completed Pre-Functional Checklists.

3.2.14.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must sign, date, and submit **Certified Certificates of Readiness** for all equipment, systems, and assemblies listed under each systems category listed under paragraph SYSTEMS TO BE COMMISSIONED as a single submittal no later than 14 calendar days prior to testing of any equipment, system, or assembly listed under respective systems category.

3.2.14.3 Commissioning Specialists Responsibilities

Commissioning Specialists must perform reviews of Certificates of Readiness for accuracy to ensure equipment, systems, and assemblies are complete and ready for testing for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.2.14.4 Installing Contractors Responsibilities

Installing Contractors must prepare, sign, and date Certificates of Readiness indicating equipment, systems, and assemblies are complete and ready for testing. Certificates of Readiness for all equipment, systems, and assemblies listed under each systems category listed under paragraph SYSTEMS TO BE COMMISSIONED must be provided to the Lead Commissioning Specialist as a single package for each systems category.

3.2.15 Construction Phase Functional Performance Testing

3.2.15.1 General Requirements

Construction Phase Functional Performance Testing must be performed for all equipment, systems, and assemblies for which a Functional Performance Test (FPT) Checklist has been prepared. Testing must be performed using approved checklists and all required Commissioning Team participants must be present prior to start of testing. Testing must be aborted and rescheduled to occur at a later date if any Commissioning Team participant is not present or if deficiencies discovered during testing prevent successful completion. Testing of any equipment, system, and assembly is prohibited until after its Certified Certificate of Readiness has been submitted and approved.

All necessary materials and system modifications to simulate actual conditions as close as is practically possible must be provided. Complete all testing by returning affected equipment, systems, and assemblies to pre-test conditions.

3.2.15.2 Complex Systems Requirements

Testing of complex systems as defined in paragraph COMPLEX SYSTEMS in paragraph SYSTEMS TO BE COMMISSIONED must be performed as specified in the following paragraphs.

3.2.15.2.1 Equipment Featuring On-Board Factory Packaged Controls

Testing of equipment featuring on-board factory packaged controls must include, but not be limited to, testing specified in the following paragraphs.

a. Equipment enabling and disabling.

b. Equipment standard and optional control points necessary to accomplish functionality whether or not control points are specified in Plans and Specifications.

c. Equipment standard and optional alarms critical to safe operation whether or not optional alarms are specified in Plans and Specifications.

d. All control points added by Controls Contractor in addition to on-board factory packaged controls whether or not control points are specified in Plans and Specifications.

3.2.15.2.2 Test Setup and Condition Simulation

Test setup must include all necessary materials and system modifications to simulate flows, pressures, temperatures, and other conditions necessary to perform testing according to specified conditions must be provided.

Condition simulation must be accomplished by temporarily changing control system setpoints. Condition simulation by over-writing control system inputs (actual) values is prohibited unless approved in advance of test by the Contracting Officer's Technical Representative.

3.2.15.2.3 Condition Simulation Methods

Condition simulation methods for equipment, systems, and assemblies listed under item entitled "Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED that are acceptable are specified in the following paragraphs.

a. When varying static pressures inside ductwork cannot be simulated within the duct, and where a sensor signals the controls system to initiate sequences at various duct static pressures, it is acceptable to simulate the various pressures with a Pneumatic Squeeze-Bulb Type Signaling Device with gauge temporarily attached to the sensing tube leading to the transmitter. It is not acceptable to reset the various setpoints, nor to simulate an electric analog signal unless approved as specified in the preceding paragraphs.

b. Dirty filter pressure drops can be simulated using sheets of cardboard at filter face.

c. Freeze-stat safeties can be simulated by packing portion of sensor with ice.

d. High outside air temperatures can be simulated with a hair blower.

e. High entering cooling coil temperatures can be used to simulate entering cooling coil conditions.

f. Do not use signal generators to simulate sensor signals unless approved by the Contracting Officer's Technical Representative, as noted above, for special cases.

g. Control setpoints can be altered. For example, to see the air conditioning compressor lockout work at an outside air temperature below lockout setpoint, when the outside air temperature is above lockout setpoint, temporarily change the lockout setpoint to be minus 0 degrees above the current outside air temperature. Caution: Setpoints are not to be raised or lowered to a point such that damage to the components, systems, or the building structure and/or contents will occur.

h. Test duct mounted smoke detectors in accordance with the manufacturer's recommendations. Perform testing with air system at minimum airflow condition in ductwork.

i. Test current sensing relays used for fan and pump status signals to control system to indicate unit failure and run status by resetting the setpoint on the relay to simulate a lost belt or unit failure while the unit is running. Confirm that the failure alarm was generated and received at the control system. After the test is performed, return the setpoint to its original setpoint or a setpoint as indicated by the Contracting Officer's Technical Representative.

3.2.15.3 Sample Strategy

3.2.15.3.1 General Requirements

100-Percent testing of all equipment, systems, and assemblies as modified in the following paragraphs. Sample strategy for testing less than 100-percent testing must be equally applied to each commissioned building in this project.

Sample strategy testing must be applied to total equipment and system counts specified for system categories under paragraph SYSTEMS TO BE COMMISSIONED.

3.2.15.3.2 Complex Systems

Sample strategy for all complex systems as defined in paragraph COMPLEX SYSTEMS must be as specified in the following paragraphs.

a. 100-Percent testing for equipment, systems, and assemblies such as chilled water systems, HVAC heating hot water systems, renewable energy systems, building automation systems, energy management and control systems, aircraft pre-conditioned air systems, air handling unit systems including all associated fans except for remote exhaust air fans, dedicated outside air systems including all associated fans except for remote exhaust air fans, computer room air-conditioning units, whole-building energy monitoring, and whole-building water utility monitoring.

b. 20-Percent sample testing for equipment, systems, and assemblies with identical control sequences such as heat pumps, split system air-conditioning units, and packaged terminal air-conditioner units. Sample must include at least one complete item such as one complete heat pump.

3.2.15.3.3 Plumbing Systems

Sample strategy for all equipment, systems, and assemblies listed under item entitled "Plumbing Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED except plumbing systems defined as complex systems in paragraph COMPLEX SYSTEMS must be as specified in the following paragraphs.

a. 100-Percent testing for equipment, systems, and assemblies such as domestic heating hot water systems, domestic solar heating hot water systems, service water pressure-booster pump systems, process water systems, natural gas systems, and irrigation systems.

b. 20-Percent sample testing for equipment, systems, and assemblies with identical features such as fixtures, semi-instantaneous water heaters, emergency eyewash stations, and vacuum systems. Sample must include at least one complete item such as one complete fixture.

3.2.15.3.4 Electrical Type II Systems

Sample strategy for all equipment, systems, and assemblies listed under item entitled "Electrical Type II Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED except electrical systems defined as complex systems in paragraph COMPLEX SYSTEMS must be as specified in the following paragraphs.

a. 100-Percent testing for equipment, systems, and assemblies such as frequency converters, emergency power off control systems, emergency response systems, electrical utility metering systems, and photovoltaic systems.

b. 20-Percent sample testing for equipment, systems, and assemblies with identical features such as audiovisual systems, automatic receptacles controls, communications systems, energy monitoring, lighting systems, lighting control devices, lighting control systems, and security systems. Sample must include at least one complete item such as lighting in one space.

3.2.15.4 Unsuccessful Testing and Re-Testing

3.2.15.4.1 Unsuccessful Testing of 100-Percent Sample

Testing must be failed for equipment or systems for which 100-percent sample size are tested if one or more of the test procedures results in discovery of a deficiency that cannot be resolved within five minutes during testing.

Re-testing must be witnessed to the extent necessary to confirm that deficiencies have been corrected without negatively impacting the performance of the rest of the system.

3.2.15.4.2 Unsuccessful Testing of Less than 100-Percent Sample

Testing must be failed for equipment or systems for which less than 100-percent sample size are tested if one or more of the test procedures results in discovery of a deficiency, regardless of whether the deficiency is corrected during the sample test.

a. Re-testing must be witnessed of only items which experienced initial failures if the failure rate is 5-percent or less, meaning that 5-percent or less of the equipment or systems had at least one deficiency.

b. Re-testing must be witnessed of items which experienced initial failures to the extent necessary to confirm that the deficiencies have been corrected if the failure rate is higher than 5-percent, meaning that more than 5-percent of equipment or systems tested had at least one deficiency. In addition, testing of another random sample must be witnessed for the same size as the initial sample for the first time. If the second random sample set has any failures, those failed items must be retested and all remaining equipment, systems, and assemblies must be tested to complete 100-percent testing of that system type.

3.2.15.5 Re-Testing Costs

Re-testing costs may be withheld by the Contracting Officer from payments to Installing Contractors. These costs may include salary, travel costs, and per diem associated with re-testing for the Lead Commissioning Specialist, Commissioning Specialists, and Government personnel.

3.2.15.6 Commissioning Team Participants

Participants of Functional Performance Testing must include, but not be limited to, participants specified in the following paragraphs. The Contracting Officer's Technical Representative, Public Works Department

Division (PWD) Representatives, and Using Agent's Representatives (Users) may, but are not required to, participate in any testing.

3.2.15.6.1 Heating, Ventilation, and Air-Conditioning (HVAC) Systems

Commissioning Team members participating in Functional Performance Testing of equipment, systems, and assemblies listed under item entitled "Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED must include, but not be limited to, participants specified in the following paragraphs.

a. Mechanical Commissioning Specialist (CxM).

b. Installing Contractors’ Quality Control Personnel (CQC).

c. Installing Contractors’ Controls Commissioning Representative (CC).

d. Installing Contractors’ TAB Commissioning Representative (TAB).

e. Installing Contractors’ Mechanical Commissioning Representative (MC).

f. Equipment Supplier (ES).

3.2.15.6.2 Plumbing Systems

Commissioning Team members participating in Functional Performance Testing of equipment, systems, and assemblies listed under item entitled "Plumbing Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED must include, but not be limited to, participants specified in the following paragraphs.

a. Mechanical Commissioning Specialist (CxM).

b. Installing Contractors’ Quality Control Personnel (CQC).

c. Installing Contractors’ Controls Commissioning Representative (CC).

d. Installing Contractors’ Plumbing Commissioning Representative (PC).

e. Equipment Supplier (ES).

3.2.15.6.3 Utility Systems

Commissioning Team members participating in Functional Performance Testing of equipment, systems, and assemblies listed under item entitled "Utility Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED must include, but not be limited to, participants specified in the following paragraphs.

a. Mechanical Commissioning Specialist (CxM).

b. Installing Contractors’ Quality Control Personnel (CQC).

c. Installing Contractors’ Controls Commissioning Representative (CC).

d. Installing Contractors’ TAB Commissioning Representative (TAB).

e. Installing Contractors’ Mechanical Commissioning Representative (MC).

f. Installing Contractors’ Plumbing Commissioning Representative (PC).

g. Installing Contractors’ Electrical Commissioning Representative (EC).

h. Equipment Supplier (ES).

3.2.15.6.4 Electrical Type II Systems

Commissioning Team members participating in Functional Performance Testing of equipment, systems, and assemblies listed under item entitled "Electrical Type II Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED must include, but not be limited to, participants specified in the following paragraphs.

a. Electrical Commissioning Specialist (CxE).

b. Installing Contractors’ Quality Control Personnel (CQC).

c. Installing Contractors’ Electrical Commissioning Representative (EC).

d. Equipment Supplier (ES).

3.2.15.7 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, perform activities specified in the following paragraphs.

a. Contact Contracting Officer's Technical Representative no later than 30 calendar days prior to performing Testing to select which systems must be tested for samples of less than 100-percent.

b. Coordinate logistics with Installing Contractors.

c. Submit **completed Final Functional Performance Test Checklists** for each systems category listed under paragraph SYSTEMS TO BE COMMISSIONED as a single submittal no later than 14 calendar days after successful testing of all systems listed under the respective systems category.

3.2.15.8 Commissioning Specialists Responsibilities

Commissioning Specialists must witness, re-witness, and document all testing for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.2.15.9 Installing Contractors Responsibilities

Installing Contractors must provide all necessary materials and perform all aspects of testing including modifications to simulate actual conditions for all equipment, systems, and assemblies related to their contract.

3.2.15.10 Equipment Suppliers Responsibilities

Equipment Suppliers must perform all aspects of testing of equipment featuring on-board factory packaged controls provided as part of their contract. Refer to paragraph COMPLEX SYSTEMS for definition of equipment featuring on-board factory packaged controls.

3.2.15.11 Installing Contractors and Equipment Supplier Coordination

Installing Contractors and Equipment Suppliers must simultaneously perform all aspects of testing of equipment featuring on-board factory packaged controls provided as part of their contract.

3.2.16 Training

3.2.16.1 Training Plans

Training Plans must be prepared for all commissioned equipment, systems, and assemblies and contents must include, but not be limited to, contents specified in the following paragraphs.

a. List of those who should receive training by position or name.

b. List of systems that require training organized by Specification Section.

c. Outline of training topics that addresses design, construction, operation, and maintenance.

d. Learning objectives and training delivery methods.

e. Planned location of the training sessions (classroom, on-site, and off-site).

f. Minimum duration of each training session.

g. Training materials requirements to be used during the instructional process.

h. Training report, records, and recording requirements.

i. Instructor qualifications.

j. Instructor name and contact information.

k. Criteria for determining whether provided training is acceptable.

l. Blank Training Attendance Rosters.

3.2.16.2 Training

Training must be performed in accordance with Training Plans and in accordance with requirements specified in other Sections.

3.2.16.3 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, provide all review comments to the Installing Contractors for all equipment, systems, and assemblies listed under each systems category as a single package no later than 45 calendar days prior to first training event for any equipment, system, or assembly in respective systems category. Review comments are not required to be submitted to the Government.

3.2.16.4 Commissioning Specialist Responsibilities

Commissioning Specialists must, at a minimum, review and prepare review comments for all Training Plans. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.2.16.5 Installing Contractors Responsibilities

Installing Contractors must, at a minimum, perform activities specified in the following paragraphs.

a. Prepare and provide all Training Plans to the Lead Commissioning Specialist for all equipment, systems, and assemblies listed under each systems category as a single package no later than 60 calendar days prior to first training event for any equipment, system, or assembly in respective systems category.

b. Incorporate all review comments.

c. Perform training as specified in the preceding paragraphs.

d. Require all training attendees to sign Training Attendance Rosters.

e. Provide all Approved Training Plans and Completed Training Attendance Rosters to the Lead Commissioning Specialist for all equipment, systems, and assemblies listed under each systems category as a single package no later than 30 calendar days after the last training event for all equipment, systems, and assemblies in respective systems category.

3.2.17 Final Commissioning Report

3.2.17.1 General Requirements

The Final Commissioning Report must be prepared and contents must include, but not be limited to, contents specified in the following paragraphs.

a. Executive summary describing the overall Commissioning Process.

b. Results of the Commissioning Process.

c. Issues And Resolution Log.

d. Commissioning Design Review Reports.

e. Approved Construction Phase Final Commissioning Plan.

f. Approved Pre-Final Test and Balance Reports.

g. Completed Pre-Functional Checklists.

h. Completed Functional Performance Test Checklists.

i. Approved Training Plans.

j. Completed Training Attendance Rosters.

k. Indication that equipment, systems, and assemblies meet requirements of the Owner's Project Requirements, Plans and Specifications, and provisions of ASHRAE 90.1 - IP.

l. Indication of plan for completion of any deferred trainings.

m. Identification of all deficiencies complete with corrective actions taken.

n. Identification of all repairs, replacements, and calibrations.

o. Identification of unresolved deficiencies complete with resolution plan and responsible Commissioning Team member.

p. List and plan for completion of any testing that must be performed during the Post-Construction (Warranty) Phase.

3.2.17.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, perform activities specified in the following paragraphs.

a. Prepare and submit the **Final Commissioning Report** no later than 30 calendar days after Beneficial Occupancy.

3.3 POST-CONSTRUCTION (WARRANTY) PHASE

3.3.1 Post-Construction Phase ASHRAE Standard 90.1 Commissioning Compliance

3.3.1.1 General Requirements

Activities must include, but not be limited to, Post-Construction Phase Commissioning Activities referred to as "Building Systems Verification and Testing" and "Building Commissioning" in ASHRAE 90.1 - IP as modified in this Section, Post-Construction Phase Commissioning Activities listed in references contained in ASHRAE 90.1 - IP as modified in this Section, and Post-Construction Phase Commissioning Activities specified in this Section.

3.3.1.2 Lead Commissioning Specialist and Commissioning Specialists Responsibilities

The Lead Commissioning Specialist and Commissioning Specialists must, at a minimum, perform all activities specified under item entitled "Post-Construction Phase ASHRAE Standard 90.1 Commissioning Compliance."

3.3.2 Post-Construction Phase Issues And Resolution Log

3.3.2.1 General Requirements

The Post-Construction Phase Issues and Resolution Log must be prepared and maintained to track all Commissioning Issues identified by any member of the Commissioning Team. The Post-Construction Phase Issues and Resolution Log must be prepared in an electronic format that can be used throughout the Post-Construction Phase such that it can be provided to the Government upon request. Post-Construction Phase Issues and Resolution Log contents must include, but not be limited to, contents specified in the following paragraphs.

a. Project title and location.

b. Lead Commissioning Specialist's contact information.

c. Issue number.

d. Date issue was discovered.

e. Name of Commissioning Team member identifying the issue.

f. Commissioning Team member responsible for resolving the issue.

g. Current status.

h. Issue description.

i. Pictures of item if available and appropriate.

j. Effects of issue on project or building operation.

k. Possible cause of issue.

l. Proposed resolution.

m. Final resolution and final resolution date.

n. Approvals of issue resolution, including approver's name.

3.3.2.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must prepare and maintain the **Post Construction Phase Issues And Resolution Log** and must submit one electronic copy no later than 14 calendar days after completion of any Post-Construction Phase Commissioning Activity.

3.3.3 Post-Construction Phase Site Observations

3.3.3.1 General Requirements

Post-Construction Phase Site Observations must be performed for each systems category specified in the following item entitled "Number of Trips" between eight and 10 months after Beneficial Occupancy to verify and document that all commissioned equipment, systems, and assemblies are operating in accordance with the Plans and Specifications.

Post-Construction Phase Site Observations are prohibited from being combined with witnessing Post-Construction Phase Functional Performance Testing.

3.3.3.2 Number of Trips

The Number of Post-Construction Phase Site Observations inclusive of all commissioned buildings in this project must be as follows

a. Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category: One eight-hour trip.

b. Plumbing Systems Category: No trips. Combine these activities with HVAC Systems Category activites.

c. Utility Systems Category: No trips.

d. Electrical Type I Systems Category: No trips.

e. Electrical Type II Systems Category: One eight-hour trip.

f. Building Envelope Systems, Components, and Assemblies Category: One eight-hour trip.

3.3.3.3 Site Observation Reports

Site Observation Reports must be prepared for each Site Observation for the purpose of documenting issues discovered during Site Observation.

3.3.3.4 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, perform activities specified in the following paragraphs.

a. Contact Contracting Officer's Technical Representative no later than 30 calendar days prior to performing Site Observation to schedule attendance of Operations and Maintenance staff and occupants and to make provisions for access to all commissioned equipment, systems, and assemblies.

b. Submit **Post-Construction Phase Site Observation Reports** no later than 14 calendar days after Site Observation.

3.3.3.5 Commissioning Specialists Responsibilities

Commissioning Specialists must, at a minimum, perform activities specified in the following paragraphs for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

a. Interview Operations and Maintenance staff.

b. Interview occupants.

c. Determine status of unresolved Commissioning Issues.

d. Compare current operations to operations identified throughout the Commissioning Process.

e. Prepare Post-Construction Phase Site Observation Reports.

3.3.4 Certified Certificates of Readiness and Certificates of Readiness

Certified Certificates of Readiness and Certificates of Readiness are not required for any Post-Construction Phase activity.

3.3.5 Post-Construction Phase Functional Performance Testing

3.3.5.1 General Requirements

Post-Construction Phase Functional Performance Testing must be performed for all commissioned equipment, systems, and assemblies that could not be fully or successfully tested during the Construction Phase. Testing must be performed using approved checklists and all required Commissioning Team participants must be present prior to start of testing. Testing must be aborted and rescheduled to occur at a later date if any Commissioning Team participant is not present or if deficiencies discovered during testing prevent successful completion.

Post-Construction Phase Functional Performance Testing requirements and activities must include, but not be limited to, requirements and activities specified for Construction Phase Functional Performance Testing as modified in paragraphs under paragraph POST-CONSTRUCTION (WARRANTY) PHASE.

3.3.5.2 Commissioning Team Participants

Participants of Functional Performance Testing must include, but not be limited to, participants specified in the following paragraphs. The Contracting Officer's Technical Representative, Public Works Department Division (PWD) Representatives, and Using Agent's Representatives (Users) may, but are not required to, participate in any testing.

a. Lead Commissioning Specialist.

b. Commissioning Specialists.

c. Installing Contractors.

d.

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3.3.5.3 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, perform activities specified in the following paragraphs.

a. Contact Contracting Officer's Technical Representative no later than 30 calendar days prior to witnessing Post-Construction Phase Functional Performance Testing to make provisions for access to all commissioned equipment, systems, and assemblies.

b. Coordinate logistics with Installing Contractors.

c. Submit completed Final Functional Performance Test Checklists for each systems category listed under paragraph SYSTEMS TO BE COMMISSIONED as a single submittal no later than 14 calendar days after successful testing of all systems listed under the respective systems category.

3.3.6 Updated Final Commissioning Report

3.3.6.1 General Requirements

The Updated Final Commissioning Report must be prepared and must be an update to the Final Commissioning Report prepared during the Construction Phase and contents must include, but not be limited to, changes resulting from Post-Construction Phase Commissioning Activities.

3.3.6.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, prepare and submit the **Updated Final Commissioning Report** no later than 30 calendar days after completion of all Functional Performance Testing.

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SECTION 01 91 19

BUILDING ENCLOSURE COMMISSIONING
05/2023

PART 1 GENERAL

1.1 SUMMARY

Use this Section in conjunction with UFGS 01 91 00.15 22 TOTAL BUILDING COMMISSIONING. Building Enclosure Commissioning (BECx) is an essential part of overall Building Commissioning. The final BECx report is to be included as a part of the Cx Firm's submittal of the Final Commissioning Report as defined in 01 91 00.15 22 TOTAL BUILDING COMMISSIONING. This Section covers Building Envelope Commissioning for all six surfaces (4 walls, roof, and floor, for a rectangular shaped zone) of a building. Additional Quality Control and testing requirements for specific components of the building envelope are described in the Division 03 thru 08 UFGS Sections of the project specifications.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced within the text by the basic designation only.

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

ANSI/ASNT CP-105 (2020) ASNT Standard Topical Outlines for Qualification of Nondestructive Testing Personnel

ANSI/ASNT CP-189 (2020) ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel

ASNT SNT-TC-1A (2020) Recommended Practice for Personnel Qualification and Certification in Nondestructive Testing

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 90.1 - IP (2019) Energy Standard for Buildings Except Low-Rise Residential Buildings

ASHRAE 202 (2024) Commissioning Process for Buildings and Systems

ASTM INTERNATIONAL (ASTM)

ASTM C1193 (2016) Standard Guide for Use of Joint Sealants

ASTM E779 (2019) Standard Test Method for Determining Air Leakage Rate by Fan Pressurization

ASTM E783	(2002; R 2018) Standard Test Method for Field Measurement of Air Leakage Through Installed Exterior Windows and Doors
ASTM E1105	(2015; R 2023) Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls, by Uniform or Cyclic Static Air Pressure Difference
ASTM E1186	(2022) Standard Practices for Air Leakage Site Detection in Building Envelopes and Air Barrier Systems
ASTM E1258	(1988; R 2018) Standard Test Method for Airflow Calibration of Fan Pressurization Devices
ASTM E1827	(2022) Standard Test Methods for Determining Airtightness of Buildings Using an Orifice Blower Door
ASTM E2813	(2018) Standard Practice for Building Enclosure Commissioning
ASTM E2947	(2021) Standard Guide for Building Enclosure Commissioning

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 6781	(1983) Thermal Insulation - Qualitative Detection of Thermal Irregularities in Building Envelopes - Infrared Method
ISO 6781-2	(2010) Performance of Buildings - Detection of Heat, Air, and Moisture Irregularities in Buildings by Infrared Methods - Part2: Equipment Requirements
ISO 6781-3	(2015) Performance of Buildings - Detection of Heat, Air, and Moisture Irregularities in Buildings by Infrared Methods - Part 3: Qualifications of Equipment Operators, Data Analysts, and Report Writers

1.3 DEFINITIONS

The definitions included here are specific to this Section. Refer to UFGS 01 91 00.15 22 TOTAL BUILDING COMMISSIONING for definitions that apply to both Sections. The following terms as they apply to this section:

1.3.1 Air Barrier Envelope

The components of building assemblies(roofs, walls, and floors) that separate the inside air from the outside air. The combination of air barrier assemblies and air barrier components, connected by air barrier accessories are designed to provide a continuous barrier to the movement of air through an environmental separator. A single building may have more than one air barrier envelope enclosing multiple environmentally separated zones. The air barrier surface includes the top, bottom, and sides of the envelope. The term "air barrier envelope" is also known as "air barrier system" or simply "air barrier".

1.3.2 Air Leakage Rate

A measurement of air infiltration or exfiltration through the building envelope over time. The air leakage rate is the rate of air flow across the air barrier per unit surface area of the envelope at a defined differential pressure.

1.3.3 Basis of Design (BoD)

A document developed by the design team which includes technical concepts, assumptions, calculations, decisions, and product selections to support the Owner's Project Requirements (OPR).

1.3.4 Bias Pressure

Also known as zero flow pressure, baseline pressure, offset pressure or background pressure. With the envelope not artificially pressurized, bias is the differential pressure that always exists between the envelope that has been prepared (sealed) for the pressure test and the outdoors. Bias pressure is made up of two components, fixed static offset (usually due to stack effect or the HVAC system) and fluctuating pressure (usually due to wind or a moving elevator). Because of pressure fluctuations many bias pressure readings are recorded and averaged for use in the calculations.

1.3.5 Blower Door

Commonly used term for an apparatus used to pressurize and depressurize the space within the building envelope and quantify air leakage through the envelope. The blower door typically includes a door fan and an air resistant fabric or a series of hard panels that extends to cover and seal the door opening between the fan shroud and door frame. The door fan is a calibrated fan capable of measuring air flow and is usually placed in the opening of an exterior door. With the air barrier otherwise sealed, air produced by the door fan pressurizes or de-pressurizes the envelope, depending on the fan's orientation.

1.3.6 Building Envelope (or enclosure)

The materials, components, systems, and assemblies intended to provide an environmental separator between the interior and exterior or between interior spaces with different environment requirements. "Enclosure" is used interchangeably with "envelope".

1.3.7 Building Envelope Commissioning (BECx)

Technical service provided on behalf of the Government to provide a quality-focused process for enhancing the performance of the building envelope by validating the design phase and verifying during the construction phase that the performance of the building enclosure materials components, assemblies, and systems are designed and installed to meet the Owner's Project Requirements (OPR).

1.3.8 Building Envelope Commissioning Specialist (CxB)

An entity that functions as part of the overall Commissioning Team and is responsible for validating the design phase and verifying during the construction phase that the performance of the building enclosure materials, components, assemblies, and systems are designed and installed to meet the Owner's Project Requirements (OPR).

1.3.9 Construction Phase

The period of a project following the award of construction Contract to project completion.

1.3.10 Design Phase

The period of the project delivery process when a design supporting the Owner's Project Requirements is developed and translated into Contract Documents.

1.3.11 Environmental Separator

The parts of a building that separate the controlled interior environment from the uncontrolled exterior environment, or that separate spaces within a building that have dissimilar environments. The term "environmental separator" is also known as the "control layer".

1.3.12 Functional Performance Test (FPT)

A systematic process to verify that controls and other elements of the building project are capable of and configured to operate or perform as required. FPT's are written and observed by the Technical Cx Specialist and performed by the Verification and Testing Provider. Tests not written by the Technical Cx Specialists, but required and defined throughout this specification manual are not described as FPTs in UFGS 01 91 00.15 22 TOTAL BUILDING COMMISSIONING.

1.3.13 Pressure Test

A generic term for a test in which the building envelope or zone is either pressurized or de-pressurized with respect to the outdoors or other building zones.

1.3.14 Negative Pressure Test (Depressurization Test)

A test wherein air inside the envelope is drawn to the outdoors. This places the envelope at a lower (negative) pressure with respect to the outdoors.

1.3.15 Positive Pressure Test (Pressurization Test)

A test wherein outdoor air is pushed into the envelope. This air movement places the envelope at a higher (positive) pressure with respect to the outdoors.

1.3.16 Verification & Testing (V&T) Provider

An entity who completes the activities needed to implement verification and testing of the building's materials, components, assemblies, and systems, including testing described throughout this specification manual and functional performance testing (FPT) activities to verify that elements of the building project meet contract document requirements.

1.4 COMMUNICATION WITH THE GOVERNMENT COMMISSIONING PROVIDER

For communication responsibilities, see Section 01 91 00.15 22 TOTAL BUILDING COMMISSIONING.

1.5 COMMUNICATION WITH GOVERNMENT'S Cx and ACCEPTANCE TESTING REPRESENTATIVES

The QC Manager must communicate directly with the Government Acceptance Testing Representatives, the government's Lead Cx Specialist (CxC), and Contracting Officer's Representative regarding Government acceptance testing activities. Inform the government's Lead Cx Specialist (CxC) andthe Contracting Officer's Representative when systems are ready for testing to be witnessed by the government's Cx Specialists and Government Acceptance Testing Representatives, and allow access to the construction site and system(s) to be tested.

1.6 ENCLOSURE COMPONENTS TO BE COMMISSIONED

1.6.1 Verification of the Installation of the Continuous Air Barrier

a. Conduct periodic field inspection of the continuous air barrier materials and assemblies, including all penetrations through the six sides of the air barrier, during construction while the continuous air barrier is still accessible for inspection and repair (after installation of all penetrations and wall cladding anchoring devices and before the installation of interior finishes) to verify and document compliance with the requirements. Conduct inspections at initial installation of each building envelope component and at in-progress construction intervals as stated in Part 3 section titled BUILDING ENVELOPE INSPECTION AND TESTING.

b. Conduct Performance Verification Testing of the entire building envelope (systems, components, and assemblies) for air tightness in accordance with Part 3 section titled BUILDING ENVELOPE AIR TIGHTNESS REQUIREMENT.

1.6.2 Inspection of Fenestration and Doors

a. Fenestration and Doors must be labeled and inspected to verify compliance with Building Envelope energy requirements, including applicable U-factors, Solar Heat Gain Coefficient (SHGC), Visible Transmittance (VT), and air leakage rates.

b. Inspect doors and windows for coordination with the insulating plane of the wall, continuity with the air/weather barrier, and flashing.

c. Inspect operation of doors, closers, and operating mechanisms for conformance with manufacturer's instructions.

d. Inspect seals and gaskets for fenestration and doors (including loading dock, sectional, and coiling doors) for proper installation and to verify that seals are in good condition and comply with ASHRAE 90.1 - IP maximum allowed leakage rates.

1.6.3 Inspection of Opaque Roof, Walls, and Floors

Opaque roof, above-grade and below-grade walls, and floors, must be subject to the following inspections during construction:

a. Use of ASHRAE 90.1 - IP-compliant materials and assemblies.

b. Insulation material meets design specifications and is continuous.

1.6.4 Fenestration Inspections

Conduct inspections of the following fenestration-related items during construction:

a. Skylights size and location in relation to the designed primary sidelighted area and secondary sidelighted area below.

b. Roof monitor size and location in relation to the designed primary sidelighted area and secondary sidelighted area below.

c. Dynamic glazing compliance with SHGC and U-factors, and testing of the operation for conformance with the manufacturer's instructions.

d. Permanent fenestration projections installation and performance in accordance with ASHRAE 90.1 - IP requirements and the Contract documents.

1.6.5 Loading Dock Weatherseal Inspections

Inspect loading dock weatherseals for proper installation and to verify the seals are in good condition.

1.6.6 Additional Building Enclosure Tests

Conduct the following building enclosure tests on mock-ups or early in the installation of the related building components:

Annex Envelop

1.7 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

Building Enclosure Commissioning Specialist Qualifications; G

Building Enclosure Testing Work Plan; G

PRELIMINARY BECx PLAN; G

SD-03 Product Data

Thermal Imaging Camera; G

Test Equipment; G

FINAL BECx PLAN; G

SD-05 Design Data

Envelope Surface Area Calculations; G

SD-06 Test Reports

Completed Building Envelope Inspection Checklists; G

Pressure Test Procedures; G

Air Leakage Test Report; G

Diagnostic Test Report; G

FINAL BECx REPORT

SD-07 Certificates

Pressure Test Agency

Thermographer Qualifications

Certificate of Readiness

SD-10 Operation and Maintenance Data

Training; G

1.8 BUILDING ENCLOSURE TESTING WORK PLAN

For each building enclosure test to be performed, submit the following not later than 120 calendar days before start of testing work, steps to be taken by the lead test technician to accomplish the required testing.

a. Memorandum of test procedure.

   (1) Proposed dates for conducting the tests.

   (2) Submit detailed test procedures prior to the test. Provide a plan view showing proposed locations (personnel doors or other similar openings) to install equipment (such as blower doors or flexible ducts for trailer-mounted fans, if used).

   (3) Submit detailed requirements for access to or pressurization of adjacent, occupied buildings or zones. Provide a plan view showing proposed locations (personnel doors or other similar openings) to install equipment (such as blower doors or flexible ducts for trailer-mounted fans, if used).

b. Test equipment to be used.

c. Scaffolding, scissor lifts, power, electrical extension cords, duct tape, plastic sheeting and other Contractor's support equipment required to perform all tests.

d. Other Contractor's support personnel who will be on site for testing.

1.9 ACCESSIBILITY REQUIREMENTS

Air barrier components, insulation, fenestration, and doors for commissioned systems must be made accessible for inspections. Contractor must make necessary modifications at Contractor's own expense if systems and enclosures are not accessible for inspections and testing. Assist commissioning team in testing and inspections by removing equipment covers, opening access panels, and other required activities that assist with visual oversight. Furnish ladders, flashlights, meters, gauges, or other inspection equipment as necessary.

1.10 SCHEDULING

Notify the Government's Lead Cx Specialist and the Contracting Officer a minimum 14 calendar days in advance of site visits and inspections by the roof system manufacturer technical representative for coordination with CxB and the Government Acceptance Testing Representative.

1.11 COORDINATION

Refer to Section 01 91 00.15 22 TOTAL BUILDING COMMISSIONING for requirements pertaining to coordination during the commissioning process. Coordinate with the Commissioning Provider in accordance with Section 01 91 00.15 22 and in accordance with the Commissioning Plan to schedule inspections as required to support the commissioning process. Furnish additional information requested by the Commissioning Provider.

Coordinate scheduling of air barrier inspections and air barrier pressure testing with the commissioning team. Upload plans, reports, notes, and other documentation to the Commissioning Provider's web-based commissioning software, or as specified in the commissioning plan, as it is completed.

1.12 QUALITY CONTROL

1.12.1 Modification of References

Perform all pressure and diagnostic tests according to the referenced publications listed in paragraph REFERENCES and as modified by this section. Consider the advisory or recommended provisions, of the referred references, as mandatory.

1.12.2 Qualifications

1.12.2.1 Pressure Test Agency

Submit, no later than 15 calendar days after Contract award, information certifying that the pressure test agency is not affiliated with any other company participating in work on this Contract. The work of the test agency is limited to pressure testing the building envelope, performing a thermography test and fog test, and investigating, through various methods, the location of air leaks through the air barrier. See Part 3 paragraph PRESSURE TEST AGENCY for additional requirements. For thermographer qualifications, see paragraph THERMOGRAPHER QUALIFICATIONS.

Use the sample TEST AGENCY QUALIFICATIONS SHEET form (Appendix C), to submit the following information.

a. Verification of 2 years of experience as an agency in pressure testing commercial or industrial buildings.

b. List of at least ten commercial/industrial facilities with building envelopes that the agency has tested within the past 2 years. Include building name, address, and name of prime construction Contractor and Contractor's point-of-contact information.

c. Confirmation of 2 years of commercial and or industrial building pressure test experience for the lead pressure test technician and the thermographer in using the specified ASTM E779 or ASTM E1827 testing standard. References from five Contracting Officers for facilities where the lead test technician has supervised commercial and or industrial building pressure tests in the last 2 years.

d. Verification that the lead pressure test technician has been employed by a building pressure testing agency in the capacity of a lead pressure test technician for not less than 1 year.

1.12.2.2 Thermographer Qualifications

To perform an infrared diagnostic evaluation, use a lead thermographer who has at least an active Level II Certification that is based on the requirements in ANSI/ASNT CP-105 or ANSI/ASNT CP-189 and is in accordance with ASNT SNT-TC-1A. The course of study is to be specifically focused on infrared thermography for building science. The thermographer must have at least two years of building science thermography experience in IR testing commercial or industrial buildings. The thermographer must also have experience in building envelopes and building science in order to make effective recommendations to the Contractor should the envelope require additional sealing. Thermographic equipment operators, data analysts and report writers must comply with the requirements of ISO 6781-3. Submit the thermographer's certificate for approval. Submit a list of at least ten commercial/industrial buildings on which the thermographer has performed IR thermography in the past two years. The thermographer is to have a current active certification. Submit certification at least 60 days prior to thermography testing.

1.12.3 Test Reports

No later than 14 days after completion of the pressure test, submit electronic copies of an organized report. The report is to contain a table of contents, an executive summary, an introduction, a results section and a discussion of the results. Submit the air leakage test report as described in paragraph AIR LEAKAGE TEST REPORT. Submit a diagnostic test report as described in paragraph LOCATING LEAKS BY DIAGNOSTIC TESTING. The diagnostic test report is to include the Thermographic Investigation Report and the Fog Test Report.

Submit field data and completed report forms found in the appendices. Use the sample forms, Test Agency Qualification Sheet, Air Leakage Test Form and Air Leakage Test Results Form to summarize the tests for the appropriate building envelope. Submit both electronically populated and field hand filled-in forms.

Report Data. Include in the report the following information for all tests:

a. Date of issue

b. Project title and number

c. Name, address, and telephone number of testing agency

d. Dates and locations of samples and tests or inspections

e. Names of individuals making the inspection or test

f. Designation of the work and test method

g. Identification of product and specification section

h. Complete inspection or test data

i. Test results and an interpretation of test results

j. Comments or professional opinion on whether inspected or tested work complies with Contract document requirements

k. Recommendations on retesting

1.13 CLIMATE CONDITIONS SUITABLE FOR A PRESSURE TEST

As the test date approaches, monitor the weather forecast for the test site. Avoid testing on days forecast to experience high winds, rain, or snow. Monitor weather forecasts prior to shipping pressure test equipment to the site. Based on current and forecast weather conditions, the Contracting Officer's representative is to grant final approval for testing to occur.

1.13.1 Rain

For safety reasons, avoid testing during rain or if rain is anticipated during testing. If pneumatic hoses are installed and exposed to rain inspect the hose to ensure rainwater has not migrated into the hose ends. Orient all exposed hose ends to keep them out of water puddles. Success in temporarily sealing outdoor ventilation components such as louvers and exhaust fans may also be compromised by rain. Don't seal roof-mounted ventilation components during times of potential lightning.

1.13.2 Wind

Because wind can skew pressure test results, test only on days and at times when winds are anticipated to be the calmest. Avoid pressure testing during gusty or high wind conditions. Avoid installing test fans on the windward side of the building if wind gusts during the test are anticipated to be greater than 10 miles per hour.

1.14 CERTIFICATE OF READINESS

Prior to scheduling the Commissioning Tests as required by this Section, submit Certificate of Readiness documentation in accordance with Section 01 91 00.15 22 TOTAL BUILDING COMMISSIONING for each enclosure-related system, certifying that inspections have been completed, open issues have been resolved, and the system is ready for the Commissioning Tests. Documentation would include Building Envelope Inspection Checklists as required within this Section, as well as any inspections or test reports required within each enclosure-related technical specification section.

Additional Quality Control and testing requirements for specific components of the building enclosure are described in the Division 03 thru 08 UFGS Sections of the project specifications.

Submit the Certificate of Readiness for each system 20 working days prior to Commissioning Tests of that system. Do not schedule Commissioning Tests for a system until the Certificate of Readiness is approved by the Government. Do not schedule air tightness testing of the building envelope system until Certificates of Readiness and supporting documentation for enclosure-related systems have been submitted, reviewed and approved by the Government.

PART 2 PRODUCTS

2.1 TEST EQUIPMENT

Provide all equipment required to perform testing for the systems and components to be commissioned. Provide all testing equipment of sufficient quality and accuracy to test and measure system performance with the tolerances specified. Provide a sufficient quantity of two-way radios for each subcontractor.

Submit a signed and dated list of instruments to be used for testing, their application, manufacture, model, serial number, range of operation accuracy, and date of most recent calibration. Calibration data applicable to fan systems must be in accordance with ASTM E1258. Also list special equipment and proprietary tools specific to a piece of equipment required for testing.

2.2 PRESSURE TEST EQUIPMENT

Depending on site conditions and size of the envelope, the test may be conducted using blower door equipment and trailer-mounted fans. The testing agency is to supply sufficient quantity of blower equipment that will produce a minimum of 75 Pa differential pressure between the envelope and outdoors using the test methods described herein. Supplying additional blower test equipment to provide additional airflow capacity or to act as a backup is highly recommended.

2.2.1 Blower Door Fans and Trailer Mounted Fans

Each air flow measuring system including blower door fans and trailer mounted fans are to be calibrated within the last 5 years. Calibrated blower door fans and trailer mounted fans must measure accurately to within plus or minus 5 percent of the flow reading. Blower door equipment and trailer mounted fans are to be specifically designed to pressurize building envelopes. Each set of blower door equipment is to include fan(s), digital gage(s), door frame, door fabric or hard panels.

2.2.2 Digital Gages as Test Instruments

Use only digital gages as measuring instruments in the pressure test; analog gages are not acceptable. The gauges must be accurate to within 1.0 percent of the pressure reading or 0.15 Pa, whichever is greater. Each gage is to have been calibrated within two years of the test. The calibration is to be checked against a National Institute of Standards and Technology (NIST, formerly National Bureau of Standards) traceable standard.

2.3 THERMAL IMAGING CAMERA REQUIREMENTS

The thermal imaging camera used in the thermography test must have a thermal sensitivity (Noise Equivalent Temperature Difference.) of +/- 0.18 degrees F at 86 degrees F or less. Ensure the camera's operating spectral range falls between 2 and 15 micrometers. Ensure the camera's IR image viewing screen resolution measures at least 320x240 pixels. Ensure the camera has a means of recording thermal images seen on the camera viewing screen. The camera is to display output as individual still frame images that also can be downloaded and inserted into an electronic Thermographic Investigation Report. All thermographic equipment must comply with the requirements of ISO 6781-2. Submit camera make and model, and catalog information that defines the camera thermal sensitivity for approval.

PART 3 EXECUTION

Coordinate with the COMMISSIONING TEAM and perform all Contractor and Equipment supplier activities and responsiblities defined in Section 01 91 00.15 22 TOTAL BUILDING COMMISSIONING and building envelope related specifications to accomplish building envelope commissioning in accordance with ASHRAE 202 and ASTM E2947 in addition to the requirements herein. Follow ASTM E2947 except where it references ASTM E2813. All verification and testing work shall be documented and provided to the government's Cx Firm in sufficient detail to demonstrate compliance of the building envelope with the design intent as defined in the Contract documents.

3.1 MEETINGS

Attend all meetings in accordance with Section 01 91 00.15 22 TOTAL BUILDING COMMISSIONING.

Provide timely updates on construction schedule changes so Commissioning Provider has scheduling information needed to execute commissioning process efficiently. Notify Contracting Officer of anticipated construction delays to commissioning activities not yet performed or not yet scheduled.

3.2 PRELIMINARY BECx PLAN

Submit the Preliminary BECx Plan no later than 14 calendar days after the Commissioning Kickoff Meeting. Submit the Preliminary BECx Plan for inclusion in the Interim Construction Phase Commissioning Plan as identified in Section 01 91 00.15 22 TOTAL BUILDING COMMISSIONING. Coordinate the development of this plan with the commissioning plans required in Section 01 91 00.15 22 TOTAL BUILDING COMMISSIONING. Outline the commissioning process, commissioning team members and responsibilities, lines of communication, inspections, and documentation requirements for BECx. Identify the Commissioning Standards chosen for the project. Establish appropriate and quantifiable enclosure related performance metrics, test standards, and test methodology in accordance with referenced standards for inclusion in the Contract documents. Provide a list of team members for systems to be commissioned with contact information, a list of tests as required to complete the scope of all commissioning and a copy of the project schedule as required by Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULE for inclusion in the Preliminary BECx Plan.

The Preliminary BECx Plan must include, at a minimum, the following:

1. Required performance of commissioned equipment, systems, and assemblies, and results of all building envelope verification and testing.

2. Summary of compliance of the building and its components, assemblies, controls, and systems with requirements.

3. Issues and resolution logs, including itemization of deficiencies found during verification, testing, and commissioning that have not been corrected at the time of report preparation.

4. Deferred tests that cannot be performed at the time of report preparation.

5. Documentation of the training of operation personnel and building occupants on commissioned systems, and a plan for the completion of any deferred trainings not completed at the time of report preparation.

6. A plan for the completion of commissioning and training, including climatic and other conditions required for performance of the deferred tests.

3.3 FINAL BECx PLAN

Submit the Final BECx Plan directly to the Government's Lead Cx Specialist (CxC) for inclusion in the Cx Firm's submittal, the Construction Phase Interim Commissioning Plan, as identified in Section 01 91 00.15 22 TOTAL BUILDING COMMISSIONING. Outline the commissioning process, commissioning team members and responsibilities, lines of communication, and documentation requirements for BECx. Identify the Commissioning Standards chosen for the project. Identify enclosure related inspections, performance metrics, test standards, and test methodologies in accordance with referenced standards for inclusion in the commissioning process.

Provide a list of team members for systems to be commissioned with contact information, a list of tests as required by this Section as well as 07 27 10 BUILDING AIR BARRIER SYSTEM, and a copy of the project schedule as required by Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULE for inclusion in the Final BECx Plan. Final BECx plan must be included in the Cx Firm's submittal of the Construction Phase Interim Commissioning Plan, and must be updated and included with the Cx Firm's submittal of the Construction Phase Final Commissioning Plan as identified in Section 01 91 00.15 22 TOTAL BUILDING COMMISSIONING.

3.4 CONSTRUCTION SUBMITTAL REVIEWS

Coordinate construction submittal document reviews for commissioned systems and assemblies with the CxC and CxB. The commissioning submittal review does not replace the designer of record (DoR) or Government submittal review, in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME).

The CxB is responsible for identifying construction submittals to be provided by the Contractor for the commissioned systems and coordinating with the CxC. The CxB is responsible for evaluating construction submittals related to BECx for compliance with the Contract documents.

3.5 TEMPLATE BUILDING ENVELOPE INSPECTION CHECKLISTS

Use the Template Building Envelope Inspection Checklists prepared by the CxC to verify the building materials and construction maintain the required air tightness of the building envelope system.

The Building Envelope Compliance Documentation Form from ASHRAE 90.1 - IP User's Manual may be used as an example. The submitted checklist is not required to match the format of the form; however, the checklist must contain the same level of detail as shown on the mandatory provisions of the sample form.

3.6 BUILDING ENVELOPE INSPECTION AND TESTING

Demonstrate that all system components have been installed, that each building enclosure component operates, and that the systems operate and perform in accordance with Contract documents. Provide all materials, services, equipment, and labor required to perform the Building Envelope Inspections.

Coordinate site observation activities with the government's CxC and participate in periodic building envelope inspections with the technical Building Envelope commissioning specialists (CxB) using the approved Building Envelope Inspection Checklists to observe and document construction of the building envelope in-progress. Coordinate inspection requirements with Section 07 27 10 BUILDING AIR BARRIER SYSTEM and other Division 7 Specifications. Complete the checklists and indicate inspection and validation of each Building Envelope Inspection Checklist item by initials at the time they are inspected. Notify the Lead Commissioning Specialist (CxC) and Contracting Officer at least 21 calendar days before checklist items are concealed to ensure inspection items can be observed before construction progresses. Submit the initialed and Completed Building Envelope Inspection Checklists no later than 14 calendar days after completion of inspection of all checklist items. Notify the CxC at least 21 calendar days prior to the building envelope pressure tests and diagnostic tests specified in this Section.

3.6.1 Additional Building Enclosure Testing

Coordinate with specification Section 01 91 19 Building Envelope Commissioning.

a. Conduct air leakage testing of fenestration in accordance with ASTM E783.

b. Conduct water penetration tests of fenestration in accordance with ASTM E1105.

c. Conduct adhesion and cohesion tests of sealants in accordance with ASTM C1193.

3.7 BUILDING ENCLOSURE AIR BARRIER PRESSURE TESTING

3.7.1 Pressure Test Agency

Employ an independent agency to conduct the pressure test on the building envelope in accordance with ASTM E779 and this specification section. The test agency is to be an independent third-party subcontractor, not an affiliated or subsidiary of the prime Contractor, subcontractors, or A/E firm. The agency is to be regularly engaged in pressure testing of commercial/industrial building envelopes. If using blower door or trailer-mounted fans, the lead test technician must have at least two years of experience in using such equipment in building envelope pressurization tests. Formal training using pressure test equipment is highly recommended. Technicians using the building's air handling system for pressure testing are to have tested at least five commercial/industrial buildings within the past two years with each building having over 50,000 square feet of floor area. Submit the name, address, and floor areas of each of these five buildings for approval.

3.7.1.1 Field Work

The lead pressure test technician and thermographer are to be present at the project site while testing is performed and is to be responsible for conducting, supervising, and managing of their respective test work. Management includes health and safety of test agency employees.

3.7.1.2 Reporting Work

The lead pressure test technician is to prepare, sign, and date the test agenda, equipment list, and submit a certified Air Leakage Test Report. The thermographer is to prepare, sign, and date the test agenda, equipment list, and submit a certified Thermographic Investigation Report. The Contractor is to prepare a final report that identifies improvements that were made to the envelope to reduce air leaks , mitigate thermal bridging, eliminate moisture migration, repair insulation voids discovered during diagnostic tests. The Verification & Testing (V&T) providers must certify completion of required verification and testing and include a plan for the completion of any deferred testing, including climatic and other conditions required for performance of the deferred tests; include the results of the testing and verification activities in the completed verification and testing documentation. Jointly submit all reports.

3.7.2 Envelope Surface Area Calculation

The architectural air barrier boundary includes the floor, walls, and ceiling. After construction of the air barrier envelope is complete, field measure the envelope to ensure the physical measurements match the design drawings and the air barrier envelope surface area calculations are generated. If the calculation result is not within 10 percent of the defined air barrier boundary calculation result as indicated, submit the envelope surface area calculation and results for review. If the air barrier was defined during design but the air barrier envelope surface area was not calculated, calculate it during construction and submit the envelope surface area calculations and result for review.

3.7.3 Preparing The Building Envelope For The Pressure Test

3.7.3.1 Testing During Construction

The pressure test cannot be conducted until all components of the air barrier system have been installed. After all sealing as described herein has been completed, inspect the envelope to ensure it has been adequately prepared. During the pressure test, stop all ongoing construction within and neighboring the envelope which may impact the test or the air barrier integrity. The pressure test may be conducted before finishes that are not part of the air barrier envelope have been installed. For example, if suspended ceiling tile, interior gypsum board or cladding systems are not part of the air barrier the test can be conducted before they are installed. Recommend testing prior to installing the finished ceilings and interior wall finishes within the envelope and immediately surrounding it. The absence of finishes allows for inspection and diagnostic testing of the roof/wall interface and for implementation of repairs to the air barrier, if necessary to comply with the maximum allowed leakage.

3.7.3.2 Sealing the Air Barrier Envelope

The Contractor is to seal all penetrations through the air barrier. Unavoidable penetrations due to electrical boxes or conduit, plumbing, and other assemblies that are not air-tight are to be made so by sealing the assembly and the interface between the assembly and the air barrier or by extending the air barrier over the assembly. Support the air barrier to withstand the maximum positive and negative air pressure to be placed on the building without displacement or damage, and transfer the load to the structure. Durably construct the air barrier to last the anticipated service life of the assembly and to withstand the maximum positive and negative pressures placed on it during pressure testing. Do not install lighting fixtures that are equipped with ventilation holes through the air barrier.

3.7.3.3 Sealing Plumbing

Prime all plumbing traps located within the envelope full of water.

3.7.3.4 Close and Lock Doors

Close and lock all doors and windows in the envelope perimeter. For doors not equipped with latching hardware, temporarily secure them in the closed position. Secure the doors in such a way that they remain fully closed even when the maximum anticipated differential air pressure produced during the test acts on them. Provide signage stating not to open the door and the time and duration of the test.

3.7.3.5 Hold Excluded Building Areas at the Outdoor Pressure Level

Keep building areas immediately surrounding but excluded from the test envelope at the outdoor pressure level during the pressure test. Maintain these areas at the outdoor pressure level by propping exterior doors open, opening windows and de-energizing all air moving devices in or serving these areas.

3.7.3.6 Maintain an Even Pressure within the Envelope

Ensure the pressure differences within the envelope are minimized by opening all internal air pathways including propping open all interior doors. Distribute test fans throughout the envelope as necessary to ensure the internal pressures are uniform (within 10 percent of the average differential pressure). Ideally, do not install suspended ceilings until after all pressure tests have been completed. If, however the envelope includes finished suspended ceiling spaces, temporarily remove approximately 5 percent of all ceiling tiles or a minimum of 1 tile from each isolated suspended ceiling space, whichever comprises the greatest surface area. Temporarily remove additional ceiling tiles during testing to allow for inspection and diagnostic testing of the ceiling/wall interface. An alternative to removing ceiling tiles is to measure the differential pressure between each isolated suspended ceiling space and the outdoors when the area below the suspended ceiling is maintained at a differential pressure of 75 Pa with respect to the outdoors. If the suspended ceiling differential pressure measurement is within ten percent of the 75 Pa pressure below the suspended ceiling no ceiling tiles need to be removed.

3.7.3.7 Maintain Access to Mechanical and Electrical Rooms

Maintain access to mechanical rooms and electrical rooms associated with the envelope to allow for de-energizing ventilation equipment and resetting circuit breakers tripped by blower door equipment, if used.

3.7.3.8 Minimize Potential for Blowing Dust and Debris

Because high velocity air will be blown into and out of the envelope during the test, debris, including dust and litter, may become airborne. Airborne debris may become trapped or entangled in test equipment, thereby skewing test results. Ensure areas within and surrounding the envelope are free of dust, litter and construction materials that are easily airborne. If pressurizing existing, occupied areas, provide adequate notice to building occupants of blowing dust and debris, and general disruption of normal activities during the test.

3.7.3.9 De-energize Air Moving Devices

De-energize all air moving devices serving the envelope to keep air within the envelope as still as reasonably achievable. De-energize all fans that deliver air to, exhaust air from, or recirculate air within the envelope. Also, de-energize all fans serving areas adjacent to but excluded from the envelope.

3.7.3.10 Installing Blower Door Equipment in a Door Opening

Where blower door fans are used, before installing blower door equipment, select a door opening that does not restrict air flow into and out of the envelope and has at least 5 feet clear distance in front of and behind the door opening. Disconnect the door actuator and secure the door open to prevent it from being drawn into the fan-by-fan pressure. Avoid installing blower door equipment on the windward side of the building.

3.7.4 Building Envelope Air Tightness Requirement

For each building envelope, perform the Architectural Only test and if noted below, the Architectural Plus HVAC System test. The purpose of the pressure (air leakage) test is to determine final compliance with the airtightness requirement by demonstrating the performance of the continuous air barrier. An effective air barrier envelope minimizes infiltration and exfiltration through unintended air paths (leaks). The tests may be performed in any desired order.

3.7.4.1 Architectural Only Test

The test envelope is the architectural air barrier boundary as defined on the Contract drawings. This boundary includes connecting walls, roof and floor which comprise a complete, whole, and continuous three-dimensional envelope. Perform both a positive pressure test and a negative pressure test on this envelope, unless otherwise directed.

3.7.4.1.1 Test Goal

Input data from the test into the Air Leakage Rate by Fan Pressurization spreadsheet as described in paragraph CALCULATION PROGRAM via the Air Leakage Test Form. Compare output from the spreadsheet against the maximum allowable leakage defined in Section 07 27 10 BUILDING AIR BARRIER SYSTEM. The envelope passes the test if the leakage rate, as calculated using the spreadsheet, is equal to or lower than the Architectural Only leakage rate goal.

3.7.4.1.2 Preparing the Envelope for the Pressure Test - Seal All Openings through the Air Barrier

Temporarily close all perimeter windows, roof hatches and doors in the envelope perimeter except for those doors that are to remain open to accommodate blower door or trailer mounted fan test equipment installation. Seal, or isolate all other intentional openings, pathways, and fenestrations through the architectural envelope prior to pressure testing. Follow the Recommended Test Envelope Conditions identified in ASTM E1827, Table 1, for the Closed Envelope condition. These openings may include boiler flues, fuel-burning water heater flues, fuel-burning kitchen equipment, clothes dryer vents, fireplaces, wall or ceiling grilles, diffusers, and other similar openings. Before sealing flues, close their associated fuel valves and verify the associated pilot lights are extinguished. Prime all plumbing traps located within the envelope full of water. In lieu of applying tape or plastic, typical temporary sealing materials include tape and sheet plastic or a self-adhesive grille wrap. Use and apply tape and plastic in a manner that does not deface or remove paint or mar the finish of permanent surfaces. Be especially aware of residue that remains from tape applied to stainless steel surfaces such as kitchen hoods or rollup doors. For painted surfaces, use tape types that do not remove finish paint when the tape is removed. If paint is removed from the finished surface, repaint to match existing surfaces. Secure dampers closed either manually or by using the building's HVAC system controls. Use the table below for further guidance in building preparation.

Building Component	Envelope Condition
Air handling units, duct fans	As found (open) or temporarily sealed as necessary
Clothes dryer	Off
Clothes dryer vents	Temporarily sealed
Dampers - intake, exhaust	Physically closed or closed using control power or temporarily sealed
Diffusers, registers, grilles within the envelope	Temporarily sealed
Doors, personnel type, at the envelope perimeter	Secured closed
Doors, personnel type, within the envelope	Secured (propped) open
Doors, roll-up type, at the envelope perimeter	Closed (no additional sealing)
Exhaust hoods	Closed* and temporarily sealed

Building Component	Envelope Condition
Fireplace hearth	Temporarily sealed *
Kitchen hoods	Temporarily sealed *
Pilot light and associated fuel valve	Extinguished and closed, respectively
Vented combustion appliance	Temporarily sealed *
Vented combustion appliance exhaust flue	Off
Windows	Secured closed
* If the building component has an associated manual or automatic damper, consider securing the damper closed in lieu of temporarily sealing.	

3.7.4.2 Architectural Plus HVAC System Test

This test envelope includes the architectural air barrier boundary as defined on the Contract drawings plus all HVAC supply, return and exhaust systems that penetrate and terminate within said architectural air barrier boundary and that extends outward from said boundary. All associated ductwork, intake and exhaust dampers, and air moving devices, including air handling units and fans, are included in this test envelope even if they are physically located outside of the architectural air barrier boundary. The boundary extends to and includes the low leakage intake and exhaust dampers. Perform both a positive pressure test and a negative pressure test on this envelope, unless otherwise indicated.

3.7.4.2.1 Test Goal

Data from the test is to be input into the Air Leakage Rate by Fan Pressurization spreadsheet as described in paragraph CALCULATION PROGRAM via the Air Leakage Test Form. If both positive and negative pressure tests were performed, both data sets are to be input in the spreadsheet. Compare output from the spreadsheet against the leakage rate goal. The envelope passes the test if the leakage rate, as calculated using the spreadsheet, is equal to or lower than the Architectural Plus HVAC System leakage rate goal.

3.7.4.2.2 Preparing the Building for the Pressure Test

In preparation for this test, de-energize all air moving devices within this envelope by putting their controls in the Unoccupied mode. This allows the building's HVAC controls to close all associated motorized intake, exhaust, and relief dampers. Make no other changes to the HVAC systems. Temporarily sealing diffusers, grilles, registers, kitchen hoods, exhaust hoods, fans, air handling units and all other HVAC system elements with tape or plastic sheeting or any other means is not allowed. If the envelope includes a fireplace hearth do not seal it with tape and plastic. Use the table below for further guidance in building preparation.

Building Component	Envelope Condition
Air handling units, duct fans	As found (open)
Clothes dryer	Off
Clothes dryer vents	As found (no preparation)
Dampers - intake, exhaust	As found (no preparation)
Diffusers, registers, grilles within the envelope	As found (open)
Doors, personnel type, at the envelope perimeter	Secured closed
Doors, personnel type, within the envelope	Secured (propped) open
Doors, roll-up type, at the envelope perimeter	Closed (no preparation)
Exhaust hoods	Closed
Fireplace hearth	As found (open)
Kitchen hoods	As found (open)
Pilot light and associated fuel valve	Extinguished and closed, respectively
Vented combustion appliance	Off
Vented combustion appliance exhaust flue	As found (open)
Windows	Secured closed

]3.7.5 Conducting The Pressure Test

Notify the Contracting Officer at least ten(10) working days before conducting the pressure tests to provide the Government the opportunity to witness the tests and to monitor weather forecasts for conditions favorable for testing. Do not pressure test until verifying that the continuous air barrier is in place and installed without failures in accordance with installation instructions. During the pressure test periodically inspect temporarily sealed items to ensure they are still sealed. Seals on temporarily sealed items tend to release more readily at higher pressures. Test data obtained after temporarily sealed items become unsealed cannot be used as input into the calculation program. Follow the Envelope Pressure Test Procedures in the paragraphs below. Submit detailed pressure test procedures indicating the test apparatus, the test methods and procedures, and the analysis methods to be employed for the building envelope pressure (air tightness) test. Submit these procedures not later than 60 days after Notice to Proceed.

3.7.5.1 Extend Pneumatic Tubes and Establish a Reference Differential Pressure

Confirm the various zones within the envelope have a relatively uniform interior pressure distribution by establishing a representative differential pressure between the envelope and the outdoors with blower door or trailer-mounted fans operating. The number of indoor pressure difference measurements (pneumatic hoses) required depends on the number of interior zones separated by bottle necks that could create significant pressure drops (e.g., doorways and stairwells). Extend at least four pneumatic hoses (differential pressure monitoring ports) to locations within the envelope that are physically opposite of each other. In multiple story buildings, especially those over three stories, extend hoses to multiple floors. Locate the hose ends away from the effects of air discharge from blower test equipment. Select one of the four (or more) interior hoses, one judged by the test agency to be the most unaffected by air velocity produced by blower test equipment, to serve as the interior reference pressure port. Extend at least one additional pneumatic hose to the outdoors (outdoor pressure port). To the end of this hose manifold at least four hoses together and terminate each hose on a different side of the building. With the envelope sealed and the blowers energized, measure the differential pressure using the interior reference pressure port and the four outdoor pressure ports. Then measure and record the differential pressure by individually using each of the remaining three interior hoses. Ensure each reading is within plus or minus 10 percent of the reference reading. Thus, at an average 75 Pa maximum pressure difference across the envelope, the difference between the highest and lowest interior pressure difference measurements should be 15 Pa or less. If this condition cannot be met, attempt to create additional air pathways within the envelope to minimize pressure differences within the envelope. If necessary, move the interior hose ends. See step 2.13 of the Air Leakage Test Form in Appendix A.

3.7.5.2 Bias Pressure Readings

With the fan pressurization equipment de-energized and the envelope sealed, obtain the differential pressure between the outdoors and the envelope. Record 12 bias pressure readings before the pressure test and 12 bias pressure readings after the pressure test. Each reading is the average of ten or more 1-second measurements. Include positive and negative signs for each reading. To help dampen bias pressures that significantly contribute to test pressure, reduce temperature differences between indoor and outdoor air. Temperature differences can be reduced by operating test fan equipment for a few minutes to replace most of the indoor air with outdoor air.

3.7.5.3 Testing in Both Positive and Negative Directions

The preferred method for testing a building envelope is to test in both the pressurized and depressurized directions. Testing in one direction is only allowed if opposite direction testing cannot logistically be performed due to test equipment limitations or restrictions. After obtaining the pre-test bias differential pressure readings, conduct the pressure test. Record the envelope pressures (in units of Pascals) from one interior pneumatic hose (monitoring port) and the outdoor pneumatic hose(s), averaged or manifolded, with corresponding flows (in units of cfm) for each fan. Record the flow rates at least 10 to 12 positive and 10 to 12 negative building pressure readings. If conducting both positive and negative pressure tests the lowest allowable test pressure is 40 Pa and the highest test pressure is 85 Pa. Keep at least 25 Pa difference between the lowest and highest test pressure readings. Include the 75 Pa pressure value between the lowest and highest readings. The 10 to 12 readings in each direction are to be roughly evenly spaced along the range of pressures and flows. After testing is complete de-energize the equipment used to provide pressurization and obtain an additional 10 to 12 post-test bias pressure readings. None of the bias pressure readings are allowed to exceed 30 percent of the minimum test pressure. If these limits are exceeded the test fails and must be repeated.

3.7.5.4 Pressure Testing - Special Cases

3.7.5.4.1 Pressure Testing a Multiple Isolated Zoned Building

Pressure test each exterior corner zone plus at least an additional 20 percent (as measured by floor area) of remaining zones. The Contracting Officer is responsible for selecting which of these additional zones to test. If all zones pass the pressure test it is assumed that all untested zones also pass and no further testing is required. If, however, any zone fails to pass the test's leakage requirements, re-seal, and re-test until it passes in accordance with paragraph FAILED PRESSURE TEST. Test an additional 20 percent of previously untested zones. If all tested zones pass, no further testing is needed. If any zone in this group fails the test re-seal and re-test the zone until it passes. Continue this process until all the tested zones pass. When testing a zone, the doors to all adjacent zones that share a common surface with the tested zone are to have their doors opened to the outdoors. The resulting leakage from the test zoned is that through all six surfaces (4 walls, roof, and floor, for a rectangular shaped zone).

3.7.5.4.2 Pressure Testing a Building Addition

If the existing building is occupied, coordinate the pressure test with building representatives. In preparation of the test, de-energize the air handling system serving that portion of the existing building that shares surfaces with the new building addition. Pressure testing a new building addition may also require pressurizing that part of the existing building that shares surfaces in common with the new building addition. If an air barrier is applied to the common surfaces separating the existing building from the new addition, prior to the test prop open a sufficient quantity of doors and windows to keep the existing building at the same pressure as the outdoors. If an air barrier is not applied to the common surfaces separating the existing building from the new addition, pressurize that part of the existing building that shares surfaces in common with the building addition to the same level as the as the addition using separate test pressurization equipment.

3.7.5.5 Failed Pressure Test

If the pressure test fails to meet the established criteria, use diagnostic test methods described in paragraph LOCATING LEAKS BY DIAGNOSTIC TESTING to discover the leak locations. Provide additional permanent sealing measures to reduce or eliminate leak sources discovered during diagnostic testing. Retest (perform another pressure test) after sealing has been completed. Repeat this sequence of documenting test results in the test report, performing diagnostic tests, documenting recommendations for additional sealing measures in the test report, sealing leak locations per recommendations, and re-testing as necessary until the building envelope passes the pressure test and is in compliance with the performance requirements.

3.7.5.6 Air Leakage Test Report

Report volumetric flow rates and corresponding differential pressures in cubic feet per minute (cfm) and Pascals (Pa), respectively, on the Air Leakage Test Form sample form found in Appendix A. Populate the accompanying spreadsheet file titled Pressure Test Data Analysis with information obtained during the test. The spreadsheet uses equations found in ASTM E779 as a basis for calculating the envelope leakage rate. Other similar leakage rate calculation programs cannot be used or submitted for review. Submit a printout of the data input and output in the report. Should any air tightness (pressure) test fail, the pressure test report is to include data and results from all previous failed tests along with the final successful test data and results. Indicate if the resulting leakage rate did or did not meet the goal leakage requirement. Identify and document deficiencies in the building construction upon failure of a test to meet the specified maximum leakage rate.

Test reports on leakage tests performed using fan pressurization must meet ASTM E779 requirements; reports on tests performed utilizing blower door testing must meet ASTM E1827 requirements. Include the Test Agency Qualification Sheet, Air Leakage Test Form and Air Leakage Test Results Form in the written report. Document every test set-up condition with diagrams and photos to ensure the tests can be made repeatable. Document all pneumatic hose termination locations. Record in detail how the building envelope was prepared for the tests. Also describe in detail which building items were temporarily sealed. Include photos of test equipment and sealing measures in the report. Include an electronic (pdf) version of all test reports on a CD. If the building envelope fails to meet the leakage rate goal, provide recommendations to further seal the envelope and document these recommendations in the test report.

3.7.6 Locating Leaks By Diagnostic Testing

Use diagnostic test methods described herein to discover obvious leaks through the envelope. Perform diagnostic tests on the building envelope regardless of the envelope meeting or failing to meet the designated leakage rate goal. Use diagnostic test methods in accordance with ASTM E1186 and in conjunction with pressurization equipment, as necessary. Use the thermography diagnostic test to establish a baseline for envelope leakage. Apply additional diagnostic tests (find, feel, fog or other tests) as necessary to further define leak locations and pathways discovered using thermography or to find additional leaks not readily detected by thermography. Using a variety of diagnostic tests may help locate leaks that would otherwise go undetected if only a single diagnostic test were used. Pay special attention to locating leaks at interfaces where there is a change in materials or a change in direction of like materials. These interfaces, at a minimum, include roof/wall, wall/wall, floor/wall, wall/window, wall/door, wall/louver, roof mounted equipment/roof curb interfaces and all utility penetrations (such as ducts, pipes, and conduit) through the envelope's architecture. Also use diagnostic tests to check for leakage between the air duct and duct damper, when the damper, under normal control power, is placed in the closed position.

3.7.6.1 Sealing and Re-Testing

Should leaks be discovered during diagnostic tests, thoroughly document the exact leak locations on a floor plan so that sealing can be later applied, if required or as directed. If the envelope passes the leakage test, use the diagnostic test procedure described above to identify obvious leakage locations. Seal the leaks at the discretion of the COR based on the magnitude, location, potential for liquid moisture penetration or retention, potential for condensation, presence of daylight through an architectural surface or if the leakage location could potentially cause rapid deterioration or mold growth of, or in the building envelope materials and assemblies. Apply sealing measures after diagnostic testing is complete and all pressurization blowers are off. To verify that the applied sealing measures are effective, re-test for leaks using the same diagnostic methods that discovered the leak. Reseal and retest until the envelope meets the leakage rate goal and all obvious leaks through the envelope are sealed.

3.7.6.2 Find Test

Use visual observation to locate daylight or artificial light streaming from the opposite side of the envelope. Observe all interfaces identified above.

3.7.6.3 Feel Test

Use the building's air handling system or blower door equipment to negatively pressurize the building envelope, to at least 25 Pa but no greater than 85 Pa, with respect to the outdoors. The larger the pressure difference, the easier discovering leaks by feeling them becomes. While inside the envelope, hand feel roof/wall, wall/wall, and floor/wall interfaces and utility penetrations (such as ducts, pipes and conduit) for leaks and note the leak locations on a floor plan. The "Feel" test may also be used to check for leaks between the ductwork and ductwork damper. To do this, positively pressurize the envelope and check for air movement from the envelope exterior.

3.7.6.4 Infrared Thermography Test

Coordinate thermography examination with the pressure test agency and the test agency's pressurization equipment. The pressure test agency is to allow adequate time for the thermographer to perform a complete thermographic examination, as described hereinafter, of the envelope interior and exterior, including readings of the roof.

3.7.6.4.1 Thermography Test Methods

Before thermographic testing, remove furniture, construction equipment, and all other obstructions both inside and outside the building as necessary to gain a clear field of view. In the Thermographic Investigation Report, document all areas where obstructions remain. For exterior thermal examination of the envelope, including readings of the roof, verify that no direct solar radiation has heated the envelope surfaces to be examined for a period of approximately 3 hours for frame construction and for approximately 8 hours for masonry veneer construction. Conduct exterior investigations after sunset, before sunrise, or on an overcast day when the influence of solar radiation can be determined to be minimal. Limit exterior examinations to times when the influence of solar radiation is minimal, such as after sunset or before sunrise or during an overcast day. Conduct thermal imaging tests only when wind speeds are less than 8 mph at the time of analysis and at the end of analysis. Document any variations in wind during the test. Document all variations of test conditions in the Thermographic Investigation Report. Test only when exterior surfaces are dry. Monitor and document ongoing test parameters, such as the temperatures inside and outside the air barrier envelope, wind speed, and differential pressure.

3.7.6.4.1.1 Thermography Testing of the Air Barrier

Test the building envelope in accordance with ISO 6781, and ASTM E1186. Perform a complete thermographic inspection consisting of the full inspection of the interior and exterior of the complete air barrier envelope, to include readings of the roof. Document envelope areas that are inaccessible for testing. Use infrared thermography technology in concert with standard pressurization methods (blower doors, trailer mounted fans or the building's own air handling systems) to locate leaks through the air barrier. Adjust the HVAC system, if possible, to create or enhance the temperature difference between the envelope interior and exterior. The minimum allowable temperature difference is 9 degrees F. Maintain this temperature difference for at least 3 hours prior to the test. Use pressurization methods to establish a minimum of +20 Pa pressure difference with respect to the outdoors while using an infrared camera to view the envelope from outdoors. When viewing with the camera from inside the envelope, keep the envelope at a pressure differential of -20 Pa with respect to the outdoors using pressure testing equipment or the building's own air handling system.

3.7.6.4.2 Thermography Test Results

Document the location of all leaks, anomalies, and unusual thermal features on a floor plan or elevation view and catalog them with a visible light picture for locating the defect for correction. The thermographer is to recommend corrective actions to eliminate the leaks, anomalies, and unusual thermal features. Where leaks are found perform corrective sealing as necessary to achieve the whole envelope air leakage rate specified. After sealing, again use thermography in concert with standard pressurization methods to verify that the air leakage has been reduced. After these leaks have been permanently sealed, note all actions taken on the drawings or in the Thermographic Investigation Report. Submit the drawings for approval as part of the Thermographic Investigation Report. Also include thermographic photos that show where leaks were discovered. Include thermograms using an imaging palette that clearly shows the observed thermal patterns indicating air leakage. The Contracting Officer's Representative is to witness all testing.

3.7.6.5 Fog Test

Before using a theatrical fog generator, disable all building smoke detectors as they may alarm when fog is issued. Coordinate fog tests and the disabling of all smoke detectors with the Contracting Officer's representative and the local fire department as necessary. Use pressure test equipment or the buildings own air handling system to positively pressurize the building envelope to at least 25 Pa but not greater than 85 Pa over the outdoors. Using a theatrical fog generator within the envelope, direct fog at suspected leakage points such as at building interfaces. Test the following interfaces: roof/wall, wall/wall, floor/wall, wall/window, roof/mounted mechanical equipment. From the vantage point immediately outside the envelope and opposite that of the interface being tested, observe the effect as the fog is issued. Detection may also be further enhanced by using a scented fog liquid or a fog liquid that produces a colored fog. Look for fog and smell for associated odor percolating through the interface. Also use smoke puffers and smoke sticks as necessary to locate leaks at these and other interface locations. If the Architectural Plus HVAC System pressure test will be/was performed introduce fog into ductwork to check for leakage between ductwork and associated dampers. After fog testing has ended, reactivate the building smoke detectors, and notify the Contracting Officer and local fire department that the test has ended. After sealing has been completed retest these areas using fog. Seal additional leaks that are found. (Note that theatrical fog may release quantities of the water vapor, glycol, mineral oil, or other media that could damage electronics and interior finishes or cause allergic reactions. Small scale smoke trace could be used as an alternative. See "Effects of Theatrical Smokes and Fogs on Respiratory Health in the Entertainment Industry" at www.nih.gov.)

3.7.6.6 Diagnostic Test Report

Once the diagnostic tests have been completed and the leakage locations identified and sealed, document these procedures, locations, and recommendations in the diagnostic test report. Submit plan and profile drawings that thoroughly identify leak locations. Describe in detail all leak locations so that the seal-up crew knows where to apply sealing measures. After sealing measures have been applied, describe the methods used along with applicable photos of the final sealed condition.

3.7.6.6.1 Thermographic Investigation Report

Submit a report of each thermographic investigation identifying the thermal discontinuities in the thermal control layer. Indicate in the final report locations to which improvements for both the air control layer and the thermal control layer were made to reduce air leaks and correct discontinuities in the thermal control layer. Include in the report some selected radiometric images of suspected failure points in the air barrier envelope that indicate before and after conditions. Devote a chapter(s) of the Thermographic Investigation Report to identifying suspected points of thermal bridging, moisture migration through roofs and walls, and insulation voids. Indicate in the final report improvements that were made to the envelope to reduce air leaks. Include the following items in the report:

a. Brief description of the building construction

b. Types of interior and exterior surface materials used in the building.

c. Geographical orientation of the building with a description of the exterior surroundings including other buildings, vegetation, landscaping, and surface water drainage.

d. Camera brand, model and serial number, and most recent calibration date; optional lenses with serial numbers (if applicable)

e. Thermographer's and Government Inspector's names

f. Date and time of tests

g. Air temperature and humidity inside the air barrier envelope

h. Outdoor air temperature and humidity

i. General information for the last 12 hours on the solar radiation conditions in the geographic area where the test is being performed.

j. Ambient conditions such as precipitation and wind direction and speed occurring with the last 24 hours, as applicable. Refer to specific requirements in each section of each thermographic inspection type for requirements in each specific area.

k. Documentation of those portions of the building envelope which were not within test conditions when the scan was performed, and which portions were obstructed by adjacent structures, interior furnishings, intervening cavities, or reflective surfaces.

l. Other relevant information, which may have influenced test results.

m. Drawings, sketches, floor plans and photographs detailing the locations in the buildings where thermograms were taken detailing possible irregularities in the components being tested.

n. Thermal images taken during the inspection with their relative locations and written or voiced recorded explanations of the anomaly listed along with visual and reference images.

o. An identification of the aspects or components of the building being examined.

p. Explanations for the type and the extent of each construction defect observed during the inspection.

q. Any results from additional measurements and investigations. Identify additional equipment used and support with type, model number, serial number, and date of most recent calibrated.

3.7.6.6.2 Fog Test Report

Document all turbulent air flow and dead air spaces within the envelope. Report fog behavior as it exits from or is entrained within the building. Include a floor plan in the report that documents the locations where fog passed through the envelope.

3.7.7 Calculation Program

To calculate the envelope leakage rate and other required outputs, input the data obtained during the pressure tests as documented in the Air Leakage Test Form (Appendix A) into the Air Leakage Rate by Fan Pressurization Excel spreadsheet. This spreadsheet can be found at the following web site:
https://www.wbdg.org/dod/ufgs/ufgs-forms-graphics-tables.

3.7.8 After Completion Of The Pressure And Diagnostic Test

After all pressure and diagnostic testing has been completed unseal all temporarily sealed items. Unless otherwise directed by the Contracting Officer, return all dampers, doors, and windows to their pre-test condition. Remove tape and plastic from all temporarily sealed openings, being careful not to deface painted surfaces. If paint is removed from finished surfaces, repaint to match existing surfaces. Unless otherwise directed by the Contracting Officer's representative, return fuel (gas) valves to their pre-test position and relight pilot lights. Return all fans and air handling units to pre-test conditions. Restore smoke/fire detectors to operating condition.

3.7.9 Repair And Protection

Repair and protection are the Contractor's responsibility, regardless of the assignment of responsibility for testing, inspection, and similar services. Upon completion of inspection, testing, or sample taking and similar services, repair damaged construction and restore substrates and finishes, protect construction exposed by or for quality control service activities, and protect repaired construction.

3.8 TRAINING

The Commissioning Provider is responsible for reviewing and approving the training plan required by Section 01 78 00.00 22 OPERATION AND MAINTENANCE (PWD ME) DATA and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel.

Provide, coordinate, schedule, and document training for all commissioned systems as required by Section 01 91 00.15 22 TOTAL BUILDING COMMISSIONING paragraph titled "Training".

3.9 FINAL BECx REPORT

Provide a final BECx report that includes information generated by the BECx process including: BECx meeting minutes, test reports, final BECx Plan, site observations, as-built drawings (provided by the Contractor), submittals, record of training, and recommended preventive maintenance actions and intervals. These documents will be combined with the design review reports, OPR, BoD, issue and resolution logs and other documents as required by Section 01 91 00.15 22 in order to complete the documentation of the BECx process.

The government's CxC will ensure the Final BECx Report is submitted as a portion of the Cx Firm's submittal of the Final Commissioning Report for the project, as defined in Section 01 91 00.15 22 TOTAL BUILDING COMMISSIONING. Submit the BECx Report within 14 calendar days of completion of all commissioning efforts.

3.10 APPENDICES

The following forms are available for download as a MS Word file at https://www.wbdg.org/dod/ufgs/ufgs-forms-graphics-tables.

Appendix A - Air Leakage Test Form
Appendix B - Air Leakage Test Results Form
Appendix C - Test Agency Qualifications Sheet

-- End of Section --

SECTION 02 41 00

DEMOLITION AND DECONSTRUCTION
08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI Guideline K	(2009) Guideline for Containers for Recovered Non-Flammable Fluorocarbon Refrigerants

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.6	(2006) Safety & Health Program Requirements for Demolition Operations - American National Standard for Construction and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety and Health Requirements Manual

U.S. DEFENSE LOGISTICS AGENCY (DLA)

DLA 4145.25	(Jun 2000; Reaffirmed Oct 2010) Storage and Handling of Liquefied and Gaseous Compressed Gases and Their Full and Empty Cylinders; https://www.dla.mil/Portals/104/Documents/DispositionS /ddsr/docs/cylinderjointpub.pdf

U.S. DEPARTMENT OF DEFENSE (DOD)

DOD 4000.25-1-M	(2006) MILSTRIP - Military Standard Requisitioning and Issue Procedures
MIL-STD-129	(2014; Rev R; Change 1 2018; Change 2 2019) Military Marking for Shipment and Storage

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 61	National Emission Standards for Hazardous Air Pollutants
40 CFR 82	Protection of Stratospheric Ozone
49 CFR 173.301	Shipment of Compressed Gases in Cylinders and Spherical Pressure Vessels

1.2 PROJECT DESCRIPTION

1.2.1 Definitions

1.2.1.1 Demolition

Demolition is the process of tearing apart and removing any feature of a facility together with any related handling and disposal operations.

1.2.1.2 Deconstruction

Deconstruction is the process of taking apart a facility with the primary goal of preserving the value of all useful building materials.

1.2.1.3 Demolition Plan

Demolition Plan is the planned steps and processes for managing demolition activities and identifying the required sequencing activities and disposal mechanisms.

1.2.1.4 Deconstruction Plan

Deconstruction Plan is the planned steps and processes for dismantling all or portions of a structure or assembly, to include managing sequencing activities, storage, re-installation activities, salvage and disposal mechanisms.

1.2.2 Demolition/Deconstruction Plan

Prepare a Demolition/Deconstruction Plan and submit proposed salvage, demolition, deconstruction, and removal procedures for approval before work is started. Include in the plan procedures for careful removal and disposition of materials specified to be salvaged, coordination with other work in progress, a disconnection schedule of utility services (outage plan), and a detailed description of methods and equipment to be used for each operation and of the sequence of operations. Identify components and materials to be salvaged for reuse or recycling with reference to paragraph Existing Facilities to be Removed. Append tracking forms for all removed materials indicating type, quantities, condition, destination, and end use. Coordinate with Waste Management Plan in accordance with Section 01 74 19.00 22 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL (PWD ME). Include statements affirming Contractor inspection of the existing roof deck and its suitability to perform as a safe working platform or if inspection reveals a safety hazard to workers, state provisions for securing the safety of the workers throughout the performance of the work. Provide procedures for safe conduct of the work in accordance with EM 385-1-1. Plan must be approved by Contracting Officer prior to work beginning.

1.2.3 General Requirements

Do not begin demolition or deconstruction until authorization is received from the Contracting Officer. The work of this section is to be performed in a manner that maximizes the value derived from the salvage and recycling of materials. Remove rubbish and debris from the project site; do not allow accumulations inside or outside the building. The work includes demolition, deconstruction, salvage of identified items and materials, and removal of resulting rubbish and debris. Remove rubbish and debris from Government property daily, unless otherwise directed. Store materials that cannot be removed daily in areas specified by the Contracting Officer. In the interest of occupational safety and health, perform the work in accordance with EM 385-1-1, Section 23, Demolition, and other applicable Sections.

1.3 ITEMS TO REMAIN IN PLACE

Comply with FAR 52.236-9 to protect existing structures, equipment, utilities, and improvements. Coordinate the work of this section with all other work indicated. Construct and maintain shoring, bracing, and supports as required. Ensure that structural elements are not overloaded. Increase structural supports or add new supports as may be required as a result of any cutting, removal, deconstruction, or demolition work performed under this contract. Do not overload structural elements or pavements to remain. Provide new supports and reinforcement for existing construction weakened by demolition, deconstruction, or removal work. Repairs, reinforcement, or structural replacement require approval by the Contracting Officer prior to performing such work.

1.3.1 Existing Construction Limits and Protection

Do not disturb existing construction beyond the extent indicated or necessary for installation of new construction. Provide temporary shoring and bracing for support of building components to prevent settlement or other movement. Provide protective measures to control accumulation and migration of dust and dirt in all work areas. Remove snow, dust, dirt, and debris from work areas daily.

1.3.2 Weather Protection

For portions of the building to remain, protect building interior and materials and equipment from the weather at all times. Where removal of existing roofing is necessary to accomplish work, have materials and workmen ready to provide adequate and temporary covering of exposed areas.

1.3.3 Utility Service

Maintain existing utilities indicated to stay in service and protect against damage during demolition and deconstruction operations. Prior to start of work, utilities serving each area of alteration or removal will be shut off by the Government and disconnected and sealed by the Contractor. The Government will disconnect and seal utilities serving each area of alteration or removal upon written request from the Contractor.

1.3.4 Facilities

Protect electrical and mechanical services and utilities. Where removal of existing utilities and pavement is specified or indicated, provide approved barricades, temporary covering of exposed areas, and temporary services or connections for electrical and mechanical utilities. Floors, roofs, walls, columns, pilasters, and other structural components that are designed and constructed to stand without lateral support or shoring, and are determined to be in stable condition, must remain standing without additional bracing, shoring, or lateral support until demolished or deconstructed, unless directed otherwise by the Contracting Officer. Ensure that no elements determined to be unstable are left unsupported and place and secure bracing, shoring, or lateral supports as may be required as a result of any cutting, removal, deconstruction, or demolition work performed under this contract.

1.4 BURNING

The use of burning at the project site for the disposal of refuse and debris will not be permitted. Where burning is permitted, adhere to federal, state, and local regulations.

1.5 AVAILABILITY OF WORK AREAS

Areas in which the work is to be accomplished will be available at contract award.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

Demolition/Deconstruction Plan; G

Existing Conditions

Qualifications of Jacking Contractor; G

Photographic Documentation; G

SD-02 Shop Drawings

Temporary shoring design drawings; G

Jacking design drawings; G

SD-05 Design Data

Temporary shoring design calculations; G

Jacking design calculations; G

SD-07 Certificates

Notification; G

SD-11 Closeout Submittals

Receipts

1.7 QUALITY ASSURANCE

Submit timely notification of demolition, deconstruction, and renovation projects to Federal, State, regional, and local authorities in accordance with 40 CFR 61, Subpart M. Notify the Regional Office of the United States Environmental Protection Agency (USEPA) State's environmental protection agency local air pollution control district/agency and the Contracting Officer in writing 10 working days prior to the commencement of work in accordance with 40 CFR 61, Subpart M. Comply with federal, state, and local hauling and disposal regulations. In addition to the requirements of the "Contract Clauses," conform to the safety requirements contained in ASSP A10.6. Comply with the Environmental Protection Agency requirements specified. Use of explosives will not be permitted.

1.7.1 Dust and Debris Control

Prevent the spread of dust and debris and avoid the creation of a nuisance or hazard in the surrounding area. Do not use water if it results in hazardous or objectionable conditions such as, but not limited to, ice, flooding, or pollution. Vacuum and dust the work area daily. Sweep pavements as often as necessary to control the spread of debris that may result in foreign object damage potential.

1.8 TEMPORARY PARTITIONS

Provide temporary partitions of wood or metal frame, heavy duty plastic or plywood sheathing as required to protect historic items to remain in place during demolition activities.

1.9 PROTECTION

1.9.1 Traffic Control Signs

a. Where pedestrian and driver safety is endangered in the area of removal work, use traffic barricades with flashing lights. Anchor barricades in a manner to prevent displacement by wind. Notify the Contracting Officer prior to beginning such work.

1.9.2 Protection of Personnel

Before, during and after the demolition and deconstruction work continuously evaluate the condition of the site specific features and structure being demolished and deconstructed and take immediate action to protect all personnel working in and around the project site. No area, section, or component of floors, roofs, walls, columns, pilasters, or other structural element will be allowed to be left standing without sufficient bracing, shoring, or lateral support to prevent collapse or failure while workmen remove debris or perform other work in the immediate area.

1.10 RELOCATIONS

Perform the removal and reinstallation of relocated items as indicated with workmen skilled in the trades involved. Repair or replace items to be relocated which are damaged by the Contractor with new undamaged items as approved by the Contracting Officer.

1.11 EXISTING CONDITIONS

Before beginning any demolition or deconstruction work, survey the site and examine the drawings and specifications to determine the extent of the work. Record existing conditions in the presence of the Contracting Officer showing the condition of structures and other facilities adjacent to areas of alteration or removal. Photographs or electronic images with a minimum resolution of 3072 x 2304 pixels, capable of a print resolution of 300 dpi, will be acceptable as a record of existing conditions. Include in the record the elevation of the top of foundation walls, finish floor elevations, possible conflicting electrical conduits, plumbing lines, alarms systems, the location and extent of existing cracks and other damage and description of surface conditions that exist prior to starting work. It is the Contractor's responsibility to verify and document all required outages which will be required during the course of work, and to note these outages on the record document. Submit survey results to the Contracting Officer.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 EXISTING FACILITIES TO BE REMOVED

Inspect and evaluate existing structures onsite for reuse. Disassemble existing construction scheduled to be removed for reuse. Dismantled and removed materials are to be separated, set aside, and prepared as specified, and stored or delivered to a collection point for reuse, remanufacture, recycling, or other disposal, as specified. Designate materials for reuse onsite whenever possible.

3.1.1 Utilities and Related Equipment

3.1.1.1 General Requirements

Do not interrupt existing utilities serving occupied or used facilities, except when authorized in writing by the Contracting Officer. Do not interrupt existing utilities serving facilities occupied and used by the Government except when approved in writing and then only after temporary utility services have been approved and provided. Do not begin demolition or deconstruction work until all utility disconnections have been made. Shut off and cap utilities for future use, as indicated.

3.1.1.2 Disconnecting Existing Utilities

Remove existing utilities, as indicated and terminate in a manner conforming to the nationally recognized code covering the specific utility and approved by the Contracting Officer. When utility lines are encountered but are not indicated on the drawings, notify the Contracting Officer prior to further work in that area. Remove meters and related equipment and deliver to a location in accordance with instructions of the Contracting Officer.

3.1.2 Paving and Slabs

Remove concrete and asphaltic concrete paving and slabs as indicated. Provide neat sawcuts at limits of pavement removal as indicated. Move, grind and store pavement and slabs designated to be recycled and utilized in this project as directed by the Contracting Officer. Remove pavement and slabs not to be used in this project from the installation at Contractor's expense.

3.1.3 Roofing

Cut existing felts along straight lines. Remove roofing system as indicated without damaging the roof deck. Sequence work to minimize building exposure between demolition or deconstruction and new roof materials installation.

3.1.3.1 Temporary Roofing

Install temporary roofing and flashing as necessary to maintain a watertight condition throughout the course of the work. Remove temporary work prior to installation of permanent roof system materials unless approved otherwise by the Contracting Officer. Make provisions for worker safety during demolition, deconstruction, and installation of new materials as described in paragraphs entitled "Statements" and "Regulatory and Safety Requirements."

3.1.3.2 Reroofing

When removing the existing roofing system from the roof deck, remove only as much roofing as can be recovered by the end of the work day, unless approved otherwise by the Contracting Officer. Do not attempt to open the roof covering system in threatening weather. Reseal all openings prior to suspension of work the same day.

3.1.4 Masonry

Sawcut and remove masonry so as to prevent damage to surfaces to remain, to removed materials being salvaged and to facilitate the installation of new work. Where new masonry adjoins existing, abut or tie the new work into the existing construction as indicated. Provide square, straight edges and corners where existing masonry adjoins new work and other locations. Salvage masonry removed in whole blocks and store for reuse. .

3.1.5 Miscellaneous Metal

Salvage shop-fabricated items such as access doors and frames, steel gratings, metal ladders, wire mesh partitions, metal railings, metal windows and similar items as whole units. Salvage light-gage and cold-formed metal framing, such as steel studs, steel trusses, metal gutters, roofing and siding, metal toilet partitions, toilet accessories and similar items. Scrap metal is the Contractor's property. Recycle scrap metal as part of demolition and deconstruction operations. Provide separate containers to collect scrap metal and transport to a scrap metal collection or recycling facility, in accordance with the Waste Management Plan.

3.1.6 Carpentry

Salvage for reuse or recycle lumber, millwork items, and finished boards as indicated. Salvage windows, doors, and similar items as whole units, complete with trim and accessories. Salvage hardware attached to units for reuse. Brace the open end of door frames to prevent damage.

3.1.7 Acoustic Ceiling Tile

Remove, neatly stack, and recycle acoustic ceiling tiles. Recycling may be available with manufacturer. Otherwise, give priority to a local recycling organization. Recycling is not required if the tiles contain or may have been exposed to asbestos material.

3.1.8 Patching

Where removals leave holes and damaged surfaces exposed in the finished work, patch and repair these holes and damaged surfaces to match adjacent finished surfaces, using on-site materials when available. Where new work is to be applied to existing surfaces, perform removals and patching in a manner to produce surfaces suitable for receiving new work. Make finished surfaces of patched area flush with the adjacent existing surface and match the existing adjacent surface as closely as possible to texture and finish. Provide patching as specified and indicated, and include the following:

a. Concrete and Masonry: Completely fill holes and depressions, caused by previous physical damage or left as a result of removals in existing masonry walls to remain, with an approved historic masonry patching material, applied in accordance with the manufacturer's printed instructions.

3.1.9 Air Conditioning Equipment

Remove air conditioning, refrigeration, and other equipment containing refrigerants without releasing chlorofluorocarbon refrigerants to the atmosphere in accordance with the Clean Air Act Amendment of 1990. Recover all refrigerants prior to removing air conditioning, refrigeration, and other equipment containing refrigerants and dispose of in accordance with the paragraph entitled "Disposal of Ozone Depleting Substance (ODS)."

3.1.10 Cylinders and Canisters

Remove all fire suppression system cylinders and canisters and dispose of in accordance with the paragraph entitled "Disposal of Ozone Depleting Substance (ODS)."

3.1.11 Locksets on Swinging Doors

Remove all locksets from all swinging doors indicated to be removed and disposed of. Deliver the locksets and related items to a designated location for receipt by the Contracting Officer after removal.

3.1.12 Mechanical Equipment and Fixtures

Disconnect mechanical hardware at the nearest connection to existing services to remain, unless otherwise noted. Disconnect mechanical equipment and fixtures at fittings. Remove service valves attached to the unit. Salvage each item of equipment and fixtures as a whole unit; listed, indexed, tagged, and stored. Salvage each unit with its normal operating auxiliary equipment. Transport salvaged equipment and fixtures, including motors and machines, to a designated storage area as directed by the Contracting Officer. Do not remove equipment until approved. Do not offer low-efficiency equipment for reuse; provide to recycling service for disassembly and recycling of parts.

3.1.12.1 Preparation for Storage

Remove water, dirt, dust, and foreign matter from units; drain tanks, piping and fixtures; if previously used to store flammable, explosive, or other dangerous liquids, steam clean interiors. Seal openings with caps, plates, or plugs. Secure motors attached by flexible connections to the unit. Change lubricating systems with the proper oil or grease.

3.1.12.2 Piping

Disconnect piping at unions, flanges and valves, and fittings as required to reduce the pipe into straight lengths for practical storage. Store salvaged piping according to size and type. If the piping that remains can become pressurized due to upstream valve failure, attach end caps, blind flanges, or other types of plugs or fittings with a pressure gage and bleed valve to the open end of the pipe to ensure positive leak control. Carefully dismantle piping that previously contained gas, gasoline, oil, or other dangerous fluids, with precautions taken to prevent injury to persons and property. Store piping outdoors until all fumes and residues are removed. Box prefabricated supports, hangers, plates, valves, and specialty items according to size and type. Wrap sprinkler heads individually in plastic bags before boxing. Classify piping not designated for salvage, or not reusable, as scrap metal.

3.1.12.3 Ducts

Classify removed duct work as scrap metal.

3.1.12.4 Fixtures, Motors and Machines

Remove and salvage fixtures, motors and machines associated with plumbing, heating, air conditioning, refrigeration, and other mechanical system installations. Salvage, box and store auxiliary units and accessories with the main motor and machines. Tag salvaged items for identification, storage, and protection from damage. Classify broken, damaged, or otherwise unserviceable units and not caused to be broken, damaged, or otherwise unserviceable as debris to be disposed of by the Contractor.

3.1.13 Electrical Equipment and Fixtures

Salvage motors, motor controllers, and operating and control equipment that are attached to the driven equipment. Salvage wiring systems and components. Box loose items and tag for identification. Disconnect primary, secondary, control, communication, and signal circuits at the point of attachment to their distribution system.

3.1.13.1 Fixtures

Remove and salvage electrical fixtures. Salvage unprotected glassware from the fixture and salvage separately. Salvage incandescent, mercury-vapor, and fluorescent lamps and fluorescent ballasts manufactured prior to 1978, boxed and tagged for identification, and protected from breakage.

3.1.13.2 Electrical Devices

Remove and salvage switches, switchgear, transformers, conductors including wire and nonmetallic sheathed and flexible armored cable, regulators, meters, instruments, plates, circuit breakers, panelboards, outlet boxes, and similar items. Box and tag these items for identification according to type and size.

3.1.13.3 Wiring Ducts or Troughs

Remove and salvage wiring ducts or troughs. Dismantle plug-in ducts and wiring troughs into unit lengths. Remove plug-in or disconnecting devices from the busway and store separately.

3.1.13.4 Conduit and Miscellaneous Items

Salvage conduit except where embedded in concrete or masonry. Consider corroded, bent, or damaged conduit as scrap metal. Sort straight and undamaged lengths of conduit according to size and type. Classify supports, knobs, tubes, cleats, and straps as debris to be removed and disposed.

3.1.14 Elevators and Hoists

Remove elevators, hoists, and similar conveying equipment including plunger, oil, and casing and dispose of accordingly. Remove the existing elevator plunger and hydraulic fluid from the existing in-ground casing. The existing in-ground casing must be completely filled with flowable concrete.

3.1.15 Items With Unique/Regulated Disposal Requirements

Remove and dispose of items with unique or regulated disposal requirements in the manner dictated by law or in the most environmentally responsible manner.

3.2 TEMPORARY SHORING

Provide temporary shoring of the building structure as indicated and as required by Contractor's means and methods. Temporary shoring must be designed by a licensed professional engineer. Submit temporary shoring design calculations and temporary shoring design drawings signed and sealed by the licensed professional engineer responsible for their preparation.

Foundations: Temporary shoring must be designed to limit movement of the existing foundation structure to less than 1/4-inch horizontal or vertical. Provide daily monitoring for vertical and horizontal movements of the foundations walls and temporary shoring system.

Construction activities that cause vibrations, such as site preparation, backfilling, or compaction may affect settlement of the existing building structure. Evaluate the effect of these vibrations for potential settlement on the existing structure and monitor the existing structure during activities that produce vibrations to ensure movement is limited.

Conduct a pre-construction survey that includes photographic documentation of the building and nearby buildings to document the existing condition of the exterior masonry and stone construction. If existing cracks exist, monitor each crack for the duration of the construction activities to ensure no change has occurred in the existing cracks. Stop work and notify the Contracting Officer if changes occur in any of the existing cracks.

3.3 WOOD FRAMING JACKING

Provide wood framing jacking of the existing wood floor structure as indicated. Jacking design and procedures must be designed by a licensed professional engineer. Submit jacking design calculations and jacking design drawings signed and sealed by the licensed professional engineer responsible for their preparation. Submit qualifications of jacking contractor.

3.3.1 Photographic Documentation

Provide photographic documentation of the existing granite veneer and existing third floor wood framing prior to floor jacking. Submit photographic documentation in digital format. The quantity of photographs must be sufficient to show all surfaces in sufficient detail of the granite veneer blocks, veneer joints, and the wood timber truss members and connections. Upon completion of the floor jacking, provide a final condition set of photographic documentation in digital format. Identify any damage that has resulted due to the floor jacking process.

3.4 DISPOSITION OF MATERIAL

3.4.1 Title to Materials

Except for salvaged items specified in related Sections, and for materials or equipment scheduled for salvage, all materials and equipment removed and not reused or salvaged, become the property of the Contractor and must be removed from Government property. Materials approved for storage by the Contracting Officer must be removed before completion of the contract. Title to materials resulting from demolition and deconstruction, and materials and equipment to be removed, is vested in the Contractor upon approval by the Contracting Officer. The Government will not be responsible for the condition or loss of, or damage to, such property after contract award. Showing for sale or selling materials and equipment on site is prohibited.

3.4.2 Reuse of Materials and Equipment

Remove and store materials and equipment indicated to be reused or relocated to prevent damage, and reinstall as the work progresses. Coordinate the re-use of materials and equipment with the re-use requirements in accordance with Section 01 74 19.00 22 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL (PWD ME). Capture re-use of materials in the diversion calculations for the project.

3.4.3 Salvaged Materials and Equipment

Remove materials and equipment that are indicated to be removed by the Contractor and that are to remain the property of the Government, and deliver to a storage site as directed by the Contracting Officer.

a. Salvage items and material to the maximum extent possible.

b. Store all materials salvaged for the Contractor as approved by the Contracting Officer and remove from Government property before completion of the contract. Coordinate the salvaged materials with tracking requirements in accordance with Section 01 74 19.00 22 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL (PWD ME). Capture salvaged materials in the diversion calculations for the project.

c. Remove salvaged items to remain the property of the Government in a manner to prevent damage, and packed or crated to protect the items from damage while in storage or during shipment. Items damaged during removal or storage must be repaired or replaced to match existing items. Properly identify the contents of containers.

d. Remove historical items in a manner to prevent damage. Deliver the historical items to the Government for disposition as indicated.

e. Remove and capture all Class I ODS refrigerants in accordance with the Clean Air Act Amendment of 1990, and turn in to the Navy by shipping the refrigerant container to the Defense Logistics Agency at the following address:

Defense Depot Richmond VA (DDRV)
SW0400
Cylinder Operations
8000 Jefferson Davis Highway
Richmond, VA 23297-5900

3.4.4 Disposal of Ozone Depleting Substance (ODS)

Class I and Class II ODS are defined in Section, 602(a) and (b), of The Clean Air Act. Prevent discharge of Class I and Class II ODS to the atmosphere. Place recovered ODS in cylinders meeting AHRI Guideline K suitable for the type ODS (filled to no more than 80 percent capacity) and provide appropriate labeling. Remove recovered ODS from Government property and dispose of in accordance with 40 CFR 82. Dispose products, equipment and appliances containing ODS in a sealed, self-contained system (e.g. residential refrigerators and window air conditioners) in accordance with 40 CFR 82. Submit Receipts or bills of lading, as specified. Submit a shipping receipt or bill of lading for all containers of ozone depleting substance (ODS) shipped to the Defense Depot, Richmond, Virginia.

3.4.4.1 Special Instructions

No more than one type of ODS is permitted in each container. Apply a warning/hazardous label to the containers in accordance with Department of Transportation regulations. Provide a tag with the following information on all cylinders including but not limited to fire extinguishers, spheres, or canisters containing an ODS:

a. Activity name and unit identification code

b. Activity point of contact and phone number

c. Type of ODS and pounds of ODS contained

d. Date of shipment

e. National stock number (for information, call (804) 279-4525).

3.4.4.2 Fire Suppression Containers

Deactivate fire suppression system cylinders and canisters with electrical charges or initiators prior to shipment. Also, safety caps must be used to cover exposed actuation mechanisms and discharge ports on these special cylinders.

3.4.5 Transportation Guidance

Ship all ODS containers in accordance with MIL-STD-129, DLA 4145.25 (also referenced one of the following: Army Regulation 700-68, Naval Supply Instruction 4440.128C, Marine Corps Order 10330.2C, and Air Force Regulation 67-12), 49 CFR 173.301, and DOD 4000.25-1-M.

3.4.6 Unsalvageable and Non-Recyclable Material

Dispose of unsalvageable and non-recyclable noncombustible material off Government property.

3.5 CLEANUP

Remove debris and rubbish from project site and similar excavations. Remove and transport the debris in a manner that prevents spillage on streets or adjacent areas. Apply local regulations regarding hauling and disposal.

3.6 DISPOSAL OF REMOVED MATERIALS

3.6.1 Regulation of Removed Materials

Dispose of debris, rubbish, scrap, and other nonsalvageable materials resulting from removal operations with all applicable federal, state and local regulations as contractually specified in the Waste Management Plan. Storage of removed materials on the project site is prohibited.

3.6.2 Burning on Government Property

Burning of materials removed from demolished and deconstructed structures will not be permitted on Government property.

3.6.3 Removal from Government Property

Transport waste materials removed from demolished and deconstructed structures, except waste soil, from Government property for legal disposal. Dispose of waste soil per Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME).

3.7 REUSE OF SALVAGED ITEMS

Recondition salvaged materials and equipment designated for reuse before installation. Replace items damaged during removal and salvage operations or restore them as necessary to usable condition.

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SECTION 02 42 91

REMOVAL AND SALVAGE OF HISTORIC CONSTRUCTION MATERIALS
11/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1000	Air Contaminants
29 CFR 1926.55	Gases, Vapors, Fumes, Dusts, and Mists
40 CFR 261	Identification and Listing of Hazardous Waste

1.2 PROJECT DESCRIPTION

The work includes removal and salvage of identified historic items and materials, and removal of resulting rubbish and debris. General demolition of non-historic materials and removal of resulting rubbish and debris must comply with the requirements of Section 02 41 00 DEMOLITION AND DECONSTRUCTION. Store salvaged or recycled materials daily in areas and in a manner specified by the Contracting Officer. In the interest of conservation, pursue salvage and recycling to the maximum extent possible. Submit a Work Plan that includes procedures proposed for the accomplishment of the work. The Work Plan procedures must provide for safe conduct of the work, careful removal and disposition of materials specified to be salvaged or recycled, dust control, protection of property which is to remain undisturbed, coordination with other work in progress, and timely disconnection of utility services. Include a detailed description of the methods and equipment to be used for each operation, and the sequence of operations in the Work Plan. Work Plan must include an inventory of all materials to be stored and a labeling plan with place of origin and the identified storage location(s). Inventory and storage locations must be maintained by the Contractor and coordinated with the Contracting Officer throughout the project. Final disposition of materials will be reviewed at the Red Zone Meeting.

1.2.1 Dust Control

Control the amount of dust resulting from removal, salvage and demolition operations to prevent the spread of dust to occupied portions of the construction site, to avoid creation of a nuisance in the surrounding area and to minimize occupational exposures. Occupational exposures cannot exceed the requirements in 29 CFR 1910.1000 and 29 CFR 1926.55. Use of water to control dust will not be permitted when it will result in, or create, damage to existing building materials and hazardous or objectionable conditions such as ice, flooding and pollution.

1.2.2 Protection

1.2.2.1 Protection of Existing Historic Property

Survey the site and examine the drawings and specifications to determine the extent of work before beginning any removal, salvage or demolition work. Take necessary precautions to avoid damage to existing historic items that are to remain in place, to be reused, or to remain the property of the Government. Repair or restore items damaged by the Contractor to original condition, or replace, as approved by the Contracting Officer. Coordinate the work of this section with all other work and construct and maintain shoring, bracing and supports, as required. Ensure that structural elements are not overloaded and provide additional supports as may be required as a result of any cutting, removal, or demolition work performed under this contract.

1.2.2.2 Protection From the Weather

Protect the interior of buildings to remain and salvaged materials from the weather at all times. Store salvaged historic materials off the ground and under weathertight covering.

1.2.2.3 Environmental Protection

Comply with the requirements of Section 02 82 00.00 22 ASBESTOS REMEDIATION (PNS PROJECTS) and Section 02 83 00.00 22 MANAGEMENT OF LEAD, CADMIUM, AND CHROMIUM (VI) DURING RENOVATION, DEMOLITION, REMOVAL, AND ABATEMENT (PNS PROJECTS). Ensure a thorough inspection has been performed for hazardous materials prior to beginning work.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Work Plan; G

1.4 QUALIFICATIONS

Provide qualified workers trained and experienced in whole-building recycling, including removal and salvage of historic materials. Submit documentation of five consecutive years of work of this type with a list of similar projects identifying when, where, and for whom the work was done. Provide a current point-of-contact for identified references.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 HAZARDOUS MATERIALS

Unforeseen hazardous materials may exist in wall cavities, beneath floors, in chases, inside various components, as well as other building materials. Exercise extreme care when performing demolition and salvage operations to ensure unexpected hazardous materials area not inadvertently disturbed creating a potential exposure concern. If suspect hazardous materials are observed or encountered, stop work and notify the Contracting Officer immediately.

3.2 SALVAGED ITEMS

Salvage items to the maximum extent possible. Remove historic items to be salvaged from the structure prior to demolition work. Remove salvageable items by hand labor to the maximum extent possible. Do not damage historic portions of the structure to remain or items identified for salvage. Remove furnishings, equipment, and materials not scheduled for salvage or recycling prior to any salvaging procedures. Keep a complete recording of all salvaged materials including the condition of such materials before, and after, salvage operations.

3.2.1 Wood

Provide survey of and protect the following materials: architectural woodwork including second floor wood strip flooring and second floor beadboard ceiling.

3.2.2 Doors and Windows

Remove doors, windows, borrowed lights, and associated hardware and operating mechanisms intact (including glass) and salvage/repair in accordance with the contract drawings.

3.2.3 Finishes

Protect the following special or historic finishes: second floor wood strip flooring and second floor beadboard ceiling.

3.3 DISPOSITION OF MATERIALS

The Contractor, upon receipt of notice to proceed, is vested with the title to materials and equipment to be demolished, except Government salvage and historical items. The Government will not be responsible for the condition, loss or damage to such property after notice to proceed.

3.3.1 Material Salvaged for the Contractor

Temporarily store salvaged material as approved by the Contracting Officer and remove from Government property before completion of the contract. Sale of salvaged material on the site is prohibited.

3.3.2 Items Salvaged for the Government

Remove salvaged items to remain the property of the Government in a manner to prevent damage, packed or crated to protect the items from damage, or as directed by the Contracting Officer. Repair or replace items damaged during removal or storage to match existing items. Properly label and identify containers as to contents. Upon completion of construction deliver salvaged items to location at PNSY as identified by the Contracting Officer.

3.4 CLEAN-UP

Upon completion of the work, clean portions of structure to remain and adjacent areas and structures of dust, dirt, and debris caused by salvage and demolition operations. Verify that debris and rubbish created by the work is non-hazardous. If any debris and rubbish is suspect hazardous waste characterize it in accordance with 40 CFR 261. If the debris and rubbish is determined to be hazardous materials notify the Contracting Officer. Remove and transport non-hazardous debris and rubbish in a manner that prevents spillage on streets or adjacent areas. Transport and dispose of all material in accordance with all local, state and Federal regulations. Provide copies of all disposal manifests to the the Contracting Officer.

-- End of Section --

SECTION 02 82 00.00 22

ASBESTOS REMEDIATION
(PNS PROJECTS)
06/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP Z9.2	(2018) Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems

ASTM INTERNATIONAL (ASTM)

ASTM C732	(2006; R 2012) Aging Effects of Artificial Weathering on Latex Sealants
ASTM D2794	(1993; R 2010) Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)
ASTM D4397	(2016) Standard Specification for Polyethylene Sheeting for Construction, Industrial, and Agricultural Applications
ASTM D522/D522M	(2014) Mandrel Bend Test of Attached Organic Coatings
ASTM E119	(2018) Standard Test Methods for Fire Tests of Building Construction and Materials
ASTM E1368	(2014) Visual Inspection of Asbestos Abatement Projects
ASTM E736/E736M	(2017) Standard Test Method for Cohesion/Adhesion of Sprayed Fire-Resistive Materials Applied to Structural Members
ASTM E84	(2018) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E96/E96M	(2016) Standard Test Methods for Water Vapor Transmission of Materials

COMPRESSED GAS ASSOCIATION (CGA)

CGA G-7	(2014) Compressed Air for Human

	Respiration; 6th Edition

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z87.1	(2015) Occupational and Educational Personal Eye and Face Protection Devices

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 701	(2019) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

NIOSH NMAM	(2016; 5th Ed) NIOSH Manual of Analytical Methods

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(Latest Revision) Safety and Health Requirements Manual

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 340/1-90/018	(1990) Asbestos/NESHAP Regulated Asbestos Containing Materials Guidance
EPA 560/5-85-024	(1985) Guidance for Controlling Asbestos-Containing Materials in Buildings (Purple Book)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.147	Control of Hazardous Energy (Lock Out/Tag Out)
29 CFR 1926.103	Respiratory Protection
29 CFR 1926.1101	Asbestos
29 CFR 1926.200	Accident Prevention Signs and Tags
29 CFR 1926.51	Sanitation
29 CFR 1926.59	Hazard Communication
40 CFR 61-SUBPART A	General Provisions
40 CFR 61-SUBPART M	National Emission Standard for Asbestos
40 CFR 763	Asbestos
42 CFR 84	Approval of Respiratory Protective Devices
49 CFR 107	Hazardous Materials Program Procedures
49 CFR 171	General Information, Regulations, and Definitions

49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements

U.S. NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

NAVFAC P-502	(2017) Asbestos Program Management
ND OPNAVINST 5100.23	(2005; Rev G) Navy Occupational Safety and Health (NAVOSH) Program Manual

UNDERWRITERS LABORATORIES (UL)

UL 586	(2009; Reprint Dec 2017) UL Standard for Safety High-Efficiency Particulate, Air Filter Units

STATE OF MAINE, DEPARTMENT OF ENVIRONMENTAL PROTECTION (DEP)

ME L&C PROGRAM	State of Maine Licensing and Certification Program http://www.mainelegislature.org/legis/statutes/38/titl 3
06 096 CMR	425 Asbestos Management Regulations http://www.maine.gov/sos/cec/rules/06/096/096c425.

1.2 DEFINITIONS

1.2.1 ACM

Asbestos Containing Materials.

1.2.2 Amended Water

Water containing a wetting agent or surfactant with a maximum surface tension of 0.00042 psi.

1.2.3 Area Sampling

Sampling of asbestos fiber concentrations which approximates the concentrations of asbestos in the theoretical breathing zone but is not actually collected in the breathing zone of an employee.

1.2.4 Asbestos

The term asbestos includes chrysotile, amosite, crocidolite, tremolite asbestos, anthophyllite asbestos, and actinolite asbestos and any of these minerals that has been chemically treated or altered. Materials are considered to contain asbestos if the asbestos content of the material is determined to be equal to or greater than one percent.

1.2.5 Asbestos Control Area

That area where asbestos removal operations are performed which is isolated by physical boundaries which assist in the prevention of the uncontrolled release of asbestos dust, fibers, or debris.

1.2.6 Asbestos Fibers

Those fibers having an aspect ratio of at least 3:1 and longer than 5 micrometers as determined by National Institute for Occupational Safety and Health (NIOSH) Method 7400.

1.2.7 Asbestos Permissible Exposure Limit

0.1 fibers per cubic centimeter of air as an 8-hour time weighted average measured in the breathing zone as defined by 29 CFR 1926.1101 or other Federal legislation having legal jurisdiction for the protection of workers health.

1.2.8 Authorized Person

Any person authorized by the Contractor and required by work duties to be present in the regulated areas.

1.2.9 Background

The ambient airborne asbestos concentration in an uncontaminated area as measured prior to any asbestos hazard abatement efforts. Background concentrations for other (contaminated) areas are measured in similar but asbestos free locations.

1.2.10 Competent Person (CP)

A person meeting the requirements for competent person as specified in 29 CFR 1926.1101 including a person capable of identifying existing asbestos hazards in the workplace and selecting the appropriate control strategy for asbestos exposure, who has the authority to take prompt corrective measures to eliminate them, and is specifically trained in a training course which meet the criteria of EPA's Model Accreditation Plan (40 CFR 763) for project designer or supervisor, or its equivalent. The competent person must have a current State of Maine asbestos contractor's or supervisor's license.

1.2.11 Contractor

The Contractor is that individual, or entity under contract to perform the herein listed work.

1.2.12 Disposal Bag

A 6 mil thick, leak-tight plastic bag, pre-labeled in accordance with 29 CFR 1926.1101, used for transporting asbestos waste from containment to disposal site.

1.2.13 Disturbance

Activities that disrupt the matrix of ACM, crumble or pulverize ACM, or generate visible debris from ACM. Disturbance includes cutting away small amounts of ACM, no greater than the amount which can be contained in one standard sized glovebag or waste bag, not larger than 60 inches in length and width in order to access a building component.

1.2.14 Encapsulation

The abatement of an asbestos hazard through the appropriate use of chemical encapsulants.

1.2.15 Encapulants

Specific materials in various forms used to chemically or physically entrap asbestos fibers in various configurations to prevent these fibers from becoming airborne. There are four types of encapsulants as follows which must comply with performance requirements as specified herein.

a. Removal Encapsulant (can be used as a wetting agent)

b. Bridging Encapsulant (used to provide a tough, durable surface coating to asbestos containing material)

c. Penetrating Encapsulant (used to penetrate the asbestos containing material encapsulating all asbestos fibers and preventing fiber release due to routine mechanical damage)

d. Lock-Down Encapsulant (used to seal off or "lock-down" minute asbestos fibers left on surfaces from which asbestos containing material has been removed).

1.2.16 Friable Asbestos Material

A term defined in 40 CFR 61-SUBPART M and EPA 340/1-90/018 meaning any material which contains equal to or greater than 1 percent asbestos, as determined using the method specified in 40 CFR 763, Polarized Light Microscopy (PLM), that when dry, can be crumbled, pulverized, or reduced to powder by hand pressure.

1.2.17 Glovebag Technique

Those asbestos removal and control techniques put forth in 29 CFR 1926.1101.

1.2.18 Government Consultant (GC)

That qualified person employed directly by the Government to monitor, sample, inspect the work or in some other way advise the Contracting Officer. The GC is normally a private consultant, but can be an employee of the Government.

1.2.19 HEPA Filter Equipment

High efficiency particulate air (HEPA) filtered vacuum and exhaust ventilation equipment with a filter system capable of collecting and retaining asbestos fibers. Filters must retain 99.97 percent of particles 0.3 microns or larger as indicated in UL 586.

1.2.20 Model Accreditation Plan (MAP)

USEPA training accreditation requirements for persons who work with asbestos as specified in 40 CFR 763.

1.2.21 Negative Pressure Enclosure (NPE)

That engineering control technique described as a negative pressure enclosure in 29 CFR 1926.1101.

1.2.22 NESHAP

National Emission Standards for Hazardous Air Pollutants. The USEPA NESHAP regulation for asbestos is at 40 CFR 61-SUBPART M.

1.2.23 Nonfriable Asbestos Material

Material that contains asbestos in which the fibers have been immobilized by a bonding agent, coating, binder, or other material so that the asbestos is well bound and will not normally release asbestos fibers during any appropriate use, handling, storage or transportation. It is understood that asbestos fibers may be released under other conditions such as demolition, removal, or mishap.

1.2.24 Permissible Exposure Limits (PELs)

1.2.24.1 PEL-Time Weighted Average (TWA)

Concentration of asbestos not in excess of 0.1 fibers per cubic centimeter of air (f/cc) as an 8-hour time weighted average (TWA).

1.2.24.2 PEL-Excursion Limit

An airborne concentration of asbestos not in excess of 1.0 f/cc of air as averaged over a sampling period of 30 minutes.

1.2.25 Personal Sampling

Air sampling which is performed to determine asbestos fiber concentrations within the breathing zone of a specific employee, as performed in accordance with 29 CFR 1926.1101.

1.2.26 Private Qualified Person (PQP)

That qualified person hired by the Contractor to perform the herein listed tasks.

1.2.27 Qualified Person (QP)

A Registered Architect, Professional Engineer, Certified Industrial Hygienist, consultant or other qualified person who has successfully completed training and is therefore accredited under a legitimate State Model Accreditation Plan as described in 40 CFR 763 as a Building Inspector, Contractor/Supervisor Abatement Worker, and Asbestos Project Designer; and has successfully completed the National Institute of Occupational Safety and Health (NIOSH) 582 course "Sampling and Evaluating Airborne Asbestos Dust" or equivalent. The QP must be qualified to perform visual inspections as indicated in ASTM E1368. The QP must be appropriately licensed in the State of Maine.

1.2.28 TEM

Refers to Transmission Electron Microscopy.

1.2.29 Time Weighted Average (TWA)

The TWA is an 8-hour time weighted average airborne concentration of asbestos fibers.

1.2.30 Transite

A generic name for asbestos cement wallboard and pipe.

1.2.31 Wetting Agent

A chemical added to water to reduce the water's surface tension thereby increasing the water's ability to soak into the material to which it is applied. An equivalent wetting agent must have a surface tension of at most0.00042 psi.

1.2.32 Worker

Individual (not designated as the Competent Person or a supervisor) who performs asbestos work and has completed asbestos worker training required by 29 CFR 1926.1101, to include EPA Model Accreditation Plan (MAP) "Worker" training; accreditation, if required by the OSHA Class of work to be performed or by the state where the work is to be performed. The worker must be appropriately licensed in the State of Maine.

1.3 REQUIREMENTS

1.3.1 Description of Work

The work covered by this Section includes the handling and control of asbestos containing materials and describes some of the resultant procedures and equipment required to protect workers, the environment and occupants of the building or area, or both, from contact with airborne asbestos fibers. The work also includes the disposal of any asbestos containing materials generated by the work. More specific operational procedures must be outlined in the Asbestos Hazard Abatement Plan called for elsewhere in this specification. The asbestos work includes the demolition and removal and as indicated on the Drawings,which is governed by MEDEP Chapter 425: Asbestos Management Regulations, 40 CFR 763, NAVFAC P-502, and other applicable Federal, State, and local requirements. Refer to the report located at the end of this Section. Under normal conditions non-friable or chemically bound materials containing asbestos would not be considered hazardous; however, this material may release airborne asbestos fibers during disturbance, demolition, and/or removal and therefore must be handled in accordance with the removal and disposal procedures as specified herein. Provide negative pressure enclosure and glovebag techniques as outlined in this specification. The building(s)work area(s) will be evacuated during the asbestos abatement work. A competent person must supervise asbestos removal work as specified herein.

1.3.1.1 Wallboard/Joint Compound

Both composite samples of the wallboard and discrete samples of the components (wallboard and joint compound) have been tested and results are attached.

1.3.2 Unexpected Discovery of Asbestos

Notify the Contracting Officer if any previously untested building components suspected to contain asbestos are impacted by the work.

1.3.3 Medical Requirements

Provide medical requirements including but not limited to medical surveillance and medical record keeping as listed in 29 CFR 1926.1101.

1.3.3.1 Medical Examinations

Before exposure to airborne asbestos fibers, provide workers with a comprehensive medical examination as required by 29 CFR 1926.1101 or other pertinent State or local directives. This requirement must have been satisfied within the 12 months prior to the start of work on this Contract. The same medical examination must be given on an annual basis to employees engaged in an occupation involving asbestos and within 30 calendar days before or after the termination of employment in such occupation. Specifically identify x-ray films of asbestos workers to the consulting radiologist and mark medical record jackets with the word "ASBESTOS."

1.3.3.2 Medical Records

Maintain complete and accurate records of employees' medical examinations, medical records, and exposure data for a period of indefinite time after termination of employment and make records of the required medical examinations and exposure data available for inspection and copying to: The Assistant Secretary of Labor for Occupational Safety and Health (OSHA), or authorized representatives of them, and an employee's physician upon the request of the employee or former employee.

1.3.4 Employee Training

Submit certificates, prior to the start of work but after the main abatement submittal, signed by each employee indicating that the employee has received training in the proper handling of materials and wastes that contain asbestos in accordance with 40 CFR 763 and 49 CFR 172; understands the health implications and risks involved, including the illnesses possible from exposure to airborne asbestos fibers; understands the use and limits of the respiratory equipment to be used; and understands the results of monitoring of airborne quantities of asbestos as related to health and respiratory equipment as indicated in 29 CFR 1926.1101 on an initial and annual basis. Organize certificates by individual worker, not grouped by type of certification. Post appropriate evidence of compliance with the training requirements of 40 CFR 763. Train personnel involved in the asbestos control work in accordance with United States Environmental Protection Agency (USEPA) Asbestos Hazard Emergency Response Act (AHERA) training criteria or State training criteria whichever is more stringent. Document the training by providing: dates of training, training entity, course outline, names of instructors, and qualifications of instructors upon request by the Contracting Officer. Furnish each employee with respirator training and fit testing administered by the PQP as required by 29 CFR 1926.1101 and 29 CFR 1926.103. Fully cover engineering and other hazard control techniques and procedures. Asbestos workers must have a current State of Maine asbestos worker's license. Comply with the ME L&C PROGRAM.

1.3.5 Permits, Licenses and Notifications

Prior to the start of work, obtain necessary permits and licenses in conjunction with asbestos removal, encapsulation, hauling, and disposition, and furnish notification of such actions required by Federal, State, regional, and local authorities. Notify the Maine Department of Environmental Protection, the NAVFAC Asbestos Program Manager, and the Contracting Officer in writing 20 working days prior to commencement of work in accordance with MEDEP Chapter 425: Asbestos Management Regulations and 40 CFR 61-SUBPART M and for approval by NAVFAC NPT. Notify the Contracting Officer and other appropriate Government agencies in writing 20 working days prior to the start of asbestos work as indicated in applicable laws, ordinances, criteria, rules, and regulations. Submit copies of all Notifications to the Contracting Officer. Notify the Shipyard fire department 3 days prior to removing fire-proofing material from the building including notice that the material contains asbestos.

1.3.6 Environment, Safety and Health Compliance

In addition to detailed requirements of this specification, comply with those applicable laws, ordinances, criteria, rules, and regulations of Federal, State, regional, and local authorities regarding handling, storing, transporting, and disposing of asbestos waste materials. Comply with the applicable requirements of the current issue of EM 385-1-1, 29 CFR 1926.1101, 40 CFR 61-SUBPART A, 40 CFR 61-SUBPART M, 40 CFR 763, 49 CFR 171, and ND OPNAVINST 5100.23. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting the work. Where the requirements of this specification, applicable laws, rules, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirement as defined by the Government apply. The following laws, ordinances, criteria, rules and regulations regarding removal, handling, storing, transporting and disposing of asbestos materials apply:

a. U.S. EPA, Title 40, CFR Part 61, National Standards for Hazardous Air Pollutants (NESHAP), Subparts A and M.

b. OSHA, 29 CFR, Part 1926.1101, Asbestos in Construction.

c. State of Maine, Department of Environmental Protection (ME DEP), 06 096 CMR Asbestos Management Regulations, Chapter 425, effective date 4-3-2011.

1.3.7 Respiratory Protection Program

Establish and implement a respirator program as required by 29 CFR 1926.1101, and 29 CFR 1926.103. Submit a written description of the program to the Contracting Officer. Submit a written program manual or operating procedure including methods of compliance with regulatory statutes.

1.3.7.1 Respirator Program Records

Submit records of the respirator program as required by 29 CFR 1926.103, and 29 CFR 1926.1101.

1.3.7.2 Respirator Fit Testing

The Contractor's PQP must conduct a qualitative or quantitative fit test conforming to 29 CFR 1926.103 for each worker required to wear a respirator, and any authorized visitors who enter a regulated area where respirators are required to be worn. A respirator fit test must be performed prior to initially wearing a respirator and every 12 months thereafter. If physical changes develop that will affect the fit, a new fit test must be performed. Functional fit checks must be performed each time a respirator is put on and in accordance with the manufacturer's recommendation.

1.3.7.3 Respirator Selection and Use Requirements

Provide respirators, and ensure that they are used as required by 29 CFR 1926.1101 and in accordance with CGA G-7 and the manufacturer's recommendations. Respirators must be approved by the National Institute for Occupational Safety and Health NIOSH, under the provisions of 42 CFR 84, for use in environments containing airborne asbestos fibers. For air-purifying respirators, the particulate filter must be high-efficiency particulate air (HEPA)/(N-,R-,P-100). The initial respirator selection and the decisions regarding the upgrading or downgrading of respirator type must be made by the Contractor's Designated IH based on the measured or anticipated airborne asbestos fiber concentrations to be encountered.

1.3.8 Asbestos Hazard Control Supervisor

The Contractor must be represented on site by a supervisor, trained using the model Contractor accreditation plan as indicated in the Federal statutes for all portions of the herein listed work.

1.3.9 Hazard Communication

Adhere to all parts of 29 CFR 1926.59 and provide the Contracting Officer with a copy of the Safety Data Sheets (SDS) for all materials brought to the site.

1.3.10 Asbestos Hazard Abatement Plan

Submit a detailed plan of the safety precautions such as lockout, tagout, tryout, fall protection, and confined space entry procedures and equipment and work procedures to be used in the removal and demolition of materials containing asbestos. The plan, not to be combined with other hazard abatement plans, must be prepared, signed, and sealed by the PQP. Provide a Table of Contents for each abatement submittal, which follows the sequence of requirements in the Contract. The plan must include but not be limited to the precise personal protective equipment to be used including, but not limited to, respiratory protection, type of whole-body protection and if reusable coveralls are to be employed decontamination methods (operations and quality control plan), the location of asbestos control areas including clean and dirty areas, buffer zones, showers, storage areas, change rooms, removal method, interface of trades involved in the construction, sequencing of asbestos related work, disposal plan, type of wetting agent and asbestos sealer to be used, locations of local exhaust equipment, planned air monitoring strategies, and a detailed description of the method to be employed in order to control environmental pollution. The plan must also include (both fire and medical emergency) response plans and an Activity Hazard Analyses (AHAs) in accordance with EM 385-1-1. The Asbestos Hazard Abatement Plan must be approved in writing prior to starting any asbestos work. The Contractor, Asbestos Hazard Control Supervisor, CP, and PQP must meet with the Contracting Officer prior to beginning work, to discuss in detail the Asbestos Hazard Abatement Plan, including work procedures and safety precautions. Once approved by the Contracting Officer, the plan will be enforced as if an addition to the specification. Any changes required in the specification as a result of the plan must be identified specifically in the plan to allow for free discussion and approval by the Contracting Officer prior to starting work.

The Plan must be accompanied by a completed Asbestos Hazard Abatement Plan Checklist included as Attachment A at the end of this specification and referenced in Paragraph entitled SUBMITTALS herein. The Checklist must be prepared and signed by the PQP.

1.3.11 Testing Laboratory

Submit the name, address, and telephone number of each testing laboratory selected for the sampling, analysis, and reporting of airborne concentrations of asbestos fibers along with evidence that each laboratory selected holds the appropriate State license and permits and certification that each laboratory is American Industrial Hygiene Association (AIHA) accredited and that persons counting the samples have been judged proficient by current inclusion on the AIHA Asbestos Analysis Registry (AAR) and successful participation of the laboratory in the Proficiency Analytical Testing (PAT) Program. Where analysis to determine asbestos content in bulk materials or transmission electron microscopy is required, submit evidence that the laboratory is accredited by the National Institute of Science and Technology (NIST) under National Voluntary Laboratory Accreditation Program (NVLAP) for asbestos analysis. The testing laboratory firm must be independent of the asbestos contractor and must have no employee or employer relationship which could constitute a conflict of interest.

1.3.12 Landfill Approval

Landfill approval is not required as Portsmouth Naval Shipyard takes responsibility for disposal of all asbestos waste generated on the Shipyard.

1.3.13 Transporter Certification

Transporter certification is not required as Portsmouth Naval Shipyard takes responsibility for transport and disposal of all asbestos waste generated on the Shipyard.

1.3.14 Medical Certification

Provide a written certification for each worker and supervisor, signed by a licensed physician indicating that the worker and supervisor has met or exceeded all of the medical prerequisites listed herein and in 29 CFR 1926.1101 and 29 CFR 1926.103 as prescribed by law. Submit certificates prior to the start of work but after the main abatement submittal.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-03 Product Data

Amended Water; G

Safety Data Sheets (SDS) for All Materials; G

Encapsulants; G

Respirators; G

Local Exhaust Equipment; G

Pressure Differential Automatic Recording Instrument; G

Vacuums; G

Glovebags; G

SD-06 Test Reports

Air Sampling Results; G

Pressure Differential Recordings for Local Exhaust System; G

Clearance Sampling; G

Asbestos Disposal Quantity Report; G

SD-07 Certificates

Employee Training; G

Notifications; G

Respiratory Protection Program; G

Asbestos Hazard Abatement Plan; G

Asbestos Hazard Abatement Plan Checklist; G

Testing Laboratory; G

Medical Certification; G

Private Qualified Person Documentation; G

Competent Person; G

Worker's License; G

Contractor's License; G

Federal, State or Local Citations on Previous Projects; G

Encapulants; G

Equipment Used to Contain Airborne Asbestos Fibers; G

Water Filtration Equipment; G

Vacuums; G

Ventilation Systems; G

SD-11 Closeout Submittals

Permits and Licenses; G

Notifications; G

Respirator Program Records; G

Rental Equipment; G

1.5 QUALITY ASSURANCE

1.5.1 Private Qualified Person Documentation

Submit the name, address, and telephone number of the Private Qualified Person (PQP) selected to prepare the Asbestos Hazard Abatement Plan, direct monitoring and training, and documented evidence that the PQP has successfully completed training in and is accredited and where required is certified as, a Building Inspector, Contractor/Supervisor Abatement Worker, and Asbestos Project Designer as described by 40 CFR 763 and has successfully completed the National Institute of Occupational Safety and Health (NIOSH) 582 course "Sampling and Evaluating Airborne Asbestos Dust" or equivalent. The PQP must be appropriately licensed in the State of Maine as a Project Monitor. The PQP and the asbestos contractor must not have an employee/employer relationship or financial relationship which could constitute a conflict of interest. **The PQP must be a first tier subcontractor.**

1.5.2 Competent Person Documentation

The Competent Person must be experienced in the administration and supervision of asbestos abatement projects including exposure assessment and monitoring, work practices, abatement methods, protective measures for personnel, setting up and inspecting asbestos abatement work areas, evaluating the integrity of containment barriers, placement and operation of local exhaust systems, ACM generated waste containment and disposal procedures, decontamination units installation and maintenance requirements, site safety and health requirements, and notification of other employees onsite. The Competent Person must be on-site at all times when asbestos abatement activities are underway. Submit training certification and a current State of Maine Asbestos Contractor's and Supervisor's License. Submit evidence that the Competent Person has a minimum of 2 years of on-the-job asbestos abatement experience relevant to OSHA competent person requirements.

1.5.3 Worker's License

Submit documentation that workers meet the requirements of 29 CFR 1926.1101, 40 CFR 61-SUBPART M and have a current State of Maine Asbestos Workers License.

1.5.4 Contractor's License

Submit a copy of the asbestos contractor's license issued by the State of Maine. Submit the following certification along with the license: "I certify that the personnel I am responsible for during the course of this project fully understand the contents of 29 CFR 1926.1101, 40 CFR 61-SUBPART M, EM 385-1-1, and the Federal, State and local requirements for those asbestos abatement activities that they will be involved in." This certification statement must be signed by the Company's President or Chief Executive.

1.5.5 Air Sampling Results

Complete fiber counting and provide results to the PQP and GC for review within 16 hours of the "time off" of the sample pump. Notify the Contracting Officer and Asbestos Program Manager immediately of any airborne levels of asbestos fibers in excess of the acceptable limits. Submit sampling results to the Contracting Officer and Asbestos Program Manager and the affected Contractor employees where required by law within three working days, signed by the testing laboratory employee performing air sampling, the employee that analyzed the sample, and the PQP and GC. Notify the Contractor, the Contracting Officer and Asbestos Program Manager immediately of any variance in the pressure differential which could cause adjacent unsealed areas to have asbestos fiber concentrations in excess of 0.010 fibers per cubic centimeter or background whichever is higher. In no circumstance must levels exceed 0.10 fibers per cubic centimeter. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

1.5.6 Pressure Differential Recordings for Local Exhaust System

Provide a local exhaust system that creates a negative pressure of at least 0.02 inches of water relative to the pressure external to the enclosure and operate it continuously, 24-hours a day, until the temporary enclosure of the asbestos control area is removed. Submit pressure differential recordings for each work day to the PQP and GC for review and to the Contracting Officer and Asbestos Program Manager within 24-hours from the end of each work day. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

1.5.7 Federal, State or Local Citations on Previous Projects

Submit a statement, signed by an officer of the company, containing a record of any citations issued by Federal, State or local regulatory agencies relating to asbestos activities within the last 5 years (including projects, dates, and resolutions); a list of penalties incurred through non-compliance with asbestos project specifications, including liquidated damages, overruns in scheduled time limitations and resolutions; and situations in which an asbestos-related contract has been terminated (including projects, dates, and reasons for terminations). If there are none, a negative declaration signed by an officer of the company must be provided.

1.5.8 Preconstruction Conference

Conduct a safety preconstruction conference to discuss the details of the Asbestos Hazard Abatement Plan, Accident Prevention Plan (APP) including the AHAs required in specification Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). The safety preconstruction conference must include the Contractor and their Designated Competent Person, NAVFAC Asbestos Program Manager, the PQP, Competent Person, Designated IH and Project Supervisor and the Contracting Officer. Deficiencies in the APP will be discussed. Onsite work must not begin until the APP has been accepted.

1.6 SECURITY

A log book must be kept documenting entry into and out of the regulated area. Entry into regulated areas must only be by personnel authorized by the Contractor and the Contracting Officer. Personnel authorized to enter regulated areas must be trained, medically evaluated, and wear the required personal protective equipment.

1.7 EQUIPMENT

1.7.1 Rental Equipment

Provide a copy of the written notification to the rental company concerning the intended use of the equipment and the possibility of asbestos contamination of the equipment.

PART 2 PRODUCTS

2.1 ENCAPSULANTS

Encapsulants must conform to current USEPA requirements, contain no toxic or hazardous substances as defined in 29 CFR 1926.59, and conform to the following performance requirements.

2.1.1 Removal Encapsulants

Requirement	Test Standard
Flame Spread - 25, Smoke Emission - 50	ASTM E84
Life Expectancy - 20 years	ASTM C732 Accelerated Aging Test
Permeability - Minimum 0.4 perms	ASTM E96/E96M
Fire Resistance - Negligible affect on fire resistance rating over 3 hour test (Classified by UL for use over fibrous and cementitious sprayed fireproofing)	ASTM E119
Impact Resistance - Minimum 43 in/lb	ASTM D2794 Gardner Impact Test
Flexibility - no rupture or cracking	ASTM D522/D522M Mandrel Bend Test

2.1.2 Bridging Encapsulant

Requirement	Test Standard
Flame Spread - 25, Smoke Emission - 50	ASTM E84

Requirement	Test Standard
Life Expectancy - 20 years	ASTM C732 Accelerated Aging Test
Permeability - Minimum 0.4 perms	ASTM E96/E96M
Fire Resistance - Negligible affect on fire resistance rating over 3-hour test (Classified by UL for use over fibrous and cementitious sprayed fireproofing)	ASTM E119
Impact Resistance - Minimum 43 in/lb	ASTM D2794 Gardner Impact Test
Flexibility - no rupture or cracking	ASTM D522/D522M Mandrel Bend Test

2.1.3 Penetrating Encapsulant

Requirement	Test Standard
Flame Spread - 25, Smoke Emission - 50	ASTM E84
Life Expectancy - 20 years	ASTM C732 Accelerated Aging Test
Permeability - Minimum 0.4 perms	ASTM E96/E96M
Cohesion/Adhesion Test - 50 pounds of force/foot	ASTM E119
Fire Resistance - Negligible affect on fire resistance rating over 3-hour test (Classified by UL for use over fibrous and cementitious sprayed fireproofing)	ASTM E119
Impact Resistance - Minimum 43 in/lb	ASTM D2794 Gardner Impact Test
Flexibility - no rupture or cracking	ASTM D522/D522M Mandrel Bend Test

2.1.4 Lock-down Encapsulant

Requirement	Test Standard
Flame Spread - 25, Smoke Emission - 50	ASTM E84

Requirement	Test Standard
Life Expectancy - 20 years	ASTM C732 Accelerated Aging Test
Permeability - Minimum 0.4 perms	ASTM E96/E96M
Fire Resistance - Negligible affect on fire resistance rating over 3-hour test (Tested with fireproofing over encapsulant applied directly to steel member)	ASTM E119
Bond Strength: 100 pounds of force/foot	ASTM E736/E736M
(Tests compatibility with cementitious and fibrous fireproofing)	

2.2 DUCT TAPE

Industrial grade duct tape of appropriate widths suitable for bonding sheet plastic and disposal container.

2.3 DISPOSAL CONTAINERS

Leak-tight (defined as solids, liquids, or dust that cannot escape or spill out) disposal containers must be provided for ACM wastes as required by 29 CFR 1926.1101. Disposal containers can be in the form of:

a. Disposal Bags

2.4 SHEET PLASTIC

Sheet plastic must be polyethylene of 6 mil minimum thickness and must be provided in the largest sheet size necessary to minimize seams. Film must be clear and conform to ASTM D4397, except as specified below.

2.4.1 Flame Resistant

Where a potential for fire exists, flame-resistant sheets must be provided. Film must be frosted and must conform to the requirements of NFPA 701.

2.4.2 Reinforced

Reinforced sheets must be provided where high skin strength is required, such as where it constitutes the only barrier between the regulated area and the outdoor environment. The sheet stock must consist of translucent, nylon-reinforced or woven-polyethylene thread laminated between 2 layers of polyethylene film. Film must meet flame resistant standards of NFPA 701.

2.5 MASTIC REMOVING SOLVENT

Mastic removing solvent must be nonflammable and must not contain methylene chloride, glycol ether, or halogenated hydrocarbons. Solvents used onsite must have a flash point greater than 140 degrees F.

2.6 LEAK-TIGHT WRAPPING

At least two layers of 6 mil minimum thick polyethylene sheet stock must be used for the containment of removed asbestos-containing components or materials such as large tanks, boilers, insulated pipe segments and other materials. Upon placement of the ACM component or material, each layer must be individually leak-tight sealed with duct tape.

2.7 VIEWING INSPECTION WINDOW

Where feasible, a minimum of one clear, 1/8 inch thick, acrylic sheet, 18 by 24 inches, must be installed as a viewing inspection window at eye level on a wall in each containment enclosure. The windows must be sealed leak-tight with industrial grade duct tape.

2.8 WETTING AGENTS

Removal encapsulant (a penetrating encapsulant) must be provided when conducting removal abatement activities that require a longer removal time or are subject to rapid evaporation of amended water. The removal encapsulant must be capable of wetting the ACM and retarding fiber release during disturbance of the ACM greater than or equal to that provided by amended water. Performance requirements for penetrating encapsulants are specified in paragraph ENCAPSULANTS above.

PART 3 EXECUTION

3.1 EQUIPMENT

Provide the Contracting Officer or the Contracting Officer's Representative, with at least two complete sets of personal protective equipment as required for entry to and inspection of the asbestos control area. Provide equivalent training to the Contracting Officer or a designated representative as provided to Contractor employees in the use of the required personal protective equipment. Provide manufacturer's certificate of compliance for all equipment used to contain airborne asbestos fibers.

3.1.1 Air Monitoring Equipment

The Contractor's PQP must approve air monitoring equipment. The equipment must include, but must not be limited to:

a. High-volume sampling pumps that can be calibrated and operated at a constant airflow up to 16 liters per minute.

b. Low-volume, battery powered, body-attachable, portable personal pumps that can be calibrated to a constant airflow up to approximately 3.5 liters per minute, and a self-contained rechargeable power pack capable of sustaining the calibrated flow rate for a minimum of 10 hours. The pumps must also be equipped with an automatic flow control unit which must maintain a constant flow, even as filter resistance increases due to accumulation of fiber and debris on the filter surface.

c. Single use standard 25 mm diameter cassette, open face, 0.8 micron pore size, mixed cellulose ester membrane filters and cassettes with 50 mm electrically conductive extension cowl, and shrink bands for personal air sampling.

3.1.2 Respirators

Select respirators from those approved by the National Institute for Occupational Safety and Health (NIOSH), Department of Health and Human Services.

3.1.2.1 Respirators for Handling Asbestos

Provide personnel engaged in pre-cleaning, cleanup, handling, encapsulation, removal and demolition of asbestos materials with respiratory protection as indicated in 29 CFR 1926.1101 and 29 CFR 1926.103. Breathing air must comply with CGA G-7.

3.1.3 Exterior Whole Body Protection

3.1.3.1 Outer Protective Clothing

Provide personnel exposed to asbestos with disposable "non-breathable," or reusable "non-breathable" whole body outer protective clothing, head coverings, gloves, and foot coverings. Provide disposable plastic or rubber gloves to protect hands. Cloth gloves may be worn inside the plastic or rubber gloves for comfort, but must not be used alone. Make sleeves secure at the wrists, make foot coverings secure at the ankles, and make clothing secure at the neck by the use of tape.

3.1.3.2 Work Clothing

Provide cloth work clothes for wear under the outer protective clothing and foot coverings and either dispose of or properly decontaminate them as recommended by the PQP after each use.

3.1.3.3 Personal Decontamination Unit

Provide a temporary, negative pressure unit with a separate decontamination locker room and clean locker room with a shower that complies with 29 CFR 1926.51(f)(4)(ii) through (V) in between for personnel required to wear whole body protective clothing. Provide two separate lockers for each asbestos worker, one in each locker room. Keep street clothing and street shoes in the clean locker. HEPA vacuum and remove asbestos contaminated disposable protective clothing while still wearing respirators at the boundary of the asbestos work area and seal in impermeable bags or containers for disposal. HEPA vacuum and remove asbestos contaminated reusable protective clothing while still wearing respirators at the boundary of the asbestos work area, seal in two impermeable bags, label outer bag as asbestos contaminated waste, and transport for decontamination. Do not wear work clothing between home and work. Locate showers between the decontamination locker room and the clean locker room and require that all employees shower before changing into street clothes. Collect used shower water and filter with approved water filtration equipment to remove asbestos contamination. Wastewater filters must be installed in series with the first stage pore size 20 microns and the second stage pore size of 5 microns. Dispose of filters and residue as asbestos waste. Discharge clean water to the sanitary system. Dispose of asbestos contaminated work clothing as asbestos contaminated waste or properly decontaminate as specified in the Contractor's Asbestos Hazard Abatement Plan. Keep the floor of the decontamination unit's clean room dry and clean at all times. Proper housekeeping and hygiene requirements must be maintained. Provide soap and towels for showering, washing and drying. Cloth towels provided must be disposed of as ACM waste or must be laundered in accordance with 29 CFR 1926.1101. Physically attach the decontamination units to the asbestos control area. Construct both a personnel decontamination unit and an equipment decontamination unit onto and integral with each asbestos control area.

3.1.3.4 Eye Protection

Provide eye protection that complies with ANSI/ISEA Z87.1 when operations present a potential eye injury hazard. Provide goggles to personnel engaged in asbestos abatement operations when the use of a full face respirator is not required.

3.1.4 Regulated Areas

All Class I, II, and III asbestos work must be conducted within regulated areas. The regulated area must be demarcated to minimize the number of persons within the area and to protect persons outside the area from exposure to airborne asbestos. Control access to regulated areas, ensure that only authorized personnel enter, and verify that Contractor required medical surveillance, training and respiratory protection program requirements are met prior to allowing entrance.

3.1.5 Load-out Unit

Provide a temporary load-out unit that is adjacent and connected to the regulated area and access tunnel. Attach the load-out unit in a leak-tight manner to each regulated area.

3.1.6 Warning Signs and Labels

Provide warning signs printed in English and at all approaches to asbestos control areas. Locate signs at such a distance that personnel may read the sign and take the necessary protective steps required before entering the area. Provide labels and affix to all asbestos materials, scrap, waste, debris, and other products contaminated with asbestos. Containers with preprinted warning labels conforming to the requirements are acceptable.

3.1.6.1 Warning Sign

Provide vertical format conforming to 29 CFR 1926.200, and 29 CFR 1926.1101 minimum 20 by 14 inches displaying the following legend in the lower panel:

Legend	Notation
DANGER	one inch Sans Serif Gothic or Block
ASBESTOS	one inch Sans Serif Gothic or Block
MAY CAUSE CANCER	one inch Sans Serif Gothic or Block
CAUSES DAMAGE TO LUNGS	1/4 inch Sans Serif Gothic or Block

Legend	Notation
AUTHORIZED PERSONNEL ONLY	1/4 inch Sans Serif Gothic or Block
WEAR RESPIRATORY PROTECTION AND PROTECTIVE CLOTHING IN THIS AREA	1/4 inch Sans Serif Gothic or Block

Spacing between lines must be at least equal to the height of the upper of any two lines.

3.1.6.2 Warning Labels

Provide labels conforming to 29 CFR 1926.1101 of sufficient size to be clearly legible, displaying the following legend:

DANGER
CONTAINS ASBESTOS FIBERS
MAY CAUSE CANCER
CAUSES DAMAGE TO LUNGS
DO NOT BREATHE DUST AVOID CREATING DUST

3.1.7 Local Exhaust System

Provide a local exhaust system in the asbestos control area in accordance with ASSP Z9.2 and 29 CFR 1926.1101 that will provide at least four air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until the asbestos control area is removed and must be leak proof to the filter and equipped with HEPA filters. Maintain a minimum pressure differential in the control area of minus 0.02 inch of water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the asbestos control area. Filters on exhaust equipment must conform to ASSP Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

3.1.8 Tools

Vacuums must be leak proof to the filter and equipped with HEPA filters. Filters on vacuums must conform to ASSP Z9.2 and UL 586. Do not use power tools to remove asbestos containing materials unless the tool is equipped with effective, integral HEPA filtered exhaust ventilation systems. Remove all residual asbestos from reusable tools prior to storage or reuse. Reusable tools must be thoroughly decontaminated prior to being removed from the regulated areas.

3.1.9 Rental Equipment

If rental equipment is to be used, furnish written notification to the rental agency concerning the intended use of the equipment and the possibility of asbestos contamination of the equipment.

3.1.10 Glovebags

Glovebag operations must be performed in accordance with the requirements of the State of Maine Department of Environmental Protection. Submit written manufacturers proof that glovebags will not break down under expected temperatures and conditions.

3.1.11 Decontamination Area Exit Procedures

Note: Whenever possible, decontamination area must be contiguous with, and connected to, the negative pressure enclosure.

Ensure that the following procedures are followed:

a. Before leaving the regulated area, remove all gross contamination and debris from work clothing using a HEPA vacuum.

b. Employees must remove their protective clothing in the equipment room and deposit the clothing in labeled impermeable bags or containers for disposal or laundering.

c. Employees must not remove their respirators until showering.

d. Employees must shower prior to entering the clean room. If a shower has not been located between the equipment room and the clean room or the work is performed outdoors, ensure that employees engaged in Class I asbestos jobs: a) Remove asbestos contamination from their work suits in the equipment room or decontamination area using a HEPA vacuum before proceeding to a shower that is not adjacent to the work area; or b) Remove their contaminated work suits in the equipment room, without cleaning worksuits, and proceed to a shower that is not adjacent to the work area.

3.2 WORK PROCEDURE

Perform asbestos related work in accordance with MEDEP Chapter 425: Asbestos Management Regulations, 29 CFR 1926.1101, 40 CFR 61-SUBPART M, 49 CFR 107, 49 CFR 172, NAVFAC P-502 and as specified herein. Use wet removal procedures and negative pressure enclosure techniques. Wear and utilize protective clothing and equipment as specified herein. No eating, smoking, drinking, chewing gum, tobacco, or applying cosmetics is permitted in the asbestos work or control areas. Personnel of other trades not engaged in the encapsulation removal and demolition of asbestos containing material must not be exposed at any time to airborne concentrations of asbestos unless all the personnel protection and training provisions of this specification are complied with by the trade personnel. Seal all rooftop penetrations, except plumbing vents, prior to asbestos roofing work. Shut down the building heating, ventilating, and air conditioning system, cap the openings to the system, and provide temporary heating and ventilation, prior to the commencement of asbestos work. Power to the regulated area must be locked-out and tagged in accordance with 29 CFR 1910.147. Disconnect electrical service when encapsulation wet removal is performed and provide temporary electrical service with verifiable ground fault circuit interrupter (GFCI) protection prior to the use of any water encapsulant. All electrical work must be performed by a licensed electrician. Stop abatement work in the regulated area immediately when the airborne total fiber concentration: (1) equals or exceeds 0.010 f/cc, or the pre-abatement concentration, whichever is greater, outside the regulated area; or (2) equals or exceeds 1.0 f/cc inside the regulated area. Correct the condition to the satisfaction of the Contracting Officer, including visual inspection and air sampling. Work must resume only upon notification by the Contracting Officer. Corrective actions must be documented. If an asbestos fiber release or spill occurs, stop work immediately, correct the condition to the satisfaction of the Contracting Officer including clearance sampling, prior to resumption of work.

3.2.1 Building Ventilation System and Critical Barriers

Building ventilation system supply and return air ducts in a regulated area must be shut down and isolated by lockable switch or other positive means in accordance with 29 CFR 1910.147. The airtight seals must consist of air-tight rigid covers for building ventilation supply and exhaust grills where the ventilation system is required to remain in service during abatement. Edges to wall, ceiling and floor surfaces must be sealed with industrial grade duct tape.

a. A Competent Person must supervise the work as required herein.

b. For indoor work, critical barriers must be placed over all openings to the regulated area.

c. Impermeable dropcloths must be placed on surfaces beneath all removal activity.

3.2.2 Protection of Existing Work to Remain

Perform work without damage or contamination of adjacent work. Where such work is damaged or contaminated as verified by the Contracting Officer using visual inspection or sample analysis, it must be restored to its original condition or decontaminated by the Contractor at no expense to the Government as deemed appropriate by the Contracting Officer. This includes inadvertent spill of dirt, dust, or debris in which it is reasonable to conclude that asbestos may exist. When these spills occur, stop work immediately. Then clean up the spill. When satisfactory visual inspectionincluding all necessary wipe and air sampling results are submitted to the Contracting Officer & the NAVFAC Abestos Program Manager directly from the PQP, work may proceed at the discretion of the Contracting Officer. All cleanup and testing must be completed in accordance with the requirements outlined herein this specification. All costs for the required clearance sampling and clean up efforts are the responsiblity of the Contractor. Any delays associated with the clean up efforts are the responsiblity of the Contractor.

3.2.3 Furnishings

Furniture and equipment will remain in the building. Cover and seal furnishings and equipment with 6-mil plastic sheet or remove from the work area and store in a location on site approved by the Contracting Officer.

3.2.4 Precleaning

Wet wipe and HEPA vacuum all surfaces potentially contaminated with asbestos prior to establishment of an enclosure.

3.2.5 Asbestos Control Area Requirements

3.2.5.1 Negative Pressure Enclosure

Removal of Asbestos Contaminated Materials (ACM) require the use of a negative pressure enclosure. Block and seal openings in areas where the release of airborne asbestos fibers can be expected. Establish an asbestos negative pressure enclosure with the use of curtains, portable partitions, or other enclosures in order to prevent the escape of asbestos fibers from the contaminated asbestos work area. Negative pressure enclosure development must include protective covering of uncontaminated walls, and ceilings with a continuous membrane of two layers of minimum 6-mil plastic sheet sealed with tape to prevent water or other damage. Provide two layers of 6-mil plastic sheet over floors and extend a minimum of 12 inches up walls. Seal all joints with tape. Provide local exhaust system in the asbestos control area. Openings will be allowed in enclosures of asbestos control areas for personnel and equipment entry and exit, the supply and exhaust of air for the local exhaust system and the removal of properly containerized asbestos containing materials. Replace local exhaust system filters as required to maintain the efficiency of the system.

3.2.5.2 Glovebag

If the construction of a negative pressure enclosure is infeasible for the removal of ACM, use alternate techniques as indicated in MEDEP Chapter 425: Asbestos Management Regulations and 29 CFR 1926.1101. Establish designated limits for the asbestos regulated area with the use of rope or other continuous barriers, and maintain all other requirements for asbestos control areas. The PQP must conduct personal samples of each worker engaged in asbestos handling (removal, disposal, transport and other associated work) throughout the duration of the project. If the quantity of airborne asbestos fibers monitored at the breathing zone of the workers at any time exceeds background or 0.010 fibers per cubic centimeter whichever is greater, stop work, evacuate personnel in adjacent areas or provide personnel with approved protective equipment at the discretion of the Contracting Officer. This sampling may be duplicated by the Government at the discretion of the Contracting Officer. If the air sampling results obtained by the Government differ from those obtained by the Contractor, the Government will determine which results predominate. If adjacent areas are contaminated as determined by the Contracting Officer, clean the contaminated areas, monitor, and visually inspect the area as specified herein.

3.2.6 Removal Procedures

Wet asbestos material with a fine spray of amended water during removal, cutting, or other handling so as to reduce the emission of airborne fibers. Remove material and immediately place in 6 mil plastic disposal bags. Remove asbestos containing material in a gradual manner, with continuous application of the amended water or wetting agent in such a manner that no asbestos material is disturbed prior to being adequately wetted. Where unusual circumstances prohibit the use of 6 mil plastic bags, submit an alternate proposal for containment of asbestos fibers to the Contracting Officer for approval. For example, in the case where both piping and insulation are to be removed, the Contractor may elect to wet the insulation, wrap the pipes and insulation in plastic and remove the pipe by sections. Containerize asbestos containing material while wet. Do not allow asbestos material to accumulate or become dry. Lower and otherwise handle asbestos containing material as indicated in MEDEP Chapter 425: Asbestos Management Regulations and 40 CFR 61-SUBPART M.

3.2.6.1 Sealing Contaminated Items Designated for Disposal

Remove contaminated architectural, mechanical, and electrical appurtenances such as venetian blinds, full-height partitions, carpeting, duct work, pipes and fittings, radiators, light fixtures, conduit, panels, and other contaminated items designated for removal by completely coating the items with an asbestos lock-down encapsulant at the demolition site before removing the items from the asbestos control area. These items need not be vacuumed. The asbestos lock-down encapsulant must be tinted a contrasting color and spray-applied by airless method. Thoroughness of sealing operation must be visually gauged by the extent of colored coating on exposed surfaces. Lock-down encapsulants must comply with the performance requirements specified herein.

3.2.6.2 Exposed Pipe Insulation Edges

Contain edges of asbestos insulation to remain that are exposed by a removal operation. Wet and cut the rough ends true and square with sharp tools and then encapsulate the edges with a 1/4 inch thick layer of non-asbestos containing insulating cement troweled to a smooth hard finish. When cement is dry, lag the end with a layer of non-asbestos lagging cloth, overlapping the existing ends by at least 4 inches. When insulating cement and cloth is an impractical method of sealing a raw edge of asbestos, take appropriate steps to seal the raw edges as approved by the Contracting Officer.

3.2.7 Methods of Compliance

3.2.7.1 Mandated Practices

The specific abatement techniques and items identified must be detailed in the Contractor's AHAP. Use the following engineering controls and work practices in all operations, regardless of the levels of exposure:

a. Vacuum cleaners equipped with HEPA filters.

b. Wet methods or wetting agents except where it can be demonstrated that the use of wet methods is unfeasible due to the creation of electrical hazards, equipment malfunction, and in roofing.

c. Prompt clean-up and disposal.

d. Inspection and repair of polyethylene.

e. Cleaning of equipment and surfaces of containers prior to removing them from the equipment room or area.

3.2.7.2 Control Methods

Use the following control methods:

a. Local exhaust ventilation equipped with HEPA filter;

b. Enclosure or isolation of processes producing asbestos dust;

c. Where the feasible engineering and work practice controls are not sufficient to reduce employee exposure to or below the PELs, use them to reduce employee exposure to the lowest levels attainable and must supplement them by the use of respiratory protection.

3.2.7.3 Unacceptable Practices

The following work practices must not be used:

a. High-speed abrasive disc saws that are not equipped with point of cut ventilator or enclosures with HEPA filtered exhaust air.

b. Compressed air used to remove asbestos containing materials, unless the compressed air is used in conjunction with an enclosed ventilation system designed to capture the dust cloud created by the compressed air.

c. Dry sweeping, shoveling, or other dry clean up.

d. Employee rotation as a means of reducing employee exposure to asbestos.

3.2.8 Class I Work Procedures

In addition to requirements of paragraphs MANDATED PRACTICES and CONTROL METHODS, the following engineering controls and work practices must be used:

a. A Competent Person must supervise the installation and operation of the control methods.

b. Place critical barriers over all openings to the regulated area.

c. HVAC systems must be isolated in the regulated area by sealing with a double layer of plastic or air-tight rigid covers.

d. Impermeable dropcloths (6 mil or greater thickness) must be placed on surfaces beneath all removal activity.

e. Where a negative exposure assessment has not been provided or where exposure monitoring shows the PEL was exceeded, the regulated area must be ventilated with a HEPA unit and employees must use PPE.

3.2.9 Specific Control Methods for Class I Work

Use Class I work procedures, control methods and removal methods for the following ACM:

a. Spray Applied Fireproofing

b. Gypsum Wallboard and Joint Compound

c. Thermal System Insulation and Mudded Pipe Fittings

d. Plaster and Textured Ceilings and Walls

e. Vermiculite

3.2.9.1 Negative Pressure Enclosure (NPE) System

The system must provide at least four air changes per hour inside the containment. The local exhaust unit equipment must be operated 24-hours per day until the containment is removed. The NPE must be smoke tested for leaks at the beginning of each shift and be sufficient to maintain a minimum pressure differential of minus 0.02 inch of water column relative to adjacent, unsealed areas. Pressure differential must be monitored continuously, 24-hours per day, with an automatic manometric recording instrument and Records must be provided daily on the same day collected to the Contracting Officer and Asbestos Program Manager. The Contracting Officer and Asbestos Program Manager must be notified immediately if the pressure differential falls below the prescribed minimum. The building ventilation system must not be used as the local exhaust system for the regulated area. The NPE must terminate outdoors unless an alternate arrangement is allowed by the Contracting Officer and Asbestos Program Manager. All filters used must be new at the beginning of the project and must be periodically changed as necessary and disposed of as ACM waste. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

3.2.9.2 Glovebag Systems

Glovebags must be used without modification, smoke-tested for leaks, and completely cover the circumference of pipe or other structures where the work is to be done. Glovebags must be used only once and must not be moved. Glovebags must not be used on surfaces that have temperatures exceeding 150 degrees F. Prior to disposal, glovebags must be collapsed using a HEPA vacuum. Before beginning the operation, loose and friable material adjacent to the glovebag operation must be wrapped and sealed in 2 layers of plastic or otherwise rendered intact. At least two persons must perform glovebag removal. Asbestos regulated work areas must be established for glovebag abatement. Designated boundary limits for the asbestos work must be established with rope or other continuous barriers and all other requirements for asbestos control areas must be maintained, including area signage and boundary warning tape.

a. Attach HEPA vacuum systems to the bag to prevent collapse during removal of ACM.

b. The negative pressure glove boxes must be fitted with gloved apertures and a bagging outlet and constructed with rigid sides from metal or other material which can withstand the weight of the ACM and water used during removal. A negative pressure must be created in the system using a HEPA filtration system. The box must be smoke tested for leaks prior to each use.

3.2.9.3 Mini-Enclosure

Single bulkhead containment, double bulkhead containment, or mini -containment (small walk-in enclosure) to accommodate no more than two persons, may be used if the disturbance or removal can be completely contained by the enclosure. The mini-enclosure must be inspected for leaks and smoke tested before each use. Air movement must be directed away from the employee's breathing zone within the mini-enclosure.

3.2.9.4 Wrap and Cut Operation

Prior to cutting pipe, the asbestos-containing insulation must be wrapped with polyethylene and securely sealed with duct tape to prevent asbestos becoming airborne as a result of the cutting process. The following steps must be taken: install glovebag, strip back sections to be cut 6 inches from point of cut, and cut pipe into manageable sections.

3.2.9.5 Class I Removal Method

Class I ACM must be removed using a control method described above. Prepare work area as previously specified. Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area. Spread one layer of 6-mil seamless plastic sheeting on the floor below the work area. Using the following techniques are applicable:

a. Remove asbestos containing spray applied fireproofing using a scraper and wet methods and immediately place into 6-mil thickness disposal bag. After removal of the material use a wire brush to clean the exposed substrate to remove residual material. Continue wet cleaning until surfaces are free of visible debris.

b. Cut manageable sections of gypsum wallboard and joint compound and immediately place into a 6-mil minimum thickness disposal bag or other approved container. Make every effort to keep the material from falling to the floor of the work area. Use a wire brush and wet clean to remove residual material from studs. Continue wet cleaning until the surface is clean of visible material and encapsulate stud walls.

c. Remove ACM thermal system insulation and mudded pipe fittings using mechanical means and wet methods and immediately place into 6-mil thickness disposal bag. Continue wet cleaning until surfaces are free of visible debris.

d. Remove ACM plaster ceilings or walls using mechanical means and adequately wet methods and immediately place into 6-mil thickness disposal bag. Make every effort to keep the material from falling to the floor of the work area. Continue wet cleaning until surfaces are free of visible debris.

e. Remove ACM textured ceiling finish using a scraper and wet methods and immediately place into 6-mil thickness disposal bag. Floors are considered contaminated from fallen textured ceiling finish. Clean up debris on floor and dispose of carpet as asbestos contaminated material. After removal of the material use a wire brush to clean the exposed concrete ceiling to remove residual material. Continue wet cleaning until surfaces are free of visible debris.

f. Remove ACM vermiculite using mechanical means and adequately wet methods and immediately place into 6-mil thickness disposal bag. Make every effort to keep the material from falling to the floor of the work area. Continue wet cleaning until surfaces are free of visible debris.

Bag all asbestos debris which has fallen to the floor as asbestos-containing debris. Place all debris in plastic disposal bags of 6-mil minimum thickness. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 6-mil minimum thickness disposal bag. Containerize asbestos containing waste while wet. Lower and otherwise handle asbestos containing materials as indicated in 40 CFR 61-SUBPART M. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the quantity of airborne asbestos fibers monitored at the breathing zone of the workers or the designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work, and immediately correct the situation.

3.2.10 Class II Work Procedures

In addition to the requirements of paragraphs MANDATED PRACTICES and CONTROL METHODS, the following engineering controls and work practices must be used:

a. A Competent Person must supervise the work.

b. For indoor work, critical barriers must be placed over all openings to the regulated area.

c. Impermeable dropcloths must be placed on surfaces beneath all removal activity.

3.2.11 Specific Control Methods for Class II Work

3.2.11.1 Suspect Fire Doors

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. A detached decontamination system may be used. Spread 6-mil plastic sheeting on the ground beneath the work area and around the perimeter of the work area extending out in all directions. Remove door intact from hinges and wrap with 6-mil plastic sheeting. Inspect the interior areas of the door to determine if ACM is present. If ACM is not present the door may be disposed of as general construction debris. If ACM is present place whole door in enclosed container for disposal. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.2 Roofing Materials

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. When removing roofing materials which contain ACM as described in 29 CFR 1926.1101(g)(8)(ii), use the following practices. Roofing material must be removed in an intact state. Wet methods must be used to remove roofing materials that are not intact, or that will be rendered not intact during removal, unless such wet methods are not feasible or will create safety hazards. When removing built-up roofs, with asbestos-containing roofing felts and an aggregate surface, using a power roof cutter, all dust resulting from the cutting operations must be collected by a HEPA dust collector, or must be HEPA vacuumed by vacuuming along the cut line. Asbestos-containing roofing material must not be dropped or thrown to the ground, but must be lowered to the ground via covered, dust-tight chute, crane, hoist or other method approved by the Contracting Officer. Any ACM that is not intact must be lowered to the ground as soon as practicable, but not later than the end of the work shift. While the material remains on the roof it must be kept wet or placed in an impermeable waste bag or wrapped in plastic sheeting. Intact ACM must be lowered to the ground as soon as practicable, but not later than the end of the work shift. Unwrapped material must be transferred to a closed receptacle. Critical barriers must be placed over roof level heating and ventilation air intakes. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.3 Cementitious Siding and Shingles or Transite Panels

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. When removing cementitious asbestos-containing siding, shingles or Transite panels use the following work practices. Intentionally cutting, abrading or breaking is prohibited. Each panel or shingle must be sprayed with amended water prior to removal. Nails must be cut with flat, sharp instruments. Unwrapped or unbagged panels or shingles must be immediately lowered to the ground via covered dust-tight chute, crane or hoist, or placed in an impervious waste bag or wrapped in plastic sheeting and lowered to the ground no later than the end of the work shift. Place debris into a 6-mil minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 6-mil minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.4 Gaskets

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. Gaskets must be thoroughly wetted with amended water prior to removal and immediately placed in a disposal container. If a gasket is visibly deteriorated and unlikely to be removed intact, removal must be undertaken within a glovebag. Any scraping to remove residue must be performed wet. Place debris into a 6-mil minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 6-mil minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.12 NOT USED

3.2.13 Abatement of Asbestos Contaminated Soil

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. Asbestos contaminated soil must be removed from areas to a minimum depth of 2 inches. Soil must be thoroughly dampened with amended water and then removed by manual shoveling into labeled containers. Place debris into a 6-mil minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 6-mil minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.14 Air Sampling

Perform sampling of airborne concentrations of asbestos fibers in accordance with 29 CFR 1926.1101, the Contractor's air monitoring plan and as specified herein. Sampling performed in accordance with 29 CFR 1926.1101must be performed by the PQP. Sampling performed for environmental and quality control reasons must be performed by the PQP. Unless otherwise specified, use NIOSH Method 7400 for sampling and analysis. Monitoring may be duplicated by the Government at the discretion of the Contracting Officer and NAVFAC Asbestos Program Manager. If the air sampling results obtained by the Government differ from those results obtained by the Contractor, the Government will determine which results predominate. Results of breathing zone samples must be posted at the job site and submitted to the Contracting Officer and NAVFAC Asbestos Program Manager. Submit all documentation regarding initial exposure assessments, negative exposure assessments, and air-monitoring results. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

3.2.14.1 Sampling Prior to Asbestos Work

Provide area air sampling and establish the baseline one day prior to the masking and sealing operations for each demolition or removal or encapsulation site. Establish the background by performing area sampling in similar but uncontaminated sites in the building.

3.2.14.2 Sampling During Asbestos Work

The PQP must provide personal sampling as indicated in 29 CFR 1926.1101.

Breathing zone samples must be taken for at least 25 percent of the workers in each shift, or a minimum of two, whichever is greater. Air sample fiber counting must be completed and results provided within 24-hours (breathing zone samples), and 8 hours (environmental/clearance monitoring) after completion of a sampling period. At the same time, the GC will provide area sampling close to the work inside the enclosure, outside the clean room entrance to the enclosure, at the waste load-out, and at the exhaust opening of the local exhaust system. If sampling outside the enclosure shows airborne levels have exceeded background or 0.010 fibers per cubic centimeter, whichever is greater, stop all work, correct the condition(s) causing the increase, and notify the Contracting Officer and NAVFAC Asbestos Program Manager immediately. Where alternate methods are used, perform personal and area air sampling at locations and frequencies that will accurately characterize the evolving airborne asbestos levels. The written results must be signed by testing laboratory analyst, testing laboratory principal and the Contractor's PQP. The air sampling results must be documented on a Contractor's daily air monitoring log. Copies of the air monitoring results must also be submitted daily directly to the Contracting Officer and NAVFAC Asbestos Program Manager. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

3.2.14.3 Final Clearance Requirements, NIOSH PCM Method

For PCM sampling and analysis using NIOSH NMAM Method 7400, the fiber concentration inside the abated regulated area, for each airborne sample, must be less than 0.010 f/cc. The abatement inside the regulated area is considered complete when every PCM final clearance sample is below the clearance limit. If any sample result is greater than 0.010 total f/cc, the asbestos fiber concentration (asbestos f/cc) must be confirmed from that same filter using NIOSH NMAM Method 7402 (TEM) at Contractor's expense. If any confirmation sample result is greater than 0.010 asbestos f/cc, abatement is incomplete and cleaning must be repeated at the Contractor's expense. Upon completion of any required recleaning, resampling with results to meet the above clearance criteria must be done at the Contractor's expense.

3.2.14.4 Final Clearance Requirements, EPA TEM Method

For EPA TEM sampling and analysis, using the EPA Method specified in 40 CFR 763, abatement inside the regulated area is considered complete when the arithmetic mean asbestos concentration of the five inside samples is less than or equal to 70 structures per square millimeter (70 S/mm). When the arithmetic mean is greater than 70 S/mm, the three blank samples must be analyzed. If the three blank samples are greater than 70 S/mm, resampling must be done. If less than 70 S/mm, the five outside samples must be analyzed and a Z-test analysis performed. When the Z-test results are less than 1.65, the decontamination must be considered complete. If the Z-test results are more than 1.65, the abatement is incomplete and cleaning must be repeated. Upon completion of any required recleaning, resampling with results to meet the above clearance criteria must be done at the Contractor's expense.

3.2.14.5 Sampling After Asbestos work Final Clean-Up (Clearance Sampling)

Provide area sampling of asbestos fibers using aggressive air sampling techniques as defined in the EPA 560/5-85-024 and establish an airborne asbestos concentration of less than 0.010 fibers per cubic centimeter after final clean-up but before removal of the enclosure or the asbestos work control area. After final cleanup and the asbestos control area is dry but prior to clearance sampling, the PQP and GC must perform a visual inspection in accordance with ASTM E1368 to ensure that the asbestos control and adjacent work area, including the Facility's HVAC Ductwork and Filters, are free of any accumulations of dirt, dust, debris or asbestos fibers. Prepare a written report signed and dated by the PQP certifying with all appropriate documenting that the asbestos control area, as well as the adjacent work areas, are free of dust, dirt, debris, asbestos fiber and all waste has been removed. Collect the appropriate number of samples in accordance with Maine DEP regulations. Use transmission electron microscopy (TEM) to analyze clearance samples and report the results in accordance with current NIOSH criteria. The asbestos fiber counts from these samples must be less than 0.010 fibers per cubic centimeter or be not greater than the background, whichever is greater. The report from the PQP must be sent directly to the Contracting Officer and NAVFAC Asbestos Program Manager for review. Do not remove the work control barriers and warning signs prior to the Contracting Officer's acknowledgement of receipt and approval of the PQP certification. Clearance sampling results must be submitted to Contracting Officer & NAVFAC Asbestos Program Manager within 72 hours after the samples are taken. The Contractor should allow fourteen (14) calendar days for the Government review period. Should any of the final samples indicate a higher value take appropriate actions to re-clean the area and repeat the sampling and TEM analysis at the Contractor's expense.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent or related to failure to comply with these specifications are entirely the responsibility of the Contractor and shall not be made the subject, or any component of any request for additional time or compensation

3.2.14.6 Air Clearance Failure

If clearance sampling results fail to meet the final clearance requirements, the Contracor is responsible to pay all costs associated with the required recleaning, resampling, and analysis, until final clearance requirements are met.

3.2.15 Lock-Down

Prior to removal of plastic barriers and after pre-clearance clean up of gross contamination, the PQP must notify the Contracting Officer and the NAVFAC Asbestos Program Manager of the date of the final visual inspection. The visual inspection of the all areas affected by the removal shall be completed in accordance with ASTM E1368. Inspect for any visible fibers and to ensure that encapsulants were applied evenly and appropriately. Spray apply a post removal (lock-down) encapsulant to ceiling, walls, floors and other areas exposed in the removal area. The exposed area includes but is not limited to plastic barriers, furnishings and articles to be discarded as well as dirty change room, air locks for bag removal and decontamination chambers.

3.2.16 Site Inspection

While performing asbestos engineering control work, the Contractor must be subject to on-site inspection by the Contracting Officer who may be assisted by or represented by safety or industrial hygiene personnel. If the work is found to be in violation of this specification, the Contracting Officer or his representative will issue a stop work order to be in effect immediately and until the violation is resolved. All related costs including standby time required to resolve the violation must be at the Contractor's expense.

3.3 CLEAN-UP AND DISPOSAL

3.3.1 Housekeeping

Essential parts of asbestos dust control are housekeeping and clean-up procedures. Maintain surfaces of the asbestos control area free of accumulations of asbestos fibers. Give meticulous attention to restricting the spread of dust and debris into other work areas beyond the asbestoc control area; keep waste from being distributed over the general area. Use HEPA filtered vacuum cleaners. DO NOT BLOW DOWN THE SPACE WITH COMPRESSED AIR. When asbestos removal is complete,all sampling & testing has been completed as required herein, including the PQP providing all required reports and inspections, all asbestos waste is removed from the work-site, and final clean-up is completed, the Contracting Officer will attest that the area is safe before the signs can be removed.

After final clean-up and acceptable airborne concentrations are attained but before the HEPA unit is turned off and the enclosure removed, remove all pre-filters on the building HVAC system and provide new pre-filters. The Contractor is to clean all HVAC Ductwork if the facility's HVAC system is operational and cannot be isolated from the asbestos control area or there is any evidence that the pre-filters were not effective. Dispose of filters as asbestos contaminated materials. Reestablish HVAC mechanical, and electrical systems in proper working order. The PQP and CP will visually inspect all surfaces within the enclosure for residual material or accumulated dust or debris. The Contractor must re-clean all areas showing dust or residual materials. If re-cleaning is required, air sample and establish an acceptable asbestos airborne concentration after re-cleaning. The Contracting Officer must agree that the area is safe in writing before unrestricted entry will be permitted. The Government must have the option to perform monitoring to determine if the areas are safe before entry is permitted.

3.3.2 Title to Materials

All waste materials, except as specified otherwise, become the property of the Contractor and must be disposed of as specified in applicable local, State, and Federal regulations and herein.

The Portsmouth Naval Shipyard Hazardous Waste Facility takes title and is responsible for all asbestos waste disposal.

3.3.3 Disposal of Asbestos

3.3.3.1 Procedure for Disposal

Coordinate all waste disposal with the Contracting Officer and the Portsmouth Naval Shipyard Hazardous Waste Facility (Code 106, Building 357). Collect asbestos waste, contaminated waste water filters,air filters, asbestos contaminated water, scrap, debris, bags, containers, equipment, and asbestos contaminated clothing which may produce airborne concentrations of asbestos fibers and place in sealed fiber-proof, waterproof, non-returnable containers (e.g. double plastic bags 6 mils thick, cartons, drums or cans). Wastes within the containers must be adequately wet in accordance with MEDEP Chapter 425: Asbestos Management Regulations and 40 CFR 61-SUBPART M. All waste materials, including, but not limited to, piping, must be less than 5 feet in length. Affix a warning and Department of Transportation (DOT) label to each container including the bags or use at least 6 mils thick bags with the approved warnings and DOT labeling preprinted on the bag. Clearly indicate on the outside of each container the name of the waste generator and the location at which the waste was generated. Prevent contamination of the transport vehicle (especially if the transport vehicle is a rented truck likely to be used in the future for non-asbestos purposes). These precautions include lining the vehicle cargo area with plastic sheeting (similar to work area enclosure) and thorough cleaning of the cargo area after transport and unloading of asbestos debris is complete.

The Contractor must carefully remove material and immediately place it in approved asbestos disposal bags/containers. If the bags/containers are not pre-marked, a Contractor furnished caution label must be permanently attached to the bags/containers. The Contractor must also furnish and affix a label permanently attached to the bags which provides the following information:

* Abatement Contractor Name
* Construction Contract Number
* Building Number
* Portsmouth Naval Shipyard, Kittery, Maine
* Date

Contractor must ensure all asbestos materials are properly packaged prior to placement into government-furnished dumpsters.

3.3.3.2 Asbestos Disposal Quantity Report

Direct the PQP to record and report, to the Contracting Officer and Asbestos Program Manager, the amount of asbestos containing material removed and released for disposal. Deliver the report for the previous day at the beginning of each day shift with amounts of material removed during the previous day reported in linear feet or square feet as described initially in this specification and in cubic feet for the amount of asbestos containing material released for disposal.

Allow the GC to inspect, record and report the amount of asbestos containing material removed and released for disposal on a daily basis.

-- End of Section --

Specification Section 1.3.10 - Asbestos Hazard Abatement Plan Checklist - Building XX
Prepared by: _____________ Date: __________

Spec Section(s)	Item Required	Included?	Section/ Page #	Notes
	Confirm that the Asbestos Hazard Abatement Plan contains the following:			
1.3.10	- Signature and seal to document that Plan was prepared by a PQP			
1.3.10	- Site sketch with all project details (control areas, critical barriers, physical boundaries, decon facilities, viewing ports, and mechanical ventilation system (negative air machines, etc.))			
1.3.10, 1.5.8	- Confirmation that meeting will be scheduled w/ ktr, supervisor, CP, PQP, CO, and APM to review Plan			
3.2.2, 3.2.3, 3.2.4	- Description of preparatory activities - protecting existing conditions, moving furniture, pre-cleaning			
1.3.10, 3.2	- Description of safety precautions including, but not limited to, LO/TO, CSE, fall protection			
1.3.10	- Description of work procedures for **each** activity			
1.3.10, 3.2.6	- Description of asbestos removal method(s) for **each** activity			
3.1.10, 3.2.5.2	- Description of glovebag equipment and process (as appropriate)			
1.3.10	- Description of controls for **each** activity			
1.3.11	- Description of job responsibilities for **each** activity			
1.3.10	- Description of respirators and personal protective equipment			
1.3.10, 1.4	- Description of equipment for **each** activity. Include all equipment spec sheets			
1.3.9, 1.3.10, 1.4	- Description of materials for **each** activity. Include all SDS			
1.3.10	- Description of hygiene facilities and sanitary procedures (including no eating, drinking, smoking)			
1.3.10	- Description of interface of trades			
1.3.10	- Description of work sequencing			
3.1.4	- Description of the method and construction of containment for **each** activity			
3.1.6	- Description of required signage for **each** activity			
1.3.10, 3.1.3.3, 3.1.11	- Description of decontamination equipment and process			
1.3.10, 3.1.7, 3.2.1	- Description of local exhaust system and critical barrier equipment and process, including:			
	- Confirmation that Pressure Differential Recordings will be submitted w/in **24 hours**			
	- Confirmation that negative pressure will be maintained continuously until work area cleared			
3.3	- Description of cleanup procedures during and at the end of work.			
3.2.15	- Description of method, material, etc. for application of lock-down encapsulant.			
3.2.14	- Description of personal sampling and analysis strategy and methodology, number of samples, and qualifications of sampling personnel, including:			
3.2.14	- Confirmation that all personal sampling results will be submitted to NAVFAC w/in **24 hours**			
3.2.14	- Specific thresholds for work stoppage, PPE upgrade, etc.			
3.2.14	- Detailed description of area sampling and analysis strategy and methodology, frequency of sampling, sample locations, duration of sampling, and qualifications of sampling personnel, including:			
3.2.14	- Confirmation that all area sampling results will be submitted to NAVFAC w/in **16 hours**			
3.2.14	- Listing of specific sampling locations. **Include locations in Notes column (to the right)**			
3.2.14	- Specific thresholds for work stoppage, PPE upgrade, etc.			
3.2.14	- Detailed description of clearance sampling and analysis strategy and methodology, number of samples, and qualifications of sampling personnel, including:			
3.2.14	- Confirmation that all clearance sampling results will be submitted to NAVFAC w/in **8 hours**			
3.2.14	- Listing of specific sampling locations. **Include locations in Notes column (to the right)**			
1.3.10, 1.4	- Detailed plan for Asbestos Waste Management, including:			
1.3.10	- Asbestos Materials			
1.3.10	- Ancillary waste, including poly sheeting			
1.3.10	- Wastewater - washwater, decon water, etc. treated and/or disposal			
3.1.5	- Detailed description of waste load-out set-up and procedure			
1.3.8	- Name, qualifications, and certification of Asbestos Hazard Control Supervisor			
1.3.10	- Copies of Fire and Medical Emergency Response Plans and AHA			
1.3.4, 1.4	- Copies of Training Certificates for Workers and Supervisors			
1.3.4, 1.4, 1.5.3	- Copies of Licenses for Workers and Supervisors			
1.3.4, 1.3.7	- Copies of Respiratory Protection Program, Including Training and Fit Testing Records			
1.3.3, 1.3.14	- Copies of Medical Examinations, Surveillance and Records			
1.3.5	- Copies of Abatement Licenses, Permits and Notifications			
1.3.11, 1.4	- Documentation of Testing Laboratory Qualifications/Certificates, including documentation that the lab is:			
1.3.11	- Maine licensed/permitted/certified			
1.3.11	- AIHA accredited			
1.3.11	- NIST/NVLAP accredited			
1.3.11	- Independent of contractor			
1.4, 1.5.1	- Documentation of Private Qualified Person Qualifications			
1.4, 1.5.2	- Documentation of Competent Person Qualifications			
1.4, 1.5.4	- Copy of Contractor's License			
1.4, 1.5.7	- Details on Federal, Sate, Local Citations w/in 5 Years (if any)			
1.4	- Documentation of Third Party Consultant Qualifications			
1.4, 3.1.9	- Copy of Statement Regarding Use of Rental Equipment			

CHECKLIST COMPLETED BY:____ DATE: _________________________________

NOTE: This checklist is not intended to list all of the Asbestos Hazard Abatement Plan components required per Specification Section 02 82 00.00 22, Asbestos Remediation. The Contractor is required to identify and meet all applicable Specification requirements.

SECTION 02 83 00.00 22

MANAGEMENT OF LEAD, CADMIUM, AND CHROMIUM (VI) DURING RENOVATION, DEMOLITION, REMOVAL, AND ABATEMENT

(PNS PROJECTS)
10/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ANSI/ASSP Z9.2	(2018) Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems

ASTM INTERNATIONAL (ASTM)

ASTM E1613	(2012) Standard Test Method for Determination of Lead by Inductively Coupled Plasma Atomic Emission Spectrometry (ICP-AES) (IH DB-DBB SoW only identifies AAS), Flame Atomic Absorption Spectrometry (FAAS), or Graphite Furnace Atomic Absorption Spectrometry (GFAAS) Techniques
ASTM E1644	(2017) Standard Practice for Hot Plate Digestion of Dust Wipe Samples for the Determination of Lead
ASTM E1726	(2001; R 2009) Preparation of Soil Samples by Hotplate Digestion for Subsequent Lead Analysis
ASTM E1727	(2016) Standard Practice for Field Collection of Soil Samples for Subsequent Lead Determination
ASTM E1728	(2016) Collection of Settled Dust Samples Using Wipe Sampling Methods for Subsequent Lead Determination
ASTM E1792	(2003; R 2016) Standard Specification for Wipe Sampling Materials for Lead in Surface Dust

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 701	(2023) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety and Health Requirements Manual

U.S. DEPARTMENT OF HOUSING AND URBAN DEVELOPMENT (HUD)

HUD 6780	(1995; Errata Aug 1996;Rev Ch. 7 - 1997) Guidelines for the Evaluation and Control of Lead-Based Paint Hazards in Housing

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1926.103	Respiratory Protection
29 CFR 1926.1126	Chromium (VI)
29 CFR 1926.1127	Cadmium
29 CFR 1926.21	Safety Training and Education
29 CFR 1926.33	Access to Employee Exposure and Medical Records
29 CFR 1926.55	Gases, Vapors, Fumes, Dusts, and Mists
29 CFR 1926.59	Hazard Communication
29 CFR 1926.62	Lead
29 CFR 1926.65	Hazardous Waste Operations and Emergency Response
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 745	Lead-Based Paint Poisoning Prevention in Certain Residential Structures
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements

49 CFR 178	Specifications for Packagings

U.S. NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

ND OPNAVINST 5100.23	(2005; Rev G) Navy Occupational Safety and Health (NAVOSH) Program Manual

UNDERWRITERS LABORATORIES (UL)

UL 586	(2009; Reprint Dec 2017) UL Standard for Safety High-Efficiency Particulate, Air Filter Units

PORTSMOUTH NAVAL SHIPYARD (PNSY)

5090.6F	(4-21) Solid Waste Operations
5090.30	(4-05) Hazardous Waste Generator Standards

1.2 DEFINITIONS

1.2.1 Abatement

Measures defined in 40 CFR 745, Section 223, designed to permanently eliminate lead-based paint hazards.

1.2.2 Action Level

Employee exposure, without regard to use of respirators, to an airborne concentration of lead of 30 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of cadmium of 2.5 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of chromium (VI) of 2.5 micrograms per cubic meter of air averaged over an 8-hour period.

1.2.3 Area Sampling

Sampling of lead, cadmium, and chromium (VI) concentrations within the lead, cadmium, and chromium (VI) control area and inside the physical boundaries which is representative of the airborne lead, cadmium, and chromium (VI) concentrations but is not collected in the breathing zone of personnel (approximately 5 to 6 feet above the floor).

1.2.4 Cadmium Permissible Exposure Limit (PEL)

Five micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.1127. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

PEL (micrograms/cubic meter of air) = 40/No. hrs worked per day

1.2.5 Certified Industrial Hygienist (CIH)

As used in this Section refers to a person retained by the Contractor who is certified as an industrial hygienist and who is trained in the recognition and control of lead, cadmium and chromium (VI) hazards in accordance with current Federal, State, and local regulations. CIH must be certified for comprehensive practice by the American Board of

Industrial Hygiene. The Certified Industrial Hygienist must be independent of the Contractor and must have no employee or employer relationship which could constitute a conflict of interest.

1.2.6 Child-Occupied Facility

Real property which is a building or portion of a building constructed prior to 1978 visited regularly by the same child, six-years of age or under, on at least two different days within any week (Sunday through Saturday period), provided that each day's visit lasts at least 3-hours, and the combined annual visits last at least 60-hours. Child-occupied facilities include but are not limited to, day-care centers, preschools and kindergarten classrooms.

1.2.7 Chromium (VI) Permissible Exposure Limit (PEL)

Five micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.1126. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

PEL (micrograms/cubic meter of air) = 40/No. hrs worked per day

1.2.8 Competent Person (CP)

As used in this Section, refers to a person employed by the Contractor who is trained in the recognition and control of lead, cadmium and chromium (VI) hazards in accordance with current Federal, State, and local regulations and has the authority to take prompt corrective actions to control the lead, cadmium and chromium (VI) hazard. The Contractor may provide more than one CP as required to supervise and monitor the work. The CP must be a Certified Industrial Hygienist (CIH) certified by the American Board of Industrial Hygiene or a Certified Safety Professional (CSP) certified by the Board of Certified Safety Professionals or a licensed lead-based paint abatement Supervisor/Project Designer in the State of Maine.

1.2.9 Contaminated Room

Refers to a room for removal of contaminated personal protective equipment (PPE).

1.2.10 Decontamination Shower Facility

That facility that encompasses a clean clothing storage room, and a contaminated clothing storage and disposal rooms, with a shower facility in between.

1.2.11 Deleading

Activities conducted by a person who offers to eliminate lead-based paint or lead-based paint hazards or paints containing cadmium/chromium (VI) or to plan such activities in commercial buildings, bridges or other structures.

1.2.12 Eight-Hour Time Weighted Average (TWA)

Airborne concentration of lead, cadmium, and chromium (VI) to which an employee is exposed, averaged over an 8-hour workday as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

1.2.13 High Efficiency Particulate Air (HEPA) Filter Equipment

HEPA filtered vacuuming equipment with a UL 586 filter system capable of collecting and retaining lead, cadmium, and chromium (VI)-contaminated particulate. A high efficiency particulate filter demonstrates at least 99.97 percent efficiency against 0.3 micron or larger size particles.

1.2.14 Lead

Metallic lead, inorganic lead compounds, and organic lead soaps. Excludes other forms of organic lead compounds. The use of the term Lead in this Section also refers to paints which contain detectable concentrations of Cadmium and Chromium (VI). For the purposes of the Section lead-based paint (LBP) and paint with lead (PWL) also contains cadmium and chromium (VI).

1.2.15 Lead-Based Paint (LBP)

Paint or other surface coating that contains lead in excess of 1.0 milligrams per centimeter squared or 0.5 percent by weight.

1.2.16 Lead-Based Paint Activities

In the case of target housing or child occupied facilities, lead-based paint activities include; a lead-based paint inspection, a risk assessment, or abatement of lead-based paint hazards.

1.2.17 Lead-Based Paint Hazards

Paint-lead hazard, dust-lead hazard or soil-lead hazard as identified in 40 CFR 745, Section 65. Any condition that causes exposure to lead from lead-contaminated dust, lead-contaminated soil, lead-based paint that is deteriorated or present in accessible surfaces, friction surfaces, or impact surfaces that would result in adverse human health effects.

1.2.18 Lead, Cadmium, and Chromium (VI) Control Area

A system of control methods to prevent the spread of lead, cadmium, and chromium (VI) dust, paint chips or debris to adjacent areas that may include temporary containment, floor or ground cover protection, physical boundaries, and warning signs to prevent unauthorized entry of personnel. HEPA filtered local exhaust equipment may be used as engineering controls to further reduce personnel exposures or building/outdoor environmental contamination.

1.2.19 Lead Permissible Exposure Limit (PEL)

Fifty micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.62. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

PEL (micrograms/cubic meter of air) = 400/No. hrs worked per day

1.2.20 Material Containing Lead/Paint with Lead (MCL/PWL)

Any material, including paint, which contains lead as determined by the testing laboratory using a valid test method. The requirements of this Section do not apply if no detectable levels of lead are found using a quantitative method for analyzing paint or MCL using laboratory instruments with specified limits of detection (usually 0.01 percent). An X-Ray Fluorescence (XRF) instrument is not considered a valid test method.

1.2.21 Personal Sampling

Sampling of airborne lead, cadmium, and chromium (VI) concentrations within the breathing zone of an employee to determine the 8-hour time weighted average concentration in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Samples must be representative of the employees' work tasks. Breathing zone must be considered an area within a hemisphere, forward of the shoulders, with a radius of 6 to 9 inches and centered at the nose or mouth of an employee.

1.2.22 Physical Boundary

Area physically roped or partitioned off around lead, cadmium, and chromium (VI) control area to limit unauthorized entry of personnel.

1.2.23 Target Housing

Residential real property which is housing constructed prior to 1978, except housing for the elderly or persons with disabilities (unless any one or more children age 6-years or under resides or is expected to reside in such housing for the elderly or persons with disabilities) or any zero bedroom dwelling.

1.3 DESCRIPTION

This specification covers all project-related activities, including, but not limited to, renovation, demolition, removal, and/or remediation, impacting PWL and/or material containing lead, cadmium, and chromium (VI) located as indicated on the drawings. Refer to the report located at the end of Section 02 82 00.00 22 ASBESTOS REMEDIATION (PNS PROJECTS). The work covered by this Section includes work tasks and the precautions specified in this Section for the protection of building occupants and the environment during and after the performance of the hazard abatement activities in accordance with State and Federal regulations and as specified herein.

1.3.1 Protection of Existing Areas To Remain

Project work including, but not limited to, lead, cadmium, and chromium (VI) hazard abatement work, storage, transportation, and disposal must be performed without damaging or contaminating adjacent work and areas. Where such work or areas are damaged or contaminated, restore work and areas to the original condition.

1.3.2 Coordination with Other Work

Coordinate with work being performed in adjacent areas to ensure there are no exposure issues. Explain coordination procedures in the Lead, Cadmium, and Chromium (VI) Compliance Plan and describe how the Contractor will prevent lead, cadmium and chromium (VI) exposure to other contractors and Government personnel performing work unrelated to lead, cadmium and chromium (VI) activities.

1.3.3 Sampling and Analysis

Submit a log of the analytical results from sampling conducted during the abatement. Keep the log of results current with project activities and brief the results to the Contracting Officer and Planning Design and Construction (PD&C) Hazardous Materials point of contact (HAZMAT POC) as analytical results are reported. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

1.3.3.1 Dust Wipe Materials, Sampling and Analysis

Sampling must conform to ASTM E1728 and ASTM E1792. Analysis must conform to ASTM E1613 and ASTM E1644.

1.3.3.2 Soil Sampling and Analysis

Sampling must conform to ASTM E1727. Analysis must conform to ASTM E1613 and ASTM E1726.

1.3.3.3 Clearance Monitoring

a. Collect dust wipe samples inside, and, as appropriate, outside, the lead, cadmium and chromium (VI) hazard control area after the final visual inspection.

b. Collect exterior bare soil samples inside the lead, cadmium and chromium (VI) hazard control area after the final visual inspection.

1.3.4 Clearance Requirements

The target clearance level is as follows:

Target Clearance Level

(1) Lead - 40 ug/sq.ft. on floors; 250 ug/sq.ft. on interior window sills; 400 ug/sq.ft. on window troughs; or below concentrations detected in wipe samples collected before commencement of work - whichever is lower.

(2) Chromium (VI) - Below concentrations detected in wipe samples collected before commencement of work.

(3) Cadmium - Below concentrations detected in wipe samples collected before commencement of work.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Competent Person Qualifications; G

Training Certification; G

Occupational and Environmental Assessment Data Report; G

Medical Examinations; G

Lead, Cadmium, and Chromium (VI) Waste Management Plan; G

Licenses, Permits and Notifications; G

Lead, Cadmium, and Chromium (VI) Compliance Plan; G

Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist; G

Waste Characterization Sampling Plan; G

Sample Results; G

SD-03 Product Data

Respirators; G

Vacuum Filters; G

Negative Air Pressure System; G

Materials and Equipment; G

Expendable Supplies; G

Local Exhaust Equipment; G

Pressure Differential Automatic Recording Instrument; G

Pressure Differential Log; G

SD-06 Test Reports

Sampling and Analysis; G

Occupational and Environmental Assessment Data Report; G

Sampling Results; G

Pressure Differential Recordings For Local Exhaust System; G

SD-07 Certificates

Testing Laboratory; G

Third Party Consultant Qualifications; G

Notification of the Commencement of Work Impacting Lead, Cadmium, or Chromium (VI); G

Clearance Certification; G

SD-11 Closeout Submittals

Turn-In Documents or Weight Tickets; G

1.5 QUALITY ASSURANCE

1.5.1 Qualifications

1.5.1.1 Competent Person (CP)

Submit name, address, and telephone number of the CP selected to perform responsibilities specified in paragraph COMPETENT PERSON (CP) RESPONSIBILITIES. Provide documented construction project-related experience with implementation of OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium (VI) standard (29 CFR 1926.1126), and Cadmium standard (29 CFR 1926.1127) which shows ability to assess occupational and environmental exposure to lead, cadmium, and chromium (VI); experience with the use of respirators, personal protective equipment and other exposure reduction methods to protect employee health. Demonstrate a minimum of 5 years experience implementing OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium (VI) standard (29 CFR 1926.1126), and Cadmium standard (29 CFR 1926.1127). Submit proper documentation that the CP is trained and licensed and certified in accordance with Federal, State and local laws. The competent person must possess the qualifications detailed in Paragraph entitled COMPETENT PERSON herein.

1.5.1.2 Training Certification

Submit a current certificate for each worker and supervisor, signed and dated by the accredited training provider, stating that the employee has received the required lead, cadmium and chromium (VI) training specified in 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 40 CFR 745 and is certified to perform or supervise deleading, lead removal or demolition activities in the State of Maine.

1.5.1.3 Testing Laboratory

Submit the name, address, and telephone number of the testing laboratory selected to perform the air, soil, and wipe analysis, testing, and reporting of airborne concentrations of lead, cadmium and chromium (VI). Use a laboratory participating in the EPA National Lead Laboratory Accreditation Program (NLLAP) by being accredited by either the American Association for Laboratory Accreditation (A2LA) or the American Industrial Hygiene Association (AIHA) and that is successfully participating in the Environmental Lead Proficiency Analytical Testing (ELPAT) program to perform sample analysis. Laboratories selected to perform blood lead analysis must be OSHA approved. Maine Department of Environmental Protection certification, National Environmental Laboratories Accreditation Conference (NELAC) certification, and certifications for all corresponding EPA test methods are required for waste determination testing, as applicable.

1.5.1.4 Third Party Consultant Qualifications

Submit the name, address and telephone number of the third party consultant selected to perform the wipe sampling for determining concentrations of lead, cadmium and chromium (VI) in dust. Submit proper documentation that the consultant is trained and certified as an inspector technician or inspector/risk assessor by the USEPA authorized State (or local) certification and accreditation program.

1.5.1.5 Certified Risk Assessor

The Certified Risk Assessor must be certified pursuant to 40 CFR 745, Section 226 and be responsible to perform the clearance sampling, clearance sample data evaluation and summarize clearance sampling results in a section of the abatement report. The risk assessor must sign the abatement report to indicate clearance requirements for the contract have been met.

1.5.2 Requirements

1.5.2.1 Competent Person (CP) Responsibilities

a. Verify training meets all Federal, State, and local requirements.

b. Review and approve Lead, Cadmium, and Chromium (VI) Compliance Plan for conformance to the applicable referenced standards.

c. Continuously inspect LBP/PWL or MCL work for conformance with the approved plan.

d. Perform (or oversee performance of) air sampling. Recommend upgrades or downgrades (whichever is appropriate based on exposure) on the use of PPE (respirators included) and engineering controls.

e. Ensure work is performed in strict accordance with specifications at all times.

f. Control work to prevent hazardous exposure to human beings and to the environment at all times.

g. Supervise final cleaning of the lead, cadmium, and chromium (VI) control area, take clearance wipe samples if necessary; review clearance sample results and make recommendations for further cleaning.

h. Certify the conditions of the work as called for elsewhere in this specification.

1.5.2.2 Lead, Cadmium, and Chromium (VI) Compliance Plan

Submit a detailed job-specific plan of the work procedures to be used in the disturbance of lead, cadmium and chromium (VI), LBP/PWL or MCL. Include in the plan a sketch showing the location, size, and details of lead, cadmium, and chromium (VI) control areas, critical barriers, physical boundaries, location and details of decontamination facilities, viewing ports, and mechanical ventilation system. Include a description of equipment and materials, work practices, controls and job responsibilities for each activity from which lead, cadmium, and chromium (VI) is emitted. Include in the plan, eating, drinking, smoking, hygiene facilities and sanitary procedures, interface of trades, sequencing of lead, cadmium, and chromium (VI) related work, collected waste water and dust containing lead, cadmium, and chromium (VI) and debris, air sampling, respirators, personal protective equipment, and a detailed description of the method of containment of the operation to ensure that lead, cadmium, and chromium (VI) is not released outside of the lead, cadmium, and chromium (VI) control area. Include site preparation, cleanup and clearance procedures. Include occupational and environmental sampling, training and strategy, sampling and analysis strategy and methodology, frequency of sampling, duration of sampling, and qualifications of sampling personnel in the air sampling portion of the plan. Include a description of arrangements made among contractors on multicontractor worksites to inform affected employees and to clarify responsibilities to control exposures. If work will occur on a historic building, then include in the plan a description of how the proposed activities will comply with other relevant specification Sections to avoid or minimize impacts to historic fabric (for example, but not limited to, specification Sections related to historic masonry).

The plan must be developed and signed by a certified Lead Project Designer in the State of Maine. The plan must include the name and certification number of the person signing the plan.

1.5.2.3 Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist

The Contractor must complete the Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist included as Attachment A at the end of this Section. The Checklist must be signed by the properly-certified person preparing the Plan and must be submitted to the Government for review and approval with the Lead, Cadmium, and Chromium (VI) Compliance Plan.

1.5.2.4 Occupational and Environmental Assessment Data Report

If initial monitoring is necessary, submit occupational and environmental sampling results to the Contracting Officer and PD&C HAZMAT POC within three working days of collection, signed by the testing laboratory employee performing the analysis, the employee that performed the sampling, and the CP.

In order to reduce the full implementation of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 the Contractor must provide documentation. Submit a report that supports the determination to reduce full implementation of the requirements of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and supporting the Lead, Cadmium, and Chromium (VI) Compliance Plan.

a. The initial monitoring must represent each job classification, or if working conditions are similar to previous jobs by the same employer, provide previously collected exposure data that can be used to estimate worker exposures per 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. The data must represent the worker's regular daily exposure to lead, cadmium, and chromium (VI) for stated work. The initial monitoring must represent each task and/or type of activity. The initial monitoring must represent each/every PWL and/or material containing lead, cadmium, and chromium (VI).

b. Submit worker exposure data gathered during the task based trigger operations of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 with a complete process description. This includes, but is not limited to, manual demolition, manual scraping, manual sanding, heat gun, power tool cleaning, rivet busting, cleanup of dry expendable abrasives, abrasive blast enclosure removal, abrasive blasting, welding, cutting and torch burning where lead, cadmium and chromium (VI) containing coatings are present.

c. The initial assessment must determine the requirement for further monitoring and the need to fully implement the control and protective requirements including the lead, cadmium, and chromium (VI) compliance plan per 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

1.5.2.5 Medical Examinations

Submit a signed letter from an officer of the company confirming each worker engaged in heavy metals related activities in the contract has current medical surveillance as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Where exposure is determined to be above a lead, cadmium, or chromium action level the contractor must provide pre-work and post-work blood levels as requested by the government. All medical records submitted shall not contain any Protected Health Information (PHI) (e.g's home address, social security number, physicians name, medical record number, etc.). Initial medical surveillance as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 must be made available to all employees exposed to lead, cadmium, and chromium (VI) at any time (one day) above the action level. Full medical surveillance must be made available to all employees on an annual basis who are or may be exposed to lead, cadmium and chromium (VI) in excess of the action level for more than 30 days a year or as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Adequate records must show that employees meet the medical surveillance requirements of 29 CFR 1926.33, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 29 CFR 1926.103. Provide medical surveillance to all personnel exposed to lead, cadmium, and chromium (VI) as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Maintain complete and accurate medical records of employees for the duration of employment plus 30 years.

1.5.2.6 Training

Train each employee performing work that disturbs lead, cadmium, and chromium (VI), who performs LBP/MCL/PWL disposal, and air sampling operations prior to the time of initial job assignment and annually thereafter, in accordance with 29 CFR 1926.21, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, 40 CFR 745, 49 CFR 172, and State of Maine and local regulations where appropriate.

1.5.2.7 Respiratory Protection Program

a. Provide each employee required to wear a respirator a respirator fit test at the time of initial fitting and at least annually thereafter as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

b. Establish and implement a respiratory protection program as required by 29 CFR 1926.103, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, and 29 CFR 1926.55.

c. Provide training and fit test certificates for each employee performing LBP/MCL/PWL work.

1.5.2.8 Hazard Communication Program

Establish and implement a Hazard Communication Program as required by 29 CFR 1926.59.

1.5.2.9 Lead, Cadmium, and Chromium (VI) Waste Management

The Lead, Cadmium, and Chromium (VI) Waste Management Plan must comply with applicable requirements of federal, State, and local hazardous waste regulations and address:

a. Identification and classification of wastes associated with the work.

b. Estimated quantities of wastes to be generated and disposed of.

c. Names and qualifications (experience and training) of personnel who will be working on-site with hazardous wastes.

d. List of waste handling equipment to be used in performing the work, to include cleaning, volume reduction, and transport equipment.

e. Spill prevention, containment, and cleanup contingency measures including a health and safety plan to be implemented in accordance with 29 CFR 1926.65.

f. Work plan and schedule for waste containment and management. Proper containment of the waste includes using acceptable waste containers, as well as proper marking/labeling of the containers. Clean up and containerize wastes daily.

g. Include any process that may alter or treat waste. Note: Altering or treatment of a hazardous waste is not allowed to include any process that dilutes the waste or renders a hazardous waste non-hazardous.

h. All HW must be placed in a Shipyard-permitted HWAA or Contractors must turn HW in to a HWAA not later than the end of the shift on which it is generated. Responsibility for compliance is upon the Contractor. All hazardous wastes generated within the confines of the Shipyard are disposed of by the Government. Accordingly, all hazardous wastes generated by the Contractor to accomplish requirements of this Contract will be considered Government-generated, and disposed of by the Government. Contractor must not bring hazardous wastes onto Government property.

1.5.2.10 Waste Characterization Sampling Plan

A sampling plan must be developed that details practices to be followed for the collection, management, and analysis of waste characterization samples to determine if waste materials generated by the referenced project will be classified as hazardous or non-hazardous to ensure proper disposal. The plan must propose sampling that meets the requirements of all applicable local, State, and Federal statutes and regulations, Navy and PNSY requirements, and project specifications. At a minimum, the sampling plan must include:

a. Confirmation that the sampling plan will be reviewed and approved by the Navy (Code 106.3) prior to conducting any sampling. Code 106.3 review period is fourteen days from receipt of the plan.

b. The timing of waste characterization sampling and analysis. To the extent feasible, waste sampling and analysis must be completed prior to the accomplishment of any project work to properly characterize waste and ensure hazardous waste is not commingled with non-hazardous waste.

c. A project summary with a brief description of the project, a description of the anticipated waste and how the waste will be generated, a map or diagram with sample locations and type, an estimate of the amount of waste to be generated, and the name and location of the disposal facility the waste will go to if waste is to be disposed of off-site, as approved by Code 106.3.

d. A statement in the plan that the sample collection, management, and analysis will be performed per all applicable local, state, and federal statutes and regulations, project specifications, Navy and PNSY requirements to include NAVSHIPYD PTSMHINST 5090.8 (latest revision) and other PNSY guidance as applicable.

e. The proposed date(s) for sampling and expected duration of the sampling event(s). All waste sampling requires Code 106.3 oversight with seven (7) calendar day notification prior to a sampling event.

f. A narrative of how the sample(s) will be taken (e.g. discrete, composite, grab), a list of all equipment to be used for sampling including PPE, sampling instruments, containers; a description of the decontamination process for reusable equipment between consecutive samples, documentation that the Environmental Field Technician (EFT) conducting the waste sampling has training in proper collection and decontamination procedures, and a statement giving rationale for why the sample(s) will be representative.

g. A listing of the constituents to be analyzed for, identifying the analytical tests methods to be performed with the corresponding EPA test methods numbers. At a minimum, the waste characterization must include laboratory analysis for RCRA 8 Metals via the Toxicity Characteristic Leaching Procedure (TCLP). Depending on the methods used in project implementation, additional waste characterization analysis may be required (for example, waste characterization analysis for corrosivity may be required if high pH solutions are used for paint removal).

h. The name and location of the laboratory the samples will be sent to and confirmation that the selected lab has Maine Department of Environmental Protection certification, National Environmental Laboratories Accreditation Conference (NELAC) certification and certifications for all corresponding EPA test methods.

i. Confirmation that the samples will be submitted, under strict chain-of-custody (COC) requirements, to the certified laboratory.

    1. Laboratory analytical results.

    2. Field log book entries.

    3. Copy of the Chain of Custody (COC).

    4. Pertinent diagrams and maps.

    5. All other supporting data.

j. A statement that all materials used in the collection of the samples (gloves, wipes, trash, etc.) and hazardous debris generated by the sampling process will be containerized in a bag and delivered to

Building 357 for disposal by the end of the shift generated.

k. Confirmation that the sampling procedure will include, but not be limited to, the following:

1. The Contractor will notify the Contracting Officer's Representative and PNSY Code 106.3 Sampling Program Manager seven (7) calendar days prior to each sampling event to ensure the area and materials to be sampled are witnessed by all parties.

2. The EFT taking the samples will inspect the work area/waste with Code 106.3 to visually and tactilely assess the materials to be sampled.

3. Using a field log, the EFT will record the locations, types, quantities and other salient details related to materials to be sampled.

4. The EFT will collect sufficient sample size of the materials from the project work area to complete all required analytical testing. Samples must be representative of all materials that comprise the waste for disposal.

5. The EFT will place the sample in a clean jar or vial, secure the jar or vial closed and label it with a unique sample number.

6. The EFT will record the sample number on the COC and note the location of the sample in the field log and on the site drawing.

7. The EFT will verify that the sample number(s) on the jar/vial are accurate and consistent with the COC.

8. The EFT will place the secured and labeled jar/vial into a cooler and arrange for prompt delivery of bulk samples to the analytical laboratory.

9. The EFT will clean up any debris generated by the sampling process.

10. The EFT will decontaminate the collection tool using wet wipes, water and paper towels after each sample is collected, or other approved method. Razor blades will be replaced on a periodic basis as determined by the EFT.

11. All samples will be collected in a manner to eliminate cross-contamination by the previous sample.

12. All materials used in the collection of the samples (gloves, wipes, trash, etc.) and hazardous debris generated by the sampling process will be containerized in a bag and delivered to Building 357 for disposal by the end of the shift generated.

13. The EFT will obtain the following sampling equipment to perform the bulk sampling.

    (a) Individual sample jars and vials;
    (b) Field log;
    (c) Site diagram;
    (d) Chain-of-Custody (COC) forms;
    (e) Nitrile gloves;

(f) Stainless steel or plastic soil scoops;
(g) Metal chisels of assorted sizes;
(h) Razor knives and replacement blades;
(i) Water and spray bottle;
(j) Trash bags;
(k) Disposable cleaning wipes;
(l) Paper towels;
(m) Duct tape;
(n) Flashlight;
(o) Personnel lifts; and
(p) Other equipment as necessary.

1.5.2.11 Environmental, Safety and Health Compliance

In addition to the detailed requirements of this specification, comply with laws, ordinances, rules, and regulations of Federal, State, and local authorities regarding lead, cadmium, and chromium (VI). Comply with the applicable requirements of the current issue of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, EM 385-1-1, and ND OPNAVINST 5100.23. Submit matters regarding interpretation of standards to the Contracting Officer for resolution before starting work. Where specification requirements and the referenced documents vary, the most stringent requirements apply. The following local laws, ordinances, criteria, rules and regulations regarding removing, handling, storing, transporting, and disposing of lead, cadmium, and chromium (VI)-contaminated materials apply:

a. Shipyard Instruction 5090.6F Solid Waste Operations (included as Attachment B at the end of this Section).

b. Shipyard Instruction 5090.30 Hazardous Waste Generator Standards.

Licensing and certification in the State of Maine is required.

1.5.3 Pressure Differential Recordings for Local Exhaust System

Provide a local exhaust system that creates a negative pressure of at least 0.02 inches of water relative to the pressure external to the enclosure and operate it continuously, 24-hours a day, until the temporary enclosure of the lead, cadmium, and chromium (VI) control area is removed. Submit pressure differential recordings for each work day to the Private Qualified Person (PQP) and General Contractor (GC) for review and to the Contracting Officer and PD&C HAZMAT POC within 24-hours from the end of each work day. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

1.5.4 Licenses, Permits and Notifications

Certify and submit in writing to the Maine Department of Environmental Protection and the Contracting Officer at least 10 days prior to the commencement of work that licenses, permits, and notifications have been obtained. All associated fees or costs incurred in obtaining the licenses, permits and notifications are included in the Contract price.

1.5.5 Pre-Construction Conference

Along with the CP, meet with the Contracting Officer to discuss in detail the Lead, Cadmium, and Chromium (VI) Waste Management Plan and the Lead, Cadmium, and Chromium (VI) Compliance Plan, including procedures and precautions for the work.

1.6 EQUIPMENT

1.6.1 Respirators

Furnish appropriate respirators approved by the National Institute for Occupational Safety and Health (NIOSH), Department of Health and Human Services, for use in atmospheres containing lead, cadmium and chromium (VI) dust, fume and mist. Respirators must comply with the requirements of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

1.6.2 Special Protective Clothing

Personnel exposed to lead, cadmium, and chromium (VI) contaminated dust must wear proper disposable protective whole body clothing, head covering, gloves, eye, and foot coverings as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Furnish proper disposable plastic or rubber gloves to protect hands. Reduce the level of protection only after obtaining approval from the CP.

1.6.3 Rental Equipment Notification

If rental equipment is to be used during PWL or MCL handling and disposal, notify the rental agency in writing concerning the intended use of the equipment.

1.6.4 Vacuum Filters

UL 586 labeled HEPA filters.

1.6.5 Equipment for Government Personnel

Furnish the Contracting Officer with two complete sets of personal protective equipment (PPE) daily, as required herein, for entry into and inspection of the lead, cadmium and chromium (VI) removal work within the lead, cadmium and chromium (VI) controlled area. Personal protective equipment must include disposable whole body covering, including appropriate foot, head, eye, and hand protection. PPE remains the property of the Contractor. The Government will provide respiratory protection for the Contracting Officer.

1.6.6 Abrasive Removal Equipment

The use of powered machine for vibrating, sanding, grinding, or abrasive blasting is prohibited unless equipped with local exhaust ventilation systems equipped with high efficiency particulate air (HEPA) filters. The use of abrasive removal equipment is prohibited unless expressly approved by the NAVFAC Cultural Resources (CR) Managers. Documentation of CR Manager approval must be included in the Lead, Cadmium, and Chromium (VI) Compliance Plan.

1.6.7 Negative Air Pressure System

1.6.7.1 Minimum Requirements

Do not proceed with work in the area until containment is set up and HEPA filtration systems are in place. The negative air pressure system must meet the requirements of ANSI/ASSP Z9.2 including approved HEPA filters in accordance with UL 586. Negative air pressure equipment must be equipped with new HEPA filters, and be sufficient to maintain a minimum pressure differential of minus 0.02 inch of water column relative to adjacent, unsealed areas. Negative air pressure system minimum requirements are listed as follows:

a. The unit must be capable of delivering its rated volume of air with a clean first stage filter, an intermediate filter and a primary HEPA filter in place.

b. The HEPA filter must be certified as being capable of trapping and retaining mono-disperse particles as small as 0.3 micrometers at a minimum efficiency of 99.97 percent.

c. The unit must be capable of continuing to deliver no less than 70 percent of rated capacity when the HEPA filter is 70 percent full or measures 2.5 inches of water static pressure differential on a magnehelic gauge.

d. Equip the unit with a manometer-type negative pressure differential monitor with minor scale division of 0.02 inch of water and accuracy within plus or minus 1.0 percent. The manometer must be calibrated daily as recommended by the manufacturer.

e. Equip the unit with a means for the operator to easily interpret the readings in terms of the volumetric flow rate of air per minute moving through the machine at any given moment.

f. Equip the unit with an electronic mechanism that automatically shuts the machine off in the event of a filter breach or absence of a filter.

g. Equip the unit with an audible horn that sounds an alarm when the machine has shut itself off.

h. Equip the unit with an automatic safety mechanism that prevents a worker from improperly inserting the main HEPA filter.

1.6.7.2 Auxiliary Generator

Provide an auxiliary generator with capacity to power a minimum of 50 percent of the negative air machines at any time during the work. When power fails, the generator controls must automatically start the generator and switch the negative air pressure system machines to generator power. The generator must not present a carbon monoxide hazard to workers.

1.6.8 Vacuum Systems

Vacuum systems must be suitably sized for the project, and filters must be capable of trapping and retaining all mono-disperse particles as small as 0.3 micrometers (mean aerodynamic diameter) at a minimum efficiency of 99.97 percent. Properly dispose of used filters that are being replaced.

1.6.9 Heat Blower Guns

Heat blower guns must be flameless, electrical, paint-softener type with controls to limit temperature to 1,100 degrees F. Heat blower must be (grounded) 120 volts ac, and must be equipped with cone, fan, glass protector and spoon reflector nozzles.

1.7 PROJECT/SITE CONDITIONS

1.7.1 Protection of Existing Work to Remain

Perform work without damage or contamination of adjacent areas. Where existing work is damaged or contaminated, restore work to its original condition or better as determined by the Contracting Officer.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

Keep materials and equipment needed to complete the project available and on the site. Submit a description of the materials and equipment required; including Safety Data Sheets (SDSs) for material brought onsite to perform the work.

2.1.1 Expendable Supplies

Submit a description of the expendable supplies required.

2.1.1.1 Polyethylene Bags

Disposable bags must be polyethylene plastic and be a minimum of 6 mils thick (4 mils thick if double bags are used) or any other thick plastic material shown to demonstrate at least equivalent performance; and capable of being made leak-tight. Leak-tight means that solids, liquids or dust cannot escape or spill out.

2.1.1.2 Polyethylene Leak-tight Wrapping

Wrapping used to wrap lead, cadmium, and chromium (VI) contaminated debris must be polyethylene plastic that is a minimum of 6 mils thick or any other thick plastic material shown to demonstrate at least equivalent performance.

2.1.1.3 Polyethylene Sheeting

Sheeting must be polyethylene plastic with a minimum thickness of 6 mil, or any other thick plastic material shown to demonstrate at least equivalent performance; and be provided in the largest sheet size reasonably accommodated by the project to minimize the number of seams. Where the project location constitutes an out of the ordinary potential for fire, or where unusual fire hazards cannot be eliminated, provide flame-resistant polyethylene sheets which conform to the requirements of NFPA 701.

2.1.1.4 Tape and Adhesive Spray

Tape and adhesive must be capable of sealing joints between polyethylene sheets and for attachment of polyethylene sheets to adjacent surfaces. After dry application, tape or adhesive must retain adhesion when exposed to wet conditions, including amended water. Tape must be minimum 2 inches wide, industrial strength.

2.1.1.5 Containers

When used, containers must be leak-tight and be labeled in accordance with EPA, DOT and OSHA standards. For containerization of hazardous waste, the

Contractor must use containers provided by Building 357.

2.1.1.6 Chemical Paint Strippers

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Chemical paint strippers must not contain methylene chloride and be formulated to prevent stain, discoloration, or raising of the substrate materials.

2.1.1.7 Chemical Paint Stripper Neutralizer

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Neutralizers for paint strippers must be compatible with the substrate and suitable for use with the chemical stripper that has been applied to the surface.

2.1.1.8 Detergents and Cleaners

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Detergents or cleaning agents must not contain trisodium phosphate and have demonstrated effectiveness in lead, cadmium and chromium (VI) control work using cleaning techniques specified by HUD 6780 guidelines.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Protection

3.1.1.1 Notification

a. Notify the Contracting Officer 20 days prior to the start of any lead, cadmium and chromium (VI) work.

b. Notification of the Commencement of Work Impacting Lead, Cadmium, or Chromium (VI)

3.1.1.2 Lead, Cadmium, and Chromium (VI) Control Area

a. Physical Boundary - Provide physical boundaries around the lead, cadmium, and chromium (VI) control area by roping off the area designated in the work plan or providing curtains, portable partitions or other enclosures to ensure that lead, cadmium and chromium (VI) will not escape outside of the lead, cadmium and chromium (VI) control area. Prohibit the general public from accessing the lead, cadmium, and chromium (VI) control areas.

b. Warning Signs - Provide warning signs at approaches to lead, cadmium, and chromium (VI) control areas. Locate signs at such a distance that personnel may read the sign and take the necessary precautions before entering the area. Signs must comply with the requirements of 29 CFR 1926.62.

3.1.1.3 Equipment

Equipment will remain in the building. Protect and cover equipment or remove equipment from the work areas and store in a location approved by the Contracting Officer. Reinstall and equipment at the completion of work.

3.1.1.4 Heating, Ventilating and Air Conditioning (HVAC) Systems

Shut down, lock out, and isolate HVAC systems that supply, exhaust, or pass through the lead, cadmium, and chromium (VI) control areas. Seal intake and exhaust vents in the lead, cadmium, and chromium (VI) control area with 6 mil plastic sheet and tape. Seal seams in HVAC components that pass through the lead, cadmium, and chromium (VI) control area.

3.1.1.5 Local Exhaust System

Provide a local exhaust system in the lead, cadmium, and chromium (VI) control area in accordance with ANSI/ASSP Z9.2, 29 CFR 1926.62, 29 CFR 1926.1126 and 29 CFR 1926.1127 that will provide at least 4 air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until the lead, cadmium, and chromium (VI) control area is removed and must be leak proof to the filter and equipped with HEPA filters. Maintain a minimum pressure differential in the lead, cadmium, and chromium (VI) control area of minus 0.02 inch of water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the lead, cadmium, and chromium (VI) control area. Filters on exhaust equipment must conform to ANSI/ASSP Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

3.1.1.6 Negative Air Pressure System Containment

a. Operate the negative air pressure systems to provide at least 4 air changes per hour inside the containment. Operate the local exhaust unit equipment continuously until the containment is removed. Smoke test the negative air pressure system for leaks at the beginning of each shift. The certified supervisor is responsible to continuously monitor and keep a pressure differential log with an automatic manometric recording instrument. Notify the Contracting Officer and PD&C HAZMAT POC immediately if the pressure differential falls below the prescribed minimum. Submit the continuously monitored pressure differential log, as specified. Do not use the building ventilation system as the local exhaust system. Terminate the local exhaust system out of doors unless the Contracting Officer and PD&C HAZMAT POC allows an alternate arrangement. All filters must be new at the beginning of the project and be periodically changed as necessary to maintain specified pressure differential and disposed of as lead, cadmium and chromium (VI) contaminated waste.

b. Discontinuing Negative Air Pressure System. Operate the negative air pressure system continuously during work activities unless otherwise authorized by the Contracting Officer and PD&C HAZMAT POC. At the completion of the project, units must be run until full cleanup has been completed and final clearance testing requirements have been met. Dismantling of the negative air pressure systems must be as presented in the Lead, Cadmium, and Chromium (VI) Compliance Plan.

Seal the HEPA filter machine intakes with polyethylene to prevent environmental contamination.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

3.1.1.7 Decontamination Shower Facility

Provide clean and contaminated change rooms and shower facilities in accordance with this specification and 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

3.1.1.8 Eye Wash Station

Provide suitable facilities within the work area for quick drenching or flushing of the eyes where eyes may be exposed to injurious corrosive materials.

3.1.1.9 Mechanical Ventilation System

a. Use adequate ventilation to control personnel exposure to lead, cadmium and chromium (VI) in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. To the extent feasible, use local exhaust ventilation or other collection systems, approved by the CP. Evaluate and maintain local exhaust ventilation systems in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

b. Vent local exhaust outside the building and away from building ventilation intakes or ensure system is connected to HEPA filters.

c. Use locally exhausted, power actuated tools or manual hand tools.

3.1.1.10 Personnel Protection

Personnel must wear and use protective clothing and equipment as specified herein. Eating, smoking, or drinking or application of cosmetics is not permitted in the lead, cadmium, and chromium (VI) control area. No one will be permitted in the lead, cadmium, and chromium (VI) control area unless they have been appropriately trained and provided with protective equipment.

3.2 ERECTION

3.2.1 Lead, Cadmium, and Chromium (VI) Control Area Requirements

Establish a lead, cadmium, and chromium (VI) control area by completely establishing barriers and physical boundaries around the area or structure where PWL or MCL removal operations will be performed that will not create airborne, dust containing lead, cadmium, chromium (VI) (such as carefully unfastening sheets containing lead, cadmium, chromium (VI) from walls).

Full containment - Contain removal operations by the use of critical barriers and HEPA filtered exhaust, or a negative pressure enclosure system with decontamination facilities and with HEPA filtered exhaust where removal practice will create airborne, dust containing lead, cadmium, chromium (VI) (such as sanding, sawing, grinding, thermal cutting, or digging or demolition activities) and as required by the CP. For containment areas larger than 1,000 square feet install a minimum of two 18 inch square viewing ports. Locate ports to provide a view of the required work from the exterior of the enclosed contaminated area. Glaze ports with laminated safety glass.

3.3 APPLICATION

3.3.1 Lead, Cadmium, and Chromium (VI) Work

Perform lead, cadmium, and chromium (VI) work in accordance with approved Lead, Cadmium, and Chromium (VI) Compliance Plan. Use procedures and equipment required to limit occupational exposure and environmental contamination with lead, cadmium, and chromium (VI) when the work is performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 or 40 CFR 745 and as specified herein. Dispose of all PWL or MCL and associated waste in compliance with Federal, State, and local requirements.

3.3.2 Paint with Lead, Cadmium, and Chromium (VI) or Material Containing Lead, Cadmium, and Chromium (VI) Removal

Manual or power sanding or grinding of lead, cadmium, and chromium (VI) surfaces or materials is not permitted unless tools are equipped with HEPA attachments or wet methods. The dry sanding or grinding of surfaces that contain lead, cadmium, and chromium (VI) is prohibited. The use of abrasive removal equipment is prohibited unless expressly approved by the NAVFAC Cultural Resources (CR) Managers. Documentation of CR Manager approval must be included in the Lead, Cadmium, and Chromium (VI) Compliance Plan. Provide methodology for removing lead, cadmium, and chromium (VI) in the Lead, Cadmium, and Chromium (VI) Compliance Plan. Select lead, cadmium, and chromium (VI) removal processes to prevent contamination of work areas outside the control area with lead, cadmium, and chromium (VI) contaminated dust or other lead, cadmium, and chromium (VI) contaminated debris or waste and to ensure that unprotected personnel are not exposed to hazardous concentrations of lead, cadmium, and chromium (VI). Describe this removal process in the Lead, Cadmium, and Chromium (VI) Compliance Plan.

Avoid flash rusting and deterioration of the substrate. Provide surface preparations for painting in accordance with Section 09 90 00 PAINTS AND COATINGS.

Provide methodology for lead, cadmium and chromium (VI), LBP/PWL removal, abatement/control and processes to prevent contamination of work areas outside the control area with lead, cadmium, and chromium (VI) contaminated dust or other lead, cadmium, and chromium (VI) contaminated debris/waste and to ensure that unprotected personnel are not exposed to hazardous concentrations of lead, cadmium, and chromium (VI). Describe this lead, cadmium and chromium (VI), LBP/PWL removal/control process in the Lead, Cadmium, and Chromium (VI) Compliance Plan.

3.3.2.1 Paint with Lead, Cadmium, and Chromium (VI) or Material Containing Lead, Cadmium, and Chromium (VI) - Indoor Removal

Perform manual or mechanical removal and thermal cutting in the lead, cadmium, and chromium (VI) control areas using enclosures, barriers or containments and powered locally exhausted tools equipped with HEPA filters. Collect residue and debris for disposal in accordance with Federal, State, and local requirements.

3.3.2.2 Paint with Lead, Cadmium, and Chromium (VI) or Material Containing Lead, Cadmium, and Chromium (VI) - Outdoor Removal

Perform outdoor removal as indicated in Federal, State, and local regulations and in the Lead, Cadmium, and Chromium (VI) Compliance Plan. The worksite preparation (barriers or containments) must be job dependent and presented in the Lead, Cadmium, and Chromium (VI) Compliance Plan.

3.3.3 Personnel Exiting Procedures

Whenever personnel exit the lead, cadmium, and chromium (VI) controlled area, they must perform the following procedures and must not leave the work place wearing any clothing or equipment worn in the control area:

a. Vacuum all clothing before entering the contaminated change room.

b. Remove protective clothing in the contaminated change room, and place them in an approved impermeable disposal bag.

c. Shower.

d. Change to clean clothes prior to leaving the clean clothes storage area.

3.4 FIELD QUALITY CONTROL

3.4.1 Tests

3.4.1.1 Air and Wipe Sampling

Conduct sampling for lead, cadmium, and chromium (VI) in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 and as specified herein. Air and wipe sampling must be directed or performed by the CP.

a. The CP must be on the job site directing the air and wipe sampling and inspecting the project work to ensure that the requirements of the Contract, local, State and Federal statutes and regulations, Navy and PNSY requirements, and project specifications have been satisfied during all work impacting lead, cadmium, and/or chromium (VI).

b. Collect personal air samples on employees who are anticipated to have the greatest risk of exposure as determined by the CP. In addition, collect air samples on at least twenty-five percent of the work crew or a minimum of two employees, whichever is greater, during each work shift.

c. Conduct area air sampling daily, on each shift in which lead, cadmium and chromium (VI) and lead-based paint removal operations are performed, in areas immediately adjacent to the lead, cadmium and chromium (VI) control area at exhaust points for negative air machines, at entrances to the negative pressure enclosure (NPE), and at other locations determined appropriate for the project and site conditions. Conduct sufficient area monitoring to ensure unprotected personnel are not exposed at or above 30 micrograms of lead per cubic meter of air or 2.5 micrograms of cadmium/chromium (VI) per cubic meter of air. If 30 micrograms of lead per cubic meter of air or 2.5 micrograms of cadmium/chromium (VI) per cubic meter of air is reached or exceeded, stop work, correct the conditions(s) causing the increased levels. Notify the Contracting Officer and PD&C HAZMAT POC immediately. Determine if condition(s) require any further change in work methods. Resume removal work only after the CP and the Contracting Officer and PD&C HAZMAT POC give approval.

d. Submit sample results of air samples, signed by the CP, to the Contracting Officer and PD&C HAZMAT POC within 72-hours after the air samples are taken.

e. Before any work begins, a third party consultant must collect and analyze baseline wipe, air, and soil (if bare soil is present during exterior remediation operations) samples in accordance with methods defined by Federal, State, and local standards inside and outside of the physical boundary to assess the degree of dust contamination in the facility prior to lead, cadmium and chromium (VI) disturbance or removal. Provide Initial Sample Results to the Contracting Officer and PD&C HAZMAT POC before work begins.

f. Surface Wipe Samples - Collect surface wipe samples on floors at a location no greater than 10 feet outside the lead, cadmium, and chromium (VI) control area at a frequency of once per day while lead, cadmium, and chromium (VI) removal work is conducted in occupied buildings. Surface wipe samples or Micro Vacuum surface sample results must meet criteria in paragraph CLEARANCE CERTIFICATION.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

3.4.1.2 Sampling After Removal

After the visual inspection, collect wipe samples according to the HUD protocol contained in HUD 6780 to determine the lead, cadmium and chromium (VI) content of settled dust in micrograms per square meter foot of surface area.

3.4.1.3 Testing of Material Containing Lead, Cadmium, and Chromium (VI) Residue

Test residue in accordance with 40 CFR 261 for hazardous waste.

3.5 CLEANING AND DISPOSAL

3.5.1 Cleanup

Maintain surfaces of the lead, cadmium, and chromium (VI) control area free of accumulations of dust and debris. Restrict the spread of dust and debris; keep waste from being distributed over the work area. Do not dry sweep or use pressurized air to clean up the area. At the end of each shift and when the lead, cadmium, and chromium (VI) operation has been completed, clean the controlled area of all visible contamination by vacuuming with a HEPA filtered vacuum cleaner, wet mopping the area and wet wiping the area as indicated by the Lead, Cadmium, and Chromium (VI) Compliance Plan. Reclean areas showing dust or debris. After visible dust and debris is removed, wet wipe and HEPA vacuum all surfaces in the controlled area. If adjacent areas become contaminated at any time during the work, clean, visually inspect, and then wipe sample all contaminated areas. The CP must then certify in writing to the Contracting Officer and PD&C HAZMAT POC that the area has been cleaned of lead, cadmium and chromium (VI) contamination before clearance testing.

3.5.1.1 Clearance Certification

The CP must certify in writing to the Contracting Officer and PD&C HAZMAT POC that air samples collected outside the lead, cadmium, and chromium (VI) control area during paint removal operations are less than 30 micrograms of lead per cubic meter of air and less than 2.5 micrograms of cadmium/chromium (VI) per cubic meter of air; the respiratory protection used for the employees was adequate; the work procedures were performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127; and that there were no visible accumulations of material and dust containing lead, cadmium, and chromium (VI) left in the work site. Do not remove the lead, cadmium, and chromium (VI) control area or roped off boundary and warning signs prior to the Contracting Officer's acknowledgement of receipt of the CP certification. Clearance sampling results must be submitted to NAVFAC within 72 hours after the samples are taken.

The third party consultant must perform surface wipe sampling for clearance certification pursuant to Paragraph entitled SAMPLING AND ANALYSIS herein. The third party consultant must certify surface wipe sample results collected inside and outside the work area are less than the thresholds detailed in Paragraph entitled CLEARANCE REQUIREMENTS herein.

For lead, cadmium and chromium (VI)-based paint hazard abatement work, surface wipe and soil sampling must be conducted and clearance determinations made according to the requirements detailed in Paragraph entitled SAMPLING AND ANALYSIS herein and to the work practice standards presented in 40 CFR 745.227.

3.5.2 Disposal

a. Dispose of material, whether hazardous or non-hazardous in accordance with local, State, and Federal statutes and regulations, Navy and PNSY requirements, and project specifications. Ensure all waste is properly characterized in accordance with the Waste Characterization Sampling Plan developed under Paragraph entitled WASTE CHARACTERIZATION SAMPLING PLAN herein. The result of each waste characterization will dictate disposal requirements.

b. Contractor is responsible for segregation of waste. Collect lead, cadmium, and chromium (VI) contaminated waste, scrap, debris, bags, containers, equipment, and lead, cadmium, and chromium (VI)-contaminated clothing that may produce airborne concentrations of lead, cadmium, and chromium (VI) particles. Label the containers in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 40 CFR 261, 40 CFR 262 and corresponding state regulations.

c. Accumulate waste materials in U.S. Department of Transportation (49 CFR 178) approved 55 gallon drums or appropriately sized container for smaller volumes. Properly label each drum to identify the type of hazardous material (49 CFR 172). For hazardous waste, the collection container requires marking/labeling in accordance with 40 CFR 262 and corresponding state regulations.

d. Manage lead, cadmium, and chromium (VI) or lead, cadmium, and chromium (VI) contaminated waste in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, and 40 CFR 265.

e. All lead, cadmium, and chromium (VI) waste generation, management,and disposal will be coordinated with the Shipyard environmental function (Code 106, Building 357) and the Contracting Officer.

All HW must be turned in to Building 357 not later than the end of the shift on which it is generated. Responsibility for compliance is upon the Contractor. All hazardous wastes generated within the confines of the Shipyard are disposed of by the Government. Accordingly, all hazardous wastes generated by the Contractor to accomplish requirements of this Contract will be considered Government-generated, and disposed of by the Government. Contractor must not bring hazardous wastes onto Government property.

Containers positioned within the work area boundaries must have covers in place whenever containers are not in use.

3.5.2.1 Disposal Documentation

For non-hazardous waste, provide a certificate that the waste was accepted by the disposal facility. Provide turn-in documents or weight tickets for non-hazardous waste disposal.

3.5.2.2 Payment for Non-Hazardous Waste

Payment for disposal of non-hazardous waste will not be made until a signed copy of the manifest from the disposal facility is received and approved by the Contracting Officer. The manifest must detail and certify the amount of non-hazardous waste delivered to the disposal facility.

-- End of Section --

Specification Section 1.5.2.3 - Pb, Cr, Cd Compliance Plan Checklist - Building XX
Prepared by: _____________ Date: __________

Spec Section(s)	Item Required	Included?	Section/ Page #	Notes
	Confirm that the Pb/Cr/Cd Compliance Plan contains the following:			
1.5.2.2	- Signature and seal to document plan prepared by Maine Lead Project Designer			
1.5.2.2	- Site sketch with all project details (control areas, critical barriers, physical boundaries, decon facilities, viewing ports, and mechanical ventilation system)			
	- Confirmation that meeting will be scheduled w/ ktr, supervisor, CP, PQP, CO, and APM to review Plan			
3.1	- Description of preparatory activities - protecting existing conditions, moving furniture, pre-cleaning			
1.5.2.2	- Description of safety precautions including, but not limited to, LO/TO, CSE, fall protection			
1.5.2.2	- Description of work practices for **each** activity, including:			
	- If abrasive removal is proposed, document approval by NAVFAC Cultural Resources MGR			
1.5.2.2	- Description of controls for **each** activity			
1.5.2.2	- Description of job responsibilities for **each** activity			
1.5.2.2	- Description of respirators and personal protective equipment			
1.5.2.2	- Description of equipment for **each** activity. Include all equipment spec sheets			
1.5.2.2	- Description of materials for **each** activity. Include all SDS			
1.5.2.2	- Description of hygiene facilities and sanitary procedures (including no eating, drinking, smoking)			
1.5.2.2	- Description of interface of trades			
1.5.2.2	- Description of work sequencing			
1.5.2.2	- Description of the method of containment of the operation to ensure that metals are not released outside of the control area. NPE or Regulated Area?			
3.1.1.2	- Description of required signage for **each** activity			
3.3.3, 3.1.1.7	- Description of decontamination equipment and process			
	- Rationale for not including shower (if applicable)			
1.5.3	- Description of local exhaust system and critical barrier equipment and process, including:			
	- Confirmation that Pressure Differential Recordings will be submitted w/in **24 hours**			
	- Confirmation that negative pressure will be maintained continuously until work area cleared			
1.5.2.2, 3.5	- Description of cleanup procedures during and at the end of work.			
3.4.1	- Description of personal sampling and analysis strategy and methodology, number of samples, and qualifications of sampling personnel, including:			
	- Confirmation that all personal sampling results will be submitted to NAVFAC w/in **XX hours**			
	- Specific thresholds for work stoppage, PPE upgrade, etc.			
3.4.1	- Detailed description of area sampling and analysis strategy and methodology, frequency of sampling, sample locations, duration of sampling, and qualifications of sampling personnel, including:			
	- Confirmation that all area sampling results will be submitted to NAVFAC w/in **XX hours**			
	- Listing of specific sampling locations. **Include locations in Notes column (to the right)**			
	- Specific thresholds for work stoppage, PPE upgrade, etc.			
3.4.1	- Detailed description of clearance sampling and analysis strategy and methodology, number of samples, and qualifications of sampling personnel, including:			
	- Confirmation that all clearance sampling results will be submitted to NAVFAC w/in **XX hours**			
	- Listing of specific sampling locations. **Include locations in Notes column (to the right)**			
	- Listing of clearance thresholds. **Include thresholds in Notes column (to the right)**			
1.4, 1.5.2.8, 3.5	- Detailed plan for Pb, Cd, and Cr Waste Management, including:			
	- Painted Materials			
	- Paint chips, dust, debris			
	- Ancillary waste, including poly sheeting			
	- Wastewater			
1.4, 1.5.1.2	- Copies of Training Certificates for Workers and Supervisors			
1.4	- Copies of Licenses for Workers and Supervisors			
1.4, 1.5.2.7	- Copies of Respiratory Protection Program, Including Training and Fit Testing Records			
1.4, 1.5.2.4	- Copies of Medical Examinations, Surveillance and Records			
1.4, 1.5, 1.5.3	- Copies licenses, permits and notifications for demolition, abatement, remediation, etc.			
1.4, 1.5.1.3	- Documentation of Testing Laboratory Qualifications/Certificates			
1.4, 1.5	- Copy of Occupant Protection Plan			
1.4, 1.5.1.5	- Documentation of Certified Risk Assessor Qualifications			
1.4	- Sample Copy of Clearance Certification			
3.1.1.1	- Copy of Notification to Contracting Officer			
3.1.1.1	- Copy of Notification to Occupants			
1.4	- Name, qualifications, and certification of Supervisor			
1.4, 1.5.1.1	- Documentation of Competent Person Qualifications			
1.4, 1.5.1.1	- Certified Industrial Hygienist?			
1.4, 1.5.1.1	- Certified Safety Professional?			
1.4, 1.5.1.1	- Licensed Lead-Based Abatement Supervisor/Project Designer in the State of Maine?			
1.3.3	- Description of approach for maintaining and submitting log of analytical results			
1.4, 1.5	- Copy of Occupational and Environmental Assessment Data Report			
1.4, 1.5.1.4	- Documentation of Third Party Consultant Qualifications			

CHECKLIST COMPLETED BY:____ DATE: ______________________________

NOTE: This checklist is not intended to list all of the Pb, Cr, Cd Compliance Plan components required per Specification Section 02 83 00.00 22, Pb, Cr and Cd Remediation. The Contractor is required to identify and meet all applicable Specification requirements.

SECTION 02 84 16.00 22
^^
HANDLING OF LIGHTING BALLASTS AND LAMPS CONTAINING PCBs AND MERCURY (PWD ME)
11/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1000	Air Contaminants
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 268	Land Disposal Restrictions
40 CFR 270	EPA Administered Permit Programs: The Hazardous Waste Permit Program
40 CFR 273	Standards for Universal Waste Management
40 CFR 761	Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions
49 CFR 178	Specifications for Packagings

1.2 REQUIREMENTS

Removal and disposal of PCB containing lighting ballasts and associated mercury-containing lamps. Contractor may encounter leaking PCB ballasts.

1.3 DEFINITIONS

1.3.1 Certified Industrial Hygienist (CIH)

A industrial hygienist hired by the contractor must be certified by the American Board of Industrial Hygiene.

1.3.2 Leak

Leak or leaking means any instance in which a PCB article, PCB container, or PCB equipment has any PCBs on any portion of its external surface.

1.3.3 Lamps

Lamp is defined as the bulb or tube portion of an electric lighting device. A lamp is specifically designed to produce radiant energy, most often in the ultraviolet, visible, and infra-red regions of the electromagnetic spectrum. Examples of common electric lamps include, but are not limited to, fluorescent, high intensity discharge, neon, mercury vapor, high pressure sodium, and metal halide lamps.

1.3.4 Polychlorinated Biphenyls (PCBs)

PCBs as used in this specification mean the same as PCBs, and all related items, as defined in 40 CFR 761, Section 3, Definitions.

1.3.5 Spill

Spill means both intentional and unintentional spills, leaks, and other uncontrolled discharges when the release results in any quantity of PCBs running off or about to run off the external surface of the equipment or other PCB source, as well as the contamination resulting from those releases.

1.3.6 Universal Waste

Universal Waste means any of the following hazardous wastes that are managed under the universal waste requirements 40 CFR 273:

(1) Batteries as described in Sec. 273.2 of this chapter;

(2) Pesticides as described in Sec. 273.3 of this chapter;

(3) Mercury containing equipment as described in Sec. 273.4 of this chapter; and

(4) Lamps as described in Sec. 273.5 of this chapter.

1.4 QUALITY ASSURANCE

1.4.1 Regulatory Requirements

Perform PCB related work in accordance with 40 CFR 761. Perform mercury-containing lamps storage and transport in accordance with 40 CFR 261, 40 CFR 264, 40 CFR 265, 40 CFR 273.

1.4.2 Training

Instruction and certification of training of all persons involved in the removal of PCB containing lighting ballasts and mercury-containing lamps is the responsibility of the Certified Industrial Hygienist (CIH). Include in the instruction: The dangers of PCB and mercury exposure, decontamination, safe work practices, and applicable OSHA and EPA regulations. The CIH must review and approve the PCB and Mercury-Containing Lamp Removal Work Plans.

1.4.3 Regulation Documents

Maintain at all times one copy each at the office and one copy each in view at the job site of 29 CFR 1910.1000, 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 265, 40 CFR 268, 40 CFR 270, 40 CFR 273 and of the Contractor removal work plan and disposal plan for PCB and for associated mercury-containing lamps.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-07 Certificates

Qualifications of CIH; G

Training Certification; G

PCB and Lamp Removal Work Plan; G

PCB and Lamp Disposal Plan; G

SD-11 Closeout Submittals

Transporter Certification of notification to EPA of their PCB waste activities and EPA ID numbers; G

Certification of Decontamination

Certificate of Disposal and/or recycling. Submit to the Government before application for payment within 30 days of the date that the disposal of the PCB and mercury-containing lamp waste identified on the manifest was completed.

DD Form 1348-1

1.6 ENVIRONMENTAL REQUIREMENTS

Use special clothing:

a. Disposable gloves (polyethylene)

b. Eye protection

c. PPE as required by CIH

1.7 SCHEDULING

Notify the Contracting Officer 20 days prior to the start of PCB and mercury-containing lamp removal work.

1.8 QUALITY ASSURANCE

1.8.1 Qualifications of CIH

Submit the name, address, and telephone number of the Industrial Hygienist selected to perform the duties in paragraph CERTIFIED INDUSTRIAL HYGIENIST. Submit training certification that the Industrial Hygienist is certified, including certification number and date of certification or re certification.

1.8.2 PCB and Lamp Removal Work Plan

Submit a job-specific plan within 20 calendar days after award of contract of the work procedures to be used in the removal, packaging, and storage of PCB-containing lighting ballasts and associated mercury-containing lamps. Include in the plan: Requirements for Personal Protective Equipment (PPE), spill cleanup procedures and equipment, eating, smoking and restroom procedures. The plan must be approved and signed by the Certified Industrial Hygienist. Obtain approval of the plan by the Contracting Officer prior to the start of PCB and/or lamp removal work.

1.8.3 PCB and Lamp Disposal Plan

Submit a PCB and lamp Disposal Plan within 45 calendar days after award of contract, complying with the applicable requirements of federal, state, and local PCB and Universal waste regulations and address:

a. Estimated quantities of wastes to be generated, disposed of, and recycled.

b. Names and qualifications of each Contractor that will be transporting, storing, treating, and disposing of the wastes. Include the facility location. Furnish two copies of EPA and state PCB and mercury-containing lamp waste permit applications and EPA identification numbers, as required.

c. Names and qualifications (experience and training) of personnel who will be working on-site with PCB and mercury-containing lamp wastes.

d. Spill prevention, containment, and cleanup contingency measures to be implemented.

e. Work plan and schedule for PCB and mercury-containing lamp waste removal, containment, storage, transportation, disposal and or recycling. Clean up and containerize wastes daily.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 WORK PROCEDURE

Furnish labor, materials, services, and equipment necessary for the removal of PCB containing lighting ballasts, and associated mercury-containing fluorescent lamps in accordance with local, state, or federal regulations. Do not expose PCBs to open flames or other high temperature sources since toxic decomposition by-products may be produced. Do not break mercury containing fluorescent lamps or high intensity discharge lamps.

3.1.1 Work Operations

Ensure that work operations or processes involving PCB or PCB-contaminated materials are conducted in accordance with 40 CFR 761, 40 CFR 262 40 CFR 263, and the applicable requirements of this section, including but not limited to:

a. Obtaining suitable PCB and mercury-containing lamp storage sites.

b. Notifying Contracting Officer prior to commencing the operation.

c. Reporting leaks and spills to the Contracting Officer.

d. Cleaning up spills.

e. Inspecting PCB and PCB-contaminated items and waste containers for leaks and forwarding copies of inspection reports to the Contracting Officer.

f. Maintaining inspection, inventory and spill records.

3.2 PCB SPILL CLEANUP REQUIREMENTS

3.2.1 PCB Spills

Immediately report to the Contracting Officer any PCB spills.

3.2.2 PCB Spill Control Area

Rope off an area around the edges of a PCB leak or spill and post a "PCB Spill Authorized Personnel Only" caution sign. Immediately transfer leaking items to a drip pan or other container.

3.2.3 PCB Spill Cleanup

40 CFR 761, subpart G. Initiate cleanup of spills as soon as possible, but no later than 24 hours of its discovery. Mop up the liquid with rags or other conventional absorbent. Properly contain and dispose of spent absorbent as solid PCB waste.

3.2.4 Records and Certification

Document the cleanup with records of decontamination in accordance with 40 CFR 761, Section 125, Requirements for PCB Spill Cleanup. Provide test results of cleanup and certification of decontamination.

3.3 REMOVAL

3.3.1 Ballasts

As ballast are removed from the lighting fixture, inspect label on ballast. Ballasts without a "No PCB" label must be assumed to contain PCBs and containerized and disposed of as required under paragraphs STORAGE FOR DISPOSAL and DISPOSAL. If there are less than 1600 "No PCB" labeled lighting ballasts, dispose of them as normal demolition debris.

3.3.2 Lighting Lamps

Remove lighting tubes/lamps from the lighting fixture and carefully place (unbroken) into appropriate containers (original transport boxes or equivalent). In the event of a lighting tube/lamp breaking, sweep and place waste in double plastic taped bags and dispose of as universal waste as specified herein.

3.4 STORAGE FOR DISPOSAL

3.4.1 Storage Containers for PCBs

49 CFR 178. Store PCB in containers approved by DOT for PCB.

3.4.2 Storage Containers for lamps

Store mercury containing lamps in appropriate DOT containers. Store and label boxes for transport in accordance with 40 CFR 273.

3.4.3 Labeling of Waste Containers

Label with the following:

a. Date the item was placed in storage and the name of the cognizant activity/building.

b. "Caution Contains PCB," conforming to 40 CFR 761, CFR Subpart C. Affix labels to PCB waste containers.

c. Label mercury-containing lamp waste in accordance with 40 CFR 273. Affix labels to all lighting waste containers.

3.5 DISPOSAL

Dispose of off Government property in accordance with EPA, DOT, and local regulations at a permitted site.

3.5.1 Identification Number

Federal regulations 40 CFR 761, and 40 CFR 263 require that generators, transporters, commercial storers, and disposers of PCB waste posses U.S. EPA identification numbers. Verify that the activity has a U.S. EPA generator identification number for use on the Uniform Hazardous Waste manifest. If not, advise the activity that it must file and obtain an I.D. number with EPA prior to commencement of removal work. For mercury containing lamp removal, Federal regulations 40 CFR 273 require that large quantity handlers of Universal waste (LQHUW) must provide notification of universal waste management to the appropriate EPA Region (or state director in authorized states), obtain an EPA identification number, and retain for three years records of off-site shipments of universal waste. Verify that the activity has a U.S. EPA generator identification number for use on the Universal Waste manifest. If not, advise the activity that it must file and obtain an I.D. number with EPA prior to commencement of removal work.

3.5.2 Transporter Certification

Comply with disposal and transportation requirements outlined in 40 CFR 761 and 40 CFR 263. Before transporting the PCB waste, sign and date the manifest acknowledging acceptance of the PCB waste from the Government. Return a signed copy to the Government before leaving the job site. Ensure that the manifest accompanies the PCB waste at all times. Submit transporter certification of notification to EPA of their PCB waste activities (EPA Form 7710-53).

3.5.2.1 Certificate of Disposal and/or Recycling

40 CFR 761. Include in the certificate for the PCBs and PCB items disposed:

a. The identity of the disposal and or recycling facility, by name, address, and EPA identification number.

b. The identity of the PCB waste affected by the Certificate of Disposal including reference to the manifest number for the shipment.

c. A statement certifying the fact of disposal and or recycling of the identified PCB waste, including the date(s) of disposal, and identifying the disposal process used.

d. A certification as defined in 40 CFR 761.

3.5.3 Disposal by the Government

Comply with disposal and transportation requirements outlined in 40 CFR 761 and 40 CFR 263. Coordinate delivery of PCBs on-site with local Environmental for subsequent disposal on Defense Logistics Agency Disposition Services (DLA DS) contracts. If the primary site is filled to capacity, contact the Contracting Officer. The transport distance to any storage site will not exceed the distance between the project site and the DLA DS storage site.

3.5.3.1 DD Form 1348-1

Prepare DD Form 1348-1 Turn-in Document (TID), which will accompany the PCB to the storage site. Ensure that a responsible person from the activity that owns the PCB signs the DD Form 1348-1.

-- End of Section --

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SECTION 02 84 33.00 22
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REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCBs)(PWD ME)
11/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.145	Specifications for Accident Prevention Signs and Tags
29 CFR 1910.1000	Air Contaminants
40 CFR 761	Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions
49 CFR 171	General Information, Regulations, and Definitions
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
49 CFR 174	Carriage by Rail
49 CFR 175	Carriage by Aircraft
49 CFR 176	Carriage by Vessel
49 CFR 177	Carriage by Public Highway
49 CFR 178	Specifications for Packagings
49 CFR 179	Specifications for Tank Cars

1.2 REQUIREMENTS

The work includes the removal and disposal of PCB containing materials as indicated . Perform work in accordance with 40 CFR 761 and the requirements specified herein.

1.3 DEFINITIONS

1.3.1 Leak

Leak or leaking means any instance in which a PCB Article, PCB Container, or PCB Equipment has any PCBs on any portion of its external surface.

1.3.2 PCBs

PCBs as used in this specification mean the same as PCBs, PCB Article, PCB Article Container, PCB Container, PCB Equipment, PCB Item, PCB Transformer, PCB-Contaminated Electrical Equipment, as defined in 40 CFR 761, Section 3, Definitions.

1.3.3 Spill

Spill means both intentional and unintentional spills, leaks, and other uncontrolled discharges when the release results in any quantity of PCBs running off or about to run off the external surface of the equipment or other PCB source, as well as the contamination resulting from those releases.

1.4 QUALITY ASSURANCE

1.4.1 Training

Instruct employees on the dangers of PCB exposure, on respirator use, decontamination, and applicable OSHA and EPA regulations.

1.4.2 Certified Industrial Hygienist (CIH)

Obtain the services of an industrial hygienist certified by the American Board of Industrial Hygiene to certify training, review and approve the PCB removal plan, including determination of the need for personnel protective equipment (PPE) in performing PCB removal work.

1.4.3 Regulation Documents

Maintain at all times one copy each at the office and one copy each in view at the job site 29 CFR 1910.1000, 40 CFR 761, and Contractor work practices for removal, storage and disposal of PCBs.

1.4.4 Surveillance Personnel

Surveillance personnel may enter PCB control areas for brief periods of time provided they wear disposable polyethylene gloves and disposal polyethylene foot covers, as a minimum. Additional protective equipment may be required if respiratory hazard is involved or if skin contact with PCB is involved.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

PCB Removal Work Plan; G

PCB Disposal Plan; G

SD-07 Certificates

Training certification

Qualifications of CIH; G

Notification

Transporter Certification; G of notification to EPA of their PCB waste activities and EPA ID numbers

Certification of Decontamination for PCB Spill

Post Cleanup Sampling Data; G

Certificate of Disposal

1.6 EQUIPMENT

1.6.1 Special Clothing

Work clothes consist of PPE as required by OSHA regulations, including, but not limited to the following:

a. Disposable coveralls

b. Gloves (Disposable rubber gloves may be worn under these)

c. Disposable foot covers (polyethylene)

d. Chemical safety goggles

e. Half mask cartridge respirator.

1.6.2 Special Clothing for Government Personnel

Provide PPE specified in paragraph SPECIAL CLOTHING to the Contracting Officer as required for inspection of the work.

1.6.3 PCB Spill Kit

Assemble a spill kit to include the following items:

ITEM	MINIMUM QUANTITY
1. Disposable gloves (polyethylene)	6 prs
2. Gloves with a high degree of impermeability to PCB	6 prs
3. Disposable coveralls with permeation resistance to PCB	4 ea
4. Chemical safety goggles	2 ea

5. Disposable foot covers (polyethylene)	6 prs
6. PCB Caution Sign: "PCB Spill--Authorized Personnel Only"	2 ea
7. Banner guard or equivalent banner material	100 feet
8. Absorbent material	5 bags
9. Blue polyethylene waste bags	5 ea
10. Cloth backed tape	1 roll
11. Area access logs, blank	10 ea
12. Brattice cloth, 6' x 6'	1 piece
13. Rags	20 ea
14. Ball point pens	20 ea
15. Herculite, 4' x 4' and 8' x 8'	2 ea and 1 ea
16. Blank metal signs and grease pencils	2 ea
17. Waste containers 55 gallon drum, may be used as container for kit	1 ea

1.7 QUALITY ASSURANCE

1.7.1 Training Certification

Submit certificates, prior to the start of work but after the main abatement submittals, signed and dated by the CIH and by each employee stating that the employee has received training. Organize certificates by individual worker, not grouped by type of certificates.

1.7.2 Qualifications of CIH

Submit the name, address, and telephone number of the Industrial Hygienist selected to perform the duties in paragraph CERTIFIED INDUSTRIAL HYGIENIST. Submit proper documentation that the Industrial Hygienist is certified, including certification number and date of certification/recertification.

1.7.3 PCB Removal Work Plan

Submit a detailed job-specific plan of the work procedures to be used in the removal of PCB-containing materials, not to be combined with other hazardous abatement plans. Provide a Table of Contents for each abatement submittal which follows the sequence of requirements in the contract. Include in the plan a sketch showing the location, size, and details of PCB control areas. Include in the plan, eating, drinking, smoking and restroom procedures, interface of trades, sequencing of PCB related work, PCB disposal plan, respirators, protective equipment, and a detailed description of the method of containment of the operation to ensure that PCB contamination is not spread or carried outside of the control area. Obtain approval of the plan prior to the start of PCB removal work.

1.7.4 PCB Disposal Plan

Submit a PCB Disposal Plan within 45 calendar days after award of contract for Contracting Officer's approval. Comply with applicable requirements of Federal, State, and local PCB waste regulations and address:

a. Identification of PCB wastes associated with the work.

b. Estimated quantities of wastes to be generated and disposed of.

c. Names and qualifications of each contractor that will be transporting, storing, treating, and disposing of the wastes. Include the facility location and a 24-hour point of contact. Furnish two copies of EPA, State, and local PCB waste permit applications, permits, and EPA Identification numbers.

d. Names and qualifications (experience and training) of personnel who will be working on-site with PCB wastes.

e. List of waste handling equipment to be used in performing the work, to include cleaning, volume reduction, and transport equipment.

f. Spill prevention, containment, and cleanup contingency measures to be implemented.

g. Work plan and schedule for PCB waste containment, removal and disposal. Clean up and containerize wastes daily.

1.7.5 Notification

Notify the Contracting Officer 20 days prior to the start of PCB removal work.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 PROTECTION

3.1.1 PCB Control Area

Isolate PCB control area by physical boundaries to prevent unauthorized entry of personnel. Food, drink and smoking materials are not permitted in areas where PCBs are handled or PCB items are stored.

3.1.2 Personnel Protection

Workers must wear and use PPE, as recommended by the Industrial Hygienist, upon entering a PCB control area. If PPE is not required per the CIH, specify in the PCB removal work plan.

3.1.3 Footwear

Work footwear remains inside work area until completion of the job.

3.1.4 Permissible Exposure Limits (PEL)

PEL for PCBs is 3.1 E-08 lb/cubic foot on an 8-hour time weighted average basis.

3.1.5 Special Hazards

a. Do not expose PCBs to open flames or other high temperature sources since toxic decomposition by-products may be produced.

b. Do not heat PCBs to temperatures of 135 degrees F or higher without Contracting Officer's concurrence.

3.1.6 PCB Caution Label

40 CFR 761, Subpart C. Affix labels to PCB waste containers and other PCB-contaminated items. Provide label with sufficient print size to be clearly legible, with bold print on a contrasting background, displaying the following: CAUTION: Contains PCBs (Polychlorinated Biphenyls).

3.1.7 PCB Caution Sign

29 CFR 1910.145. Provide signs at approaches to PCB control areas. Locate signs at such a distance that personnel may read the sign and take the necessary precautions before entering the area.

3.2 WORK PROCEDURE

Furnish labor, materials, services, and equipment necessary for the complete removal of PCBs located at the site as indicated or specified in accordance with local, State, or Federal regulations. Package and mark PCB as required by EPA and DOT regulations and dispose of off Government property in accordance with EPA, DOT, and local regulations at a permitted site.

3.2.1 No Smoking

Smoking is not permitted within 50 feet of the PCB control area. Provide "No Smoking" signs as directed by the Contracting Officer.

3.2.2 Work Operations

Ensure that work operations or processes involving PCB or PCB-contaminated materials are conducted in accordance with 40 CFR 761 and the applicable requirements of this section, including but not limited to:

a. Obtaining advance approval of PCB storage sites.

b. Notifying Contracting Officer prior to commencing the operation.

c. Reporting leaks and spills to the Contracting Officer.

d. Cleaning up spills.

e. Maintaining an access log of employees working in a PCB control area and providing a copy to the Contracting Officer upon completion of the operation.

f. Inspecting PCB and PCB-contaminated items and waste containers for leaks and forwarding copies of inspection reports to the Contracting Officer.

g. Maintaining a spill kit as specified in paragraph PCB SPILL KIT.

h. Maintaining inspection, inventory and spill records.

3.3 PCB TRANSFORMERS

3.3.1 Draining of Transformer Liquid

Perform work in accordance with 49 CFR 171, 49 CFR 172, 49 CFR 173, 49 CFR 174, 49 CFR 175, 49 CFR 176, 49 CFR 177, 49 CFR 178, and 49 CFR 179, Subchapter C and as specified herein. Drain the transformer, switches, and regulators of free flowing liquid prior to transportation. Place the drained liquids in DoT Spec drums. The drums must not contain more than 50 gallons of oil. If the equipment cannot be drained, then place it in DoT Spec drums. DoT drum specifications must meet UN 1A1/Y1.6 (steel drum) or UN 1H1/Y1.6 (plastic drum).

3.3.2 Markings

Provide drums and drained PCB-contaminated electrical equipment with caution label markings as specified in paragraph PCB CAUTION LABEL.

3.3.3 Laboratory Analysis

Provide all transformers with a laboratory analysis for turn-in. DLA DS prefers a gas chromatograph test. The only two exceptions to this rule are:

a. The transformer is hermetically sealed (solder sealed or fusion sealed. No access ports or openings).

b. The name plate states that the transformer contains Pyranol, Interteen, etc.

Attach a copy of the lab analysis to both the DD 1348-1 and the transformer itself.

3.3.4 Markings

3.3.4.1 Transformers, Less Than 50 ppm

Add absorbent material to absorb residue oil remaining after draining. Write the date drained on the transformer. Turn in transformers to DLA DS.

3.3.4.2 Transformers, 50-499 ppm

Same procedure as transformers in the less than 50 ppm range.

3.3.4.3 Transformers, Greater Than 500 ppm

Stencil date drained on the transformer. Turn in transformer to DLA DS.

3.3.4.4 Drums

Stencil on DOT-approved 55 gallon drums containing PCB liquid the following:

a. ppm

b. Date drum filled

c. Serial number of transformer liquid came from

d. National Stock Number

(1) "9999-00-OIL" for <50 ppm

(2) "9999-00-CONPCB" for 50-499 ppm

(3) "9999-00-PCBOIL" for >500 ppm

Do not mix different ppms in the same drum. Drums must have a 2 inch ullage space from the top of the drum.

3.4 PCB REMOVAL

Select PCB removal procedure to minimize contamination of work areas with PCB or other PCB-contaminated debris/waste. Handle PCBs such that no skin contact occurs. PCB removal process should be described in the work plan.

3.4.1 Confined Spaces

As feasible, do not carry out PCB handling operations in confined spaces. A confined space means a space having limited means of egress and inadequate cross ventilation.

3.4.2 Control Area

Establish a PCB control area around the PCB item as specified in paragraph PCB CONTROL AREA. Only personnel briefed on the elements in the paragraph TRAINING and on the handling precautions are allowed into the area.

3.4.3 Exhaust Ventilation

If used, discharge exhaust ventilation for PCB operations to the outside and away from personnel.

3.4.4 Temperatures

As feasible, handle PCBs at ambient temperatures and not at elevated temperatures.

3.4.5 Solvent Cleaning

Clean contaminated tools, containers, etc., after use by rinsing three times with an appropriate solvent or by wiping down three times with a solvent wetted rag. Suggested solvents are Stoddard solvent or hexane.

3.4.6 Drip Pans

Drip pans are required under portable PCB transformers and rectifiers in use or stored for use. Provide the pans with a containment volume of at least one and one-half times the internal volume of PCBs in the item.

3.4.7 Evacuation Procedures

Provide written procedures for evacuation of injured workers. Do not delay aid for a seriously injured worker for reasons of decontamination.

3.5 PCB SPILL CLEANUP REQUIREMENTS

3.5.1 PCB Spills

Immediately report to the Contracting Officer any PCB spills on the ground or in the water, PCB spills in drip pans, or PCB leaks.

3.5.2 PCB Spill Control Area

Rope off an area around the edges of a PCB leak or spill and post a "PCB Spill Authorized Personnel Only" caution sign. Immediately transfer leaking items to a drip pan or other container.

3.5.3 PCB Spill Cleanup

40 CFR 761, Subpart G. Initiate cleanup of spills as soon as possible, but no later than 48 hours of its discovery. To clean up spills, personnel must wear the PPE prescribed in paragraph SPECIAL CLOTHING of this section. If misting, elevated temperatures or open flames are present, or if the spill is situated in a confined space, notify the Contracting Officer. Mop up the liquid with rags or other conventional absorbent. Properly contain and dispose of spent absorbent as solid PCB waste.

3.5.4 Records and Certification

Document the cleanup with records of decontamination in accordance with 40 CFR 761, Section 125, Requirements for PCB Spill Cleanup. Provide certification of decontamination.

3.5.5 Sampling Requirements

Perform post cleanup sampling as required by 40 CFR 761, Section 130, Sampling Requirements. Do not remove boundaries of the PCB control area until site is determined satisfactorily clean by the Contracting Officer.

3.6 STORAGE FOR DISPOSAL

3.6.1 Storage Containers for PCBs

49 CFR 178. Store liquid PCBs in Department of Transportation (DOT) Specification 17E containers. Store nonliquid PCB mixtures, articles, or equipment in DOT Specification 5, 5B, or 17C containers with removable heads.

3.6.2 Waste Containers

Label with the following:

a. "Solid (or Liquid) Waste Polychlorinated Biphenyls"

b. The PCB Caution Label, paragraph PCB CAUTION LABEL

c. The date the item was placed in storage and the name of the cognizant activity/building.

3.6.3 PCB Articles and PCB-Contaminated Items

Label with items b. through c. above.

3.6.4 Approval of Storage Site

Obtain in advance Contracting Officer approval using the following criteria without exception.

a. Adequate roof and walls to prevent rainwater from reaching the stored PCBs.

b. An adequate floor which has continuous curbing with a minimum 6 inch high curb. Such floor and curbing must provide a containment volume equal to at least two times the internal volume of the largest PCB article or PCB container stored therein or 25 percent of the total internal volume of all PCB equipment or containers stored therein, whichever is greater.

c. No drain valves, floor drains, expansion joints, sewer lines, or other openings that would permit liquids to flow from the curbed area.

d. Floors and curbing constructed of continuous smooth and impervious materials such as portland cement, concrete or steel to prevent or minimize penetrations of PCBs.

e. Not located at a site which is below the 100-year flood water elevation.

f. Post each storage site with the PCB Caution Sign, paragraph PCB CAUTION SIGN.

3.7 CLEANUP

Maintain surfaces of the PCB control area free of accumulations of PCBs. Restrict the spread of dust and debris; keep waste from being distributed over work area.

Do not remove the PCB control area and warning signs prior to the Contracting Officer's approval. Reclean areas showing residual PCBs.

3.8 DISPOSAL

Comply with disposal requirements and procedures outlined in 40 CFR 761. Do not accept PCB waste unless it is accompanied by a manifest signed by the Government. Before transporting the PCB waste, sign and date the manifest acknowledging acceptance of the PCB waste from the Government. Return a signed copy to the Government before leaving the job site. Ensure that the manifest accompanies the PCB waste at all times. Submit transporter certification of notification to EPA of their PCB waste activities.

3.8.1 Certificate of Disposal

40 CFR 761. Submit to the Government within 30 days of the date that the disposal of the PCB waste identified on the manifest was completed. Include in the certificate for the PCBs and PCB items disposed:

a. The identity of the disposal facility, by name, address, and EPA identification number.

b. The identity of the PCB waste affected by the Certificate of Disposal including reference to the manifest number for the shipment.

c. A statement certifying the fact of disposal of the identified PCB waste, including the date(s) of disposal, and identifying the disposal process used.

d. A certification as defined in 40 CFR 761, Section 3.

3.8.1.1 Payment Upon Furnishing Certificate of Disposal of PCBs

Payment will not be made until the certificate of disposal has been furnished to the Contracting Officer.

3.8.2 Disposal of PCB Waste

3.8.2.1 Procedure for Disposal

Coordinate all waste disposal with the Contracting Officer and the Portsmouth Naval Shipyard Hazardous Waste Facility (Code 106, Building 357). Collect PCB waste and other items contaminated with PCBs and place in Government provided containers and deliver to Building 357 at the end of each day waste is produced as directed and approved by the Contracting Officer and Code 106 personnel. Affix a warning and Department of Transportation (DOT) label to each container with the approved warnings and preprinted DOT labeling. Clearly indicate on the outside of each container the name of the waste generator and the location at which the waste was generated. Prevent contamination of the transport vehicle (especially if the transport vehicle is a rented truck likely to be used in the future for non-asbestos purposes). These precautions include lining the vehicle cargo area with plastic sheeting (similar to work area enclosure) and thorough cleaning of the cargo area after transport and unloading of PCB waste is complete. The Contractor must carefully remove material and immediately place it in approved containers. If the containers are not pre-marked, a Contractor furnished caution label must be permanently attached to the containers. The Contractor must also furnish and affix a label permanently attached to the containers which provides the following information:

* Abatement Contractor Name
* Construction Contract Number
* Building Number
* Portsmouth Naval Shipyard, Kittery, Maine
* Date

Contractor must ensure all PCB waste is properly packaged prior to placement into Government-furnished containers.

3.8.2.2 PCB Disposal Quantity Report

Direct the PQP to record and report, to the Contracting Officer and Code 106 personnel, the amount of PCB waste removed and released for disposal. Deliver the report for the previous day at the beginning of each day shift with amounts of material removed during the previous day.

Allow the GC to inspect, record and report the amount of PCB waste removed and released for disposal on a daily basis.

3.8.2.3 DD Form 1348-1

Prepare DD Form 1348-1 Turn-in Document (TID), which will accompany the PCBs to the storage site. Ensure that a responsible person from the activity that owns the PCBs signs the DD Form 1348-1.

3.8.2.4 Payment Upon Furnishing DD Form 1348-1

Payment will not be made until a completed DD Form 1348-1 has been furnished to the Contracting Officer.

-- End of Section --

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